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TRANSFORMATIONAL GRAMMAR

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Contents

1 *The Subject* 1

1.1 *Linguistics and learning theory* 2

1.2 *The evidential basis of syntactic theory* 8

1.3 *Where is syntax?* 18

2 *Phrase Structure* 21

2.1 *Substitution Classes* 24

2.2 *Phrases* 31

2.3 *Recursive Phrases* 44

2.4 *Arguments and Modifiers* 63

2.5 *Deriving Phrase Structure Rules* 73

3 *Positioning Arguments* 81

3.1 *Small Clauses and the Derived Subjects Hypothesis* 95

3.2 *The EPP* 104

3.3 *Argument Movement* 107

3.3.1 *Passive* 107

3.3.2 *Raising* 111

3.4 *Quantifier Float* 112

3.5 *Infinitives* 121

3.5.1 *The CP/IP status of infinitives* 124

3.5.2 *On the Control relation* 129

3.5.3 *Argument movement and* PRO 134

3.5.4 PRO *and Control: A Summary* 148

3.6 The Theta Criterion Reconsidered 150
3.7 Theta-roles and Case 161
3.7.1 Unaccusatives 167
3.7.2 Burzio's Generalization 178

4 Features 199
4.1 The Classic account of Verbal Inflection 199
4.2 Movement to C° 211
4.3 Verb Second word order 215
4.4 The Pollockian revolution 226
4.5 AGREE 239
4.6 AGREE *and θ-roles* 256
4.7 AGREE *and Case* 262
4.7.1 Low Subjects: English 268
4.7.2 Low Subjects: Subject Inversion Languages 270
4.7.3 Case and Agreement 277
4.7.4 A locality condition on Agreement 281
4.8 Summary 285

5 Complement Structure 287
5.1 Accusatives Move 287
5.2 Split VP 301
5.2.1 Hidden states and "again" 302
5.2.2 Double Objects 315
5.3 Argument Structure of Nominalizations 333
5.3.1 How the Theta Criterion applies to nominals 333
5.3.2 A gap in clausal complementation 349

6 Theories of Movement 359
6.1 Trace Theory 363
6.1.1 Deep and Surface anaphors 363

6.1.2 Sloppy Anaphora 371

6.1.3 Antecedent Contained Anaphora 379

6.1.3.1 Traces, Referential Indices, and Quantifiers 385

6.1.3.2 Relative Clauses and Wh Movement 403

6.1.3.3 Antecedent Contained Anaphora Resolved 413

6.1.4 The Syntax of Quantificational DPs 421

6.2 Copy Theory 434

6.2.1 Reconstruction 434

6.2.1.1 Quantifier Lowering 435

6.2.1.2 Head Movement and Derivations: a problem 443

6.2.1.3 Reconstructed Variables 451

6.2.1.4 Achieving reconstruction with traces 456

6.2.2 Copies 476

6.2.2.1 Copies and Reconstruction 480

6.2.2.2 Pied-Piping 491

6.2.2.3 Problem: Antecedent Contained Anaphora 497

6.3 Summary 510

7 Bibliography 513

Preface

These are the always evolving notes from an introductory course on syntactic theory taught at the University of Massachusetts at Amherst. Its target audience is first-year graduate students. No background knowledge of syntax is presupposed.

1
The Subject

Syntactic theory is charged with the duty of modeling a certain aspect of an individual's knowledge of their language. This enterprise has been heavily influenced by the general framework of linguistic theory that Noam Chomsky developed in the last half of the twentieth century. It has also been heavily influenced by Noam Chomsky's specific analyses of syntactic phenomena. These lecture notes are an attempt to introduce some of Chomsky's ideas about syntax as well as the many developments of those ideas in the contemporary literature on syntactic theory.

My aim is to introduce the reader to a handful of syntactic phenomena and the models of these phenomena that have a certain general currency. I have largely ignored differences in how these phenomena have been approached by theorists, and so I have also ignored the important process of evaluating competing analyses. In addition, I have left out many of the phenomena present syntactic theory engages. Some of the areas that drive a lot of the literature right now—agreement, ellipsis, anaphora, island phenomena—are almost completely ignored. Moreover, I have chosen topics that I feel I have enough mastery of to present. These notes are consequently a slim slice of what contemporary syntax includes. My hope is that this slice is nonetheless sufficiently large to illustrate an image of syntax that many syntacticians have, and to reveal the reasons for having that image. I have tried to aim for explicitness and consistency throughout, and this means that I have occasionally made up ways to connect results from different literatures. Not only is the material in these notes woefully incomplete, then, it is also burdened with parochial, idiosyncratic, theory. This is not a complete list of the shortcomings and flaws herein, but I hope that what follows will furnish the reader with a good starting point for further study nonetheless. The model I will sketch here is wrong, but hopefully it will be wrong in ways that are useful to figure out.

In this first chapter, I will sketch a setting for syntactic theory and discuss some of the methodological problems that face the syntactician. The first scene-setting decision is that syntactic theory is part of the enterprise of cognitive science. We are studying a facet of an individual's cognition. Syntax,

as it is conceived of here, is not charged with studying group behavior. This is not a branch of sociology or anthropology. This places certain constraints on the enterprise. Syntax must be a form of knowledge or ability, broadly defined. It must be the kind of thing that human cognition is. That means it must guide behavior and interact with thought. It must emerge from the way the human brain works.

It must also be an ability or knowledge that can be learned. The developmental sequence that we go through as children confers upon us the ability to master one or more syntaxes, and disables that ability for all the rest. One of Noam Chomsky's contributions to linguistic theory has been to emphasize this shaping role on the models linguists come up with. The knowledge an individual has about their language—the abilities that enable them to use a language—must be acquirable by any normal infant exposed to speakers of the language over seven or so years. A number of considerations combine to make the task of acquiring this knowledge look very difficult indeed: the complexity of the acquired grammar, the amount of information that needs to be acquired, and the attenuated nature of the information available to the child. It is made even more puzzling by the fact that children appear to complete this task with relative ease in a comparatively short period of time and that the course of acquisition appears to go through a set schedule of stages. If languages are as complex as we think they are, then how can these impossibly complex objects possibly be learned? The parts of our cognition that encapsulate our linguistic knowledge and ability will have properties that are consistent with their learnability. They will hold clues to the learnability puzzle. This fact can help guide our inquiry.

1.1 Linguistics and learning theory

Chomsky proposed that models of syntax should contribute to solving the learnability problem. The classical formulation of this idea (see *Aspects* and *The Sound Pattern of English*) characterizes the situation as follows. Think of a grammar of L (G_L) (this is what Chomsky (1986b) calls "I-Language") as a set of rules that generates structural descriptions of the strings of the language L (Chomsky (1986b)'s E-language). Our model of this grammar is descriptively adequate if it assigns the same structural descriptions to the strings of L that G_L does. We can think of the learning process as involving a selection from the Universe of G_Ls the very one that generates these structured strings of L.

The learning problem can now be stated in the following terms: how is it that the learning procedure is able to find G_L when the universe of Gs is so huge and the evidence steering the device so meager.

One step towards solving this problem would be to hypothesize that the universe of Gs has a structure that enables convergence on G_L given the sort of information that the child is exposed to. This is Chomsky's proposal. It amounts to the claim that there are features of Gs which are built-in: certain properties which distinguish the natural class of Gs from the rest. There is a kind of meta-grammar of the Gs, then, which is sometimes referred to with the label Universal Grammar. Chomsky further hypothesizes that these properties are biologically given: that it is something about the construction of the human brain/mind that is responsible for the fact that the class of Gs are the way they are. This argument, the one that leads from the observation that G_Ls have features that are too complex to be learned by an unstructured system to the conclusion that the universe of Gs is constrained is often called "The Poverty of the Stimulus" argument. It is a classic from Epistemology, imported with specific force by Chomsky into linguistics.

Chomsky proposes, then, that a goal of syntactic theory should be to contribute towards structuring the universe of Gs. He makes some specific proposals about how to envision this in *Aspects of The Theory of Syntax*. He suggests that syntactic theory should include an evaluation metric which "ranks" Gs. A syntactic theory that has this feature he calls explanatory. Thus "explanatory theory" has a specific, technical, sense in linguistic theory. A theory is explanatory if and only if it encapsulates the features that rank Gs in such a way that it contributes to the learnability problem, distinguishing the learnable Gs from the unlearnable ones. This criterion can help the syntactician decide whether the model of G_L they've proposed corresponds exactly to G_L. In particular, the many descriptively adequate models of G_L can be distinguished on this basis: we should select only those that are ranked highly by the evaluation metric. These grammars meet the criterion of explanatory adequacy.

A very important role, therefore, is played by the evaluation metric. At the time of *Aspects*, the learning procedure was conceived of as a process very much like that which the linguist goes through. The child builds a battery of rules which generate the strings of L. The evaluation metric steering this process was thought to have essentially two parts: a simplicity metric, which guides the procedure in its search through the space of grammars, and inviolable constraints, which partitions the set of Gs into the learnable ones and the unlearnable ones. Thus, for example, we might imagine that rules which used fewer symbols could be defined as "simpler" than ones that used a greater number of symbols. Inviolable constraints might be those, for example, expressed as part of the principles which place constraints on the way that strings can be partitioned into groups, and therefore simply removes from the universe of Gs a great many possible Gs. Let's call these models of Gs "rule based," because the simplicity metric is defined as a procedure that constructs rules, and the companion picture of the acquisition process the "Little Linguist" model.

Let's take a concrete example. We will see that certain substrings in a sentence have a special status. They make units, called phrases, that have certain properties. These phrases are the building blocks of sentence structure and most of what we will investigate here are how those building blocks are assembled. When a child learns a language, one of the things that needs to be learned is where the phrases of that language are. It appears that the criteria that are relevant for defining phrases are quite limited. So far as is presently known, phrases, in all of the world's languages, are defined by virtue of what the phrasal substrings are adjacent to in the larger sentence. The general recipe seems to be something of the sort in (1).

(1) For a string σ to be a phrase, it must:
   a. be contiguous, and
   b. be identifiable by looking at what is adjacent to σ.

(We will see that (1) is not quite sufficient, but what replaces it continues to make reference to adjacent positions.) What (1) says is that for a child to determine where the phrases of her language are, she needs to look just at what the contiguous substrings of the sentences in her language can be surrounded by. This is not a logically necessary way in which phrases could be identified, of course. It is entirely possible for phrases to be defined by the criteria in (2) or (3).

(2) For a string σ to be a phrase, it must:
   a. be contiguous,
   b. start with n words followed by at most one phrase, and
   c. be identifiable by determining n.

(3) For a string σ to be a phrase, it must:
   a. be contiguous,
   b. end with n words preceded by at most one phrase, and
   c. be identifiable by determining n.

A language that built phrases according to the recipe in (2), would have sentences with the structure in (4a), when n is 1 and (4b), when n is 2.

(4) a. [w [w [w [w]]]]
   b. [w w [w w [w w]]]

   phrases are included in "[]," and "w" represents a word.

And a language that built phrases according to the recipe in (3) would produce sentences with the phrasing in (5a), when n is 1, and (5b), when n is 2.

(5) a. [[[[[w] w] w] w] w]
   b. [[[w w] w w] w w]

Both (2) and (3) are simple recipes, and are plausibly easier to use in acquiring the phrase structure of a sentence. A candidate inviolable constraint is that phrases are defined by a recipe along the lines of (1), and not by any number of other possible recipes. The fact that (1), and not (2) or (3), seems to be true of the languages of the world would be explained if the acquisition process used only (1) in discovering phrase structure. It would be explained if the space of possible grammars could never fail to use (1).

While distinguishing an evaluation metric, which ranks the grammars, and inviolable constraints, which delimits the universe of possible grammars, is useful for a theoretician in evaluating the facts of languages, there is no necessary difference between them. To see this, let's sketch out how the acquisition process might be modeled. A child starts out in a state of relative ignorance about the language she is to acquire, and upon listening to a certain number of sentences of that language, she makes a guess as to its grammar. We can diagram that process with (6).

(6) $\boxed{G_0} \longrightarrow \boxed{G_1}$

In the state that has her first grammar, G_1, we can imagine the acquisition device repeating the process that got her to G_1 from her initial state, G_0. She tests the predictions of G_1 against the sentences she is hearing, and when she discovers that the sentences she hears do not fulfill the predictions of G_1, she will modify G_1 to G_2, so that they do.

(7) $\boxed{G_0} \longrightarrow \boxed{G_1} \longrightarrow \boxed{G_2}$

This process repeats itself until she finds herself in a state where the grammar she is testing makes predictions that do not differ from the sentences she is hearing.

(8) $\boxed{G_0} \longrightarrow \boxed{G_1} \longrightarrow \boxed{G_2} \longrightarrow \ldots \boxed{G_n}$

While this final grammar, G_n, might not entirely match the grammars of the speakers around her, it will be undetectably different from them and the learner will remain in G_n. The grammars of the world's languages are the set of G_ns that reside in the minds of each language speaker. By this way of counting, there are at least as many grammars (=languages) as there are language speakers. The poverty of the stimulus argument amounts to the observation that for convergence to a G_n to be guaranteed, the possible paths from G_0 to G_n must be finite. For convergence to be guaranteed in the short period of time it seems to happen, and with the limited amount of information available to a child about G_n, the possible paths from G_0 to G_n are plausibly small in number.

If the initial state is the same for every language acquirer, as seems plausible, the evaluation metric is a model of each of the "$\longrightarrow$"s in (8). These rank the Gs in the acquisition sequence, making some of them "easier" to get to from the initial state. We might define inviolable constraints as those that

characterize the grammars that the sequence in (8) never converges to. These are grammars that are either not in the series of Gs, or are Gs the acquisition process cannot stay in, or are Gs so far into the series that there is an earlier G out of which the acquisition process will never move. An evaluation metric, in other words, also provides a model of inviolable constraints.

One of the goals of syntax is to discover the properties of grammars that reflect the shaping role that the acquisition process has. As we model the grammars of a speaker's syntax, we should stay alert to the ways in which it is constrained. These constraints, some of them at least, will be due to the ways in which the $\longrightarrow$s function. Properties of a syntax that do not seem likely to be acquired in the acquisition process are candidates for such a constraint. (1) is such a property, given its abstract, arbitrary, nature.

There is another kind of criterion that can be used to identify constraints on the acquisition of a syntax. This one comes to us through the narrowly linguistic study of language typology, and only tangentially from learnability considerations. I will call it "Humboldt's argument," though it no doubt has an earlier champion. Humboldt's argument is based on the observation that there are certain properties that appear to hold true of all G_Ls. This can be explained, Humboldt argues, only if the universe of Gs is constrained to just those which have the relevant, universal, properties. Like Chomsky, Humboldt relates this to the construction of the mind, and uses the language of learnability in his account. He puts it this way:

> Since the natural inclination to language is universal to man, and since all men must carry the key to the understanding of all languages in their minds, it follows automatically that the form of all languages must be fundamentally identical and must always achieve a common objective. The variety among languages can lie only in the media and the limits permitted the attainment of the objective.
>
> (von Humboldt 1836)

(One might read the last sentence of this passage as making the distinction, touched on above, between aspects of Universal Grammar ("the media") and the limits our cognition places on exploiting UG ("the limits permitted the attainment of the objective").) So, like Chomsky, he supposes that there is a Universal Grammar, a feature of the mind, which constrains the form that languages may have. But his perspective is different from Chomsky's. He expresses the notion of Universal Grammar not in terms of learning theory, or through the glass of the Poverty of the Stimulus argument, but from the perspective of language variability. He links limits on language variability to a universal ability he sees in human psychology to acquire a language.

Humboldt's goal is an explanation for the observed limits in variability of the grammars of extant languages. One might imagine that there are explanations for these limits that do not involve, as Humboldt proposes, constraints imposed by human psychology. Similarities in extant languages might reflect their common ancestry, for instance. if all languages descend from a common

one, then features that are shared among them could simply be vestiges of the ancestral language that historical change has left untouched. I think it's possible to read Sapir as advancing this alternative. Sapir is commonly associated with the position exactly opposite to Humboldt's; in Sapir's words:

> Speech is a human activity that varies without assignable limit as we pass from social group to social group, because it is a purely historical heritage of the group, the product of long-continued social usage.
>
> (Sapir, 1921, p. 4)

But, perhaps because of his vagueness, it's possible to credit Sapir with a more complex view. One that assigns the universal properties of languages to the limits of historical change:

> For it must be obvious to any one who has thought about the question at all or who has felt something of the spirit of a foreign language that there is such a thing as a basic plan, a certain cut, to each language. …Moreover, the historical study of language has proven to us beyond all doubt that a language changes not only gradually but consistently, that it moves unconsciously from one type towards another, and that analogous trends are observable in remote quarters of the globe.
>
> (Sapir, 1921, pp. 120-121)

Perhaps the common properties of known languages are a function of two facts: all languages descend from a common language, and the forces that cause languages to change are not fully random—they preserve certain features and change others only according to some "basic plan." If historical relatedness is to explain the common traits that extant languages have, some limit must be placed on how languages change and diverge. Otherwise, language change would act as a kind of randomizer that, over time, would destroy the limits in variability that we observe. Monogenesis needs to be coupled, then, with a theory of diachrony that characterizes the limits it imposes on change. Could it be, then, that the similarities in languages are all due to these laws of diachrony?

This seems to me to be a coherent account for language variability. But it may be just a disguised version of the Chomsky/Humboldt hypothesis that the limits of human cognition are responsible for the constraints on linguistic variation. The thesis of monogenesis entails that language variation is solely the product of historical change, as Sapir's quotes makes clear. So we expect that languages vary in features which historical change can affect, but will remain similar in those ways that are immutable. Which of the features appear as language universals, then, is determined by the internal mechanisms of historical change, and the limits thereon. What are the internal mechanisms of historical change? The only proposal I know of is that historical change is a by-product of language acquisition. It is the accumulation of the small mismatches in G_Ls that successive generations of language acquirers select. Language acquisition, the poverty of the stimulus argument tells us, is guided by Universal Grammar. So even granting the diachronic argument for language

universals, we see that as historical change weeds out the mutable properties from the immutable ones, the properties it leaves are those that characterize Universal Grammar. The antidote for the argument I have blamed on Sapir, then, involves bringing the poverty of the stimulus argument into play. I don't know if Humboldt's argument can stand against this alternative unaided.

But even if it can't, it provides us with another way of viewing how to factor out the components of the evaluation metric. Following the logic of Humboldt's argument, what we expect is that language comparison should give us a means of separating inviolable constraints from the evaluation metric. The inviolable constraints will be (among) those things found in all languages; the differences in languages are to be credited to the evaluation metric. The recipe for characterizing phrases in (1), for instance, seems to be inviolable; languages have not been discovered that build their phrases along a different plan. Put somewhat differently, an explanatory theory is to give us both how languages cannot be constructed, and how their construction can vary. The data it must fit, then, emerges only once languages are compared: for not only does this allow the universals to be clearly discerned, but it is only through this means that the particulars of language variation are known.

1.2 The evidential basis of syntactic theory

If linguistics is one part of the study of human cognition, in the sense just described, then syntax can be defined as that subdiscipline of linguistics which seeks to discover what speakers know about how to arrange the words of their language into meaningful sentences. Because speakers are not conscious of the principles that characterize this knowledge, the syntactician must make recourse to indirect means of determining these principles. The syntactician's first task, then, is to determine how to find evidence that reveals the nature of this knowledge.

One plausible source of relevant information comes from observing how speakers put this knowledge to use. We could, for instance, collect the utterances from some speaker and look for generalizations in these utterances from which evidence about the underlying knowledge-base can be gleaned. This is rarely done, however, as there are few instances of such collections that arise naturally, and to assemble them from scratch is onerous enough to have been avoided. With the exception of studies of prodigious literary figures, there are vanishingly few attempts at linguistic studies that go this route.

More common is to study the linguistic utterances of a group of speakers. This is standardly done by using the lexicographer's device of combing texts and newspapers for examples. There are several excellent "parsed" corpora of this sort, and even corpora of spoken utterances can be found. With the advent of the World Wide Web, it has become possible to search a very large collection of sentences, and more and more linguists are availing themselves of this resource. This technique has the unique advantage of allowing one to

See Marcus et al. (1993) and Godfrey et al. (1992), for example.

determine frequencies of certain kinds of sentences as well. It is possible, for example, to judge how rare some particular arrangement of words is relative to some other, or to find statistically significant correlations between, say, the position of an argument relative to its predicate and the person or number marked on that argument. Some linguistic theories are specifically designed to model these sorts of frequency data.

The papers in Bod et al. (2003) are examples of statistically based corpora studies, and the work of Paul Boersma (e.g., Boersma and Hayes 2001) contains a theory that is designed to model statistical data of this sort.

There are some serious pitfalls to using group corpora, however. One is simply that it obliterates differences among speakers and treats the data as if it were all manufactured by the same grammatical system. Since nothing is known about the producers of these sentences—they may include speakers of different dialects and speakers for whom the language in question is non-native or has been influenced by another language, for instance—this could be a serious source of error. Without some measure of the heterogeneity of the speakers who produced the corpus, it is very difficult to judge how faithfully it represents the syntactic knowledge of any one of those speakers.

Another shortcoming is that linguistic behavior, even of one individual, is not a faithful projection of the knowledge that that individual has of his or her language. People say sentences whose syntactic form is at odds with what they would otherwise deem well-formed. A significant proportion of any corpus could be made up of such "mistakes," and indeed it would be prudent to assume so, given the degree to which misshapen sentences populate the utterances of such well-placed contributors to corpora as Donald Trump. There is a distinction between a speaker's linguistic "performance" and his or her linguistic "competence," to use the names Chomsky gives to this distinction. Corpora level this distinction.

For these reasons, then, group corpora contain an unknown amount of data that should be weeded out. They contain examples of sentences that are produced by speakers whose grammatical systems differ, and they contain sentences that are not representative of any grammatical system. But group corpora are not only noisy with error, they are also silent about certain kinds of information.

One important piece of evidence that corpora cannot provide concerns where speakers draw the line between impossible and possible forms in their language. This distinction is easiest to elicit in linguistic domains where there are a comparatively small number of relevant forms. For example, the morphological and phonological inventories of any one speaker at any one time is reasonably small and it is therefore salient when a novel morphological or phonological form is introduced. For many such novel forms, speakers are capable of distinguishing those that are admissible members to their languages and those that are not. Most English speakers I have asked, for instance, can tell that *blick* ([blɪk]) is an admissible addition to their lexicon but that *bnick* ([bnɪk]) is not. Presumably this ability to distinguish admissible from inadmissible forms is due to the knowledge speakers have of their language, and so it is an important piece of information about how that knowledge is con-

stituted. A typical way of characterizing this distinction goes as follows. The phonology of a language permits many forms that are not exploited by the lexicon of that language (e.g., [blɪk]). Which of these forms are used and which are not is completely extra-grammatical. By contrast, because the phonology of a language limits the forms that are available to that language (e.g., English prevents the onset cluster [bn]) these forms (e.g., [bnɪk] in English) will be blocked from its lexicon. The absence of these forms is determined by the grammar; they are said to be "ungrammatical," and when they are cited, they are prefixed with the diacritic "*" to indicate their status.

The same distinction can be elicited for sentences, although because of the larger number of forms involved it is more difficult to recognize a novel sentence. Consider, by way of illustration, the pair of sentences in (9).

(9) a. Whenever the earth revolves around its equator, the moon begins to rotate about its axis.

b. Whenever the earth revolves around its equator, the moon begins itself to rotate about its axis.

I judge (9b) to be an impossible English sentence, and (9a) to be a possible one. Because I don't read enough science fiction, I think it's likely that both sentences are novel for me, but I do not have the certainty about this that I have about *blick* and *bnick*. I recognize that there are considerably more sentences that I have encountered than there are words I've encountered, and consequently I also recognize that it is likelier that I will mistake a sentence as novel than it is that I will mistake a word as novel. Nonetheless, most linguists would agree that the contrast in (9) is of the same kind that distinguishes *blick* from *bnick*. It does seem unlikely that the distinction could be reduced to one of novelty. After all, I am roughly as certain of the novelty of (9a) as I am of the novelty of (9b) and yet this does not affect the strength of my judgement concerning their Englishness. It seems probable that my ability to judge the difference between (9a) and (9b) traces back to an ability my syntactic knowledge gives me to judge well-formedness.

This distinction between grammatical and ungrammatical forms is important because it seems to tap directly into a speaker's linguistic knowledge. Studying corpora cannot provide what is needed to see this distinction; corpora conflate ungrammatical and grammatical but non-occurring forms. For this reason, and because of its noisiness, I will not use data from corpora in these lectures. But do not forget that corpus studies, and so far as I know only corpus studies, can provide statistical data, for this might be an important resource in forming a complete model.

Instead, the central piece of evidence used in these lectures will be elicited grammaticality judgments. The linguist will ask a speaker of the language under investigation to judge whether the string of interest is or is not a well-formed sentence in their grammar. This has become the standard tool for syntactic analysis, and much of the literature relies on it. Elicited grammati-

cality judgments have their own shortcomings, and it is worth spending some time considering them.

One of these problems is that the speaker's report about the string presented to them is not guaranteed to be about the well-formedness of the sentence. There are many ways a string can fail to make a speaker classify it as a sentence, and not all of these are of relevance to our task. Some sentences, for instance, are more difficult to perceive than others and a speaker can report the difficulty of perception as a failure of the string to be a sentence. To see how this might happen, consider the task faced by the cognitive machinery tasked with perceiving a sentence in real time. This machinery— I'll call it a "parser"—must deliver an analysis of the series of words coming into the hearer's ears at speed. Studies have shown that this is done incrementally as the words come in. We also know that the parser builds expectations about what will follow the words it's already computed, and that these guesses are partly what makes the parser able to keep up with the rapid flow of incoming words. There is also evidence that these guesses are based on a limited block of incoming words; there is a limit to the size of the groups of words the parser can work with at any given moment. Let's call the group of words that the parser is using to make its guesses about upcoming material, the "buffer." The picture we have of the process is (10).

Julie Sedivy puts this beautifully: "Leisurely introspection of carefully crafted example sentences in scholarly articles masks the intensely frenetic nature of everyday real-time language processing." (Sedivy, 2007, p. 476)

(10) $w_1\ w_2\ w_3\ w_4\ \boxed{w_5\ w_6\ w_7} \longrightarrow \begin{bmatrix} 1^{st} : w_8\ w_9 \ldots \\ 2^{nd} : w_8\ w_9 \ldots \\ 3^{rd} : w_8\ w_9 \ldots \end{bmatrix}$

The box represents the buffer, and each w_i is a word. Following the "$\longrightarrow$" are a list of possible continuations that the parser has rank ordered, from most probable to least. Those that are at the top of the list will be preprocessed, so that should w_8 meet the expectations of the more probable continuations, it will already have been assigned a position in the resulting parse.

There are well-formed sentences that create problems for the parser because of the way they interact with the architecture in (10). A famous example of this comes from strings that are ambiguous with respect to the placement of some late occurring phrase. When this happens, the buffer's size is too limited to maintain the correct menu of expectations. The pair of sentences in (11) illustrates.

(11) a. I decided to marry on Tuesday.
b. I decided that my daughter should marry on Tuesday.

Most speakers will recognize that (11a) has two meanings. It can assert that the time of my decision to marry was Tuesday, or it can assert that what my decision was, is to marry on Tuesday. As we will see, this ambiguity reflects the fact that (11) maps onto two sentences, whose difference in phrase structure is responsible for the two meanings. The first meaning corresponds to a structure which groups the words as sketched in (12a), whereas the second interpretation corresponds to the syntactic structure shown in (12b).

(12) a. I [decided [to marry] on Tuesday]
b. I [decided [to marry on Tuesday]]

Unlike (11a), (11b) seems to have only the second of these two meanings. It can assert that my decision was for my daughter to marry on Tuesday, but it does not seem to say that the time of my decision was Tuesday. This, at least, is the typical outcome of querying English speakers about the meanings of (11b). A conclusion one might reach, then, is that our syntax should allow the string in (11b) to be grouped into phrases in the way that (**??**a) indicates, but not the way that (13b) indicates. That is, we should assign to the two sentences in (13) the grammaticality outcomes indicated.

(13) a. yes: I [decided [that my daughter should marry on Tuesday]]
b. no: I [decided [that my daughter should marry] on Tuesday]

At present, however, this difference in (11a) and (11b) is thought to be due to constraints of sentence processing, and not the well-formedness conditions of sentences. The relevant difference between these examples is the number of words between the word *decided* and the prepositional phrase *on Tuesday*. As that number grows beyond what can be held in the buffer, the processor is forced to start making decisions about how to parse the initial portions of the string and this reduces the number of possible ways the sentence can continue. When the parser is inputting the words *should marry*, it has effectively removed from expected continuations those that group *on Tuesday* with *decided*. Thus, in the case of (11b) the parser favors the structure in (**??**a) over that in (13b).

On this account, then, it is not that there is a difference in the syntactic well-formedness conditions which causes speakers' differing judgments about (11a) and (11b). Instead, because of the relative difficulty that (11b) presents to the on-line processor, one of the syntactic representations associated with this string (i.e., (**??**a)) becomes difficult to perceive. This effect of the on-line processor is what Kimball called "right association."[1] There is considerable evidence that the judgment speakers deliver on these sentences is influenced by their ease of perception. Changing the distance between the word *decided* and *on Tuesday* so that the buffer's size is honored seems to change speakers' reports, for instance. And so does manipulating other factors about the sentence that control the expectations generated by the parser. Indeed, most speakers can perceive both of the meanings in (11b) with a bit of time. The verdict, then, is our syntax should classify both of the representations in (13) as grammatical.

[1] See Kimball (1973), Frazier (1978) and Gibson (1998).

Let me describe another case where it seems to me plausible that a judgment of ill-formedness might actually arise because of difficulty in perception. While the evidence for a role in misparsing in this case is less clear, the effect is sharper than that in (11b) and so it illustrates better how difficult it is to tell, a priori, whether the badness of a sentence is due to its construction or its difficulty to perceive.

The example I have in mind is found in the contrast in (14).

(14) a. I know the man who John likes and we hope will win.
b. I know a man who Bill saw and likes Mary.
(Williams, 1978, 34)

English speakers report that (14a) is fine but that (14b) is not. The good example in (14a) instantiates what is called "across-the-board movement." This is a syntactic operation that arises in coördinations—here two sentences joined by the conjunct *and*—and allows some material outside the coördination to be shared by each conjunct. In (14a), that material is the relative pronoun *who*.

(15) …who [[John likes] and [we hope will win]]

Relative pronouns have a syntactic and semantic relationship with the clause that follows them that targets a certain position in that clause. In (16), for instance, the relative pronoun *who* is syntactically and semantically related to the position marked <gap>.

(16) The man who [John likes <gap>] (has arrived).

We will study this relationship in some detail in these lectures, but what we need to know now is just that it exists. To put the words together semantically and syntactically requires a way of relating a relative pronoun like *who* with a "gap" in the clause following that relative pronoun. In the "across-the-board movement" construction, there are two gaps that the relative pronoun is related to:

(17) The man who [[John likes <gap>] and [we hope <gap> will win]] (has arrived)

What we learn from (14) is that where those gaps are inside the conjoined clauses matters. The pattern shown in (18) yields the ugliness that is (14b).

(18) The man who [[Bill saw <gap>] and [<gap> likes Mary]]

It's not that the gaps cannot be in these positions when they are connected to relative pronouns normally. Both of the cases in (19) are fine.

(19) a. The man who [Bill saw <gap>]
b. The man who [<gap> likes Mary]

It's something that involves the coördination and accompany across-the-board movement. Many people have tried to find a syntactic condition on across-the-board movement that will account for this contrast. (A recent attempt, and perhaps the best, is in Citko and Gračanin-Yuksek 2020.)

It seems likely to me, however, that the badness of (14b) could be due to the parser. Consider what the parser might be faced with at the point in which the following words have been input:

(20) the man who Bill saw and likes

The parser will be expecting to find a gap because of the presence of *who* in its buffer, and it will have two possibilities open:

(21) a. who [Bill saw <gap> and <gap> likes...] (prediction: future words will tell what is liked.)
b. who [Bill [saw and likes]...] (prediction: future words will include <gap> and will tell what is liked and seen)

If the online processor is, for whatever reason, prone to choose (21b) and reject (21a), then when *Mary* is encountered this will be taken to name the thing liked and seen. If the clause then ends, the expected gap will be missed. This could give the percept of ungrammaticality.

That it is the competition between the two parses in (21) that is relevant to the badness of (14b) is argued by Anderson (1983), who notes that the ungrammaticality of these examples is affected by the likeliness of the competing parse in (21b). Examples like (22) have the same general syntactic shape as (14b) but are much better. (The relative pronouns have been underlined, since they are not always obvious.)

(22) a. Nancy Reagan was wearing a gown <u>that</u> Galanos designed <gap> and <gap> cost over $5000.
b. We went to see a movie <u>which</u> the critics praised <gap> but <gap> was too violent for my taste. Here is a book <u>which</u> I haven't read <gap> but <gap> was recommended by several professors.

(Anderson, 1983, (2): 3)

This would follow from an account of the badness of (14b) that depended on the architecture of the parser. But it is difficult to see how this effect could be explained by what is otherwise known about the syntax of across-the-board movement.

In general, judgments of well-formedness will not be able to distinguish those sentences that do not conform to the constraints of the grammar from those that do conform to those constraints but present problems for the online processor.[2] There is no simple way of distinguishing these cases; they can be separated only through analysis. In the case of (11), the decision that the effect is not grammatical but, instead, the result of the parser comes partly from finding no good grammatical way of distinguishing the cases and partly from finding that manipulating factors relevant for the parser's architecture determines whether the effect materializes. If (14b) turns out to also be due to a parsing effect, then the evidence from (22) was required to see that the relevant factors in its badness are perception based and not grammatical.

One way eliciting grammaticality judgments can go wrong, then, is for the speaker to report the percept of unparsability as ungrammaticality. Another similar difficulty involves the fact that the meanings which sentences convey

[2] Chomsky and Miller (1963) is an early, and still useful, examination of this distinction.

are typically bound to the context of a larger discourse. Let me illustrate this by describing a place where this has played a role in shaping theory. There is a process in English syntax that allows certain strings of words to go unpronounced. An example is (23).

(23) Mary eats nattoo but you shouldn't △.

The sentence following *but* is understood to mean: *you shouldn't eat nattoo*. Part of the meaning of that sentence is found spoken in the material before *but*. One way of modeling this to imagine that the words *eat natto* are indeed part of the sentence that follows *but*, and there is a process that allows them to go unspoken. The name for this process is "ellipsis."

(24) Ellipsis
The pronunciation of the string of words α is optional if they match a string β spoken elsewhere.

A better formulation of Ellipsis would limit what α can be and define "match." This way of formulating Ellipsis requires that there be somewhere in the signal spoken words that match the ones that go unspoken, and this predicts that ellipsis shouldn't be possible if those words are not present. In Hankamer and Sag (1976), this consequence was investigated and their verdict was that it is correct. For instance, they raised the following kind of case.

(25) *context: Sag raises a cleaver and prepares to hack off his left hand.*
   a. Hankamer: *Don't worry, he never actually does △.
   b. Hankamer: Don't worry, he never actually does it.

Miller (2011) did a corpus study of ellipsis and discovered that where it was used seemed to have a pattern that put it in opposition to the use of *do it*. In his analysis of that pattern, he pulled out some descriptions of the contexts where these two forms of anaphora are used, and he suggested that these contexts defined which was used. For example, ellipsis, he claims, is used in places where the sentence with the ellipsis is used to resolve possibilities raised in the sentence from which the spoken material is found. This happens in questions, for instance:

(26) A: Senator Brown, does this bill end too-big-to-fail?
   B: Yes it does △.
   B': *Yes it does it.

(27) A: Can I go now △?
   B: Of course you can △.
   B': *Of course you can do it.

There seems to be a choice between using *do it* and ellipsis when a speaker is trying to make reference to a previously occurring set of words, and this choice is made in connection with a context. Miller argues that the contexts

that choose for ellipsis are not ones that can be set up easily without using linguistic means, and that could be why it looks like ellipsis requires matching linguistic material elsewhere. If one looks carefully for contexts of the sort in which ellipsis is chosen, one can find examples of ellipsis without matching linguistic material:

(28) The aisles at the Lakewood Wal-Mart are surprisingly packed at 11 p.m. 'Can we? Can we?' Vanessa tugs at her mother, pointing to a rack of 'Lady and the Tramp' DVDs. Diaz shrugs. OK.

For this reason, it looks like a formulation of ellipsis should not require matching linguistic material. Instead, it should require something that ensures ellipsis is available only in certain, yet to be defined, contexts. Those contexts generally, though not always, correlate with providing the ellipsis with matching linguistic material.

This, by the way, is an illustration of the sort of thing that studying corpora is good for. We can find patterns of distribution with corpora. And we can simply find examples that we aren't able to think up cold. Corpora studies, then, are an important tool. For English, there are convenient online tools for doing it:

(29) https://www.english-corpora.org

Results like Miller's should encourage the use of corpora in studying syntactic knowledge. But it's important to keep in mind that the theories we will be building are not theories of corpora. They are instead theories about the linguistic knowledge that individual speakers possess. Only to the extent that the speakers who've contributed to the corpora we study have similar syntaxes will we be able to use data from corpora to reach conclusions about those syntaxes.

The problems posed by the contingencies of the real-time perception of sentences, and the function sentences have in a discourse, are similar. In both, the difficulty is in distinguishing judgments of well-formedness from other types of ill-formedness. The effect of these difficulties can be lessened if the following two practices are used in eliciting judgments.

First, embed the sentences whose well-formedness you wish to determine in discourse contexts that make the meaning these sentences should have available and salient. This helps remove the second problem.

Second, for every sentence you suspect is ungrammatical, present your informant with a matching sentence which you suspect to be grammatical. These two sentences—the suspected grammatical and the suspected ungrammatical one—should differ minimally. Your aim should be to remove all differences between these two sentences except for the factor that you suspect is responsible for the ungrammaticality. This will help mitigate processing effects, as the two sentences will end up matched in length and close to matched in complexity. It will also help remove any other confounds which

might be responsible for the ungrammaticality of the sentence you wish to test.

These practices are rarely used, unfortunately. As a result, the history of syntactic research is littered with dead ends and wrong turns that have resulted from errors in the empirical base. Don't fall victim to these errors. Wherever you can, follow the Two Laws of Elicitation.

(30) Two Laws of Elicitation

a. Every suspected ungrammatical sentence should be part of a minimal pair, the other member of which is grammatical.

b. The sentences for which you elicit a grammaticality judgement should be embedded in a discourse that makes the meaning that sentence would have salient. You can determine this by finding a context in which your matched minimal pair is grammatical.

A minimal pair are two sentences that differ only with respect to the feature you are trying to investigate. We want to control for as many of the unknown variables that contribute to speakers' judgement of unacceptability and focus in on those that are the result of ungrammaticality.

In these lectures, I will sometimes violate (30a) whenever I haven't found a context that improves an ill-formedness judgement. In these cases, my guess is that the ungrammaticality of the sentence is not tied to its context dependency. Similarly, I will occasionally fail to give a minimal pair when I feel that the ungrammaticality of the sentence involved is dramatic enough to be obvious. For instance, examples such as (31) are so clearly violations of English sentence structure, that I cannot imagine a discourse context that could improve them, nor would minimally contrasting grammatical examples help remove the possibility of a processing effect.

(31) a. * Many happy the puppies barked.

b. * She talked people to.

c. * He ate should apples.

I do this partly because it will make the exposition cleaner, but obviously also because I am lazy. It would be wise to maintain a healthy skepticism about the data I present when I've taken these shortcuts.

I will also violate (30) when I am reporting data from the literature in which (30) have not been followed.

There is one last danger in relying on elicited grammaticality judgments, and it is the mundane and familiar one of introducing bias. It is a commonplace among experimental psychologists that eliciting psychological data can involve very subtle ways of introducing bias. Whenever the judgments are less clear than obvious cases like (31), the syntactician should clearly not rely on her or his own judgments. In these cases only judgments elicited from naïve informants will do. And in eliciting those judgments, the syntactician should adopt some of the techniques developed by experimental psychologists. Produce a survey of examples that include the sentences you wish to find judgments for but include irrelevant "fillers" as well. Those sentences

should be crafted in accordance with the Two Laws of Elicitation. Then present this survey to a number of speakers native in the relevant language, controlling as best as possible for dialect variation. Finally, present the items in the survey in a randomized order, mitigating any bias that the order of presentation might introduce. When reporting data, you should also report the number of informants you have used, and make a note of any variation in the judgments you have encountered. While these safeguards wouldn't satisfy the rigorous criteria of the experimental psychologist, they will go a long way towards removing error and making the data you report comparable to the data someone else gathers. Should the sentences being investigated elicit weak or highly variable judgements, or should they harbor unusual complexity, it is worth considering bringing more of the experimental psychologist's toolkit to bear.

For a good discussion of the problems of using grammaticality judgments, their reliability, and ways of improving methodology see Sprouse 2008, Sprouse 2009, Sprouse and Almeida 2012, Sprouse and Almeida 2017, and Sprouse and Almeida 2018.

Grammaticality judgments, then, will be the central evidence used here in uncovering the principles that constitute a speaker's syntactic knowledge.

1.3 Where is syntax?

In my discussion of the forces that might play a role in delivering judgments about the well-formedness of a sentence, I have highlighted two activities that speakers and perceivers of language are engaged in. Because sentences are used to convey information, their meanings are derived in full by considering the conversational contexts they are in. The context in which a sentence is uttered is relevant to an evaluation of its meaning and form. This means that speakers planning their sentences will consider how they fit into the conversation, and hearers perceiving those sentences will similarly calculate what message is intended by the speaker through an evaluation of that context. The evaluation of sentences, then, involves the cognitive mechanisms used by interlocutors to evaluate conversations. These mechanisms are not usually deemed part of what the syntactician is tasked with modeling, and so we regard the contribution made by these mechanisms as noise to be controlled for. Similarly, the mechanisms that are responsible for the fast, real-time, processing of the structure of sentences—the tasks that the parser undergoes—are also not what syntacticians are usually charged with modeling. The effects that trace back to the architecture of the parser are also usually regarded as noise by the syntactician. The Laws of Elicitation are partly aimed at reducing these sources of noise.

One might reasonably ask what syntacticians are modeling, if it is not the processes that are involved in producing and perceiving sentences. A common frame of reference is the learning theoretic one discussed above. The syntactician is looking for information from the structure of sentences about how the learning process is constrained. But the models a syntactician comes up with aren't models of the acquisition process either. Models of acquisition should include mechanisms that extract information from the

environment that informs the learning mechanism's guesses about the syntax being learned. It should include procedures that test the present guess against that information, and an engine that generates new guesses based on the outcome of those testing procedures. The models that a syntactician produces don't have these ingredients.

What the syntactician models is not, strictly speaking, linguistic behavior. We might think of it as a model of knowledge of a certain domain: the knowledge humans have about their sentence structure. That such a domain even exists is merely a guess, I think we must admit. And what "knowledge" means is far from clear as well. The "thing" that syntacticians model is not easily identified. Indeed, identifying that thing turns largely on the success of the modeling. Because of this, it is easy to mistake a syntactician's model for one or the other of the models of language behavior described above. As we'll see, many of the models of syntax we will look at are framed in terms of a procedure, rather than a static formula or function. This encourages the misperception that syntactic models are, say, models for producing or perceiving a sentence. Similarly, because syntactic models include a map of language variation, they frequently have components that define how languages can, and cannot, vary. It's easy to mistake these components as a theory of language acquisition, when they are nothing more than a model of variation. One must guard against these misperceptions.

It might be useful to think about what a syntax models in terms of its relation to the more straightforward models of linguistic behavior. The linguists who study how sentences convey information need to include in their models information about how interlocutors reason about the goals of a conversation. The linguists and psychologists who study how sentences are produced and perceived need to include in their models information about how human memory and attention are deployed, and how the human perceptual system works. To study the learning process involves studying child development, on the one hand, and formal models of learning, on the other. Models of acquisition need to include all of the kinds of information that the other linguistic enterprises described here involve, as well as the parochial mechanisms of guess-testing and revision. One way of viewing what the syntactician is modeling is in terms of what is shared in these other enterprises. A model of syntax is a model of the information about sentence structure that is found at the intersection of all these other modeling tasks. A successful model of syntax is one that reveals the relations that are required to determine, for any given language, which strings of its words are a sentence or not. For instance, a successful syntactic model will have identified the criteria by which strings of words form groupings. It will define the procedure that relates relative pronouns to their <gap> in a way that characterizes when that is possible in a natural language. The relations and procedures that a successful model of syntax provides should be found as a component of all the models of linguis-

tic behavior that involve sentences: learning, processing, production, and information.

It is entirely possible that there will be no thing "out there" that syntax models. It might be nothing more than a way of extracting information about the shared pieces of these other models of linguistic behavior. It might be that syntax is a result of how these other parts of language behavior interact, and nothing more. In this respect, "syntax" may be no different than "English." Both might be words whose referent is a thing in our model, not a thing that exists in what is being modeled.

Syntactic models can be useful even if it turns out there is no thing "external" that they model. They partition the unwieldy mishmash of linguistic behavior into a piece that can be studied with precision. Syntax is tasked with finding some of the laws behind linguistic behavior, not with finding their source. When a linguist proposes sources for the discoveries of syntax, they are no longer doing syntax. Syntax is part of the supply chain to models of linguistic behavior. Syntacticians provide an image of the structure of sentences. That image provides a goal, and hopefully offers clues, to all the other members up the chain. Everything, in other words, depends on us.

2

Phrase Structure

Now that we have something of a setting for our inquiry, it is time to carve out that portion of linguistic theory that belongs to syntax. I think a rather conventional definition of syntax might be (1).

(1) Syntax is the study of those principles and processes responsible for organizing words into sentences.

Like many definitions of areas of study, this one uses terms that are the subject matter of the study. We need definitions of "word," "sentence" and "organizing," and syntax is partly responsible for delivering these definitions. Nonetheless, we can start with some informal definitions of what these terms refer to and work from there.

For "word," let us start with the definition that orthographic practice provides. If it's an entry in a dictionary, it'll be a word. We'll soon see that this is too narrow a definition, but it will get us started. For "sentence," we should aim for those sequences of words that discourses are made up of. Again, orthographic practice gives us a starting point. Those strings of words that a language pundit would acknowledge as being able to punctuate with a period will count. There are a variety of examples that have an uncertain status. For example, the utterances by "B" in the following exchanges are incomplete in other contexts.

(2) A: Jon has probably read this book.
B: In fact, he hasn't.

(3) A: Has Jon read this book?
B: No.

(4) A: What has Jon read?
B: This book.

But in these contexts we will treat them as sentences. We'll need to understand, of course, why the conversations they are part of allow them to be understood as complete sentences. And finally, for "organizing" we can begin with simple linear order. We will start, therefore, with the goal of modeling how words are linearly ordered to produce grammatical sentences.

Beyond simple grammaticality judgments, there is one other kind of datum that is important to the syntactician. As we witnessed in the previous chapter, the syntax of sentences is intimately tied to the meanings they convey. It is the semanticist's job to discover the principles that allow users of language to extract these meanings from sentences, but their task and ours can be usefully done in tandem. Over a wide range of cases, the meanings conveyed by sentences appear to be derived, perhaps by general processes, from the meanings of the words they contain. This is a trivial observation. The kind of individual that each of the sentences in (5) is about, depends on the meaning of the second word.

(5) a. A woman jumped.
b. A child jumped.
c. A puppy jumped.
d. A grasshopper jumped.

But what these sentences say about that kind of individual stays the same. It would vary if the last word varied, as you can see by replacing *jump* with some other verb. This observation about the meanings of sentences is known as the Law of Compositionality.

(6) The Law of Compositionality
If the meaning of a collection of formatives, α, is derivable from the meanings of those formatives then α's meaning is compositional. The meaning of a sentence is compositional.

A working hypothesis is that there is a regular and productive relationship between the syntactic structure of a sentence and the ways its meanings are composed. To the extent that this is correct, then, we can use evidence about how meanings compose to investigate our hypotheses about how the syntax arranges words.

It should be noted that the "formatives" named in the Law of Compositionality do not match up with the "words" over which we have defined syntax. There are cases where strings of words seem to have a meaning that is not compositional. One famous example of that sort are idioms, such as (7).

(7) Professor Maribel Romero is barking up the wrong tree.

bark up the wrong tree: "to promote or follow a mistaken course (as in doing research)," www.merriam-webster.com.

The string *barking up the wrong tree* has two meanings. One is compositional, and involves an action and a tree and is thoroughly uncharacteristic of Professor Romero. The other is not compositional, invokes a particular kind of mistake but no tree, and is also thoroughly uncharacteristic of Professor Romero. Here, then, we have words that syntax has arranged but from which a compositional meaning is not derived. Similarly, there are certain morphemes that do not fit our criteria for words but which compose semantically in a way that fits the Law of Compositionality. One of these is the plural suffix found on many English nouns. The meaning of *bees* is, apparently, derivable

from the meanings of *bee* and *s*, for instance. A more complete theory than will be attempted in this chapter would cover these cases too. We'll set them aside for the moment.

Here, then, is our specific task: discover the principles that underlie a language user's ability to recognize grammatical arrangements of words as sentences. What constitutes a grammatical sentence will vary from language to language, of course, and, less saliently, from one individual to another. We'll want a model of this variation as well, for the reasons sketched in the previous chapter. We must start with one grammar, of course, and because I can assume the readers of these notes to have some command of English, I will start with that language.

Our first observation is that we can get far in this task using very little particular information about the words involved. A great deal about the processes which determine well-formed arrangements of words can be characterized using nothing more than syntactic position. To see this, let me introduce you to the word [blowɹsnɪk] ('bloresnick'). Here is an example of *bloresnick* in a sentence.

(8) The bloresnick should be grey.

If I tell you that (8) is grammatical, you will also be able to determine that only (9b) of the examples in (9) is grammatical too.

(9) a. It bloresnick ran.
    b. He removed the long bloresnicks.
    c. She finds Sammy bloresnick.
    d. He let his face bloresnicks.

The position that *bloresnick* has in (8) is enough for you to know a lot about which positions it can, and cannot have, in English. Part of our knowledge of English syntax gives us this ability. We know that positions in sentences of English are connected to other positions in English sentences. If a word can stand in one of these positions in a grammatical English sentence, then we know it is possible for that word to stand in all the other connected positions, but no others. There are other forces that determine the well-formedness of a sentence, and sometimes these other forces will intervene. For instance, once the meaning of *bloresnick* is discovered, we might find that (9b) is also ungrammatical. What we're looking at here is a part of our syntactic knowledge that is independent of other influences that come with the particular word involved, like its meaning. If all you know is that a word can stand in position X, then this confers information about all the positions connected to X; namely, they are positions where that word might be expected to be able to stand. You know, then, that this word cannot stand in any of the positions that aren't connected to X. This is what we are seeing in the inference from (8) to (9).

We will model this part of syntax by dividing the words of our vocabulary into kinds. The rules of syntax that are responsible for conveying this

position-based piece of information refer to those kinds. Our syntactic rules will therefore define a set of positions in a sentence as being reserved for words of a particular kind. Thus, if all you know about a word is what position it can occupy in a sentence, you will know from your syntax what kind that word belongs to. This knowledge will block this word from being in any position that isn't reserved for words of this kind. These kinds are known as morphosyntactic categories, and many of them are familiar from the folk grammar we learn in school: noun, verb, preposition, adjective, and adverb. The sentence in (8) is apparently sufficient for a knower of English to deduce that *bloresnick* is a noun. And knowing that it is a noun is sufficient to know it cannot be in the position it occupies in the sentences of (9), except (9b).

Our first task, then, will be to model this very small part of syntax. We will model what it is we know about the possible arrangements of morphosyntactic categories that make sentences in some language.

2.1 Substitution Classes

This way of thinking about what morphosyntactic categories are, makes them "substitution classes." When a word is of morphosyntactic category "noun," this means nothing more than that this word can be replaced by any other word in that morphosyntactic category and the resulting string will have the same grammaticality status (modulo whatever information the particular word brings). We can see this by considering the sentences in (10)-(15).

This way of conceiving of morphosyntactic categories, along with much of the particulars of this section, come from Harris (1946).

(10) The lamp exists.
The girl exists.
The sky exists.
The streetcar exists.
⋮

(11) * The happy exists.
* The blue exists.
* The short exists.
* The flat exists.
⋮

(12) * The in exists.
* The out exists.
* The from exists.
* The on exists.
⋮

(13) * The swim exists.
* The have exists.
* The ate exists.
* The broke exists.
⋮

(14) * The slowly exists.
* The apparently exists.
* The always exists.
* The decidedly exists.
⋮

(15) * The every exists.
* The much exists.
* The no exists.
* The a exists.
⋮

The grammaticality status of the sentence *the lamp exists* is preserved when *lamp* is replaced by a word that belongs to the same morphosyntactic category, but changes when *lamp* is replaced by a word belonging to another morphosyntactic category. We want to find a way of saying that the position *lamp* occupies in (10a) is defined for nouns, and not words of other morphosyntactic categories.

Since we are thinking of the syntax of a sentence as being a particular linear arrangement of words, the notion of "position" that is relevant for defining where a word of a particular category can be should be linear as well. A simple definition of position is (16).

(16) The position of a word, α, in a sentence is defined as the word or words that α is string adjacent to.

(16) expresses the hypothesis that the most information one needs to define a category's position is very local: adjacent terms. We could define noun as occupying the position in (17).

(17) the ___ exists

As can be seen, this technique picks out a list of words that match what the grammar school curriculum calls nouns, and segregates them from the others. A similarly discriminating environment can be devised for each category. For each (major) word class, I've given a distinguishing environment in (18).

(18) a. have ___ eaten: Adverb (Adv)
b. the ___ thing: Adjective (A)
c. dance ___ it: Preposition (P)
d. in ___ orange: Determiner (D)
e. must ___ there: Verb (V)

Understand (18) as abbreviating the following claim: there is a sentence that is grammatical which contains "X ___ Y," and for which replacing a word of category category into "___" preserves grammaticality. So, for instance, (18a) should be understood as claiming that all the ways of completing the sentence in (18a) involve filling "___" with an adverb.

(19) They have ___ eaten rutabagas.

On this view, morpho-syntactic categories are simply partitions of the vocabulary into equivalence classes. The labels "noun," "verb" and so on are merely convenient names for the resulting subsets of vocabulary items.

It should be noted that other linguistic processes also recognize these same partitions of the vocabulary. The suffix *able*, for instance, combines with verbs but not other morpho-syntactic classes, and the suffix *ness* targets adjectives and not other categories.

(20) a. *readable*, *doable*, *washable*,...
b. * *happiable*, *brightable*, *inable*, *carable*,...

(21) a. *happiness*, *brightness*, *sadness*,...
b. * *readness*, *doness*, *inness*, *carness*,...

Verbs in English also have a different stress pattern than do nouns and adjectives, and this leads to couplets like those in (22).

(22)

Verb	*Noun*
transpórt	tránsport
compóund	cómpound
permít	pérmit
overláp	óverlap
segmént	ségment
compréss	cómpress

The division of the vocabulary into these subclasses is not merely a syntactic thing, then. There are syntax-independent linguistic processes that also indicate that lexicons are structured into subclasses, and moreover these processes are sensitive to the same subclasses as the syntax is.

I have defined morphosyntactic categories with an environment that is defined in terms of other words. The class *noun*, for instance, is made up of those words in English that can stand between *the* and *exists* and result in a grammatical sentence. It is easy to see that if every word belongs to a morpho-syntactic category, this process could be translated into defining

morpho-syntactic categories in terms of other morpho-syntactic categories. After all, replacing *the* in (10) with another word that belongs to the same morpho-syntactic class often has no effect on the result:

(23) *a* lamp exists.
some lamp exists.
no lamp exists.
that lamp exists.
⋮

And, similarly, replacing *exists* with another word of the same morph-syntactic category doesn't change the outcome:

(24) the lamp *remains*.
the lamp *persists*.
the lamp *falls*.
the lamp *disappeared*.
⋮

If *the* belongs to the morpho-syntactic category determiner, and *exists* is a verb, then we can define noun as that set of words that can grammatically stand between a determiner and a verb. The case of *the* and *exists* is merely a particular example of this more general pattern. We can make a similar generalization for each of the other environments that define a morpho-syntactic category in (18). The environment defining adjective, for instance, can go from "*the* ___ thing" to "D ___ N."

Moving from defining categories from particular words, to defining them on the basis of other categories introduces errors that arise because of information that is brought into the mix by the particularities of the specific word chosen. For instance, the Determiner class picked out by (18d) does not include *much* or *many*. There aren't grammatical sentences that contain *in much thing* or *in many thing* as a substring. But the more generalized definition of noun, does include *much* and *many*, because it is possible to find other nouns that fit into the template.

(25) a. Many things existed.
b. Much stuff existed.

If we look at a larger set of strings, we will discover that the class of words that can stand in the "noun" position of triples that start with *much* or *many* have a pattern. Just like our definition of morpho-syntactic category, the information concerning the position of *many* and *much* is not idiosyncratic; there is a pattern that conveys information about position. For instance, if I know that (26) is grammatical, then I can with confidence guess that (26a) is too, but that (26b) is not.

(26) Many bladel existed.
  a. Some bladel existed.
  b. Every bladel existed.

And if I know that (27) is grammatical, I can predict that (27a) will be grammatical and (27b) won't be.

(27) Much bladel existed.
  a. Some bladel existed.
  b. Many bladel existed.

What we are seeing here then is the same kind of information that the *blores-nick* phenomenon is, and we could employ the same tool to capture it. We could divide up the lexicon so that *many* is in the same set of words as *some*, *the*, etc. and separate that set from that which includes *every*, *a*, etc. Then there will be a third set of words that includes *much*, and this set will intersect the set that includes *many*; the words *some* and *the* will be in both, for instance.

There is another reaction to these data, however, and it is the one I shall follow. It's clear by comparing the environments that distinguish these cases that they have something to do with the meaning of the nouns and determiners involved. The difference in grammaticality between (28a) and (28b) concerns whether the noun describes a group or an item.

(28) a. Many cats existed.
  b. * Many cat existed.

A difference between the meaning of *cats* and *cat* is that the second describes a set objects that necessarily has a cardinality of 1, and the first can, and usually does, describe a set of objects whose cardinality is greater than 1. More technically, if some X is *cats*, then there can be parts of X that are *cat*. But if X is *cat*, there can be no part of X that is also *cat*. We say that *cats* is plural and *cat* singular.

A similar sort of observation can be made about the contrast in (29).

(29) a. Much water existed.
  b. * Many water existed.

There is a sense in which *water* is inherently plural.If X is *water*, then there is a part of X that is also *water*. Unless you learn otherwise in a physics class, this will always be true of *water*. Thus, *water*, from the standpoint of its meaning, isn't singular. Nouns whose meanings either have, or are treated as having, this inherent plurality are known as mass nouns. The determiners are distributed in a way that is sensitive to this information about the nouns that follow them. The determiner *much* can be followed only by a mass noun; and the determiner *many* cannot be followed by a mass noun. The determiner

This way of thinking about what makes mass nouns special is from Quine (1960).

Nouns that aren't mass are called count nouns.

many must be followed by a plural noun, and cannot be followed by a singular noun (hence the contrast in (28)). Some determiners, like *every* can only be followed by singular nouns.

(30) a. Every cat existed.
b. * Every cats existed.
c. * Every water existed.

Note that ***water*** in (30c) is grammatical, but with a shift in the meaning of ***water***. When a mass noun is forced to be singular, it shifts in meaning to refer to a kind. In (30c), the meaning is roughly the same as "every kind of water existed."

The generalization we are seeing here might, then, be best described semantically rather than syntactically. That is, the information that seems relevant to these patterns has to do with the meanings of the words involved. We could capture these patterns by letting the semantic rules that combine adjacent words (or strings of words) be sensitive to the status of a determiner and noun's plurality. We could give to plural nouns a meaning component that is shared across the class, and demand that a determiner which semantically combines with a noun do so in a way that is sensitive to this meaning component. We could do a similar thing for the distinction between mass and non-mass (count) nouns. We could then capture these sorts of patterns by way of semantic categories, and not morpho-syntactic categories.

Is there really a distinction of this sort between semantic and morpho-syntactic category? Couldn't we imagine, for instance, that the fact that a noun is the kind of word that can be right-adjacent to a determiner a fact about the meanings of determiners and nouns? Could we not seek a similar common meaning component to nouns and rely on a semantic rule that combines words with that meaning component to define determiner? This is, in principle, possible and if successful, we could define morpho-syntactic categories in wholly semantic terms. The *bloresnick* phenomenon would emerge as a fact about semantic categories. At present, however, no method of reducing morpho-syntactic categories like "noun" and "verb" to semantic ones is known. There are minimal contrasts, like (31), that make the semantic difference between noun and verb ephemeral.

I've described what makes mass nouns different from count nouns in a way that makes it sound like it's a physical distinction; i.e., it has to do with how our folk physics understands the world. Unlike our "real" physics, our folk physics requires every part of water also be water, and it's this attribute that makes it mass. But that can't be the way we should understand what it means to be semantically mass. We can see this by considering the noun ***furniture***, which is mass in English (****many furniture*** vs. ***much furniture***) but our folk physics clearly lets parts of the group of things that are furniture be things that aren't furniture: an attribute that count nouns alone have. If the referent of ***furniture*** is three chairs, for instance, we can divide that group into single individual chairs (each of which is furniture), but no further. A chair leg is not furniture. Our folk physics and our semantic classification differ. We should understand semantics as characterizing a certain set of affairs in a particular way. That ***furniture*** is mass means that its denotation makes it refer to stuff in the same way that water does. It doesn't require that our folk physics of furniture is the same as our folk physics of water. See Chierchia (2010), and the papers in Massam (2012) and Lin et al. (2018) for an examination of this issue.

(31) a. Erika's periodic examination of the problem (drove her partner mad.)
b. Erika periodically examined the problem

The difference between *examination* and *examine* is the difference between noun and verb; but there is very little difference in what these words mean in (31). I will proceed as if there are morpho-syntactic categories that cannot be defined by their meanings. If this turns out to be incorrect, then it should be possible to translate our system into one that rewrites "noun" into the appropriate semantic terms, and so on with the other categories.

Our morpho-syntactic categories, then, are names for sets of words whose positions in a grammatical sentence can be defined by adjacent words. What this delivers is a set of triples, each showing the possible coöccurrence of word categories.

(32) DNV, VDN, PDN, DAN, VPD, …

The set of possible sentences in English, then, could be represented by a set of strings that are made from these triples, each corresponding to a possible sentence form.

(33) List of Potential Sentences (LOPS)

DNV, DANV, DNVDN, DNVPDN, DANVDN, DANVDAN, DNVPDN, DANVPDN, DANVPDAN, …

I will call this the List of Potential Sentences (LOPS). These are merely potential sentences, as our system is designed only to capture the *bloresnick* phenomenon. It merely models how we know something about where a word can be in a grammatical sentence if we know just one of those positions. There is knowledge about what constitutes a grammatical sentence that is determined by other kinds of information that accompany a word, but that isn't yet in our system.

LOPS gives a kind of outside limit for what grammatical sentences can be. It will be very long. We should seek a more compact way of representing it. Indeed, there are reasons for thinking that there must be a more compact way of encoding the set of strings in (33). Consider the series in (34).

(34) The dog barked.
The big dog barked.
The big bad dog barked.
The big brown bad dog barked.
The big hairy brown bad dog barked
⋮

(34) represents a particular way of rendering the series in (35).

(35) DANV
DAANV
DAAANV
DAAAANV
DAAAAANV
⋮

This series is part of the list in (33). The series in (34) is finite, there is no doubt. No human can speak or perceive an infinity of sentences. But the reason for that is arguably not syntactic. The point at which the series in (35) ends is not determined by English syntax. This can be appreciated by picking any adjacent pairs of sentences in (34) and determining their grammaticality status. If syntax were responsible for the finiteness of the series, there should be a pair somewhere in the series that spans the grammatical/ungrammatical divide. It seems likely that no such pair exists. Imagine having mustered the cognitive strength to understand a sentence with n adjectives, where n is

large. Is it likely that adding one more adjective to that sentence would do anything more than merely demand more cognitive strength to understand it? The standard conclusion is that the list of strings of morphosyntactic categories that make grammatical sentences is infinite in length. If what we are modeling is a cognitive ability, then this list cannot be represented in our minds in this form. We need a method of generating this list from a more compact—at a minimum, finitely determinable—procedure. It is to this more compact procedure that we now turn.

2.2 *Phrases*

We have an intuition that sentences contain subgroupings of words. That intuition lives on several observations. One of these is that the pronunciation of sentences comes with a prosody which, like syllables, gives the impression of subgroupings. These prosodic groupings have an effect on various aspects of the phonology of sentences. They seem to influence the pitches and volume, and also the rate, of speech. For instance, a natural pronunciation of (36) can introduce a slight pause after *cracker*, and slow the relative duration of [kɹæk̚ɹ̩].

(36) I can't eat every cracker today.
  a. I can't eat every CRACKER | today.
  b. * I can't eat EVERY | cracker today.

I've indicated that pronunciation in (36a) by expressing the relatively slowed articulation of *cracker* with all caps, and the pause with a pipe. Note that (36) does not permit a natural pronunciation of the sort indicated by (36b), where *every* is slowed and followed by a pause. The slowdown plus pause is thought to be permitted only at the right edge of prosodic phrases. There is a prosodic phrase that ends with *cracker*, but not one that ends with *every*. Phenomena like these, then, suggest that there are subgroupings within sentences.

To get a feel for what is involved in locating prosodic groupings, see Liberman and Prince (1977), Nespor and Vogel (1986), Beckman and Pierrehumbert (1986), Pierrehumbert (1980), Selkirk (1980, 1986) and Ladd (1996).

A second observation that speaks on behalf of the existence of subgroupings is semantic. Certain substrings of a sentence can be associated with a meaning in isolation, and others cannot. For instance, there is a difference in the meanings attached to the two strings in (37), both parts of (36).

(37) a. cracker today
  b. every cracker

(37a) is just a (short) list of words. (37b) has a meaning. Indeed, it seems to obey the Law of Compositionality. Some groups of words seem to be compositional in this way while others do not.

A working hypothesis is that the subgroupings which the Law of Compositionality is sensitive to and the subgroupings that prosody is sensitive to are one and the same. There is some superficial evidence in support of this view. Consider, for example, the sentence in (38).

(38) She ate the cracker in the box.

There are two ways this string can be semantically partitioned. In one, *the cracker in the box* is associated with a meaning; it refers to an object that (38) reports was eaten. In the other grouping, *the cracker* and *in the box* are independent; their meanings are not composed into the meaning that *the cracker in the box* has. Under this second grouping, (38) reports that the cracker was eaten in the box. These two semantic groupings can be associated with two prosodic groupings. If *the cracker in the box* makes a semantic unit, (37) can be spoken in the way indicated in (39a) but not (39b). If, by contrast, *the cracker* and *in the box* make independent semantic units, the prosody in (39b) is preferred instead.

(39) a. She ate the cracker in the BOX |.
b. She ate the CRACKER | in the box.

The prosody in (39b) corresponds to one in which a prosodic phrase ends at *cracker*, whereas the one in (39a) does not. If the prosodic and semantic groupings are the same, then this is what's expected. For *the cracker in the box* to form a semantic unit, it cannot end at *cracker* as in (39b). But if *the cracker* and *in the box* do not form a semantic unit, then the juncture between these groupings in (39b) is expected.

In addition to these criteria for subgroupings, there are wholly syntactic phenomena that suggest a similar subgrouping of the words in a sentence. Like words, the strings that make subgroupings are typed, and their distribution in sentences is controlled by that typing. This can be discovered in the same way that *bloresnick* informed us that our vocabulary is divided into classes. If you come to believe that (40) is a grammatical sentence in English, you will have derivative beliefs about the grammaticality of the sentences in (41).

(40) should be pronounced with prosody that is consistent with ***bel droresnick*** being a subgrouping. If a marked pause is put between ***bel*** and ***droresnick***, there is another way of grouping the words in (40), and a different set of inferences emerges.

(40) Bel droresnick has arrived.

(41) a. I stood on bel droresnick.
b. I am happy bel droresnick.
c. The discussion bel droresnick is here.
d. Bel the droresnick is here.

Knowing that the string *bel droresnick* ([bɛl ʤɹɑowɹsnɪk]) can be the initial string of a grammatical sentence is enough to know that (41b-d) are unlikely to be grammatical, but that (41a) could be grammatical. Our model of syntax should capture this piece of our knowledge as well. For those substrings that constitute phrases, knowing the position it occupies in a grammatical sentence is all that's required to know that it cannot occupy certain other positions in a grammatical sentence.

There is one syntactic environment that can be useful in discovering subgroupings. It involves constructions in which *or*, or, *and* are found.

(42) a. The crackers or the cheese are here.
b. The crackers and the cheese are here.

In rough outline, the words *or* and *and* must have a string left-adjacent to it that is a subgrouping of the same kind as the substring right-adjacent to it. In (42), the string *the cracker* and the string *the cheese* are subgroupings of the same kind. They meet the prosodic and semantic criterion for subgroupings sketched above, and they have the same syntactic distribution. None of the other positions for *or* or *and* shown in (43) are grammatical.

(43) a. * Or/and the crackers the cheese are here.
b. * The or/and crackers the cheese are here.
c. * The crackers the or/and cheese are here.
d. * The crackers or/and are here.
e. * The crackers are or/and here.
f. * The crackers are here or/and.

I'll call these constructions "coördinations." They have the general shape indicated by (44).

(44) α and/or α, where α is a subgrouping of a particular type and the entire string "α and/or α" is itself a subgrouping of the same type as α.

As predicted, the knowledge that (40) is grammatical confers the ability to make the grammaticality judgments indicated in (45).

(45) a. The cracker or/and bel droresnick are here.
b. * The or/and cracker bel droresnick are here.
c. * The cracker bel or/and droresnick are here.
d. * The cracker bel droresnick or/and are here.

We will use coördinations in what follows to discover subgroupings, and to determine their types as well, but there is an important caveat that needs to be placed on the utility of coördinations. Coördinations invoke processes that obscure what the strings they bring together are. This can be seen from the examples in (46).

(46) a. A UMass professor knows syntax and teaches phonology.
b. Some professors know syntax and others phonology.

In these examples, the string that *and* has to its left is a sentence. That sentence is a subgroup of the larger sentence, and it is the grouping that *and* is joining with the string to its right. But, as we can see from the ungrammaticality of (47), the strings to the right of *and* are, neither of them, grammatical sentences.

(47) a. * Teaches phonology.
b. * Others phonology.

Interestingly, the strings to the right of *and* in (46) have a meaning that seems to be more than the meanings delivered by the words in those strings. The sentences in (46) are synonymous with the sentences in (48).

(48) a. A UMass professor knows syntax and a UMass professor teaches phonology.
b. Some professors know syntax and others know phonology.

In fact, (46a) is ambiguous. On one of its meanings, the group ***knows syntax*** and the group ***teaches phonology*** are what are brought together by ***and*** in the coördination. This is the way (44) leads us to expect the sentence to be constructed. The meaning derived from this way of putting (46a) together is paraphrasable by: "There is a UMass professor who both knows syntax and teaches phonology." (Depending on how stringent the requirements imposed by ***know*** are, this sentence is false at the time of writing.) We are interested here in the problematic syntax for (46a), the one in which two full sentences are brought together by ***and***. This delivers the meaning that (48a) has. A sentence that is more likely to be true.

What we are seeing here is that the meaning conveyed by some of the words in the first group are also present in the second group. One way of accounting for this is to use the process of Ellipsis described in the previous chapter. Ellipsis allows certain strings of words to go unpronounced if they are pronounced elsewhere. On this account, there are words in the second group that are not pronounced; those words match ones in the first group. On this view, the sentences in (46) are actually just the sentences in (48), but with the words *a UMass professor* and *know* unspoken.

This kind of ellipsis is called "Gapping," and it is found primarily in coördinations. It doesn't form sentences like (49), for instance.

(49) a. * A UMass professor knows syntax because teaches phonology.
b. * Some professors know syntax because others phonology.

The name comes from the first paper that examined this process: J. R. Ross."Gapping and the order of constituents." In M. Bierwisch and K. E. Heidolph, editors, ***Progress in Linguistics***, pages 249–259. Mouton, The Hague, 1970.

Because of Gapping, one must be alert to the possibility that the rightmost group in coördination might have elided material. Gapping allows for material in the second group to be unspoken, but not material in the first group; (50) are ungrammatical.

(50) a. * Knows syntax and a UMass professor teaches phonology.
b. * Some professors syntax and others know phonology.

So one must be diligent in discovering the true nature of the rightmost group in the coördination. This can be done, often, by paying close attention to the meaning of that group.

Another construction that problematizes coördination is exemplified by (51).

(51) Jo-Wang talked to, and Tohru talked about, the philosopher.

In cases like these, there is material apparently in the second conjunction that seems to be part of the first conjunct. In (51), that material is *the philosopher*. These constructions are often called Right Node Raising, and they have a characteristic intonation which helps identify them.It's possible to think of Right Node Raising as involving a process that makes the first conjunct contain unspoken material that is found in the second conjunct. In (**??**), then, the left conjunct could be an entire clause, with an unspoken *the philosopher*

See Phillips (2003) for a good survey of Right Node Raising, and Hartmann (1998) for an account that focuses on the intonation.

following *to*. The existence of Right Node Raising means that we need to be careful about deciding what the string to the left of a coördination is; it might be more than it appears to be.

We have three criteria that determine the groupings of words in a sentence, then. Those groupings have an effect on prosody. They have a compositional meaning. And they have a distribution in sentences that shows that they are units, and of a particular type. Discovering the principles that determine these syntactic phrases will be one of our central tasks.

As with the groupings that the semantic and prosodic phenomena pick out, a working hypothesis is that the groupings that the syntactic criterion select are the same thing. I'll adopt this hypothesis too.

(52) The subgroupings that the words of a sentence divide into are the same for its syntactic, prosodic, and semantic organization.

For some of these problems, see Selkirk (1984, 1996), Truckenbrodt (1995, 1999), Zec and Inkelas (1990) and the papers in Inkelas and Zec (1990).

It is not at all clear that (52) is true; there are many apparent mismatches to understand if it is true. Nonetheless, this is a useful starting position. Indeed, we will adopt the slightly stronger hypothesis in (53).

(53) The syntax of a sentence produces phrases, and the phonological and semantic groupings make use of these phrases.

(53) expresses a strategy for explaining (52), should it be true. Syntax will include a procedure that forms phrases, and these phrases are used by the phonological and semantic components to assign a prosody and a meaning to the sentence. That image of syntax as a intermediary between the phonological and semantic interpretations of a sentence will recur throughout these lectures. A presently popular thesis is that many of the properties that syntax has can be explained by its role as an intermediary between the phonological and semantic manifestations of a sentence. In the case of phrases, for instance, one might imagine that the pronunciation of a sentence requires an ordering of the words it contains into phonological groupings. This is a requirement that the principles of phonology (and phonetics) alone impose. Similarly, one might imagine that for a sentence to convey a meaning, its words must be grouped for the semantic rules to be able to deliver that meaning. This requirement is a wholly semantic one; perhaps it is whatever underlies the Law of Compositionality. The "function" of syntax, we might imagine, is to deliver these groupings. Syntax could be defined as that portion of the phonological and semantic components that is shared. This is a view of the place of syntax in the larger linguistic panoply that Noam Chomsky has advanced in the last thirty years. He sometimes suggests that syntax is only those portions of the other "interpretive" components that are shared. This is a guiding idea behind the so-called "Minimalist Program," one of several current trends in syntactic theorizing.[1] This will be an idea that figures largely in what we will do in these lectures as well.

[1] Chomsky, Noam. 1995. ***The minimalist program***. Cambridge, Massachusetts: MIT Press

(54) The Minimalist Hypothesis
Syntax is made up only of those mechanisms necessary to mediate between the Phonology and Semantics of sentences.

What, then, are the syntactic criteria for recognizing phrases? One proposal is to extend the process used to locate the morphosyntactic classes of words. Like nouns, verbs and the like, certain strings of categories also have a distribution within sentences that can be defined in terms of adjacent items. For example, the string D(eterminer)+Adj(ective)+N(oun) can appear immediately after a preposition and immediately preceding the *'s* which marks the "possessive."

(55) ___ *'s* & P ___ : D+Adj+N

a. I talked to **the happy woman**.

b. **the happy woman**'s friend

This string can also be coördinated with another identical string of categories. Thus, we find examples like (56) but not (57).

(56) the happy woman and an unhappy man

(57) a. * the angry and an unhappy man

b. * the and an unhappy man

Finally, with respect to all these distributional tests, the strings D+N+P(reposition)+N, N+P+N, Adj+N, N, and (infinitely) many others also pass. We need some way of describing the fact that these strings are "the same," and different from, say, P+N which has a distinct distributional pattern. That is, this family of strings is a substitution class in the same sense that morpho-syntactic categories are.

Families of strings like this are called "phrases," and we can write Phrase Structure rules to describe which strings belong to such a family. In the case at hand, these rules might look like (58).

(58) a. αP $\rightarrow$ N

b. αP $\rightarrow$ Adj N

c. αP $\rightarrow$ D Adj N

Phrases are a model of the groupings that the prosody, semantics and syntax recognize.

This leaves out the strings D+N+P+N and N+P+N. But these strings involve another phrase, made up of the string P+N. This string, along with any string that conforms to the template P or P+αP or P+P or P+P+αP has the defining distribution in (59).

(59) A ___ & P ___

a. I stood **around**.

b. I knew the man **by Mary**.

c. I remain disliked **by Mary**.

d. I stood next **to Mary**.

Like αPs, βPs may be coördinated with other βPs, but not with other αPs, as the following examples illustrate.

(60) a. Under the bed and behind the sofa are usually good places to find money in my house.

b. * The dining room table and behind the sofa are usually good places to find money in my house.

Hence, just as with αPs, this family of strings constitutes a substitution class.

Putting these observations together, we come up with the Phrase Structure rules in (61).

(61)

αP $\rightarrow$ N	αP $\rightarrow$ N βP
αP $\rightarrow$ Adj N	βP $\rightarrow$ P
αP $\rightarrow$ D Adj N	βP $\rightarrow$ P αP

Note that (61a) and (61d) together have the property of being recursive. This is an important aspect of phrase structure rules for it is the primary means by which we describe the indefinite length, and cardinality, of sentences. These two phrase structure rules are able to characterize infinitely many and infinitely long strings of words. This is a correct result, for we are, as far as our linguistic abilities go, capable of forming grammaticality judgments about an infinity of sentences and about sentences of infinite length. We have other properties that prevent us from doing this, of course. Our memories and attention are too ephemeral for even very long sentences; and even if we were able to overcome these cognitive limitations, our death will bring an eventual end to any sentence, or series of sentences, that we are evaluating. But there is no reason to think that this limitation is a linguistic one. There is not particular point at which a sentence's length become ungrammatical. We should let our model of grammaticality judgments characterize an infinity of sentences, as well as permit sentences of infinite length, and let the actual limits on the lengths and numbers of sentences that we evaluate be determined by other factors. The recursiveness of phrase structure rules is a step in that direction.

Still another phrase structure rule is required to account for the fact that the family of strings that include V, V+αP, V+βP, V+αP+βP, and an infinite set of other such strings is a substitution class. The environment that defines them is (62).

(62) Infl ___

a. I should **eat rutabagas**.

b. I will **talk to Mary**.

c. I will **tell Mary about rutabagas**.

"Infl" is a morpho-syntactic category that includes *should*, *will*, *must*, *would*, *can*, *could* and a few other words. It's an abbreviation for "inflection." The words that belong to it are tied to the inflectional classes that verbs belong to, as we shall see. Like αPs and βPs, coördination treats members of this family as equivalent and distinct from αPs and βPs.

(63) a. Mary walked and talked.

b. Mary visited Paul and kissed Barry.

c. Mary talked to Paul and met with Barry.

These facts call for Phrase Structure rules like the following:

(64) $\gamma P \rightarrow V$
$\gamma P \rightarrow V\ \alpha P$
$\gamma P \rightarrow V\ \alpha P\ \beta P$
$\gamma P \rightarrow V\ (\alpha P)\ (\beta P)$

We have now arrived at the rules in (65).

(65) a. $\alpha P \rightarrow N$

b. $\alpha P \rightarrow Adj\ N$

c. $\alpha P \rightarrow D\ Adj\ N$

d. $\alpha P \rightarrow N\ \beta P$

e. $\beta P \rightarrow P$

f. $\beta P \rightarrow P\ \alpha P$

g. $\beta P \rightarrow P\ \beta P$

h. $\gamma P \rightarrow V$

i. $\gamma P \rightarrow V\ \alpha P$

j. $\gamma P \rightarrow V\ \alpha P\ \beta P$

There is a common property to all these rules. In each case, all of the constituents are optional, except one. Thus, a verb is the only necessary member of a γP, a noun the only requisite member of an αP and a preposition is all that's required to make a βP. This is just another way of observing that the environments that define these phrases include the environments in which a word class is defined. This relationship is sometimes called "Endocentricity," and I will define it with (66).

The earliest discussions of the need to make Phrase Structure rules Endocentric that I have found are in Bach (1964) and Schachter (1962).

(66) Endocentricity
If $\mathcal{P}$ is a position in a string that a kind of phrase, XP, can occupy, then $\mathcal{P}$ is a position that the kind of word X can occupy.

A present hypothesis is that Endocentricity is a universal feature of phrases. This predicts that there will never be more phrase types than there are word types in a language. If a language doesn't have a kind of word that corresponds to English "preposition," for instance, that language will not have a phrase that corresponds to the English βP. I don't know that this prediction has been systematically tested, and it is not trivial to do so, but I am not aware of counter-examples to it. Let us hypothesize that Endocentricity is a universal feature of phrase structure rules.

Although we have not seen enough to warrant concluding that taking Endocentricity to be universal is valid, it is useful to behave as if it is. That will force the rules we discover to be scrutinized for their relevance to this hypothesis. If making a phrase structure rule for a phrase we discover in a way that is contrary to Endocentricity seems to be necessary, we will have discovered the reasons for jettisoning (66). If the phrases we discover seem capable of being characterized with rules that are Endocentric, then we will have discovered that there is no reason to abandon (66). Prematurely raising patterns we detect in our data to hypothesized laws enables us to discover actual laws by guiding our inquiry towards counter-examples to those patterns. Doing so might also blind us to patterns that are visible only after a certain number of cases are in front of us as well, and so we will want to be alert to the prejudiced image that this methodology delivers. But it has proven to be an effective method of guiding an inquiry away from spurious patterns. I will adopt it throughout.

The word type that is defined by the same criteria that define a phrase type will be a member of every such phrase. Every γP, for instance, will have a verb in it. The word that is in the relationship that Endocentricity defines for the phrase containing that word, is called the phrase's *head*. It is common practice to name the phrases after their heads, so we'll rename αP, NP, βP PP and γP VP. Thus, we now have the rules in (67).

(67) a. NP $\rightarrow$ N
b. NP $\rightarrow$ Adj N
c. NP $\rightarrow$ D Adj N
d. NP $\rightarrow$ N PP
e. PP $\rightarrow$ P
f. PP $\rightarrow$ P NP
g. PP $\rightarrow$ P PP
h. VP $\rightarrow$ V
i. VP $\rightarrow$ V NP
j. VP $\rightarrow$ V NP PP

In addition to these Phrase Structure Rules, we'll need quite a few others. Indeed, the principle of endocentricity leads us to expect that for every category, there could be a Phrase Structure rule that builds a phrase headed by

that category. For example, corresponding to the category Adjective, there could be rules that build adjective phrases; (68) is a good first approximation.

(68) a. AP → A PP
b. AP → A

The presence of PPs within Adjective phrases is supported by the existence of strings like:

(69) a. She is **interested in syntax**.
She is **interested**.
b. He seems **happy with linguistics**.
He seems **happy**.

The coördination test treats A and A+PP strings as being the same, as (70) indicates.

(70) a. She is interested in syntax and happy.
b. He seems happy with syntax and bored.

We'll also need a Phrase Structure rule that tells us how these various phrases are put together to form a sentence. (71) looks roughly right.

(71) S → NP Infl VP

The morpho-syntactic category that sentences are in a one-to-one relation with is Infl, and so in keeping with the convention of naming phrases after their heads, we should change (71) to (72).

(72) IP → NP Infl VP

In this case, however, unlike what we found for the VP, NP and AP rules, Infl is not the only obligatory member of a sentence. It is presently controversial whether sentences are the only phrases that have this property. We will see in later chapters that there are syntactic reasons for the obligatory presence of NP within IP, and likely semantic ones for the obligatory presence of VP.

With this rule we have finally come to the task of characterizing the grammaticality judgments of English speakers. For any speaker of English whose vocabulary has been partitioned into noun, verb, preposition, adjective, determiner and Infl, (72), with the rules in (67), characterizes those strings of words that will be judged grammatical.

(72) says that no sentence can fail to have an Infl between NP and VP, but if Infl are just words such as *can*, *could*, *will*, and so on, this is obviously false. There are grammatical sentences aplenty that fail to have these words in them. (73) is an example.

(73) Zahra walked.

Where is the Infl between *Zahra* and *walked* in this sentence?

If we look hard, we find that sentences are, in fact, in a one-to-one correlation with a category, but that category includes not just words, but bound morphemes as well. Consider the sentences in (74).

Morphemes are "bound" if they are only found within larger words, and "free" if they aren't bound.

(74) a. Zahra leave**s**.
b. Erika lef**t**.
c. Seoyoung ha**s** left.
d. Deniz ha**d** left.
e. Brandon **should** leave.
f. Rong **might** have left.
g. Carolyn desires [Chris **to** leave].
h. Caroline remembers [Coral leav**ing**].

The boldfaced terms have similar distributions: they are found either immediately preceding the verb (if they are free) or affixed onto the following verb (if they are bound). Every sentence has one of these, and so these terms meet the criteria of being the head of a sentence. To explain how it is that those Infls which are bound morphemes materialize affixed onto the following verb, we will have to invoke a process that goes beyond phrase structure rules. Let us put off doing this.

Let's add a couple of additional phrases to our inventory. One of these is a sort of "sentence" found in examples like (75).

(75) a. Zahra said **that Brandon likes chocolate**.
b. Mary recalled the rumor **that Brandon likes chocolate**.
c. **That Brandon likes chocolate** bothers Zahra.
d. Seoyoung is angry **that Brandon likes chocolate**.

Note that the strings following the word *that* meet the conditions imposed by the rule that builds IPs. The word *that* is called a "complementizer" and it is the head of the phrase found in these sentences. This phrase, or clause as sentence-like phrases are often called, is a "Complementizer Phrase" (CP). CPs conform to the requirements of the following Phrase Structure rule.

(76) CP → C IP

Other complementizers are *if* and *whether*, as found in the following examples.

(77) a. I wonder if Zahra likes chocolate.
b. I asked whether Zahra likes chocolate.

Having introduced this constituent, we will now need to revise our previous Phrase Structure rules to include the positions where they may lie. Let me take this opportunity to introduce a new notational convention. Because one of our symbols for a category type—namely, "I"—also is the way the first person singular pronoun is spelled in English, I'd like to find a way of dodging this potential confusion. I will, from now on, adopt the convention of representing terminals with a superscripted "o." Thus the terminal corresponding to the head of a sentence is I°, and the first person singular pronoun is *I*. With this convention in hand, our new battery of phrase structure rules is (78).

The conclusion that sentences are headed by Infl, and that the subordinate sentences that include ***that*** are headed by complementizers, is reached in Chomsky (1986a) and connected to the pioneering work on English clause types in Bresnan (1972). In the generative literature that precedes this work, these two types of sentences were given various treatments. Bresnan (1972) suggested that ***that***-clauses were of the same category as sentences; she called them $\bar{S}$s. Jackendoff (1977) argued that sentences are larger VPs, headed by a verb. The IP/CP model is now standard, though the issue is periodically revisited.

(78) a. IP → NP I° VP
b. IP → CP I° VP
c. CP → I° IP
d. NP → N°
e. NP → Adj N°
f. NP → D° Adj N°
g. NP → N° PP
h. NP → N° CP
i. PP → P°
j. PP → P° NP
k. PP → P° PP
l. VP → V°
m. VP → V° CP
n. VP → V° PP CP
o. VP → V° NP CP
p. VP → V° NP
q. VP → V° NP PP

Note that the option of having a CP in place of an NP at the beginning of a sentence.

The second phrase we'll need are ones headed by adverbs. Adverbs are a word class that is sometimes defined on the position left adjacent to adjectives, as in (79).

(79) a. A **deliberately** angry child
b. The **noticeably** large boat

They can also be found left adjacent to verbs, as in (80).

(80) a. I have **deliberately** misled.
b. I have **noticeably** erred.

We should modify our rules for VPs and APs to reflect these possibilities, then.

(81) a. AdvP → Adv°
b. AdvP → Deg° Adv°
c. AP → Deg° AP
d. VP → AdvP VP

The "Deg" symbol here refers to another morpho-syntactic class of words in English that includes *very* and *so*, and a number of others. It is found in examples like (82).

(82) a. She very quickly walked to town.
b. She is so smart.

These words indicate an "extent" or "measure" of the thing they are modifying. So, for instance, *very* indicates that the angriness of the angry child is high. If the adjective that is to be modified is not "gradable" it cannot combine with these kinds of modifiers.

(83) * a very absolute position

There are some other changes to the VP rule that are necessary. Note, for instance, that VPs may occur immediately following a verb, as in (84).

(84) a. Zahra has walked.
b. Zahra has talked to Shay.
c. Zahra has visited Seoyoung.

We can allow for these options by adding to our rules the one in (85).

(85) $VP \rightarrow V^{\circ}\ VP$

Further, it is possible to find APs embedded within VPs; (86) provides some examples.

(86) a. Alex remains [$_{AP}$ angry at Chris].
b. Kaden is [$_{AP}$ happy with himself].

When APs follow verbs, they may be preceded by, at most, a PP, as in (87).

(87) Kaden seems [$_{PP}$ to Anissa] [$_{AP}$ happy with his rutabagas].

So we change the rule that characterizes VPs to:

(88) a. $VP \rightarrow V^{\circ}\ VP$
b. $VP \rightarrow V^{\circ}\ AP$
c. $VP \rightarrow V^{\circ}\ PP\ AP$

Consider now strings like those in (89).

(89) a. Zahra's book
b. the man's toy
c. the man on the table's nose

These examples involve a possessive or genitive phrase: *Mary's*, *the man's* and *the mon on the table's*. This phrase is an NP with the morpheme *'s* appended to the end. Further, note that this genitive phrase never co-occurs with a determiner, as (90) illustrates.

(90) a. * the Zahra's book
b. * the the man's toy
c. * a the man on the moon's nose

One very typical explanation for this is to understand determiners and possessives as competing for the same position. That can be done by rigging the NP phrase structure rule in such a way that it either produces a DP or a genitive phrase in the same position.

(91) NP → NP's N°

One final Phrase Structure rule is required by the sorts of examples we've so far reviewed. This is the Phrase Structure rule that generates coördinated phrases. This can be done with the following.

(92) $\alpha \rightarrow \alpha$ Conj α

This rule says that a phrase of any category can be made up of two other such phrases with a conjunct stuck between them. I'll call the morphosyntactic class of words that *and* and *or* belong to "conjuncts." Another word that belongs to this category is *but*.

Summarizing, we've now introduced the following battery of Phrase Structure rules:

(93)

IP → NP I° VP	NP → N°	VP → V°	AdvP → Adv°
IP → CP I° VP	NP → AP N°	VP → V° CP	AdvP → Deg° Adv°
CP → C° IP	NP → D° AP N°	VP → V° NP CP	AP → Deg° A°
	NP → N° PP	VP → V° NP	AP → AdvP A°
PP → P°	NP → N° CP	VP → V° NP PP	AP → A°
PP → P° NP	NP → NP's N°	VP → V° VP	
PP → P° PP		VP → V° AP	
		VP → V° PP AP	
		VP → AdvP V°	

2.3 *Recursive Phrases*

The rules in (93) miss many phrases that are the result of combining these rules. For instance, they allow for the VP in (94), but not (95).

(94) She will [$_{VP}$ quickly run].

(95) She will [$_{VP}$ quickly run the race].

We have a rule that allows an Adverb Phrase to precede a verb, but not one that allows that option when the VP contains anything other than one verb. In fact, any VP can start with an Adverb Phrase, so we should re-write these rules so that this option is available for any way in which a VP is formed.

When we do this we should also make it so that there can be more than one adverb that begins a VP, since sentences like (96) are possible.

(96) a. She will probably quickly run the race.
b. She will probably carefully quickly run the race.
c. She will probably carefully never quickly run the race.

This series, one that is formed by adding an adverb to the beginning of a VP that already has an adverb, stops for reasons that are probably not syntactic. We might run out of meaningful adverbs, we might wish to lessen the burden on our hearer, we might get bored; there are many reasons we might decide not to make a sentence with a string of adverbs at the beginning that exceed some certain number. But that number will be determined by these extra-syntactic reasons. We don't want to make the syntax decide that a VP that starts with n adverbs is grammatical, but one that starts with n+1 adverbs is not. For this reason, the usual decision is to allow the phrase structure rules to allow an indefinite number of adverbs in this position, and let other forces determine how this option is exploited. We can achieve this goal, and also connect the rule that builds a VP with an initial adverb with the other rules that build a VP by changing the rule we presently have to (97).

(97) VP → AdvP VP

This rule is recursive, and this is what allows it to make an indefinite number of adverbs at the beginning of a VP.

We will want to do the same thing for PPs inside VPs. They too can be found in VPs indefinitely.

(98) a. Alex danced.
b. Brandon danced Chris Tuesday.
c. Brandon Kaden on Tuesday with Shay.
d. Brandon danced on Tuesday with Shay in the ballroom.
e. Alex danced on Tuesday with Shay in the ballroom around the tables.

We can achieve this with (99).

(99) VP → VP PP

Similar conclusions can be reached about the rules that put APs and CPs within VPs. We have series of sentences that differ only in the number of APs that are in the VP:

(100) a. Rong found her [$_{AP}$ laughing].
b. Rong found her [$_{AP}$ laughing] [$_{AP}$ happy with the findings].
c. Rong found her [$_{AP}$ laughing] [$_{AP}$ happy with the findings] [$_{AP}$ drunk].

This can be captured with (101).

(101) VP → VP AP

There are also series of this kind with CPs, though to see this requires having a better view of what a CP can look like. We won't develop that view for awhile, but when we do we will see that the bracketed strings in (102) are recursive CPs.

(102) a. Brandon thought [$_{CP}$ it had rained].
b. Brandon thought [$_{CP}$ it had rained] [$_{CP}$ when he noticed the wet streets].
c. Brandon thought [$_{CP}$ it had rained] [$_{CP}$ when he noticed the wet streets] [$_{CP}$ as had everyone else].

We can capture this with (103).

(103) VP → VP CP

With these new rules, we can integrate our present rules for VPs so that they better express what strings make a VP. A revision to the VP rules is in (104).

(104) VP → AdvP VP
VP → VP PP
VP → VP AP
VP → VP CP
VP → V° NP
VP → V° VP
VP → V°

A similar situation arises in Adjective Phrases. We've seen that they can include degree words and Adverb Phrases as well as the head adjectives, but as the series in (105) indicates, they may also include an indefinite number of PPs.

(105) a. She was happy with Rong.
b. She was happy with Rong after the meeting.
c. She was happy with Rong after the meeting because of the agreement.

This calls for the rule in (106).

(106) AP → AP PP

An indefinite number of CPs are also possible within an AP:

(107) a. She was happy that it had snowed.
b. She was happy when it started raining that it had snowed.
c. She was happy when it started raining that it had snowed like many people had said.

This can be captured with the rule in (108).

(108) AP → AP CP

And APs can contain an indefinite number of adverb phrases as well.

(109) a. She is usually happy.
b. She is never usually happy.
c. She is probably never usually happy.

We need (110).

(110) AP → AdvP AP

Together, we have the rules for APs in (111).

(111) AP → A°
AP → Deg° AP
AP → AP CP
AP → AP PP
AP → AdvP AP

Our rules for NPs need a similar revision. They presently do not integrate well, and they fail to express where recursion is present. One thing we will need to do is allow NPs to contain an indefinite number of APs, as (112) shows.

(112) a. the big dog
b. the big ugly dog
c. the big ugly hairy dog
d. the big ugly hairy brown dog

We could capture this with the rule (113).

(113) NP → AP NP

Note that, unlike our earlier rule, this one places APs rather than just As inside NPs. That is made necessary by examples like (114).

(114) the [$_{AP}$ very big] dog.

Degree words like *very* are part of adjective phrases, and not noun phrases, so we know that it is an AP that precedes the noun *dog* in (114). This rule also predicts, wrongly, that every other AP can be in this position. For reasons that I do not understand, that isn't always the case. Examples like (115) are ungrammatical, for example.

(115) the happy with Mary boy

Instead, in these sorts of cases, a different word order is preferred:

(116) the boy happy with Mary

This is a fact that will go unexplained. (I leave it to you to solve this.)

The rule in (113) is wrong for another reason, however. It predicts that APs can precede anything that is an NP, and that would allow strings like (117).

(117) happy the dog
happy Mary's dog

The problem with the rule in (113) is that it shouldn't characterize APs, but instead some phrase within APs. These phrases should follow determiners and possessive NPs (like the *Mary's* in (117)), and they should have as their head a Noun. These phrases are often called [en bɑɹz] and represented with the symbol: $\overline{\text{N}}$. We change our rules to include these $\overline{\text{N}}$s.

(118) NP → D° $\overline{\text{N}}$
NP → NP's $\overline{\text{N}}$
NP → $\overline{\text{N}}$
$\overline{\text{N}}$ → AP $\overline{\text{N}}$ $\overline{\text{N}}$ → N°

In addition to APs, PPs and CPs can also occur indefinitely within NPs.

(119) a. the dog behind Smith
b. the dog behind Smith in the room
c. the dog behind Smith in front of Jones in the room

(120) a. the fact that it's raining
b. the fact that it's raining that we disputed
c. the fact that it's raining that we disputed which no one else could believe

I will speculate that these phrase are within $\overline{\text{N}}$s, leading to the rules in (121).

(121) a. $\overline{\text{N}}$ → $\overline{\text{N}}$ PP
b. $\overline{\text{N}}$ → $\overline{\text{N}}$ CP

Our rules for NPs now look like (122).

(122) NP → $\overline{\text{N}}$
NP → D° $\overline{\text{N}}$
NP → NP's $\overline{\text{N}}$
$\overline{\text{N}}$ → AP $\overline{\text{N}}$
$\overline{\text{N}}$ → $\overline{\text{N}}$ PP
$\overline{\text{N}}$ → $\overline{\text{N}}$ CP
$\overline{\text{N}}$ → N°

There is another solution to this problem that we will see below.

Finally, let's reconsider the rules we have for building sentences. These are presently (123).

(123) a. IP → NP I° VP
b. IP → CP I° VP

There is a recursive phrase inside IPs as well that these rules do not capture. This can be seen by the ability of IPs to contain an indefinite number of Adverb Phrases right before the head I°.

(124) a. They probably will run.
b. They apparently probably will run.
c. They occasionally apparently probably will run.
d. They occasionally apparently probably never will run.

To capture this requires that there be a phrase within an IP that contains the head I° and the VP, and that this phrase be recursive. The rules should have the form in (125) and be wedded to new IP rules like those in (126).

(125) XP → AdvP XP
XP → I° VP

(126) IP → NP XP
IP → CP XP

XP has I as its head, but it shouldn't be IP because that would make the rules in (126) recursive and lead to the possibility of sentences with an indefinite number of NPs at the beginning.

(127) * [$_{IP}$ Rong [$_{IP}$ Michael [$_{IP}$ Seoyoung [$_{IP}$ will dance]]]].

The solution here is to allow for a different kind of phrase—a smaller IP—that is called [aɪ baɹ], and symbolized with $\bar{\text{I}}$. The new rules for sentences are (128).

(128) IP → NP $\bar{\text{I}}$
IP → CP $\bar{\text{I}}$
$\bar{\text{I}}$ → AdvP $\bar{\text{I}}$
$\bar{\text{I}}$ → I° VP

Together, these revisions deliver the body of phrase structure rules in (129).

(129) a. IP → NP $\bar{\text{I}}$
IP → CP $\bar{\text{I}}$
b. $\bar{\text{I}}$ → AdvP $\bar{\text{I}}$
$\bar{\text{I}}$ → I° VP
c. NP → $\bar{\text{N}}$
NP → D° $\bar{\text{N}}$
NP → NP's $\bar{\text{N}}$
d. $\bar{\text{N}}$ → AP $\bar{\text{N}}$
$\bar{\text{N}}$ → $\bar{\text{N}}$ PP
$\bar{\text{N}}$ → $\bar{\text{N}}$ CP
$\bar{\text{N}}$ → N°
e. VP → AdvP VP
VP → VP PP
VP → VP AP
VP → VP CP
VP → V° NP

VP → V° VP
VP → V°

f. AP → A°
AP → Deg° AP
AP → AP CP
AP → AP PP
AP → AdvP AP

g. AdvP → Adv°
AdvP → Deg° Adv

h. PP → P°
PP → P° NP
PP → P° PP
PP → P° IP

I've added to our battery of PP rules one that allows PPs to include an IP so that we can capture cases like (130).

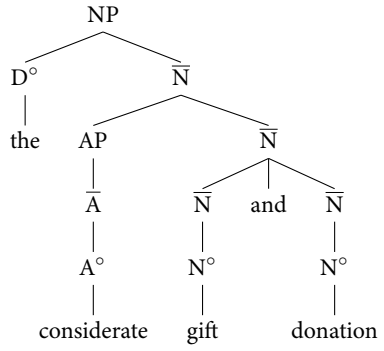
(130) Mike danced [$_{PP}$ before [$_{IP}$ Rodica danced]].

There is a feature of this method of representing these subfamilies that I would like to draw attention to now. It allows for two separate parses of examples such as (131).

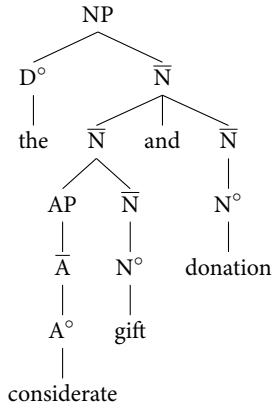
(131) the considerate gift and donation

It is possible to produce this string either by grouping *considerate* and *gift* into one $\overline{N}$ and conjoining that with an $\overline{N}$ consisting of just *donation*, or it is possible to conjoin *gift* and *donation* into one $\overline{N}$ and then group that phrase with *considerate* into an $\overline{N}$. It is easy to represent these two parses by way of "phrase marker trees," which graphically elucidate the constituent structure of strings. The two ways of producing (131) are represented by the trees in (132).

(132) a.



b.



Phrase marker trees have two interpretations in the literature. They can be regarded as a record of how the rules were used to produce the string. When interpreted this way, they are called "derivational trees." But they can also be understood as objects created by the rules. On the second interpretation, they are part of the syntax, not merely a method of showing how the syntax created a particular string. It is this second interpretation that is the more common, and it is how we will understand phrase markers. See Lasnik and Kupin (1977) for discussion.

We might note that there are two meanings attached to this string as well, having to do with how the meaning of *considerate* is combined with the meanings of the rest of the parts. A loose paraphrase of these two meanings might be as given in (133).

(133) a. the things which are considerate and which are, first, a gift and, second, a donation

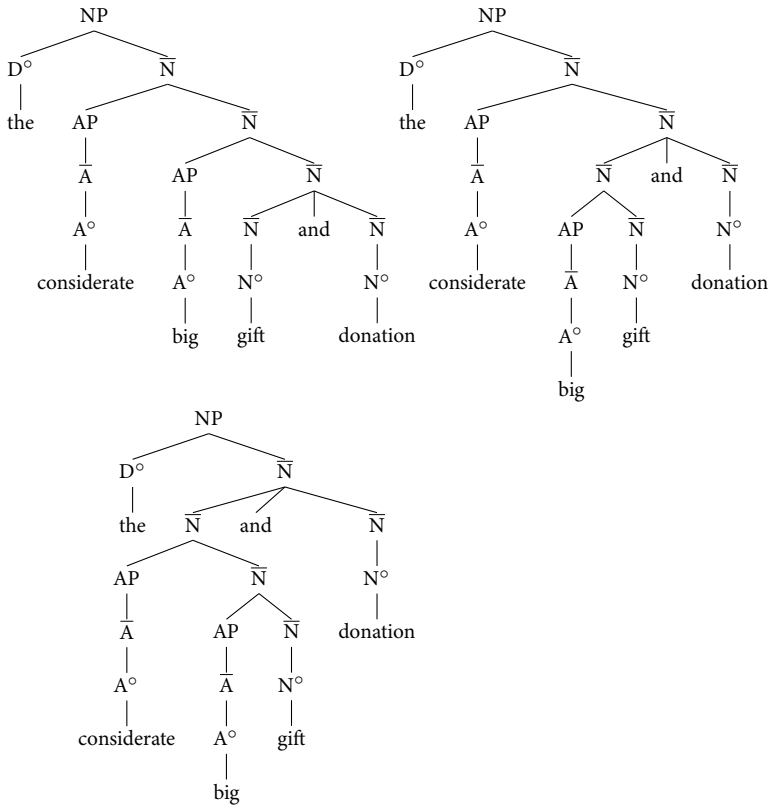
b. the things which are, first, a considerate gift and, second, a donation

The difference in these meanings tracks the semantic groupings we discussed earlier. In (133a), the meaning of *considerate* is combined with the meaning of *gift and donation*. In (133b), the meaning of *considerate* instead combines with the meaning of *gift*. These semantic groupings mirror the syntactic groupings in (132).

There are a variety of reasons for thinking that syntactic phrasing always maps onto semantic groupings in the way that (132) and (133) illustrate. For one thing, the number of meanings and the number of parses matches. For instance, if we add one more adjective to the left of the coördinated nouns, as in (134), our rules allow for a total of three parses, shown in (135), and there are three meanings as well, as indicated in (136).

(134) the considerate big gift and donation

(135)



(136) a. the things which are considerate and big and are also a gift and a donation.

b. the things which are considerate and are also a big gift and a donation

c. the things which are a considerate big gift and a donation

There is, then, a correspondence between the number of groupings that these rules afford and the number of meanings that can be associated with the relevant strings. This suggests that the syntactic phrases these rules produce correspond to the semantic groupings. This is what's expected from our thesis that phrases are just those strings that make compositional meanings.

Furthermore, the meanings vary in a predictable way with the linear order that these terms are arranged in. Thus, for instance, putting the second adjective to the left of the coördinated nouns creates the three meanings listed in (136), whereas putting the second adjective to the left of just the rightmost noun, as in (137), produces just two readings: they are paraphrased in (138).

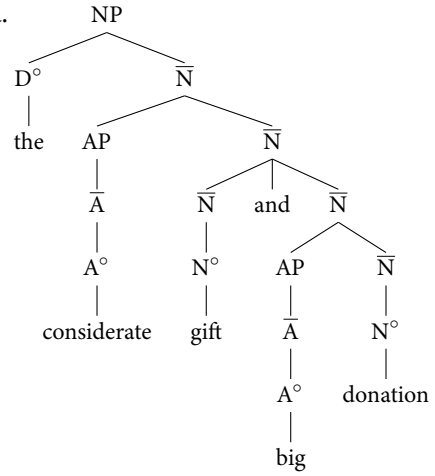
(137) the considerate gift and big donation

(138) a. the things which are considerate and both a gift and a big donation

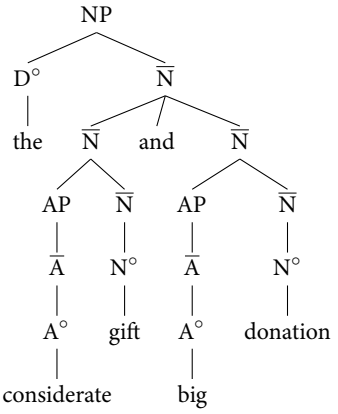
b. the things which are a considerate gift and a big donation

This is predictable in the sense that our syntactic characterization of these strings would deliver just the two parses for (137) shown in (139).

(139) a.



b.



This correspondence should give us some courage that we are on the right track in characterizing the infinite strings under discussion in terms of self-recursive phrases. It provides a set of structures that are in correspondence with what look like a parallel set of meanings. Our next step should be to flesh out this correspondence by spelling out the semantic rules that deliver the compositional meanings. But we have some work still to do in characterizing these basic facts about that portion of a syntax that concerns just the morpho-syntactic types of words and phrases. Let's return to that task.

All the phrase structure rules we've come up with (they're back in (129)) meet the requirement of Endocentricity. Every phrase is matched to a head. That is, every phrase type is defined by criteria that are also part of the criteria that define a word type. Because our goal is to explain the *bloresnick* phenomenon, those criteria are just "position in a string." So, for example, the phrase VP is defined as the set of strings that can be in these positions: immediately following an Infl, immediately following a verb, immediately preceding a PP, and so on. The set of strings that make VP will always be found in one of these positions, and no others (according to our rules). The set of positions that define a verb are larger. They include, for instance, immediately preceding an NP. But they also include all of the positions that define a VP: immediately following an Infl, immediately following a verb, immediately preceding a PP, etc. The law of Endocentricity says that this will always be true of phrases in natural language syntax, and so far our rules have not needed to stray from this requirement. For many of the word type/phrase type pairs in (129), another correspondence holds. For almost every word type, there is a distinct phrase type. For instance, there is a set of strings that has this Endocentric relationship to the word type verb—those are the VPs. And there is a set of strings that has this Endocentric relationship to adjectives: the Adjective Phrases. There are two word types, however, that do not have accompanying phrase types. These are determiners and degree words. We might ask why this is. What makes degree words and determiners special?

One popular answer is that the specialness of determiners and degree words isn't that they don't have accompanying phrases, but that their phrases are just harder to recognize. Let's see how this answer plays out in detail for determiners. I'll merely gesture to how it works for degree words; I won't work through the details for degree words since it would require diving into a construction type (comparatives) that we don't need to understand.

For determiners, the proposal is to take the head of the strings I've been calling NPs to be the determiner rather than the noun. This proposal for determiners is credited to Steven Abney's dissertation.[2] We won't have time to examine all his arguments for this view. To see them requires knowing more about how rules relating sentences work. I'll present those argument in a later chapter, when we've gotten to the point where they will make sense. At this stage, I just want you to see this proposal as another way of parsing nominals and to get a feel for some of the considerations that doing so turns on. This DP Hypothesis gives us rules like (140).

[2] Abney, Steven. 1987. The English noun phrase in its sentential aspect. Doctoral Dissertation, Massachusetts Institute of Technology

(140) a. $DP \rightarrow DP's\ \overline{D}$
b. $DP \rightarrow \overline{D}$
c. $\overline{D} \rightarrow D^{\circ}\ NP$
d. $\overline{D} \rightarrow D^{\circ}$
e. $NP \rightarrow \overline{N}$
f. $\overline{N} \rightarrow \overline{N}\ PP$
⋮

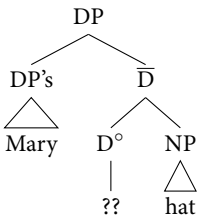
These rules predict, in line with the stronger form of Endocentricity, that every DP will have a determiner in it and that the smallest DP is just a determiner. Both of these predictions are not obviously true. Maybe the best case that a DP can be made up of just a determiner are those like (141).

(141) a. Many have wondered how syntax can be interesting.
b. Few understand why phonology belongs to linguistics.
c. (looking at a list of phrase structure rules) Each has its own flaw.

For the strings *many*, *few*, and *each* to be NPs and still obey Endocentricity in these examples would require there to be a silent noun in them. But in these cases there isn't any evidence, so far as I am aware, for this silent noun. Normally, silent pieces of a sentence share with the ellipsis process that we saw in the previous chapter the requirement that the silent words be spoken somewhere else in the conversation. That isn't what's happening in (141). If determiners are the heads of these phrases however, then Endocentricity is satisfied by these strings.

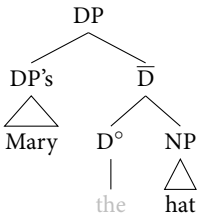
There are several examples which appear to counter-exemplify the prediction that every DP has a determiner in it. One such is found with DPs whose specifiers are filled, as in (142).

(142)



These are semantically roughly equivalent to *the hat that belongs to Mary*. There is an element to the meaning of these phrases that we cannot identify with any of the words spoken within it. We can identify the meaning "that belongs to," perhaps, with the possessive *s*, and of course *Mary* and *hat* make their contributions to the phrase *Mary's hat*. But in addition, we need to understand where the meaning associated with *the* comes from. *Mary's hat* doesn't mean "a hat belonging to Mary" or "some hat belonging to Mary," it means "THE hat belonging to Mary." There is a specific determiner meaning associated with this phrase, and this suggests that there is a determiner within this phrase. Perhaps we could credit the "the" part of this meaning to a silent determiner? Let's take that view. This will give to these kinds of phrases the structure in (143).

(143)



The need for silent determiners might be indicated elsewhere in English too. Many DPs cannot fail to have a determiner.

(144) * hat fell.

But there are exceptions. One salient group of exceptions are plurals and the "mass" nouns we encountered earlier.

(145) Hats fell.
Rice poured.

Mass nouns are grammatically singular, but somehow semantically inherently plural. The noun *rice*, for instance, refers to a group of kernels, even though it does so as a singular. And *water*, another mass noun, refers to a group of water parts even though it is singular. You can see that the thing *water* refers to contains a plurality of things that are also *water* by noticing that if you half a quantity of water, the two things you have are each still called *water*. Plurals—whether grammatically plural or inherently plural—are capable of being used without expressing a determiner along with them.

There is evidence, however, that there is a silent determiner in these cases. As in the case of possessives, this can be seen by playing close attention to the meanings of such expressions. To see this, consider first the meaning of the DP *no crackers* in (146).

(146) I have brought no crackers.

Like determiners generally, *no* has a meaning that bears on the amount of the things that are denoted by the NP that follows. In the case of *no*, that quantity is zero. The information that (146) conveys is that there isn't anything that qualifies as "crackers" that I have brought. If, by contrast, (146) were *I have brought many crackers*, the determiner *many* would convey the information that the amount of crackers this sentence is talking about is numerous.

(147) I have brought many crackers.

(147) says that there are crackers that I have brought and that they are numerous. We can learn something about the meaning that *crackers* has by comparing *no crackers* to *many crackers*. The contribution *crackers* makes to these two DPs is merely to describe the things that the determiner is quantifying. Note in particular that the meaning of *crackers* can't include information about the quantity of cracker things; that job is entirely performed by the determiner. A way of paraphrasing the meanings of these DPs that makes salient what the contribution made by *crackers* is, is (148).

(148) a. *no crackers* = there is no thing that is crackers
b. *many crackers* = there is a numerous thing that is crackers

The meaning of *crackers* is the part expressed by "that is crackers." When these DPs are semantically integrated into the larger sentence, we get meanings that can be expressed in stilted English with (149).

(149) a. *I have brought no crackers* = there is no thing that is crackers which I have brought.
b. *I have brought many crackers* = there is a numerous thing that is crackers which I have brought.

The "there is X" part corresponds to the contribution made by the determiner, the "that is crackers" part corresponds to the contribution made by *crackers*, and the "which I have brought" part corresponds to the contribution made by *I have brought*.

Consider now what the meaning of (150) is.

(150) I have brought crackers.

A paraphrase of (150) that uses the same kind of recipe I used to produce (149) is (151).

(151) *I have brought crackers* = there is a thing that is crackers which I have brought.

The *that is crackers* part corresponds to *crackers*, and the *which I have brought* part corresponds to *I have brought*, but there is also in this paraphrase a "there is X" part. This is the meaning that is associated with the so-called indefinite determiner *a*, as we can see by considering (152).

(152) *I brought a cracker* = there is a thing that is cracker which I have brought.

There appears to be an element of the meaning associated with (150) that corresponds to that found for the indefinite determiner *a*.) This could be explained if there is in fact an indefinite determiner in (150), but obviously one that isn't pronounced.

A perhaps relevant fact about determiners in English is that some of them are sensitive to the plurality of the noun they combine with. The determiners *every* and *a*, for instance, are grammatical only with singular nouns, and the determiner *many* is grammatical only with plural nouns.

(153) a. every hat
b. * every hats
c. a hat
d. * a hats
e. * many hat
f. many hats

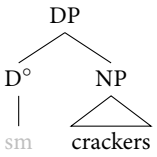
In some languages related to English, there is a plural version of the indefinite determiner *a*. Perhaps we could view English as also having a plural indefinite determiner, one whose morpho-phonology gives it no phonological material. A common idea is that there is plural version of *a* in DPs with plural nouns that otherwise don't seem to have a determiner. Let's adopt that view. It will

But see Carlson (1977) for a different way of thinking about these facts.

give to DPs of this kind, the parse in (154), where "sm" represents the silent plural indefinite determiner.

I choose the symbol "sm" to remind one of the reduced, unstressed, version of the determiner ***some***, which has a very similar meaning.

(154)



Pronouns are also expressions which seem to be DPs—they have the distribution that DPs have, at any rate—but which do not seem to come with determiners. These are another counterexample, then, to the view that every DP must include the determiner that is its head. Postal (1969) argued that pronouns simply *are* determiners, however, noting that under certain conditions, personal pronouns can combine with common nouns:

(155) a. We linguists

b. Them knuckleheads

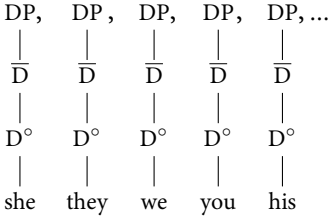
Outside of these specialized expressions, pronouns cannot coöccur with any of the material normally associated with DPs.

(156) a. * Unhappy he left the classroom.

b. * He unhappy left the classroom.

Perhaps this can be explained if pronouns are determiners which, except in the special cases that (155) discovered, are determiners that do not like to combine with an NP. Let us adopt this view about pronouns; they are determiners which head DPs in the way that (157) illustrates.

(157)



Finally, proper names do not appear with determiners in English. Interestingly, they do in many closely related languages, and in those languages the determiner that appears is the equivalent to English's *the*. Perhaps English has a silent *the* in these scenarios too, but the evidence for this is considerably more ephemeral. When a name is expressed in English, there is even less that can accompany that name than there is with a pronoun. We have very little information about what the internal shape of the DPs that contain names is. This case is presently just a mystery. I will adopt the convention of expressing my ignorance about DPs with names in them by hiding the structure of those DPs with a triangle:

Though see Longobardi (1994) for an illuminating idea about the problem.

(158)



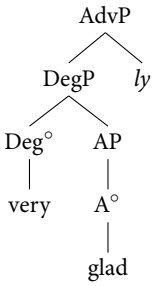
If these arguments are correct, and bracketing the problem raised by names, then every nominal phrase in English is accompanied by a determiner. As we've seen, however, there are nomninal phrases that do not come with a noun. This speaks on behalf of the conclusion that nominal phrases are in the Endocentric relationship with the category determiner, and not noun. We should henceforth know them, then, as DPs.

The problem posed by the absence of degree phrases has a similar solution. We should see (some) APs as actually being Degree Phrases that contain AP. This can be achieved with the rules in (159).

(159) a. DegP → Deg° AP
b. AP → A°
c. AP → AP PP
d. AP → AP CP
e. AP → AdvP AP

This too is a standard solution for degrees and adjectives, but because the literature on degree phrases isn't as large as nominals, it's not as well known. The case of AdvPs has not been widely examined, but I suspect that the AdvPs that have degree words in them are actually APs converted to AdvPs. To put this idea in a diagram, it would involve embracing something like (160).

(160)



Let's set aside the cases of AdvPs and APs that have degree words within them.

If we adopt these innovations, our rules become (161).

(161) a. CP → C° IP
b. IP → DP $\overline{\text{I}}$
IP → CP $\overline{\text{I}}$
c. $\overline{\text{I}}$ → AdvP $\overline{\text{I}}$
$\overline{\text{I}}$ → I° VP
d. DP → DP $\overline{\text{D}}$
DP → $\overline{\text{D}}$
$\overline{\text{D}}$ → D° NP
$\overline{\text{D}}$ → D°
e. NP → $\overline{\text{N}}$
f. $\overline{\text{N}}$ → AP $\overline{\text{N}}$
$\overline{\text{N}}$ → $\overline{\text{N}}$ PP

$\overline{N} \rightarrow \overline{N}$ CP
$\overline{N} \rightarrow N^{\circ}$

g. VP → AdvP VP
VP → VP PP
VP → VP AP
VP → VP CP
VP → V° DP
VP → V° VP
VP → V°

h. DegP → Deg° AP

i. AP → AP CP
AP → AP PP
AP → AdvP AP
AP → A°

j. AdvP → Adv°

k. PP → P° NP
PP → P° PP
PP → P° IP
PP → P°

Notice that because NPs no longer contain determiners and possessives, the need we had to invoke $\overline{N}$s is no longer present. It was the way in which determiners and possessive are positioned within nominals that led us to embrace the existence of $\overline{N}$s, but once these terms are now recognized as part of DPs, $\overline{N}$s are no longer needed. That is reflected in the rules in (161) by seeing that the only NP rule left is the one that makes it an $\overline{N}$. This amounts to saying that $\overline{N}$s are always just NPs. We could, then, change the rules for NPs so that they no longer include $\overline{N}$s, and we would capture the same set of phrases. I haven't done that renovation, however, for reasons that will be clear soon.

Consider how our structural method of stopping the recursion relates the terms that are within some phrase. We expect that those terms which are introduced in the terminal expansion "$\overline{X} \rightarrow X^{\circ}$ …" or XP → X° …(that is, the non-recursively introduced terms) will form the most inclusive substitution class of the phrase involved. There are some kinds of phenomena which suggest that this expectation is fulfilled. There are processes, for example, in which a rather surprisingly short string can substitute for one or another of the families we have discovered. This happens under conditions of anaphora.

For example, the VP family can be anaphorically connected to other VPs with the words *do so*. When a VP is made up of *do so*, its meaning derives from a VP spoken elsewhere. This is illustrated in (162).

"Anaphora" refers to processes in which a phrase in one position refers, in some fashion or other, to the same things that another phrase, in a different position, refers to. For instance, in the sentence: ***Mary knows that she is smart***, it is possible for ***she*** to refer to the same individual that ***Mary*** refers to. In such a case, we say that ***she*** is anaphoric to, or with, ***Mary***.

(162) a. Kaden must leave town, but Shay mustn't **do so**.
do so = "leave town"

b. Kaden should eventually read *Aspects*, and Shay should immediately **do so**.
do so = "read *Aspects*"

From the examples in (163), we learn that DPs cannot be a sister to **do so**.

(163) a. * Kaden must leave Chicago and Shay must **do so** New York.

b. * Kaden must understand *Aspects* and Shay must **do so** *Syntactic Structures*.

This follows if *do so* cannot stand in place of a V°, but can stand in place of a VP. Because DPs are never sisters to VP—only sisters to a V°—they cannot combine with *do so*

(164) A VP can be pronounced as *do so*, if its meaning is the same as a VP spoken elsewhere.

This way of distinguishing the recursively available parts and the non-recursively available parts of phrases also predicts that the non-recursive parts will always come between the head of their phrase and the recursive parts. This seems true sometimes, as in (165) and (166).

(165) a. Jill ate it at noon.

b. * Jill ate noon it.

(166) a. Jill ate spätzle at noon.

b. * Jill ate at noon spätzle.

But for other cases it seems uncertain, or downright wrong, as in (167) and (168).

(167) a. Jill ate the rotting kumquats.

b. Jill ate at noon the rotting kumquats.

(168) a. ?? Jill said [that you shouldn't eat kumquats] at noon.

b. Jill said at noon [that you shouldn't eat kumquats].

This then, is the difficulty in trying to place certain DPs within VP (and other phrases too, as we'll see). Let's set this problem aside momentarily. It will be the focus of a lot of our work in the chapters that follow.

There is a similarity to the organization of the family of substitution classes that make up DP and IP. Abstractly, they both have one of the forms in (169).

(169) a. $XP \rightarrow YP\ \overline{X}$
b. $XP \rightarrow \overline{X}$
c. $\overline{X} \rightarrow ZP\ \overline{X}$
d. $\overline{X} \rightarrow \overline{X}\ WP$
e. $\overline{X} \rightarrow X^{\circ}\ MP$
f. $\overline{X} \rightarrow X^{\circ}$

In addition to these rule forms, the other rules we have include these possibilities.

(170) a. $XP \rightarrow X^{\circ}$
b. $XP \rightarrow X^{\circ}\ YP$
c. $XP \rightarrow XP\ WP$
d. $XP \rightarrow YP\ XP$

We might speculate that these are the only kinds of phrases allowed in a syntax. If so, this would be a major contribution to the issue of understanding how the learning sequence succeeds. If the only kinds of phrases that a child entertains as possible are those that can be produced by rules of the forms in (169) or (170), then the space of possible guesses the child makes when updating her grammar will be very usefully constrained. Although we have only looked at a small set of sentences in one language, it is useful to prematurely raise the patterns we detect to hypothesized principles of this sort because doing so forces us to find and confront counter-examples. We will formulate patterns we think we detect in our rules into laws. As we look at new data to be characterized, we will use these laws to shape our analysis; we will try to formulate our models so that the conform to those laws. When that becomes impossible, or drives us to implausible conclusions, then we will have discovered how that original pattern was mis-analyzed. We will discover not only how the law we formulated is wrong, but also exactly how it is wrong. That will, with luck, allow us to find a different way of formulating the law so that it is more accurate, or, to recognize that the original pattern we fashioned our law for didn't exist. Prematurely raising generalizations to laws like this, gives us an ordered way of exploring the data we need to account for. It steers how we parse the facts that we are trying to model. Without this order, we have no way of navigating the search for patterns and the things that are responsible for those patterns. Our search through data become just an exercise in cataloguing.

Of course, this heuristic brings with it the danger that the laws we are trying to conform to will bias how we look at the data we gather. It imposes a prior bias that is entirely a function of where our investigation started. So we will have to be diligent about facing this bias as we go forward. We should periodically reassess what our laws look like and how well that fit to the patterns we are detecting.

At this juncture, however, let us adopt the hypothesis that all of the phrase structure rules of a syntax must look like one of the set in (169) or (170). In fact, we can do even better with a simple modification to the rules we've seen so far. We can translate all of the rules that have the forms in (170) into rules that have the form in (169), and lose none of the facts we have explained. The rewriting is simple: we simply let every XP be an $\overline{X}$, and then use that $\overline{X}$ to introduce the head or the recursion. Rewriting the rules along these lines gives us the set in (171).

(171)
a. $CP \rightarrow \overline{C}$
   $\overline{C} \rightarrow C^{\circ}\ IP$
b. $IP \rightarrow DP\ \overline{I}$
   $IP \rightarrow CP\ \overline{I}$
c. $\overline{I} \rightarrow AdvP\ \overline{I}$
   $\overline{I} \rightarrow I^{\circ}\ VP$
d. $DP \rightarrow DP\ \overline{D}$
   $DP \rightarrow \overline{D}$
   $\overline{D} \rightarrow D^{\circ}\ NP$
   $\overline{D} \rightarrow D^{\circ}$
e. $NP \rightarrow \overline{N}$
f. $\overline{N} \rightarrow AP\ \overline{N}$
   $\overline{N} \rightarrow \overline{N}\ PP$
   $\overline{N} \rightarrow \overline{N}\ CP$
   $\overline{N} \rightarrow N^{\circ}$
g. $VP \rightarrow \overline{V}$
h. $\overline{V} \rightarrow AdvP\ \overline{V}$
   $\overline{V} \rightarrow \overline{V}\ PP$
   $\overline{V} \rightarrow \overline{V}\ AP$
   $\overline{V} \rightarrow VP\ CP$
   $\overline{V} \rightarrow V^{\circ}\ NP$
   $\overline{V} \rightarrow V^{\circ}\ VP$
   $\overline{V} \rightarrow V^{\circ}$
i. $DegP \rightarrow \overline{Deg}$
j. $\overline{Deg} \rightarrow Deg^{\circ}\ AP$
k. $AP \rightarrow \overline{A}$
l. $\overline{A} \rightarrow \overline{A}\ CP$
   $\overline{A} \rightarrow \overline{A}\ PP$
   $\overline{A} \rightarrow AdvP\ \overline{A}$
   $\overline{A} \rightarrow A^{\circ}$
m. $AdvP \rightarrow \overline{Adv}$
   $\overline{Adv} \rightarrow Adv^{\circ}$

n. $PP \rightarrow \overline{P}$

o. $\overline{P} \rightarrow P^{\circ}\ NP$
$\overline{P} \rightarrow P^{\circ}\ PP$
$\overline{P} \rightarrow P^{\circ}\ IP$
$\overline{P} \rightarrow P^{\circ}$

This seems wrong in the sense that many of the XPs in this list amount to always just being $\overline{X}$s. If it turns out that VPs, for instance, are indeed always the same thing that $\overline{V}$s are, then this move is unwarranted. We should allow for the forms in (170). We'll see soon, however, that there is indeed a kind of phrase that a VP can be that a $\overline{V}$ cannot be. And we'll discover the same thing for APs and PPs as well. Adverb Phrases are too small—they never have many words in them—and NPs are too difficult to find, for us to be able to see that they differ from their $\overline{X}$s. But the bulk of evidence suggests that in fact only the forms in (169) are what the syntaxes of English use. And that hypothesis also seems to be reasonably well supported in the syntaxes of other languages that have been well studied. At present, a good hypothesis is that these are the only kinds of phrase structure rules found.

Sometimes this constraint on phrase structure rules is described by saying that they must conform to the $\overline{X}$ Skeleton.

(172) $\overline{X}$ Skeleton: $XP \rightarrow \overline{X}$
$XP \rightarrow ZP\ \overline{X}$
$\overline{X} \rightarrow QP\ \overline{X}$
$\overline{X} \rightarrow \overline{X}\ WP$
$\overline{X} \rightarrow X^{\circ}\ YP$
$\overline{X} \rightarrow X^{\circ}$

The symbols "X," "Y," "Z," "Q" and "W" should be thought of as variables that range of morphosyntactic category names.

There are names for the positions that the $\overline{X}$ Skeleton creates. ZP is called the *Specifier* of XP; WP, QP are called *Adjunct*(s); and YP is called the *Complement* of X.

These definitions of "Specifier," "Adjunct" and "Complement" come from Jackendoff (1977), as do most of the basic ideas sketched in this section about how phrases are organized. Jackendoff's work is a detailed working out, and considerable extension, of proposals in Chomsky (1970), where there is also a proposal about the underlying building blocks of morpho-syntactic category. Chomsky's work has its seeds in Harris (1946).

It should be said that these rules leave out considerable detail. If you have a native English speaker handy, it won't take much time to discover many other kinds of combinations of words that form grammatical sentences. I will not provide an exhaustive list of the phrase structure rules for English. Instead, I will develop guidelines that you can use to make up those phrase structure rules when you confront the need to. One of those guidelines is that the rules you come up with must conform to the $\overline{X}$ Skeleton.

2.4 Arguments and Modifiers

Charting substitution classes in the way we've been doing characterizes grammatical arrangements of words solely in terms of the words' categorial status. It throws strings of words together according to their status as nouns, verbs,

adjectives and the like and ignores all other properties of the particular verbs, nouns, adjectives, etc. chosen. It forms some pretty odd sentences:

(173) a. Kaden danced at noon at midnight.
b. Kaden slowly stood still.
c. a green idea

These are odd because the meanings they deliver are so bizarre. But they are still recognizable as grammatical strings of words.

But some combinations which these rules allow seem to go bad in a very different way. Consider (174).

(174) a. Kaden laughed Zahra.
b. Hotze gave it at Anissa.
c. Shay died that you should eat better.
d. Rong claimed to Chris.
e. Kaden slapped.

These don't go together into weird meanings; they don't go together at all. These examples, arguably, should be marked as syntactically ill-formed. They are ungrammatical, and so our syntax should be revised.

What's wrong here is that we've matched up verbs with the material that follows them incorrectly. As can be seen by comparing (174) with (175), if the verbs are followed by different material, the results are grammatical.

(175) a. Kaden laughed.
b. Hotze gave it to Anissa.
c. Shay died.
d. Rong claimed that you should eat better.
e. Kaden slapped his thigh.

Here, then, is something more particularly about the words themselves that seems to be relevant to the procedure that recognizes grammatical strings. To capture what goes on here, we must do more than know what the category of the words being combined is.

There's another respect in which the particular choices of words seems to play a role in the syntax. Consider the different semantic contributions the DP *Tuesday* makes in (176).

(176) a. I danced Tuesday.
b. I remember Tuesday.

In the first case, *Tuesday* indicates when *I danced* happened. We say in this case that *Tuesday* is a modifier. It modifies the sentence's meaning by restricting the events denoted by *I danced* to just those that transpire on Tuesday. But this is not the role it has in the second case. Here *Tuesday* refers to the

thing remembered. We say in this case that it is an argument of the relation that *remember* denotes.

A similar contrast can be seen in the pair in (177).

(177) a. I kissed her on the bus.
b. I put her on the bus.

Again, *on the bus* is a modifier in the first case. It locates the event described by *I kissed her*; it indicates that this event took place on board the bus. In the second case, by contrast, it names a location related to the referents of *I* and *her* by *put*. It is an argument.

The semantic role an argument has in a sentence is determined by the word for which it is an argument. The meaning that modifiers contribute to the sentence they're part of is much less dependent on the meanings of the words and phrases nearby. For a kissing to be on the bus is the same thing as a running, a playing, a singing, etc. is to be on the bus. All VPs combine semantically with in the same way. Moreover, whether can combine with a VP is not determined by the verb in that VP. Modifiers of VPs are not chosen by the verbs heading those VPs in the way that arguments are. However, modifiers do seem to be choosey about the kind of phrases they modify. Adjectives, for instance, can semantically combine with the projections of nouns but not verbs, and adverbs can semantically combine with projections of verbs, but not nouns.

When the arguments are DPs or, sometimes, PPs, there's a way of talking about argumenthood that is commonplace, and which we inherit from Gruber (1965) and Jackendoff (1972). Gruber was concerned with the problem of verb meanings, and in particular with finding a theory that restricted the kinds of argument types that verbs permit. He speculated that there was a finite, in fact quite small, set of argument types, or 'roles', that could be put together by verbal meanings. He argued that the roles which verbs related were always ones that have to do with the metaphysics of motion. For example, a verb like *send* involves three terms, one that can be seen as indicating the source of the motion, another that denotes the moved term and a third that names the goal, or endpoint, of that motion. Gruber called the role borne by the term undergoing motion 'Theme.'

Fillmore (1968) makes a very similar proposal to Gruber's.

(178) Jason sent his book to Masashi.
Source *Theme* *Goal*

On Gruber's thesis, then, verbs are just words that name different kinds of motion. The arguments of verbs will therefore always be the objects, locations, and manners of movement that are involved in that motion. Arguments will have roles in the motion relation named by a verb, and what those roles are will be determined by the verb. Those roles Gruber called "theta roles," after the Theme role. You will often see "theta role" expressed as "θ-role." We say, then, that verbs assign θ-roles to their arguments.

There are cases where it is not obvious that motion underlies the relation named by a verb. Nothing has to move in the events described by (179), for example.

(179) a. Jason showed his book to Masashi.
b. Jason pleases Masashi.

Perhaps in these cases, one might imagine that there is a kind of abstract motion involved. The image of *his book* might be transferred from Jason to Masashi in (179a), for example; and pleasure might be thought to have moved from Jason to Masashi in (179b). Gruber often resorted to this sort of analysis to maintain the thesis that all verb meanings could be related to the physics of motion. But there are cases which even metaphorical extensions of the logic of motion look unlikely to characterize. (180) are some.

(180) Jason finds Masashi upleasant.
Masashi is uhappy.
'Londres' refers to London in Portuguese.
This means that something is wrong.
Jason became unhappy.
They know something.

That the way humans conceive of motion is employed in constraining the range of verb meanings has been pretty firmly abandoned. But the idea that the possible range of verb meanings emerges from the ways in which a finite set of elemental predicates, like CAUSE, MOVE, and BECOME, as well as θ-roles, remains a live idea. On this conception, the roles that arguments play in a sentence can be viewed as resulting from being the arguments of these more basic predicates. We will continue to talk about verbs and the θ-roles they assign, but we won't be constrained by Gruber's theory of what a θ-role can be. And we should allow that the relations that verbs have with their arguments might go beyond what the stock of θ-roles that Gruber, Jackendoff, and others have imagined.

Dowty (1979, 1989) are good points of entry into this literature.

The relation between verbs and their arguments that is expressed by θ-roles can be seen as a special instance of a more general relationship which goes under the name "selection," or sometimes "s-selection" (for "semantic" selection). This refers to the connection between a verb's (or other similar term's) meaning and the semantic values that its arguments deliver. θ-roles express a similar function: they name the meaning that an argument's semantic value must be compatible with. But the relation holds for other cases too, where the language of θ-roles does not so easily extend. One of those places is where verbs connect with clauses of various types. So, a verb's meaning determines somehow whether the clause it combines with must have the meaning of an interrogative or a declarative, for example.

There is a subtle difference between what θ-roles are and what s-selection is. When a predicate s-selects an argument, it specifies what kind of meaning its argument must have. When a predicate assigns a θ-role to an argument, it determines how the meaning of its argument will be composed into the meaning of the sentence. In practice, these different notions cover nearly the same phenomena, since how a term's meaning is composed into a sentence is closely tied to what meanings that term may have.

(181) a. Erika denied that Kaden has left.
b. Erika said that Kaden has left.
c. * Erika wonders that Kaden has left.

(182) a. * Erika denied whether Kaden has left.
b. Erika said whether Kaden has left.
c. Erika wonders whether Kaden has left.

We say of these cases that verbs select or s-select a question or declarative. Note that some verbs are compatible with either, as is *say*.

Now that we have in view this distinction between arguments and non-arguments, let's return to the contrast between (174) and (175), repeated below.

(174) a. * Kaden laughed Zahra.
b. * Hotze gave it at Anissa.
c. * Shay died that you should eat better.
d. * Rong claimed to Michael.
e. * Kaden slapped.

(175) a. Kaden laughed.
b. Hotze gave it to Anissa.
c. Shay died.
d. Rong claimed that you should eat better.
e. Kaden slapped his thigh.

In all these examples, I've chosen phrases that cannot get an interpretation as a modifier. The contrast between (174) and (175), then, hinges on giving the verbs involved the right argument. The post-verbal phrases differ with respect to their category. What we discover from these examples is that verbs are picky with respect to what category their arguments belong to. This can also be seen in examples, such as those in (183) and (184), where the difference in the category of an argument is not tangled up so much with the meaning of that argument.

(183) a. Andrew pleases Rong.
b. * Andrew pleases to Rong.

(184) a. * Andrew talks Rong.
b. Andrew talks to Rong.

We say that verbs are *subcategorized* by the category of their argument. Or—this term has been re-lexicalized—that verbs subcategorize their arguments. Sometimes this is also described as a verb *c-selecting* its argument.

The terms ***s-select*** and ***c-select*** come from Pesetsky (1982).

Jackendoff (1977) argues that arguments (when they follow the head they are an argument of) are necessarily in complement position. This is supported by contrasts like those in (185) and (186), which make use of the *do so* construction.

These sentences should be pronounced with contrastive focus on the terms in small caps.

(185) a. Erika will talk about this TODAY and Shay will do so TOMORROW.

b. * Erika will remember TODAY and Shay will do so TOMORROW.

(186) a. Erika will talk after DINNER and Shay will do so after LUNCH.

b. * Erika will talk to ANISSA and Shay will do so to KADEN.

Because the phrase following the verb is an argument in (185b) and (186b), Jackendoff's thesis requires it to be in complement position. This means that it must be within the $\overline{\text{V}}$ which gets mapped onto *do so*. In (185a) and (186c), the phrase following the verb is a modifier and can therefore remain outside the $\overline{\text{V}}$ that *do so* stands in place of.

Recall that we saw that ***do so*** cannot stand in the head position of a VP. Thus, the $\overline{\text{V}}$ it stands in for would, according to Jackendoff's thesis, necessarily have to "contain" any arguments that the verb whose meaning ***do so*** also "contains" takes.

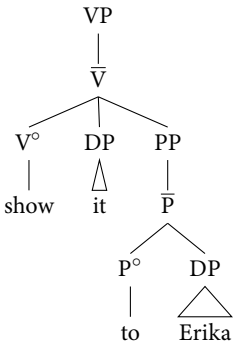
A problem for Jackendoff's thesis is that a verb may have two internal arguments, and in these cases neither of those arguments can stand outside *do so*.

(187) a. * Erika showed it to ANISSA and Shay will do so to KADEN.

b. * Erika put it on the TABLE and Shay will do so on the SOFA.

Jackendoff thought that there could be two complement positions, and that a parse for a VP headed by *show* might be as in (188).

(188)



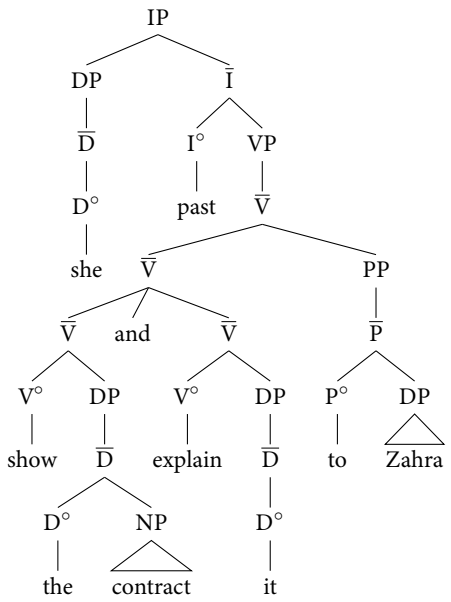
This isn't countenanced by the $\overline{\text{X}}$ Skeleton we've adopted, which allows for only one complement position. (Indeed, our $\overline{\text{X}}$ Skeleton allows no phrase to have more than two daughters.) There is evidence that the structures our parses deliver are correct, however. We can discover with coördination, for instance, that the second argument can be a sister to a $\overline{\text{V}}$ that contains the first argument.

(189) a. She both showed the contract and explained it to Zahra.

b. She first put it and then rolled it under the table.

The parse for (189a) is (190). (I will leave out *both*.)

(190)



If V̄s with two arguments in them are parsed in the way our rules require—rather than Jackendoff's—we cannot give a simple structural account of the conditions on *do so* anaphora. Instead, the law that determines what *do so* can stand in place of will have to be given by something like (191).

(191) *do so* Anaphora
do so stands in the position of a verbal projection, X, and gets its meaning from some verbal projection, Y spoken elsewhere. Y must contain a verb and all of its arguments.

Okay, to summarize: we're looking for a way to factor into our procedure for recognizing grammatical sentences enough of the meanings of the words involved to guarantee that verbs and nouns (and perhaps other words) combine with the arguments they select and subcategorize for. Moreover, when these arguments follow them, we must find a way of guaranteeing that they are in some X̄ headed by that argument taking verb, noun, what have you.

We can ensure that these arguments are in one of the X̄s if we force them to bear a θ-role, and allow θ-roles to be assigned only to positions within the X̄. We need also to describe the fact that when a verb has a θ-role, there must be an argument present in the syntax which bears that θ-role. It is customary to divide this task into two parts, which can be expressed as follows:

(192) Theta Criterion

a. For every θ-role associated with a word, there is a syntactic position to which that θ-role is assigned.

b. For every θ-position, there is something with an appropriate semantic value that occupies that position (i.e., the argument).

It is usual to strengthen the Theta Criterion to a bijection between arguments and θ-roles, because of cases like (193).

Remember my warning about making arguments from coördination: there are ellipses process that arise in coördinations that can obscure what the two coördinated phrases actually are. One of those ellipses processes is what's called Right Node Raising, and it is the one that threatens to undercut my argument here. In Right Node Raising, material in the first of two conjuncts can be elided when it matches something in the second conjunct. This, for instance, is what's happened in (i).

i. Janina talked to, and Winnie talked about, Pius.

A diagnostic property of Right Node Raising is that material at the end of the first conjunct must be contrasted with something in a parallel position in the right conjunct. In (i), for instance, ***to*** is contrasted with ***for***. (See Hartmann (1998) and Féry and Hartmann (2005).) I have tried to construct an example that doesn't have this property with the hope that it will block a Right Node Raising analysis.

The Theta Criterion comes from Chomsky's important ***Lectures on Government and Binding***.

(193) Anissa showed Chris *doesn't mean* Anissa showed Chris himself.

Without constraining the Theta Criterion to a bijection, we might expect (193) to get the disallowed interpretation since presumably the DP *Chris* could name the object which bears both the Theme and Goal θ-roles. So I'll change (192) to (194).

(194) The Theta Criterion

a. For every θ-role associated with a word, there is a unique syntactic position to which that θ-role is assigned.

b. For every θ-position, there is something with an appropriate semantic value that occupies that position (i.e., the argument).

Although this principle is stated in terms of θ-roles, it is meant to hold of all arguments, even those for which the language of θ-roles is not often used. Thus, an equivalent, though less usual, way of formulating the Theta Criterion is (195).

(195) Theta Criterion
For every argument specified in a word's lexical semantics, there is a unique syntactic position where that argument must reside.

We need to worry about cases like the following, of course, in which there appears to be an optional argument.

(196) a. Hotze ate (pie).

b. It seems (to me) that Shay left.

Here we might imagine either that there actually is an object in these cases that bears the θ-role, but that argument is unpronounced. Alternatively, we might conjecture that something relaxes the condition which forces every θ-role to be assigned to a position holding an argument. The common wisdom is that both possibilities exist—we will return to this issue in the next chapter. For now, let us imagine that there is a lexically determined process which allows θ-roles for certain predicates to avoid the Theta Criterion.

The Theta Criterion glues the positions that arguments can be in to the positions that θ-roles are assigned to. Our next step should be to specify where those positions are. But we might also linger for a moment and ask whether we should write the Theta Criterion in terms of c-selection instead of θ-roles, or "s-selection." Might we say, for instance, that if a word c-selects a certain kind of phrase, there must be a single position that is occupied by a phrase of that type. There is an intimate connection between the kind of phrase c-selected, and the semantics of the θ-role it is assigned, so it is not easy to tell which of these is relevant.

In fact, so intimate a connection is there between s-selection and c-selection, that some have suggested that they should be collapsed. It is sometimes suggested that the categorial type some argument has can be determined from its

θ-role. Jane Grimshaw[3] has provided a way of viewing this hypothesis which has gained some popularity. Her idea is that one of the functions that makes up the learning device assigns a categorial status to arguments on the basis of their θ-role. She calls this function "Canonical Structural Realization" (CSR). She sketches how this function might work in the special case of CPs and DPs.

[3] Grimshaw, Jane. 1979. Complement selection and the lexicon. ***Linguistic Inquiry*** 10:279–326

So let's look as some of the facts she considers. Note first that CPs may distinguish themselves as according to whether they have meanings that we might describe as "Propositions," "Exclamatives" and "Questions." DPs can denote things of these types too, sometimes. Some examples are in (197).

(197) a. Rong asked me { what the time is / the time } (Question)

b. I'll assume { that he's intelligent / his intelligence } (Proposition)

c. Michael couldn't believe { how hot it is / the heat } (Exclamative)

In these cases, then, the verbs s-select either Q(uestion), P(roposition) or E(xclamative) and c-select either an DP or CP.

There are other verbs, however, which s-select these very same meanings, but c-select only CPs. Some examples are in (198).

(198) a. Michael wondered { what the time was / *the time } (Question)

b. I'll pretend { that he's intelligent / *his intelligence } (Proposition)

c. Rong complained { how hot it was / *the heat } (Exclamative)

Here then, we have a special instance of the difference in s-selection and c-selection that needs to be overcome if one is to be derived from the other.

Grimshaw's suggestion is that the CSR of Questions, Propositions and Exclamatives is CP and that those verbs which c-select DPs with these meanings are learned on a case-by-case basis. Thus, this is a partial collapse of c-selection to s-selection. It predicts that every verb which s-selects a Question, Proposition or Exclamative will c-select a CP; that is, there should be no verbs that c-select only DPs with this meaning. This seems to be correct.

Whether or not this project can be maintained for the situation involving the relation between CPs and DP and the meanings they have, I don't think a parallel story holds for the complements of other categorial type. Moreover, the scheme Grimshaw proposes won't help determine which verbs select non-finite as opposed to finite clauses, which also seems to be a rather language particular fact. The problem we face here is the same problem we face when we try to understand whether morphosyntactic word-class can be expressed

in solely semantic terms. It remains unknown what the relationship between word-class and semantic type is, and consequently, we are not in a position to yet understand what the connection, if any, is between c-selection and s-selection. From now on let us assume that c-selection is at least in part independent of s-selection, and determined on a verb-by-verb basis.

Interestingly, however, it looks like the thesis that c-selection can be derived from s-selection fares better when external arguments are concerned. To begin with, the range of categories that serve as external arguments looks somewhat less varied; to a large extent, only DPs and CPs seem to be clausal subjects in English. And second, when a θ-role is consistent with either DP and CP, any kind of CP is possible as is an DP:

With the exception of cases like "Under the bed is a slipper," plausibly instances of impersonal constructions with inversion; see Stowell (1981), Rochemont and Culicover (1990), and Bresnan (1994).

(199) { That Coral left / To have to leave / Leaving / The fact } { bothers Hsin-Lun. / makes Ivy happy. }

By contrast, when the subject θ-role is incompatible with the meanings that CPs yield they are banned from Specifier position:

(200) { ∗That Coral left / ∗To have to leave / ∗Leaving / Rodica } { kisses Hsin-Lun. / likes Ivy. }

Subject arguments do not seem to be c-selected by the verbs involved. The θ-role they bear is sufficient to determine what category they can be. Only complements are c-selected.

This is a reason, then, for writing the Theta Criterion in terms of θ-roles, and not c-selection. We want to link both subject and object arguments to positions relative to the word they are arguments of, but only the object argument will be c-selected.

We can summarize what we've discovered so far with (201).

(201) a. If a verb has a θ-role, then there is a unique syntactic position to which that θ-role is assigned.

b. A θ-marked position must be occupied by something with the appropriate semantic value.

c. A verb c-selects only its object arguments.

The statements in (201a) and (201b) are the Theta Criterion, whereas (201c) concerns the relation between c-selection and s-selection which we've just reviewed. The Theta Criterion insists that for every θ-role that some term has, there will be a unique position occupied by an argument bearing that θ-role. (201c) determines whether that argument will be c-selected or not.

We now must specify where the θ-positions are for the subject and object θ-roles. For object θ-roles, we've already seen that those positions are inside

the $\overline{\text{X}}$ projection of the word that is assigning θ-roles. There is evidence that those θ-marked positions must be closer to the θ-role assigner than positions which are not θ-marked. This can be seen by the fact that it is the first DP or PP in (202) that is understood as the object argument.

(202) a. She remembered Tuesday Wednesday.
b. She put them on it in today's class.

It is not possible to understand *Wednesday* to be the object of *remember* in (202a); and it is not possible to understand *Tuesday* to be the modifier of the VP in (202b). The θ-role that *remember* assigns to its object is occupied by *Tuesday* and not *Wednesday*. Similarly, it is not possible to understand *in today's class* to be an argument of *put* in (202b), nor is it possible to understand *in it* to be a modifier in (202b). The θ-role that *put* assigns to its second argument it must assign to the position that *in it* occupies, and not the position that *in Today's class* occupies. We want to force object arguments to be in the smallest $\overline{\text{X}}$ possible.

To force arguments to be within the smallest $\overline{\text{X}}$, it will now be sufficient to force the θ-position for that argument to be within the smallest $\overline{\text{X}}$. We want this effect for object arguments only—we don't want to force "subject" arguments into $\overline{\text{X}}$—so one way of doing this would be to restrict those positions that are c-selected to just those within the smallest $\overline{\text{X}}$.This would mean that we'd have two principles: one that determines the c-selected position for verbs, and another, yet to be determined, which locates the s-selected position for subjects.

Distinguishing "subject" argument from direct and indirect "object" arguments in this structural way is defended in Williams (1980).

We will follow a popular view of these principles that is first found in Chomsky's *Lectures on Government and Binding*. He formulates there what he calls "The Projection Principle," which is responsible for mapping the argument structure of a verb—or argument taking word more generally—into a syntactic representation. I will formulate his principle as (203).

(203) The Projection Principle
a. If β is c-selected by α, then there must be a projection of α, π, that immediately dominates β and if π dominates some other phrase, γ, then γ must be another projection of α or part of another c-selected argument of α.
b. If α assigns an external θ-role to β, then β must be in the Specifier of IP closest to α.

Chomsky's own formulation of this condition builds in various other properties that we will encounter later on; compare (203) to Chomsky (1981, pp. 34–48). An "internal θ-role" is one that is assigned to a c-selected term. An "external θ-role" is the one assigned to the subject argument.

This formulation of the Projection Principle is informal, and must be made more precise for it to be satisfied in examples like (190). (203a) should read something like this: "If α assigns an internal θ-role to β, then there must be a projection of α that immediately dominates β and doesn't dominate any modifiers." (The goal here is to let coördinators, like ***and***, be inside the projection.)

The Projection Principle, in concert with the Theta Criterion, forces object arguments of a verb to be in the lowest $\overline{\text{V}}$.

2.5 *Deriving Phrase Structure Rules*

An interesting consequence of the Projection Principle is that it factors into the lexical specification of the verbs everything needed to know what sort of phrases will be found in the non-recursive part of $\overline{\text{X}}$s. Tim Stowell, in his

dissertation,[4] argues that the information about what resides in the lowest $\overline{\text{V}}$ should only be found as part of the verb's lexical content. This sort of information doesn't properly reside in the phrase structure rules, since it is information that is tied to the particular choice of word and not the pure form that sentences may take. In fact, the information phrase structure rules give about the contents of the smallest $\overline{\text{X}}$ can now be seen as merely a summation of what is possible across particular lexical items filling the head slot. Thus, we should factor out of the phrase structure rules information which concerns the categorial nature of the complements involved. We can do the same for the subject arguments as well, since their categorial nature is derived from their θ-role.

[4] Stowell, Timothy. 1981. Origins of phrase structure. Doctoral Dissertation, Massachusetts Institute of Technology
A very similar conclusion is reached in Ann Farmer's dissertation.

What we have seen, then, is that the phrase structure rules can be stripped of a great deal of their information. Indeed, what is left is largely what the $\overline{\text{X}}$ skeleton expresses and the categorial specification of non-arguments. We could complete factoring out the phrase structure rules into general statements, then, by simply listing the class of phrase-types that can be non-arguments inside other phrases. We could add to the $\overline{\text{X}}$ Skeleton a language particular set of statements of the form in (204).

(204) a. If α modifies $\overline{\text{N}}$, then α must be ...
b. If α modifies $\overline{\text{V}}$, then α must be ...
⋮

I'll say that a phrase, α modifies another phrase, β, if the semantic composition rule that composes a compositional meaning from them is not of the argument/argument taking type.

The "..." will carry lists of category types. We might imagine doing the same thing with respect to Specifiers. We could envision the grammar of a language having a set of statements like (205).

(205) a. If α is in the Specifier of IP, then α is a DP or CP.
b. If α is in the Specifier of DP, then α is a DP's.

If this project is successful, then the Phrase Structure rules of English collapse in full to the $\overline{\text{X}}$ Skeleton, the c-selection requirements of heads and statements like (204) about what kind of phrase can modify another. Some have argued for a picture of language variation that makes the hierarchical arrangements of constituents that the $\overline{\text{X}}$ Skeleton, together with the Projection Principle and Theta Criterion and whatever yields (204), completely immutable.[5] All that varies across languages is the linear order in which the terms that follow the arrows in the $\overline{\text{X}}$ Skeleton may have. So, the phrase structure component of the grammar might have nothing more than (206) in it, where "$\{\alpha, \beta\}$" should be understood as representing both the string α+β and the string β+α.

[5] See Travis (1984), for example.

(206) a. $\text{XP} \rightarrow \overline{\text{X}}$
b. $\text{XP} \rightarrow \{\alpha\text{P}, \overline{\text{X}}\}$
c. $\overline{\text{X}} \rightarrow \{\overline{\text{X}}, \beta\text{P}\}$
d. $\overline{\text{X}} \rightarrow \{\text{X}^{\circ}, \gamma\text{P}\}$
e. $\overline{\text{X}} \rightarrow \text{X}^{\circ}$

What the morpho-syntactic category of α, β and γ are is fully determined by the c-selection properties of X° and the language particular principles governing modifier types (i.e., (204)). And we'll understand "X" to be a variable over morpho-syntactic category which is determined by the morpho-syntactic category of the morpheme that is inserted into the X° position. Thus, for instance, if *hat* is inserted into X° position, then X is N, and if *run* is inserted into X°, then X is V.

The expression "X°" refers to the position in a phrase-marker that only a lexical item, a morpheme, can be fit into. It is the position of the "head" of the phrase.

The linear arrangements of these constituents must then be determined by the language particular part of the grammar. There is evidence from language typology that whatever it is that determines the order of Specifier and $\overline{\text{X}}$ is independent of what determines the order of heads and their complements. There is no widely agreed upon account of what is responsible for this factor, so let's leave this for the future. There is also typological evidence[6] that the order or complement and head correlates with the order of modifier and $\overline{\text{X}}$. One way of expressing this is with a "headedness parameter" which specifies whether the head of a phrase may come initially or finally in its (immediate) projection. This predicts that complements will either all precede or follow their heads, and not come among them. While this is not superficially true (German/Dutch, for example, look like counterexamples), it does look like this could be true of the underlying arrangements of these constituents and it does seem to capture a tendency that languages have.

[6] See Greenberg (1963) and Dryer (1992).

As a starting point, then, let's take the view that languages linearize their phrasal constituents by way of setting separately the linear order of the immediate constituents of XP and $\overline{\text{X}}$. This can be achieved by letting languages pick the values "first" and "last" for the terms in (207).

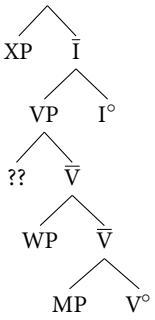
(207) If σ is an immediate daughter of π, π a projection of α, then choose a value in "$< >$" for:
   a. σ = Specifier of π: $<$first in π, last in $\pi>$
   b. σ = the term π is a projection of: $<$first in π, last in $\pi>$, modulo (207a)

β is a "projection of α" iff β dominates α and they have the same head.

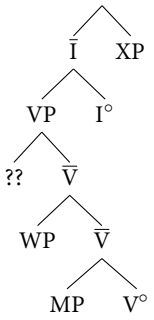
This connects the linear order of head and complement to the linear order of head and adjunct, which Dryer's work suggests might be correct.[7] So, for instance, a language that sets Specifier to "first" and Projection-of-α to "last" will license sentence structures like those in (208a), whereas a language that sets both terms to "last" will produce structures like those in (208b).

[7] See also Saito (1985) and Saito and Fukui (1998).

(208) a.



b.

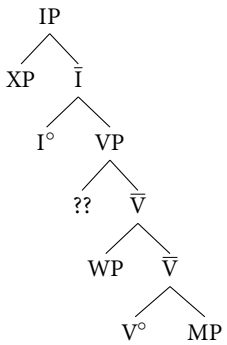


The categorial values for MP will be determined by the c-selection specification of the verb involved. The categorial values for XP will be determined by the θ-role it receives. And the categorial values for WP will be whatever (206) for the language in question allows to modify $\overline{\text{V}}$s. We haven't yet discovered what sits in the Specifier of VP, so this spot is marked with "??."

The linearization parameters in (207) produce these phrase markers in the following way. Setting Specifier to "first" in (208a) linearizes XP and ?? so that they precede $\bar{\text{I}}$ and $\overline{\text{V}}$ respectively. Setting Projection-of-α to "last" makes every other $\bar{\text{I}}$ and $\overline{\text{V}}$, as well as I and V, follow the phrase they are sisters to. As a consequence WP, MP and VP precede the phrase they are complements to or modifiers of. In (208b), by contrast, Specifier is set to "last," which linearizes XP and ?? so that they follow $\bar{\text{I}}$ and $\overline{\text{V}}$ respectively. As with (208), Projection-of-α is set to "last" in (208b) and the consequence for the position of WP, MP and VP is the same.

Restricting the linearization options to just those in (207) blocks certain phrase markers. It blocks languages, for instance, in which the complement to a verb falls on a different side of that verb than does a complement to a noun (or any other category). That is, it forces languages to unify the linearization of Specifier, Complement and modifier across phrase types. It is not hard to find languages that seem to violate this restriction, but as Greenberg and Dryer find, there is a tendency for languages to avoid this type. Similarly, (207) prevents all languages that put modifiers to one side of the $\overline{\text{X}}$ they modify but put complements to the other side. For instance, phrase markers like (209) are prevented.

(209)

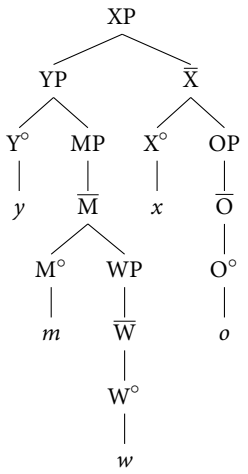


This phrase marker linearizes $\overline{\text{V}}$ (a projection of V°) "last" relative to WP, but linearizes V° (which is also a projection of V°) "first" relative to its complement. Clearly there are languages of this unexpected type. English, for instance, seems to look precisely like (209).

This proposal, then, seems clearly too restrictive. Nonetheless, it will be our starting point. In the chapters that follow we will explore ways of loosening this model so that it is enabled to account for the range of language types we do see without losing the trends in linear organization that Greenberg and Dryer have discovered. What we have now is not yet complete enough to really engage this problem.

In fact, the linearization scheme in (207) is itself not yet complete enough to generate the strings we want to associate with the phrase markers it allows, for example those in (208a) and (208b). All (207) does is linearize the phrases within a sentence. It does not determine how the strings of words within those phrases are linearized relative to the other phrases. To see this, consider a phrase marker like that in (210), in which lower-cased letters should be understood as representing words.

(210)



This phrase marker arises by imposing the $\overline{\text{X}}$ Skeleton and setting Specifier to "first" and Projection-of-α also to "first." What we would like is for this to be sufficient to generate the string $ymwxo$. Instead, however, all that these settings give us is the information in (211).

(211) a. Y precedes MP
b. M precedes WP
c. YP precedes $\overline{\text{X}}$
d. X precedes OP

What's required is something to determine how the information in (211) determines the linear order of y relative to the words within MP and $\overline{\text{X}}$, the linear order of m relative to the words in WP, and the linear order of x relative to the words within OP. Let's turn to that now.

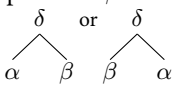
The relationship between phrase markers and the strings they map onto is investigated in a vast literature, where a wide array of different formalizations can be found. Two widely read classics are Chomsky (1985) and Lasnik and Kupin (1977). For a simple introduction to the formalisms of graph theory (from which phrase markers are drawn) and their projection onto strings, see Partee et al. (1990).

Recall that in defining morpho-syntactic category, we entertained the hypothesis that looking at only adjacent terms would be sufficient for defining the relevant substitution classes. As it happens, in defining phrases we have also obeyed this constraint. As a result, phrases are always strings of adjacent terms. Let's elevate this too to an hypothesis:

(212) Contiguity
Let $\tau = \{ \alpha_1, \alpha_2, ..., \alpha_n \}$ be terminals dominated by ζ. The string formed from τ cannot contain β if β is not dominated by ζ.

If Contiguity holds, it is possible to determine from (211) what the consequent linearization for all the words in (210) is. If the words in MP must be adjacent to each other, then (211a) is enough to know that y precedes all those words (i.e. m and w). Similarly, if the all the words in YP must form a contiguous string, and all the words in $\overline{\text{X}}$ must too, then from (211c) it is possible to deduce that every word in YP (= y, m and w) must precede every word in $\overline{\text{X}}$ (= x and o). All that is required is an explicit statement that the words within a phrase, α, are linearized with respect to the words in phrase β in the same way that α is linearized to β. This can be done with (213).

(213) $\alpha < \beta =_{\text{def.}}$ α precedes β.
$\{_\delta\ \alpha, \beta\} =_{\text{def.}}$



a. For all words, x and y, within a phrase marker, either $x < y$ or $y < x$.

b. Let X and Y be points on a phrase marker. If X < Y, then $x < y$ for all x dominated by X, and all y dominated by Y.

(213a) merely makes explicit that all the words in a phrase marker must have a linear relation to every other word in a phrase marker. (213b) determines how these linear relations are derived from the language particular orderings imposed upon phrases. It also derives Contiguity.

On this view, then, the phrase structures of languages are the result of four fixed universals—the $\overline{\text{X}}$ Skeleton, the Theta Criterion, the Projection Principle, and the linearization principles in (213)—plus the language particular pieces of information in (214).

(214) a. Specifying the categories of adjuncts (i.e., (204)).
b. Setting the "headedness parameter" (i.e., (207)).
c. A vocabulary of lexical items that s-select and c-select arguments.

There's a sense, then, in which languages do not actually have Phrase Structure rules. They are merely the epiphenomena that emerge when the various factors of Universal Grammar and language particular information are combined. This theory, if correct, meets the criterion of explanatory adequacy. It provides both inviolable constraints (i.e., $\overline{\text{X}}$ Theory, the Theta Criterion, the Projection Principle and (213)) and an evaluation metric (i.e., (214) and the language particular vagaries of vocabulary).

In moving from a battery of English specific phrase structure rules to the more explanatory interaction between $\overline{X}$ Theory and the c-selection requirements of verbs, language particular settings of modifier types, etc., we have lost some information. Because that transition removed any reference to categories, it is no longer possible to order complements in situations, like (215), where there are more than two.

(215) Ivy put this on the table.
compare: *Ivy put on the table this.

This information was conveyed in the phrase structure rules by way of referencing category type. We had rules such as (216) for instance which ensure that if a verb is followed by two complements, the first will be the DP.

(216) $\overline{V} \rightarrow V^{\circ}$ DP
$\overline{V} \rightarrow \overline{V}$ PP

Because the $\overline{X}$ Skeleton does not have information about category type in it, it is not possible to use the $\overline{X}$ Skeleton to order complements. Nor would we want to rely on the c-selection requirements of verbs to do this. That would amount to the claim that the order of complements varies as a function of the verb involved. But the fact of English is that no matter what verb is selected, the complements line up in the way that (216) requires.

There must be another component of the grammar which expresses this information. Indeed, we have several outstanding problems concerning the position that arguments take. For instance, we have the paradoxical behavior of CP objects yet to understand. How are object CPs at once within the lowest $\overline{X}$ with respect to constituency tests like *do so* anaphora, but also linearized so that they appear after modifiers that are not within this $\overline{X}$? Among the loose-ends fraying at the edges of the proposals presented in this chapter, the correct characterization of argument placement looms large. This will be the subject of the next chapter.

3
Positioning Arguments

Let's turn now to the question of what fixes the order of complements. We'll focus here on the question of what determines the linear position of the internal arguments to verbs. Recall that the way in which internal arguments to verbs are ordered does not vary with the choice of verb. We will not want to encode the order of arguments on a verb by verb basis, then. Instead, the relevant factor in determining the order of objects seems to be the categorial type that the objects are. If we focus on DP, PP and CP, the pattern our initial phrase structure rules encoded as follows.

(1) DP PP
DP CP
PP PP
PP CP

(2) a. I told [$_{DP}$ her] [$_{CP}$ that she should join].
b. * I told [$_{CP}$ that she should join] [$_{DP}$ her].
c. Amy Rose explained [$_{PP}$ to me] [$_{CP}$ that I should join].
d. ?* Amy Rose explained [$_{PP}$ that I should join] [$_{PP}$ to me].
e. Taka told [$_{DP}$ it] [$_{PP}$ to Della].
f. ?? Taka told [$_{PP}$ to Della] [$_{DP}$ it].

As a simplifying measure, we consider only finite CPs, that is: CPs whose IP contains tense inflection or a modal.

I have not considered cases where a verb's internal arguments are both CPs. These cases are either very rare or non-existent. I've also not considered cases where a verb's internal arguments are both DPs. That scenario does arise, but it has some rather special properties. We will come back to consider those cases later.

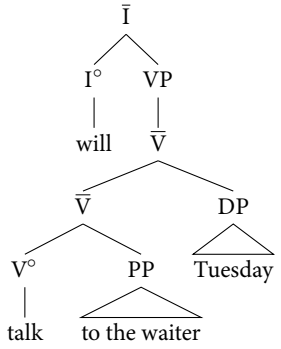
In his dissertation, Tim Stowell argued for an answer to this question that has been influential. He suggested that systems of "Case assignment" are responsible for ordering these terms. DPs differ from other kinds of phrases in Indo-European in being able to host Case morphology. In English this happens with pronouns only: the pronouns *she* and *her*, for instance, have the same meaning but have a different "Case" form. The particular Case borne by a pronoun is determined by its syntactic position. In languages

that are richer in Cases than English is, we can see that the Case borne by a DP is determined by a term proximate to the DP. In German, for instance, a certain class of prepositions and verbs determine Accusative Case for their complements, while others determine Dative Case. It is also often the case in other Indo-European languages that DPs other than just pronouns can bear Case morphology—sometimes on the head of the DP, sometimes on the determiner, sometimes spread across modifiers of various sorts within the DPs. Let's imagine then that, in general, DPs must be related to Case assigners. We'll call this the Case Filter.

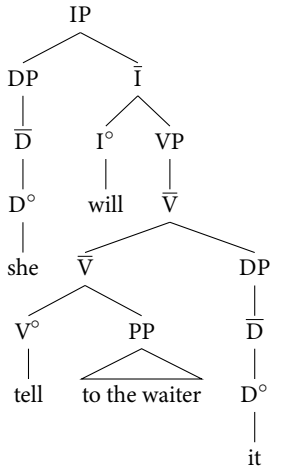
(3) Case Filter
A DP must be assigned Case if it is an argument.

I've restricted the Case Filter to argument DPs because modifying DPs do not seem to be positioned in a way that suggests they are sensitive to Case assignment. In (4), for instance, the DP *Tuesday* is not in a position that an argument DP, like the pronoun *it*, can be.

(4)



(5)



Moreover, typically the Case morphology that modifying DPs bear in languages where Case morphology is found on them is fixed, and not sensitive to nearby Case assigners. The Case filter, then, requires that some parse in a sentence's derivation puts every argument DP which that sentence contains in a Case marked position.

As I noted before, Case marked positions are ones that are close to terms that are responsible for assigning (i.e, determining) the Case. So, all we have to do now is know what those terms are and what "close" means, and we'll be able to use the Case filter to rule out certain strings as sentences. The "object" Cases—so-called Accusative, Dative, Locative, and a host of others—are assigned by particular lexical items. In English there are just three cases: accusative, nominative and genitive. The form of pronouns in English depends on this Case information, as well as the number, person, and, in some cases, gender of the referent of the pronoun.

	Nominative	*Accusative*	*Genitive*
1st, sing	I	me	my
1st, plur	we	us	our
2nd, sing	you	you	your
2nd, plur	you	you	your
3rd, sing, fem	she	her	her
3rd, sing, masc	he	him	his
3rd, sing, neut	it	it	its
3rd, plur	they	them	their

Our immediate task, then, is to determine what terms are responsible for assigning these Cases. The genitive appears on DPs that are in the Specifier of DP in English, as in (6).

(6)
a. **your** hat *compare:* ***your** left.
b. **his** marbles *compare:* *I talked to **his**.
c. **our** problems *compare:* *She met **our**.

We put off for a couple chapters investigating what the item is that assigns Genitive; there are facts about the shape of DPs that we have yet to discover that are relevant.

Nominative Case is found on the "subjects" of finite IPs, as in (7).

(7)
a. **She** left. *compare:* *I left **she**.
b. **We** arrived before **he** left. *compare:* *I talked to **he**.
c. **They** thought **I** would stay. *compare:* ***they** hat

Finite IPs are distinguished from other phrases by the presence of a finite I°: tense/agree inflection or a modal. Moreover, the nominative Case-marked DP appears in Specifier of IP, at least in the examples we have examined so far. (We will encounter soon cases that don't meet this description.) An initial stab at the conditions on nominative Case assignment, then, is (8).

(8) Finite I° assigns Nominative Case to its Specifier position.

Accusative Case is found on DPs that immediately follow some verbs and prepositions, as in (9).

(9)
a. We met **him**. *compare:* ***Him** left.
b. They talked to **me** *compare:* *They are happy **me**.
c. She stood near **us**. *compare:* ***us** hats.

It appears that verbs and prepositions are the terms that assign accusative Case. Tim Stowell argued that one of the conditions on accusative Case assignment is adjacency. We will see soon that additional conditions are required too. An initial stab at the conditions on accusative Case assignment, then, is (10).

(10) Accusative Case is assigned to the position adjacent to a verb or preposition.

It perhaps seems sensible to say that pronouns are subject to the Case filter. Pronouns simply have no pronunciation that isn't sensitive to the syntactic position they are in. To say a 3rd person singular female pronoun, for instance, requires choosing the nominative form (*she*) or the accusative form (*her*) or the genitive form (again: *her*), and those forms depend entirely on whether that pronoun is in one of the positions assigned Case. But all other DPs can be pronounced in English without morphology that depends on where that DP is in a sentence. So the evidence that all (argument) DPs are subject to the Case filter is less obvious. Let me sketch here an argument for taking all DPs to be subject to the Case filter, even those that do not have a morphology that signals that they are Case marked.

The argument is based on a fact about adjectives in English. Adjectives in English can have internal arguments, just like verbs can, but they are never DPs. We have contrasts like (11).

(11) a. * She might be fond this sentence.
b. She might be fond of this sentence.

This was recorded in our phrase structure rules. No rule for APs put a DP inside an $\overline{\text{A}}$. If we abandon putting information of this kind in phrase structure rules, then there must be some other source of the fact that DPs are never internal arguments of adjectives.

One possibility is that adjectives, for whatever reason, simply never c-select an internal argument that is a DP. Perhaps there is some force that prevents adjectives from having that kind of argument structure. I want to show you this isn't correct. We have evidence that adjectives can c-select DPs. It's only that those DPs cannot appear inside the $\overline{\text{A}}$ they project. To see this, we first have to learn something about how some adjectives can express their arguments. Consider the adjective *obvious*. It c-selects a CP argument, forming sentences like (12).

(12) It is obvious [$_{\text{CP}}$ that pigs fly].

Notice that *obvious* has no external θ-role. The Specifier of IP in (12) is occupied by an expletive pronoun. Interestingly, the CP argument of *obvious* can also be found in Specifier of IP, as in (13).

(13) That pigs fly is obvious.

We will see an idea about how to account for this flexibility of expression of the argument of *obvious*, but for now we can simply observe that there are two positions that the argument of *obvious* can be positioned. Unsurprisingly, *obvious*, like all adjectives, cannot take a DP argument that shows up in its $\overline{A}$. But it can take a DP argument that shows up in Specifier of IP. This is indicated by the contrast in (14).

(14) a. This fact is obvious.
b. * It is obvious this fact.

There is a way of pronouncing (14b) in which it is grammatical. On that pronunciation, ***it*** is the argument of ***obvious***, and ***this fact*** is a kind of after-thought that spells out what ***it*** refers to. On that way of construing (14), ***this fact*** isn't an argument of ***obvious*** and ***it*** is. Tha alternative way of getting (14) is not available for the sentence ***it is obvious it***, because ***it*** is not the kind of DP that can express an identifying after-thought.

What we learn from (14) is that the DP argument of *obvious* exists, but must be found in one of the positions that we have determined assigns Case: Specifier of finite IP. This would make sense if the thing that makes adjectives different from verbs and prepositions is their ability to assign Case, and if all DPs are subject to the Case filter.

If it is correct that all argument DPs are subject to the Case filter, then we have two forces that determine where argument DPs can be. They must be c-selected by some head, and they must be able to be in one of the positions that is assigned Case. Only when both of these requirements can be met, will a grammatical sentence have an argument DP.

Stowell (1981) proposes to derive the ordering of complements using precisely this logic. If accusative Case is only assigned to positions adjacent to a verb or preposition, then of the many places a c-selected argument DP can be, only those that are adjacent to a verb or preposition will do. Because argument DPs are subject to the Case filter, when they get a θ-role from a verb, they will necessarily be positioned adjacent to this verb. By contrast, PPs are not subject to the Case filter and they are free to appear anywhere in the VP that contains their θ-role assigner. This will ensure that when a DP and PP share a VP, the DP will come closer to the verb than does the PP. In a language like English, in which the linearization parameters cause the verb to be first in its VP, this will mean that an object DP will precede an object PP.

The Case filter, and the adjacency condition on Accusative Case assignment, together will ensure that DP internal arguments end up positioned before other internal arguments in a $\overline{V}$. Let's next consider some of the other cases where a verb has two internal arguments. One of these is when a verb c-selects two PPs as arguments. So far as I have been able to tell, there is no ordering on these PPs. Either order is equally grammatical. There are few verbs with this argument structure, so it is not easy to investigate the generality of this case. One such verb is *talk*, and (15) indicates that either order is grammatical.

(15) a. Anna talked about Angela to Matt.
b. Anna talked to Angela about Matt.

Finally, let's consider what happens when one of the two internal arguments is a finite CP. In these cases, the CP comes last.

(16) a. Meredith explained to her that pigs are excellent fliers.
b. ?? Meredith explained that pigs are excellent fliers to her.

(17) a. Meredith told her that pigs are excellent fliers.
b. * Meredith told that pigs are excellent fliers her.

The contrast in (16) is much weaker than the one in (17). It might appear that this is because (17b) is a Case filter violation, since the argument DP *fliers* is not adjacent to the verb. But there is more going on in this example than meets the eye, and it will take us some time before we're ready to face it. We can simply note at this point that there must be something additional going on in (17b) that is responsible for the fact that it is worse than (16b).

There is some evidence that what is relevant for the contrast in (16) has to do with factors that are not those we've seen relevant for syntax. If, for instance, the relative lengths of the PP and CP are manipulated, the contrast in word-order changes. Many speakers report finding no, or little, contrast in (18).

(18) a. Meredith explained to the tourists crowded along the guardrail that pigs fly.
b. Meredith explained that pigs fly to the tourists crowded along the guardrail.

One way of thinking about this is that the syntax allows either order but one is preferred over the other when it reduces an online processing load. On this view, the weak badness of (16b) arises because the online processor is tasked with holding in its buffer all of the words that make up the smallest $\overline{V}$ before being able to properly place the PP that follows. If while transitioning through this $\overline{V}$, the processor's buffer size gets tasked too heavily, or if the guesses the parser makes online lead it to decide that it has finished parsing that $\overline{V}$, there will be a boggle upon encountered the PP at the end of the $\overline{V}$. There is evidence that all of these ingredients play a role in the strength of the perceived goodness of these sentences. This, then, might be one place where speakers' reports about the goodness or badness of a sentence should not be modeled with a syntax.

Interestingly, these same factors that we might credit to the parser's behavior show up in the alternations in (19).

(19) a. i. Meredith said [$_{PP}$ on Tuesday] [$_{CP}$ that pigs are excellent flier].
ii. ?? Meredith said [$_{CP}$ that pigs are excellent fliers] [$_{PP}$ on Tuesday].
b. i. Meredith said [$_{PP}$ on every day I showed up] [$_{CP}$ that pigs fly].
ii. Meredith said [$_{CP}$ that pigs fly] [$_{PP}$ on every day I showed up].

The PP *on Tuesday* is not an argument in these examples, and so the Projection Principle leads to the expectation that it should necessarily follow the argument CP. In fact, however, the argument CP can follow the PP and, in some cases, is even preferred to be in that position. These examples look like counter-examples to the Projection Principle.

A very similar paradigm of contrasts arises in cases with DP internal arguments. We've seen that there is a preference for putting an internal argument that is a DP adjacent to the verb that it is an argument of. But this preference is lost when the relative length of the DP and the rest of the material in the $\overline{V}$ is manipulated. When the DP argument is a pronoun or is a one or two word phrase, the ordering preferences we have given an account of emerge:

(20) a. i. Mako explained it to her classmates.
ii. * Mako explained to her classmates it.
b. i. Mako explained stuff to her classmates.
ii. ?* Mako explained to her classmates stuff.
iii.
c. Mako explained Endocentricity to her classmates.
d. ?* Mako explained to her classmates Endocentricity.

The contrasts in (20b) and (20c) aren't as strong as the contrast in (20a), and can be overcome if the DP is given prosodic stress. But if the relative length of the DP and PP is manipulated, even this isn't necessary to get the rogue word-orders.

(21) a. Mako explained the problems kept in her files to them.
b. Mako explained to them the problems kept in her files.

These should be Case filter violations. In fact, these same forces seem to control the relative order of argument DPs and non-argument PPs. The PP *on Tuesday* is a non-argument in (22), and yet its relative order with the argument DP depends on the same factors that we've seen controlling the relative orders of arguments.

(22) a. i. Mako explained it on Tuesday.
ii. * Mako explained on Tuesday it.
b. i. Mako explained stuff on Tuesday.
ii. ?* Mako explained on Tuesday stuff.
c. i. Mako explained the utility of Endocentricity on Tuesday.
ii. Mako explained on Tuesday the utility of Endocentricity.
d. i. ? Mako told the story I'd made up about pigs on Tuesday.
ii. Mako told on Tuesday the story that I'd made up about pigs.

These are violations of the Projection Principle.

All of these cases don't fit our present grammar, so we should diagnose what the problem with our grammar is and then fix it. One diagnosis is that the principles which are violated by these grammatical examples should be loosened; perhaps we should replace the Projection Principle with something looser and remove the Case filter, or the adjacency condition built into it which forces argument DPs to be next to the verb. This would permit all of these word orders, and we could let other forces—perhaps ones having to do with ease of parsing—choose among them. This would involve revising the conditions on Accusative Case assignment and the Projection Principle to (23).

(23) a. Accusative Case is assigned by a verb or preposition to a position inside their highest $\overline{\text{X}}$ projection.

b. An internal θ-role is assigned by X° to a position inside its highest $\overline{\text{X}}$.

This revision will handle the cases we've just seen, if it can be wedded to something that chooses correctly from the range of word order options the looser conditions in (23) permit.

This is not the direction usually taken, however, and I will show you why in the next section. We have to learn about another construction type for the problems attendant with (23) to become visible. The solution usually taken involves a more involved modification to the grammar, one that is a bit of overkill for the problems we are now facing. But, as we shall see, this modification handles a large range of additional cases; there is an independent need for the modification I am about to develop. The leading idea behind this modification to our grammar is that the DPs and CPs that we see in the positions that the Projection Principle and Case filter predict they should not be are also in the positions that the Projection Principle and Case filter predict they are. Or, putting it differently, in these examples there is a violation of Contiguity. The DPs and CPs that are in the "wrong" positions are actually in the right positions syntactically, but they are spoken in a position different than where Contiguity says they should be. We will see many cases of this type. Our evidence about the phrase structure is at odds with the strings that those phrases produce. We are going to need a general model of such cases.

One popular way of capturing what's happening in such cases is to let a single sentence get mapped onto not just one parse, but a set of parses. This set of parses can differ with respect to where one, or more, phrases are positioned. The linearization algorithm, and so Contiguity, sees only one of those parses, but the syntax and semantics can see the others. The picture is (24).

(24) Sentence $=_{\text{def.}} \{ \text{P}_1, \text{P}_2, \ldots, \text{P}_n \}$, where

a. Each P_i is a structure produced by the $\overline{\text{X}}$ Skeleton, and

b. One P_i is subject to the linearization algorithm.

We will need some way of defining the set of parses that are assigned to a sentence. We want sentences to be given a syntactic representation that gets interpreted semantically in one way. Our goal, recall, is to characterize which substrings of a sentence have a compositional meaning. So we will want this set of parses to all be related to one parse which characterizes the way in which these semantic building blocks are assembled. To capture this, I'll add another criterion to the list that defines a sentence.

(25) Sentence $=_{\text{def.}}$ { P_1, P_2,. . .,P_n}, where
   a. Each P_i is a structure produced by the $\overline{X}$ Skeleton, and
   b. One P_i is subject to the linearization algorithm.
   c. One P_i is the result of lexical insertion and is semantically interpreted.

Part of semantic interpretation is assigning θ roles, so the parse that is semantically interpreted will also be the one where the linking rules and Theta Criterion are satisfied. Now that we have one designated parse that captures the structure of the semantically interpreted phrases, we can define the other parses in the set with relations to that designated parse. We'll say that the parses in the set are just those that could be produced by applying a rule to the parse that gets semantically interpreted. The parse that is semantically interpreted is sometimes called a "D-structure," and the parse that is linearized is called an "S-structure."

"D-structure" is meant to evoke "deep structure," and "S-structure" is meant to evoke "surface structure." The framers of this theory must have imagined that meanings bubbled up from the depths and became visible only once they were afloat.

(26) Sentence $=_{\text{def.}}$ { P_1, P_2,. . .,P_n}, where
   a. Each P_i is a structure produced by the $\overline{X}$ Skeleton, and
   b. One P_i is subject to the linearization algorithm. (S-structure), and
   c. One P_i is the result of lexical insertion and is semantically interpreted (D-structure)j, and
   d. Each $P_i \neq$ D-structure is formed from the D-structure by a rule.

We will see that the rules might not create structures that are producible by the $\overline{X}$ Skeleton. One hypothesis is that the only deviations from what the $\overline{X}$ Skeleton allows are those created by the rules. We can capture this by imposing the requirements of the $\overline{X}$ Skeleton on D-structure alone.

(27) Sentence $=_{\text{def.}}$ { P_1, P_2,. . .,P_n}, where
   a. P_1 is a structure produced by the $\overline{X}$ Skeleton, and lexical insertion, and it is semantically interpreted (D-structure), and
   b. One P_i is subject to the linearization algorithm (S-structure), and
   c. Each $P_{i\neq 1}$ is formed from P_{i-1} by a rule.

What (27) defines is a cloud of parses that are different from a D-structure in the way that one rule specifies. In the rest of this book, we will be investigating what the rules that form this set look like, and we'll discover that some

rules can apply to the output of other rules. We will need to relax (27c) so that it allows a rule to apply to parses other than the D-structure. The set of parses, then, will have structure: there will be chains of parses in it that are formed by serial application of rules. These chains are called "derivations," and we will think of them as a series.

(28) $\mathcal{D} = (P_1, P_2 \ldots, P_n)$ is a derivation iff:
   a. P_1 is formed by the $\overline{X}$ Skeleton and Lexical Insertion, and it is semantically interpreted, and
   b. Each P_{i+1} is formed from P_i by a rule, and
   c. P_n is subject to the linearization algorithm.

(29) A sentence is the set of derivations which have the same D-structure and S-structure.

With these additions to our model, we can now formulate a rule that will characterize the cases where argument DP and CPs are in positions that the Projection Principle and Case filter don't predict. The rule is one that Ross (1967) devised.

In fact, Ross devised two rules, one for DPs and one for CPs, which is called "extraposition." It's not clear to me that they need to be distinguished, so I've collapsed them.

(30) Complex NP Shift
Move a DP or CP so that it is right adjoined to the smallest IP containing it.

Ross's work predates the move to giving nominal phrases a determiner as the head, and so he named his rule after NPs, one of the terms the rule can move. Reference to "complex" in the name is intended to reference the fact that this rule is sensitive to the size of the DP or CP being moved. We should define a few terms that the rule makes reference to. First "smallest" and "right adjoined."

(31) XP is smaller than YP, if XP dominates YP.

(32) Adjoining XP to YP involves the following steps.
   a. YP is replaced by another YP, YP′
   b. YP′ has the old YP as one daughter and XP as the other daughter.

We say that XP right adjoins to YP if the linearization algorithm makes YP<XP. And we say that XP left adjoins to YP if the linearization algorithm makes XP<YP. Finally, we should define "move."

The "<" symbol means precedes.

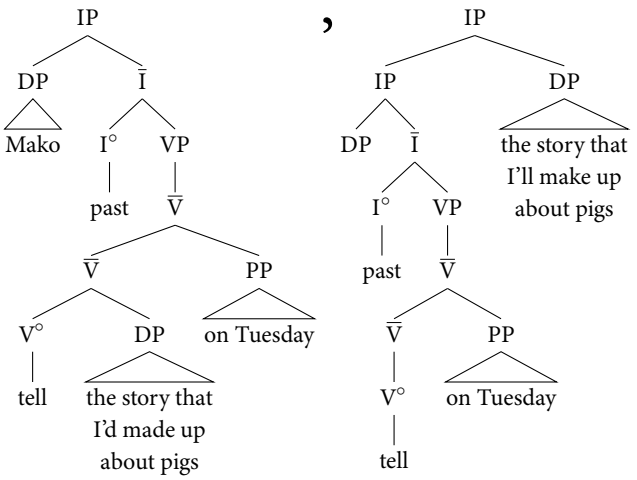
(33) α, a node in a phrase marker tree, moves to β, a position in a phrase marker tree, means:
   a. α is put in position β, and
   b. α is removed from its original position.

We will see that there are a variety of rules which make use of move.

Ross's rule allows (22d-ii) to have the derivation in (34).

(22d-ii) Mako told on Tuesday the story that I'd made up about pigs.

(34)



In this simple derivation, the D-structure is the first parse. It corresponds to the parses our system has been able to produce up to now. It satisfies the Projection Principle, the Theta Criterion, and the Case filter. It is the parse that is semantically interpreted, so it represents the phrases whose meanings are compositional. The second parse is derived from this D-structure by applying Complex NP Shift, and it is the S-structure (in this case). It is this phrase structure that the linearization algorithm produces a string from. As a consequence, the internal argument DP is pronounced in a position different than where its meaning is computed. In this simple case, the sentence (22d-ii) has only this one derivation in its set. This is the only series of parses that have the D-structure and S-structure shown in (34).

Stowell uses Ross's Complex NP Shift operation to control the relative position of PP and CP arguments too. But he suggests that Case plays a role in making Complex NP Shift relatively favored. He speculates that finite CPs, like DPs, must receive Case but that unlike DPs they cannot sit in Case-marked positions at S-structure. He calls this the "Case Resistance Principle."

(35) Case Resistance Principle
A (finite) CP may not be in a Case marked position at S-structure.

The Case Resistance Principle requires finite CPs to be moved by S-structure to some non-Case marked position. NP Shift is capable of doing this, and thus, in cases where a complement CP shows up string finally in a VP, it has satisfied Stowell's injunction against surfacing in Case marked positions by undergoing NP Shift. There is, surprisingly, a certain amount of evidence for this picture.

Consider, for example, situations where the finite clause is the "subject" of the clause. Jan Koster[1] argued that a subject finite CP isn't actually in the Nominative Case-marked, Specifier of IP position. One of Koster's arguments makes reference to a process that is found in certain question-types in

[1] Koster, Jan. 1978. Why subject sentences don't exist. In ***Recent transformational studies in European languages***, ed. Samuel Jay Keyser. Cambridge, Massachusetts: MIT Press

English. Normally, in these question contexts, it is possible for I° to be at the front of a sentence, as in (36).

(36) Maria will put the book on the table. →
Will Maria put the book on the table?

Koster's conclusions, and a modernization of some of his arguments, can be found in a more thorough form in Alrenga (2005). See also Elliott (2017) for critical discussion.

How precisely this happens is the subject of the next chapter. What's relevant here is that this word order is blocked if it interacts with another process that moves something to the left edge of IP, as in (37).

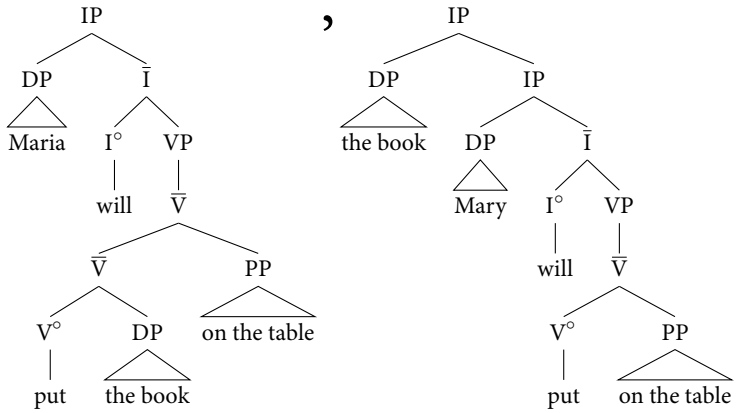
(37) Maria will put the book on the table. →
On the table, Maria will put the book.

In (37) we are seeing the action of another movement operation, sometimes called Topicalization, which we can formulate as (38).

(38) Topicalization
Move an XP so that it is left adjoined to the smallest IP containing it.

The derivation for (37) is (39).

(39)



Topicalization cannot apply to a sentence that has the word-order of a question, that is, a word order that puts the I° first. (40) illustrates.

(40) a. Maria will put the book on the table →
b. On the table, Maria will put the book. →
c. * Will on the table, Maria put the book?

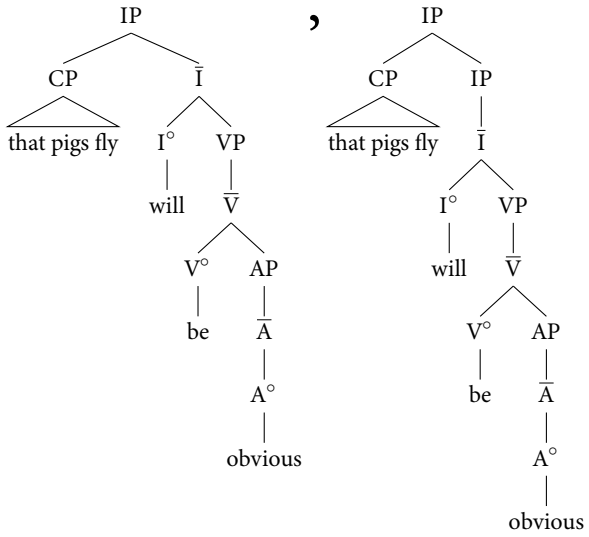
Now, interestingly, for many speakers of English the presence of a finite CP as a subject of a sentence is also incompatible with this I° first kind of sentence.

(41) a. * Should that Maria has left be obvious?
b. * Does that Paula sleep late bother Andries?

This would follow if finite CPs are driven from the nominative Case marked Specifier of IP, and adjoin to the left of IP in these cases. Stowell's method of forcing Complex NP Shift to apply to complement CPs would extend to

this scenario as well. CPs start out in the nominative Case-marked position, but are driven from there in order to form an S-structure that meets the Case Resistance Principle. This is shown in the derivation in (42).

(42)



This evidence all points to the fact that finite CPs move either to the left or to the right linear edges of the sentences they are part of. If CPs are prevented from being in Case marked positions in the S-structure, this would get us close to deriving this fact. But is there evidence for the motivation Stowell proposes for this movement? In particular, is there motivation for the claim that finite CPs, like DPs, require Case?

One piece of suggestive evidence comes from the class of verbs that permit both DPs and finite CPs. These are only just those verbs that already exceptionally allow two DPs. These include verbs like *promise*, *tell*, and *show*; they are known as "double object" verbs. Some examples indicating that double object verbs also take DP and CP are in (43).

(43) a. Karen promised me that she would sing.
Karen promised me the ring

b. Wendell told me that he can't stand Karen's singing.
Wendell told me the story.

c. Annahita showed me that she cares.
Annahita showed me her concern.

This isn't completely the case, as (44) is a counterexample.

(44) a. Shai persuaded Adam that he should go.

b. * Shai persuaded Adam the fact

But, so far as I know, (44) is the only counterexample. To the extent that there is a match in the verbs which accept DP+CP and those which accept DP+DP complements, there are grounds for believing that their surface positions are governed by the same, or similar, principles. And to the extent that the

dominant principle is the Case Filter, then there is reason to conclude from these data that CPs are subject to the Case Filter as well.

This match between the distribution of DPs and CPs should be completely general if Stowell is correct. Indeed, finite CPs are distributed in sentences much like DPs are. We find them as complements to verbs, as we have seen, and in the subject position of other finite clauses, but not in the subject position of infinitives (as we shall see shortly). These are just the Case-marked positions.

But there are several differences in their distribution. In English, finite CPs are probably never found as the complements to a preposition, though of course DPs are. The only potential counterexample comes from temporal prepositions, as in (45).

(45) a. I left before Cherlon arrived.
b. * I left before that Cherlon arrived.

Similarly, both adjectives and nouns can take CP complements, but not DP complements.

(46) a. Shigeto is unhappy that he had to sing.
b. * Shigeto is unhappy that.

(47) a. the proof that lemons cure cancer
b. * The proof this fact.

If this has a Case Theoretic explanation, then Stowell's theory is in trouble. To my knowledge, an account of the distribution of finite CPs has not yet been formulated. We will see in the next chapter that there are verbs which do not c-select DPs and do not assign accusative Case but do c-select CPs. When this is coupled with the fact that CPs can be the internal arguments of nouns and adjectives—words that never support accusative Case—it seems unlikely that Stowell's explanation for the obligatoriness of CP movement is correct. They move not because the positions they originally occupy are Case-marked ones, and they are somehow incompatible with Case. At the time of writing, there is no consensus about what is responsible for the special placement of CPs.

See Moulton (2009, 2015) for a good proposal.

Though the evidence suggests that Stowell's method of deriving the linear ordering of verbal complements meets with some success, there are problems too. One which threatens Stowell's Adjacency Condition on Case assignment, and its use in fixing the order of complements, concerns so-called "double object" constructions, as in:

(48) a. Shigeto showed Luis the picture.
b. Shigeto baked Luis a cake.

How can the second DP in these examples receive Case? We will have to develop some of the rest of the system that is responsible for linearizing arguments before we can engage this difficulty. So, let me ask for your indulgence and postpone examining this case until chapter 3.7.

3.1 Small Clauses and the Derived Subjects Hypothesis

In the last chapter, I decided that we should adopt an $\overline{X}$ Skeleton that forced every XP to contain an $\overline{X}$. This narrows the class of possible kinds of phrases that are allowed to just those types that we have evidence for in IPs and DPs. In IPs and DPs, we discovered that there are two kinds of projections: one which contains a phrase that the other cannot. That phrase is the thing that resides in Specifier position. For IPs, the Specifier position can be occupied by either a DP or CP, and the Specifier of DP can be occupied by a DP marked with the genitive "s." $\overline{I}$s and $\overline{D}$s don't have those phrases, and that's what makes it necessary to distinguish IP from $\overline{I}$ and DP from $\overline{D}$.

Forcing every phrase to have an XP and $\overline{X}$ does no harm, but it does raise the puzzle of what distinguishes the XP from the $\overline{X}$ for the majority of phrases. We haven't found any reason to think that, say, VPs are different from $\overline{V}$s and this is puzzling if there really is a distinction between these phrases. Let's face this puzzle now. Let's see if we can identify what distinguishes the XP from $\overline{X}$ of most phrases. If we can find nothing that does, then we should transit to a different $\overline{X}$ theory, one that doesn't require every phrase to have an $\overline{X}$.

(49) What resides in the Specifiers of:

a. Adjective Phrase
b. Verb Phrase
c. Prepositional Phrase
d. Determiner Phrase
e. Adverb Phrase

The structure of Adverb Phrases is not well-studied, partly because of the apparent poverty of material that can be in adverb phrases. We will generally ignore the structure of these phrases as a result. It's time, though, to see what occupies the Specifier positions of APs, VPs and PPs.

Let's begin with VPs and APs. What are the terms, if any, that stand in the Specifiers of VP and AP? One hypothesis is that VPs and APs can have external arguments in them, and these are the designated Specifiers. This can be seen in certain special circumstances, of which (50) are examples.

(50) a. She will let [them make a cabinet].
b. This will make [them angry at me].

As indicated by the brackets, it is thought that the strings following *let* and *make* form a single phrase which is serving as the complement to these verbs.

Why don't we instead believe that these verbs are followed by two complements, the first of which is *them*? Because *them* is not an argument of the verb. We can see from other examples that this isn't the case. In (51), for example, this position can be occupied by an expletive.

(51) a. I will let [it seem that there is a mistake on the handout].
b. This will make [it obvious that there is a mistake on the handout].

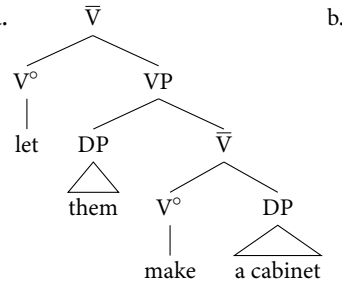
Expletive *it* can only stand in a Specifier position. Thus, the positions following *let* and *make* in these cases must be Specifier positions.

Note that the bracketed strings in (50) are, indeed, an AP and VP, and not IPs. This, we know, because the strings which follows these specifier positions are characterized by the rules which generate VPs and APs, and not those which characterize IPs. As (52) indicates, an I° is not found in these strings, for instance.

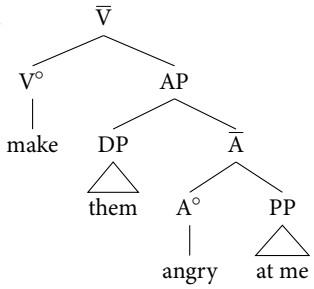
(52) a. * I will let [her should make a cabinet].
b. * I will let [her to make a cabinet].
c. * I will let [her makes a cabinet].

From these observations we will want to give the examples in (50) the parses in (53).

(53) a.



b.



These complements to *let* and *make* are sometimes called "small clauses," and it is to Stowell[2], once again, that we owe this way of viewing them. If this is correct, we have now discovered what stands in the Specifier of AP and VP. These positions can contain a "subject" argument.

A similar conclusion can be reached for PPs, though here the evidence is somewhat more meager. Cases such as (54) plausibly involve small clause PPs.

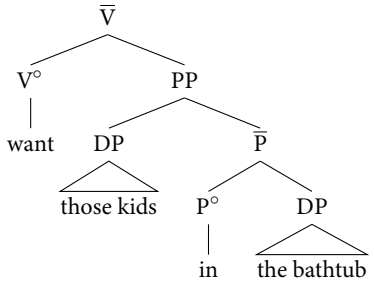
(54) a. I want those kids in the bathtub!
b. She saw them in the next room.

Both of these examples can have a meaning in which it seems that the verb has just one object. In (54a), the object of *want* seems to be, roughly, what *them in the bathtub now* describes. On this meaning, the content of my desire is for those damn kids to be in the tub. Under our present understanding of how θ-roles are assigned, this requires that (54a) have the parse in (55).

[2] Stowell, Tim. 1978. What was there before there was there. In ***Proceedings of the Fourteenth Regional Meeting of the Chicago Linguistic Society***, ed. D. Farkas, volume Proceedings of the Fourteenth Regional Meeting of the Chicago Linguistic Society. Chicago, Illinois: The University of Chicago

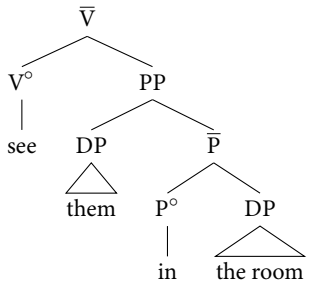
Of course, (54a) has another meaning, one in which the content of my desire is children, and more particularly, those children in the tub. That meaning makes the object of ***want*** a DP headed by ***kids***.

(55)



Similarly, among its several meanings, (54b) can describe a situation in which what was seen was *them in the room*. This reading arises when *in the room* does not describe where the seeing event took place—that reading is manufactured by letting *in the room* modify *see*—but instead describes the location of the referent of *them*. To produce such a reading, we want *them in the room* to be the object of *see*, and this is achieved by a parse like (56).

(56)



In both of the examples in (54), then, there is reason to believe that the verbs are followed by one PP, whose Specifier is filled with the DP argument.

Small clauses, incidentally, furnish evidence that the Complex NP Shift solution to the problem we faced in the previous section about complement order is better than relaxing the Projection Principle and Case filter. Recall that we considered as a way of accounting for (57) that argument DPs to be positioned higher than non-argument DPs within V̄, and let whatever forces care about the relative lengths of the DP and non-argument decide which of the two possible positions is best.

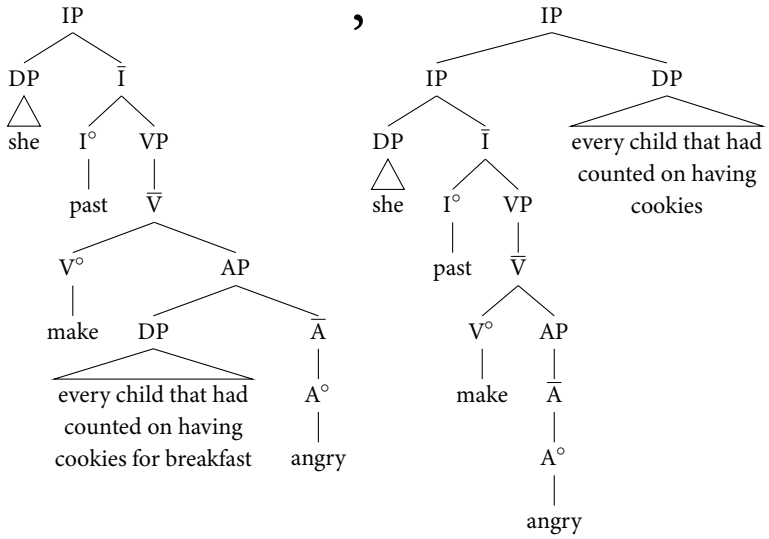
(57) She ate on Tuesday all the cookies she had brought.

But we can see from (58) that whatever allows argument DPs to have the exceptional VP-last position in (57), should allow argument DPs that aren't arguments of the relevant V̄ to have that position too.

(58) She made angry every child that had counted on having cookies for breakfast.

(58) can have the derivation shown in (59), if Complex NP Shift is responsible for these word orders.

(59)



Simply relaxing the Projection Principle, or Case filter, would not explain (58). But Complex NP Shift does.

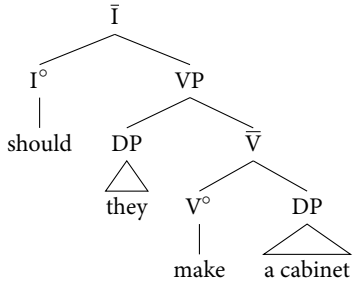
Consider now the examples in (60).

(60) a. They should make a cabinet.
b. They are angry at me.
c. They are in the next room.

These sentences contain the same VP, AP and PP that constituted small clauses in our earlier examples. Moreover, the θ-role borne by the subject in (60) seems to be the same as that borne by the DP in the examples with small clauses. Indeed, the meaning of the sentences in (60) seems to be identical to those of the small clauses in the previous examples, except for that meaning introduced by the modal *should* (in (60a)) and the verb *be* (in (60b,c)). A number of linguists independently noted that there could well be a small clause within (60), but that this arrangement is obscured by a movement transformation that relocates the subject.[3] Indeed, the Case filter based theory about where argument DPs must be predicts that (60a) could have a small clause in it at D-structure. Suppose that the θ-role that *they* bears in (60a) is assigned to the same position that it is in our small clause example in (50a), that is to the Specifier of VP. Then the Theta Criterion would require the D-structure of (60a) to be (61).

[3] See Kuroda (1988), Kitagawa (1986), Fukui and Speas (1986), Bowers (1993), and Koopman and Sportiche (1991) for proposals along these lines that fit into the general theoretical assumptions defended here, and Fillmore (1968) and especially McCawley (1970) for early antecedents to the idea.

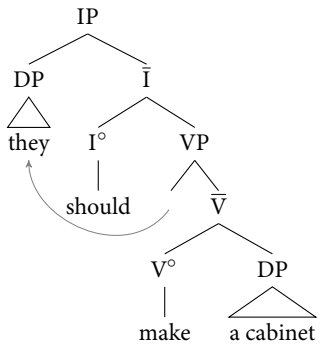
(61)



The subject argument, *they*, bears nominative Case, and yet in this phrase marker, it is not in the position our system assigns nominative Case to. That position is the Specifier of a finite IP, which is the position that *they* is, in fact, spoken in. Imagine, then, that there is a movement transformation that relocates *they* into the Specifier of IP, as in (62).

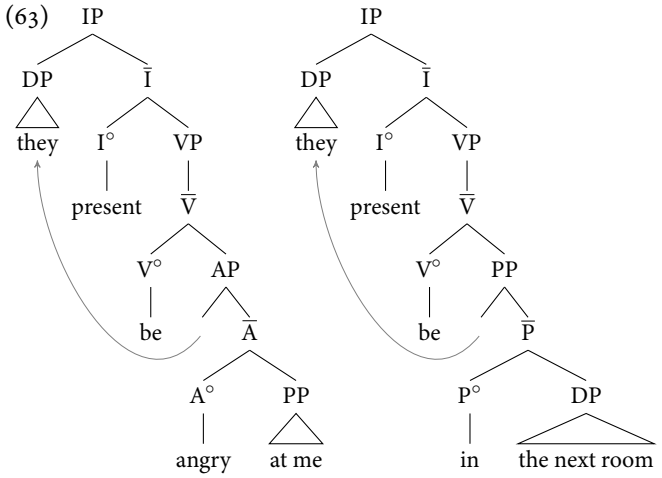
I will use this notation involving arrows to represent adjacent parses in a derivation. The tail of the arrow shows where the moved phrase is in the first of the two parses, and the head of the arrow shows where the phrase is after the relevant rule has applied, forming the second parse in the pair.

(62)



The same derivations could be present in (60b) and (60c) as well. They would be as indicated in (63).

(63)



The conjecture that the subject arguments of IPs start out inside small clauses and move into the position they are spoken goes by the name of "the derived subjects hypothesis." The movement transformation responsible for relocating the subjects is known as "Argument Movement," or more commonly: A-Movement.

(64) Argument Movement
Move an argument XP to a Specifier position.

We shall examine some evidence in support of the derived subjects hypothesis in the section on "Quantifier Floating," later in this chapter. There are other environments in which A Movement is alleged to occur, which we'll need to examine as well. But before we turn to those tasks, we need to make some modifications to our system in the light of this hypothesis.

The first modification is to sharpen the conditions on Case assignment in order for them to produce the desired effects under the derived subjects hypothesis. Our present conditions on Case assignment are (65).

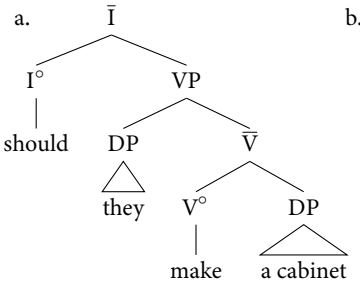
(65) Conditions on Case assignment

a. Nominative Case is assigned by finite I° to its Specifier position.

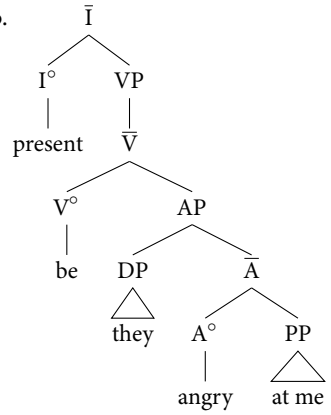
b. Accusative Case is assigned by verbs and prepositions to an adjacent position.

The description of where accusative Case is assigned does not produce the right results when coupled with the derived subjects hypothesis. To see this, consider the D-structures of some of the examples we've looked at.

(66) a.

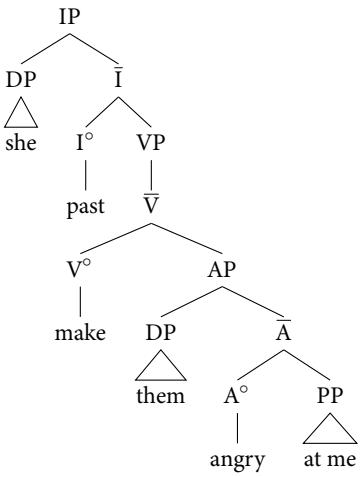


b.



The subjects in these examples cannot be assigned accusative Case; they only appear with nominative. But the statement of accusative Case assignment in (65b) does not ensure this. The subject is adjacent to the verb *make* in (66a), and in (66b) it is adjacent to the verb *be*. The case in (66b) contrasts minimally with (67).

(67)



In (67), *make* assigns accusative Case to *them*. Because of the parallel structures of (67) and (66b), the usual conclusion is that the difference in availability of accusative Case must devolve to the difference in the verbs involved.

Unlike *make*, *be* does not assign accusative Case. If that's the case, then not every verb can assign accusative Case. Instead, it appears that (68) is true.

(68) The ability to assign accusative Case is part of the lexical specification of a verb.

The response to (66a) needs to be more general. No matter what verb is involved, it is not able to assign accusative Case to its Specifier position. The usual answer given to this problem, one that extends to other cases, as we shall see, is based on the structural relationship between the verb and the positions it can assign Case to. A verb is able to assign accusative Case to its sister, as in (66a), or to a position within its sister, as in (67). But it's not able to assign Case to anything outside of the phrases it is dominated by. To capture this range of positions, we define the following relation.

(69) α c-commands β iff
   a. every phrase dominating α also dominates β, and
   b. α does not dominate β.

α dominates β if β is part of α and $\beta \neq \alpha$. Or, alternatively, β is dominated by α if the phrase structure rule expanding α leads to a rule that introduces β. The "c-command" relation comes from Reinhart (1976), whose definition is slightly different from the one produced here. (The central difference is that she uses "branching node" in place of my "phrase.") The name "c-command" is short for "constituent command." Reinhart presents her proposal as a modification of the relation "command" in Langacker (1969). It is close to the notion "in construction with" that is found in Klima (1964), which Langacker cites as his model.

The c-command relation is useful in describing a wide range of syntactic phenomena, and so it will arise in several other contexts in these lectures. The definition in (69) anticipates these later uses, and so goes a bit beyond what is necessary for the Case assignment relation. Only (69a) is needed here. (69b) prevents things from c-commanding themselves or their contents, and neither of these scenarios are relevant for the Case assignment relation. Armed with c-command, we can restrict the condition on accusative Case assignment to (70) and, thereby, prevent verbs from assigning accusative Case to their Specifier position.

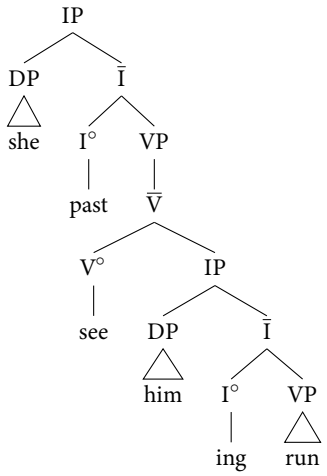
(70) Accusative Case is assigned by some verbs and prepositions to positions that they c-command and are adjacent to.

We need one additional change to the condition on accusative Case assignment. This change is made necessary by the contrast in grammaticality of examples like (71).

(71) a. * She made him running easy.
   b. She saw him running.

The complement to *saw* in (71b) is a kind of non-finite IP called a "gerund." These IPs are headed by the inflectional morpheme *ing* that appears on the following verb. The subject of this IP in (71b) receives its accusative Case from *see*; its S-structure parse is (72).

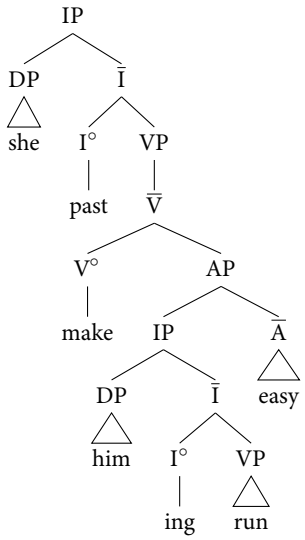
(72)



The condition on accusative Case assignment in (70) correctly captures the grammaticality of this example; *see* c-commands and is adjacent to *him*.

Consider now the contrasting (71a). In this example, the gerund is the subject of a small clause, as indicated in (73).

(73)



The condition on accusative Case assignment in (70) would allow *make* to assign Case to *him* in (73): *make* c-commands, and is adjacent to, *him*. And yet, the ungrammaticality of this example traces back to the availability of Case for *him*, as can be appreciated by considering the fact that this example becomes grammatical without *him*.

(74) She made running easy.

The ungrammaticality of (71a) should arise because it violates the Case filter, and this indicates that *make* cannot assign accusative to *him*.

A salient difference between (71a) and (71b) is the depth of embedding of the DP to be assigned Case. There appears to be a condition on how far into its sister a verb can assign Case. The precise statement of this condition

is still under investigation; we will consider some of the proposals in what follows. I will add this condition to the formulation of Case assignment in a way that allows us to modify it as we go along. An initial reformulation of the conditions on accusative Case assignment, then, is (75).

(75) Accusative Case is assigned by some verbs and prepositions to positions that they c-command, are klose to, and are adjacent to.

α is KLOSE to β iff there is no more than one phrase that dominates β but not α.

This definition allows a verb to assign Case to its sister and to the immediate daughters of its sister, but prevents it from assigning Case to any position more deeply in its sister. In *Lectures on Government and Binding*, and much subsequent work, a term—"government"—is defined that collapses the c-command and klose relations. The condition on accusative Case assignment is often formulated as (76).

(76) Accusative Case is assigned by some verbs and prepositions to positions that they govern and are adjacent to.

α GOVERNS β iff α c-commands β and α is klose to β.

Finally, what we see from the derivations involving Argument Movement is that the Case filter will have to be satisfied if an S-structure meets its requirements but a D-structure doesn't. We saw in the Complex NP Shift examples that the Case filter could also be satisfied if its requirements were met by a D-structure but not an S-structure. It looks like the Case filter is satisfied if either a D-structure or an S-structure meet its requirements.

(77) Case filter
An argument DP must be in a Case marked position at either D-structure or S-structure.

With these changes, then, the system of Case assignment and the Case filter correctly work with the derived subjects hypothesis to ensure that subjects of finite clauses show up with nominative Case in the Specifier of IP at S-structure. The existence of small clauses, and the accompanying thesis that subjects of IPs start out in small clauses, requires that the subject θ-role be assigned to the Specifier of the phrase headed by the term that has the θ-role. This brings us to the second change to our system made necessary by the Derived Subject Hypothesis. The Projection Principle determines where θ-roles are assigned, and so we should modify this condition to capture where subject θ-roles are assigned. At present, the Projection Principle requires that words assign their θ-roles to nearby positions within their projection. The Derived Subjects Hypothesis requires that this be broadened to include Specifier positions as well.

(78) The Projection Principle

a. If β is c-selected by α, then there must be a projection of α, π, that immediately dominates β and if π immediately dominates some other phrase, γ, then γ must be another projection of α or part of another c-selected argument of α.

b. An external θ-role of α must be assigned by α to a phrase in the Specifier position of α.

This view of how θ-roles are assigned makes most D-structures different than their S-structures. It requires that Argument Movement frequently be engaged to bring the subject argument of a verb from the position it gets its θ-role from to the position that determines its Case. In section 3.3, we'll examine more closely the rule that Argument Movement has in determining the form of sentences. But in the next section, we'll examine a consequence this hypothesis has for the D-structure status of IPs.

3.2 The EPP

When we were discovering the phrase structure rules for English, we hit upon a couple oddities of IPs. One of those oddities is that the smallest IP is not a string that consists of just its head. This is a possibility for every other phrase we saw: the smallest NP is just a noun, the smallest VP is just a verb, and so on. But the smallest IP consists of both a phrase in Specifier position and a phrase in complement position, in addition to the head. We recorded this fact by giving to IPs the following phrase structure rules.

(79) a. IP $\rightarrow$ DP $\bar{\text{I}}$

b. IP $\rightarrow$ CP $\bar{\text{I}}$

c. $\bar{\text{I}} \rightarrow \text{I}^{\circ}$ VP

Every IP, according to these rules, must have a DP or CP in its Specifier position, as well as a VP in its complement position.

If the derived subjects hypothesis is correct, this isn't correct. Instead, as the derivations we've seen in the previous section indicate, IPs have no Specifier at D-structure, and instead gain a Specifier by way of Argument Movement. We need D-structures to allow for the shape indicated in (80).

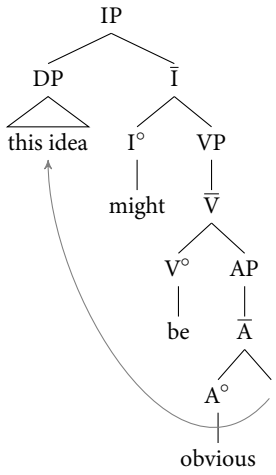
(80) IP $\rightarrow \bar{\text{I}}$

This predicts that, in principle, an IP should be able to correspond to an S-structure that has nothing in its Specifier position. This, it turns out, is incorrect.

We have an account for this for every sentence that has a DP argument which does not reside in a Case-marked position at D-structure. In those scenarios, the Specifier of IP will, when it is finite, be the Case-marked position to which that argument DP must move. Thus, for instance, when a DP is an

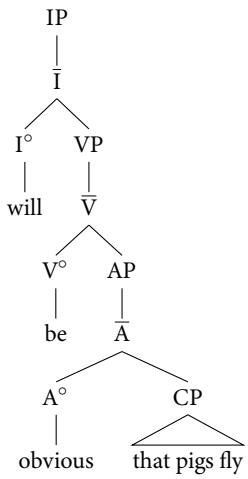
argument of an adjective like *obvious*, it must move in order to satisfy the Case filter in the way indicated in (81).

(81)



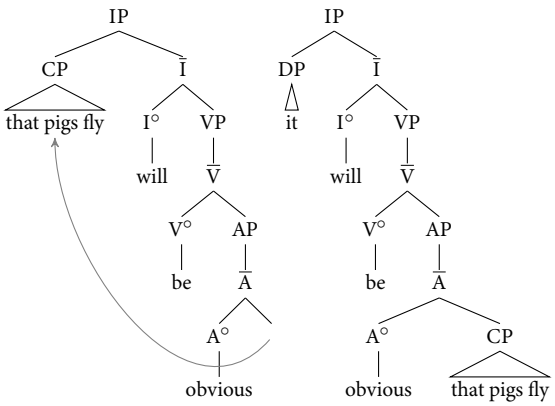
But what of IPs in which the D-structures do not violate the Case filter. If CP arguments are not subject to the Case filter, for instance, why is the D-structure shown in (82) not a grammatical S-structure.

(82)



Instead, the D-structure in (82) is associated with one of the two S-structures in (83).

(83)



The pronoun *it* in the second of these S-structures is not an argument. In fact, it doesn't refer to anything in the same way that an argument pronoun typically does. If it has a denotation, I don't know what it is. Terms that have no denotations are known as "expletives," and the *it* in (83) is one.

In a syntax that embraces derivations, we see from these cases that the requirement that an IP have a DP or CP in its Specifier position is a fact about S-structures. IPs can exist without something in their Specifier position, but the S-structures of IPs cannot. We'll recognize this fact with the following description.

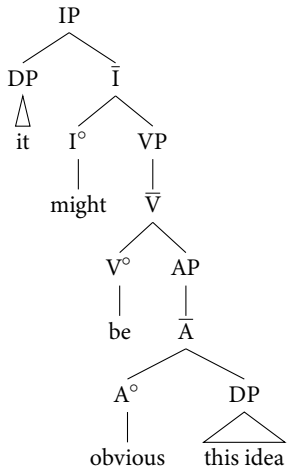
(84) The S-structure of an IP must have a DP or CP in its Specifier position.

We will investigate some of the ideas about what is responsible for this fact. Chomsky (1981) proposed that (84) is connected to the Projection Principle—the condition that determines where arguments must be at D-structure. He called (84) the "Extension to the Projection Principle" to signal this hypothesis. This proposal has been abandoned, and so the name no longer makes sense. Nonetheless, the name persists, though in the more cryptic acronym: "EPP."

The EPP can be satisfied by using Argument movement to form an S-structure from a D-structure that does not put a CP or DP in Specifier of IP. This is what happens in (82) and the first case of (83). It can also be satisfied by creating a D-structure in which the IP's Specifier is occupied by the expletive *it*. This is what is illustrated by the second case in (85), which we can take to be both the D-structure and the S-structure of that sentence. Forming this D-structure for a sentence in which the argument of the adjective is a DP, as in (**??**), leads to ungrammaticality because of the Case filter.

The EPP is a specific version of what David Perlmutter and Paul Postal, in their Relational Grammar framework, call the "Final 1 Law." See Perlmutter (1983). The earliest statement of this condition that I know of is Perlmutter (1971).

(85)



If there is only one Specifier position for any XP, then the sole Specifier position for the IP in (85) is occupied by the expletive *it*. Let's assume that this prevents Argument movement from forming an S-structure from (85) in which *this idea* occupies the Specifier of IP. There is consequently no derivation that includes (85) in which the argument DP *this idea* occupies a Case-marked position. This violates the Case filter.

In addition to the Case filter, we'll need to embrace the EPP. Together, they control the kinds of S-structures that are permitted.

3.3 Argument Movement

The derived subjects hypothesis conjectures that subjects of garden variety finite IPs are spoken in positions different from their D-structure position. Their D-structure position is conjectured to be the Specifier of the VP belonging to that IP, a position that the existence of small clauses indicates is a possible position for subject arguments. It relies, then, on the existence of a mechanism that gives arguments a different position than their θ-marked one, and we have introduced Argument Movement as that mechanism.

There is independent support for the existence of a mechanism like Argument Movement. There are situations in which we are more confident about where the relevant θ-marked position is, and the argument bearing that θ-role is spoken somewhere else.

3.3.1 Passive

Consider, for instance, the pair of examples in (86).

(86) a. He will make [her angry at me].
b. She will be made [angry at me].

The θ-role that *she* bears in (86b) is the same one that it bears in (86a), and yet *she* shows up in a different Case and in a different position in (86b). The

VPs are different in these examples. The verb in (86b) is said to be in the "passive voice," while the verb in (86a) is in the "active voice." Passive verbs in English are embedded under the auxiliary verb *be*. It is a fairly widespread generalization about Passive verbs that they do not assign Accusative Case, even if the Active verbs they are related to do. English conforms to this generalization. An informal way of thinking about the passive is that it is as an operation that derives a passive predicate from an active one by way of the changes described in (87).

(87) Passive

a. Add passive morphology to the active verb and embed it under *be*, and

b. Rob the verb of its Accusative Case, and

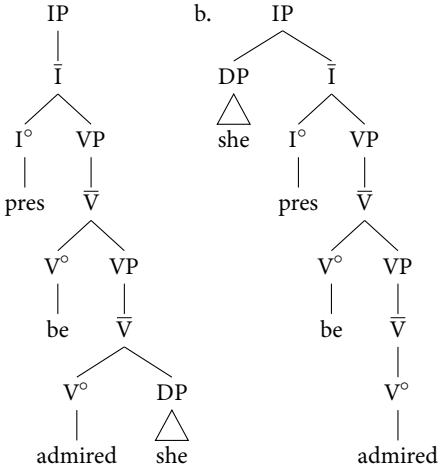
c. Rob the verb of its external θ-role.

This takes a garden variety "transitive" verb and produces a verb which looks "intransitive," as indicated in (88).

(88) Jerry admires her. → She is admired.

The D-structure representation of the Passive sentence in (88) is (89a) from which the S-structure in (89b) is produced.

(89) a.

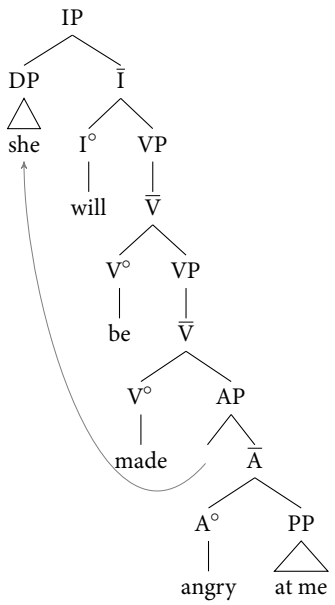


b.

Because Passive has robbed *admired* of its ability to assign Accusative Case, its object will seek out the Nominative Case marked position to satisfy the Case filter. This will cause (89a) to be mapped onto an S-structure in which *she* is positioned in the Specifier of IP.

The passive in (86b) has the derivation in (90).

(90)



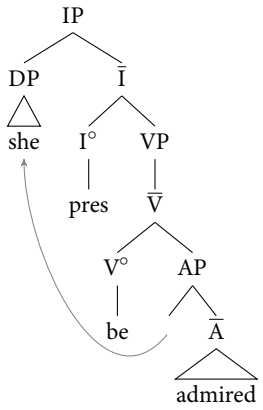
This mirrors the kind of derivation that the derived subjects hypothesis gives to every finite clause. The subject argument is put into a D-structure position from which its θ-role derives, but cannot remain there and satisfy the Case filter. Argument Movement kicks in and gives it a position in which it is Case marked. Passive Constructions, then, are ones in which we have independent support for displacement of arguments from their θ-marked positions. They even manufacture cases which resemble the situation the derived subjects hypothesis claims to be widespread.

There's a confusing aspect to English morphology which clouds how the Passive works in English. English has a morphological rule by which transitive verbs can be converted into adjectives, and these adjectives look just like Passive participles. I'll call these adjectives, "adjectival participles." This process is responsible for created the adjectival participle in (91), for instance.

(91) the admired student

It could be, then, that what we have in (89b) is an adjectival phrase headed by *admired* rather than a Passive VP. It might have a derivation like that in (92).

(92)



Over a large class of cases, then, we will not be able to tell whether we are looking at a Passive sentence or a *be*+AP (a so-called "copular construction") sentence.

In some cases, however, it is possible to tell. This is because the morphological process that derives adjectives from verbs operates on the arguments of the verb. In particular, it makes the θ-role that a verb assigns to the DP it c-selects become the external θ-role of the resulting adjective. For instance, the adjective in (92) assigns to its Specifier position the same θ-role that the verb it was derived from assigns to its complement. This is quite general. We can see this by considering the relation these adjectives have to the nouns they modify. The pre-nominal adjective in (91) is related to the noun it modifies in the same way that the verb it derives from is related to its object. When we figure out how the modification relation works semantically, we will need to derive the fact that the modification relation for adjectives roughly matches the relation these adjectival participles have to their subjects in the copular construction. We can use this correlation, then, to see which θ-role is "externalized" by the adjective formation process. What we find when we look is that, in fact, it's always the "direct object" θ-role that gets externalized. In each of the cases in (93), for instance, the modified noun is related to its adjective in the same way that the direct object is to the verb.

The ***locus classical*** for how to distinguish adjectives from verbal passives is Wasow (1977).

See Levin and Rappaport (1986) for arguments for this view of the rule.

(93) a. Ben introduced the man to Min-Joo.
the introduced man

b. Ben placed the book on the table.
the carefully placed book

c. Min-Joo baked the cake for Ben.
the baked cake

In no case is this relation like the one that holds between the verb and its subject argument or its "indirect object" argument.[4]

Because of this restriction on the adjective formation process, we can be certain that the example in (86b) is a Passive and not an adjective. This is because the argument that shows up as the nominative argument in this example is not an argument of the verb at all. It is the subject of the complement of

[4] For more details about this process, see Levin and Rappaport (1986).

the verb. If *made* were an adjective in this example, then it's the complement itself that would have appeared as subject, as can be confirmed by considering alternations like those in (94).

(94) a. Someone made this table
b. this (recently) made table.
c. Someone made him sad.
d. * the (recently) made man

What we see in the Passive, then, is that an argument DP has a different position than where its θ-role is assigned. It's not that the Passive causes a θ-role to be assigned by a verb to a different position, because sometimes the relocated argument DP isn't even an argument of the relevant verb. A syntax that embraces Argument movement can account for this.

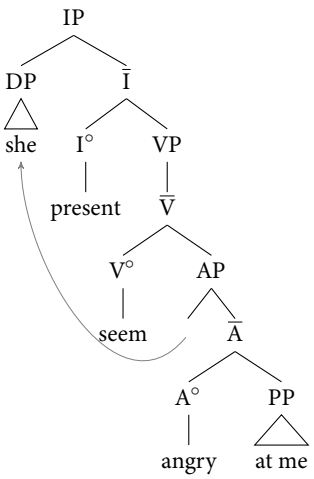
3.3.2 Raising

There are verbs which behave much like the Passive verb in (86b), but without needing to be Passivized. An example is (95).

(95) She seems [angry at me].

Just as in the Passive example, *she* is an argument of *angry at me*, not *seems*. And yet, just as in the Passive, the subject argument of *angry at me* shows up in a higher Specifier of IP and bears the nominative Case assigned to this position. This example seems to require the derivation in (96).

(96)



Verbs like *seem* are called "raising" verbs. They have properties that force an argument in their complement to "raise" into their subject position. Under the proposals we are examining, one of those properties is that it does not assign accusative Case. Raising verbs, then, are like *be* in being among those verbs that do not have an accusative Case associated with them. As a result, in situations like (96) the argument of the embedded small clause must Argument

Move into a position where it may satisfy the Case filter. In these circumstances too we see cases like those that the derived subjects hypothesis claims are common.

3.4 Quantifier Float

The derived subjects hypothesis is, presently, a kind of null hypothesis. It rests on the observation that the existence of small clauses indicates that external θ-roles can be assigned to Specifier of VP, and so the null hypothesis is that they always are. The existence of movement Transformations, or some other means of overcoming violations of Contiguity (i.e., resolving mismatches between constituent structure and linear order), makes it understandable why the surface position of the subject of finite clauses is not the Specifier of VP. And the Case filter provides a reason why a movement transformation would be invoked in these contexts: it gives an account of why subject arguments remain in Specifier of VP sometimes, and move in others. The derived subjects hypothesis, then, fits together independently needed pieces—movement transformations, the Case filter, and a Projection Principle that produces small clauses—into a simple picture of how subject arguments are placed. Nonetheless, it would be encouraging to find empirical support that directly speaks on behalf of the derived subjects hypothesis.

Sportiche (1988) argues that the phenomenon of "Quantifier Float" provides such evidence. Quantifier Float arises when a quantificational term is not in the determiner position normally reserved for quantifiers, but is instead to the right of DP it is associated with. It can be illustrated by the English and French examples in (97).[5]

(97) Les enfants (*tous) ont (tous) vu ce film (tous).
The kids (all) have (all) seen this film (*all).

Sportiche argues that there is a violation of Contiguity in these examples of the sort that movement Transformations are designed to account for. He suggests that the floated quantifier and the DP it is related to start out together at D-structure and then are separated at S-structure by Argument Movement. Note that there are language particular differences in this process that are reflected in the differing grammaticality judgments for some of the French and English examples. We'll return to these differences in the next chapter.

Sportiche's argument relies on the observation that the relation between the Floated Quantifier and the argument it is related to mimics the relation that Argument movement is designed for. Once this is appreciated, the fact that the floated quantifier appears in just those positions the derived subjects hypothesis predicts it should provides the evidence we are looking for. To see the force of his argument, we will first have to learn some generalizations about Argument Movement.

[5] Miyagawa (1989) provides a similar analysis for Quantifier Float based on Japanese data.

One of those generalizations is that the moved phrase can only be relocated to a c-commanding position. I'll call this condition "Upwards."

(98) Upwards
If α moves to the position β, then β must c-command α.

This is a generalization about all movement rules: all clear cases of a term moving to a non-c-commanding position are ungrammatical.For instance, there is another movement rule in English that moves certain DPs into Specifier of CP. This rule is part of a suite of operations that, among other things, is responsible for making question sentences in English. Some examples of this rule's operation are in (99).

We'll encounter examples in the next chapter which requires a reformulation of Upwards, however. The version in (98) isn't entirely accurate.

(99) a. I wonder who you talked to.
b. I asked which problem you will talk about.
c. I asked about which problem you will talk.

The DPs that this rule moves are called "wh-phrases," and an initial, though not complete, definition of wh-phrases is (100).

(100) A wh-phrase is an XP that:
a. is headed by a wh-determiner (e.g., *which*)
b. is headed by a wh-pronoun (e.g., *who*, *what*)
c. is a PP with a wh-phrase in complement position.

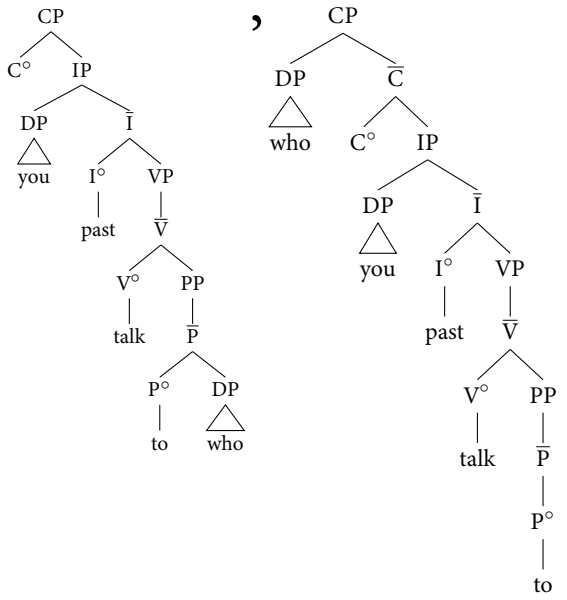
We will look closer at the details of this rule later, but an initial formulation of it is in (101).

(101) Wh Movement
Move a wh-phrase to a Specifier of CP.

This produces a derivation for (99a) like that shown in (102).

One of the peculiarities of CPs that make questions in English is that there is no obvious word in C° position. We will return to this mystery in a couple of chapters.

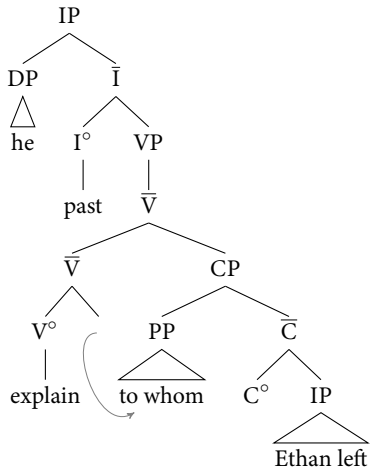
(102) I asked...



Now, to see that Wh Movement is also subject to (98), consider the sentence in (103). (103) is ungrammatical because the rule of Wh Movement has moved the wh-phrase to a Specifier of CP that does not c-command the position the wh-phrase occupies at D-structure.

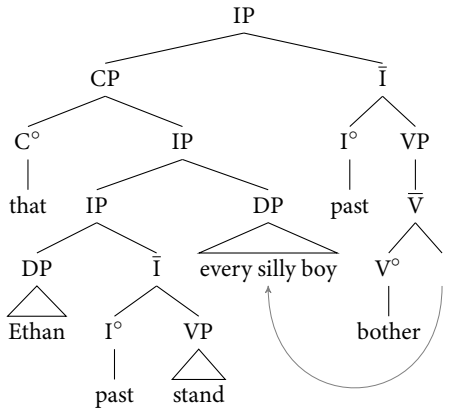
(103) * He explained to whom Ethan left.

I have placed the CP complement to ***explain*** inside the VP in (103), but we have now concluded that CPs can be Complex NP Shifted to the right of VPs. I leave out this detail here just to simplify the derivation involved. This example would still violate (98) even if I'd given the correct S-structure parse.



Complex NP Shift also obeys (98), as the ungrammaticality of (104) demonstrates.

(104) * That Ethan stood every silly boy bothered.



These are a small sample of the many examples where movement rules would create ungrammatical sentences, if they were not subject to (98).

Sportiche points out that Quantifier Float is subject to the parallel constraint in (105).

(105) If α is a floated quantifier related to the DP β, then β must c-command α.

It isn't possible for (106b) to have an interpretation like that which (106a) gets, presumably because it is not possible for Quantifier Float to relate these two structures. And because (106b) can have no other interpretation—there are no other plural DPs for *tous* to be construed with—the sentence is ungrammatical.

(106) a. [L'auteur de tous ces livres] a vu ce film.
the author of all these books has seen this film

b. * [L'auteur de ces livres] a tous vu ce film.
the author of these books has all seen this film

The ungrammaticality of (106b), then, can be credited to the fact *ces livres* fails to c-command *tous*.

What the c-command requirement on Quantifier Float establishes is that however the semantic connection between a floated quantifier and the DP it involves is worked out, it should follow that the DP c-commands the quantifier. There are a variety of ways that could be ensured, but because movement Transformations obey this relation, they provide one. We'll see what Sportiche's suggestion is about the semantic connection between the floated quantifier and the DP in a moment. But first, let's consider how Quantifier Float seems to be diagnostic of Argument movement.

We have seen in the previous section that the subject argument, *the kids*, in examples like (107) moves from a position within the small clause that follows the main verb.

(107) a. The kids seem happy.
b. The kids were made happy.
c. The kids were on the table.

And, as expected on Sportiche's proposal, a floated quantifier within the small clause can be related to *the boys*, as in (108).

Some speakers find (108b) marginal; I don't know why.

(108) a. The kids seem both happy.
b. ? The kids were made both happy.
c. The kids were both on the table.

The relationship between the floated quantifier and the subject is like that in examples such as (109), then.

(109) a. She made the kids both happy.
b. She found the kids both on the table.

In (109), unlike (108), the DP is able to remain within the small clause and satisfy the Case filter.

Consider now (110).

(110) The parents found the kids both on the table.

This is grammatical, and has an interpretation in which *both* is construed with *the kids*. It doesn't have an interpretation, however, in which *both* is construed with *the parents*. Unlike (108), the floated quantifier in the small clause of this example cannot be construed with the argument in the higher subject position. The difference between (108) and (110) is that the argument in the higher subject position is θ-marked inside the small clause in (108), but in (110) it is θ-marked outside the small clause. In fact, (110) is part of the following generalization about the relationship between a floated quantifier and the DP it is construed with.

(111) A floated quantifier construed with a DP cannot be lower in the phrase marker than the position that DP gets its θ-role.

Because *the parents* in (110) is θ-marked by some phrase that contains *find*, the floated quantifier is necessarily contained within this projection and (111) prevents it from being construed with *the parents*. Sportiche's proposal that Argument Movement relates a floated quantifier's position to the argument it is construed with derives (111). The phrases that Argument Movement will apply to must start out in their θ-marked positions; this is a consequence of the Theta Criterion. Therefore, because Argument movement brings arguments to c-commanding positions only, there will be no way for an argument to be placed within the XP that θ-marks it.

It's clear, now, why examples like (97) provide evidence for the derived subjects hypothesis.

(97) The kids have all seen this film.

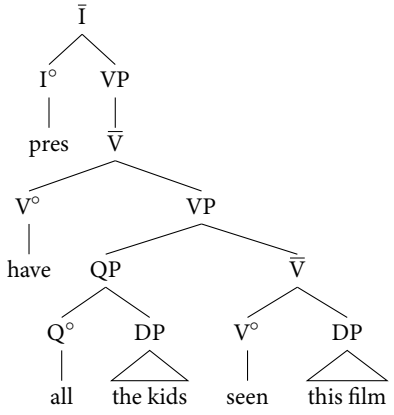
For (97) to conform to (111), the position that *the kids* gets its θ-role from must be within the phrase containing *have*. That is precisely what the derived subjects hypothesis claims.

Let's consider now how Sportiche proposes to make the semantic connection between the floated quantifier and the DP it is construed with. His leading idea is to use whatever semantic rules of interpretation derive a meaning from (112) to give what appears to be the same meaning for (97).

(112) All the kids have seen this film.

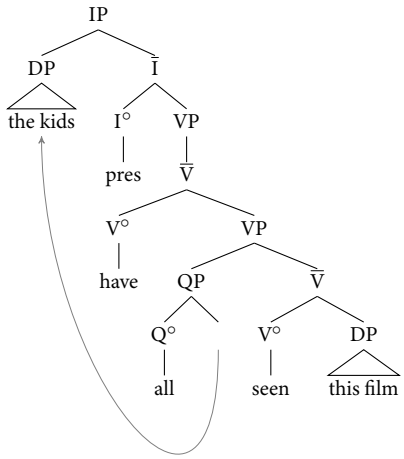
Imagine that the semantic connection between *all* and the DP it combines with in (112) is established at D-structure. This connection, then, is produced from the representation in (113).

(113)



This D-structure yields the S-structure representation for (112) when Argument movement relocates the entire QP *all the kids* into the Specifier of IP. It yields the S-structure for (97) when Argument movement applies to just the DP part of this QP, leaving the quantifier behind, as in (114).

(114)

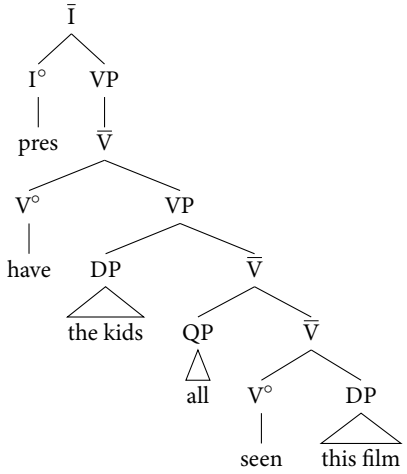


There's a popular alternative to Sportiche's account which you should keep in mind. I'll introduce it now, and offer a problem for it, but keep it at hand because it may begin to look more appealing when we discover some of the problems for Sportiche's analysis. This alternative collapses the conditions on so-called "subject oriented" (see Jackendoff 1972) adverbs with those of Q-float. Note that these adverbs have a distribution very much like floated quantifiers.

(115) The kids (deliberately) have (deliberately) seen (*deliberately) this film (*?deliberately).

To get the interpretation right, we would need a semantics that lets a floated quantifier 'distribute' the predicate that follows it over the parts of the subject. This approach to Q-float has many advocates: Bowers (1993), for example, and Kayne (1975). Roberts (1987, chapter 3) discusses the semantics for such an analysis. This alternative can be wedded to Sportiche's proposal that Argument movement is involved; it can be seen as a different way of producing the semantic composition. If we are to maintain Sportiche's suggestion that Argument movement is implicated, we should see the D-structure representation for (97) as being something like (116).

(116)



On this alternative view, Argument movement could still be credited with determining the relationship between an argument DP and a floated quantifier that Sportiche's proposal offers. What we might consider altering is how the quantifier combines syntactically with the DP it is construed with. Sportiche's (best) argument against this alternative is that the distribution of subject-oriented adverbs and floated Q are not always the same cross-linguistically.

There's another kind of argument against this alternative found in Giusti (1990b,a) which is worth looking at. Her argument is typological, and based on where floated quantifiers can be found related to objects across languages. First, note that floated quantifiers related to objects are, by themselves, something of a puzzle for an approach like the alternative sketched above. That account demands that the floated quantifier combine with a VP, or simi-

lar predicate, and steer how that predicate combines semantically with the neighboring DP. The existence of floated quantifiers related to objects, then, is by itself an argument for Sportiche's imagine of the underlying connection between quantifier and DP. What Giusti observes is that the languages which allow quantifiers to be related to objects are those for which we have independent evidence that objects can move.

Thus, German/Dutch and Icelandic, but not English or Danish, can have a floated quantifier related to an object, as shown in (117) and (118).

(117) a. Der Lehrer hat die Schüler alle gelobt.
the teacher has the students all praised
'The teacher has praised all the students.' (German)

b. De leraar heeft de kinderen allen geloofd
The teacher has the children all praised.
'The teacher has praised all the students.' (Dutch)

c. Stúdentarnir lasu greinina allir.
students-the read article-the all
'The students read all the articles. (Icelandic)

(Giusti, 1990a, (1): 137, (22a) & (25b): 144)

(118) a. * The teacher has praised the children all.

b. * Laereren roste eleverne alle
teacher-the praised children-the all
'The teacher praised all the students.' (Danish)

(Giusti, 1990a, (2c): 137)

She argues that this can be related to the fact that definite objects in Dutch, German and Icelandic can move leftwards, whereas this is not possible for objects in English and Danish. That leftwards movement is possible in these languages is, at least partly, indicated by the fact that they can precede adverbs which are normally found at the left edge of VPs in these languages. We have contrasts, then, between examples such as (119) and those in (120).

(119) a. Der Lehrer hat die Schüler ohne Zweifel gelobt.
the teacher has the students without a doubt praised.
'The teacher has without a doubt praised the students.' (German)

b. De leraar heeft de kinderen gisteren geloofd.
the teacher has the children yesterday praised.
'The teacher has yesterday praised the students.' (Dutch)

c. Stúdentarnir lasu greinina ekki.
students-the read articles-the not
'The students didn't read the articles.' (Icelandic)

(Giusti, 1990a, (2): 136, (22a): 144, (13): 141)

(120) a. * The teacher has praised the children not.

b. * Laereren roste eleverne uden tvivl.
teacher-the praised students-the without a doubt.
'The teacher without a doubt praised the students.' (Danish)
(Giusti, 1990a, (2): 137)

This is just what we would expect on a Argument Movement account of Quantifier float, of course, because on that account movement is what is responsible for separating a DP from the quantifier it is construed with. But this is also an argument for Sportiche's particular way of putting the quantifier together with the DP. On Sportiche's proposal, the only way a quantifier can appear after a DP it is construed with is by that DP's movement. The alternative account, by contrast, allows the D-structure arrangement of quantifier and DP to put the quantifier in a position that follows the DP. It's not clear how such an account could capture Giusti's correlation.

On the other hand, there are examples which Sportiche's proposal does not explain. As he notes, his model of Quantifier Float should allow quantifiers to be related to objects in English in just those cases where the object has undergone Argument movement. We've seen that one situation where this happens is in the Passive: an object moves into a higher Case-marked position when the verb it is an object of is Passivized. And yet, Quantifier Float is not possible in these contexts, as the ungrammaticality of (121) indicates.

(121) * The kids were praised all.

For some steps towards a solution, see Bošković (2001, 2004).

Another problem concerns examples like (109).

(109) She made the kids both happy.

In this example, *the kids* does not undergo movement. Its D-structure position is also the one where it gets assigned Case. If quantifiers follow the DPs they are construed with only when they move, how is it that the quantifier manages to follow the DP in this example? We will see a solution to this problem in Chapter 6. Keeping these problems in mind, let's nonetheless adopt Sportiche's account of Quantifier Float.

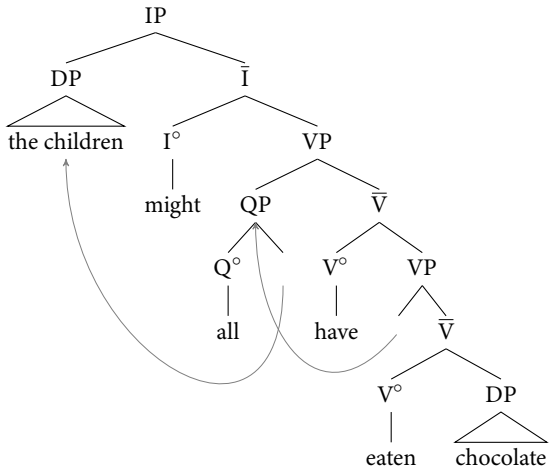
For other work on Quantifier Float, see Jaeggli (1982), Belletti (1982), Déprez (1989), Doetjes (1995) and Bobaljik (2003).

One interesting consequence of Sportiche's analysis is that floated quantifiers can be seen as indicators of the positions that arguments occupy not only in the D-structure element of the derivation, but in all the non-S-structure parses of the derivation. It gives us, in other words, a useful glimpse at these otherwise invisible parses. We learn from this that phrases can take smaller steps than necessary in their journey from a θ-marked position to a Case marked position. So in (122), for example, it appears that *the children* has made an intermediary stop in the Specifier of the VP headed by *have*.

(122) The children might all have eaten chocolate.

Argument Movement is employed twice, as indicated in (123).

(123)



We learn from these sorts of examples that it's possible for Argument movement to act on a structure that is itself the product of Argument movement. Such derivations are said to involve "successive cyclic" movement. The existence of such derivations means that complex interactions of movements are conceivable, and, as a result, the mapping from D-structure to S-structure is potentially quite complex. There are problems that lurk here, but they are for another time.

3.5 Infinitives

We have seen that "raising verbs" select small clauses but do not assign Case to the DP argument within that small clause. As a result, Argument movement brings that DP argument into a Case marked position. In (124), for instance, *Helen* has moved from within the small clause where its θ-role is assigned, into its surface, nominative Case marked, position.

(124) Helen seems happy.

Raising verbs are (some of them) also capable of selecting an IP headed by *to*, as in (125).

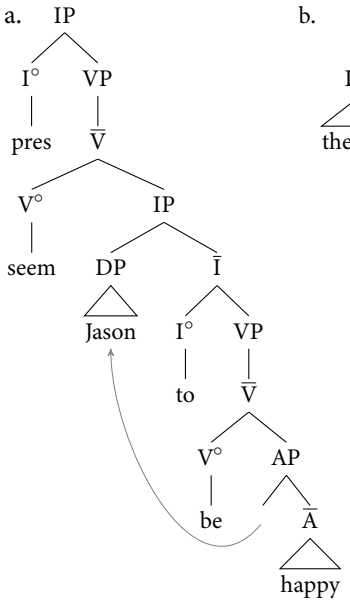
(125) Helen seems to be happy

In this example too, *Helen* has moved from a position within the small clause into its surface, nominative Case marked, position.

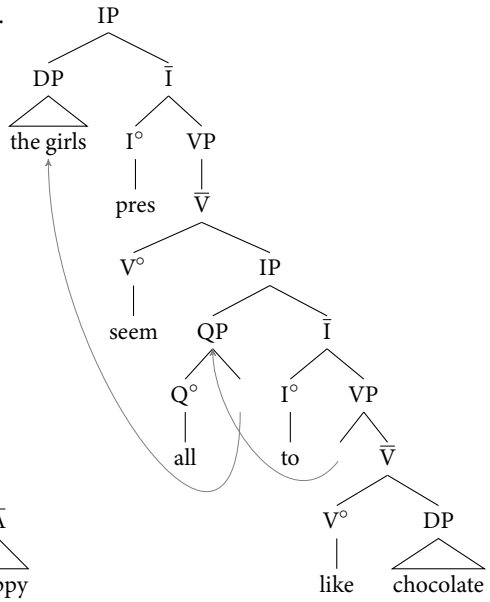
But in this case, there is a question about how the EPP is satisfied.It's clear how the highest IP satisfies the EPP: its Specifier is filled with the subject argument that resides in it at S-structure. But what of the Specifier of the IP headed by *to*? One possibility is that (125) has a successive cyclic derivation like that in (126a).

The EPP, recall, is the requirement that a Specifier of IP be filled with a DP or CP.

(126) a.



b.



From this representation, A-movement could apply again to form the S-structure we see in (125). We might imagine, then, that (125) has a successive cyclic derivation. If this is correct, we should understand the EPP to be a condition that can be satisfied by any phrase marker in a derivation.

That a successive cyclic derivation of this sort is possible is indicated by (127), whose derivation is indicated in (126b).

(127) The girls seem all to like chocolate.

We see in this example that Argument movement has moved *all the girls* into the Specifier of the infinitive before the DP part is moved into its surface position. The presence of the floated quantifier in the Specifier of the infinitival IP is the tell-tale sign that the subject argument has passed through the Specifier of the infinitival clause.

Infinitival clauses come in a variety of types that can be distinguished from those that follow raising verbs, and we will look at some of these in this section. The example in (128), for example, though it has a superficial similarity to (125), is different in some ways.

(128) She tried [to be happy]

In (125), the count of θ-roles and arguments matched, as expected. But in (128), this isn't the case. Both *tried* and *happy* assign an external θ-role, and yet there is only one argument which seems to bear them both: *she*. This is precisely what the Theta Criterion prohibits. Recall, that the Theta Criterion is designed in this way in an attempt to capture the fact that (129a) doesn't have the meaning that (129b) does.

(129) a. I showed Chris.

b. I showed Chris himself.

How the Theta Criterion applies to cases where an argument moves from one θ-marked position to another θ-marked position shouldn't be clear, since the Theta Criterion is a condition that holds just of D-structures. We come back to look at this more closely in a later section.

So, either we should find another explanation for the fact that (129) illustrates, or we should find a way of resolving (128) with the Theta Criterion.

Note also that (128) seems to be a violation of the Extension to the Projection Principle. Unless we determine that (128) involves Argument movement of *she* through the Specifier of the infinitival IP, as in the raising case, there appears to be nothing that occupies this Specifier position at any point in the derivation. Moreover, it's not just that (128) seems to counterexemplify these principles, it perversely can't satisfy them. If the Specifier of the embedded IP is given content, and in so doing hosts an argument for the θ-role that *angry* assigns, the result is ungrammatical.

(130) * She tried [him to be angry].

And if a predicate that does not have an external θ-role to assign is put into these infinitives, the result is also ungrammatical.

(131) * She tried [to seem that pigs fly].

Infinitives with this peculiar mix of properties are called control infinitives, and a wide assortment of verbs select infinitives of this type. Some are in (132).

(132) a. Jennifer prefers to eat chocolate.
b. Jennifer managed to eat chocolate.
c. Jennifer attempted to eat chocolate.
d. Jennifer wished to eat chocolate.
e. Jennifer asked to eat chocolate.
f. Jennifer proposed to eat chocolate.
g. Jennifer neglected to eat chocolate.
h. Jennifer promised to eat chocolate.
i. Jennifer decided to eat chocolate.

In all of these examples, there is an argument—here it is *Jennifer*, the subject of the higher verb—that appears to bear the θ-role assigned to the subject of the infinitival clause. We say in these cases that the subject argument "controls" the infinitival clause, or "controls" the embedded subject. There are instances of control infinitives where it's the object of the higher verb that is the controller; (133) provides some examples.

(133) a. Bernhard told Cecilia to eat chocolate.
b. Bernhard asked Cecilia to eat chocolate.
c. Bernhard required Cecilia to eat chocolate.
d. Bernhard forced Cecilia to eat chocolate.

In these examples, it is the object, *Ceclia*, not the subject, *Bernhard*, that is understood to bear the external θ-role assigned by *eat*.

From what we have seen about control infinitives, we can draw the following conclusions.

(134) a. A control infinitive must have a subject argument.
b. The subject argument of a control infinitive cannot be pronounced within the infinitive.
c. The subject argument of a control infinitive can be understood as the same as an argument in the higher clause: its "controller."

Our goal is to understand what is responsible for these properties of control infinitives. Our hope should be that it is consistent with, maybe even follows from, the forces that are responsible for positioning arguments. We'll look at a few ideas in this section that might bring us within range of these desires. I'll start with the question of whether these infinitives are CPs or not, and then turn to issues that more transparently bear on (134).

3.5.1 The CP/IP status of infinitives

Despite the lack of a complementizer, there is some evidence that control infinitives can be a CP. Of the verbs that select a control infinitive, those that also s-select a question can combine with a control infinitive that begins with a wh-phrase; (135) are some examples.

The status of complementizers in infinitival clauses, and the whole array of differences in kinds of infinitival arguments, originates with Bresnan (1972), who built upon Rosenbaum (1967).

(135) a. Caroline told Helen how to eat chocolate.
b. Caroline asked when to leave.
c. Caroline decided when to leave.
d. Caroline wondered what to eat.

(136) a. Mariko told Paul whether to eat chocolate.
b. Mariko asked whether to leave.
c. Mariko decided whether to leave.
d. Mariko wondered whether eat chocolate.

Recall that wh-phrases are moved into the Specifier of CP by Wh movement. We should conclude, then, that the infinitives in (135) are CPs. And in these very contexts, the complementizer *whether* is, as expected, also possible as (136) shows.

In many, not too distantly related, languages, even the non-question form of a control infinitive comes with a complementizer. Icelandic is a particularly clear example of this. It has subject and object control infinitives that closely resemble those of English. And yet these control infinitives appear with the same complementizer that is found in finite clauses in Icelandic: *að*.

(137) María lofaði **að** lesa bókina.
Mary promised that read the book
'Mary promised to read the book.'

compare:

María segir **að** θú hafir lesið bókina.
Mary says that you have read the book
'Mary says that you have read the book.'

We should conclude, then, that a control infinitive can be a CP.

By contrast, the same kind of evidence regarding raising infinitives suggests that they are never CPs. There are no known examples of raising infinitives that have the syntax of questions.

(138) a. * It seems whether to have left.
compare: Nancy seems to have left.

b. * It appears when to eat chocolate.
compare: Nancy appears to eat chocolate.

c. * It is likely who to dance.
compare: Nancy is likely to dance.

This could be because infinitival clauses are incapable of being CPs, and therefore do not support the CP-dependent syntax of questions. But it could also be because raising verbs (and adjectives, such as *likely*) do not semantically select questions. Whether a verb can take a question as an object depends on the semantics of the individual verb. A verb like *believe*, for instance, cannot take a question complement, whereas the similar *know* can.

(139) a. * Ania believes whether Eva has left.

b. Ania knows whether Eva has left.

The absence of question complements to raising verbs could, then, simply reflect a fact about the semantics of these predicates, and nothing about the syntax of their infinitival complements. Indeed, raising predicates do not seem able to combine with question complements even when those complements are finite clauses.

(140) a. * It seems whether Elliot has left.
compare: It seems that Elliott has left.

b. * It appears when María has eaten the chocolate.
compare: It appears that María has eaten the chocolate.

c. * It is likely who will dance.
compare: It is likely that Anna will dance.

Finite clauses can always be CPs (it seems), and so we should conclude that the absence of question complements to raising verbs has to do with their meaning.

Although we cannot conclude that raising infinitives are prevented from being CPs, we also cannot learn from their inability to be questions whether they are allowed to be CPs. Icelandic, however, provides a clue. Raising infinitives in Icelandic cannot include the complementizer *að*.

(141) * María hafði virst að hafa vaskað upp diskana.
Mary had seemed that have washed up the dishes
'Maria had seemed to have washed up the dishes'

compare:

María hafði virst hafa vaskað upp diskana.
Maria had seemed have washed up the dishes
'Mary had seemed to wash up the dishes.'

This would be explained if raising infinitives are IPs, but cannot be CPs.

In Icelandic, then, we can conclude that one difference between raising and control infinitives is their categorial status. Raising infinitives are IPs but cannot be CPs, whereas control infinitives are CPs. In English, the picture isn't as clear. There is no direct evidence that raising infinitives cannot be CPs. And though there is evidence that some control infinitives are CPs—the questions—other control infinitives do not differ outwardly from raising infinitives. English internal evidence supports the conclusion that control infinitives can be CPs, but nothing more.

If we consider the evidence from Icelandic and English together, however, and keep in mind the criterion of explanatory adequacy, we can find some help in making decisions about the proper treatment of English infinitives. We conclude from Icelandic that the mapping in (142) exists in at least one grammar.

(142) raising infinitive → IP
control infinitive → CP

Assuming that this is not a capricious relationship, we can assume that there are forces, still to be discovered, that produce it. If the mapping in (142) does not materialize in English, then we can conclude that those forces do not operate in the same way in English. When we model these forces, our model should allow them to vary from language to language. Consequently, we must credit the child with the ability to detect that variation as that child is acquiring his or her language. Let's call this "hypothesis 1."

Hypothesis 2 is that the forces responsible for the existence of (142) in Icelandic are also found in English. On this hypothesis, Icelandic and English should be identical with respect to the categorial status of their infinitives. Any differences in the appearance of infinitives in these languages will flow from other ways these languages differ. What are the differences between Icelandic and English that would have to be explained on hypothesis 2? As we've seen, Icelandic and English raising infinitives look the same; nothing would have to be said under hypothesis 2 about how raising infinitives differ

in these languages. Control infinitives in Icelandic and English also look the same in all ways except one: declarative control infinitives can appear with a complementizer in Icelandic but not in English. Hypothesis 2 requires that this difference be made compatible with the mapping in (142). One way of doing that is with (143).

(143) The complementizers of infinitival clauses in English have no phonetic realization.

It seems difficult to avoid a conclusion like (143) under hypothesis 2. Hypothesis 2 is likely to require, then, that words can be silent. It must credit the child with the ability to detect the existence of silent words in a sentence.

So we have before us two alternatives. English does not conform to the mapping in (142) and the forces that are responsible for (142) can vary in ways that are detectable by children acquiring English. Or, English does conform to the mapping in (142) but has a silent infinitival complementizer, and the existence of this silent complementizer is detectable by children learning English. Framing the analysis of English infinitives in this way allows us to make decisions based on criteria of language acquisition. If the forces behind (142) turn out to be too distant from the information available to a child, then we can conclude that they cannot vary in detectable ways and we're stuck with hypothesis 2. If, by contrast, the existence of words in a sentence cannot be detected without their phonetic manifestation, then we're stuck with hypothesis 1. Hypothesis 2 is, perhaps, the most popular in the subsequent literature. One way of ensuring that the silent complementizer is detectable to the language-acquiring child is to elevate the mapping in (142) to a language universal. If the child can rely on (142), then it can deduce that there is a silent complementizer in the declarative control infinitives of English. Let's adopt hypothesis 2 as our working hypothesis as well.

Before leaving the question of whether infinitives are CPs or not, we should look at one last "kind" of infinitive. Some predicates that take control infinitives as objects can take an infinitive which does have an overt complementizer, and in which the subject argument is visible inside the infinitive. Some examples of these predicates and the two types of infinitives they can combine with, are given in (144).

(144) a. Misato wanted to eat durian.
Misato wanted for him to eat durian.

b. Junko preferred to eat natto.
Junko preferred for me to eat natto.

c. It is possible to eat poi.
It is possible for some to eat poi.

The subject of the infinitival clauses appears in the accusative Case, as is clear from the form of the pronouns in (144a) and (144b). The complementizer is *for*, which might easily be mistaken for the homophonous preposition found

in sentences like *She wanted something for Susan.* We think that the *for* in (144) is a Complementizer for several reasons. First, it forms a constituent with the entire clause that follows it. This is indicated, among other things, by the behavior of these phrases in the context of cleft constructions.

(145) a. It's **for him to eat chocolate** that Jason would prefer.
b. * It's to him how to eat chocolate that Jason should explain.

The ungrammaticality of (145b) derives from the fact that there is only room for one phrase between the *it's* and the *that* of these clefts, and in (145b) two things, a PP and an infinitive, have been shoved into this spot. Thus the grammaticality of (145a) would seem to argue that we do not want to parse the boldfaced string as a PP followed by an infinitive, but instead as something that makes a single phrase. To the extent that an infinitive is an IP, and that the *him* in (144) is its subject, then one of the few ways of doing this is to let *for* be a complementizer.

Moreover, the *for* that comes with these infinitives has a property that no preposition in English has, but which the complementizer *that* does have. It is optional. Under a certain set of circumstances, the complementizer *that* can go unspoken in English, as in (146).

(146) a. Jason said that he eats chocolate.
b. Jason said he eats chocolate.

This is also possible for the *for* of those infinitives that are objects to verbs, as (147) shows.

(147) a. Misato wanted him to eat durian.
b. Junko preferred me to eat natto.

Furthermore, the circumstances controlling the optionality of the complementizer *that* are mimicked by the optionality of *for*. In general, the complementizer *that* can go unspoken only in contexts where the CP it heads is a complement to a verb. That is why *that* cannot go missing when it heads a CP that has been clefted, as in (148). A parallel constraint on *for* ellipsis is indicated by (149).

(148) a. It's that he eats chocolate that Jason said.
b. * It's he eats chocolate that Jason said.

(149) a. It's for him to eat chocolate that Jason would prefer.
b. * It's him to eat chocolate that Jason would prefer.

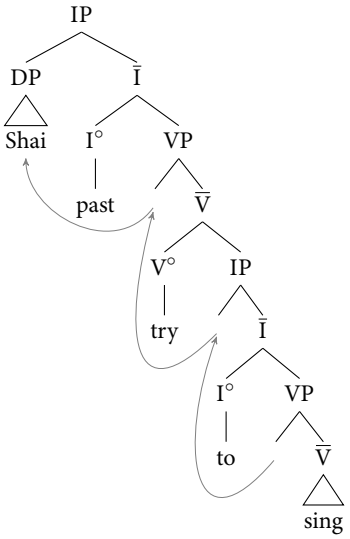
In general, the constraints on where *that* can go missing are parallel to those on where *for* can, suggesting that they are both subject to some process, restricted to complementizers, that allows them to go silent.

3.5.2 On the Control relation

Let's return now to how control infinitives express their arguments. In the cases we've looked at so far, they are rather like raising infinitives in that an argument of their main predicate surfaces in a position in the higher clause. One possibility, then, is that control infinitives use the same mechanism that raising infinitives do to put their arguments in this position. We might imagine that Argument movement relocates the argument DP of (150) in the way indicated in (151).

(150) Shai tried to sing.

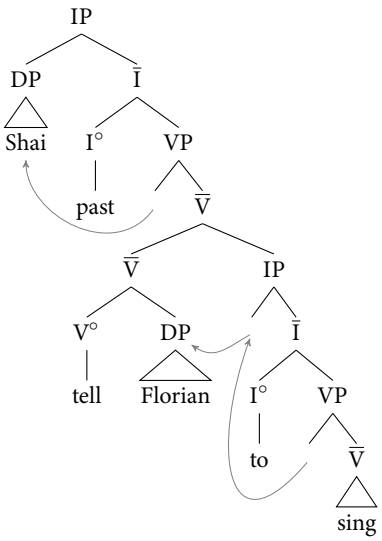
(151)



An instance of object control, like that in (152) could have the derivation indicated.

(152) Shai told Florian to sing.

(153)



If this turns out to be correct, then we will either have to change the restriction on movement that prohibits movement to a non-c-commanding position, or we will have to jettison the view that phrases have no more than two daughters. With these two assumptions, the movement, as indicated here violates the c-command requirement.

These derivations differ from the ones we've posited up to now for Argument movement in that they move arguments into θ-marked positions. If this is the correct account of control infinitives, then, we should adjust our definition of D-structure as the sole parse in a derivation that gets semantically interpreted. That definition prevents a θ-marked position from being empty at D-structure, and also prevents an argument from gaining a θ-role by moving into a θ-marked position. But if we were to make it possible, somehow, to let parses in a derivation other than the D-structure be semantically interpreted, we would have a transparent account of the Control relation. We would also provide an account for how control infinitives satisfy the EPP and that part of the Theta Criterion that requires every θ-role to be assigned to an argument. On this view, the Control relation is just the one established by Argument movement between an argument and its θ-marked position. This view of the Control relation is offered in Hornstein (1999) and its consequences explored in Hornstein (2000) and Boeckx and Hornstein (2003, 2004, 2006a,b).

There are circumstances where control infinitives do not involve control, however; in these circumstances the subject argument is not an argument expressed elsewhere. This is the case when, for instance, the verbs taking a control infinitive are passivized, as in (154).

(154) a. To eat natto shouldn't be tried.
b. To be content is often wanted.
c. To be honest should be promised in a wedding vow.

In fact, whenever a control infinitive stands in subject position, its understood subject does not have to be an argument found elsewhere; (155) provides some other examples.

(155) a. To remind him of Amherst Winters bothered Pius.
b. To eat durian shouldn't be possible.
c. To leave your shoes untied is a danger to those around you.
d. To be the largest prime number is to be very large indeed.

In these examples the subject of the infinitive is understood to have a meaning close to that which *one* gets in sentences like (156), as can be appreciated by comparing this example to (155b).

(156) For one to eat natto shouldn't be possible.

There is a subtle, and indescribable, difference in the meanings of (156) and (155b), but it doesn't seem to me that this difference resides in the meaning of the subjects.

The understood subject of these infinitives is said to be "arbitrary."

In (154) and (155) there is no argument that could be the understood subject of the infinitive and so it is easy to see that it must get another meaning. But there are also examples where a potential controlling argument is present and yet an arbitrary reading for the understood subject of the infinitive is available. That is the case for the subject argument in the examples in (157), for example, which may all have the same arbitrary interpretation that is found in (155).

(157) a. Elena discovered that to remind Pius of Amherst winters bothers him.
b. Jeff thought that to eat durian shouldn't be possible.
c. Bart remembered that to leave his shoes untied is a danger to those around him.
d. Jill knew that to be the largest prime number is to be very large indeed.

When the infinitive is a question, it may also sometimes support an arbitrary interpretation for its understood subject, as in (158).

(158) a. Jon knew how to eat live shrimp.
b. Satoshi can tell when to add vinegar to rice.
c. Julie decided where to put the *Dicentra*.

These can get interpretations roughly parallel to those in (159), in which the subject is arbitrary.

(159) a. Jon knew how one can eat live shrimp.
b. Satoshi can tell when one should add vinegar to rice.
c. Julie decided where one should put the *Dicentra*.

The precise conditions under which the arbitrary interpretation are available are imperfectly understood. See Williams (1980, 1989) and Landau (2004) for some ideas.

Control infinitives with an arbitrary reading pose the same problem for the EPP and Theta Criterion that normal control infinitives do. They seem to violate the EPP—there is nothing in the Specifier of their IP—and because the θ-role normally assigned to the subject has no argument to bear it, they seem to violate the Theta Criterion as well. Moreover, just like normal control infinitives, they are ungrammatical if there is no θ-role for a subject argument, as (160) shows.

(160) * To seem that pigs fly is worrisome.
compare: For it to seem that pigs fly is worrisome.

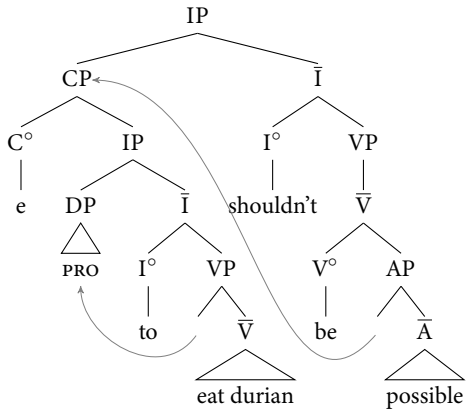
Just as for normal control infinitives, then, these facts taken together suggest that there is a subject argument present in these infinitives. If such an argument is present, then the EPP and Theta Criterion could be maintained. Moreover, the ungrammaticality of examples like (160) would be explained: if there is a subject argument, then the Theta Criterion requires that there also be a θ-role assigned to it. In these cases, however, this alleged subject argument cannot be found in some unexpected position. We cannot here finger the controller as that argument because there is no controller.

A common reaction, and the one I shall adopt, is that there is a subject argument syntactically present in these examples but that it is silent. This

For an approach that does not countenance a silent subject, see Chierchia (1984).

argument is indicated by putting "PRO" in the position our syntax suggests it should be. The representation for (155b), then, would be (161).

(161)



I've decided that this infinitival clause is a CP with a silent complementizer, represented here with "e." I've also shown PRO moving from the position that the external θ-role is assigned to, Specifier of VP, to the Specifier of IP where it satisfies the EPP. I also show the infinitival CP itself moving from Specifier of AP into Specifier of IP; this claims that the infinitival gets an external θ-role from *possible* and moves in order to satisfy the EPP.

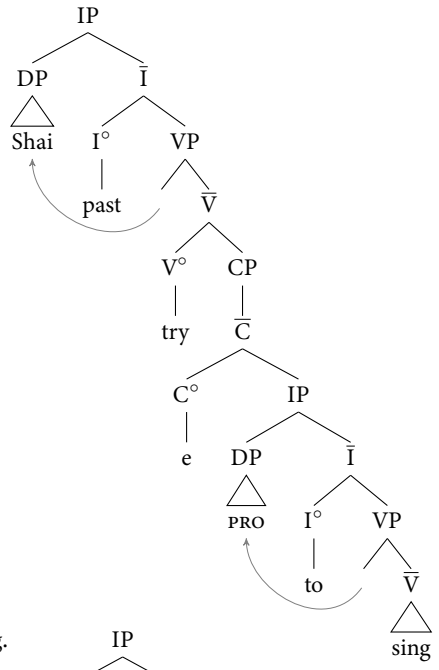
We can think of PRO as being a kind of pronoun (hence its name), maybe along the lines of a silent *one*. Of course, if the grammar of English includes a silent pronoun, then where it can be must be heavily constrained. Otherwise, sentences such as (162) should be grammatical, with Theta Criterion satisfied by the presence of PRO in subject position or object of preposition.

(162) a. * Ran into the garden.
b. * Andre talked to.

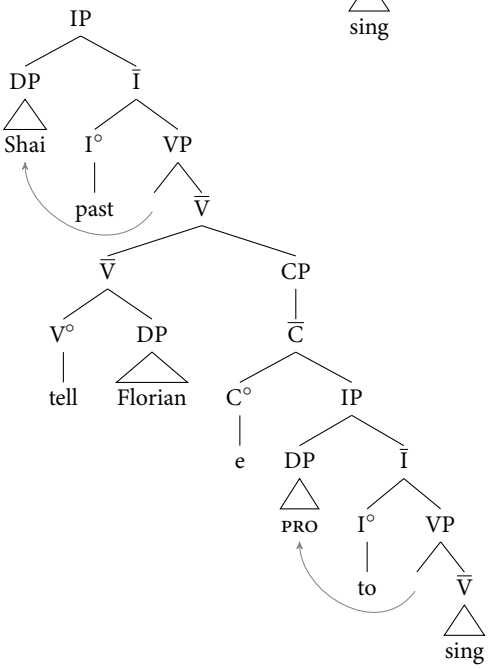
The problem of restricting PRO's distribution remains unsolved. In fact, the question of where PRO can be is still controversial. There is no controversy over the question of whether PRO can be found in complement positions; it seems clear that it cannot. And it also seems clear that PRO cannot be in subject position of finite clauses, at least not in English. But it's not clear yet in which non-finite clauses PRO can be a subject.

Perhaps the most popular view presently is that PRO can stand as subject to any control infinitive, not just those that get an arbitrary interpretation. On this view, the Control it induces does not reduce to a special instance of Argument Movement, but is instead an interpretation that PRO can receive. The representation for sentences involving Control would not be as indicated in (150) and (152), but would instead include PRO, as in (163) and (164).

(163) Shai tried to sing.



(164) Shai told Florian to sing.



Unlike the movement-based analysis of Control that (150) and (152) illustrate, the representations in (163) and (164) obey the Theta Criterion as it is now formulated. What we would need to do if these representations are correct is produce a non-movement account of Control, an account that would emerge as part of a more general account of how PRO gets its semantic interpretation.

What we've discovered in this section, then, is that there is a silent pronominal argument in English and that this silent argument is prevented from appearing in a variety of positions otherwise normally available to arguments.

Keeping in mind that this argument, PRO, cannot be in complement positions, we can partially describe its special distribution with (165).

(165) The PRO Restriction
PRO can only be in certain Specifier positions at S-structure.

Which Specifier positions PRO can be in is controversial, but it certainly doesn't include the Specifiers of finite IP.

The PRO Restriction describes something that is peculiar to the placement of the silent argument: PRO. Let's now look at whether the conditions we've been developing to capture the placement of the non-silent arguments also hold of PRO.

3.5.3 Argument movement and PRO

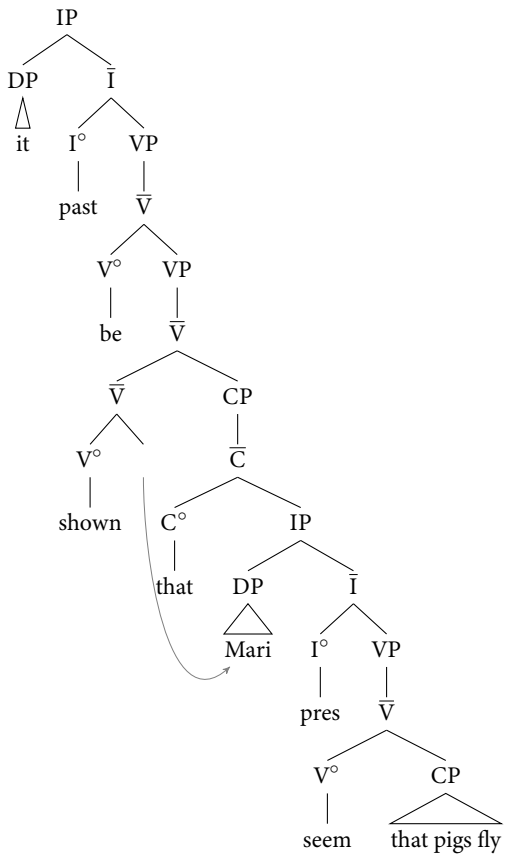
To this point, Argument movement has been invoked in situations where a DP is θ-marked in a position that is not Case marked. Because argument DPs must be both Case marked and receive a θ-role, there will be no single parse that can satisfy both of these conditions in such a case. Argument movement therefore produces a series of parses: one of which will satisfy the Theta Criterion and Projection Principle and another one of which will satisfy the Case filter. I have expressed the Theta Criterion and Projection Principle as well-formedness conditions on D-structures, and the Case filter as a well-formedness condition on derivations. This places certain conditions on how Argument movement will arise because it imposes requirements on what the input and output parses can be. If Control is an instance of Argument movement, then certain changes to the Theta Criterion will have to be made, and there will be consequent changes to how Argument movement behaves.

In addition to the constraints imposed on Argument movement by these conditions, we have seen that there is a condition imposed directly on Argument movement, and other movement operations: the moved term can only be relocated to a position that c-commands its starting position. This condition is responsible for blocking (166).

(166) * It was shown that Mari seems that pigs fly.

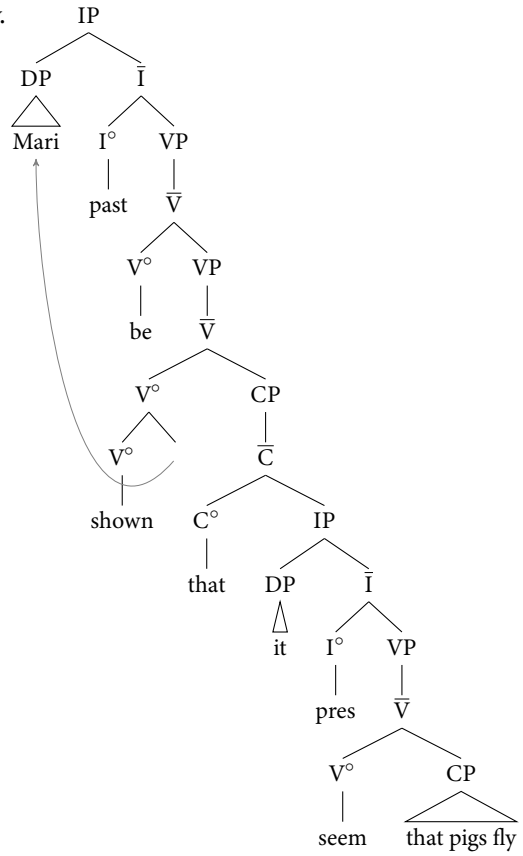
If Argument movement could move DPs to non-c-commanding positions, then (166) should be able to have the derivation indicated in (167).

(167)



The derivation satisfies every other condition we've discovered. The EPP is satisfied for both Specifier of IPs. The number of θ-roles and arguments is the same, so the Theta Criterion is satisfied. And every argument stands as a sister to the term that assigns it a θ-role at D-structure, and so the Projection Principle is satisfied as well. In particular, the argument *Mari* stands as a sister to the verb *shown* at D-structure, and there gets the object θ-role from *shown*. Because *show* has been Passivized, however, it does not have an accusative Case to assign to *Mari*, and so Argument movement must move this DP to a Case marked position. The Specifier of the finite IP that follows *Mari* is such a Case marked position, and so movement to this position should satisfy the Case filter. If this position c-commanded the D-structure position of *Mari*, every other condition would be satisfied and this sentence should be grammatical. It's the c-command requirement on Argument movement, then, that is responsible for preventing sentences such as (166). This can be appreciated by comparing the grammatical (168), in which the c-command requirement is satisfied.

(168) Mari was shown that it seems that pigs fly.



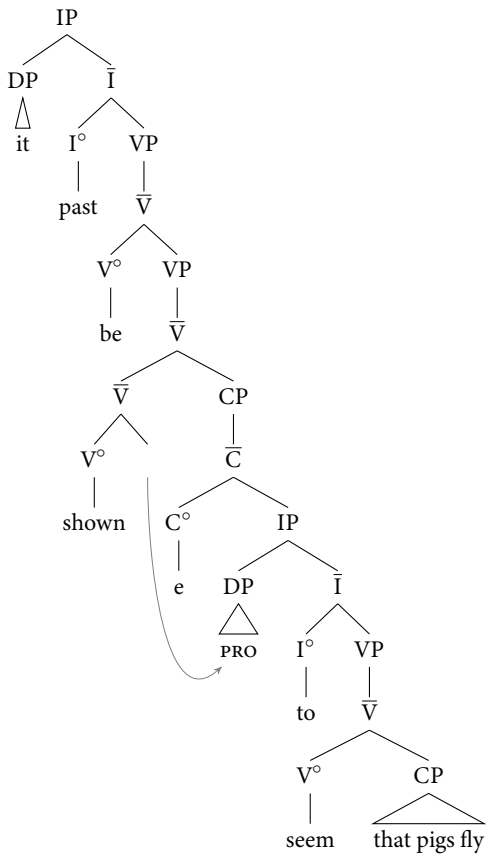
If PRO is an argument DP, then we should expect that Argument movement will be responsible for fixing its position and, consequently, we should find a parallel pattern of grammatical and ungrammatical sentences that the successful or unsuccessful application of Argument movement is responsible for. In the case of overt DPs, the success of Argument movement hinges on being able to find an S-structure that satisfies the Case filter that can be produced from a grammatical D-structure through an application of Argument movement that satisfies the c-command requirement. In the case of PRO, the grammatical sentences should be ones in which PRO can reach the Specifier of an appropriate infinitival IP from its θ-marked D-structure position by way of an application of Argument movement that satisfies the c-command requirement.

This match in grammaticality patterns shows up. The ungrammaticality of (169), for instance, parallels the ungrammaticality of (166).

(169) * It was shown to seem that pigs fly.

If the infinitival clause in (169) is of the sort in which PRO is permitted, then this example satisfies every condition except the c-command constraint on Argument movement, as (170) indicates.

(170)

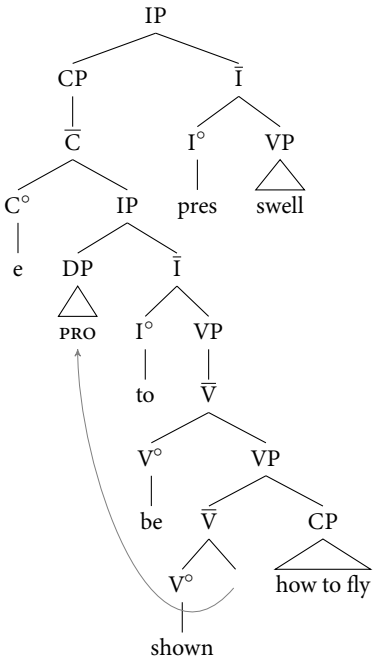


That this infinitive is of the kind that permits PRO in its Specifier position is indicated by (171), which permits an arbitrary reading for the subject of its infinitive.

(171) Mari was shown how to fly.

When the c-command requirement on Argument movement is obeyed, then examples parallel to (170) are grammatical, as (172) indicate.

(172) To be shown how to fly is swell.



Examples like these, then, suggest that the position of PRO is controlled by the same set of constraints that we've seen determine the position of overt argument DPs. To the extent that this is true, it speaks on behalf of the existence of PRO. If the ungrammaticality of sentences like (166) are due to conditions on where DP arguments may appear in S-structures, and those conditions are mimicked by arguments which are semantically present but not spoken, then we should conclude that the syntactic position of those unspoken arguments matters. They must be present in the syntactic representation for that to hold.

There is another condition on Argument movement of overt DPs that we can seek in the Argument movement of PRO. This condition we have not yet encountered; it concerns how the Case filter is satisfied. It appears that DPs cannot be Argument moved from positions that are Case marked.

This condition can be found in Chomsky (1981), and for a more recent formulation: Chomsky (2000) and Chomsky (2001).

(173) If Argument movement moves something from position α, then α cannot be a Case-marked position.

This condition is thought to be responsible for the ungrammaticality of (174).

(174) * Elena seems to that pigs fly.
(*compare:* It seems to Elena that pigs fly.)

The argument DP *Elena* resides in its θ-marked position, which is the complement position of *to*, and from there moves into the Case marked Specifier of IP position, thereby producing a representation that satisfies the EPP. The D-structure and S-structure representations meet the well-formedness requirements that they need to meet, and the position that *Elena* moves into

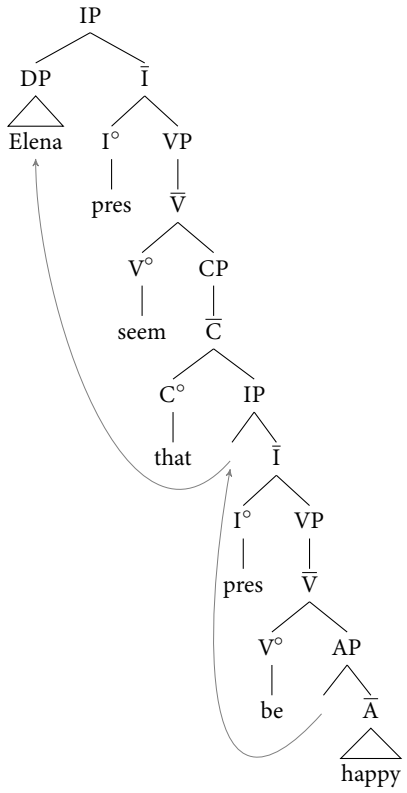
c-commands the position it moves from. Even though it meets all the other requirements on a grammatical sentence, (174) is ungrammatical because it violates (173). The position that *Elena* resides in at D-structure is assigned accusative Case by *to*, and (173) prohibits Argument movement from a Case-marked position.

Another example whose ungrammaticality relies on (173) is (175).

(175) * Elena seems that is happy.

The derivation that needs to be blocked here is shown in (176).

(176)

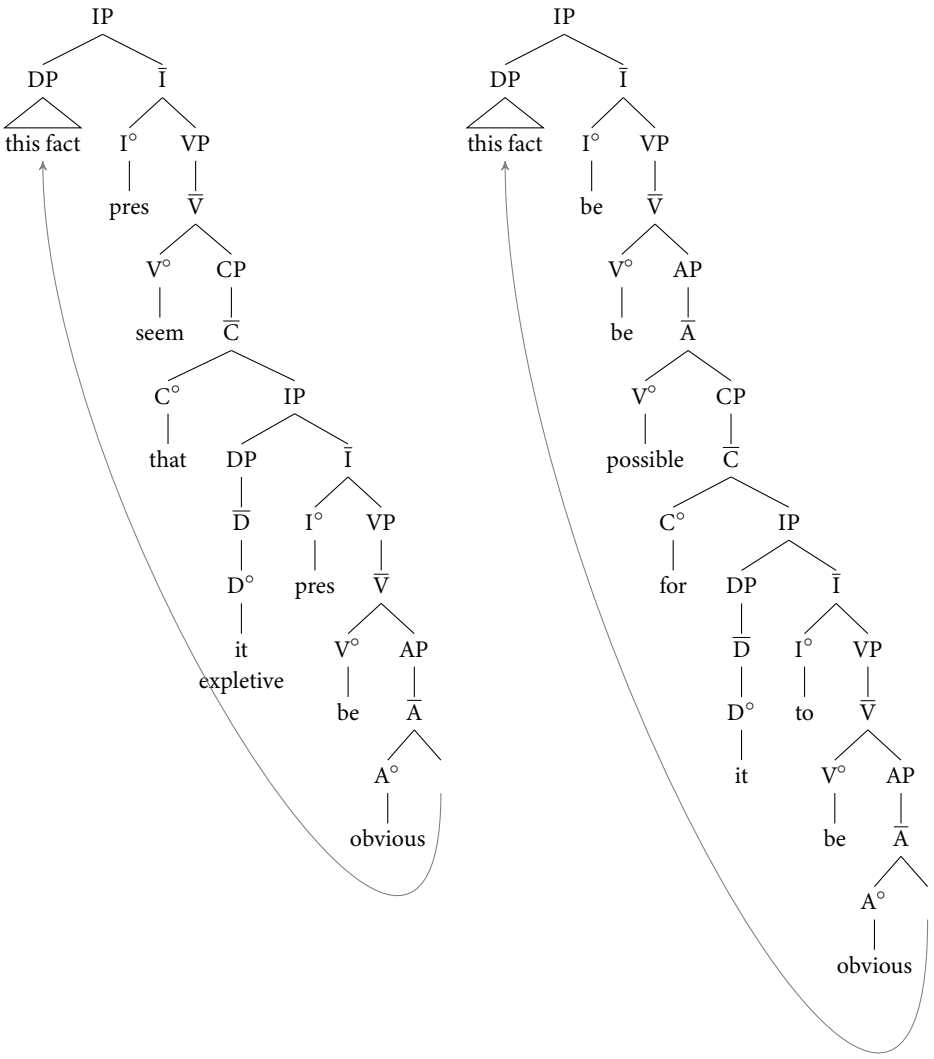


For this sentence to satisfy the EPP, it is necessary for Argument movement to bring *Elena* into the Specifier of the embedded IP. But this position is assigned nominative Case by the finite I, and so movement from this position is prohibited by (173). There is consequently no way to derive (175) that does not violate either the EPP or (173).

There is another feature of the derivation in (175) that might be responsible for its ungrammaticality. The argument has moved out of a CP in (175), and this is the first example we have considered in which that is true. In grammatical examples of Argument Movement out of embedded clauses, the clauses have been infinitival IPs. There is some evidence that Argument Movement out of CPs is independently prevented, and so that could also

be the source of the ungrammaticality of (175). That evidence comes from consideration of cases like (177).

(177) ⋆This fact seems that it is obvious. *and* ⋆ This fact is possible for it to be obvious.



On the face of it, these derivations should be grammatical. The D-structures violate the EPP and the Case filter, but both of these maladies are remedied by using Argument Movement to move into the Specifier of the largest IP. The fact that these examples are ungrammatical suggests that there must be something that prevents Argument Movement from applying in this way.

Cases of Argument Movement out of CPs in English are not attested; (178) seems true.

(178) If Argument Movement moves XP from position α to position β, then there cannot be a CP that dominates α but not β.

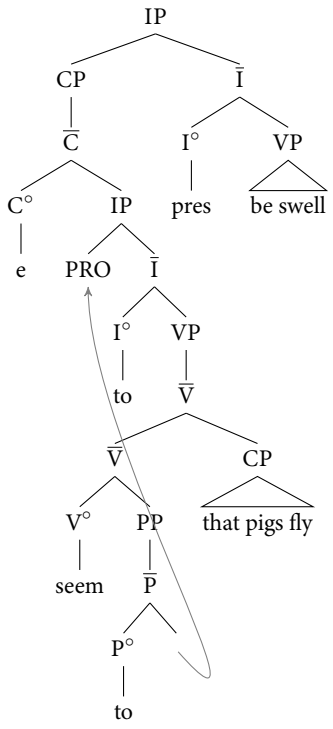
This constraint, along with (173), limit where Argument Movement can occur.

There are examples parallel to the ones supporting (178) and (173) that involve PRO, and they too speak on behalf of the existence of PRO and letting Argument movement play a role in positioning PRO. Parallel to (174) is (179).

(179) * To seem to that pigs fly is swell.

This example would be grammatical if it were possible to Argument move PRO from its D-structure position following *to* into the Specifier of the infinitival clause, as shown in (180).

(180)



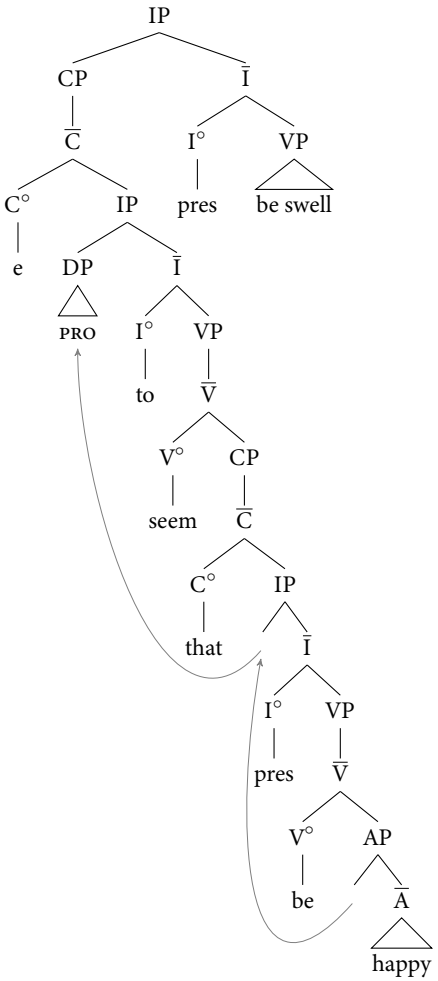
To explain its ungrammaticality in a way that relates it to (174), we should credit the ban against Argument movement from Case marked positions.

Parallel to (175) is (181).

(181) * To seem that is happy is swell.

The derivation we should be considering for (181)—the one that satisfies the EPP—is (182).

(182)



Just like (176), this violates the ban against Argument movement out of a CP.

A final condition influencing Argument movement that might be relevant for the placement of PRO is the Case filter. If PRO is an argument DP, then we should expect it to obey the Case filter. Is Argument movement invoked to put PRO into Case marked positions at S-structure just as it is to put overt DPs into Case marked positions?

It's not easy to determine the answer to this question because of the confound that the PRO Restriction presents. The PRO Restriction describes the fact that PRO seems to occur at S-structure in the Specifiers of control infinitives and nowhere else. If PRO is also subject to the Case filter, then the narrower distribution that the PRO Restriction allows it will make it difficult to detect. We are reduced, in fact, to the question of whether the positions that the PRO Restriction allows PRO to be in are Case marked positions.

We presently have two ways of determining whether a position is a Case-marked one. If an overt DP, bearing Case, is capable of surfacing in some position, then that position in assigned Case. And if Argument movement is

prohibited from some position, then through a process of elimination we can determine whether the ban against moving from a Case-marked position is to blame and in this way discover whether a position is Case marked. By the first criterion, we cannot conclude that the Specifier positions in which PRO resides are Case-marked. The Specifiers of control infinitives cannot contain overt DPs.

(183) a. * Her to run seems unlikely.
b. * She to sing is swell.

From this, we might conclude that the positions PRO resides in are not Case marked positions. This would explain why overt DPs cannot surface in this position: the Case filter would be violated if so. This is the conclusion reached in *Lectures on Government and Binding*, and it informs most of the literature on PRO from the 1980s and 1990s.

The second criterion gives a somewhat more mixed image. If the movement account of Control is correct, then the ban against moving from a Case-marked position allows us to deduce that the Specifier of the control infinitive that a controller moves from is not Case marked. So, for instance, we can conclude that the Specifier position of the infinitival complement to a verb like *try* is not Case-marked if there is Argument movement of the sort indicated in (184).

(184) Hotze tried $[_{CP}\,[_{IP}$ to eat durian]].

If the infinitive that controllers move from is the same kind that can host PRO in non-control environments, then the simplest assumption would be that the Specifier of these infinitives is not Case-marked when they host PRO as well. So, for instance, if the infinitive clause that stands in the subject position of a Passivized *try* is the same as that which stands as the object of an Active *try*, as in (185), then the PRO this infinitive contains is standing in a non-Case-marked position.

(185) $[_{CP}\,[_{IP}$ PRO to eat durian]] shouldn't be tried.

On the other hand, if the movement account of Control is incorrect, then we cannot use the prohibition against movement from a Case-marked position to deduce that the subject position of a control infinitive is Case-marked. And there are examples whose ungrammaticality could be derived from this prohibition if the subject position of control infinitives are Case-marked. One of these is (186).

(186) * Winnie was tried $[_{CP}\,[_{IP}$ to eat durian]].

If the prohibition against Argument movement from a Case-marked position is not responsible for the ungrammaticality of (186), then it's not clear why it should be bad. There is only one θ-role for the argument *Winnie*, and so the

Theta Criterion is satisfied, as is the Case filter and the c-command restriction on movement. Indeed, the contrast between (186) and (184) is problematic for the movement account of Control. The only relevant difference between these examples is the number of θ-roles involved: in (186), Passive has removed *try* of its external θ-role, whereas in (184) this θ-role remains. The movement account of Control relies on an as yet undiscovered force that will allow Argument movement from a control infinitive into a θ-marked position but prevent Argument movement from the same position that is not into a θ-marked position. A theory of Control that makes it a special reading for PRO, however, can make sense of this contrast with the tools at hand. Let the Specifier of a control infinitive be a Case-marked position, and everything follows. The Theta Criterion as well as the ban against movement from a Case-marked position will prevent Argument movement from a control infinitive into a θ-marked position. The movement that is illustrated in (184) will be blocked, and instead the subject of the embedded infinitival clause will be PRO. The ban against Argument movement from a Case-marked position will continue to prevent Argument movement from a control infinitive even when that movement is not to another θ-marked position, as in (186).

On this set of assumptions, then, what makes a control infinitive different from, say, the infinitival complement to a verb like *believe* is how the Specifier of the infinitive is Case-marked. The subject position of the infinitive that follows *believe* is accusative Case-marked by *believe*, and so Argument movement is possible from this position when *believe* is Passivized and thereby robbed of its accusative Case.

(187) a. Bart believed [$_{IP}$ him to be happy].
b. He was believed [$_{IP}$ to be happy].

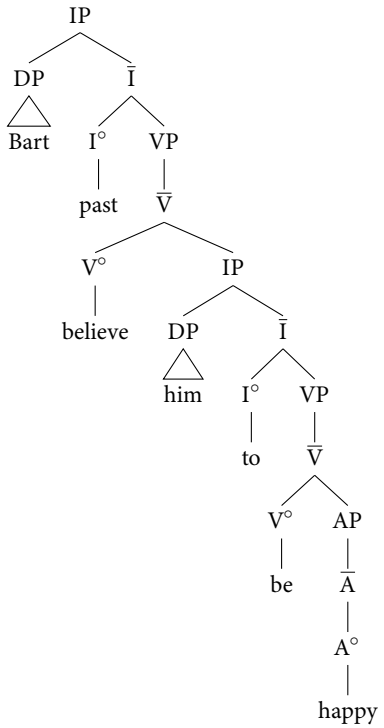
The S-structure parse for (187a) is (188).

Configurations like (187) are called instances of "exceptional Case-marking," or "Raising to Object." The second name references an analysis of these cases that we will examine in the next chapter. There are a variety of ways in which this construction can manifest itself. The verb ***wager***, for instance, does not allow sentences like (187a), but it does allow sentences like (187b).

i. * Bart wagered him to be good.
ii. He was wagered to be good.

See Postal (1974) for a thorough examination of the kinds of constructions that verbs of this type allow.

(188)



If the infinitival clause that is c-selected by *believe* is an IP, then *believe* will govern the Specifier of that IP and thereby be close enough to assign accusative Case to it. If the correspondence between the IP/CP distinction and the clauses that can and can't house PRO is correct, then the hypothesis that the infinitival complement of *believe* is an IP, not a CP, will also explain why PRO cannot appear in these infinitives.

Note, however, that ***believe*** c-selects a CP finite clause: ***She believed that pigs fly.***

(189) * Bart believed [$_{IP}$ PRO to be happy].

Unlike *try*, then, passivizing *believe* causes the position that *him* resides in in (187a) to no longer be Case marked, and this is why A Movement is possible.

The subject position of the control infinitive following *try*, by contrast, is Case marked by something other than *try* and so Argument movement from that position will be blocked in both the Active and Passive versions of *try*.

(190) a. * Bart tried [$_{CP}$ [$_{IP}$ to be happy]].

b. * Bart was tried [$_{CP}$ [$_{IP}$ to be happy]].

If we adopt this account, the subject position of control infinitives is Case-marked and since these are the positions that PRO resides in, we can let PRO be subject to the Case filter.

What's mysterious under this account, though, is why overt DPs cannot stand in the subject position of control infinitives. If these positions are Case-marked, then we should expect not just PRO to be able to reside in them; overt DPs should be able to as well. But, as we've seen, this is not possible; examples like (191) are ungrammatical.

(191) a. * She to sing is swell.
b. * Elena tried her to sing.

A popular solution to this problem is proposed in Chomsky and Lasnik (1993). They suggest that the Specifier of a control infinitive is assigned a Case that only PRO may bear. English, they speculate, has a fourth Case, one whose morphological manifestation only PRO is capable of achieving. They call this the "null Case." This would have the desired effect of blocking overt DPs from the subject position of control infinitives. It could also be made to derive the PRO Restriction. If the only place that null Case is assigned is to the Specifiers of control IPs, and if we not only restrict null Case to PRO but prevent PRO from bearing any other Case, then the Case filter alone will properly segregate the positions of PRO and overt DPs. Overt DPs will have to surface in a nominative, accusative or genitive Case-marked position, and PRO will have to surface in a position marked with null Case. Martin (1992, 2001) argues that the availability of null Case can be tied to the semantics of the infinitival *to* that comes with control infinitives. He argues that the *to* which heads control infinitives has an interpretation as a kind of tense, whereas the *to* that heads the infinitives that PRO cannot appear in does not have a tense interpretation. If correct, this would suggest that null Case is assigned under conditions rather like those that nominative Case is assigned. Nominative Case is assigned to the Specifier position of a finite I°; on Martin's view, null Case is assigned to the Specifier of another type of tense I°, the one that appears in control infinitives.

There are problems with both the idea that there is a special Case reserved for PRO and the view that this special Case is tied to a particular interpretation that distinguishes control infinitives from others. Pires (2001, 2006), Terzi (1992), and Roussou (2001) (among others) have shown that the tensed interpretation Martin claims to correlate with the presence of PRO doesn't. Therefore, the supposed Case that only PRO can receive does not seem to be predictable from other properties of the infinitives it is alleged to be present in. Moreover, there is evidence that the subject position of control infinitives is capable of being assigned the normal Cases that can be borne by overt DPs. This evidence comes from Icelandic.

In Icelandic, floated quantifiers can be marked for Case and when they are, they have the same Case that the subjects they are related to do. In (192), for instance, the subject *Strákarnir* ('the boys') is inflected for nominative Case, and so is the floated quantifier *allir* ('all').

(192) Strákarnir komust allir í skóla.
the boys.nom got all.nom to school
'The boys all managed to get to school.'
(Sigurðsson, 1991, (6a): 331)

The rules of Case assignment differ interestingly from English. Sometimes, the DPs in Specifiers of finite IPs can appear in something other than nom-

inative Case. This seems to depend on the verb involved. For instance, the subject argument of the verb *vantaði* ('lacked') appears in the accusative Case, and the subject argument of *leiddist* ('bored') shows up in the dative Case. In these situations, a floated quantifier related to the subject will inflect for the Case that the subject bears. This is illustrated in (193).

(193) a. Strákana vantaði alla í skólann.
the boys.acc lacked all.acc in the school
'The boys were all absent from school.'

b. Strákunum leiddist öllum í skóla.
the boys.dat bored all.dat in school
'The boys were all bored in school.'

(Sigurðsson, 1991, (6b,c): 331)

In Icelandic, then, we can tell what Case a subject DP has by looking at its floated quantifier; and subject arguments have a wider range of Case options than they do in English. We can use this feature of Icelandic to determine what Case, if any, PRO bears. What we find is that PRO seems able to bear whatever Case an overt DP would bear in that position. The examples in (194) show this correspondence.

But PRO is capable of bearing other Cases as well, and so they do not behave precisely like overt DPs in this regard.

(194) Strákarnir vonast til …
the boys.nom hope for …
'the boys hope …'

a. [$_{CP}$ að PRO komast allir í skóla].
[$_{CP}$ that PRO get all.nom to school]
'to all manage to get to school.'

b. [$_{CP}$ að PRO vanta ekki alla í skólann].
[$_{CP}$ that PRO lack not all.acc in school]
'to all not be absent from school.'

c. [$_{CP}$ að PRO leiðast ekki öllum í skóla].
[$_{CP}$ that PRO bore not all in school]
'to not all be bored in school.'

(Sigurðsson, 1991, (8a–c): 331)

The controller—*Strákarnir* ('the boys')—is marked with nominative Case in each of these examples. But the quantifier floated in the control infinitive inflects in a way that depends on the verb of the infinitival clause. When the verb is the kind that makes its subject appear in the accusative or dative Case, in finite clauses, it makes the floated quantifier in the control infinitives inflect for the accusative or dative Case. If the match between the Case borne by a floated quantifier and the Case borne by the subject it is related to holds in infinitival clauses as well, then we can conclude from these examples that PRO bears the nominative, accusative or dative Cases assigned to Specifiers of finite IP in Icelandic.

Using null Case, then, to segregate PRO from overt DPs does not seem attractive. The evidence we have does not suggest that PRO and overt DPs differ with respect to the Cases they can bear. At present, it seems that the PRO Restriction, and the ban against overt DPs in the subject position of control infinitives, must be sought elsewhere.

3.5.4 PRO *and Control: A Summary*

The evidence we've reviewed in this section, and the explanations of it that we've entertained, are complex and open-ended. It might be useful to arrange in one place some of the conclusions we've reached and hypotheses we've looked at.

We've discovered that there are silent argument DPs—"PRO"—which have a semantic interpretation rather like *one*. It shows up in infinitival clauses that function as subjects and as indirect questions; (195) are examples.

(195) a. PRO to eat durian is ill-advised.
b. I wondered how PRO to eat durian.

These infinitival clauses can be CPs, as the presence of the wh-phrase in (195b) shows. In closely related languages, like Icelandic, it's possible to show that they are always CPs. We've entertained the hypothesis that these infinitives are always CPs in English as well, and this requires the conclusion that the complementizers of these CPs are phonetically silent. Let's call these infinitives: "PRO-infinitives."

The existence of PRO means that there are forces which restrict its occurrence to just the places where we have evidence for it. This seems to be the Specifier of the IPs in PRO-infinitives. These same positions do not seem to allow overt DPs, as the contrast between (196) and (195) indicates.

(196) a. * She to eat durian is ill-advised.
b. * I wondered how him to eat durian.

We have entertained the hypothesis that what makes the Specifier positions of PRO-infinitives special in this way has to do with the Case they are assigned. But so far as I can tell, there is no evidence that these Specifiers get assigned a special Case, one that would make it different from the Cases assigned to other Specifiers of IPs. I suggest we abandon this hypothesis, therefore.

So, putting these parts together, we have:

(197) a. There is a silent DP: PRO.
b. PRO can get a semantic interpretation like that assigned to *one*: this is called its "arbitrary" interpretation.
c. PRO can only appear in the Specifiers of PRO-infinitives.
d. Overt DPs cannot appear in the Specifiers of PRO-infinitives.
e. PRO-infinitives can be CPs. It's possible that they are always CPs.

We've also encountered infinitival clauses that do not contain a PRO with an arbitrary reading, but instead have a silent subject that refers to the same thing that an argument spoken elsewhere does; (198) is an example.

(198) Andre tried to sing.

In these cases, the overt DP 'controls' the understood subject of the infinitive. We have two ideas about how to characterize these situations.

On the first view, control is a special instance of Argument movement, and so the syntax of (198) is identical to that posited for (199).

(199) Andre seemed to sing.

In both cases, *Andre* moves from the position *sing* assigns a θ-role to. In (199) this is the only θ-role *Andre* receives, as the verb *seem* does not assign an external θ-role. In (198), by contrast, *Andre* also receives a θ-role from *try*, presumably by moving into the position this θ-role is assigned to. If this is the correct treatment of Control, then our present formulation of the Theta Criterion will have to be changed so that it allows on argument to bear more than one θ-role. This account of Control will also have to find a way of capturing the difference between (198) and (200).

(200) * Andre was tried to sing.

And, if the Specifiers of these infinitives are Case-marked, as the evidence from Icelandic suggests, then (198) would seem to be a violation of the prohibition against Argument movement from Case-marked positions. So, this characterization of Control includes the points in (201).

(201) If Control is Argument movement, then:

a. PRO-infinitives are prevented from arising in contexts where Control is obligatory.

b. The Theta Criterion requires changing.

c. There is an unknown force responsible for the contrast in:

   i. Andre tried to sing.

   ii. * Andre was tried to sing.

d. If the Specifiers of the IPs from which Control arguments move is Case marked, then the generalization that Argument movement from Case-marked position is ungrammatical is lifted in these contexts.

On the second view, Control is a special meaning that PRO is able to receive. If this is correct, those infinitives in which Control arises are PRO-infinitives. This characterization of Control brings with it the following points.

(202) If Control is found in PRO-infinitives, then:

a. PRO cannot get an arbitrary reading in those contexts where Control is obligatory.

b. There is a semantic rule independent of Argument movement which gives the Control reading.

c. If the Specifiers of PRO-infinitives are Case-marked, then the ban against Argument movement from Case-marked positions can be used to account for the contrast between:

i. Andre tried to sing.

ii. * Andre was tried to sing.

These two approaches to the Control relation are presently hotly contested. I will try to stay neutral between the two views and their consequences in what follows. But if it becomes necessary to choose, I will pick the second view.

3.6 The Theta Criterion Reconsidered

In the previous chapter, we encountered the need to design a syntax that guides, or at least describes, how the semantics of arguments and argument-takers happens. Following the language used in Chomsky (1981), we divided this task into two parts. One part simply records the fact that the connection between the arguments in a sentence and the word that these phrases are arguments of must be made. This is the goal of the Theta Criterion.

(203) Theta Criterion

a. Every θ-role must be assigned to a unique syntactic position.

b. Every argument XP must be related to exactly one position that is assigned a θ-role.

The second part is a linking theory, which determines where the syntactic positions are that the θ-roles assigned by some word are. This is the goal of the Projection Principle.

(204) The Projection Principle

a. If β is c-selected by α, then there must be a projection of α, π, that immediately dominates β and if π immediately dominates some other phrase, γ, then γ must be another projection of α or part of another c-selected argument of α.

b. The Specifier of α is the position to which the external θ-role of α is assigned.

What we have seen in this chapter is that sentences may satisfy these conditions even when argument XPs are not spoken in θ-marked positions. These cases arise when movement rules put an argument XP into a different position than the one to which their θ-role is assigned. The Theta Criterion and

Projection Principle are presented as conditions on D-structure parses, then, and these can differ from S-structure parses in just the ways that Complex NP Shift and Argument Movement permit or require.

What we've just seen in the discussion of the differences in the raising and control infinitives, however, requires a slightly different view of the second clause of the Theta Criterion. If control infinitives do not arise as a result of A-movement, then this is blocked by preventing an Argument from occupying two θ-marked positions in a derivation. On this way of describing things, then, the Theta Criterion is a condition on derivations, not on D-structures. When I presented the Theta Criterion, I presented it as a description of a semantic effect. Its source might come from how the meanings of arguments and argument-takers are combined. Our model makes D-structures the parse that the semantics interprets. There is something inconsistent about the way the Theta Criterion's effect on constraining derivations is described in the previous sections and our model of derivations presently.

I want to resolve that problem in this section, and I want to take a few steps in the direction of understanding better what the Theta Criterion describes. There are some other problems that were left hanging when this condition was introduced that I'd like to address here.

The idea behind θ-roles is that they represent specialized relations that are part of the mix of meanings that a verb brings to a sentence. That these specialized relations exist has sometimes been argued to afford a handle on a linking theory that can predict where an argument of some verb will be based on the kind of relation it has to the verb's meaning. Recall, for instance, that Ed Keenan argues that there is a cross-linguistic correlation between being an "agent" of some action and being the external argument of the verb that describes that action. Claims of this kind require, of course, an independent definition of the relations that θ-roles encode. To test Keenan's hypothesis requires knowing when some argument is an "agent" of an action in a way that is independent of its being an external argument. Working definitions of θ-roles, however, have proven largely unsuccessful, and so determining that they have a contribution to the meanings of sentences is still not established. Nonetheless, the existence of θ-roles remains a popular working hypothesis. Let us assume their existence.

See Jackendoff (1972) for an initial, and searching, attempt to establish an inventory of θ-roles, and Dowty (1979) for an excellent examination of the problems attendant with defining θ-roles.

When we learn the meaning of a verb, we can decompose that meaning into two parts. There is the type of argument that the verb selects, and there is the action or state of affairs involving that argument that the verb describes. The language of θ-roles is generally reserved for the method by which verbs combine with DP, or sometimes PP, arguments. I'll represent the meaning of a verb as a list of these components. To do that, I'll introduce very informally a metalanguage that we'll use to express meanings.

The central ingredient of this metalanguage is a method of building a function. A function relates one set to another set by mapping each element of the first set to exactly one element of the second. So, for instance, we can

define a function, let's call it g, that relates the set {a, b, c} to the set {1, 2, 3} in the way indicated in (205).

(205) The function g:
a $\rightarrow$ 1
b $\rightarrow$ 2
c $\rightarrow$ 3

We can represent this function g with the set of ordered pairs in (206), where the first member of each pair is from the first set (the "domain" of the function) and the second member of each pair is the element of the second set that g relates the first element to. (The second set is known as the "range" of the function.)

(206) $g = \{ \langle a,1\rangle, \langle b,2\rangle, \langle c,3\rangle \}$

A third way of representing this function is in (207).

(207) $g(a) = 1$
$g(b) = 2$
$g(c) = 3$

We say that g applies to its argument a and gives 1, applies to its argument b to give 2, and applies to its argument c to give 3.

We can use the language of mathematics to describe functions that relate numbers, as well as other mathematical objects, to other numbers. For instance, the expression in (208) describes a function that accepts numbers—let's restrict the domain to positive whole numbers—and maps that number onto its square.

(208) x^2, where x is a positive whole number

We understand x here to be an element of the domain set and the "^{2}" to be a function that maps x onto its square. The use of the variable x is a compact way of representing a function that relates large sets. In (208), the domain and range are sets of infinite cardinality so representing the function by way of ^{2} is the only feasible method. This method requires an understanding of what a variable is and a way of defining "^{2}." Informally, "^{2}" is defined as a function that multiplies its argument, x, to itself. The function expressed by (208) can be expressed in the format that (207) illustrates with (209).

Of course, to understand this definition of ^{2} requires defining "multiplies," and this will in turn require a definition of addition and whole numbers. But we wish to get back to linguistics, so here is where we stop.

(209) $\textsc{sqr}(x) = x \times x$, where x is a positive whole number

It is possible to define functions whose range is another function. For instance, (210) is a compact way of representing a function whose domain is the set $\{ 2, 3, 4, \ldots \}$ and whose range is a function x^n where n is the argument of the function defined in (210).

(210) x^n, where n is a positive whole number greater than 1

Informally, "^{n}" can be defined as the function that multiplies its argument, x, to itself $n - 1$ times.

With this understanding, the function in (210) can be represented by the series in (211).

(211) $x^n(2) = x^2 = x \times x$
$x^n(3) = x^3 = x \times x \times x$
$x^n(4) = x^4 = x \times x \times x \times x$
$\vdots$

When functions return other functions, we will want to keep track of how the values of the variables within them are assigned. This gives us a way of determining the function in the range returned by the first function. A way of doing this involves an operator, λ, defined informally in (212).

(212) $\lambda x\ f(x)$ applied to a = f(a), where f is a function.

We say that the variable x is bound by the operator λx. With this notation, the function described in (208) is rendered as (213).

(213) $\lambda x\ x^2$, where x is a positive whole number

This gives us the domain to range relations in (214).

(214) $[\lambda x\ x^2](1) = 1^2 = 1$
$[\lambda x\ x^2](2) = 2^2 = 4$
$[\lambda x\ x^2](3) = 3^2 = 9$
$\vdots$

The function in (211) is rendered as (215).

(215) $\lambda n\ \lambda x\ x^n$, where x is a positive whole number and n is a positive whole number greater than 1

This gives us the domain to range relations in (216).

(216) $[\lambda n\ \lambda x\ x^n](2) = [\lambda x\ x^2]$
$[\lambda n\ \lambda x\ x^n](3) = [\lambda x\ x^3]$
$[\lambda n\ \lambda x\ x^n](4) = [\lambda x\ x^4]$
$\vdots$

The function that results from applying (215) to the number "2" is the one that (213) defines. One thing the λ notation allows us to do is order the way the variables in a complex function are assigned to elements of the domain. This will be key to expressing the compositionality of syntactic phrases.

I'll adopt a way of presenting compactly cases where the range of one function is another function. To illustrate this method of presentation, consider the functions in (217).

(217) a. $\lambda x\ P(x)$
b. $\lambda y\ Q(y)$

Suppose that (217a) has the domain of positive whole numbers (x ranges over {1,2,3...}) and (217b) has the domain of lower case roman letters (y ranges

over {a,b,c,...}). Now consider the function in (218), which puts these two functions together.

(218) $\lambda x\ \lambda y\ Q(y) + P(x)$

This is a function whose domain is positive whole numbers and whose range is the function in (217b). It could apply to a positive whole number, such as "2," and yield the function shown in (219).

(219) $[\lambda x\ \lambda y\ Q(y) + P(x)](2) = \lambda y\ Q(y) + P(2)$

This function has letters as its domain, so it can be fed something like "b" and give the resulting expression in (220).

(220) $[\lambda y\ Q(y) + P(2)](\text{b}) = Q(\text{b}) + P(2)$

Our present notation allows us to represent this situation with (221).

(221) $[\ [\lambda x\ \lambda y\ Q(y) + P(x)](2)\ ](\text{b})$

The function in (218) applies to "2" and forms a new function that applies to "b." The compact way of representing this that I will adopt here is (222).

(222) $[\lambda x\ \lambda y\ Q(y) + P(x)](2)(\text{b})$

Another common convention, found often in the writing of logicians, is to represent (221) with (i).

i. $[\lambda x\ \lambda y\ Q(y) + P(x)](\text{b},2)$

Note that the order of arguments is reversed in this notation. Logicians frequently get things backwards.

The domain of a function can also be a set of functions. The function in (223) illustrates.

(223) $\lambda P\ \lambda y\ 3 + P(y)$, where y is a positive whole number and P a function whose domain is positive whole numbers and whose range is positive whole numbers.

The function defined in (208) has a domain and range of positive whole numbers, so it is in the domain of (223). Thus, (224) is well-formed.

(224) $[\lambda P\ \lambda y\ 3 + P(y)](\lambda x\ x^2) =$
$\lambda y\ 3 + [\lambda x\ x^2](y) =$
$\lambda y\ 3 + y^2$

Note that the second line of these equalities is equal to the last line because the y is an argument of "$\lambda x\ x^2$," and so the function can apply to that argument to yield y^2. The domain of this function is the set of positive whole numbers, so this describes the relations in (225).

(225) $[\lambda y\ 3 + y^2](1) = 3 + 1^2 = 4$
$[\lambda y\ 3 + y^2](2) = 3 + 2^2 = 7$
$[\lambda y\ 3 + y^2](3) = 3 + 3^2 = 12$
$\vdots$

Because functions can take other functions as arguments, it is necessary to legislate how the variables in the function that is an argument can be manipulated by the function that takes that argument. The law is that no variable in a function can be bound by the λ operator of another function. So neither of the following cases are well-formed.

(226) a. $[\lambda P \lambda x\ P]\ (x^2) \neq \lambda x\ x^2$
b. $[\lambda P \lambda x\ P]\ (\lambda x\ x^2) \neq \lambda x\ \lambda x\ x^2$

The unbound variable (should such things exist) in the argument of (226a) cannot become bound in the function it is an argument of. And the bound variable in the argument of (226b) cannot acquire an additional binder in the function it is an argument of. To get a variable inside one function to be bound by the λ operator of another function, the technique illustrated by (224) must be employed.

Let me illustrate that method more generally. Consider the two functions in (227).

(227) a. $\lambda x\ P(x)$
b. $\lambda M\ \lambda y\ M(y)$

Suppose that (227a) is in the domain of (227b). This allows (227b) to apply to (227a) as in (228) .

(228) $[\lambda M\ \lambda y\ M(y)](\lambda x P(x)) = \lambda y\ [\lambda x\ P(x)](y)$

If y is in the domain of $\lambda x\ P(x)$ (i.e., the function in (227a)), then it can apply to y and this results in (229).

(229) $\lambda y\ P(y)$

In this way, then, the variable inside the function in (227a) has become bound by λy in the function in (227b).

Now that the λ operator has been introduced, we are at last ready to describe how the component parts of the meaning of verbs can be modeled. Recall that we have decided that there are two components to the meaning of a verb. One component describes the state of affairs that the verb refers to and the other lists the relations denoted by the θ-roles, if any, associated with that verb. Let's call the states of affairs that verbs refer to "eventualities." Using this idea, the meaning of (230a) can be paraphrased with (230b).

I am describing here a way of rendering the meaning of verbs that originates with Davidson (1967). Davidson called the states of affairs that verbs refer to "events." The term "eventualities" comes from Bach (1986), who offers this term to include both events, which can be conceived of as dynamic actions, and states, which are static. For an early argument on behalf of importing events into compositional semantics, see Higginbotham (1985).

(230) a. Kiyomi danced.
b. Kiyomi was the agent of a dancing eventuality

We'll begin by defining functions that correspond to the "agent" and "dancing" parts of the paraphrase in (230).

(231) a. AGENT$(x)(\epsilon)$ is a function from things, x, to a function whose domain is eventualities, ϵ. The range of AGENT is {True,False}. When x is the agent of ϵ, then AGENT$(x)(\epsilon) \rightarrow$ True. Otherwise, AGENT$(x)(\epsilon) \rightarrow$ False.
b. DANCE(ϵ) is a function from eventualities, ϵ, to {True,False}. When ϵ is a dancing, then DANCE$(\epsilon) \rightarrow$ True, otherwise DANCE$(\epsilon) \rightarrow$ False.

In the semantic literature, the more neutral word "entities" is used instead of "things."

Loosely speaking, what AGENT and DANCE do is describe the conditions under which an eventuality has an agent or is a dancing. They do that by spelling out when the AGENT and DANCE functions map to truth or falsity. A perhaps more transparent way of rendering this informally is with the paraphrases in (232).

(232) a. AGENT$(x)(\epsilon)$=T $\approx$ x is Agent of ϵ

b. DANCE(ϵ)=T $\approx$ ϵ is a dancing

In reading the formulas that represent denotations in what follows, you can safely make this substitution.

The meaning of the word *dance* should describe those eventualities that make the functions in (231) true. We can model this with the function in (233).

I abbreviate "Truth" with "T" in (233). Another common abbreviation is "1"; "1" abbreviates Truth and "0" abbreviates False.

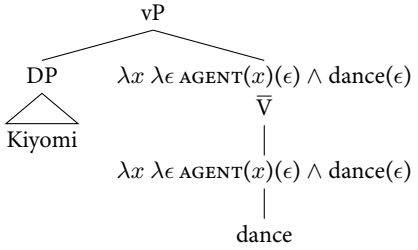
(233) $[\![\text{dance}]\!] = \lambda x\ \lambda\epsilon\ [\underline{\text{AGENT}}(x)(\epsilon) \wedge \text{DANCE}(\epsilon)]$=T, where x is a thing and ϵ is an eventuality.

The underline signals which θ-role is external; the only θ-role *dance* provides is external. The brackets "$[\![\ ]\!]$" represent a function that takes the material the brackets enclose and returns the denotation of that material. In (233), the denotation brackets apply to *dance* and return the meaning of *dance*, which is represented by the function to the right of the "=" sign.

Consider how this meaning of *dance* can be used to form the meaning of the VP in (230a), which is given in (234).

I have dropped the "=T" part of the function. Let's understand the functions we are interested in to always be those that map to T.

(234) $\lambda\epsilon$ AGENT$([\![\text{Kiyomi}]\!])(\epsilon) \wedge$ DANCE(ϵ)



To achieve this, we will need a set of semantic rules that take the meanings of the parts of a phrase and produce from them the meaning of that phrase. Adopt the following two semantic rules.

These are the rules that ensure semantic compositionality for phrases.

(235) Rules of Semantic Composition

a. If X is the sole daughter of Z, then $[\![\text{Z}]\!] = [\![\text{X}]\!]$

b. If X and Y are daughters of Z, then $[\![\text{Z}]\!]$ is the result of applying the function $[\![\text{X}]\!]$ to its argument $[\![\text{Y}]\!]$.

I've written on the nodes of (234) the denotations derived from $[\![\text{dance}]\!]$ by these rules. The meaning assigned to the VP is a function from eventualities that are dancings and that have Kiyomi as agent to truth. This is roughly what we want. To get from this the meaning of (230a), we'll have to work

However, I've not written out what the denotation of ***Kiyomi*** is. If I had done so, I'd have to find a way of expressing how the name ***Kiyomi*** allows this expression to refer to a particular person whose name is Kiyomi.

out the semantic contribution of the past tense, and figure out how to get this sentence to refer to an eventuality of the sort that this VP describes. These parts of the meaning of a sentence do not come from the verb and its arguments, however, so we can safely set them aside.

Consider next the meaning of a verb that has two θ-roles, like *eat*.

(236) $[\![\text{eat}]\!] = \lambda x\ \lambda y\ \lambda \epsilon\ \underline{\text{AGENT}}(y)(\epsilon) \wedge \text{PATIENT}(x)(\epsilon) \wedge \text{EAT}(\epsilon)$=T, where

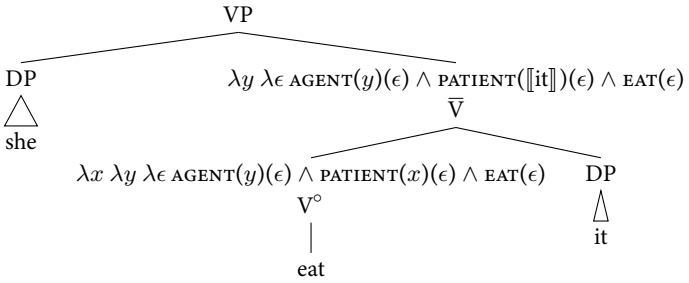
a. x and y are things and ϵ is an eventuality

b. PATIENT c-selects DP

Recall that internal arguments are c-selected by the verbs that take them, so I have added this information in (236). I've expressed this as a requirement imposed by the θ-role whose denotation is the function that takes the relevant expression as its argument. The c-selection requirement of PATIENT can be thought of as part of the definition of this function's domain. It specifies what the morpho-syntactic properties of the argument for PATIENT must be.

Our system derives the parse, and meanings, for a VP headed by *eat* as indicated in (237).

(237) $\lambda \epsilon\ \text{AGENT}([\![\text{she}]\!])(\epsilon) \wedge \text{PATIENT}([\![\text{it}]\!])(\epsilon) \wedge \text{EAT}(\epsilon)$



The meaning assigned to this VP is a function from eventualities that maps those which are eating eventualities that have the referent of *she* as its agent and the referent of *it* as its patient to True. This is roughly what we want: the VP describes eating eventualities in which she consumed it.

Now that we have a semantics in front of us for verbs and their arguments, let's reconsider the Theta Criterion.

(203) Theta Criterion

a. Every θ-role must be assigned to a unique syntactic position.

b. Every argument XP must be related to exactly one position that is assigned a θ-role.

Let's first focus on the first clause of the Theta Criterion. This clause has two consequences: it requires every θ-role to be given an argument and it prevents any θ-role from being given more than one argument. A problem for the first of these consequences is that there are numerous counter-examples. For instance, (238) is perfectly grammatical and it has all the appearances of containing the same verb that (237) does.

(238) Shai ate.

Indeed, the meaning of the verb *eat* in (238) should entail that something was eaten—that there is a Patient of the eating eventuality—because the existence of this Patient can be presupposed in a conversation that includes (238). This might be seen, for instance, by the appropriateness of B's response in (239).

(239) A: Shai ate at the function last night.
B: Yeah, and he told me nothing tasted good.

Compare this to a conversation that starts with a verb that doesn't entail the existence of a Patient:

(240) A: Shai danced at the function last night.
B: Yeah, and he told me nothing tasted good.

B's response in this conversation is a non-sequitur because no Patient of an eating eventuality is invoked by *dance*. By contrast, the word *nothing* in B's response in (239) is understood to quantify over the Patients of the eating event invoked by *eat*. Considerations such as these push towards the conclusion that there is a Patient θ-role associated with the *eat* in (238). And yet, there is no argument associated with this θ-role, in violation of the Theta Criterion.

This ability for some θ-roles to not be assigned to arguments seems to vary with the verb. So whereas the Patient θ-role need not be matched with an argument for *eat*, the Patient θ-role associated with *build* must be matched with an argument, as the ungrammaticality of (241) indicates.

(241) * Cherlon built.

compare:

(242) Cherlon built them.

Moreover, this ability to violate the Theta Criterion seems to be available only for internal arguments. Alternations like (243) are common, but so far as I know, alternations like (244), where the Agent θ-role remains but doesn't get assigned to an argument, aren't possible.

(243) a. Shai ate nattoo.
b. Shai ate.

(244) a. Shai danced.
b. * It danced.
(*it* an expletive)

There are alternations of the sort that (244) illustrates where the external θ-role is lost. And there are also alternations of the sort (244) exemplifies in which the verb has undergone some morphological change. Both of these are taken up in the next section. But finding the same verb with an external θ-role it contains both assigned and not assigned doesn't seem to happen. We return to this issue in the final section of this chapter.

One way of describing this situation is to allow for a process that creates denotations which existentially close one of the argument positions of a verb. This process would create from (236), for instance, (245).

(236) $[\![\text{eat}]\!] = \lambda x\ \lambda y\ \lambda \epsilon\ \underline{\text{AGENT}}(y)(\epsilon) \wedge \text{PATIENT}(x)(\epsilon) \wedge \text{EAT}(\epsilon)$=T

Understand $\exists x\ P(x) = T$ to be equivalent in meaning to "there is an x that is P." "$\exists x\ P(x)$" maps to T if and only if there is an a in its domain for which P(a)=T. "$\exists$" is called an "existential" operator.

(245) $[\![\text{eat}]\!] = \lambda y\ \lambda\epsilon\ \exists x\ \underline{\text{AGENT}}(y)(\epsilon) \wedge \text{PATIENT}(x)(\epsilon) \wedge \text{EAT}(\epsilon)$=T

Let's call this operation "Existential Closure." As noted, Existential Closure doesn't seem to be able to apply to the external θ-role of a verb.

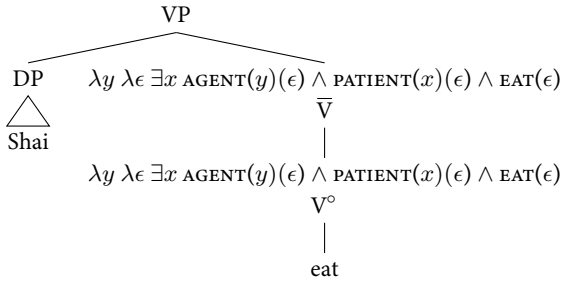
Existential Closure is modeled after an analysis in Chierchia (2004).

(246) Existential Closure cannot apply to an external θ-role.

Let's assume that Existential Closure is, like Passive, a rule that makes new verbs from existing ones. As we've seen, the choice of verb determines whether Existential Closure can apply. If there is a pattern that determines which verb is susceptible to Existential Closure or not, I don't know it.

The verb created by Existentially Closing *eat* will produce a meaning for the VP it heads indicated in (247).

(247) $\lambda\epsilon\ \exists x\ \text{AGENT}([\![\text{Shai}]\!])(\epsilon) \wedge \text{PATIENT}(x)(\epsilon) \wedge \text{EAT}(\epsilon)$



This is the right meaning for the VP; it says that this VP describes eventualities that are eatings, have a Patient, and have Shai as their Agent.

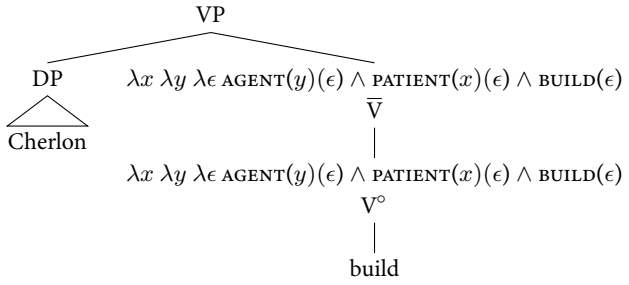
Now we have a way of explaining how the first clause of the Theta Criterion can be made consistent with cases where a θ-role is present but there is no argument for that θ-role in the syntax. These cases arise because of Existential Closure. When Existential Closure cannot apply—as with external θ-roles and the internal θ-roles of verbs like *build*—the first clause of the Theta Criterion still has the consequences that were highlighted above: (i) there must be an argument for the θ-role in the syntax, and (ii) there must not be more than one such argument. Irene Heim and Angelika Kratzer[6] argue that the first of these consequences—that there must be an argument in the syntax matched to the θ-role of a verb—can be described in semantic terms under these assumptions. One way of doing that would be with (248).

[6] Heim, Irene, and Angelika Kratzer. 1998. ***Semantics in generative grammar***. Malden, MA: Blackwell

(248) The denotation of a VP must be a function whose domain is a set of eventualities.

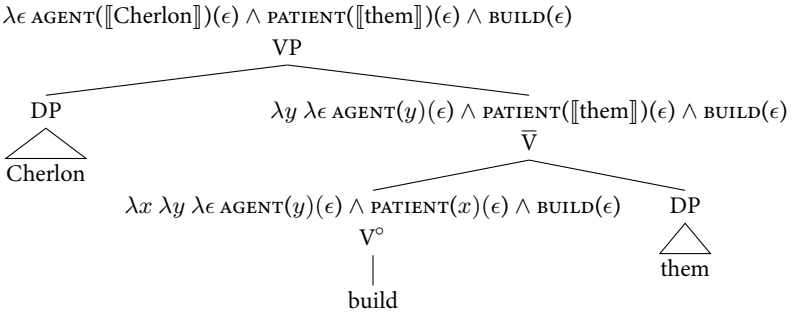
To see how this works, consider what happens if *build* is put into a VP that does not contain an argument for its Patient θ-role.

(249) $\lambda y\ \lambda\epsilon\ \textsc{agent}(y)(\epsilon) \wedge \textsc{patient}(\llbracket\text{Cherlon}\rrbracket)(\epsilon) \wedge \textsc{build}(\epsilon)$



This violates (248), and gets the meaning wrong. By contrast, when the VP headed by *build* supplies an argument of the right kind, the rules of semantic composition can achieve (248).

(250)



Let's now consider how we might achieve the second consequence of the first clause of Theta Criterion: that a θ-role cannot be matched to more than one argument. Heim and Kratzer argue that this also follows from a semantics like the one given here. Consider an imaginary verb, *schkiss*, which assigns the same θ-role to its external and internal arguments.

The semantics I've laid out here follows Heim and Kratzer (1998), but is slightly different. Their semantics does not include eventualities. I've made some adjustments to the way they derive the Theta Criterion as a consequence. See Chung and Ladusaw (2004) for a similar system.

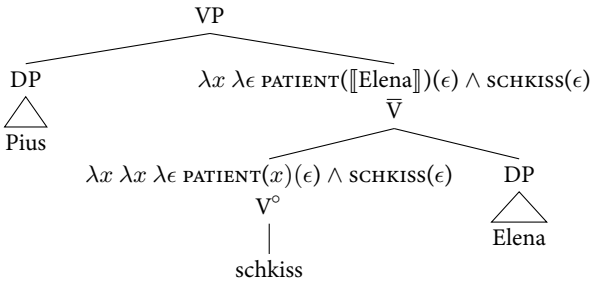
(251) $\llbracket\text{schkiss}\rrbracket = \lambda x\ \lambda x\ \lambda\epsilon\ \textsc{patient}(x)(\epsilon) \wedge \textsc{schkiss}(\epsilon)$, where

a. x is a thing and ϵ an eventuality

b. PATIENT c-selects DP

Suppose that a schkissing is an eventuality just like a kissing, but in which neither participant is an Agent. The function in (251) is broken because it binds the variable x from two λ operators. Consider how our semantic rules will apply this meaning to a VP headed by *schkiss*:

(252)



Because there is no rule of semantic composition that can bring the denotation of the $\overline{V}$ and *Pius* together in (252), this will fail to get an interpretation and the result is ungrammatical.

If this thinking is correct, then we can see the first clause of the Theta Criterion as deriving entirely from constraints on what the denotations of verbs can be and how the meanings of the pieces of a phrase compose. What's left of the Theta Criterion, then, is (253).

(253) Theta Criterion
Every argument XP must be related to exactly one position assigned a θ-role.

This follows from our thesis that the only phrase marker in a derivation that is semantically interpreted is D-structure. The $\overline{X}$ Skeleton requires every XP to occupy just one position in a phrase-marker. Our thesis that D-structure is the only phrase-marker that is semantically interpreted requires that the semantic rule of composition that puts an argument together with its θ-role look at just one phrase-marker. Together, these derive (253). These are steps, then, towards understanding how the Theta Criterion might be derived from a semantics.

Note that this way of deriving (253) is incompatible with the movement theory of control. That theory requires a syntax that allows more than just a D-structure to be semantically interpreted. There is some evidence that such a syntax is indeed required for other reasons, and we will eventually develop such a syntax. But our present syntax does not allow for movement from one θ-marked position to another for the simple reason that one or the other of the phrase-markers in such a derivation will not be semantically interpretable.

3.7 Theta-roles and Case

One can think of Argument Movement as a function from a parse that satisfies the Projection Principle and Theta Criterion to a parse that satisfies the Case Filter and EPP. θ-roles are determined by lexical items and the phrases they project, and where Case is assigned is also controlled by the these lexical items. Lexical items, and verbs in particular, play a big role in setting up the environments in which Argument movement is called upon. The conditions on Argument movement place a cap, then, on the kinds of situations that verbs can create. They won't allow verbs to exist which create conflicts that Argument movement can't fix.

Verbs set up these situations by influencing where the θ-marked and Case marked positions for arguments are. Thus, for example, we have seen that there are verbs which assign one external θ-role, and verbs which assign an external and an internal θ-role and Accusative Case. These are sometimes called intransitive and transitive verbs, respectively; examples are in (254).

(254) a. Martin slept. (intransitive)
b. Martin likes kiwis. (transitive)

A kind of verb related to transitives are exemplified by *believe*, *consider* and *make*, which assign their Accusative Case to something different than the argument they c-select. As we've seen, these verbs can take a "clausal" complement—sometimes these are "small clauses"—and assign their Accusative Case to a DP within this clause.

(255) a. She believes [$_{IP}$ him to be unhappy].
b. She considers [$_{AP}$ him happy].
c. She made [$_{VP}$ him dance].

We might ask whether there is a connection between a verb having accusative Case and c-selecting a DP internal argument. Presently our theory makes these two independent attributes that a verb might have. However, empirically they are confounded. If a verb has accusative Case, then it is capable of c-selecting a DP object that we see with accusative Case. And if a verb does not have accusative Case, then it will not be able to c-select a DP object that shows up with accusative Case. One way we can tell that a verb has accusative Case, then, is to discover that it c-selects a DP object that appears with accusative Case. And if a verb does not c-select a DP object, it will be difficult to tell whether that verb can assign accusative Case since it will not be able to occur with an accusative Case-marked object. So we should ask: are assigning accusative Case and c-selecting a DP object independent properties of a verb?

I will say that a word c-selects an argument if one of its θ-roles c-selects an argument.

The answer seems to be yes. We have already seen that passive creates verbs that c-select DP objects but do not assign accusative Case.

(256) a. Martin was believed [$_{AP}$ unhappy].
b. Martin was believed.

From (256), we see that the passive *believe* combines with the argument it c-selects—here a small clause AP. Because it does not assign accusative Case, however, the argument DP *Martin* inside that small clause AP cannot get Accusative in its D-structure position, and must move to receive Nominative case. In (256b), then, we should conclude that the same set of attributes adhere to the passive *believed*. It continues to c-select its DP argument—here *Martin*—but no longer assigns Accusative Case. Thus, just as in (256a), the DP that would have gotten accusative from the active *believe*, namely *Martin*, must move to a position where it gets another case. In (256b), that position is Specifier of IP, and the case it receives is Nominative. From these examples, then, we can learn that not all verbs that c-select DPs need be verbs that assign Accusative Case.

We can reach this same conclusion by considering the internal arguments of English adjectives. Adjectives are a class of word that never assigns accusative Case, and so it is no surprise that no adjective c-selects a DP that shows up in the accusative Case. We find contrasts like those in (257).

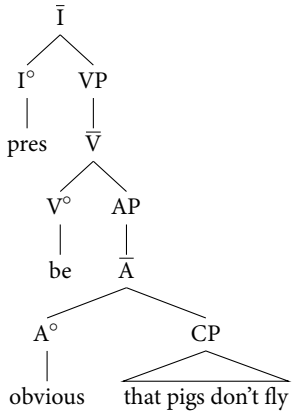
(257) a. * Elena is happy the fact.
b. Elena is happy that pigs don't fly.
c. Elena is happy about the fact that pigs don't fly.

Adjectives can c-select PPs and CPs that stand in complement position, partly because these phrases do not need to be assigned Case. However, there is some evidence that suggests that adjectives can c-select a DP if it is possible for that DP to get Case from some term other than the adjective. To see this, reconsider the alternation in (258).

(258) a. It is obvious [$_{CP}$ that pigs don't fly].
b. [$_{CP}$ That pigs don't fly] is obvious.

These sentences have a D-structure that contains the $\bar{I}$ in (259).

(259)



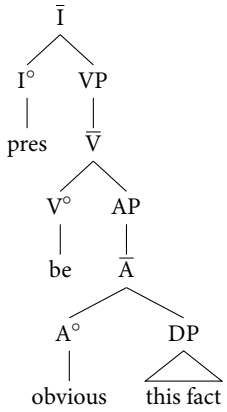
In (258a), the IP that this $\bar{I}$ is embedded in contains an expletive in its Specifier position, and this is how (258a) satisfies the EPP. In (258b), by contrast, the IP that embeds this $\bar{I}$ does exist (or is empty), and the EPP is satisfied by A Movement creating an S-structure in which the CP *that pigs don't fly* occupies this position.

Consider now the contrast in (260).

(260) a. * It is obvious this fact.
b. This fact is obvious.

This alternation is understood if both of the examples in (260) have the $\bar{I}$ in (261) at D-structure.

(261)



If the D-structure has an IP that embeds this Ī whose Specifier is filled with *it*, then there is a Case filter violation and no way of forming an S-structure that remedies that violation. But if the D-structure containing this Ī embeds it in an IP without an *it* in Specifier position, then A Movement can form an s-structure that satisfies the Case filter; this s-structure corresponds to the grammatical (260b). If this thinking is correct, then the class of words that c-select DPs cannot be just the same class that assigns accusative Case. Adjectives seem to be able to c-select DPs, and they never assign Accusative case.

Let's consider one last scenario that leads to this same conclusion. Consider the alternations in (262).

(262) a. * She hopes the outcome.
b. She hopes that this will be the outcome.
c. * She insisted the claim.
d. She insisted that this is true.
e. * She felt the hypothesis.
f. She felt that this is likely.

These are verbs that, like the adjective *obvious*, can have a c-selected CP in complement position, but not a c-selected DP. There are two ways of describing these verbs that will account for their inability to show up with an argument DP in complement position. It could be that these verbs don't assign accusative Case, but do c-select DPs. Or it could be that these verbs don't c-select a DP internal argument. One way we can tell is to look at what happens when these verbs are passivized. If, like *obvious*, they c-select DPs and all that's gone wrong is that those DPs are not able to get Case from these verbs, then the passive versions of these verbs should be fine. In the passive, the Specifier of IP will no longer be occupied by the subject argument, and A movement should be able to put the putative DP object into this Case-marked position. As it happens, however, the passive versions of these verbs still do not allow for DP arguments.

(263) a. * The outcome was hoped.
b. * The claim was insisted.
c. * The hypothesis was felt.

This follows if these verbs do not c-select a DP complement. The passive is possible of these verbs when their complements are CPs, as in (264).

(264) a. It was hoped that this will be the outcome.
b. It was insisted that this is true.
c. It was felt that this is likely.

So the problem with (263) isn't something about whether the passive can apply to these verbs. A reasonable conclusion, then, is that these verbs do not c-select DPs. In general, whether a word – verb or adjective – assigns accusative Case or c-selects a DP argument appears to be independent.

One interesting mystery that arises with the verbs *hope*, *insist*, and *felt* is that A movement cannot apply to complement CPs in the way we would expect. Examples like (265), which should be parallel to (264), are ungrammatical.

(265) a. * That this will be the outcome was hoped.
b. * That this is true was insisted.
c. * That this is likely was felt.

What distinguishes (265) from (264) is merely that the EPP is satisfied in (265) by using A Movement to move the CP into Specifier of IP. The ungrammaticality of these examples is part of a larger pattern concerning A movement in English, which can be described with (266).

(266) A movement can only move an XP that is in a position that a DP could stand in.

Because the complement position of these verbs is not a position that a DP is c-selected for, A movement from this position is prevented by (266). By contrast, because the adjective *obvious* does c-select a DP, A movement from the complement position of *obvious* is possible, and examples like (258b) are grammatical. Why (266) is true strikes many syntacticians as mysterious. An important point about accusative Case, then, is that its assignment depends entirely on the syntactic proximity that some position has to a word that licenses accusative. If a position is adjacent to and governed by a word with accusative, then that position is assigned accusative Case. In this respect, accusative is like the other Cases of English. Nominative and genitive are similarly Cases that are assigned to positions that are in a certain syntactic configuration relative to a term that licenses them. Their occurrence is determined entirely by the syntactic structures they are in.

Not all Cases of the world's languages have this property. For instance, the dative Case in German seems to be tied more directly to the meaning of a word. The dative in German shows up on DPs that are c-selected by

That CPs can only move from DP positions is argued in Alrenga (2005), who builds on Koster (1978). (The observations about where CPs can and can't move from originate, largely, from Emonds 1976.) For recent discussions of the situation involving CPs, see Moulton (2009) and Elliott (2017). And for a discussion of the general issues concerning (266), see Poole (2017).

certain verbs. For instance, the internal argument of the verb *helfen* ('help') is in the dative Case. Compare the situation with *sehen* ('see'), whose internal argument is in the accusative.

(267) a. Sie hilft mir.
she.NOM help.PRES me.DAT
'She helps me.'

b. Sie sieht mich.
she.NOM sees.PRES me.ACC
'She sees me.'

Unlike the accusative Case, the dative Case in German does seem to correlate with c-selection. If a DP bears dative Case in German, then it will get it from the very verb that it is c-selected by. Cases that are tied to the c-selection properties of the word that assigns that Case are sometimes called "inherent Case." And the Cases that English has, which are assigned in a way that is independent of the c-selection properties of the Case assigner, are called "structural Cases." In German, as in many languages, this distinction in Case can be revealed by seeing whether A movement can affect the Case borne by the relevant DP. For instance, the verb that passive creates from *helfen* does not cause the DP it c-selects to "change" its Case. This is what happens with the passivized version of *sehen*, however. Just as in English, passive makes from *sehen* a verb that no longer supports Accusative case, and consequently its c-selected DP shows up in the nominative Case. This is seen in (268).

(268) Ich wurde gesehen.
I.NOM become.PRES see.PASS
'I was seen.'

The verb that embeds a passive participle in German is not the same as the verb used to embed adjectives in German, i.e. ***sein*** ('be'). Instead the verb that embeds passive participles is ***werden***, which is often glossed with the English ***become***.

We will learn about this in the next chapter.

German syntax differs from English in a variety of ways, and the movement of *Ich* in (268) is achieved by a different set of rules than in English. What's key now is that the object of *sehen* shows up in the nominative Case, and that is required. Just as in English, it cannot be accusative. By contrast, the passive version of *helfen* leaves the DP c-selected by *helfen* in the dative Case; it cannot show up in the nominative.

(269) a. Mir wurde geholfen.
I.DAT become.PRES help.PASS
'I was helped'

b. * Ich wurde geholfen.
I.NOM become.PRES help.PASS
'I was helped.'

In view of these differences, we can render our rule for the passive with (270).

(270) Passive
  a. Add passive morphology to the active verb and embed it under *be*, and
  b. Rob the verb of its structural Case, and
  c. Rob the verb of its external θ-role.

The earlier version of Passive had (270b) rob the verb of its Accusative Case. If Accusative is always structural, and it's the only structural Case, then this new version will do what the old version did.

In IndoEuropean languages, there is a strong correlation between whether a Case is structural or inherent on the one hand and whether the Case is nominative or accusative on the other. Quite generally, accusative and nominative are structural Cases and the others—dative, genitive, instrumental, etc.—are inherent Cases. But there are corners of IndoEuropean where this correlation appears to break down. For instance, in Faroese it is possible to find passive versions of dative Case assigning verbs whose c-selected DPs show up in the nominative. Einar Sigurðsson provides the example in (271).

(271) a. Teir róstu henni altið.
they.NOM praise.PAST her.DAT always
'They always praised her.'
  b. Hon varð altið róst.
she.NOM be.PAST always praise.PASS.
'She was always praised.'

One way of accounting for these cases is to allow Passive to refer to structural Case, as it does in the new passive rule in (270), and let which Cases are structural or inherent vary.

The so-called "insular" North Germanic languages (Icelandic and Faroese) are notorious for having a complicated relationship between the morphological forms of Case and their status as structural or inherent. See Zaenen et al. (1985), and references therein.

3.7.1 Unaccusatives

We have seen two types of cases, then, in which a term c-selects a DP but does not assign accusative Case. There are certain adjectives of this kind (e.g., *obvious*) and there are verbs created by the passive operation from transitive verbs. There is evidence that some verbs which might otherwise look like intransitive verbs fall into this last class too. For instance some verbs have two versions, if you like, one in which they are transitive and the other in which they are intransitive. Some examples are (272).

(272) a. She bounced them.
  b. They bounced.
  c. She melted them.
  d. They melted.
  e. She opened them.

f. They opened.
g. She grew them.
h. They grew

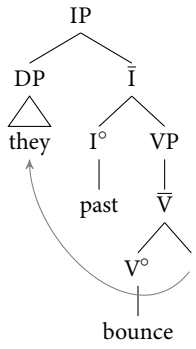
In their transitive guise, they c-select a DP and assign a certain θ-role to that object, and they also assign Accusative Case to it. In their intransitive guise, that very same c-selected argument DP appears in the Nominative Case and consequently appears in Specifier of IP. Pairs of sentences like (272) are sometimes said to involve a "causative/inchoative" alternation. The name "causative" expresses a hypothesis about the underly semantics of the transitive guise: ⟦she melted them⟧ ≈ "she caused them to become melted." And the name "inchoative" references a hypothesis about the intransitive guise: ⟦they melted⟧ ≈ "they became melted." I'll use this term to refer to this alternation, even though the underlying semantics of the transitive and intransitive guises is probably not equally well suited to it.

Perhaps this is done first in Hall (1965).

It certainly seems as if the θ-role borne by *they/them* is the same in the transitive and intransitive versions, and a simple way of understanding this is to imagine that the verb has assigned that θ-role to the complement position in both of its guises. The difference in the intransitive and transitive versions is whether or not the verb has also assigned Accusative Case and also comes with an external θ-role. That is, the intransitive versions might involve a derivation like that indicated in (273).

For instance, ***she bounced*** them doesn't correspond to "she caused them to become bounced" and ***they bounced*** doesn't correspond to "they became bounced."

(273)



If this is correct, then, it appears that there might be two kinds of single argument verbs—ones that assign their single θ-role externally and another that assign their sole θ-role internally.

David Perlmutter[7] was the first to conjecture that some intransitive verbs could assign just an internal θ-role, and no external θ-role. In Perlmutter and Rosen (1984), the name "unaccusative" is given to verbs of this kind. The intransitives whose sole argument is external are known as "unergative." I will use the term "intransitive" to refer to verbs whose sole argument gets either an external θ-role or an internal θ-role.

[7] Perlmutter, David. 1978. Impersonal passives and the unaccusative hypothesis. In ***Berkeley Linguistic Society***, volume IV, 157–189. University of California, Berkeley

(274) Intransitives

a. A ghost should sleep. (unergative)

b. A ghost should bounce. (unaccusative)

Burzio (1986), who gives an extended defense of the view that Italian has both unaccusative and unergative intransitives, uses an alternative terminology which is also common. He calls those verbs that have just an external θ-role "intransitive," and those verbs whose sole θ-role is internal "ergative." Some of the literature follows Burzio's terminology.

You will find different ways of referring to these single argument verbs, but this terminology I believe is the most common. In this book, I'll use Burzio's term ("unergative") for the intransitives whose argument is external, and Perlmutter's term ("unaccusative") for the intransitive whose argument is internal.

There are a variety of phenomena that suggest this is the right way of thinking about the two guises of the verbs in (272). These phenomena also suggest that there are intransitive verbs that don't have a transitive guise which are nonetheless unaccusatives. I'll describe some of these processes here.

Another early defense of the syntax of intransitives described here is in Rosen (1984). Levin and Hovav (1995) is recommended as an introduction.

One is the morphological rule that forms adjectival participals from verbs discussed in section 3.3.1. This rule forms an adjective whose relation to the noun it modifies is the same as that given by the internal θ-role of the corresponding verb. (275) illustrates.

(275) a. a smoothed blanket

i. = a blanket which something smoothed

ii. $\neq$ a blanket which smoothed something
(compare: *She smoothed the blanket.*

b. some mixed nuts

i. = some nuts which someone mixed

ii. $\neq$ some nuts which mixed something
(compare: *She mixed the nuts.*)

c. the painted man

i. = the man which someone painted

ii. $\neq$ the man which painted something
(compare: *She painted the man.*)

d. the known astronaut

i. = the astronaut someone knows

ii. $\neq$ the astronaut who knows someone
(compare: *She knows an astronaut.*)

As expected, adjectives formed in this way from intransitives are largely not possible.

(276) a. * the sung alto

b. * the barked dog

c. * the slept woman

d. * the cried child

If the intransitive verbs these adjectives are formed from have just an external θ-role, this is expected. The verbs that participate in the causative/inchoative alternation in (272) are among the verbs that can become adjectives with this morphology.

(277) a. a bounced ball
b. a melted car
c. an opened door
d. a grown child

But this shouldn't be surprising, as it's conceivable that the adjectives in (277) are formed from the transitive versions of the verbs.

There's a small class of intransitive verbs which are able to become adjectival participles, even though they have no transitive version. Some cases are in (278).

See Hoekstra (1984) and Levin and Rappaport (1986) for this generalization.

(278) a. the dead roach
b. the recently arrived package
c. the risen divers
d. the surfaced submarine

The verbs these adjectives derive from (*die*, *arrive*, *rise*, and *surface*) could be unaccusatives. If the verbs that adjectival participles are formed from must have an internal θ-role, then we should conclude that the verbs from which the adjectives in (278) derive from are unaccusative.

Another morphological process which could be used to distinguish the two intransitives is the suffix *er* which forms nouns from verbs. Some examples are in (279).

(279) a. (the) singer
b. (the) maker
c. (the) baker
d. (the) manufacturer
e. (the) stapler

Suffixation of *er* onto a verb creates a noun that refers to things that are whatever the external θ-role of the verb is. Thus, a *singer* is something that sings, and a *stapler* is something that staples. What we might expect, then, is that *er* can only attach to verbs that have an external θ-role. That means we should find nouns formed from transitive verbs and unergative verbs, but not from unaccusative and raising verbs. Note first that *er* cannot form a noun from the intransitive guise of the verbs that participate in the causative/inchoative alternation in (272). This is what (280) shows.

Chapter 5.3 sketches an anaylsis.

(280) a. the bouncer
= the one who bounced something
≠ the one who is bounced
(*she bounced it* ∼ *it bounced*)

b. the melter
= the one who melted something
≠ the one who is melted
(*she melted it* ∼ *it melted*)

c. the grower
= the one who grows something

i. ≠ the one who is grown
(*she grew it* ∼ *it grew*)

This supports the view that the intransitive versions of these verbs are unaccusative. If the intransitive version of these verbs doesn't have an external θ-role, then they should not be able to form nouns with *er*. Instead, the nouns can only be formed from the transitive versions of these verbs. (280) is just what we should expect on that view.

Note that *er* can suffix onto verbs that are intransitive.

(281) a. (the) runner
b. (the) dancer
c. (the) sleeper

The reason *er* cannot attach to the intransitive version of verbs which participate in the causative/inchoative alternation is not because it cannot attach to intransitives, then. Presumably, the difference in the intransitives that can and can't attach to *er* is whether the sole argument of that intransitive gets an external θ-role.

As (282) indicates, there are other intransitive verbs that cannot be made into nouns with *er*.

(282) a. * (the) seemer
b. * (the) appearer
c. * (the) arriver
d. * (the) dier
e. * (the) surfacer
f. * (the) riser

These include the raising verbs *seem* and *appear*, and this is expected as we've already seen evidence that these verbs do not have an external θ-role. But they also include the intransitive verbs we saw could form adjectival participles in (278). To the extent that there is a correlation between the intransitive verbs that cannot form nouns with *er* but can form adjectival participles, there

is evidence for the existence of unaccusative verbs because their existence predicts this correlation.

Consider next the prefix *re*, which forms verbs from other verbs. Some examples are in (283).

Using ***re*** prefixation to diagnose unaccusatives comes from Horn (1980).

(283) a. I resanded the table.
b. She repainted the wall.
c. He rethought the problem.

Roughly speaking, what *re* does is create a verb whose meaning is identical to the one it attaches to, but adds that whatever the V containing the verb and object describes is something that happened previously. Thus, for instance, (283a) communicates that *said the table* is something that was true previously. I cannot say (283a) if I am sanding a table that has never been sanded before. If, however, it was previously sanded, even if that sanding was by someone other than me, I can say (283a). Note in particular that (283a) does not communicate that I sanded something other than the table previously. It's only if the previous sanding were of the table that (283a) can be used. So very rough paraphrases of the meanings that are delivered by in (283) are given in (284).

See Chapter 5.2.1 for an analysis of the similar meaning that again has.

(284) a. ⟦I resanded the table⟧
≈ I sanded the table and there was a prior resanding of it
b. ⟦She repainted the wall⟧
≈ She painted the wall and there was a prior painting of it
c. ⟦He rethought the problem⟧
≈ He thought about the problem and there was a prior thinking about it.

The meaning of *re*, then, targets the meaning of the phrase that is formed by putting the internal argument together with the verb. If there is no internal argument, using *re* is odd. Consider (285).

(285) a. I resang.
b. I rebaked.
c. I reslept.

This matches the oddness of the paraphrases that we might expect their meanings to correspond to:

(286) a. I sang and there was a prior singing
b. I baked and there was a prior baking
c. I slept and there was a prior sleeping

The oddness of (286) might have to do with how trivial the meaning that *re* provides is. It might be taken for granted that I am not the first one to sing, bake or sleep, but this is all that *re* is adding to the sentences in (285).

Now, interestingly, there are some intransitive verbs that *re* can attach to without complaint, and the resulting meaning treats the argument of that intransitive as an internal argument. The intransitives that participate in the causative/inchoative alternation are among these.

(287) a. The ball rebounced.
≈ the ball bounced and there was a prior bouncing of the ball
b. The ice remelted.
≈ the ice melted and there was a prior melting of the ice
c. The flowers regrew.
≈ the flower grew and there was a prior growing of them
d. The door reopened.
≈ the door opened and there was a prior opening of it

Note that the referent of the flowers in (287c) cannot be the same in the two conjucts of its meaning. We need to let a definite description like ***the flowers*** be able to refer to different tokens of the same thing.

This is another piece of evidence, then, that the intransitive use of these verbs involves an internal θ-role, not an external θ-role. If has a meaning that references the phrase which combines verb and direct object, then we should treat the arguments in (287) as direct objects.

The prefix *re* can also attach to the intransitives that *er* suffixation and the rule that forms adjectival participles suggest are unaccusative. This is what (288) shows.

(288) a. She reappeared.
≈ She appeared and there was a prior appearance of her
b. It rearrived.
≈ It arrived and there was a prior arrival of it
c. It redied.
≈ It died and there was a prior death of it
d. It resurfaced.
≈ It surfaced and there was a prior surfacing of it
e. It rerose.
≈ It rose and there was a prior rising of it

(288c) describes situations that are more frequent in fiction than they are in reality.

Again, to the extent that there is a correlation between the class of intransitives that *re* can attach to, producing a meaning that includes the argument of that intransitive, and the class of verbs that cannot form nouns with *er* but can form adjectival participles, we have support for the presence of unaccusatives. This correlation is just what would be expected if this class of verbs were unaccusatives.

I'll describe one last phenomenon in English which is relevant to determining whether an intranstivie is an unaccusative or an unergative. This phenomenon involves a syntactic construction called a "resultative," of which (289) is an example.

(289) The rod pounded the lump flat.

The adjective *flat* in this example is understood to describe the result of the pounding. And the property it describes must be understood as holding of the object argument, not the subject argument. From (289) we can conclude that the pounding changed the lump into a flat lump, but not that the pounding changed the rod into a flat rod. This sensitivity to being an external or internal argument in the resultative seems to be quite general. In all of the examples of (290), the adjective describes a state that the object acquires, not a state that the subject acquires.

This generalization about resultatives is introduced in Simpson (1983). See Levin and Hovav (1995, chapter 2) for a discussion of its utility in diagnosing unaccusatives. Some possible counter-examples are in (i), from Wechsler (1997).

i. a. The wise men followed the star out of Bethlehem.
   b. The sailors caught a breeze and rode it clear of the rocks.
   c. He followed Lassie free of his captors

(290) a. She stabbed me bloody.
   b. She spun me dizzy.
   c. The grill cooked the meat black.
   d. The machine shook the bottle empty.

See Rothstein (2004, chapter 3.7) and Levin and Hovav (1995, chapter 5) for some discussion.

As expected, then, resultatives aren't felicitous unless there is an internal argument for it to describe. This is why resultatives are generally not possible with intransitive predicates.

Resultatives should be distinguished from constructions which look similar but have a different meaning. In the sentence ***Kaden met Shay drunk***, the adjective ***drunk*** can describe a property that either Kaden or Shay have. But this isn't a resultative because that property is not understood to be the result of the meeting. These constructions are what Halliday (1967) calls "depictives."

(291) a. * I walked exhausted
   b. * I sang hoarse.
   c. * It flew unscared.
   d. * The child cried tired.
   e. * I baked sweaty.
   f. * I slept awake.

These are grammatical only as depictives.

But——perhaps unsurprisingly by now——resultatives are possible with the intransitive version of the sentences that participate in the causative/inchoative alternation.

(292) a. It bounced flat.
   b. It melted away.
   c. It opened wide.
   d. It grew tall.

Like the behavior of the prefix *re*, the suffix *er*, and adjectival participles, this supports the conclusion that these intransitives are unaccusatives.

Unforuntatly resultatives do not combine with all of the other verbs we've seen could be unaccusative. The difficulty forming resultatives of these verbs is indicated by (293).

(293) a. The roach died quiet.
   b. The man arrived warm.
   c. The submarine surfaced lower than the plane.
   d. The bird rose dizzy.

To the extent that any of these sentences are grammatical, it's because the adjective is not clearly understood to be a result of the event described by the verb. Thus, for instance, (293a) is a grammatical sentence if it describes a scenario in which the roach died while quiet. It cannot describe a scenario in which the roach's death resulted in it being quiet. The adjectives in (293) are depictives, not resultatives.

The events described by *die* and *arrive* are punctual—–that is they describe events that have vanishingly short durations—–and this might make them ineligible to have results. Intuitively, resultatives describe a state that unfolds as part of the process that the verb describes. Perhaps a state cannot be part of something as punctual as an arrival or a death. That reasoning would not extend to *rise* and *surface*, though. And this brings us to a problem lurking in these various arguments for the existence of unaccusatives. All of the processes I've presented as relevant for determining when an intransitive is unergative or unaccusative aren't productive. That is, there are other criteria that determine whether these processes can apply. This is shown below, where each process and the enironment it (putatively) requires are paired with examples of that process in that environment that are nonetheless ungrammatical.

(294) a.

generalization:	adjectival participles are formed from verbs that have a direct object.
counter-examples:	**the hoped fact*
	**the costed amount*
	**the worded letter*

b.

generalization:	*er* nouns are formed from verbs that have an external argument.
counter-examples:	**(the) meaner* (from *she means it*)
	**(the) equaler* (from *this equals that*)
	**(the) coster* (from *this costs a lot*)

c.

generalization:	*re* prefixes to verbs that have a direct object.
counter-examples:	**reknow this fact*
	**recost this amount*
	**reëntertain the idea*

d.

generalization:	resultatives combine with verbs that have a direct object.
counter-examples:	**She carried the cake stale.*
	**She threw the child dizzy.*
	**She studied this conjecture understood.*

Unless we know how to give both necessary and sufficient conditions for these processes, we can expect that they will not necessarily pick out the same class of intransitives. In particular, then, we should not expect to find that any given unaccusative will form an adjectival participle, be amenable to *re* prefixation, and license a resultative. Nor should we expect any given unergative

to be able to form a noun with *er*. The other factors relevant to determining when these processes can occur might apply. Since the argument that being unaccusative or unergative relies on these phenomena correlating—if *re* and resultatives picked arbitrarily different verbs, we would have no reason to think that they are referencing the same attribute—–it is weakened when the correlations aren't perfect. For this reason, identifying the verbs that are unaccusative in any particular language has been difficult. And, consequently, finding generalizations about the kinds of verbs that can be unaccusative has also been slow-going. Perhaps the best candidates are the intransitives of the causative/inchoative, and other similar, alternations.

That there are generalizations to be found in which intransitives are unergative and which are unaccusative is suggesed by the fact that the verbs in the causative/inchoative alternation do not always permit an intransitive version. There are instances of the transitive version which have no matching intransitive version, as (295) shows.

(295) a. He broke his promise.
b. * His promise broke.
c. He broke the world record.
d. * The world record broke.
e. This book will open your mind.
f. * Your mind will open.
g. The waiter cleared the table.
h. * The table cleared.
compare:
i. The wind cleared the sky.
j. The sky cleared.
(Levin and Hovav, 1995, (9), (11)-(13): 85-6)

So far as I know, this asymmetry doesn't go in the other direction. Levin and Hovav (1995) argue that this indicates that the intransitive versions of these pairs are derived from the transitive versions. The process that forms the intransitive from the transitive is sensitive to the particulars of the relation that holds between the verb and its internal argument. That sensitivity, however it might turn out to be formulated, might reflect constraints on the kinds of intransitives that can be unaccusative.

See also Dowty (1991), Haspelmath (1993), Hoekstra and Mulder (1990), and Goldberg (1994).

What I've laid out here are some of the ways in which we might discover the existence of unaccusative verbs in English. Many of these ways exist in other IndoEuropean languages as well. In each of these cases, the phenomena I've used involves some syntactic/semantic property that singles out direct objects or internal θ-roles. The prefix *re* and the resultative, for instance, require a direct object for its interpretation. And the adjectival participal requires a verb with an internal θ-role while the nominalizing suffix *er* requires

a verb with an external θ-role. There is another common method of distinguishing unaccusative and unergative verbs that, like *er*, targets verbs with external θ-roles, but which isn't found in English. This is the passive.

The passive in English only applies to transitive verbs, but in many languages it can apply to intransitive verbs as well. This is the case in German, for instance. The intransitive verb *tanzen* ('dance'), for instance, can be turned into a passive verb. The alternations in (296) illustrate.

These data are all from Grewendorf (1989). Note that German forms the passive with the aid of a verb (***werden*** 'become') that is different than that used in situations where adjectival participles arise. With adjectival participles, German uses the verb ***sein*** ('be'), just as English does.

(296) a. Sie tanzt.
she.NOM dance.PRES
'she dances.'

b. Es wurde getantzt.
it.NOM become.PRES dance.PASS
'it was danced.'

c. Sie arbeitet.
she.NOM work.PRES
'she works.'

d. Es wurde gearbeitet.
it.NOM become.PRES work.PASS
'it was worked.'

e. Sie schläft.
she.NOM sleep.PRES
'She sleeps.'

f. Es wurde geschlafen.
it.NOM become.PRES sleep.PASS
'it was slept.'

The *es* ('it') in these examples is an expletive. Passives of intransitive verbs are sometimes called "impersonal passives." Not all intransitives can undergo the passive, however. Some cases are (297).

(297) a. * Es wurde angekommen.
it.NOM become.PRES arrive.PASS
'It was arrived.'

b. * Es wurde gewaschen.
it.NOM become.PRES grow.PASS
'It was grown.'

c. * Es wurde gefallen.
it.NOM become.PRES fall.PASS
'It was fallen.'

The intransitives that can undergo the passive in German are classed as unergatives by independent criteria, and those which cannot undergo the passive are classed as unaccusatives by those criteria. For instance, German has the same rule that forms nouns from suffixing *er* to verbs, and just as in

English, it attaches only to verbs with an external θ-role. In German, this suffix can attach to the intranstivies that passivize, but cannot attach to the intransitives that don't.

(298) a. Arbeiter
'worker'

b. Tänzer
'dancer'

c. Schläfer
'sleeper'

(299) a. * Ankommer
'arriver'

b. * Wachser
'grower'

c. * Faller
'faller'

In the (many) languages that allow intransitives to passivize, there is a contrast of this sort between those intransitives that seem to be unaccusatives, and those that seem to be unergatives. David Perlmutter and Paul Postal[8] have suggested that it is a language universal that unaccusative verbs cannot passivize, and at the time of writing this remains a live possibility.

[8] Perlmutter, David, and Paul Postal. 1984. The 1-advancement exclusiveness law. In ***Studies in relational grammar volume 2***, ed. David Perlmutter and Carol Rosen. University of Chicago Press

3.7.2 Burzio's Generalization

Now that we've established that accusative Case assignment and c-selection of DPs is independent, let's consider the predictions this makes about the types of verbs we should expect to find. We should find, and have, verbs that c-select a DP but do not assign accusative Case: passive participles and unaccusatives. We should also find, and have, verbs that c-select DPs and assign accusative Case: transitive verbs. We should expect to find verbs that do not c-select DPs but do assign accusative Case. One candidate for this type of verb is *let*, which cannot combine with an accusative Case-marked DP complement, but can, plausibly, assign accusative Case to the subject of a small clause that it c-selects.

(300) a. * She let the dance.

b. She let [$_{VP}$ the dance happen].

If *let* is responsible for assigning accusative Case in (300b), then the ungrammaticality of (300a) cannot be credited to the Case filter. We should conclude that *let* does not c-select DPs. Other possible examples of verbs of this kind are ones like *hope*. Recall that these verbs do not have DP arguments, but they can be passivized.

(301) a. * This wish was hoped.
b. It was hoped that you'd understand this section.

Since passivization generally happens in English only to verbs that assign accusative Case, we might be warranted to conclude from (301b) that *hope* assigns accusative Case. Like *let*, then, this might be a verb that assigns accusative Case but does not c-select a DP.

Finally, we should expect to find verbs that do not c-select DPs and do not assign accusative Case. A candidate for this type of verb is *seem*. We have decided that *seem* does not assign accusative Case, since this gives us an explanation for the contrast in (302).

(302) a. * It seems them to be happy.
b. They seem to be happy.

The table in (303) shows why structural accusative Case and c-selection of a DP are independent.

(303)

	Accusative Case	No Accusative Case
c-select DP	*like* (transitives)	*melt*, *liked* (unaccusatives/passives)
¬ c-select DP	*let, hope*	*seem*

Interestingly, if we look for the same independence of accusative Case assignment and external θ-role assignment, it does not emerge.

(304)

	no external θ-role	external θ-role
accusative Case and c-select DP	??	*like* (transitives)
accusative Case and ¬ c-select DP	??	*let, hope*
no accusative Case and c-select DP	*melt* (unaccusative)	??
no accusative Case and ¬ c-select DP	*seem*	*sleep*?

Some cells of this table are empty, suggesting that the availability of an external θ-role type is not independent of the way that verbs can take "objects." The cell occupied by *sleep?* is not clear. We don't see *sleep* combine with a DP complement nor do we, like *let*, see it combine with a small clause whose subject argument is assigned accusative Case. This is consistent with it neither c-selecting DP nor assigning accusative Case. But it's also consistent with *sleep* not c-selecting a DP complement but being able to assign accusative Case. In general, it's very difficult to tell if verbs of this kind exist. It could be that some unergative verbs are of this type; but unergatives could also be verbs that assign accusative Case, but do not c-select DP. They could occupy the cell occupied by *let* and *hope*. I'll ignore the status of this cell, given the difficulty in determining whether verbs of the type it describes exist.

The other gaps, however, indicate that there is a correlation between having accusative Case and having an external θ-role. Burzio discovered these gaps in this paradigm where there appear to be no verbs, and formulated generalizations which describe these absences.

(305) Burzio's Generalization

a. If a verb has structural Case, then it has an external θ-role.

b. If a verb has an external θ-role (and c-selects a DP?), then it has structural Case.

Why aren't these verbs easy to find? If they exist, they have been very difficult to identify. Putative counter-examples are extremely rare. Among the best, perhaps, are German sentences like (306).

(306) Dich hungert.
you.acc hunger.pres
'you are hungry.'

There are a small handful of German verbs that allow their single argument to be accusative like *hungern* does, in violation of the first clause of Burzio's Generalization. One conjecture is that in these situations the normal relationship between accusative and structural Case is lost; the accusative in (306) is instead an inherent Case.

Burzio's first generalization might reflect a relationship between Accusative Case and θ-role assignment for which we should find a source.

The second of these generalizations might be derivable from the conditions we have seen on Argument movement, at least when it involves verbs that c-select DPs. Consider the sentences in which such a verb could exist. One of those is (307).

(307) $[_{IP}$ should $[_{VP}$ Smith V Jones]].

If V assigns a θ-role, but no Case to *Jones*, then there is no way both of the DPs it selects are going to be able to satisfy the Case Filter. There are more DP arguments in (307) than there are Case-marked positions. So, if such a verb is to survive, the only environment it will be able to appear in are sentences which have two Case marked positions. These two positions will both have to be Specifiers, because these are the only positions reachable by Argument movement. Thus, we're looking for positions like V_2 in (308) for our verb to appear.

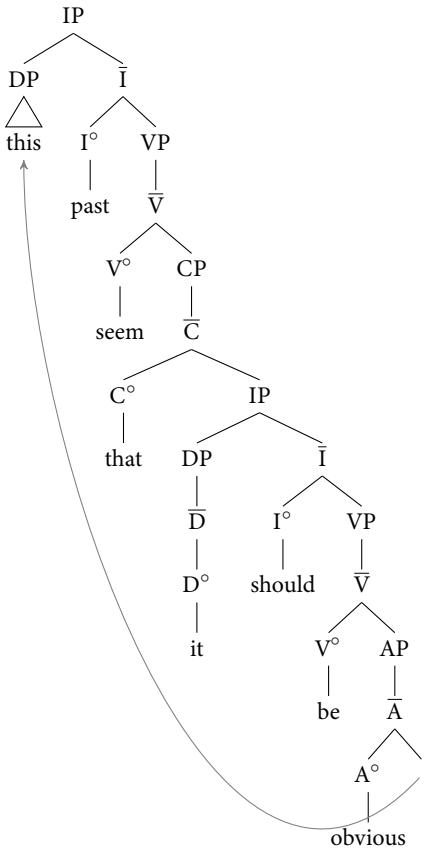
(308) $[_{IP}$ – should V_1 $[_{XP}$ – $[_{VP}$ Smith V_2 Jones]],
where both "–" positions are Case marked.

Now the first part of Burzio's Generalization tells us that V_1 cannot assign Accusative Case. If it did, then it would also assign an external θ-role, and that's going to bring the count of things that need Case to one more than there are Case marked positions. As a consequence, the Case marked position inside XP is going to also have to get its Case from some place other than V_1. So far, the only ways we have seen for this to be done are if XP is in a CP, as in (309).

(309) $[_{IP}$ – should V_1 $[_{CP}$ for $[_{IP}$ – to $[_{VP}$ Smith V_2 Jones]]]].
$[_{IP}$ – should V_1 $[_{CP}$ that $[_{IP}$ – I° $[_{VP}$ Smith V_2 Jones]]]]
$[_{IP}$ – should V_1 $[_{CP}$ e $[_{IP}$ – to $[_{VP}$ PRO V_2 Jones]]]]

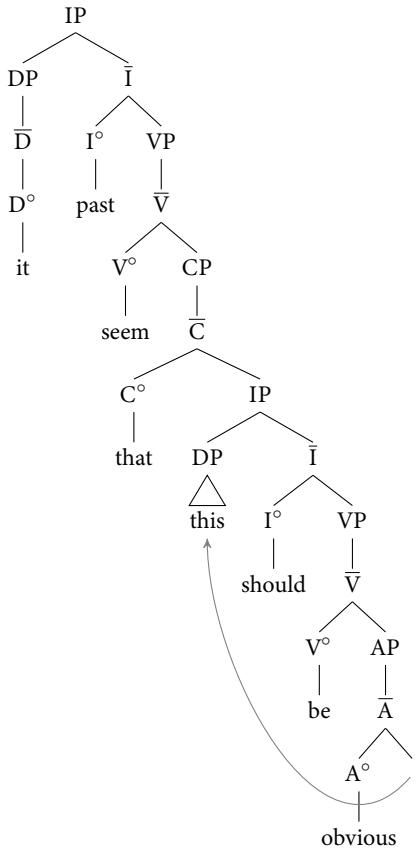
For these configurations to lead to a grammatical S-structure, one of the arguments in the embedded CP is going to have to A-move into the Specifier of the higher IP. Recall, however, that there is evidence that suggests that A-movement in English is blocked from taking an argument out of the CP it resides in. Consider, for instance, the derivation in (310).

(310)



We've seen that *obvious* c-selects a DP argument, but does not assign it accusative Case. In (310), A-movement has taken the c-selected DP (=*this*) into the nominative Case marked position in the higher IP. This should lead to a grammatical S-structure, but it doesn't. This contrasts with the grammatical derivation in (311).

(311)



The difference between (310) and (311) is whether the A-moved DP has been moved out of the embedded CP. It's facts like these that led us to posit (178).

(178) If Argument Movement moves XP from position α to position β, then there cannot be a CP that dominates α but not β.

This condition will prevent any of the scenarios in (309) from having a grammatical S-structure. So this is also an environment in which a verb of the sort we are missing could not survive.

Finally, consider the situation that arises if such a verb were to select a CP external argument:

(312) $[_{IP}$ should $[_{VP}$ CP V Jones]].

Here, only the argument DP *Jones* needs to find a Case-marked position. From what we presently know about A movement, this should be possible: *Jones* should be able to move into the Specifier of IP by A Movement and there satisfy the Case filter. If there is a place that such verbs could exist, this is where they would survive. We will see in the next chapter evidence for a constraint on A movement that would prevent these environments from being grammatical as well. That constraint prevents one argument from moving past another by A Movement under certain circumstances. That constraint might be responsible for blocking this environment as well.

Hartman (2012) argues that configurations like (312) arise for adjectives, however. He argues that sentences like ***Marcin is annoyed that pigs fly*** involve an external argument CP and a internal argument DP that has moved to the nominative Case-marked position. There is more about the syntax of these sentences needed to get the word-order right, of course. But if he is correct, then the absence of verbs of this kind cannot be credited to constraints on A Movement in the way sketched here.

For this reason, the usual conclusion is that the second of Burzio's generalizations derives from constraints on Argument movement in concert with the Case filter. What remains is (313).

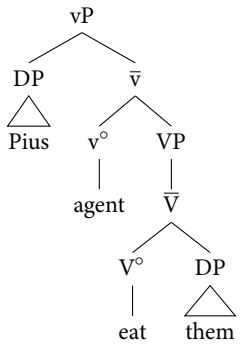
(313) Burzio's Generalization
If a verb assigns structural Case, then it assigns an external θ-role.

If Burzio's Generalization cannot be derived from constraints on the syntactic form that such verbs would have to live in, then it suggests that the assignment of Case and θ-roles is not independent in the way that we presently think it is.

Indeed, one way of describing Burzio's Generalization is that the external θ-role is the assigner of structural (accusative) Case. It isn't, as we originally thought, that the categories of words that assign accusative Case in English are prepositions and verbs. The accusative Case assigners of English are instead prepositions and whatever part of speech that external θ-roles belong to. A way of giving a syntax for that thesis involves making the external θ-role syntactically external to the verb that it is associated with. This is the position taken by Angelika Kratzer,[9] who uses the symbol "v" ('little [vij]') to indicate the category of an external θ-role.

[9] Kratzer, Angelika. 1996. Severing the external argument from its verb. In ***Phrase structure and the lexicon***, ed. Johan Rooryck and Laurie Zaring, 109–137. Dordrecht, The Netherlands: Kluwer Academic Publishers

(314)



This syntax puts the external θ-role higher than the verb it is associated with, and consequently puts the argument the external θ-role combines with higher too. By placing the external θ-role higher than the VP, this syntax also allows it to c-command the DP it allegedly assigns accusative Case to. In these ways, the proposed syntax in (314) fits with the syntax we've developed.

That external θ-roles are related to a head that is higher than the VP is an idea can also be found in Williams (1980), Hornstein and Lightfoot (1987), Bowers (1993), and Dikken (1992).

But there are several ways it does not. Our present understanding of accusative Case assignment is that it requires the Case assigner to not only c-command the position assigned Case, but to also be adjacent to it and no farther from that position than (315) describes.

(315) The position assigned accusative Case by X° cannot be dominated by more than one phrase that does not also dominate X°.

(315) is one of the requirements of "government." In (314), v° meets neither of these conditions: it is not adjacent to *them* and there is more one phrase that dominates *them* that doesn't also dominate v°. If (314) is correct, then we will have to revise our Case theory.

The semantics of θ-role assignment, and how it interacts with the syntax, also does not fit (314). Our semantics for *eat* gave it the denotation in (316).

(316) $[\![\text{eat}]\!] = \lambda x\ \lambda y\ \lambda\epsilon\ \underline{\text{AGENT}}(y)(\epsilon) \wedge \text{PATIENT}(x)(\epsilon) \wedge \text{EAT}(\epsilon)$=T, where
  a. x and y are things and ϵ is an eventuality
  b. PATIENT c-selects DP

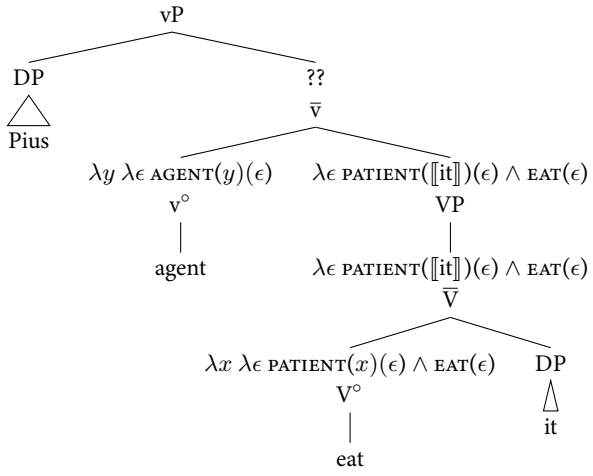
On this view, θ-roles are functions from things to functions that describe eventualities. If we are to separate the Agent θ-role from (316) in the way that (314) requires, then the denotations we get from simply doing this to (316) is (317).

(317) a. $[\![\text{agent}]\!] = \lambda y\ \lambda\epsilon\ \text{AGENT}(y)(\epsilon)$=T, where
  i. y is a thing and ϵ is an eventuality
  b. $[\![\text{eat}]\!] = \lambda x\ \lambda\epsilon\ \text{PATIENT}(x)(\epsilon) \wedge \text{EAT}(\epsilon)$=T, where
  i. x is a thing and ϵ is an eventuality
  ii. PATIENT c-selects DP

The denotation of Agent in (317) remains a function from things, x, to functions that describe eventualities. But if we plug these denotations into (314), the rules of semantic composition we adopted in (235) do not work.

(235) a. If X is the sole daughter of Z, then $[\![\text{Z}]\!] = [\![\text{X}]\!]$
  b. If X and Y are daughters of Z, then $[\![\text{Z}]\!]$ is the result of applying the function $[\![\text{X}]\!]$ to its argument $[\![\text{Y}]\!]$.

(318)



There is no rule in (235) that can combine the meanings of v° and VP in (318).

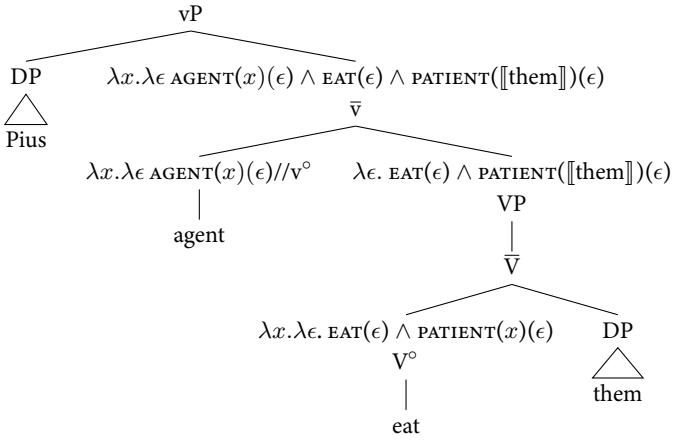
Kratzer's solution to this is a new rule of semantic composition, which she calls "Event Identification."

(319) Event Identification
In 

if $[\![\text{f}]\!]$ is $\lambda x.\lambda\epsilon.P(x)(\epsilon)$ and $[\![\text{g}]\!]$ is $\lambda\epsilon.Q(\epsilon)$, then $[\![\text{h}]\!]$ is $\lambda x.\lambda\epsilon.[P(x)(\epsilon) \wedge Q(\epsilon)]$.

Making use of this rule allows for the semantics for (318) indicated in (320).

(320) $\lambda\epsilon.$ AGENT([[Pius]])(ϵ) $\wedge$ EAT(ϵ) $\wedge$ PATIENT(Jones)(ϵ)



Another possibility is that we could imagine that θ-roles which are put into the syntax independently of the verb have a different denotation. One that is found often in the literature is (321).

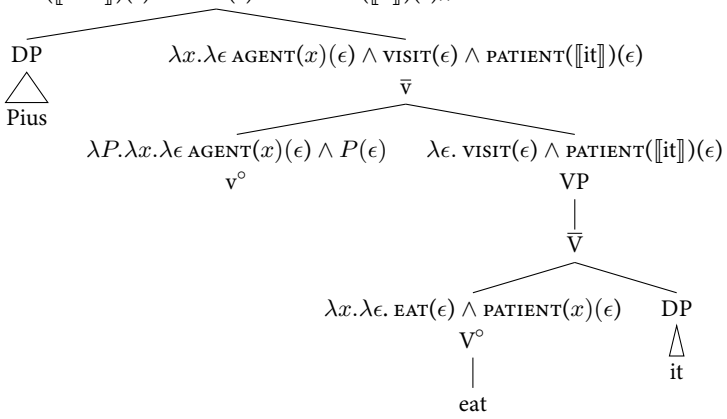
(321) [[agent]] = $\lambda P.\lambda x.\lambda\epsilon$ AGENT(x)(ϵ) $\wedge$ $P(\epsilon) = T$, where

a. P is a function whose domain is eventualities, x is a thing, and ϵ is an eventuality

b. agent c-selects VP.

With this denotation, things compose up as indicated in (322)

(322) $\lambda\epsilon.$ AGENT([[Pius]])(ϵ) $\wedge$ VISIT(ϵ) $\wedge$ PATIENT([[it]])(ϵ)// vP

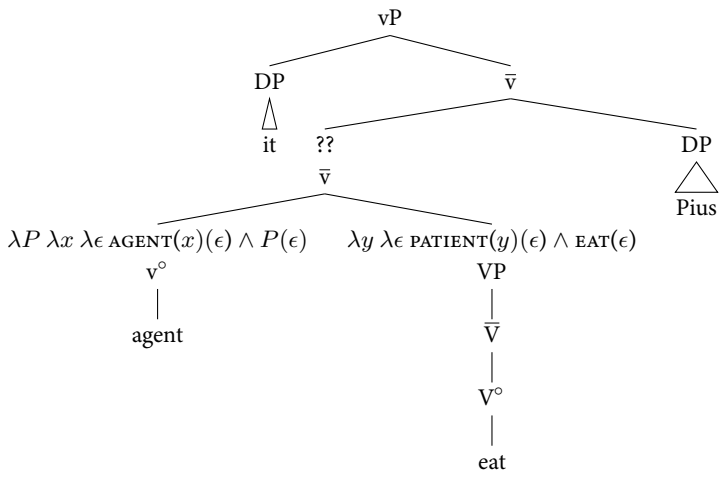


This view doesn't require Event Identification; the rule of function composition is sufficient. Both of these techniques yield the same denotation for vP, and both are found in the literature.

Giving external θ-roles denotations like those in (317a) or (321) has certain interesting consequences when combined with this new syntax. It gives an explanation for why external θ-roles bearers are found higher than other arguments. Consider, for instance, what would happen if v° were to be placed

between a verb and another DP it assigns a θ-role to and its θ-role has a denotation like that in (321). A way of doing that which the $\overline{\text{X}}$ Skeleton permits is (323).

(323)



The denotation for VP is not of the right kind to be an argument of v°, nor is the denotation of v° the right kind to be an argument of VP. In general, if the verb heading VP has a θ-role that hasn't yet been assigned, then it won't be of the right semantic kind to be an argument of v°. The same consequence arises if we use the denotation for external θ-roles of the sort given in (317), though that case is complicated by the existence of other semantic composition rules we have not yet encountered. Let's somewhat arbitrarily adopt the denotation for external θ-roles given in (321).

With this semantics, then, we can now simplify the Projection Principle. We need no longer stipulate that the external θ-role is assigned in a different way than all the others. This will follow from the thesis that external θ-roles are placed syntactically in a different position than the verbs they are associated with, and the rules of semantic composition we've adopted. The Projection Principle can now be rendered as (324).

(324) The Projection Principle
If β is c-selected by α, then there must be a projection of α, π, that immediately dominates β and if π immediately dominates some other phrase, γ, then γ must be another projection of α or part of another c-selected argument of α.

Noam Chomsky[10] argued that the Projection Principle is part of the process of lexical insertion. Lexical Insertion puts into X° positions the morphemes that will be used to make the sentence. In our system, Lexical Insertion also determines the category of the X° position it is inserted into, as well as all of the projections of that X°. Let's derive this and the Projection Principle, then, by writing out how Lexical Insertion has these effects.

[10] Chomsky, Noam. 1965. ***Aspects of the theory of syntax***. Cambridge, Massachusetts: M.I.T. Press

(325) Lexical Insertion
Put morpheme μ into X°, only if

a. The morpho-syntactic category of X° and its projections is the same as the morphosyntactic category of μ, and

b. Every projection of X° satisfies the c-selection requirements of μ.

Let c be a c-selection requirement of μ. α satisfies c, iff

i. α doesn't have an adopted daughter whose morpho-syntactic category is different from what c requires, or ii.

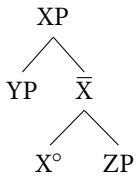
ii. c is satisfied by the phrase that projects α.

α is an adopted daughter of β if α is a daughter of β and β is not a projection of α.

These conditions have effects which are frequently assumed in the literature. But the conditions themselves are completely parochial. They are my own modernization of Chomsky (1965).

What an "adopted daughter" is can be graphically described. Consider a case like (326).

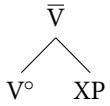
(326)



The adopted daughter of $\overline{X}$ is ZP; X° is not an adpted daughter of $\overline{X}$ because $\overline{X}$ is a projection of X°. Similarly, of the two daughters that XP has, only YP is an adopted one because XP is a projection of $\overline{X}$.

The definition of "satisfaction" in (325b) has complex consequences. It lets the c-selection requirements that comes with a verb's θ-role be satisfied in two ways. One way is to make the phrase that is c-selected an adopted daughter of one of the verb's projections. Thus, if XP in (327) is the category that the c-selection requirement of one of the verb's θ-roles calls for, then that c-selection requirement is satisfied.

(327)



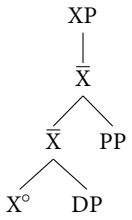
The definition of satisfaction in (325) requires that if a projection of a verb has an adopted daughter, here XP, then it cannot be a different category than what the c-selection requirements call for. If, by contrast, XP isn't the category that the c-selection requirements of that θ-role have, $\overline{V}$ will not satisfy the c-selection requirements. Since (325b) requires that all projections of the verb satisfy the c-selection requirements, if XP is not the category that one of the θ-roles c-selects, the $\overline{V}$ will be ungrammatical. This has the effect of allowing XP to be a phrase that isn't the category required by a θ-role of the verb only if the phrases within $\overline{X}$ have satisfied those requirements. This derives the generalization we had previously about arguments of a verb: they must be found in a projection of that verb that contains no non-arguments.

But the definition of satisfaction in (325) permits another way for the c-selection requirements of a θ-role associated with a verb to be satisfied. This way is illustrated in (328).

(328)


Suppose that V° has one $theta$-role that is accompanied by a c-selection. $\overline{\text{V}}$ satisfies this c-selection requirement because it has no adopted daughter whose category differs from what that c-selection requirement calls for. In this scenario, the c-selection requirements are satisfied because $\overline{\text{V}}$ has no adopted daughter. Because this is a way that the c-selection requirements can be satisfied, and because the graph in (328) is always available, this means that c-selection requirements are not responsible for ensuring that verbs (and other argument takers) are supplied with the arguments that c-select. What c-selection requirements do is merely ensure that when those arguments are provided, they be in the right positions and of the right category. That the arguments must be provided will come from the semantics, not the c-selection requirements.

Let's run through how this works in a couple of cases. Our first case involves inserting *put* into the phrase built by the $\overline{\text{X}}$ Skeleton in (329).

(329)


Assume that the denotation of *put* is (330).

(330) $[\![\text{put}]\!] = \lambda x\ \lambda y\ \lambda \epsilon$ THEME$(x)(\epsilon) \wedge$ LOCATION$(y)(\epsilon) \wedge$ PUT(ϵ)=T, where

a. x and y are things and ϵ is an eventuality

b. THEME c-selects DP

c. LOCATION c-selects PP

Lexical Insertion of *put* into the X° position of (329) is allowed if all the X's are re-written as V. Because *put* c-selects a DP and a PP, the sisters of X° and the smallest $\overline{\text{X}}$ must be DP and PP, if they exist, and this requirement is met in (329). If either DP or PP were different morpho-syntactic categories, the c-selection requirements of *put* would not be satisfied and lexical insertion would be blocked.

(329) is also a structure that *eat* could be lexically inserted into.

(331) $[\![\text{eat}]\!] = \lambda x\ \lambda\epsilon$ PATIENT$(x)(\epsilon) \wedge$ EAT(ϵ)=T, where

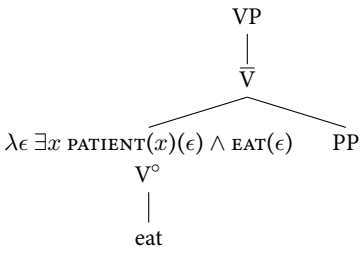
a. x is a thing and ϵ an eventuality

b. PATIENT c-selects DP

The c-selection requirement of *eat* is DP, and this is satisfied by virtue of DP being a sister to X°. Because the DP satisfies the one c-selection requirement of *eat*, the sister of the smallest $\overline{\text{X}}$ need not satisfy the c-selection of *eat*. As a consequence any kind of phrase, including PP, can reside in this position. Indeed, nothing need reside in this position.

Consider next what happens if *eat* is inserted into the structure in (332).

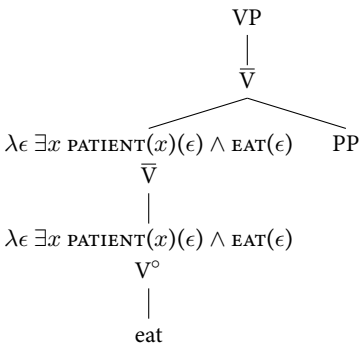
(332)



Because Existential Closure is a rule that can change the denotation of *eat*, we can create a VP headed by *eat* that satisfies the Theta Criterion without including a phrase that gets *eat*'s PATIENT θ-role. I've applied Existential Closure to *eat* in (332) so that we can consider how things work under this scenario. As it happens, Lexical Insertion is violated in (332). The c-selection requirement of *eat* is not satisfied by the XP that is immediately dominated by $\overline{\text{V}}$, and this is what (325b) requires.

Consider, by way of contrast, what happens if the Existentially Closed *eat* is inserted into the structure in (333).

(333)



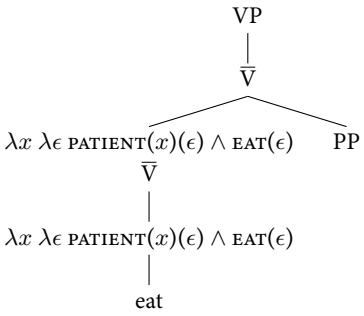
This does satisfy the conditions on Lexical Insertion. The immediate projection of V° does not contain an XP whose morpho-syntactic category differs from the c-selection of *eat*, and this meets the requirements of (325b). In general, this definition of Lexical Insertion derives the following generalization.

(334) The sister to the X° that a morpheme, μ, is lexically inserted into, will have to meet a c-selection requirement of μ if there is one.

It is common practice to regard only phrase markers that satisfy (334) to be well-formed. Because the difference between (332) and (333) is extremely difficult to detect, I do not know if this practice is sound. I will adopt it in this book.

If Existential Closure does not apply in (333), the result will still satisfy the conditions on Lexical Insertion, but the semantics will fail.

(335)



The PP will become the argument of PATIENT and we might safely assume that the denotations of PPs do not allow them to be patients.

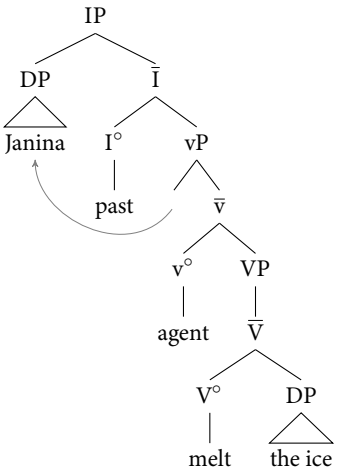
With a semantics that has the external θ-role c-select a VP, we have been able to simplify the Projection Principle. And this simplified Projection Principle can be derived from the conditions on Lexical Insertion in (325). Let's therefore regard the semantic issues that arise with the new syntax of external θ-roles solved, and consider some its consequences. We now have an account of alternations like that in (336).

(336) a. Janina melted the ice.

b. The ice melted.

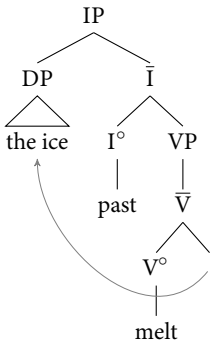
This is a situation where the external θ-role associated with *melt* is simply optional. There are two D-structure configurations available, and they give rise to the two derivations in (337).

(337) a.



There may be some evidence for this conclusion from prosody. In particular, the verb before an argument PP has the prosodic status of a word, whereas the verb before a non-argument PP has the prosodic status of a phrase. See Gussenhoven (1984).

b.



In (337a), the D-structure comes equipped with an Agent θ-role, and this v° assigns structural (accusative) Case. For this reason, a subject argument (=*Janina*) must be included in the D-structure and the object DP sits at D-structure in a Case-marked position. In (337b), by contrast, the D-structure comes without an external θ-role, and consequently there is no subject argument in this D-structure. Without the external θ-role, there is no structural Case available to the DP object, and so it will not satisfy the Case filter in its D-structure position. This is remedied by A Movement bringing that DP into a Case-marked position, in this example that is the nominative Case-marked Specifier of IP.

Consider next the Passive alternation in (338).

(338) a. Janina melted the ice.
b. The ice was melted (quickly).

Recall that forms like (338b) are ambiguous. The word ***melted*** can be a passive verb or it can be an adjective derived morphologically from the verb ***melt***. When an adjective, the reading it gives is stative. The eventualities that adjectives describe are states. We are interested here only in the Passive version of (338b). The passive has an eventive reading – a non-stative interpretation – and I have tried to help disambiguate in that direction with the adverb ***quickly***.

Our present conception of Passive is that it is a rule that builds a lexicon. It creates new verbs from old ones in the way that (339) describes. (I've now expressed the Passive as a function whose domain and range are verbs.)

(339) Passive(V) = V′
a. Put V into the morphological form appropriate for passive, and
b. Suppress the external θ-role of V, and
c. Remove the structural Case assigned by V.

In languages like English, the change of morphological form must somehow be tied to the fact that the Passive forms of verbs are always embedded inside a VP headed by *be*. This no longer fits our syntax. Under the present hypothesis, verbs never assign structural Case and they do not come with external θ-roles either. We will need to revise our view of the Passive.

Passives, like the one that English has, that involve more than a simple change to the morphology of the verb are sometimes called "periphrastic" Passives.

In doing this, let's focus on what the "suppression" of the external θ-role part of the rule consists of. It is useful here to compare the passive version of *melt* to its intransitive version in (336b).

(340) a. The ice melted. (=(336b))
b. The ice was melted (quickly).

Our account of the intransitive *melt* involved giving it a D-structure with no external θ-role. The passive *melt*, however, arguably does contain an external θ-role. One difference between the sentences in (340) is that the passive (340b) entails that there is an Agent of the melting eventuality it describes, whereas that isn't the case for the intransitive *melt* in (340a). We can see this by considering the sentences in (341) and (342).

(341) a. All the ice melted yesterday because everyone worked hard.
b. All the ice was melted yesterday because everyone worked hard.

(342) a. The ice melted yesterday, but I didn't do it.
b. The ice was melted yesterday, but I didn't do it.

The second clause in (341a) and (342a) sound disconnected from the first clause. But in (341b) and (342b), they sound connected to the first clause. This is because *everyone* can be understood to be quantifying over the Agents of the melting eventualities in (341b) but not (341a), and because *do it* can be understood to refer to an eventuality with an Agent in (342b) but not (342a). We can also see this difference by considering how adverbs that modify Agents work. One of those adverbs is *intentionally*; in(343), *intentionally* describes the attitude the Agent, but not the Patient, has to the hugging eventuality.

(343) Elena hugged Pius intentionally

This adverb also detects an Agent θ-role in the passive, but not the intransitive, use of *melt*.

(344) a. * The ice melted intentionally.
b. The ice was melted intentionally.

To the extent that (344a) is grammatical, it requires an interpretation in which ***the ice*** is an Agent of the melting eventuality; it means something roughly like ***the ice melted itself intentionally***. That is not the intransitive ***melt*** we are concerned with.

We do not want the Passive to remove the Agent θ-role, in the same way that has been done in forming the intransitive. This is the point made by Mark Baker and Ian Roberts in a paper from the 1980's.[11] Instead, what it seems we need is a way for the Agent θ-role to undergo Existential Closure. Recall that Existential Closure is an operation that takes certain verbs and makes them "intransitive" by binding the argument position of their internal θ-role with an existential operator. From transitive *eat*, for instance, it can create intransitive *eat*.

(345) a. $[\![\text{eat}]\!] = \lambda x\ \lambda\epsilon\ \textsc{patient}(x)(\epsilon) \wedge \textsc{eat}(\epsilon)$ transitive
b. $[\![\text{eat}]\!] = \lambda\epsilon\ \exists x\ \textsc{patient}(x)(\epsilon) \wedge \textsc{eat}(\epsilon)$ intransitive

We noted at the time that this operation must be limited to just internal θ-roles. We don't want to create intransitive verbs with this process by letting it apply to external θ-roles. If that were possible, we would expect to find transitive versions of *melt* with their Agent θ-role suppressed. These verbs would appear in sentences like (346).

[11] Baker, Mark, Kyle Johnson, and Ian Roberts. 1989. Passive arguments raised. ***Linguistic Inquiry*** 20:219–251

(346) * It melted the ice.
NB: *it* is an expletive.

These are, in fact, just the verbs that Burzio's Generalization claims are missing.

One way this goal could be achieved using the present system is to rely on the difference between the denotation of verbs that have θ-roles and the denotation of external θ-roles that is offered in (321).

(321) $[\![\text{agent}]\!] = \lambda P.\lambda x.\lambda \epsilon\ \textsc{agent}(x)(\epsilon) \wedge P(\epsilon)$, where
a. P is a a function whose domain is eventualities, x is a thing, and ϵ is an eventuality;
b. P is VP.

External θ-roles, on this view, are functions whose domains are the meanings that VPs have—that is, external θ-roles have VPs as their objects—while internal θ-roles make the meanings of the verbs containing them functions whose domains are things—that is, verbs take DPs as objects. If this is the right denotation for external θ-roles, then Existential Closure won't apply to them because they are not of the right semantic type. Let's adopt this account of how Existential Closure is restricted to internal θ-roles.

Here, then, are desiderata for a good theory of Passive.

(347) a. The Passive makes the external θ-role susceptible to Existential Closure
b. The Passive prevents the external θ-role from assigning accusative Case to the complement of VP.

I don't know of an account of the Passive that meets these desiderata. I'll make a proposal here, however, that is not too distant from some of the proposals one can find in the literature. What I propose is that the Passive morpheme attaches to external θ-roles—things of morphosyntactic type "v"—and changes the way that the external θ-role it attaches to combines with its arguments. I'll express this idea under the assumption that external θ-roles have denotations like that illustrated by (321). Here's the denotation for the passive morpheme, PASS.

So far as I know, there is no proposal in the literature about passive like this one. What I have done here is try to derive a common description of what the passive does in the framework that has been developed in this chapter.

(348) $[\![\textsc{pass}]\!] = \lambda\mathscr{P}\ \lambda y\ \lambda Q\ \lambda\epsilon\ \mathscr{P}(Q(\epsilon))(y)(\epsilon)$, where
$\mathscr{P}$ are functions whose domain is a function whose domain is things, y is things, and Q is a function whose domain is things.

How this formula applies to the denotations of external θ-roles is somewhat complex, and I won't step through it here. Instead, I'll indicate the change it creates with (349).

(349) Passive (PASS)
If $[\![\theta\text{-role}]\!] = \lambda P\ \lambda x\ \lambda\epsilon\ \theta\text{-ROLE}(x)(\epsilon) \wedge P(\epsilon)$=T, where
θ-ROLE c-selects DP
then
$[\![\text{PASS}(\theta\text{-role})]\!] = \lambda x\ \lambda P\ \lambda\epsilon\ \theta\text{-ROLE}(x)(\epsilon) \wedge P(\epsilon)$=T, where
θ-ROLE c-selects DP

From this point forward, I will not include in the denotations what the domain of the functions are. Instead, I will adopt the following conventions. Lower case letters (x, y,…) will indicate variables over things. Upper case letters (P, Q,…) will indicate variables over functions whose domain is the set of things or eventualities. And ϵ will indicate a variable over eventualities. I will also suppress the c-selection information when I think it is irrelevant.

What the Passive does it takes a θ-role whose first argument is a VP and whose second argument is the θ-marked subject, and creates a new θ-role that takes the θ-marked subject argument first and the VP argument second. The θ-roles that have this kind of meaning are just the external θ-roles.

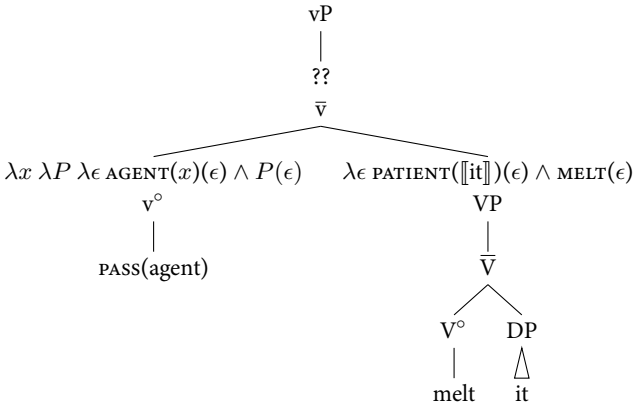
If Passive in general is a process that applies to external θ-roles, then we have an explanation for the fact that Passives do not seem to be able to apply to unaccusative verbs in those languages in which the Passive can apply to intransitive verbs. Passive doesn't seem to be able to apply to intransitive verbs in English, but in many languages this is not the case. If Passive is an operation that applies to external θ-roles, and not the verbs that those θ-roles combine with, this is explained. Unaccusative verbs are just those verbs that do not find themselves in a syntax with external θ-roles. This, then, is a fact which supports the view that the Passive is sensitive to the presence of external θ-roles, and this is built directly into (349).

The Passive makes an external θ-role become semantically the kind of thing that Existential Closure can apply to. So if we allow Existential Closure to apply to v° as well as V°, then we have an explanation for why the Passive allows Existential Closure to apply to external θ-roles. This is shown for the Agent θ-role in (350).

(350) a. $[\![\text{agent}]\!] = \lambda P\ \lambda x\ \lambda\epsilon\ \text{AGENT}(x)(\epsilon) \wedge P(\epsilon)$=T
b. $[\![\text{PASS(agent)}]\!] = \lambda x\ \lambda P\ \lambda\epsilon\ \text{AGENT}(x)(\epsilon) \wedge P(\epsilon)$=T.
c. $[\![\text{EXIST CLOSURE(PASS(agent))}]\!] = \lambda P\ \lambda\epsilon\ \exists x\ \text{AGENT}(x)(\epsilon) \wedge P(\epsilon)$=T.

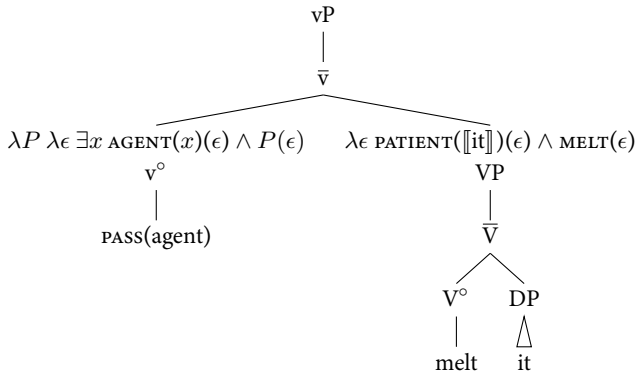
Consider, now, how an external θ-role fits into a D-structure when a passive morpheme has combined with it.

(351)



Our semantic composition rules do not allow v° and VP to semantically combine – neither is a function that can take the other as its argument. This could be remedied if we applied Existential Closure to the external θ-role, as in (352).

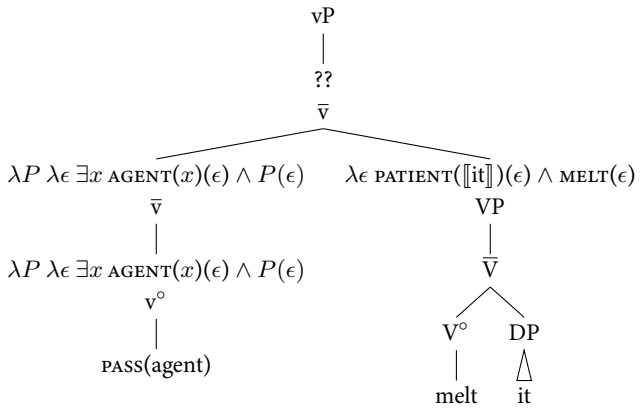
(352)



But this structure also violates the c-selection restrictions of "PASS(agent)." The AGENT θ-role c-selects a DP, in both its active and passive guises. The conditions in Lexical Insertion prevent "PASS(agent)" from being inserted in the position it occupies in (351).

A structure that Lexical Insertion does allow a Existentially closed, passivized θ-role to occupy, however, is (353).

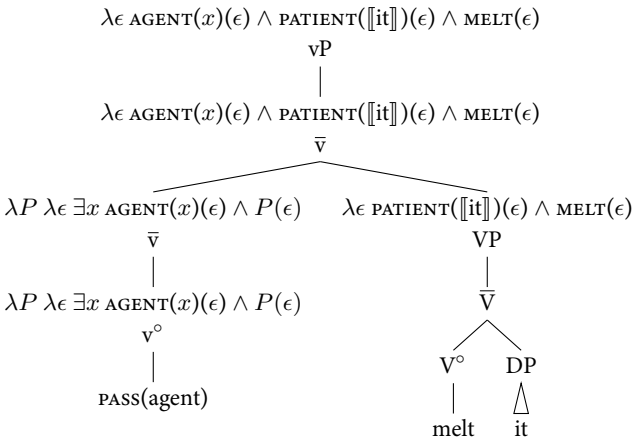
(353)



Recall that the c-selection requirements of a morpheme can be satisfied if the X° position it is inserted in has no sister. This happens, however, only when Existential Closure occurs, and that is required in (353) as well for otherwise the highest $\overline{\text{V}}$ will get no denotation. Hence, applying Existential Closure to a passivized external θ-role produces VPs like (354).

It may be useful to return to the discussion of (325) on page 187.

(354)



This is the right outcome.

Note that in (354) v° does not c-command *it*. This, I suggest, is why there is no accusative Case available to this DP in the Passive. As a consequence, A Movement must apply, causing the object DP to relocate into a Case marked position.

We have so far only looked at English Passive sentences in which the external argument is not expressed, but it is also possible to express the external argument. This is shown in sentences like (355).

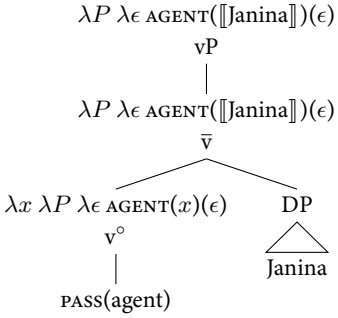
When a Passive construction includes an expression that contains the external θ-role bearer, as the ***by***-phrase does in English, it is called a "long passive." The ones with no expression of the external θ-role bearer are called "short passives." Not all languages have both.

(355) a. The ice was melted by Janina.
b. We were bothered by the news.
c. It was seen by the boy.
d. This is entailed by that.

As this series indicates, the θ-role that the active versions of these verbs sanction on the subject argument are borne by the DP inside the *by*-phrase. This correspondence is completely regular in English: the external θ-role of an active sentence is the same that is assigned to the DP in a *by*-phrase in the corresponding Passive sentence. Chris Collins[12] has suggested that this is because the external θ-role, v°, is pronounced in the Passive as *by*. Let's adopt this idea. In these sentences, then, the external θ-role does not undergo Existential Closure. Instead it combines with the DP argument that it c-selects. This is illustrated by (356).

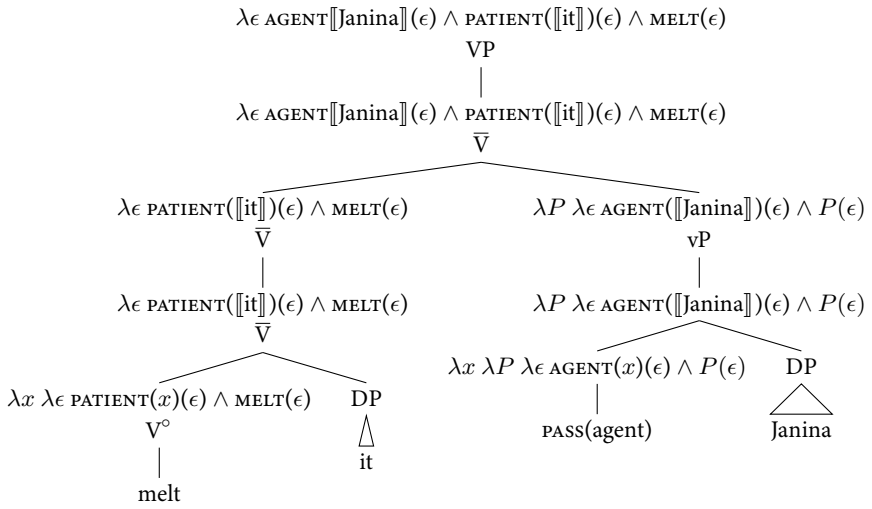
[12] Collins, Chris. 2005. A smuggling approach to the passive in English. ***Syntax*** 8:81–120

(356)



In this scenario, v° cannot embed the VP it combines with—here, the v° combines with the DP it c-selects. Another way of parsing a sentence with (356) will be have to be found. One parse that works arises if the vP is put inside the projection of the verb; (357) illustrates.

(357)



In (357), v° c-commands *Janina*, and it is to this DP that v° assigns accusative Case. Here too, then, the DP *it* does not get accusative Case in its D-structure position and it must move to satisfy the Case filter.

This sketch of how the Passive works captures how the external θ-role survives in the passive construction, and meets the desiderata of a theory of Passive laid out in (347). It leaves unexplained why the Passive v° is pronounced as *by* when it combines with a DP, and not apparently pronounced at all otherwise. I have nothing insightful to offer for this problem.

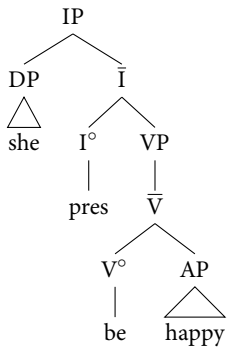
With the adoption of a syntax that puts external θ-roles outside the VP their meaning is part of, we capture a variety of facts. With suitable adjustments to the denotations of verbs and θ-roles, we can improve the Projection Principle. It is no longer a strange disjunction of conditions – one for external θ-roles and another for internal θ-roles. Instead, it is a simple condition that applies to all θ-roles alike. We also have a way of modeling Burzio's Generalization: external θ-roles assign accusative Case, not verbs. And it is possible to design an account of Passive that captures some of its puzzles.

But putting the external θ-role outside of the VP it is construed with raises problems as well. As noted above, it is inconsistent with our conditions on accusative Case assignment. And it raises the salient puzzle of how to connect the external θ-role with the verb that heads its complement. Which external θ-role is found seems to depend on the selection of the verb. Indeed, whether an external θ-role is found also seems to depend on the choice of verb. By syntactically separating the verb from its external θ-role, we can no longer explain this dependence by assuming that the external θ-role is part of the verb that is inserted in V° position. If this is the correct direction, we need to find

a way for the external θ-role in v° position to communicate morphologically with the verb in V° position.

We have already encountered a problem of a similar type. When a finite IP is headed by present or past tense, as in (358), this information must communicate morphologically with the verb inserted in the neighboring V° position.

(358)

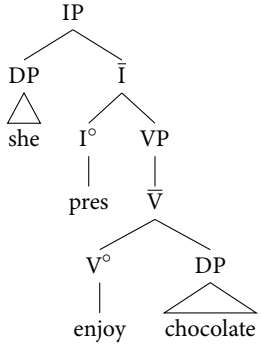


We need a theory of the morphology/syntax interface that achieves those goals. This is the job of the next chapter.

4
Features

One problem our grammar suffers from presently has to do with the fact that the heads of sentences are often not words. Many finite clauses are headed by bound morphology, as in (1).

(1)



We reached this conclusion because it is morphology of this sort—agreement/tense morphology—that is in the one-to-one relation with the family of strings that we call IP. What we need to understand is how it is that this bound morpheme manages to find itself expressed on the verb that follows.

There is some evidence that in certain cases of this sort, the verb moves to the position our rules put the inflectional morphology. The rule responsible is, therefore, sometimes called Verb Movement. Its classic description is found in Chomsky (1957).

4.1 The Classic account of Verbal Inflection

Chomsky uses a type of reasoning that I will dub a "Correlation Argument," because it's based on the correlation that holds between a verb's syntactic position and its inflection. The generalization that accounts like his hope to derive is the dependence between a verb's syntactic position on its inflectional class. This dependence can be seen from the relative positions of auxiliary verbs in English finite clauses. An auxiliary verb cannot follow *not, so*, *too* (which I'll call "polarity terms") if it is inflected with the tense/subject-agreement morpheme. We find paradigms like those in (2).

(2) a. Andre has not been eating.
b. Andre is not eating.
c. * Andre not has been eating.
d. * Andre not is eating.
e. Andre has too/so been eating.
f. Andre is too/so eating.
g. * Andre too/so has been eating.
h. * Andre too/so is eating.

But when an auxiliary verb is not inflected with tense/subject-agreement morphology it can follow these terms.

(3) a. Andre must not have eaten.
b. Andre must not be eating.
c. Andre will too/so have eaten.
d. Andre will too/so be eating.

In these contexts we find that the polarity item must be found immediately before the auxiliary verb; it cannot precede the contents of I°, the modal, for instance.

(4) a. * Andre not must have eaten.
b. * Andre not must be eating.
c. * Andre must have not eaten.
d. * Andre must be not eating.
e. * Andre too/so will have eaten.
f. * Andre too/so will be eating.
g. * Andre will have too/so eaten.
h. * Andre will be too/so eating.

Some English speakers may judge (4c,d) grammatical. For these speakers, I suspect that *not* is being used differently than it is in situations where *not* appears between the modal and auxiliary, as in (5).

(5) a. Andre will not have eaten.
b. Andre will not be eating.

In (5), *not* negates the entire sentence. The speaker of these sentences means to deny the positive versions of these sentences, i.e. the sentences in (6).

(6) a. Andre will have eaten.
b. Andre will be eating.

By contrast, the sentences in (2c) and (2d) are positive assertions. The speaker of these sentences means to assert that Andre must have done something—namely not eaten—or must be doing something—namely not eating. This

difference between (2c,d) and (5) can be brought out by considering how they combine with so-called question "tags." These tags, illustrated in (7), have the opposite polarity as the clause they are appended to.

(7) a. Andre will eat, won't he?
b. Andre will not eat, will he?

If the polarity of the tag is the same as the clause it's appended to, the result has a rhetorical flavor:

(8) a. Andre will eat, will he?
b. Andre will not eat, won't he?

When *not* stands between the modal and a following auxiliary, adding a positive question tag gives a question of the sort illustrated by (7), as in (9).

(9) Andre will not have eaten, will he (have)?

But when *not* follows both modal and auxiliary, adding a positive question tag yields the rhetorical sorts of questions in (8), as in (10).

(10) Andre will have not eaten, will he (have)?

On the other hand, when the auxiliary is finite, and so in the I° position, a *not* that follows it behaves the same as the one that follows a modal:

(11) Andre has not eaten, has he?

To the extent that there are real judgments here—it's not easy to be clear about what the difference between the "real" and "rhetorical" questions is—it suggests that we do not have the same use of *not* when it stands before and after a non-finite auxiliary. If these uses of *not* correspond to the syntactic position it occupies, then we want to conclude that *not* is in the same position in (12), but a different position in (13).

(12) a. Andre will not have eaten.
b. Andre has not eaten.

(13) Andre will have not eaten.

These data, then, lead us to conclude that polarity items lie between I° and VP. And from this we can conclude from the contrast between (2) and (3) that verbs are in I° when they bear the inflection that resides there. There is, in other words, a correlation between a verb's inflectional class and its syntactic position.

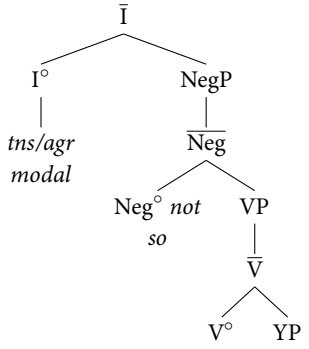
That auxiliary verbs have a different position when they are inflected is confirmed by the action of rules that affect VP. These rules seem to necessarily strand the verb inflected for agr/tense. This is indicated by the following examples of VP deletion and VP topicalization.

(14) a. Aynat is eating pickles because Emily is △.
b. Aynat should be eating pickles because Emily should △.
c. * Aynat is eating pickles because Emily △.
d. I claimed that Aynat is eating pickles, and [$_{VP}$ eating pickles] she is.
e. I claimed that Aynat has to be eating pickles and [$_{VP}$ be eating pickles] she has to.
f. * I claimed that Aynat is eating pickles, and [is eating pickles], she.

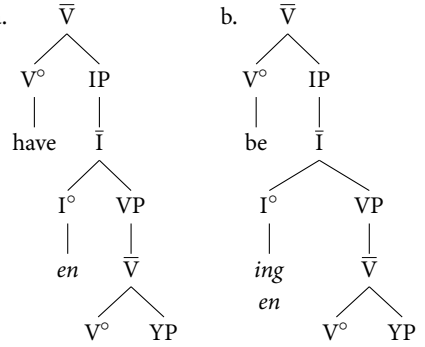
If VP Ellipsis is capable of eliding any VP, then the ungrammaticality of (14c) indicates that the finite form of *be* is not within a VP. Similarly, (14f) indicates that the movement rule responsible for moving a VP at the beginning of a clause cannot relocate a finite form of *be*. This would follow if finite forms of *be* are not within the VP that is being moved in these constructions.

A Verb Movement account of these phenomena lets phrase structure rules fix the location of the inflectional morphemes involved, and moves the appropriate verb to this location by way of a transformational rule. Thus, simple finite clauses are assigned a D-Structure representation like that in (15), and clauses with a complex tense (i.e., an auxiliary verb), have D-Structure representations like those in (16).

(15)



(16) a.



b.

In these parses I've decided to position polarity items as heads that embed the following VP. Moreover, I've renamed these items, and the phrase they head, "Negation." This renaming is common in the literature, as it is typical

In the previous chapter we saw that there is evidence that verbs which have an external θ-role are accompanied with a "v°" that expresses that θ-role. One of our tasks in this chapter is to understand how to capture the correlation between the θ-role in v° and the verb that follows. We take up this task in 4.6. Until then, I will ignore the possible presence of v° in the parses, and revert to parses in which VPs involve V° and not v°.

that only the negative item among the polarity terms is talked about. The representation in which Neg° embeds VP is argued for in Pollock (1989), and we will consider his reasons soon. There are other possibilities which are consistent with what we've seen so far, however. One is to treat polarity items as heading a phrase that is a left-linearized daughter of the highest $\overline{\text{V}}$, like an adverbial modifier can be.

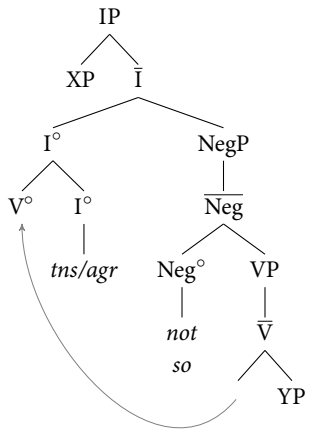
The representations in (16) claims that auxiliary verbs select IPs as their complements. This gives us a way of capturing the fact that auxiliary verbs determine the inflection that is borne by the VPs they embed. The auxiliary verb *have*, for instance, determines that the VP it embeds will be inflected as what's called a "perfect participle" in traditional grammars. These forms frequently look very much like the simple past form of a verb: *heard*, *liked*, and *dreamed* can be either perfect participles or past tense. But not all perfect forms look like simple pasts, for example: *taken* is the perfect of *take*, and *took* is the past; *eaten* is the perfect, and *ate* the past. If we imagine that auxiliary *have* selects an Inflectional Phrase headed by the morphology that is responsible for inflecting a verb as a perfect participle, we can derive that the perfect form of a verb follows auxiliary *have*. Similarly, there is an auxiliary verb *be* that causes the VP it is embeds to be headed by a verb in the so-called "progressive" or "present participle" inflectional form, for example: *eating*, *running*, etc. Another auxiliary verb *be* embeds the passive participle, which we've encountered in our discussion of the passive rule. For all three of these auxiliary verbs, we might speculate that they select an Inflectional Phrase headed by the morphology responsible for inflecting the embedded verb.

The Verb Movement rule could be formulated with (17).

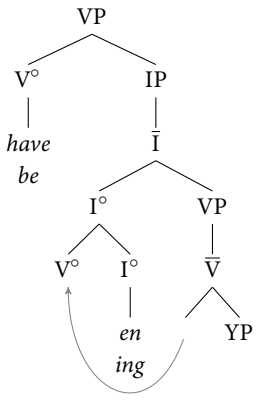
(17) Verb Movement
   a. Remove a verb, α, from the position it occupies and
   b. Adjoin α to I°.

This will produce from the D-structures in (15) and (16) representations like those in (18).

(18) a.



b.



This gives a straightforward account of the correlation described above. Verbs stand in the positions determined by the inflections they bear because they are moved to the positions that the phrase structure rules assign to these affixes.

To force Verb Movement in these contexts, we could posit a requirement that an affix must share an X° with its stem at S-structure. Lasnik (1981), where this idea is proposed, calls this condition the "stray affix filter."

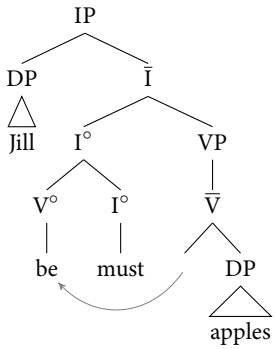
(19) Stray Affix Filter
An affix must co-exist with its stem under a common X° at S-Structure.

To prevent Verb Movement in contexts where the I° contains a word, such as a modal, we could strengthen the Stray Affix Filter to (20).

(20) The Word Criterion
A maximal X° must be parsed as one complete word at S-structure. α a maximal X° only if it is immediately dominated by $\overline{\text{X}}$.

The Word Criterion, then, will force Verb Movement when I° contains something less than a full word, and block Verb Movement in cases such as (21), where the result would put into I° more than one word.

(21) * Jill be must eating apples.



What this case shows us is that there is more to providing a syntactic representation with words than populating the X° positions in a D-structure. It looks like words can be formed by derivations. In the ase of auxiliary verbs in English, for instance, it is Verb Movement that brings the two morphological pieces together. What is needed, then, is to distinguish the morphemes that are inserted at D-structure from the actual words that make up the string that derivations result in. We will continue to let Lexical Insertion be a process that forms D-structures, but we will understand the things that Lexical Insertion inserts to include the pieces that make up a word. Let's call the things that Lexical Insertion matches to X° positions *morphemes*.

(22) Lexical Insertion
Insert morphemes into X° position.

Lexical Insertion, then, can populate a D-structure with the morphemes that a derivation can assemble into a word. In addition to Lexical Insertion, we'll

need a process that matches words to the positions which the syntax has assembled the morphemes in. This process is sometimes called *vocabulary insertion*.

This method of letting the syntax control how sub-word parts are assembled is the image that Harris (1946) had and is associated now with what is known as "Distributed Morphology." See Halle and Marantz (1993) and Marantz (1997).

(23) Vocabulary Insertion
Insert a vocabulary item into an X° containing the morphemes it expresses.

We'll start with the idea that Vocabulary Insertion happens at S-structure. We can view the Word Criterion as a result of the conditions on Vocabulary Insertion. It will be responsible for inserting an inflected auxiliary verb into an I° position that carries both the auxiliary and the inflection that makes it tensed.

These are the cases, then, where it appears that the morphology in I° and the verb combine by way of movement. They all involve the auxiliary verbs *have* and *be*. But this process doesn't seem to be involved for English "main" verbs. In these cases, the verbs do not occupy the positions determined by the inflections they bear, but remain instead in the location given to main verbs by the phrase structure rules. Examples like (24) are ungrammatical.

(24) *Andre likes not/too/so apples.

And the processes that indicate whether a verb is in its VP, i.e., VP ellipsis and VP fronting, indicate that finite main verbs are indeed positioned within the VPs they head.

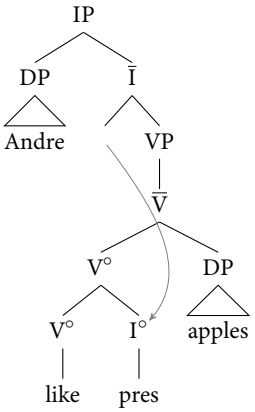
(25) a. * Andre put apples on the table because Jill put △.
b. * I said that Andre put apples on the table and [$_{VP}$ apples on the table] Andre put.

The process that brings inflection together with the verb on which it is expressed must have a different outcome when main verbs are involved. The challenge is to find a way of conceiving of this process so that both kinds of outcomes are possible. There have been a number of attempts to capture this flexibility, none of them entirely successful.

One account, found in Chomsky (1957), posits a transformational rule that moves the inflection down onto the verbal root when main verbs are involved. A sentence like (26), for example, would enter into the derivation in (27).

(26) Andre likes apples.

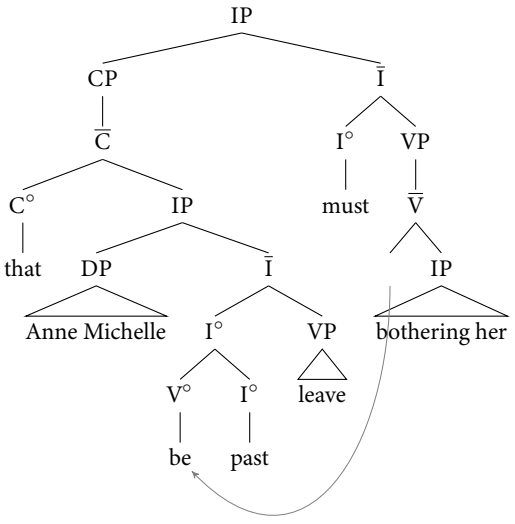
(27)



On this view, then, verbs and their inflections always combine by way of a movement rule. The difference between auxiliary and main verbs concerns what moves: the verb, in cases involving an auxiliary, and the inflection, in all other cases.

A problem with this proposal is that it requires movement transformations to be able to move things downwards. We have seen that for other movement transformations this does not seem to be possible. Outside of these contexts where main verbs get inflected without moving, the same generalization holds with respect to the rule that moves verbs around. It looks like it is worth preserving the generalization that movement rules do not move things to downwards. This can be seen, for example, in (28).

(28)



= That Anne Michelle was leave must bothering her.

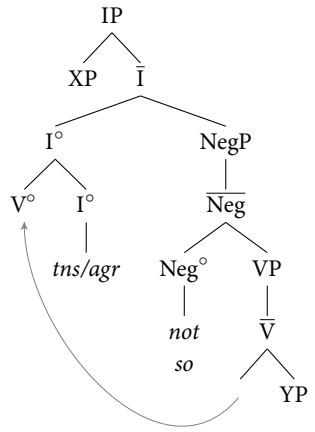
In (28), a verb, *be*, has moved into an I° that is not "higher" than where that *be* starts out. This is the kind of movement that we saw was blocked for the other examples of movement: Argument Movement, Wh Movement, and Complex NP Shift.

To block downwards movement, we formulated the constraint in (29).

(29) Upwards
For α to move to position β, β must c-command α.

This condition will correctly block (28), but it would also block Chomsky's Affix Hopping rule, and this is the problem faced by Chomsky's proposal. Interestingly, it would also block good examples of Verb Movement. This can be seen by looking at the structure formed by Verb Movement in (18a), repeated here.

(18)



If we are to use Upwards to prevent (28), then there is a problem with the way Upwards is conceived. There are several ways of capturing the idea that Verb Movement shares with the other movement rules the property that it is always "upwards." Some of those avenues involve preserving the formulation of Upwards in (29), and changing either how c-command is defined or changing the structures that result from Head Movement. None of these revisions, however, make Affix Hopping possible, and for this reason the proposal in Chomsky (1957) has been largely abandoned. In these notes, I will preserve the definition of c-command and also keep the (admittedly somewhat *ad hoc*) decision about the structure that Verb Movement creates. To bring Verb Movement in line with Upwards, I shall redefine Upwards.

See Matushansky 2006 for some discussion of these alternatives.

(30) Upwards
If α moves from position T to position P, then the sister of P must have a projection that dominates T.

By "projection," is meant the relation between heads and their phrases that the $\overline{X}$ Skeleton defines. $\overline{X}$ is a projection of its head X°, and XP is a projection of them both.

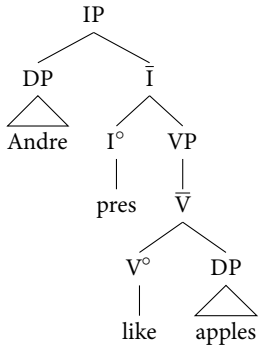
In (18), the sister of the moved V° is I°. The projections of I° include the I° immediately dominating it, the $\bar{I}$ dominating that I° and, finally, IP. Both IP and $\bar{I}$ dominate the position that this moved V° came from. For this reason, Verb Movement in this scenario satisfies the formulation of Upwards in (30). By contrast, (28) violates Upwards. There is no projection of the sister to *be* in the position it occupies after moving that dominates the position *be*

moved from. The definition of Upwards in (30) applies to the other cases of movement—Argument Movement, Complex NP Shift, and Wh Movement—we've seen in the same way that the earlier formulation of Upwards did. It also prevents Affix Hopping. We cannot use Affix Hopping to bring tense inflection and main verbs together in English.

A second account lets the mechanism that brings a main verb together with its inflection be a modification of how "lexical insertion" works. Presently, our formulation of lexical insertion matches morphemes to a single X° positions at D-structure by inserting them into the X° positions. One word cannot, consequently, span more than one terminal position in a phrase marker. Jonathan Bobaljik[1] argues that this should be weakened so that a lexical item can be put into a position that does not map directly onto the one position that a terminal position in a phrase marker occupies. In particular, he proposes to let an inflected form of a main verb be inserted into a position occupied by two X°s, just in case those two X°s are linearly adjacent. This account requires a different architecture than the one we have adopted, since "adjacency" is not a relation that can be evaluated at D-structure. It is only after the linearization algorithm has interpreted S-structures that it will be known which positions are adjacent to another. An implementation of Bobaljik's proposal, then, requires some changes to our present architecture. Under his proposal, a representation like (31) could be directly mapped onto the string *Andre likes apples*.

[1] Bobaljik, Jonathan. 1994. What does adjacency do? In ***The morphology-syntax connection***, ed. Heidi Harley and Colin Phillips, 1–32. MIT; and Bobaljik, Jonathan. 1995. Morphosyntax: The syntax of verbal inflection. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge

(31)



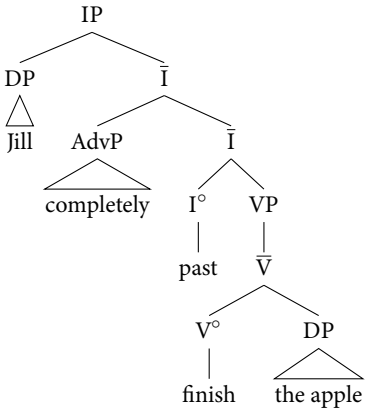
This proposal has the benefit of explaining why finite sentences containing polarity items in English are ungrammatical if they do not contain an auxiliary verb.

(32) a. * Andre not likes apples.
b. * Jill not ran down the street.
c. * Winnie so read that encyclopedia.
d. * Bart too ran into the street.

Because polarity terms are linearized between I° and the verb that follows, it will prevent an inflectional morpheme residing in I° from being adjacent to that verb and block lexical insertion from rendering them as a single inflected verb. Indeed, this proposal predicts that a finite clause with no auxiliary verbs

in it should not allow any material to stand between I° and the main verb. The only grammatical parse for an example like (33) below should be one in which the adverb is to the left of I°, as indicated.

(33) Jill completely finished the apple.



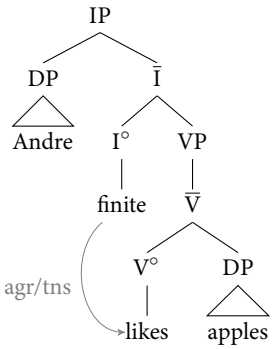
This is not an obviously correct prediction, however. Contrasts like those in (34) indicate that *completely* must be placed between I° and the following VP.

(34) a. Jill will completely finish the apple.
b. * Jill completely will finish the apple.

This account, then, has difficulties making sense of data like (33). Unlike Chomsky's proposal, it does address the problem of why the mechanism that inflects main verbs is sensitive to the presence of polarity items; but it doesn't explain how these items differ from other material that appears to be able to stand between the main verb and finite inflection.

The final idea we will consider makes the relationship between an I° that is associated with inflectional morphology and the verb that expresses that morphology to be akin to the relationship we have posited between a DP and the term that assigns that DP Case. We can think of I° as assigning the relevant inflectional morphology to the verb that follows, as in (35).

(35)



Chomsky (1995), where this proposal can be found, calls the assignment relationship between an inflected verb and the I° where that inflection is

determined: AGREE. The precise formulation of AGREE will be taken up later; but it should be clear from (35) that it doesn't match perfectly the condition we arrived at for Case assignment in the previous chapter. While the finite I° does c-command the verb it assigns tense/agreement inflection to in (35), it is not klose to that verb: there is more than one phrase that dominates the verb but not I°. This would block Case assignment. One desirable consequence of the formulation of AGREE is to make it account for the inability of a polarity item to stand between I° and a main verb. That would be a natural place for this account to attempt an explanation for this fact.

If finite I° assigns inflectional morphology to a main verb that follows, then there are a couple ways of characterizing the different outcome with auxiliary verbs. One possibility is that auxiliary verbs are inflected in the same way, and that there is an independent reason that they must move to the I° that assigns them their inflection. Another possibility is that auxiliary verbs are prevented from getting their inflectional morphology by way of AGREE, and instead make recourse to Verb Movement. Hopefully, a better understanding of what makes an auxiliary verb different from a main verb will help us understand how to express the difference in how their inflection influences their surface position. We can learn something about this question by considering how these processes arise in other languages. We'll look at other languages in the sections that follow, and then return to the question of how to give a picture of the main verb/auxiliary verb distinction in English.

We should also note here the special status of the "auxiliary" *do*. This provides a way of speaking the bound tense morphemes in English alongside whatever mechanism allows these morphemes to be pronounced on the main verb that follows. In general, we find *do* in contexts where the tense morphemes cannot otherwise find a way of being expressed. For instance, *do* appears in contexts where there is no auxiliary verb, but there is the presence of *not*, which blocks the mechanism by which main verbs inflect.

There is a main verb ***do*** in English as well. It is found in sentences like ***Mary will do this***. The thing I'm calling an auxiliary ***do*** doesn't have the distribution of a verb, so it is not the same kind of thing that the auxiliary verbs ***have*** and ***be*** are. The auxiliary ***do***, for instance, is not found in infinitival clauses, like ***have*** or ***be*** are. There are contrasts between examples like ****I will try to do eat nattoo.*** and ***I will try to have eaten nattoo.*** It appears that ***do*** only shows up in I°.

(36) a. Smith did not eat nattoo.
b. Smith did so eat nattoo.
c. Smith did too eat nattoo.

If *not* isn't present, however, *do* cannot be employed:

These examples are grammatical if there is focus on ***did***. We might imagine that this is the case because to focus-mark I° requires a way of pronouncing I° independently of the verb.

(37) a. * Smith did eat nattoo.
b. * Smith did eat nattoo.
c. * Smith did eat nattoo.

And *do* is not present when there is an auxiliary verb that can move to form a word with I°.

(38) a. * Smith did have eaten nattoo.
b. * Smith did be eating nattoo.

A way of describing this distribution is to say that *do* appears when no other way of pronouncing the bound morphemes in I° are possible. Finding a way of expressing that condition isn't possible in the syntax we presently have. There are proposals in the literature about how to achieve that, but to look at these now would take us too far off track. I will ignore the distribution of auxiliary verb *do* in these lectures.

We will, however, look at how to model the competing ways in which auxiliary and main verbs in English inflect. That is best done, as mentioned above, by looking at how these systems of inflecting verbs vary cross-linguistically. But first, there is one last piece to the picture of English verb placement that we should see.

4.2 Movement to C°

The rules we've looked at that relate verbs to I° can be embellished with another that plays a role in forming questions. One sort of question involves relocating the finite verb to the beginning of the sentence, as in (39).

(39) a. Have you eaten pickles?
b. Should you eat pickles?

These are called Yes/No questions. Another sort of question which involves a relocation of the finite verb seeks more specific information and is called a Wh-Question. In these questions, as we saw in the previous chapter, there is also movement of a phrase into Specifier of CP. Some examples are in (40).

(40) a. Which pickles have you eaten?
b. What should you eat?

In both of these types of questions, the contents of I° has moved to some sentence-initial position. As a consequence of this rule's targeting I°, only material which can stand in I° can move. Therefore, main verbs in English do not undergo this process, and instead the nearly meaningless auxiliary verb *do* is used.

(41) a. * Eat you pickles?
* Which pickles eat you?
b. Do you eat pickles?
Which pickles do you eat?

Where does I° move to? The common answer to this question, based on work of the Dutch linguists Jan Koster[2] and Hans den Besten,[3] is that it is C°. The reason for this speculation is that the rule involved here seems to be in complementary distribution with complementizers. That is, its effects are blocked in those situations where we believe that complementizers are present in this position. Thus, there is a distinction between (42a) and (42b).

[2] Koster, Jan. 1975. Dutch as an SOV language. ***Linguistic Analysis*** 1:111–136

[3] den Besten, Hans. 1983. On the interaction of root transformations and lexical deletive rules. In ***On the formal syntax of the Westgermania***, ed. W. Abraham. Amsterdam, The Netherlands: John Benjamins Publishing Company

(42) a. Have you eaten?
b. * I remember (that) have you eaten.

The same contrast is found in the position of finite auxiliaries in wh-questions.

(43) a. Which pickles have you eaten?
b. * I remember that which pickles have you eaten.

Instead, Yes/No questions that are embedded are marked with a special complementizer: *whether*. And wh-questions in embedded contexts involve no evidence of a C° at all.

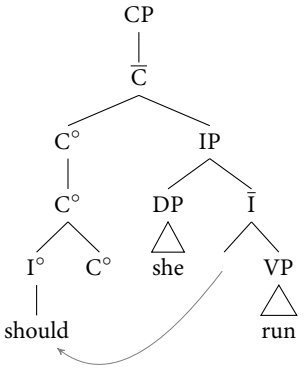
Under this view, the rule involved in these questions, then, might be formulated as in (44).

(44) I-to-C Movement
a. Remove an I°, α, from the position it occupies and
b. Adjoin α to C°.

This will produce derivations like (45).

Note that this model of questions means that every root sentence that is a question in English must be a CP. By contrast, non-question root sentences in English are IPs.

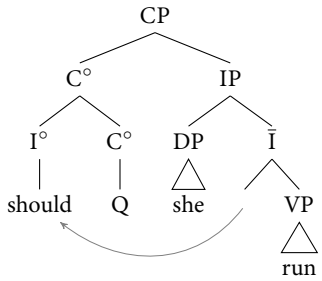
(45)



The Word Criterion will correctly block movement of I° to C° in those cases where C° is filled with a complementizer, assuming that a complementizer and the modal or inflected verb in I° cannot together form one word.

Okay, so this is, roughly, a sketch of the rules that go into making up our knowledge of this fragment of English syntax. We've got three rules, one that moves auxiliary verbs to I°, another than moves I° to C° and a third that inflects main verbs. The first and last are driven by the Stray Affix Filter, or some parallel condition on the view that inflection is assigned under AGREE, and the second arises in questions. We can make the movement of I° to C° obligatory with the Stray Affix Filter if we imagine that there is an inflectional morpheme in the C° position of questions as well. This morpheme has no phonology associated with it, apparently, but expresses itself instead by forcing movement of I°. We can think of this morpheme as an affix on words that belong to the part of speech that I° names, just as tense/agreement inflection is an affix on the words that belong to the part of speech that V° names. I'll represent this abstract question affix with "Q."

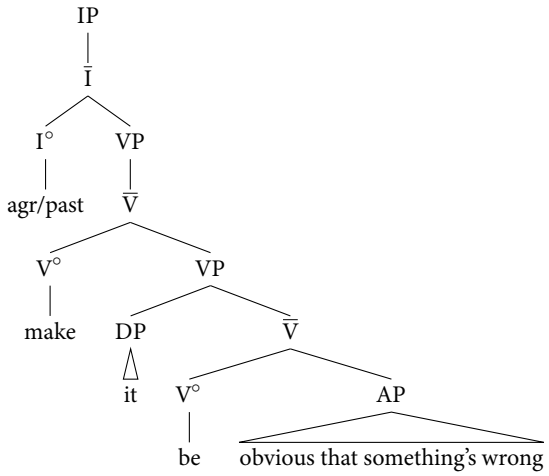
(46)



With this amendment, both I°-to-C° movement and Verb Movement are controlled by the Word Criterion, which prevents I°, C° or V° from combining whenever they cannot form a word and forces it when there is an affix that needs a stem to make a complete word.

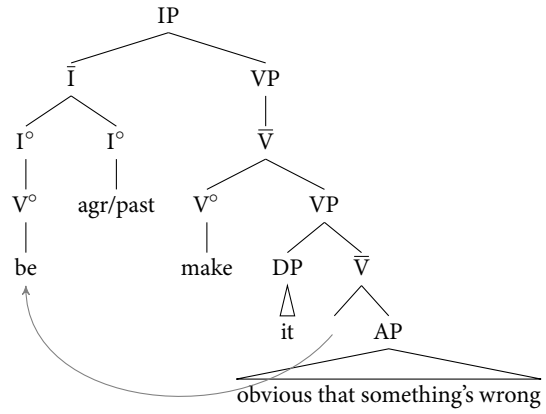
There's one last feature of these rules that we should consider. In all the cases so far examined, the V° that moves to I° is always the one that is closest to it. It appears that this is not just an accident of the examples we have chosen. Instead it looks like it is a necessary feature of the verb movement rule. Consider, for example, how these rules might combine to apply to a representation like (47).

(47)



Suppose that we move *be* to *agr/past* and form thereby the past tense form: *were.*

(48)



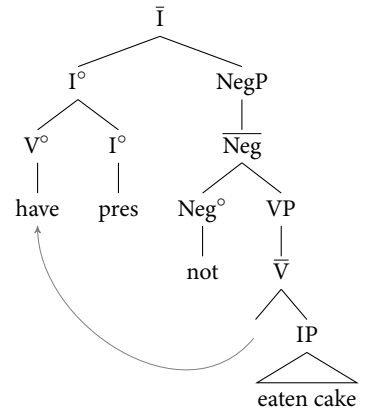
But this is an ungrammatical outcome. Instead, the only grammatical outcome is one in which the past tense morphology is expressed on *make*. We need to find a way of preventing derivations that include the movement in (48).

Lisa Travis[4] suggested that there is a constraint on head movement rules that blocks them from moving past other X°s. She calls this condition the "head movement constraint," and I will formulate it as (49).

(49) The Head Movement Constraint
If X° adjoins to Y°, then there can be no Z° that c-commands X° and is c-commanded by Y°.

If the structure of sentences with *not* in them is correct, then the Head Movement Constraint is violated by cases of verb movement to inflect auxiliary verbs, as in (50).

(50)



This tension between the Head Movement Constraint and this syntax for negation has a payoff, which we will examine later. For the moment, let's bracket this issue and develop a better understanding of the rest of the verb movement system.

[4] Travis, Lisa. 1984. Parameters and effects of word order variation. Doctoral Dissertation, Massachusetts Institute of Technology
Travis formulates it differently, using the notion of government described in chapter 3. For the cases that we will look at, (49) will do.

4.3 Verb Second word order

Let's now consider verb placement phenomena in languages closely related to English. We'll start with examples from the Germanic family. In German we find that the position of the finite verb depends, roughly speaking, on whether the clause it is in is embedded or not. In embedded clauses, the verbs stack up at the end of the sentence in the mirror image of their arrangement in English: the finite verb comes last.

(51) a. …daß Hans das Buch kauft.
…that John the book buys

b. …daß Hans das Buch gekauft hat.
…that John the book bought has

c. …daß Hans das Buch gekauft haben muß.
…that John the book bought have must.

From this we can conclude, perhaps, that VPs are head-final. Further, if there are IPs and a method of combining verbs with the inflection associated with the I° of these IPs that employs Head Movement, they too are head final in German.

Interestingly, we find a different word-order in root, or independent, clauses. Here, the inflected verb no longer comes finally in the series. Instead it appears immediately following the subject. So unlike (51), we find only the word order in (52).

(52) a. Hans kauft das Buch.
John buys the book

b. Hans hat das Buch gekauft.
John has the book bought

c. Hans muß das Buch gekauft haben
John must the book bought have

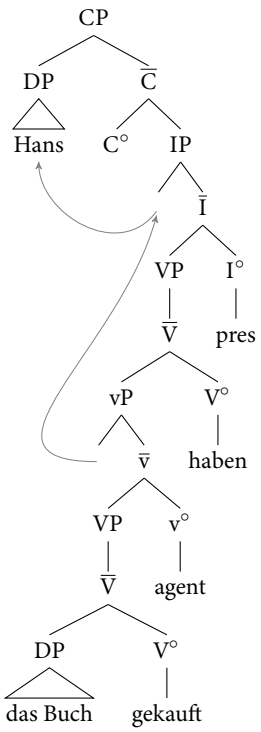
This seems to suggest that there is a movement rule which relocates finite verbs into the position immediately following the subject. Using the logic of the correlation argument, we might imagine that the position where finite inflection is in German immediately follows the subject, and it's into this position that finite verbs are driven in German.

But this would miss the fact that the position of the finite verbs differs for embedded and independent clauses. What we want is some way of forcing verbs to move into the post-subject position in root clauses only. This suggests that it is not the finite distinction that is responsible for verbs' position in root clauses, but something else. Something that distinguishes root from embedded clauses.

We've already seen a similar difference in the grammar of English: recall that I°-to-C° movement is restricted to root clauses in English. Perhaps the verbs are moving through I° into C° in cases like (52), then. This would

credit German and English with two differences. On the one hand there is the difference in headedness that we see most directly in embedded contexts. And then there is something that allows/forces subjects to move past C° in embedded clauses. We might imagine that this second force, whatever it is, is like the process that moves wh-phrases in English into Specifier of CP. Thus, the examples in (52) might get a parse like that in (53). The subject moves into Specifier of IP, to satisfy the EPP and get Case, and then into Specifier of CP.

(53)

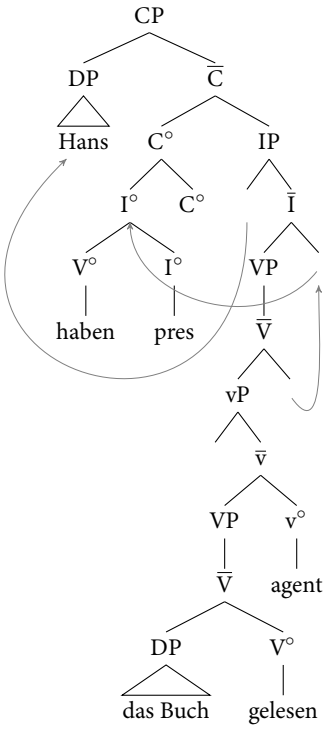


The finite verb follows the subject because it moves through I° into C°, as indicated in (54).

I have put the I° in this parse to the right of the VP it embeds. We will see evidence that C° is linearized so that it is to the left of the IP it embeds, and we have just seen evidence that V° is linearized so that it follows whatever phrase it embeds. German appears to be a language, then, that allows the linearization algorithm to choose different orderings for its heads: CPs (and DPs) are head initial, but VPs are head final. It is not easy to determine how the linearization algorithm linearizes IPs. I have decided, as an initial hypothesis, that I°s are linearized to the right of their complement. We should be alert to the possibility that the other ordering is correct and seek evidence that helps decide between these two.

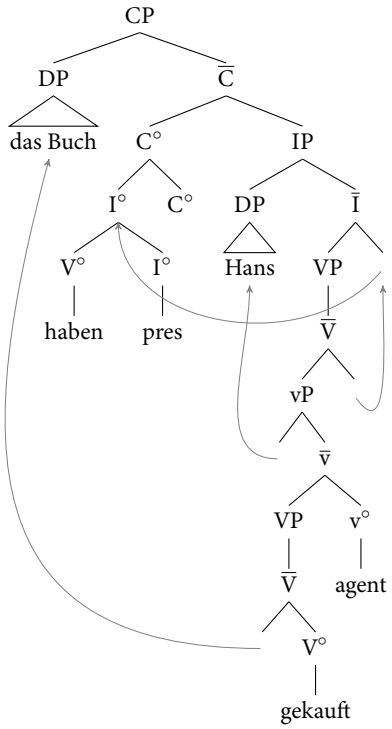
I have also made the complement of the auxiliary verb, ***haben***, a VP rather than an IP headed by the inflection that causes the embedded verb to be a perfect participle. The relationship between auxiliary verbs in German and the VPs they embed is the same as is found in English, so a first guess should be that the syntax of both languages involves making the complement to an auxiliary verb an IP. I will ignore this, hopefully irrelevant, detail in my parses of German.

(54)



This leads to the expectation that we might find other phrases preceding the finite verb; and we do. It is not just the subject that may immediately precede the finite verb in root contexts, any phrase can. When some other phrase comes before the verb, the subject (typically) immediately follows the finite verb and the phrase that shows up at the front is understood to be "topicalized." We can capture these word-orders by letting the finite verb move into C° in all these clauses, but letting the Specifier of CP be filled by any phrase through a movement rule. Thus, alongside (53) we also find (55).

(55)



The generalization about German word-order can be described this way: any phrase may be initial, but exactly one must be. German is sometimes described as having "Verb Second" word-order for this reason. The account we've just sketched of this captures Verb Second by way of a rule that moves the verbs that have moved into finite I° into C°, and by moving anything, but always something, into the single position that exists to the left of C°.

That the verb in these cases has in fact moved into C° is further substantiated by cases where Verb Second word-order is found in embedded clauses. Though there is considerable dialectal variation here, in the standard dialects, Verb Second is possible in embedded clauses just in those cases where a complementizer may go missing. As in English, it is possible in German to unexpress the complementizer when the CP it heads is the complement to a certain class of verbs. The verb *say*, for example, can go without a complementizer—as in (56)—and when it does so in German we find Verb Second word-order, see (57).

(56) Kathryn said (that) Martin has a dime.

(57) a. Er sagt, daß die Kinder diesen Film gesehen haben.
He says that the kids this film seen have
'He says that the kids have seen the film.'

b. Er sagt, diesen Film haben die Kinder gesehen.
he says this film have the kids seen
'He says that, this film, have the kids seen.'

Despite the similarities in verb movement that German and English have, note that one place where they differ is whether main verbs fall under the scope of verb movement. Unlike English, German main verbs can undergo movement. This is seen by their ability to move into C° in main clauses, as in (52a).

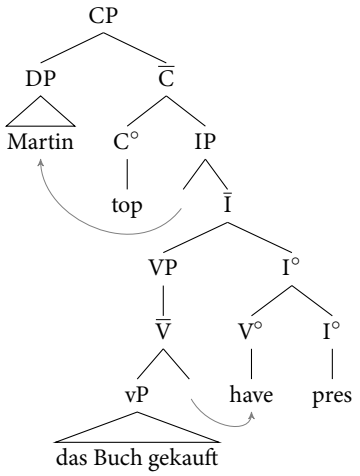
There are a host of mysteries about the grammar of verb placement in German that we will not attempt to solve here. What, for instance, is responsible for forcing movement of something into the Specifier of CP in German root clauses. And why isn't the movement of a phrase into Specifier of CP allowed in embedded clauses, as in (58)?

(58) * Ich sagte [$_{CP}$ das Buch [daß [$_{IP}$ Hans [$_{VP}$ gekauft] hat]]].

Vikner (1995) discusses some of the ideas that have been offered for answers to these questions. One thing we can model, however, is the force that requires I° to move to C° in root declarative clauses in German. We see a very similar thing happening in English questions: I° must move to C° in questions of English, but only when they are root sentences. We captured that fact by imagining that root questions in English were CPs headed by a silent question morpheme. That morpheme, represented by "Q," we can imagine is a suffix on words of category I°, and the Word Criterion will therefore require it to share an X° with its root I° at S-structure. This is what forces movement of I° to C° in English root questions. We can give a parallel model to the obligatory movement of I° to C° in root declarative sentences in German. Imagine that root sentences of German must be CPs, and that when they are declaratives, the head of that CP is a silent suffix that requires an I° stem. I'll call this silent suffix "top."On this view, the structure in (59) violates the Word Criterion, because its C° is not occupied by a complete word.

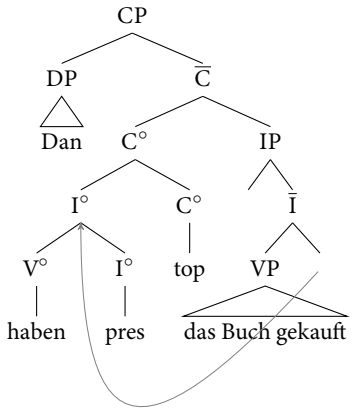
A common description of the conditions that accompany movement of a phrase into Specifier of CP is that the phrase is "topicalized." I choose "top" as the name for the complementizer to evoke this description.

(59)



This is remedied by moving I° to C° to form (60).

(60)



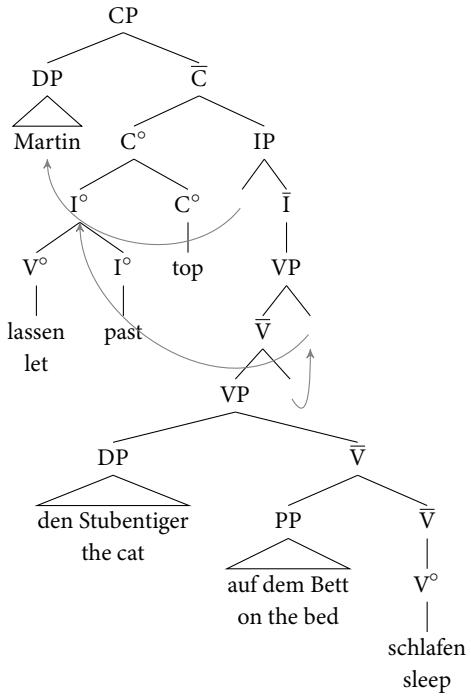
There are some things that we might notice about the verb movement processes in German which are similar to the parallel processes in English. For example, the Head Movement Constraint is preserved in the grammar of verb placement in German. Sentences like the following, where the verb from an embedded VP moves into the higher clause, are ungrammatical.

(61) * Martin schlief den Stubentiger auf dem Bett lassen.
Martin slept the cat on the bed let
'Martin let the cat sleep on the bed.'

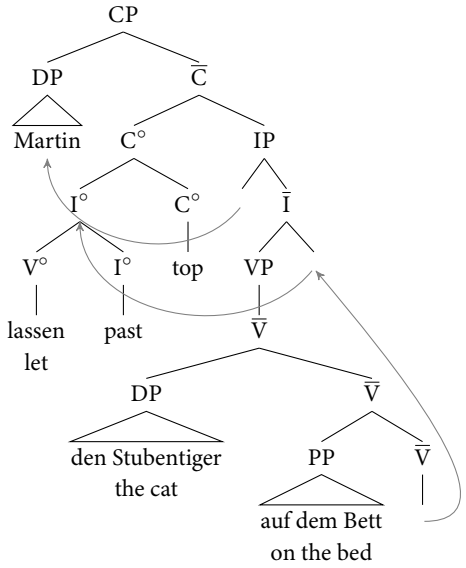
The grammatical way of expressing the sentence that underlies (61) is (62), whose syntax is in (63). The ungrammatical string in (61) is produced by letting the embedded verb move in violation of the Head Movement Constraint in the way that (64) indicates.

(62) Martin läst den Stubentiger auf dem Bett schlafen.
Martin let the cat on the bed sleep
'Martin let the cat sleep on the bed.'

(63)



(64)



The system that moves heads around in German seems to be constrained by the Head Movement Constraint, in the same way that the system that moves heads around in English is.

Moreover, we can observe that there is a pattern to all these rules in German and English: the moved term always adjoins to another head. Because this appears to be something that is the same across German and English, it is a candidate language universal. Indeed, over a large range of cases examined across languages, this generalization appears to hold up. This suggests a constraint on movement rules that is sensitive to the phrase/head distinc-

tion. One of the first proposals for such a constraint is found in Baltin (1982), who argues that this is part of a more general constraint on movement rules. He proposes that the position to which a phrase or head is moved always has the same phrasal status as the phrase or head being moved. So a head adjoins to another head position only, and a maximal projection can only move to another maximal projection position, and so on. Baltin calls this the Like-attracts-Likes Condition:

(65) Likes Attracts Likes
An X° may only adjoin to another X°, and an XP may only move to a Specifier position or adjoin to another XP.
(Baltin, 1982, roughly (1), p. 2)

One can view the Likes Attracts Likes constraint as a special instance of constraint that Joe Emonds[5] defended in his dissertation. Emonds suggested that no parse in a derivation can have a shape that couldn't have been made by phrase structure rules. In our terms, this amounts to saying that every parse in a derivation must meet the requirements of the $\overline{\text{X}}$ Skeleton. The $\overline{\text{X}}$ Skeleton, recall, sanctions the following shapes.

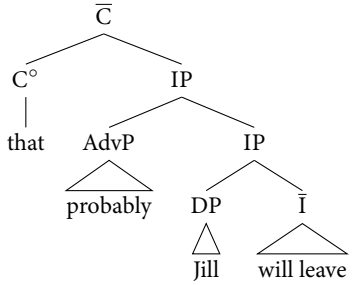
(66) $\overline{\text{X}}$ Skeleton
a. $\text{XP} \rightarrow \overline{\text{X}}$
b. $\text{XP} \rightarrow \alpha\text{P}\ \overline{\text{X}}$
c. $\overline{\text{X}} \rightarrow \{\ \alpha\text{P}, \overline{\text{X}}\}$
d. $\overline{\text{X}} \rightarrow \{\ \text{X}^\circ, \alpha\text{P}\}$
e. $\overline{\text{X}} \rightarrow \text{X}^\circ$

For Emonds' condition to be correct, we'd have to add to (66) the shapes in (67).

(67) a. $\text{XP} \rightarrow \{\ \text{XP}, \alpha\text{P}\}$
b. $\overline{\text{X}} \rightarrow \{\ \text{X}^\circ, \alpha^\circ\}$

Evidence that we need to add to our stock of licit shapes for a D-structure the one described by (67a) comes, perhaps, from examples like (68).

(68) She said that probably Jill will leave.



We'll see evidence for adding to our stock of possible shapes for a D-structure the form in (67b) below (see page 243). If we add (67) to the $\overline{\text{X}}$ Skeleton,

[5] Emonds, Joseph E. 1970. Root and structure-preserving transformations. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge, Massachusetts

Emonds' proposal was slightly more complex. He suggested that there was a distinction between transformational rules that applied to root nodes only, and all the others. The root transformations had special properties. This distinction remains in contemporary theories—I°-to-C° movement is restricted to root clauses (mostly)—but there isn't a well-developed explanation of it.

then Likes Attract Likes is a special instance of Emonds' condition. Emonds called his condition "Structure Preservation," and I'll formulate it in terms appropriate for our grammar.

(69) Structure Preservation
Every element of a derivation must conform to the $\overline{X}$ Skeleton.

In what follows, I'll refer to this idea about how derivations are constrained variously as Likes Attracts Likes or Structure Preservation.

The derivations that involve moving V° or I° in both German and English conform to the Head Movement constraint and Structure Preservation. Finally, note that these movement rules both languages all have the feature that the verb or I° that has moved has moved up. We don't find cases where Verb Movement has relocated the verb downwards, as in examples like (70).

(70) a. * …daß Hans das Buch gehabt kauft.
…that Hans the book had buys
b. * John must had buy the book.

This is just a consequence of Upwards, which we've seen holds of all the movement rules we've seen so far. This too looks like a good candidate for a universal condition.

Let's take a look now at what we find with respect to verb placement in some of the other Germanic languages. In the Scandinavian languages we find a situation similar to German, as Vikner (1995) reviews. I will use Danish as a guide; much of what we see for Danish is found in Norwegian and Swedish as well. The same sensitivity to embedding is found in the placement of the finite verb in these languages. As the contrast in (71) indicates, Danish is like German in allowing any constituent, but only one, to precede the finite verb in independent clauses. That is, it shares with German the trait of being "Verb Second."

(71) a. Børnen har set denne film
kids-the have seen this film
'The kids have seen this film.'
b. Denne film bar børnen set.
this film have kids-the seen
'The kids have seen this film.'
c. * Denne film børnen har set.
this film kids-the have seen
'The kids have seen this film.'

But, as (72) shows, these traits are not shared by embedded clauses, where, instead, the subject must precede the finite verb.

(72) a. * Jeg ved ikke hvor i går har koen stået.
I know not where yesterday has cow-the stood
'I don't know where the cow stood yesterday.'

b. Jeg ved ikke hvor koen har i går stået.
I know not where cow-the has yesterday stood
'I don't know where the cow stood yesterday.'

Moreover, as the contrasts in (73) show, the placement of the finite verb relative to negation is sensitive to the embedded/non-embedded context.

(73) a. Børnen har ikke set denne film.
kids-the have not seen this film
'The kids haven't seen this movie.'

b. * Børnen ikke har set denne film
kids-the not have seen this film
'The kids haven't seen this movie.'

c. Jeg ved hvor børnen ikke har set denne film
I know where kids-the not have seen this film
'I know where the kids haven't seen this movie.'

d. * Jeg ved hvor børnen har ikke set denne film
I know where kids-the have not seen this film
'I know where the kids haven't seen this movie.'

This indicates that the finite verb has moved out of VP, past negation, into C° in independent clauses, just as it does in German.

It looks, therefore, like Danish has the word-order of English—placing verbs before their objects—but the syntax of verb movement that German has. Note in particular that main verbs are able to move as well as auxiliary verbs, as we see partly in (72) (look at the root clause word-order). If we were to look further, we would see that Danish (and the remainder of Germanic) also have the range of constraints we have seen on verb movement operations.

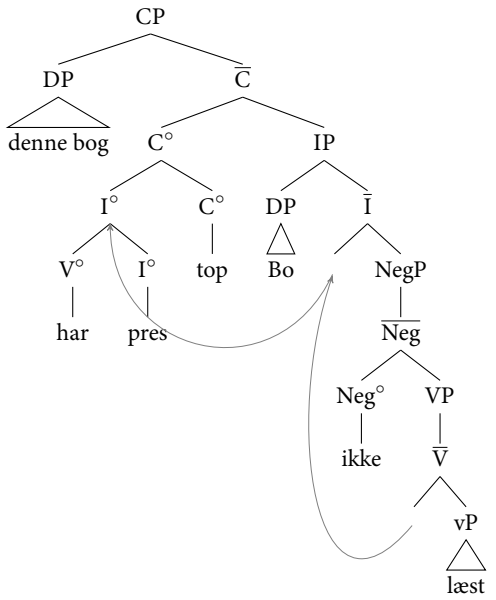
There are a couple differences, however. One at least potential difference between Danish and German is the existence of V°-to-I°. Whereas we cannot easily discern whether such an operation exists in German, it apparently does not in Danish (nor does it in standard Norwegian and Swedish), since when the verb has not relocated into C°, it remains to the right of negation. (Of course, we have made the perhaps incorrect assumption that negation occupies the same place in Danish as it does in English.)

Another difference concerns the range of contexts where verb second is found in embedded contexts. Recall that in German the verb moves into C° in embedded clauses only in those contexts where a complementizer is not required to fill that C°. But in Danish (as in the other Scandinavian languages), V2 word-order is possible even in those embedded clauses which have a complementizer associated with them, as in (74).

(74) Vi ved at denne bog har Bo ikke læst.
we know that this book has Bo not read
'We know that Bo hasn't read this book.

A variety of hypotheses about this situation have been offered. Let us adopt for concreteness the hypothesis that there is a CP "shell" that can be embedded within a regular CP. This CP shell provides the C° into which verbs move in embedded clauses in Scandinavian. This would give to the embedded CP in (74), the representation in (75).

(75)



In this derivation, the verb moves to I° across Neg° in violation of the Head Movement Constraint. We noted earlier that a similar situation arises in English. We will return to this feature of the derivation.

So, let's see what we've got so far. If we factor out the parts to the set of rules in German, Danish and English that are common to the syntax of verb placement, we have the following:

(76) Universals
a. Structure Preservation
b. Upwards Constraint
c. The Head Movement Constraint (with some counterexamples)
e. Word Criterion

The differences that are left can be boiled down to the following four parts.

(77)
a. German VPs (and IPs?) are head final.
b. English and Danish VPs (and IPs?) are head initial.
c. German and Danish: verbs (and I°?) are movable.
d. English: auxiliary verbs and I° are movable.
e. German and Danish: Root C° and Specifier of CP must be filled. (i.e., Verb Second)
f. English: Only wh-phrases move to Specifier of CP, and only questions trigger I°-to-C° movement.

Putting this in the terms of language acquisition which we started with, the differences in (77) constitute what must be learned in order to determine whether the grammar he or she is acquiring is German, Danish or English.

By contrast, the Universals in (76) are candidates for how the acquisition trajectory is limited. The guesses that children make about the syntax they are acquiring don't seem to include ones that violate one of the constraints in Universals.

4.4 The Pollockian revolution

Let us now add to our family of languages: French. In French we also find evidence of rules that move verbs. Just as in English, French verbs distribute themselves relative to polarity items, like sentential negation, on the basis of their inflectional class. Thus, for example, we find that finite verbs in French, just like their counterparts in English, only appear before negation.

(78) a. Jean n'a pas lu livres.
John n'have not read books
'John hasn't read books.'

b. * Jean ne pas a lu livres.

A word about negation in French is needed before the contrast in (78) will be interpretable. Unlike English, sentential negation in French is expressed by way of two words, one (*ne*) appears to be in a position like that we have put inflectional morphemes in English. We might imagine that like the contracted form of *not* in English (*n't*) it is contracted onto whatever word has moved into I° (as in (78a)). The second word (*pas*) is the one that has a distribution more like that of English not. And, as (78) illustrates, the finite form of have (*ha*) must precede this part of sentential negation.

French differs from English (but is like German and Danish) in moving main verbs. This can be seen in French by the fact that they are placed to the left of sentential negation when they are finite, as (79) demonstrates.

(79) a. Jean n'aime pas Marie.
John ne'love not Mary
'John doesn't love Mary.'

b. * Jean ne pas aime Marie.

As the contrast between the position of main verbs in (79a) and (78a) indicates, it is the inflectional class of the verb that determines its position. That is, just as in English, there is a correlation between the inflectional class of the verb and its syntactic position—a correlation that is captured by fixing the position of inflectional morphemes with the phrase structure rules and driving the verbs to these positions with the verb movement operation.

That main verbs may move is also responsible for another difference between French and English. This derives from the ordering that the Projection Principle places on arguments and non-arguments. As we have had occasion to discuss several times, now, the Projection Principle forces arguments to be closer to their θ-marking head than non-arguments. In verb initial languages,

like French and English, this means that complements should come before non-complements. This is roughly true for English, as we have seen; but it is not the case for French.

(80) a. Jean embrasse souvent Marie.
b. * John kisses often Mary.

This apparent difference in the function of the Projection Principle can be credited to verb movement. Since we already know that main verbs in French but not English move into I°, it makes sense that the main verbs in French, but not English, should be able to be separated from their objects by all sorts of material, including adverbs.

But now consider what happens in French non-finite clauses.

(81) a. **Comprendre** *à peine* l'italien apres cinq ans d'étude
understand barely the-Italian after five years of study
dénote un manque de don pour les langues.
shows a lack of gift for the languages
'To barely understand Italian after five years of study shows a lack of talent for languages.'

These data all from Pollock (1989).

b. **Perdre** *complètement* la tête pour les belles étudiantes c'est
lose completely the head for the pretty students it is
dangereux.
dangerous
'To completely lose your head for pretty students is dangerous.'

(82) a. ne *pas* **sembler** heureux est une condition pour écrire
ne not seem happy is a condition for writing
des romans
novels
'To not seem happy is a (pre?)condition for writing novels.'

b. * ne **sembler** *pas* heureux est une condition pour écrire
ne seem not happy is a condition for writing
des romans.
novels
'To not seem happy is a (pre?)condition for writing.'

Here it appears that main verbs may move past adverbs but not negation. (The infinitival verbs are in bold face, in these examples, and the term that separates them from their objects is in italics.) In (81), we see examples of main verbs with an adverb between it and its object, and this word order is grammatical. In (82), by contrast, we see that the main verb *sembler* ('seem') cannot be separated from its object (here the adjective *heureux* ('happy')) by negation.

Auxiliary verbs behave differently.

(83) a. ne *pas* **être** heureux est une condition pour écrire des romans
ne not be happy is a condition for writing novels
'To not be happy is a condition for writing novels.'

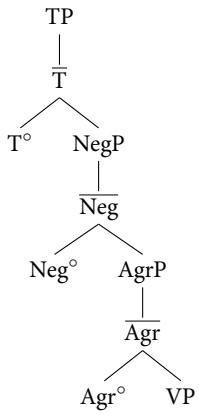
b. N'**être** *pas* heureux est une condition pour écrire des romans.
ne'be not happy is a condition for writing novels
'To not be happy is a condition for writing novels.'

They may optionally appear to the left of negation.

We learn two important things from these facts. First, that the contrast between auxiliary and main verbs that seems to distinguish English from German/Danish is actually not a distinction in languages. Instead, it is a difference which is found within a single language: French. That is, this is not a parameter along which languages vary, then, but rather a parameter along which clause-types vary. We need to express the distinction between English, and these other languages, in terms of the clause types these languages host, not in terms of the targets for their verb movement rules. So, we're looking for something that distinguishes English finite clauses and French non-finite clauses on the one hand from French finite clauses on the other.

The second lesson of this paradigm is the main point of Pollock's paper. He suggests that the two different positions that main verbs may occupy across finite and non-finite clauses warrants giving a structure like that in (84) to clauses, where the higher I° is equated with Tense and the lower one with Agreement. This answers to the correlation that appears to hold for main verbs between whether they bear tense morphology or not: in finite clauses they do, and in non-finite clauses they do not.

(84)



Pollock also introduces the idea that Neg heads a phrase which stands between the two other Inflectional phrases, as indicated in this pares. There are several reasons for wanting to treat negation differently from other adverbials. One is that the syntactic position of negation is comparatively rigid when compared to other adverbs. Another is that only negation blocks the process that inflects main verbs in English, as we saw earlier; other adverbs

don't. Pollock suggests distinguishing negation from adverbs structurally, and then making reference to this structural difference to capture these ways in which negation behaves uniquely. I won't examine in detail how this idea works, partly because we will eventually go in a slightly different direction than does Pollock. But let us nonetheless adopt—at least as an intermediary hypothesis—the thesis that negation does head a phrase as shown in (84). It should be noted, however, that this introduces a problem: how is it that Agr° can move to T° past negation without violating the Head Movement Constraint. Pollock offers a solution to this problem that I will come to soon.

Note that though the difference between main and auxiliary verbs in French non-finite clauses that we've just reviewed speaks on behalf of two head positions to which verbs may move, it doesn't really indicate what the identity of these two head positions might be. While most of Pollock's proposal involves examining how the hypothesis that there are two inflectional positions within a sentence can be used to explain the differences in verb position across clause types, he also assigns values to these inflectional phrases. But, in fact, it is extremely difficult to tell what the value of these heads is. Pollock decides in favor of giving the higher I° the value of Tense and the lower one the value of Agreement. This involves two decisions: first that these positions should be equated with inflectional categories, and second that they should be equated with these particular ones. The first decision, note, is a bit of a leap here as there is no convergent evidence of the headedness sort to anchor this conclusion. That is, part of the reason we decided that the landing site of verb is I° came in response to the evidence that I° was a head.

There is, however, a sort of reasoning which suggests that the first decision is correct, but that the second isn't. Consider the following verbs, drawn from a variety of IndoEuropean languages.

The argument I'm reporting here comes from Belletti (1990).

(85) a. Legg - eva - no
read - Imperfect - 3plur

b. Parl - er - ò
speak - Future - 1sing

c. Buhl - t - en
woo - past - plur

In each of these cases, the morpheme which encodes tense information precedes that which encodes agreement. Is there a way of determine the syntactic arrangement of inflectional terms from their morphological arrangement? Mark Baker has made famous an argument for thinking that the answer to this question is yes. Following proposals by Pieter Muysken and Donna Gerdts, Baker argues that there are correlations between the order that inflectional affixes have relative to the verbal stem and the order of the syntactic operations that these affixes encode. Thus, for example, in Chamorro, the passive morpheme necessarily comes closer to the verbal stem than does the subject agreement affix, as (86) shows.

(86) Para.u.fan - s - in - aolak l famgu'un gi as tat-n-niha
irr.3pS - pl - pass - spank the children obl father.their
'The children are going to be spanked by their father.'

(86) comes from Gibson (1980). The principle formulated in Baker (1985) has its roots in many earlier works, among which are: Muysken (1979, 1981) and Gerdts (1981).

This ordering on affixation mimics the order in which the syntactic operations of Passive and subject agreement take place—the passive must bring the underlying object into the subject relation before the agreement process can apply.

Baker calls correlations of this type, which he suggests are widespread enough to be considered law-like, the "Mirror Principle."

(87) Mirror Principle
Morphological derivations must directly reflect syntactic derivations (and vice versa).
(Baker, 1985, p. 375)

Baker argues that the Mirror Principle will emerge once the correct theory of the syntax-morphology interface is found. In particular, (87) can be derived on models that posit for each of the agreement, passive, and other such operations one rule underlying both the morphological and syntactic effects. Baker tentatively concludes that these meta-rules apply in the Syntax—that is, after the verbal roots are inserted into phrase-markers. The resulting picture is sketched in (88).

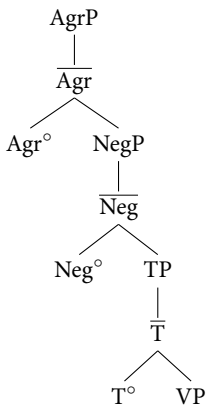
(88)

D-structure:	V_{root}	
	$\downarrow$	$\leftarrow$ Metarule α
	$V_{\text{root}} + Af_{\alpha}$	
	$\downarrow$	$\leftarrow$ Metarule β
S-structure:	$V_{\text{root}} + Af_{\alpha} + Af_{\beta}$	

Each metarule brings about the relevant syntactic operation and adds to the verbal root the corresponding affix. Note in particular that this picture correlates syntactic operations with affixes; the correspondences that Baker summarizes with the Mirror Principle only concern the relation between affixal orderings and the orderings of syntactic operations. Indeed, Baker's argument leads to the conclusion that the metarules in (88) necessarily involve affix-syntactic operation pairs. It is the order of affixes that correlates with the relevant syntactic operations, and not some more abstract information, such as morphological features or the like. We shall have an opportunity to revisit this point.

Now if Baker's conclusions from the Mirror Principle are imported into the domain of Verb Movement, then the arrangement of morphemes in (85) suggests that the higher of Pollock's positions should be associated with agreement morphology, and the lower with tense morphology. Then, the fact that tense morphology comes closer to the verbal stem will follow from the Head Movement Constraint. So we should adopt, perhaps, a picture like that in (89).

(89)



We can reach this conclusion, perhaps, through a different route. If we compare Icelandic with Danish, we see a difference in verb placement that suggests that only Icelandic has movement of verbs into the I° that stands above negation. In embedded Icelandic clauses the finite verb must precede negation, as in (90).

(90) a. * Ég spurði af hverju Helgi ekki hefði lesið þessa bók
I asked whether Helgi not had read this book
'I asked whether Helgi hadn't read this book.'

b. Ég spurði af hverju Helgi hefði ekki lesið þessa bók
I asked whether Helgi had not read this book
'I asked whether Helgi hadn't read the book.'

But as we have seen, Danish finite verbs cannot precede negation. Another example illustrating this fact is (91).

(91) a. Jeg spurgte hvorfor Peter ikke havde læst den.
I asked why Peter not had read it
'I asked why Peter hadn't read it.'

b. * Jeg spurgte hvorfor Peter havde ikke læst den.
I asked why Peter had not read it
'I asked why Peter hadn't read it.'

If we assume that the position of negation is constant across these languages, then this indicates that the verb has moved past this position in Icelandic, but not Danish. Now, interestingly, Icelandic has a full paradigm of subject agreement, but Danish has no subject agreement. Comparing Icelandic with Danish, then, what we find is a correlation between subject agreement and position to the left of Neg°. Indeed, this correlation holds throughout the Scandinavian languages.Standard Swedish and Norwegian are like Danish in lacking subject agreement morphology, and they are like Danish as well in placing their finite verbs to the right of negation. This is just the correlation that (89) predicts.

Adopting this modified Pollockian parse for clauses, we are left with a series of questions about how verbs distribute themselves over the two head

This correlation holds only partially. There are some cases in which agreement morphology exists, though in a very impoverished form, and verb movement past negation nonetheless appears to be blocked. (This happens in certain dialects of Faroese, in a register of Norwegian, and in a Swedish dialect heavily influenced by Finnish.) Some of these facts are reported in Vikner (1995), where references to the relevant literature can also be found.

positions. Why can main verbs not move to the "agreement" X° in non-finite clauses in French, for instance; and why are they unable to occupy this position even in finite clauses in English? We don't have satisfactory answers to these questions. Pollock relies on an observation due originally to Ian Roberts: The X°s in which main verbs cannot surface are devoid or close to devoid of morphological content. English agreement is much less "robust," somehow, than is French agreement. And subject agreement in French non-finite clauses is considerably less pronounced than it is in French finite clauses. (In fact, it is thoroughly absent.) Now, this difference in agreement morphology Pollock exploits to control whether or not a verb can move into the lower of the two inflectional X°s—this is because Pollock assigned agreement to this lower head. But we can copy his idea into the assignment we have given to these positions. Tense morphology is also richer in French than it is in English, in both finite and non-finite clauses; and agreement is richer in finite clauses than it is in non-finite clauses too. In fact, there is no evidence of agreement at all in infinitival clauses.

Even if this "robustness" in the morphology associated with the T° and Agr° positions is not what is relevant (and further cases suggests that it isn't), let's continue to exploit the language that Pollock adopts for whatever the relevant difference turns out to be. Those heads that are associated with morphology that allows main verbs to occupy them, let's call "strong." And those head positions that are not associated with morphology that allows main verbs to sit in them, let's call "weak." We can now describe in these terms what we've seen so far. In particular, we will ascribe to the Tense and Agreement positions different "strengths," and in this way describe how they control the Move X° operation. For English and French, we might assign the values as shown in (92) and (93).

(92) **English:**

a. *Finite:*
Tense: weak
Agreement: weak

b. *Infinitive:*
Tense: *to*
Agreement: empty?

(93) **French:**

a. *Finite:*
Tense: can't tell
Agreement: strong

b. *Infinitive:*
Tense: strong
Agreement: empty

In these value assignments, I've decided to treat the infinitival marker in English, i.e., *to*, as something that can optionally move from T° to Agr°. On this view it is rather like an auxiliary verb in French, and should behave similarly, therefore, in infinitival clauses. For some speakers of English there is, in fact, an alternation that resembles that we've seen for French auxiliary verbs:

(94) a. Jerry tried [not to eat candy].
b. Jerry tried [to not eat candy].

Note that we cannot determine whether finite T° is strong or weak in French because verbs are always driven up to Agr° in this context.

Turning now to Scandinavian, we have something like:

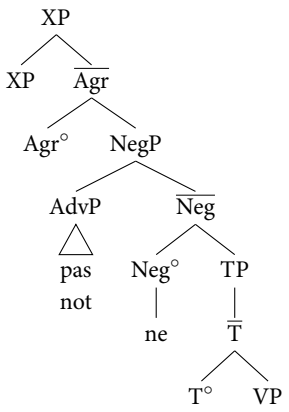
(95) **Danish, Norwegian and Swedish:**
a. *Finite:*
Tense: weak
Agreement: not there
b. *Infinitive:*
haven't looked yet (it varies across the group, as it turns out)

(96) **Icelandic:**
a. *Finite:*
Tense: can't tell
Agreement: strong
b. *Infinitive:*
haven't looked yet

My decision to classify the Mainland Scandinavian group as having weak tense morphology is based on the fact that this language group behaves like English, and not like French, with respect to the positioning of main verbs relative to their complements. That is, in this group, non-complements cannot intervene between main verb and DP objects (except in contexts where the verb has relocated into C°).

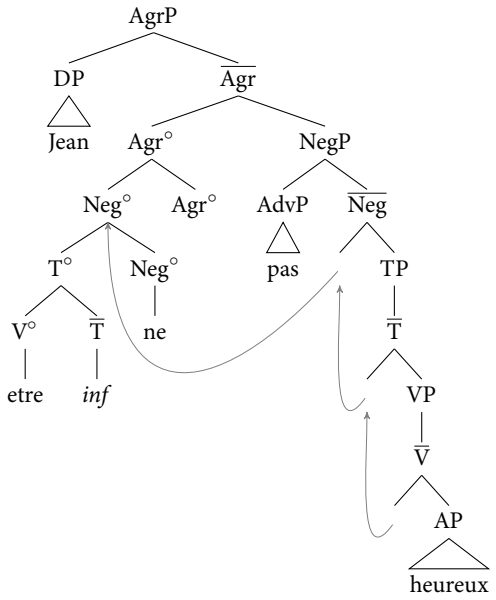
Okay, this is one part of the story. The morals are: (i) there are two inflectional heads between C° and VP in some languages, and (ii) it is properties of those heads that determine what must and can move into them. Let's now address the problem that treating negation as a head poses. Recall that we should not expect the Head Movement Constraint to allow movement of T° to Agr° past negation, if negation is a head that lies between them. Pollock's suggestion is that *not* and *pas* are in the Specifier of NegP, and that *ne* is what heads NegP in French. In English, we might imagine that this head is silent; I'll represent this silent morpheme with *neg*. Thus, the correct way of parsing sentences with sentential negation in them is as in (97).

(97)



On this view, then, when T° moves to Agr° it doesn't have to skip Neg°, it can move through this position and pick up the head (if there is one) on the way. In a finite French clause, then, the surface parse might look roughly as in (98) below.

(98)



The S-structure in (98) is derived as follows (ignoring the movement of the subject). The main verb, ***aime***, is moved into T°, where it combines with the tense morpheme. The resulting tensed verb (that is, T°) moves and adjoins to ***ne***. the resulting negated verb (that is, Neg°) moves into Agr°, where the agreement morpheme combines with the verb. This surface parse, then, claims that negation—***ne***—is closer to the verbal stem than is the agreement morpheme, and this is somewhat surprising.

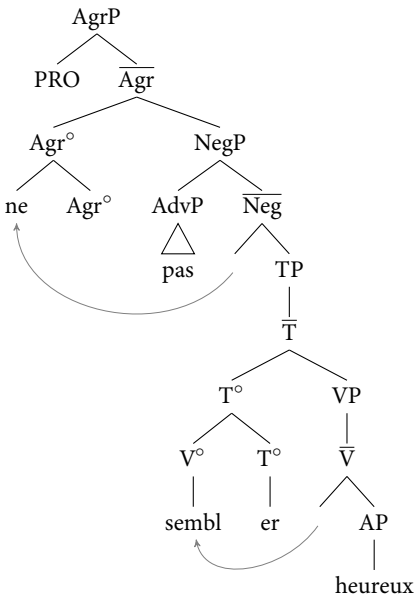
As this parse indicates, *ne* gets carried to the position before *pas* by the tensed verb as it moves into Agr°. Note that we have to let something force the bound morphemes, that is, the tense and agreement suffixes, to linearize to the right of the term that has adjoined to them, but force *ne* to linearize to the left of the term that has adjoined to it.

This can't be the only way that *ne* manages to get to the left of *pas*, because *ne* shows up before *pas* in infinitival clauses as well, even when no verb has moved into Agr°. In these situations, we might imagine that *ne* moves on its own into Agr°. This, in fact, is what Pollock suggests. He hypothesizes that *ne* adjoins to Agr° on its own in much the same way that so-called "clitics" do. "Clitics" are prosodically weak pronouns that, in the Romance languages, surface in construction with a verb; more particularly, they form a prosodic word

with a verb and are therefore adjacent to that verb (or adjacent to something else that is part of the verb). In finite clauses, it is the finite verb that clitics appear with, and in infinitival clauses, it is typically the infinitival verb that the clitic appears on. One way of capturing this distribution is let clitics adjoin to Agr°, where they will come close enough to the finite or infinitival verb to be parsed with that verb prosodically.

Clitics, then, instantiate another instance of movement. Object clitics, for instance, satisfy the Theta Criterion and Projection Principle by way of occupying a position that is sister to the verb they are arguments of, but surface in Agr° position. This type of movement is called "Cliticization." Cliticization doesn't have the same properties that Head Movement has—it isn't subject to the Head Movement Constraint, for instance—and therefore we can treat it is as being independent of the rules we are investigating here. With this assumption about *ne*, then, an infinitive that doesn't have verb movement to Agr°, but is negated, might have an S-structure like (99).

(99)



If we adopt Pollock's suggestion, that *ne* cliticizes to Agr°, we might be better poised to understand why it is linearized to the left of the verb+ tense+-agreement complex. Indeed, we could impose a general constraint on adjunction that would linearize the moved item to the left of the phrase it adjoins to. Something along the lines of (100), perhaps.

(100) Adjunction Linearization
If α° adjoins to β°, then α° precedes β°.

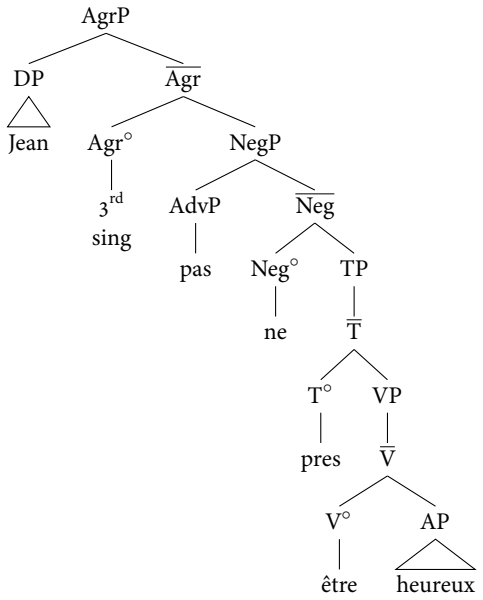
That tense morphemes are linearized to the right of their verbal stem is the predominant order in the world's languages. Julien (2002), who surveyed 530 languages, found tense suffixes far outnumbered tense prefixes. She divided

Julien equated tense and aspect morphemes in her survey. These are often difficult to distinguish, and in some languages aspect is the only means of expressing temporal information.

her sample of languages into 280 genera—groups of languages that are historically related, and counted how many of those genera used tense prefixes or tense suffixes. (If languages in a genera had either order, the genera was counted as illustrating both types.) What she found was that 202 genera had tense suffixes but only 60 genera had tense prefixes. Most of the languages that had tense prefixes in her survey are head initial, and Julien argues that they have a different morphological status than tense suffixes.

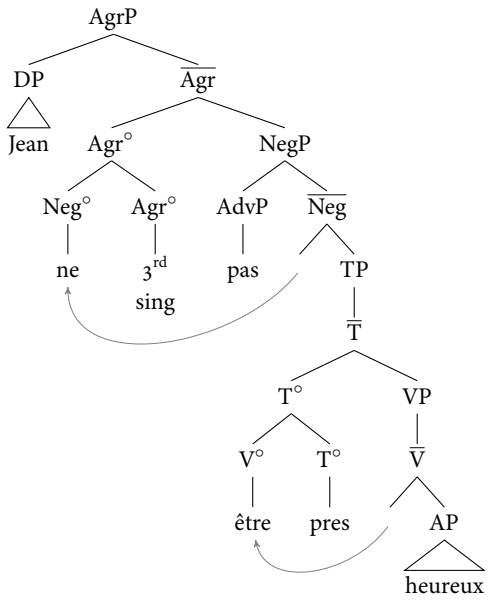
With these preliminaries, let's reconsider how the derivation in (98) might go for French. The D-structure representation would be as in (101).

(101)



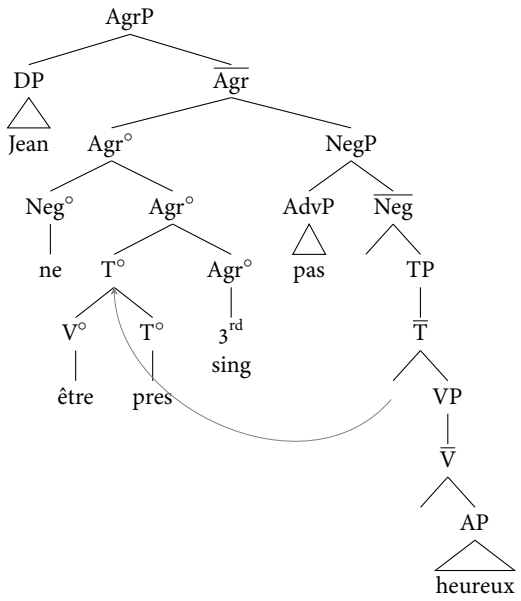
The first two steps of the derivation could involve Verb Movement to T° and cliticization of *ne*, giving (102).

(102)



At this step in the derivation, Neg° no longer stands between Agr° and T°, and this means that using movement to bring them together no longer invokes a violation of the Head Movement Constraint. To bring T° together with the Agr° that contains the agreement morphology, but not *ne*, could be achieved by using Head Movement to produce (103).

(103)



Note that Upwards permits this movement. The sister to the moved T° in the position it moves to is Agr°. And Agr° has a projection, namely Agr, that dominates the position that T° moved from.

Let us assume that these are the derivations that French uses to avoid violation of the Head Movement Constraint. We will return to look at how English manages this.

So, now what we would like to do is see if we can't get a better understanding of what it is about this "strong"/"weak" contrast that makes main verbs susceptible to it, but not auxiliary verbs. Pollock makes a proposal in this direction. He suggests that we think of main verbs as differing from auxiliary verbs with respect to their ability to assign a θ-role. In particular, he suggests that we define main verbs as ones that have a θ-role to assign, and auxiliary verbs as ones that don't. He then crafts a proposal that would prevent verbs from assigning a θ-role if they move to a position associated with a "weak" inflectional head. And this will essentially prevent main verbs from both moving to and moving through the positions we have assigned the "weak" value.

There are some immediate problems with this idea that Pollock himself addresses. It wouldn't seem able to account for the British (105), nor for the contrast in (104).

(104) Mary hasn't a dime.

(105) a. Mary isn't unhappy.

b. * Mary seemsn't unhappy.

Why doesn't the *be* in (104a) have the same number of θ-roles as (104b)? And it sure looks like *has* in (105) is capable of moving past negation, though it

appears equally like it assigns θ-role. Pollock suggests that the θ-role assigned in (105) comes not from the verb, but from some hidden term. Something similar might be attempted in distinguishing the cases in (104).

But there is another kind of problem, specifically for the prediction that movement through a weak X° is unavailable to main verbs, which seems to me insurmountable. Danish, as we have seen, and the other Mainland Scandinavian languages, show Verb Second word-order; (106) is an example, for instance, where the main verb has relocated to C° position.

(106) Købte han bogen
bought he book-the
'Did he buy the book?'

But we have also just discovered that Danish has weak tense morphology. Thus, if we preserve the Head Movement Constraint, this entails that Danish main verbs are capable of moving into T° and from there moving into C°. If Danish T° is weak, this means, in turn, that moving through a weak X° is possible for main verbs.

For this reason, I will interpret the weak/strong typology to determine only whether a main verb may surface in the relevant X°. This seems to be the way these values are interpreted most commonly in the literature. It is, in fact, on this basis that I gave the assignments of "weak" and "strong" to Agr° and T° in English, French and the Scandinavian languages in (92), (93), (95) and (96). In these assignments, then, I have given a "weak" value to those X° positions for which there is evidence that main verbs cannot surface, and to those X° positions where there is evidence that main verbs can surface I have given the "strong" value. I have also indicated certain positions—namely, French and English infinitival Agr°—as empty positions. This is my, somewhat idiosyncratic, way of expressing the optionality of movement to these spots. Because they have no morpheme in them, these infinitival Agr°s do not cause the Word Criterion force a verb stem to move into this position. But, because it is present, it is still a possible landing site for X° movement. It's in this respect, then, that the infinitival Agr°s of these languages are different from the finite Agr° of, say, Danish. Movement of a verb to the position we've identified for finite Agr° is not possible in Danish—neither auxiliary nor main verbs may surface in this position—and I've coded that here by making Agr° completely absent in Danish.

Under this typology, X° positions identified with inflectional morphemes come in four strengths. In their weakest manifestation, they are completely absent, and nothing, as a consequence, may surface in their position. In their next strongest manifestation, they are present but are associated with no morpheme. In this guise we find optional movement of the infinitival marker *to* and auxiliary verbs, but main verbs are still not capable of surfacing in them. (This is what we see for the infinitival Agr° in French.) In their next strongest manifestation, they are associated with a "weak" morpheme. At this level of

strength, the Word Criterion kicks in and requires that something express the morphology they are associated with. The "weak" property, however, prevents main verbs from doing this by adjoining to the morpheme. Finally, there are the X° positions occupied by a "strong" morpheme. These positions allow either auxiliary or main verb to surface in them, and the Word Criterion forces one or the other to. Because we want to prevent main verbs from surfacing in all but the last of these positions, the constraint that controls the auxiliary/main verb distinction should be stated as (107).

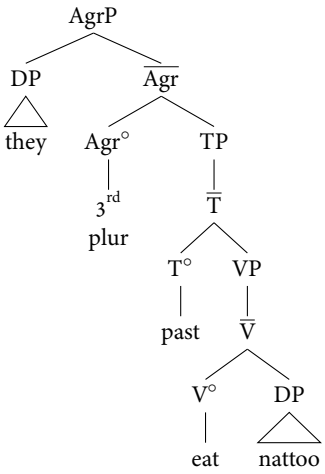
(107) A main verb merged to a morpheme, α, at S-structure creates ungrammaticality unless α is "strong."

(107) is going to have certain consequences for the syntax of a sentence in which the verb has moved into C°. In these cases—questions and, in the Germanic languages other than English, root finite clauses—we must imagine that the ***top*** morpheme that occupies C° is strong.

4.5 AGREE

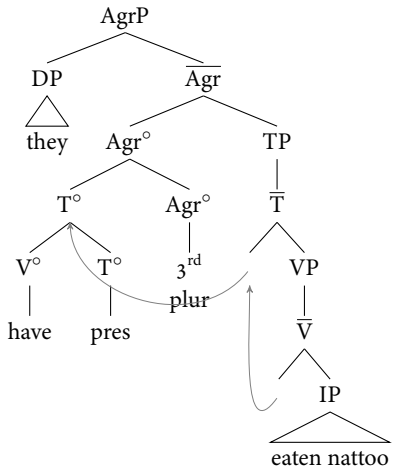
We've developed a rough description of how main and auxiliary verbs are driven into I°s that uses the "strong"/"weak" vocabulary. We should now look at how this matter interacts with the way in which inflectional heads assign their morphology to the relevant verbs when verb movement is not involved. The conception we have adopted of how verbs inflect is that it parallels the kind of relation that we have seen in connection with Case. The tense and agreement heads assign the relevant inflection to the verb in configurations like (108).

(108)



When auxiliary verbs are involved, the verb assigned inflection moves to the heads responsible for that inflection, as in (109).

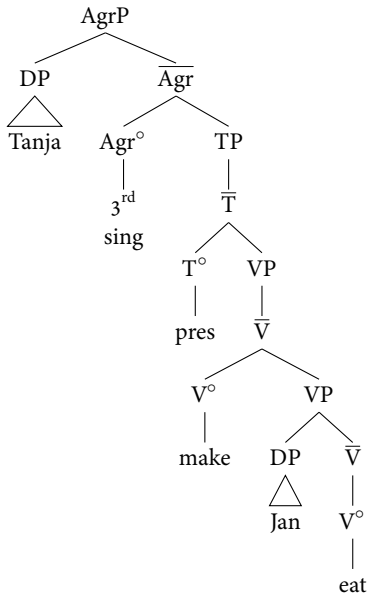
(109)



The syntactic results of inflecting the verbs in these two cases is obviously different, and so a reasonable first question is: could the method of inflecting auxiliary and main verbs be different? Could it, for instance, be that main verbs get inflected by assignment while auxiliary verbs get inflected by moving?

There are some things that weigh against this direction. The conditions we have placed on verb movement are mirrored by conditions on the assignment relationship. Movement of V° and I° can move the term to just c-commanding I°, and never past a c-commanding X°. And, similarly, the assignment relation always goes from an I° (or C°) to a term that it c-commands, and it never crosses a c-commanding X°. This is indicated by the fact that the parse in (110) cannot produce the outcome in (111).

(110)

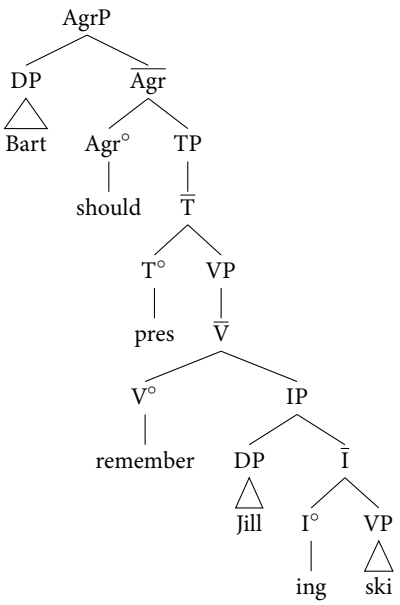


(111) * Jerry make Mary eats.

This would be possible if the Agr and T heads of the root clause were able to assign their inflection to the verb that heads the embedded VP. This would be prevented if these heads are blocked from assigning their inflection past the "closer" head: *make*. This is quite general: cases where an I° assigns the inflection it determines on a head that is embedded below another possible target for its inflection do not seem to arise. This looks completely parallel to the Head Movement Constraint, which proscribes movement of a head past a c-commanding head.

That inflectional heads assign the inflection they determine only to verbs they c-command can be appreciated by considering examples like (112) below.

(112)



The grammatical outcome of this structure is one in which the inflectional head determining the suffix *ing* assigns that inflection to the verb it c-commands: *ski*. What can't happen is for this I° to assign this inflection to *remember*, a verb it does not c-command, but is otherwise a good candidate for being inflected. (112) does not lead to the outcome in (113).

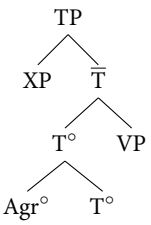
(113) * Bart should remembering Jill ski.

In general, examples of this sort are not found. If a verb bears inflection that is associated with an I°, that I° c-commands the verb. We don't want to design our theory so that the method by which main verbs in English get their inflection is completely divorced from the movement operation that inflects auxiliary verbs. If we did that, we would lose the fact that both Upwards and the Head Movement Constraint seem to govern how both processes work. We should design the system that inflects main verbs in English so that it is, somehow, the same process that is found governing verb movement, the process relevant for inflecting auxiliary verbs.

Note, by the way, that (112) confronts us with a question that our move to the Pollockian framework has produced. The root clause in (112) contains a modal, and this is a word that we have mapped onto the position that heads sentences. But we now have two heads, and two consequent phrases, that sentences are made from: which of these two should we map modals onto? Modals in English necessarily precede negation, and so this is evidence that modals are in the highest of projections that make up tense/agreement inflection. The evidence we have seen that distinguishes Agr° from T° in French doesn't exist in English. We have seen no cases where a verb surfaces in a position higher than its D-structure place but lower than negation in English. When modals or the infinitival *to* appear, we see that both tense and agreement morphology are absent. This might be evidence that tense and agreement morphology in English in fact are mapped to the same X°. This would be consistent with the fact that English verb placement doesn't have the pattern we've seen in French, and that tense and agreement morphology necessarily co-occur. Höskuldur Thraínsson[6] has argued that a difference between French and English is how the languages have decided to distribute their morphology across syntactic positions. In French, as we've seen, tense morphology and agreement morphology are inserted into different D-structure positions and brought together in the syntax. Maybe English decides to put tense morphology and agreement morphology together into one X° position at D-structure, and let the syntax only determine on which verb that morphology gets pronounced. I'll take this view from now on. We can represent English finite clauses as in (114).

[6] Thraínsson, Höskuldur. 1996. On the (non)-universality of functional projections. In ***Minimal ideas***, ed. Werner Abraham, Samuel D. Epstein, Höskuldur Thraínsson, and C. Jan-Wouter Zwart, 253–281. Benjamins

(114)



This is just like the model of English sentences that we started with, except that I've renamed "I°" "T°." And to keep the structures of English T°s parallel to the inflectional head (= Agr°) that holds tense and agreement morphology in French, I've shown T° has having these two parts.

Bobaljik and Thráinsson (1998) argues that whether a language maps tense and agreement onto one, or two, syntactic positions correlates with a number of other differences in their clausal syntax in a way that confrims the idea that this can be the locus of language variation. That languages can differ depending on whether they put two features in different X° positions at D-structure, or "bundle" them together in one X° has found application in a variety of places. See, for examples, Giorgi and Pianesi (1996), Bianchi (1999), Pylkkänen (2008), Harley (1989), Legate (2014), and Erlewine (2018).

For structures like (114) to be possible D-structures, we will have to add to our inventory of $\overline{\text{X}}$ rules one that allows for an X° position to have two daughter X° positions. Let's therefore embellish the rules that generate D-structures so that they include (115g).

Recall that "$\{\alpha, \beta\}$" stands for a linear-order-free relation of sisterhood.

(115) a. $XP \rightarrow \{ \alpha P, XP \}$
b. $XP \rightarrow \overline{X}$
c. $XP \rightarrow \alpha P\ \overline{X}$
d. $\overline{X} \rightarrow \{ \alpha P, \overline{X} \}$
e. $\overline{X} \rightarrow \{ \alpha P, X^\circ \}$
f. $\overline{X} \rightarrow X^\circ$
g. $X^\circ \rightarrow \{ \alpha^\circ, X^\circ \}$

With this amendment behind us, let's return to the task of understanding the difference in how main and auxiliary verbs inflect in English. We should endeavor to capture the parallel constraints on the ways that main and auxiliary verbs are inflected in English. It is not clear how that could be done if the means by which they get their inflection is independent. If, on the other hand, the process is the same for both cases, then we can place these constraints on that process and in this way explain why they are shared. This is the direction we will take.

The c-command requirement and the Head Movement Constraint are presently formulated as constraints on the movement relation. That is where they were originally discovered and that is how they were initially framed, as a result. But we can now see that there is reason to rethink this decision. Main verbs do not move, and yet the Head Movement Constraint and the c-command condition nonetheless hold between a main verb and the I° its inflection derives from. We want the Head Movement Constraint and the c-command conditions to hold, therefore, on the inflection process itself. We then want those constraints to be reflected in how the inflected verbs moves to the I° that determines their inflection. I will sketch a proposal here that puts together some of the ideas in chapters 3 and 4 of Chomsky (1995). It is not a perfectly faithful a reproduction of those ideas, however.

The system I will lay out here is modeled after Adger (2003).

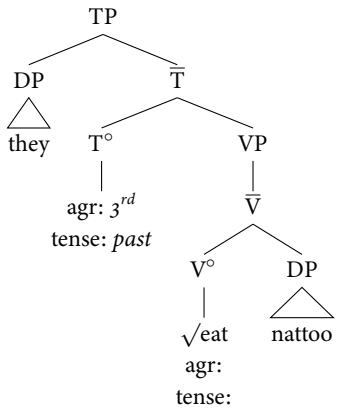
First, let us formulate the condition under which inflectional heads assign verbal inflection: the relation known as AGREE. We begin by expressing the inflectional morphology involved in terms of features. Let verbs come with features that will determine how their morphological form is determined, and let Agr° and T° also come with those features. This allows us to express the assignment of inflectional morphology in terms of matching the relevant features on a verb with the features on Agr° and T°. The AGREE relation specifies the conditions required for this matching of features to occur.

(116) AGREE (provisional)
Let X° and Y° dominate features, and let a feature in X° have a value and a feature in Y° be unvalued. A feature in Y° can get the value of a matching feature in X° just in case Y° AGREEs with X°. X° and Y° AGREE iff:
a. X° c-commands Y°, and
b. there is no Z° such that X° c-commands Z° and Z° c-commands Y°.

The head with the unvalued features is called the probe and the head with the valued features is called the goal. Chomsky (1995, 2000), where this system is laid out, requires that the probe be the c-commanding term, not the c-commanded one. When the probe is the lower of the two in the agree relation, we say that we have "upwards" agree. The unvalued features look upwards for their values. See Zeijlstra (2012) and Bjorkman and Zeijlstra (2019) who make the case that all agree relations involve at least an upwards component. In later sections we will see the need for downwards agree as well.

A simple transitive clause would get a representation like that in (117).

(117)

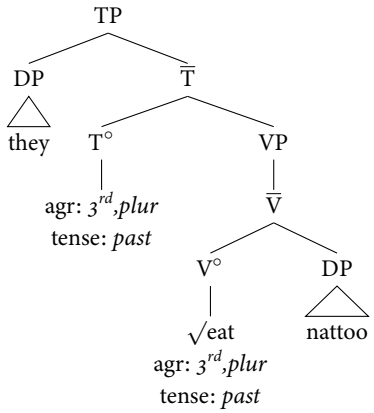


I have collapsed the internal structure of T° in this representation, and simply listed the two feature bundles—those associated with T° and those associated with Agr°—inside the highest T° node. I will do this when this space-saving doesn't create problems.

The verb in this example has features for agreement and tense which, when they are given values, will let the process of lexical insertion know which lexical item to put into this position. Similarly, the T° position is occupied by something that has features for agreement and tense and these are each given a value.

Because T° c-commands V°, and there is no head between them, the values for the agreement and tense features can value the agreement and tense features on V°. This produces (118).

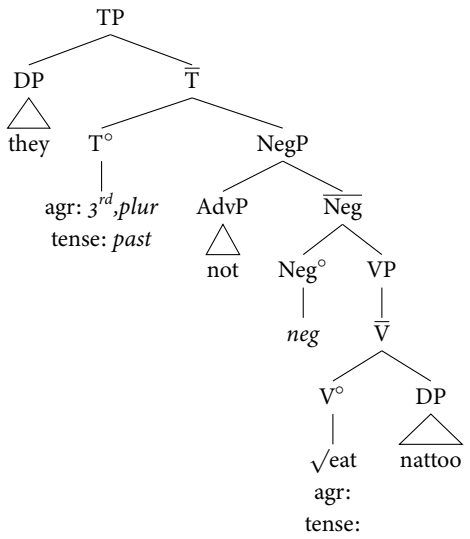
(118)



The values of inflectional features are assigned down to the verb, which then expresses those feature values morphologically.

We can now model the disruptive effect that NegP has on inflecting main verbs in terms of the "Head Movement Constraint," which is now expressed as a condition on AGREE. If we assume that the head of NegP does not come with the tense or agreement features, a sentence with NegP would get a representation like (119).

(119)



The values on the tense and agreement features that T° has cannot be given to the unvalued agreement and tense features on V° because Neg° is a head that intervenes. As a consequence, the agreement feature on the verb will also not be valued. The ungrammaticality of (119), then, can be credited to the inability of this system to inflect the verb with both tense and agreement when NegP is present.

To make explicit the requirement that values of the features on T° get expressed on a verb, let's adopt (120).

(120) The Stray Feature Filter
The values of a feature must be morphologically expressed. The morphologically interpreted features must be valued.

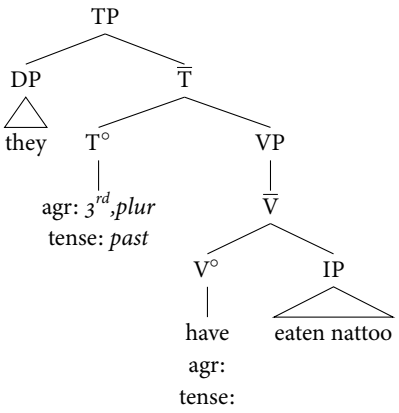
The Stray Feature Filter requires, first, that features with values must have a morphological expression. In general terms, this can be seen as the requirement that every morpheme in a sentence have the exponent which expresses that morpheme in the string of words that results. And, second, the Stray Feature Filter requires that morphemes that come with features must have the information in them that determines how those features should be expressed. That information is the values that features control.

The ungrammaticality of (119), then, comes from the fact that the value for the agreement feature is not found on a term that can express that value: it is not assigned to a verb, and only verbs in English are equipped with the morphology to express agreement. Moreover, the features on the verb, the ones that will express tense and agreement information, are not valued. The condition in (120) is just a restatement of Lasnik (1981)'s Stray Affix filter. His condition said that the bound morphology placed in an inflectional head could not be expressed without the aid of a verbal stem in that very same inflectional head. The Stray Feature Filter does nothing more than translate that into a framework which expresses inflectional morphology with features;

(120) says that the inflectional feature values carried by some inflectional head must find a way of being assigned to a verbal stem that can express them. I've named it the Stray Feature Filter to remind us of this connection.

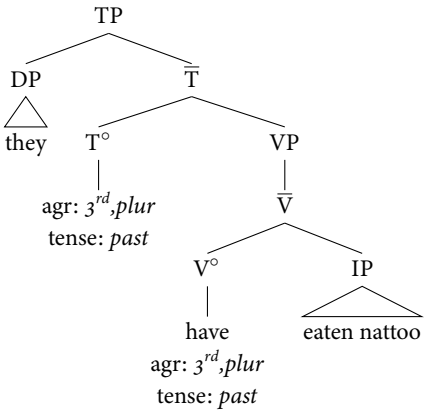
Let's now consider how this system will work with auxiliary verbs. A simple clause will have the structure in (121).

(121)



The AGREEment relation that we saw in the main verb case will apply here too, producing the representation in (122).

(122)



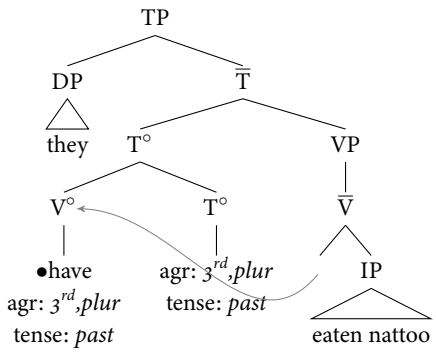
This representation satisfies all of our present conditions and so should lead to a grammatical string. As we have seen, however, this isn't what we want. Auxiliary verbs surface in T° in these contexts. There must be some property that auxiliary verbs have that prevents them from surfacing in their D-structure position when they are inflected. I don't know what this property is, but it is possible, nonetheless, to characterize the consequences it has. It requires the auxiliary to adjoin to the inflectional heads which are responsible for giving values to the features the auxiliary contains. If we call this mysterious property: ●, then we can state this requirement with (123).

(123) If a morpheme or feature has ●, then the X° containing that morpheme or feature must be pronounced inside a Y° that also contains whatever X° AGREEs with.

(123) says the same kind of thing that the Word Criterion expresses. The Word Criterion combines the Stray Feature Filter with the condition on ●s expressed here. The Word criterion requires that all of the parts of a word that have been put in separate places at D-structure find themselves in one X° by S-structure. The ● expresses the same requirement: the two features that are in an AGREE relation find themselves within one X° by S-structure. What we learn from the inflection of main verbs in English, is that not all affix/stem pairs are subject to this requirement. The ● expresses which pieces of a word are subject to this condition. A feature that has ● is what was called a "strong" feature in the previous section.

Auxiliary verbs have •, and so representations like (122) will not be well-formed S-structures. Instead, the head movement operation must kick in to produce (124).

(124)



The fact that auxiliary verbs differ from main verbs in needing to move is now coded in the • property. We should hope to find what the • property is, and it seems likely that it has something to do with how the inflectional morphology is expressed since it is a requirement that makes reference to the features expressed by that morphology. One possibility would be to let auxiliary verbs differ from main verbs in the kinds of features they come with.If, for instance, auxiliary verbs are not equipped with the tense and agreement features, but the morphological forms they have nonetheless express those features, then the Stray Feature Filter could not be satisfied by way of AGREEment. There will be no way of passing the values of tense and agreement onto the auxiliary verbs on this view because auxiliary verbs do not come with those features. If the Stray Feature Filter can be satisfied by letting the syntax form a word in which the tense and agreement feature values are expressed, then we could see head movement as the sole agent responsible for forming that word in the case of auxiliary verbs. The T° in (124), on this view, would satisfy the Stray Feature Filter. This interpretation of the • property would amount to letting the features and the morphology that expresses those features be packaged in two different ways. In one, the features and the morphology are separated in D-structures and are brought together by syntactic means. This is what happens with auxiliary verbs in English. In the other, the morphology and the features are packaged together into one lexical item at D-structure, but the values those features acquire are determined by AGREE with another morphological item that resides elsewhere in the syntactic representation. This is what happens with main verbs in English.

While this seems to me a natural way of expressing what the • property is in these scenarios, I'll not commit to that account in these notes. Instead, I'll continue to leave this property unaccounted for, and rely just on the descriptive consequences it has.

The • property can be used to describe what it is that makes auxiliary verbs, unlike main verbs, move. But we must still explain why main verbs

To my knowledge, this method of forcing auxiliary verb movement in English is not found in the literature. It seems to me, however, a natural way of expressing this fact about auxiliary verbs with a technology that involves •s.

See Lasnik (2003) for this idea.

are unable to move. One idea that has had found some favor is to posit a force that disfavors movement.Chomsky (1995) suggests that something of this sort is responsible for blocking movement of main verbs. He observes that we could relate the fact that main verbs are not required to move, that is, they don't have the • property, to the fact that they cannot move if we could build into the grammar a force that allows movement only when it is required.Chomsky's own proposal is built upon a different account of how main verbs get inflected, and so will not work for us. But David Pesetsky[7] has proposed a way of expressing this notion that is very close to Chomsky's and fits our account of verbal inflection.

To see Pesetsky's idea, it'll be useful to recall what our definition of a sentence is. When we first encountered movement operations in the last chapter, I laid out how we can define the notion of a sentence that involves the collection of parses that we've been using ever since. We allowed for a sentence to involve a derivation, which we think of as a series of parses created by the transformational rules we've been developing. The definition of a sentence is the set of such derivations for a particular D-structure/S-structure pair. The definition is repeated here.

(125) $\mathscr{D}$ = (P_1, P_2...,P_n) is a derivation iff:

a. P_1 is formed by the $\overline{X}$ Skeleton and Lexical Insertion, and it is semantically interpreted,

b. Each P_{i+1} is formed from P_i by a rule,

c. P_n is subject to the linearization algorithm.

(126) A sentence is the set of derivations which have the same D-structure (P_1) and S-structure (P_n).

When more than one rule is used to convert a D-structure parse into an S-Structure parse there are, in principle, more than one way of applying those rules. For instance, if a D-structure requires Rule 1 and Rule 2 to be applied to create a licit S-Structure, then that sentence has potentially two derivations: one that applies Rule 1 before Rule 2, and another that applies Rule 2 before Rule 1. What (125b) allows is for us to think of the sentence with that D-structure as including both of these derivations. A sentence, according to (125) is all the ways the rules of syntax can be applied to a given D-structure so that a particular S-structure is achieved.

Pesetsky's idea is to put a constraint on this set of derivations. Because the aim of this constraint is to control which S-structures are possible, it requires a slight modification of (125b). Rather than require a sentence with a certain D-structure to have a specific S-structure, Pesetsky's condition on derivations will determine which S-structures are possible from a given D-structure. So first we change (126) to (127).

(127) A sentence is the set of derivations which have the same D-structure and some well-formed S-Structure.

Chomsky assumes that main verbs inflect by moving "covertly" to the inflectional heads that hold the relevant morphology. He then assigns different weights to covert and overt movement so that covert movement is favored. A difficulty with his proposals that has not yet been overcome is defining a notion of "covert" movement that can do what is required of it in other contexts. Covert movement is thought to arise in other contexts, but in those contexts it always has semantic consequences. Covert main verb movement, by contrast, does not. It appears that covert movement should be designed in such a way that it must have semantic consequences, and it is therefore not a suitable mechanism for inflecting main verbs. See Stechow (1996) and Fox (2000) for some discussion.

See also Lasnik (1995) and Stroik (1999) for an exploration of the idea that movement cannot be optional.

[7] Pesetsky, David. 1989. The earliness principle. Paper presented at GLOW

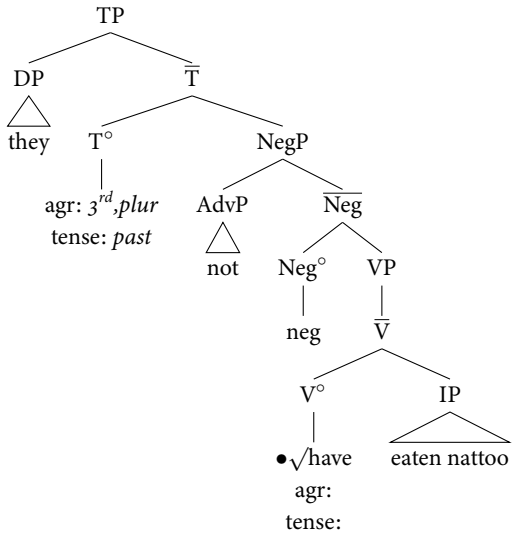
Now we can give Pesetsky's condition, which will select from the various possible S-structures some D-structure can produce, ones that are allowed. Pesetsky calls his condition "Earliness," and I will formulate it with (128).

(128) Earliness
Let the set of derivations, $\mathscr{D} = \{D, D_1, \ldots, D_n\}$, be a sentence. There can be no $D_i \in \mathscr{D}$ if there is a D_j in $\mathscr{D}$ such that $|D_i| > |D_j|$, where "$|D|$" is the number of elements in D.

What Earliness requires is that derivations from D-structure to S-structure minimize the number of parses in the derivation. If it is possible to satisfy the constraints on S-structure without moving, for instance, then movement is blocked. This is precisely what happens in the inflection of main verbs. They are able to satisfy the Stray Feature Filter by way of AGREE, and this makes movement of main verbs to T° superfluous.

This is a popular account of the failure of main verb movement, but it is not entirely successful. This becomes apparent when the cases in which an auxiliary verb combines with NegP are considered. In such cases, we have the D-structure representation in (129).

(129)



As we have seen in the case of main verbs, AGREE is prevented from giving the value that T° has for the agreement feature to the verbs beneath T°. Just as in the case with main verbs, then, this representation does not satisfy the Stray Feature Filter. Instead, as we know, the auxiliary verb moves to the T° position. It seems reasonable to assume that it's the ability of the auxiliary verb to move that allows it to get inflected. We want movement and AGREE to have different outcomes in this circumstance.

I don't know of a working solution to this question that has gained currency in the literature. I will make one up here, just so that we have something that works. We are looking for something that distinguishes auxiliary verbs from other verbs that allows just auxiliary verbs to move to T° when Neg° is

Earliness comes from the unpublished Pesetsky (1989), which can be found at his website. This is not his formulation of the condition, but I think it expresses his idea. See also Pesetsky (1998) and Pesetsky and Torrego (2001) for other formulations. A derivation that is not yet blocked but should be involves moving a main verb to Neg°, where it will be close enough to AGREE with T°. That should allow a main verb to move past negation into T° in the same way that auxiliary verbs can (see below). One way of blocking that derivation would be to adopt a condition that requires that every step in a derivation to repair a violation. The movement of a main verb to Neg° would not repair a violation; it would be a step that puts the verb into a position in which a violation (of the Stray Feature Filter) could be repaired. See Takahashi (2006) for a version of Earliness that has this consequence.

present. One thing that distinguishes auxiliary verbs in English from others that might be relevant is that auxiliary verbs have negative forms, and others don't.

(130) a. Tanja hadn't run today.
b. Karen isn't running today.

(131) a. * Tanja ran't today.
b. * Karen runsn't today.

One might be tempted to think that [n̩t] that is found on these auxiliaries is mere *not* in some prosodically reduced form. If that were the case, then one could explain why only auxiliary verbs in English have these forms with the fact that only auxiliary verbs in English are capable of showing up to the left of *not*, thereby feeding the reduction process. But as Stephen Anderson[8] notes, the [n̩t] forms of auxiliary verbs move as heads in contexts where T° moves to C°.

[8] Anderson, Stephen R. 2008. English reduced auxiliaries really are simple clitics. ***Lingue e linguaggio*** 7:169–186

(132) Haven't you finished this chapter yet?
Wasn't she happy with the solution?

And this distinguishes these cases from other cases of reduction, which don't have this property of being able to feed movement. For example, auxiliary verbs themselves can be reduced onto certain DPs in Specifier of TP position; this is illustrated in (133).

(133) a. She said that you've been tense.
b. Who thought that who's pleasant?
c. She thinks that he's pleasant.

The forms *you've*, *he's* and *who's* prosodically behave like a single unit. But the syntax does not recognize them as a constituent, as can be seen by noting that Wh Movement isn't able to move them.

(134) a. * who've did you think ⌐ been tense?
↑_____┘
b. * who's did she say ⌐ pleasant?
↑_____┘

If we treat forms like *isn't* and *haven't* as single X°s, this is explained. Under our present set of assumptions, for something to be an X° requires that they be a word.

Imagine, then, that auxiliary verbs can come with a feature that expresses its polarity, as indicated in (135).

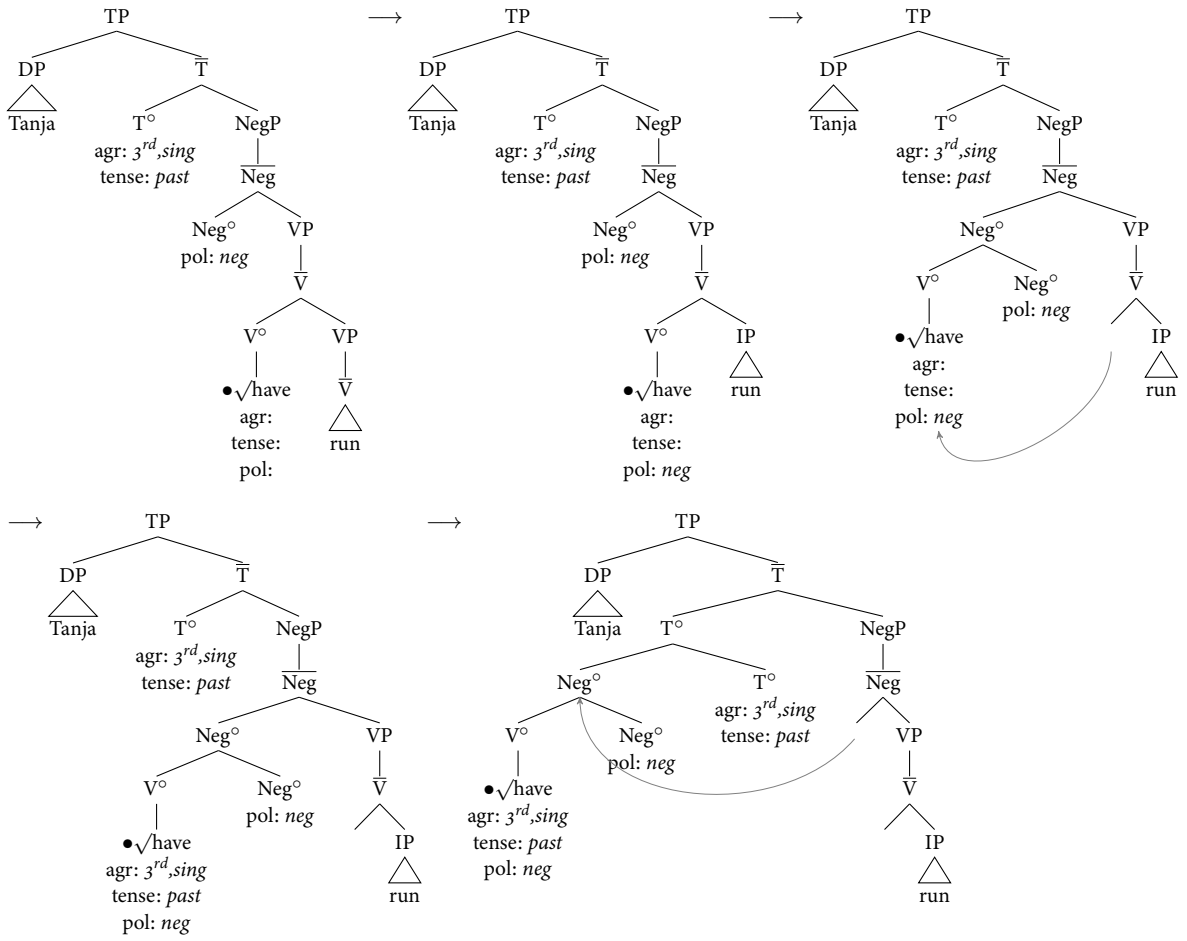
I'll use the "⇒" symbol to represent Vocabulary Insertion, the S-structure process that matches a word to clusters of morphemes and features in an X°.

(135)

V°	⇒ *has*	V°	⇒ *hasn't*	V°	⇒ *isn't*
√have		√have		√be	
tense: *pres*		tense: *pres*		tense: *pres*	
agree: 3rd,*sing*		agree: 3rd,*sing*		agree: 3rd,*sing*	
pol: *pos*		pol: *neg*		pol: *neg*	

Consider now how the syntax for negated auxiliaries proceeds.

(136)



In the first step, AGREE values the pol feature on *have*, and this allows for the second step in which the V° containing *have* moves to Neg°. This movement is required by the • feature on *have*. In Neg° position, *have* is now close enough to T° for AGREE to hold, and this what allows for the fourth parse in this derivation, where the tense and agreement features on *have* are valued. This is again recapitulated by movement of Neg° into T°, which is again required by the • on *have*.

This syntax will also underlie (137).

(137) a. Smith had not run.

b. Jones is not running

What we see here is that the polarity feature on *have* and *be* can be valued to *neg* and not result in Vocabulary Insertion producing *haven't* and *isn't*. Instead, *not* expones negation. The morphological machinery developed here doesn't provide a way of capturing this. For this proposal to be successful, then, it requires some process that is capable of capturing the dependency

between pronouncing *not* and pronouncing n̩t. I gamble that such a process exists, and suggest that we adopt this account of the ability auxiliary verbs have in English to move past Neg°.

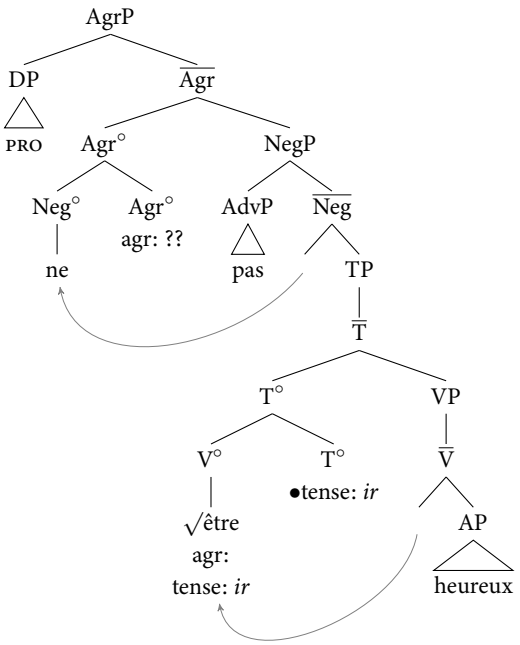
There are problems with Earliness from examples in which head movement appears to be optional. We've seen something with that appearance in two contexts. In French infinitival clauses, we've seen that auxiliary verbs are able to surface in one of two positions: to the left of *pas* or to the right of *pas*. This is illustrated by (83), repeated here.

(83) a. ne *pas* **être** heureux est une condition pour écrire des romans
ne not be happy is a condition for writing novels
'To not be happy is a condition for writing novels.'

b. N'**être** *pas* heureux est une condition pour écrire des romans.
ne'be not happy is a condition for writing novels
'To not be happy is a condition for writing novels.'

Consider the representation given to (83a), in (138).

(138)

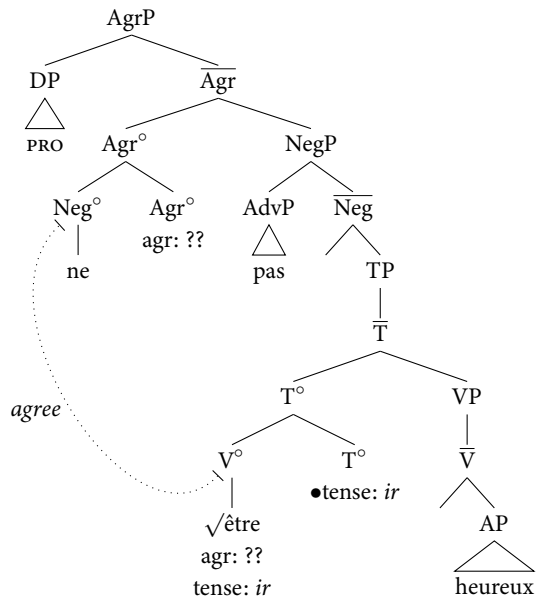


I have given the infinitival inflectional feature inside T° the • property, as this head forces head movement of both auxiliary and main verbs to it. (Recall: we saw that main verbs in French infinitival clauses move to T° and are thereby separated from their DP complements by adverbs.) In general, we can characterize those inflectional heads that allow main and auxiliary verbs to move to them, i.e., those heads that are described as being "strong," in Pollock's paper, as having the • property.

Consider now the status of Agr° in this representation. The inflectional material this head is associated with is obscure; there is nothing in infinitival

clauses that looks like the agreement phenomenon we see in finite clauses. I will indicate the value of the agreement morphology in non-finite Agr° with "??," as a consequence. In (83a), we don't want this morphology to force movement, so I have placed no • on it or the auxiliary verb. Note that *ne* has moved to Agr°, and this now allows Agr° to AGREE with T°. The conditions on AGREE do not allow it to hold between two heads if there is another head intervening. The position Neg° has at D-structure causes it to intervene between Agr° and T°, so at D-structure AGREE would be blocked between Agr° and T°. But once Neg° moves to Agr° in the way indicated in (83), this blockage is alleviated and AGREE is possible. From (83) we can create an S-structure that satisfies the Stray Affix filter by AGREE.

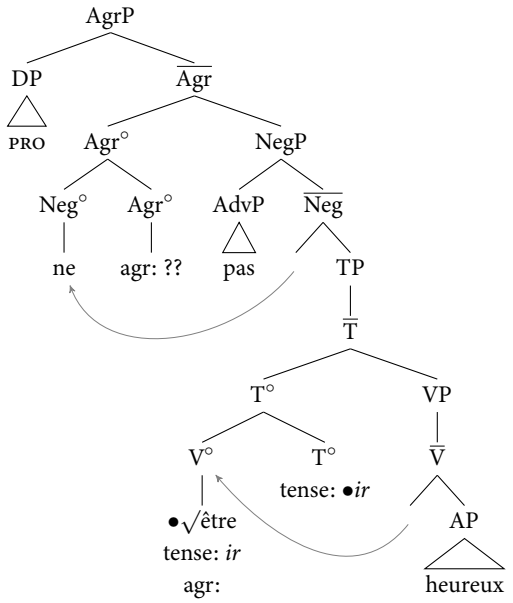
(139)



This is how (83a) is derived.

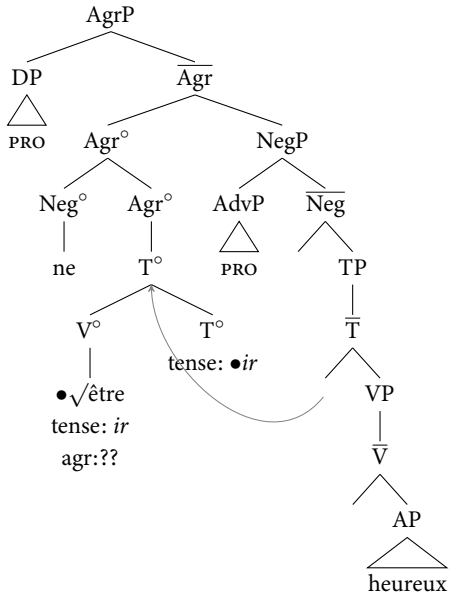
But the outcome in (83b) requires a different set of assumptions. In that circumstance, an auxiliary verb that Agr° AGREEs with moves to Agr°. We want this option to only be available to auxiliary verbs. The technology we have allows that to be expressed by giving to auxiliary verbs the •. For (83b), then, our representation should be (140).

(140)



Once AGREE holds between Agr° and T° in this configuration, the • value on *être* will require movement, forming the representation in (141).

(141)



The • property requires that a verb expressing the value the agr feature be adjoined to Agr°, and this results in the auxiliary verb moving to Agr°. Since this is the only derivation in which the • property gets its requirements satisfied, Earliness permits it. To characterize "optional movement," then, a system which embraces Earliness requires that there be two possible representations: one that forces movement and another in which it is blocked. Under the particulars of the account I am sketching here, that would mean that French speakers who allow the alternation in (83) allow for two kinds of auxiliary verbs: one with • and the other without. Some French speakers report that

these two word orders correspond to two registers of French. Perhaps this difference in registers, if it is indeed systematically associated with these two word orders, could be expressed as a kind of difference in vocabulary. In one register, auxiliary verbs have • and in the other, auxiliary verbs do not have •.

If these methods for characterizing the flexible position of certain heads can be confirmed, then we can continue to entertain the hypothesis that Earliness exists, and rely on its consequences to block main verbs from moving in English (non-negated) sentences. Earliness has the interesting property of relating the absence of main verb movement in English to the presence of obligatory auxiliary verb movement in English. It, along with the • attribute, expresses the idea that the controls on movement will allow for just two categories: obligatory movement or no movement. With the possible exceptions found in infinitives, this does seem to be the case for all the instances of head movement we've investigated. An open question is whether this holds more generally. All the cases of Argument Movement that we've looked at are obligatory, so those cases seem to fit the prediction. We've also briefly encountered Wh Movement, which is responsible for producing constituent questions of the sort that (142) exemplifies.

(142) She asked what Sam gave to him.

In simple questions, this seems to be a movement that is also obligatory. (142) would be ungrammatical if *who* did not move. But in questions where there is more than one moveable term, as in (143), it is not obligatory for all those terms to move.

(143) She asked what Sam gave to whom.

In these kinds of questions, the terms that are not required to move are also prevented from moving; (144) is ungrammatical.

(144) * She asked what whom Sam gave to .

So, these cases fit the prediction as well. We need only say that the CPs which are interpreted as constituent questions invoke a requirement that a wh-phrase move to their Specifier position, and Earliness will permit only one wh-phrase to move, since any additional movements will not be needed to satisfy that requirement.

For Argument Movement, Wh Movement and Head Movement in English, then, a grammar that embraces Earliness seems to be consistent with the facts. But it's not hard to find cases that don't conform to Earliness. In many languages, for instance, Wh Movement appears to be optional and there are also languages in which Argument Movement appears to be optional. And Heavy NP Shift in English is not transparently consistent with Earliness either. It is difficult to classify either of the examples in (145) as ungrammatical, for instance.

(145) a. She gave a lot of books to me.
b. She gave to me a lot of books.

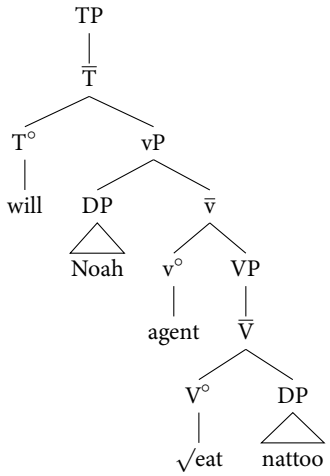
But if (145b) is derived from (145a) by movement, then Earliness should block it. There are analyses of all these counter-examples that make them consistent with Earliness, or with some modified version of Earliness, but no consensus has formed about whether these analyses are successful. The validity of Earliness, then, is an open question, under active investigation.

In any case, let's adopt this system. Let's assume that verbs inflect by way of the AGREE relation, and that the Head Movement Constraint is a constraint on this relation and not movement. As we've seen, in order to spread this constraint on AGREE to cases where head movement arises, we need a method of explaining how head movement is tied to AGREE. Head movement needs to be seen as one outcome of the AGREE relation, and therefore an outcome that inherits the constraints on AGREE. I've sketched a way of doing that in this section, but it's clear that this is not yet a full account. It relies on finding a source for the • property, and for understanding what prevents main verbs from moving through Neg° into higher heads in English. And it relies on Earliness, which has effects across the grammar that have yet to be confirmed.

4.6 AGREE *and θ-roles*

Now that we've developed a vocabulary for describing how morphological information is spread across the heads of a syntactic representation, let's revisit the issue we left at the end of the previous chapter concerning θ-roles. We discovered there that some things fell into place, Burzio's Generalization for instance, if we took the external θ-role associated with the meaning of a verb to be syntactically separated from rest of the ingredients in the meaning of a verb. The syntactic representation we assigned to verbs that have external θ-roles is like that in (146).

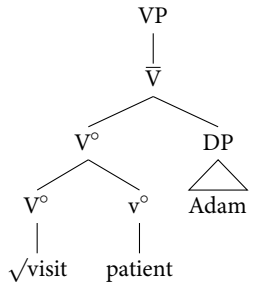
(146)



The choice of verb that gets inserted into V° position is tied to the choice of θ-role that is inserted into v° position, and this connection seems to be arbitrary. There has not been a pattern established between the nature of the verb in V° and the nature of the θ-role in v°. So far as I know, this is simply learned: *eat* is obligatorily accompanied by an Agent θ-role in v°, whereas, *like* is obligatorily accompanied by an Experiencer θ-role in v°, and so on. To the extent that this connection is arbitrary, it suggests that there is a morphological connection between the θ-role in v° and the verb in V°. We want the connection between the morphology in v° and the morphology in V° to be the one that a word makes. Now that we have technology for that, let's see how it might be applied to this case.

When we first looked at the denotation of verbs, we embraced the idea that verbs have two pieces to their meaning. One of those pieces describes the activity, state of affairs, or what have you. The other piece describes the θ-roles that verb uses to relate its arguments to that state of affairs. We've since spelled out this idea in terms of an event semantics. Verbs include information that determines what sort of event they describe and also the θ-roles that relate the arguments the syntax will supply to that event. Imagine that these are independent pieces to the vocabulary item that correspond to the words *eat*, *like*, etc. Imagine, that is, that lexical insertion populates D-structures that have a position for a verb with these pieces. Setting aside the features that represent tense/agreement inflection for a moment, the proposal is that V° positions like (147) can be produced at D-structure.

(147)



I've called the X° position that PATIENT is inserted into v°, because that's what we've called the head position that external θ-roles appear in. I've represented the portion of the verb that corresponds the event description with $\sqrt{}$visit. This is often called the "root" of the verb. If we give to the Patient θ-role the same kind of denotation we adopted by external θ-roles, then the semantic contribution of each of the pieces in the maximal V° position of (147) is (148).

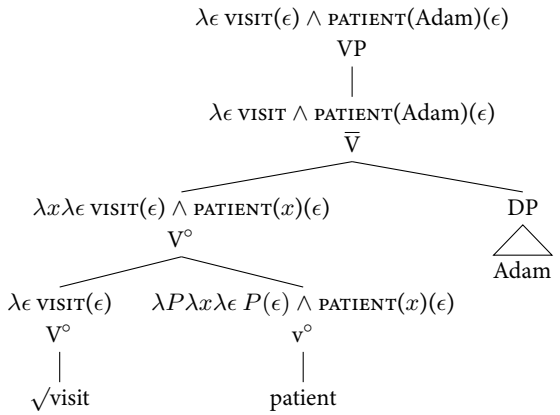
(148) a. $[\![\sqrt{\text{visit}}]\!] = \lambda\epsilon\ \textsc{visit}(\epsilon)$

b. $[\![\text{patient}]\!] = \lambda P\ \lambda x\ \lambda\epsilon\ P(\epsilon) \wedge \textsc{patient}(x)(\epsilon)$

Notice that these denotations, along with our rules of semantic composition, produce the correct meaning with this syntax.

If we take the meanings of verbs to be derived syntactically in the way that is illustrated here, we will have to change our conception of the rule of Existential Closure introduced in the previous chapter. That rule was offered as a rule that changed lexical items—verbs—by binding off their internal argument with an existential operator. The view introduced here is that verbs are the way in which morphologically complex V°s are pronounced. They are not themselves morphemes with idiosyncratic denotations, and so they cannot be the input of a rule that builds new words from old ones. Instead, we'll have to take Existential Closure to be a syntactic rule that applies to V°s.

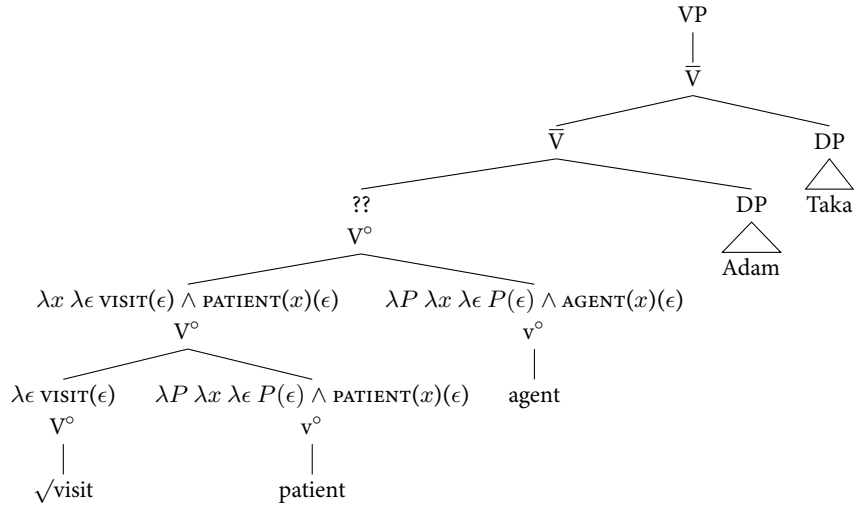
(149)



It's possible to see the meanings of verbs, then, as being constructed in the syntax and matched to the vocabulary items we recognize as verbs. The idea I'll sketch here is that external θ-roles are simply a special case of the situation illustrated in (147).

Notice that when there are two θ-roles, they cannot semantically combine with the rules we adopted, if they do so within one X°. (150) is illicit.

(150)



The denotations for both Patient and Agent requires that they combine with something whose denotation is a function whose domain is the set of eventualities. That requirement is satisfied for the Patient, but not for Agent. The V° Agent combines with is a function whose domain is the set of things, not the set of eventualities. These two denotations cannot combine. When a θ-role combines with a root, it creates something whose meaning can combine with the argument of the θ-role – it creates a verb whose meaning is a function from things to descriptions of eventualities. That isn't the right kind of meaning to combine with another θ-role.

What this demonstrates is that if we take verbs to be matched to a X° position in which the θ-roles and eventuality descriptions have been assembled by

Lexical Insertion, we can use the rules of semantic composition to constrain the meanings that verbs have. If we have the denotation for θ-roles right, and there are no other relevant rules of semantic composition, one prediction is that a verb can assign no more than one θ-role. We have taken θ-roles to mediate the connection between verbs and their DP and PP arguments, so the fact that verbs can combine with more than one DP and PP argument shows that something about these assumptions is incorrect. If θ-roles only mediate the combination of verbs with DPs, however, we are closer to the mark. Setting aside specialized constructions, English verbs allow for only one internal DP argument.

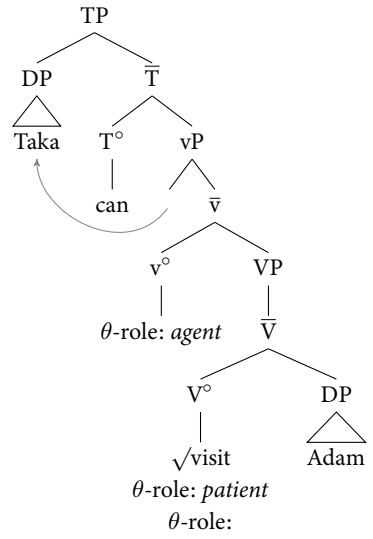
One of the specialized constructions is the so-called double object construction: "Helen gave Matt a book." We return to look at these cases in chapter 5.

Let us provisionally adopt the view, then, that PPs, along with CPs and other clause-like objects, do not semantically combine with their verbs by way of θ-roles. If that is correct, the syntax-semantics given to θ-roles derives the fact that for a verb to combine with more than one DP argument, one of those θ-roles will have to be external. In the case of *visit*, that θ-role is Agent.

What is required, then, is to find a syntax in which the denotation of the Agent θ-role is put in a place different than where the other other pieces of a verb's meaning are assembled. This is precisely the same situation that arises with tense inflection. The denotation of tense needs to be in a place different than where its exponent is, and AGREE is the way this is achieved. We'll allow θ-roles to be features, then. A verb that comes with two θ-roles will allow one of those θ-roles to be inserted into the V° position valued. The other θ-role will be placed in a position that gives it the right kind of semantic argument, but distances it from the verb—its exponent. Thus, in the V° position will be an unvalued θ-role feature that gets into the AGREE relation with the external θ-role, thereby allowing the external θ-role to be exponed, and satisfy the Stray Feature Filter.

Reverting back to the truncated way I've represented X° positions with complex contents, we can represent the D-structure scenario just described with (151).

(151)

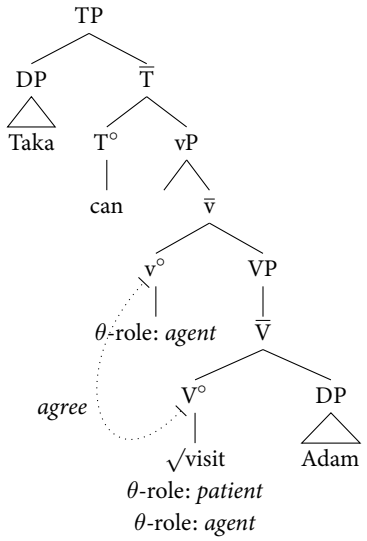


Imagine that the Spell Out rules for *visit* are (152).

(152) $[_{V^\circ}\ \sqrt{}\text{visit}+\textit{patient}+\textit{agent}\ ] \Rightarrow /\text{VIZIT}/$

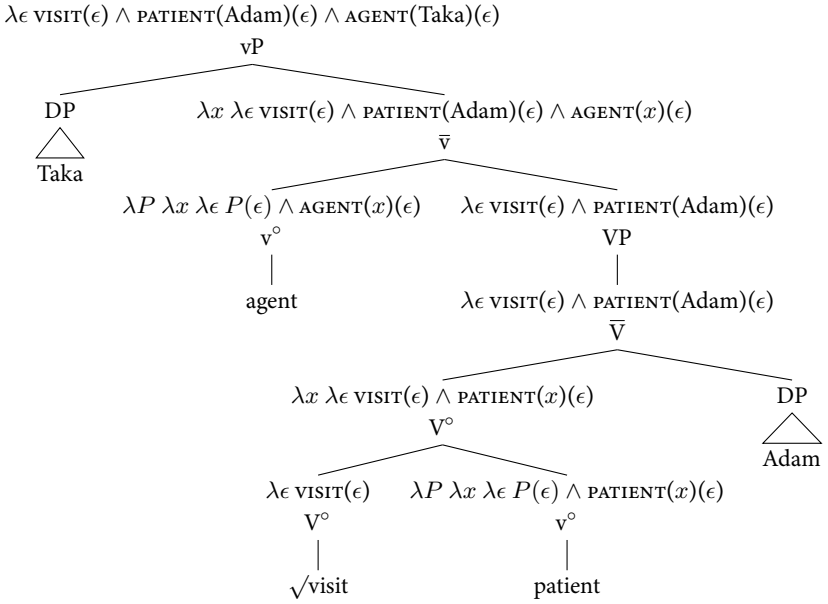
This will require AGREE to produce from (151), (153).

(153)



This representation satisfies the constraints on S-structures – in particular the Stray Feature Filter – and connects the presence of the verb's external θ-role in a position outside of the VP headed by that verb with the verb. And, as we saw in the previous chapter, our semantics correctly combines the denotations of the morphemes in (153). We can see this by considering what the D-structure of the vP in (153) looks like:

(154)

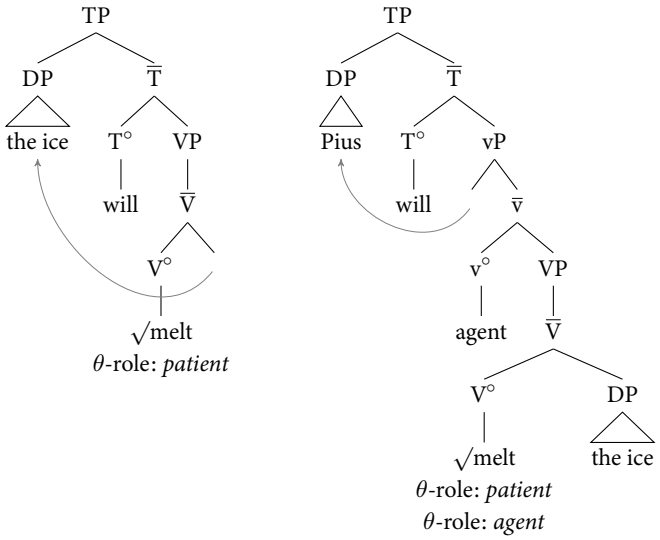


We can also use the rules for Vocabulary Insertion to account for optional external θ-roles. Consider, by way of illustration, the verb *melt*. We've seen that this verb has both a transitive and intransitive guise. This can be expressed by letting the vocabulary item mɛlt be able to spell out a V° that contains √melt, whether or not it also contains Agent.

(155) a. [$_{V°}$ √melt+*patient*+*agent*] ⇒ /mɛlt/
b. [$_{V°}$ √melt+*patient*] ⇒ /mɛlt/

This permits either of the representations in (156) to contain *melt*.

(156)



4.7 AGREE *and Case*

We have now encountered two systems in which inflectional morphology is assigned by one term to another. In addition to the cases examined in this chapter, where an inflectional head assigns feature values to verbs that they c-command, we made use of a system of Case assignment in the previous chapter to determine the position of argument DPs. These two systems do something very similar: they both determine the inflectional morphology on a term by way of the proximity that term has to an item whose presence seems to be responsible for giving a value to that inflectional morphology. In the scenarios involving Case, the values range over nominative, accusative and genitive; and the terms that seem to be responsible for fixing those values are finite (and maybe some infinitival) T° (for nominative), a determiner for genitive Case, and the external θ-role assigner, v, and prepositions (for accusative). Given the similarities in what these two systems are doing, it is reasonable to hypothesize that they are the same system. We'll consider in this section how the two might be collapsed.

I'll set aside how genitive Case is assigned, and restrict attention to just nominative and accusative. Presently, the two systems are as defined in (157) and (158).

(157) Verbal Inflection

a. AGREE (provisional)
Let X° and Y° be terms that dominate features, and let a feature in X° have a value and a feature in Y° be unvalued. A feature in Y° can get the value of a matching feature in X° just in case Y° AGREEs with X°. X° and Y° AGREE iff:
   i. X° c-commands Y°, and
   ii. there is no Z° such that X° c-commands Z° and Z° c-commands Y°.

b. Stray Feature Filter
A valued feature must be morphologically expressed and an unvalued feature must be valued at S-structure.

c. If a term has •, then the X° containing that term must be inside a Z° that X° AGREEs with at S-structure.

d. Feature assignments
   i. An agreement feature is part of certain Agr°s and verbs. It comes with a value in Agr°/T°.
   ii. A tense feature is part of certain T°s and verbs. It comes valued in T°s.

The statement that unvalued Case features are a part of argument DPs is cryptic. Our model requires that all of the features and morphemes come into a sentence through lexical insertion. For a feature to be a part of a phrase amounts to saying, then, that there is a head in that phrase whose lexical item contains that feature. What that head is, and why it is found in argument DPs, but not non-argument DPs, is unknown.

(158) Case Assignment

a. Nominative Case is assigned by finite T°/Agr° to the Specifier of the phrase it heads.

b. Accusative Case is assigned by v° and some prepositions to positions they:

i. c-command, and

ii. are klose to.

c. The Case Filter
An argument DP must be in a Case-marked position in some parse in a derivation.

We saw evidence that Accusative Case assignment is subject to a locality condition, and this is what the "klose to" clause in (158) refers to. We haven't yet arrived at a working definition of this locality condition, however.

To bring the inflectional feature system and the Case system together, I'll start by translating Case into the feature based system that we've expressed verbal inflection with. We'll let certain T°s, prepositions, and v come with a value for the Case feature, and we'll assume that every DP has an unvalued Case feature. We'll model Case assignment, then, as fixing values for the Case feature under AGREE. Taking this step will create the system in (159)–(163).

(159) Let X and Y be terms that dominate features, and let a feature in X have a value and a feature in Y be unvalued. A feature in Y can get the value of a matching feature on X just in case Y AGREEs with X. X and Y AGREE iff:

a. X c-commands Y, and

b. X is klose to Y.

(160) X is klose to Y iff:

a. For Y a head, there is no Z° such that X c-commands Z° and Z° c-commands Y.

b. For Y a phrase, <something yet to be determined>

(161) Stray Feature Filter
Every valued feature must be morphologically expressed and every unvalued feature must get a value at S-structure.

(162) If a lexical item has •, then it must be pronounced adjoined to the heads whose features it is in an AGREE relation with at S-structure.

(163) Feature assignments

a. An agreement feature is part of verbs and certain T°/Agr°s. It comes valued in T°/Agr°.

b. A tense feature is part of certain T°s and verbs. It comes valued on some T°s.

c. A unvalued Case feature is part of every argument DP.

d. A valued Case feature is part of v and certain prepositions.

The following changes have been made. First, the terms that can enter into an AGREE relation have been widened to include phrases as well as heads. The locality condition imposed on the term that has the value to assign and the term whose feature gets assigned that value has been relativized so that it discriminates between the cases in which a head or phrase are involved. If a feature is in a head, then the Head Movement Constraint applies; if a features is part of a phrase, then the definition there is a different locality condition invoked, one whose particulars we have yet to figure out.

Next, the Case Filter has been stated as a condition on features. The requirement is now that unvalued features must get a value before the term that bears those features can be pronounced. This condition has been added to the Stray Feature filter, which requires that values get assigned to terms that can express them. The combined condition is renamed the Feature Valuation Requirement. Finally, valued Case features are stated to be part of our former roster of Case assigners, and unvalued Case features are stated to be part of the DPs that the Case filter holds of.

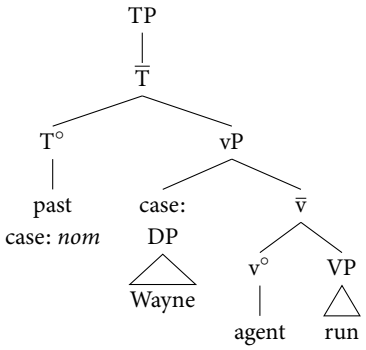
This reformulation leaves out the condition under which nominative Case assignment was formerly thought to apply. That condition had nominative Case assigned to the Specifier position of TP by certain agreement morphemes that reside in T°. It seems that it is the agreement that determines nominative Case because it appears that whether nominative Case is available depends on the kind of clause involved, and the "kinds" can be distinguished by their heads. The assumption that nominative Case is assigned to the Specifier of TPs, is assumed because it allows for an explanation of why argument DPs seem to be forced to move into that position when no other Case is available to them. Together, these two assumptions require that nominative Case be assigned by T° to their Specifier positions. To fold nominative Case assignment into this system requires we abandon one or the other of these assumptions.

In fact, both assumptions are suspect. As you might have noticed, the fact that Argument DPs must move to Specifier of TP is often redundantly captured by the EPP. We might conjecture, then, that nominative Case is assigned to some other position, and let Argument movement to Specifier of TP be driven by the need to satisfy the EPP. On the other hand, we've just seen that where nominative Case is available seems to depend on the nature of T°. There is a way of keeping both these ideas.

Consider what would be expected if the features in finite T° were to assign Case under the conditions that prepositions and vs do. In examples in which no NegP is present, we would expect the Specifier of vP to be the position that nominative Case is assigned to. A sentence like (164) would have the D-structure indicated.

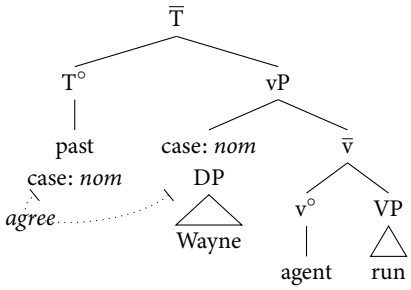
Only the Case features are indicated in (164).

(164) Wayne ran.



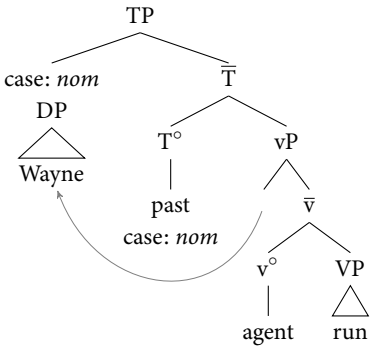
The DP bears an unvalued Case feature, a feature that must be valued before the Stray Feature Filter will be satisfied. We can invoke AGREE to remedy this.

(165)



This representation satisfies the Stray Feature Filter (the former Case Filter), but it does not yet satisfy the EPP, which requires further movement of *Wayne* to Specifier of TP. The result is (166) below, which corresponds to the grammatical S-structure outcome for this sentence.

(166)



Under this interpretation of how nominative Case is assigned, Argument movement must be allowed to move arguments from positions in which they get Case marked. There are cases which suggest Argument movement should be disallowed from moving from Case marked positions; (167) is one such example.

(167) * Wayne seemed to that pigs fly.

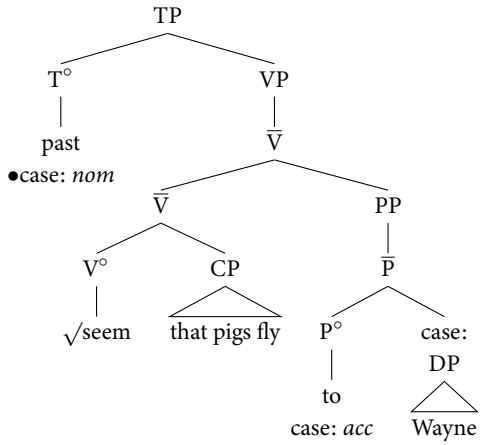
Adopting this view of nominative Case assignment therefore requires resolving this conflict.

One proposal is to allow movement from Case marked positions only when the argument moves to the Specifier of the head responsible for assigning that Case. Indeed, one proposal is to reduce the EPP to the • property. Imagine that the nominative feature associated with T° gives T° the • property. The • property would then require that T° be put together with the DP whose Case feature it values. The Likes-Attracts-Likes Constraint, however, prevents phrases from adjoining to heads, so it will prevent the DP from merging directly with T°. If we loosen the statement of the requirement that the • property invokes so that the Likes-Attracts-Likes Constraint is left to determine how the moving term adjoins to the head with the • property, then we could see the EPP as just another instance of the • property. The requirement invoked by the • property could be expressed as (168).

(168) If a term, α, has •, then a projection of the X° that contains α must be pronounced as a sister to a projection of a Z° that X° AGREEs with.

This will explain why Argument movement is blocked in examples like (167), but allowed in (166). Consider (169): the D-structure representation (167) gets.

(169)



The Case feature associated with *Wayne* can get a value only from the preposition *to*, as this is the only term with a valued Case feature that is klose enough to it. This means that T° cannot assign its Case feature value to this, or any other, DP. Its valued nominative Case feature will therefore not get morphologically expressed, and this is a violation of the Stray Feature Filter. What makes nominative Case different from the others, then, is the EPP, which independently requires that Specifier of TP be occupied. A working hypothesis is that the EPP is just •, whose consequences in this domain is movement of the Nominative Case marked DP rather than movement of a head. Let us add, then, (170).

Indeed, the common way of referring to • is as an "EPP feature," though technically it's an attribute of a feature rather than a feature.

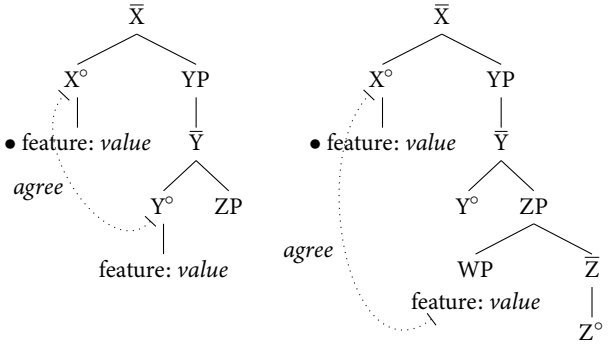
(170) A valued • Nominative feature resides within T°.

Presently the requirement that • imposes is written so that it applies correctly to cases in which one head moves to another. We need to widen this so that its requirements can be satisfied by moving a phrase. This is what (171) does.

(171) If a term, α, has •, then a projection of the X° containing α must be pronounced as a sister to a projection of a Z° that X° AGREES with.

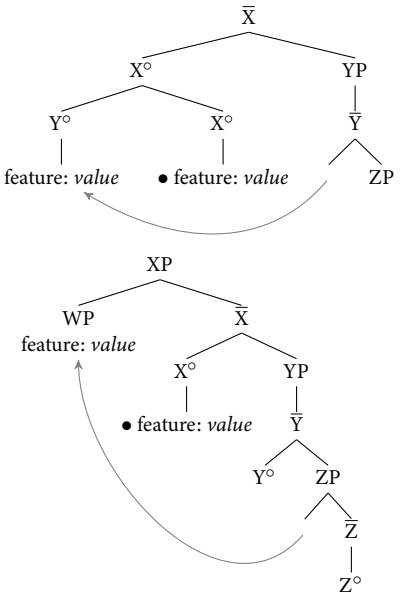
To see how (171) works, consider the two scenarios in (172).

(172)



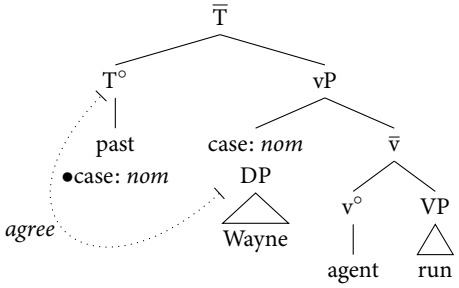
For the requirement in (171) to be satisfied, movement must create a structure in which Y° or WP is a sister to a projection of Y°. The Likes-Attracts-Likes constraint allows that to happen by adjoining Y° to X° with Head Movement. But the Likes-Attracts-Likes constraint doesn't allow that to happen by adjoining WP to Y°. Instead, the Likes-Attracts-Likes compliant way of satisfying (171) is for WP to move to Specifier of YP. The •, then, requires the structures in (172) to map onto the representations in (173).

(173)



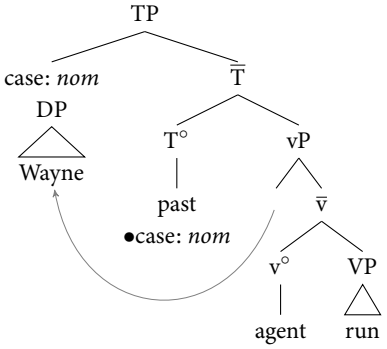
Let's consider how this will apply to a situation in which nominative Case is applied. The sentence will have a representation like (174) upon agreement.

(174)



And though this isn't a licit S-structure, since the requirements attendant with the • aren't met, it can be made so by Argument Movement.

(175)



One problem in this method of collapsing the EPP with the • attribute comes when examples like (176) are considered.

(176) [$_{CP}$ that mice squeak] seems [$_{AP}$ obvious].

In (176), the EPP is satisfied by letting Argument movement move the CP *that mice squeak* from its θ-marked position within the AP headed by *obvious* into the root AgrP. For this example to fit the proposal, it would require that the finite Agr° pass the Nominative value to the Case feature on this CP. That is, we would have to let this CP be assigned Nominative Case. In our discussion of Stowell's proposals about the order of verbal complements, we saw that it doesn't seem that CPs are subject to the Case filter, despite the evidence adduced on behalf of this by Stowell. We'll return to this problem in 4.7.2.

4.7.1 Low Subjects: English

This reworking of the conditions under which nominative Case assignment happens claims that nominative Case is assigned to a position lower than where the DP bearing that Case surfaces. There are examples which can be construed as evidence for this claim. In these examples, the nominative Case marked DP surfaces in a position c-commanded by the Case-assigning Agr°, and the EPP is satisfied with *there*. One of these examples is (177).

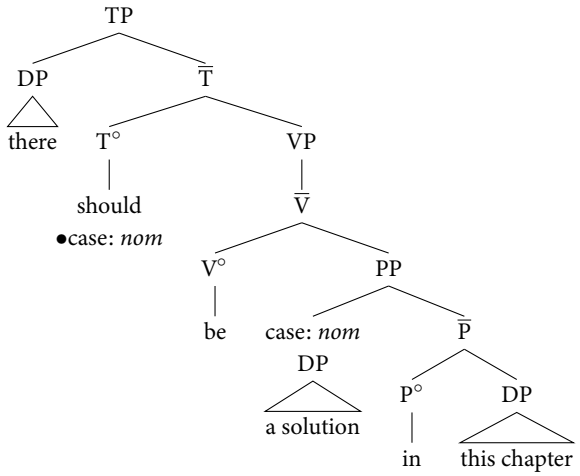
(177) There should be a solution in this chapter.

It doesn't seem that *there* in this example has the same meaning that the locative *there* has in (178).

(178) A solution stands there.

Indeed, the distribution of the *there* that is found in (177) is what is expected if it is an expletive. Just as in the scenarios in which the expletive *it* appears, we might conjecture that the EPP (or • property) is satisfied by virtue of *there* standing in Specifier of AgrP. We know from other examples that the copular verb *be* does not assign accusative Case, so we can also conjecture that the Case which *a solution* bears is nominative. Under these assumptions, an S-structure parse for (177) might be (179).

(179)



If this is the right interpretation of these examples, then it confirms the hypothesis that nominative Case assignment occurs under c-command, just as accusative Case assignment does. But it presents a host of problems for the particular proposals we have arrived at. If *there* satisfies the EPP in this example, then collapsing the EPP and the • property becomes difficult. Further, the distance between T° and the DP it Case marks in (179) does not fit to our definition of klose: there is clearly more than one phrase that dominates *a solution* but not Agr°. To fit this analysis to our system would require a different definition of klose, then.

Moreover, examples of this type are governed by a set of constraints that remain little understood. The semantic type of the nominative Case marked DP matters, for instance. That argument must be an indefinite of a certain type, as the contrasts in (180) illustrate.

(180) a. There shouldn't be any solutions in this chapter.
b. There should be solutions in this chapter.
c. There should be no solutions in this chapter.
d. There should be many solutions in this chapter.
e. * There should be every solution in this chapter.
f. * There should be the solution in this chapter.
g. * There should be most solutions in this chapter.
h. * There should be all solutions in this chapter.
i. * There should be both solutions in this chapter.
j. * There should be each solution in this chapter.
k. * There should be your solution in this chapter.

And the kind of predicate involved also matters. While PPs support this construction, APs and DPs do not.

(181) a. * There is a man a fink.
compare: A man is a fink.
b. * There is a man happy.
compare: A man is happy.

VPs support the construction, but for many verbs, only when they are embedded under *be*.

(182) a. There was a woman given a book.
b. There was a woman drinking milk.
c. * There has a woman given me a book.
d. * There has a woman drunk milk.
e. * There a woman drank milk.

The exception are some (but not all) unaccusative verbs, as in (183).

(183) a. There arrived a woman.
b. There appeared an elf.

And in these cases, as can be seen, the nominative Case marked DP remains in its θ-marked object position.

A better understanding of how to assimilate this construction to the general theory of Case assignment will probably appear once the source of these constraints is found.

For some discussion of the constraints on the semantic type of the DP, see Milsark (1974), Safir (1985, 1987), Heim (1987) and Diesing (1992). For discussion of the constraints on the predicates, see Levin (1993). There is a large, and heterogeneous, literature that addresses the problem of assimilating the way ***there*** satisfies the EPP to a more general theory of the EPP. Among many others, see Chomsky (1995), Lasnik (1992, 1995), Moro (1997), Chomsky (1998, 2000, 2001), Bošković (2002b,a), Groat (1999, 1995), Hazout (2004), and Epstein et al. (2005). A good introduction to the language variation in expletives is Vikner (1995).

4.7.2 Low Subjects: Subject Inversion Languages

There are languages in which Nominative Case-marked DPs surface in S-structure positions that, like those we've just seen in English, are c-commanded by T°. In these languages, interestingly, there is no expletive like *there* involved. Spanish, Italian and Greek are such languages. Examples from Spanish and Greek are in (184).

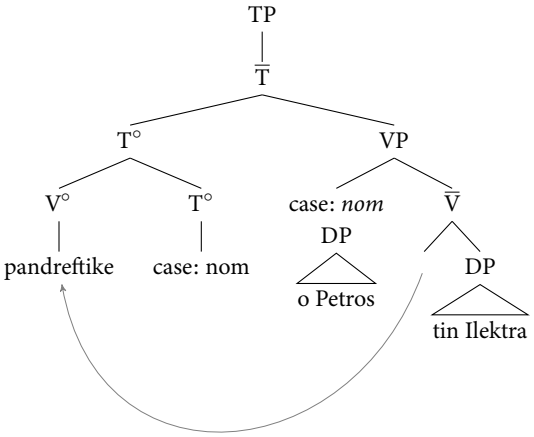
My description of these languages will assume that, as in English apparently, the agreement and tense features are bundled into the T° position at D-structure. If, as in French, they are inserted into different positions at D-structure, then I should call Agr° the node that assigns Nominative Case.

(184) a. Leyo Juan el libro.
read John the book
'John read the book.'

b. Pandreftike o Petros tin Ilektra.
married Peter Ilektra
'Peter married Electra.'
(Alexiadou and Anagnostopoulou, 1998, (3c,d): 492)

We might venture a parse for (184b) like that in (185).

(185)



I've suppressed "v°" in this parse.

These examples don't pose the same problem for our definition of klose that the ones in English did, but in these cases there appears to be a problem for expressing the EPP with the • property. That no expletive is required in the Specifier of TP in these examples suggests that these are languages that have no EPP. Maybe that means there is no • on their nominative case.

Note that in (185), the main verb has moved to T°, thereby putting it in a position that the linearization principles of Greek will cause it to precede the nominative Case marked DP. This is a very common trait of languages that allow their nominative DPs to remain low; as in Greek those DPs follow the verb (when they are not moved by other processes like wh-movement). In particular, the implications in (186) seem to hold.

(186) If a language has (186a)–(186b), then its nominative DPs can follow the finite verb.

a. Allows nominative DPs to surface c-commanded by T°.

b. Linearizes VPs and the heads that verbs move to so that they precede their complements

c. Has no expletive

These languages are said to display "subject inversion." Perlmutter (1971) and Rizzi (1982) proposed that subject inversion is always found in languages that don't have the EPP, and are verb initial. One influential attempt to explain this

correlation is in Alexiadou and Anagnostopoulou (1998, 1999). They argue that, in fact, the EPP does hold in these languages, but that it is satisfied by moving the verb instead of the nominative Case-marked DP.

For that to make sense under an interpretation of the EPP that collapses it with the • property, we cannot express the EPP as a property of Case. Verbs do not carry morphology that expresses Case features, and so they will not form an AGREE relation with the Case feature associated with T°. If the EPP is the • attribute associated with T°'s Case feature, then our present formulation of what the • attribute requires would not be satisfied by moving the verb to T°. For the proposal in Alexiadou and Anagnostopoulou (1998) to make sense, we will have to see the • attribute as being associated with some feature that both the verb and the nominative Case marked DP express. That feature, as it turns out, is agreement. We have already built into our system that verbs express the values of agreement features. But, as the label "agreement" hints at, those features are determined by the nominative Case marked DP. As the table in (187) indicates, the number and person of the nominative Case marked DP in English controls the morphological form of the finite verb.

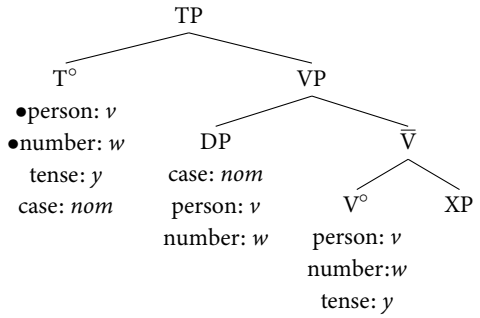
(187) For present tense *be*

	singular	plural
1st person	*am*	*are*
2nd person	*are*	*are*
3rd person	*is*	*are*

Indeed, all other verbs merely distinguish the 3rd singular forms from all others. But the existence of the paradigm in (187) indicates that both person and number features are present in the agreement morphology of English. We should therefore decompose the agreement feature into number and person features and let AGREE fix the values of these features in a way that involves not just T° and the verb, but the nominative Case-marked DP as well. We'll take up that task in the next section.

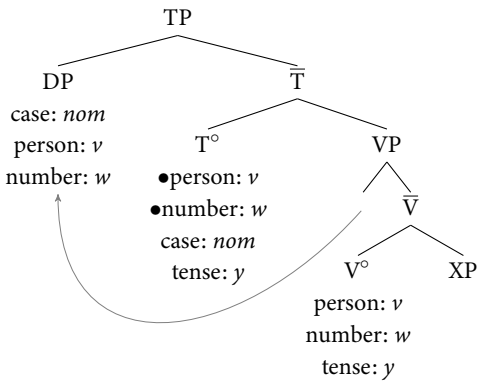
The proposal in Alexiadou and Anagnostopoulou (1998), then, requires that it be the agreement feature that has the • property responsible for manufacturing EPP effects. Schematically, then, the relevant verb initial languages will have a D-structure representation like that given in (188).

(188)



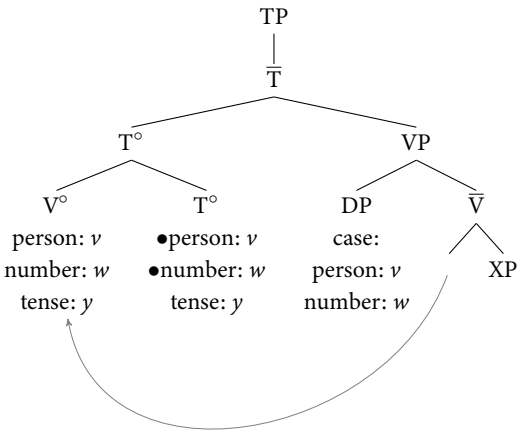
In one class of language, the requirements of the • attribute are satisfied by Argument moving the DP into Specifier of TP. This produces the typical word-order in English sentences, illustrated by (189).

(189)



In the other class of language, the verb's movement satisfies the • property on the person and number features, resulting in an S-structure like (190).

(190)



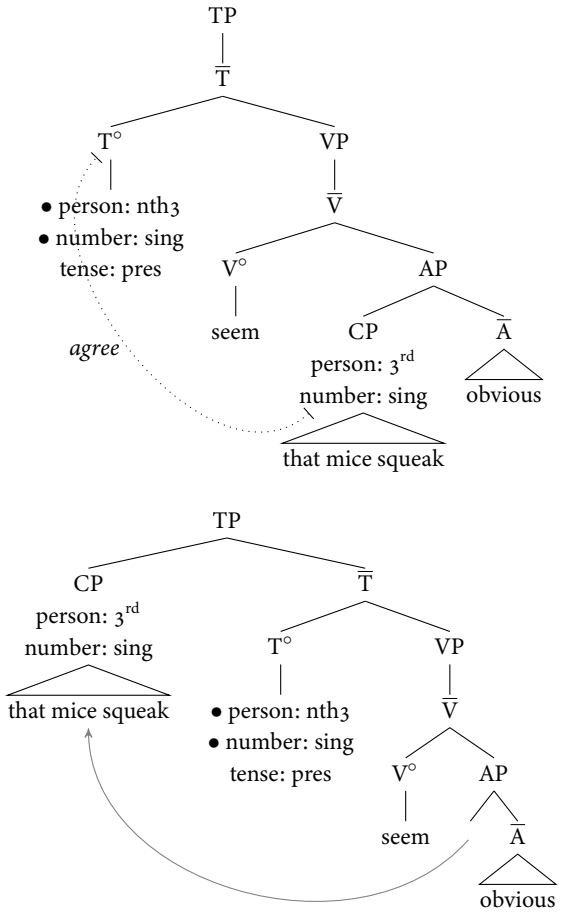
As can be seen, the correlative properties of high verb and low nominative DP are captured.

The cases in English involving expletive *there* seem to form a special case. Like the languages with the syntax in (190), the movement of the nominative Case marked DP is not required by the • property. But unlike the languages in which this is generally possible, the movement of verb does not satisfy the • property; instead, the presence of *there* does. We will not examine how *there* has this ability, nor will we try to unpack the causes for the constraints that govern this word-order in English. With these problems bracketed, however, we can see the *there*-construction in English as a special instance of the syntax that gives subject inversion elsewhere.

If the EPP arises because of • being on agreement features, rather than Case features, we might have a solution to the problem raised at the end of 4.7. That problem is how the EPP is satisfied by the movement of a CP, as well

as a DP. If • is on Nominative Case, then we should expect its requirements to be satisfied only by moving things that AGREE in Case with the Nominative feature. We have evidence that CPs are not subject to the Case filter, and so it's unlikely that they could host an unvalued Nominative feature. But it's conceivable that CPs have the person and number features that English agreement morphology houses. This would permit derivations like that in (191).

(191)



A fact that might be related to this is peculiar restriction on A-movement of CPs discussed in section 3.7. We discovered there evidence that CPs can A-move only if they stand in positions where argument DPs can. Thus, for example, though (192) are grammatical instances of the passive, (193) are ungrammatical.

(192) a. It was hoped that this will be the outcome.
b. It was insisted that this is true.
c. It was felt that this is likely.

(193) a. * That this will be the outcome was hoped.
b. * That this is true was insisted.
c. * That this is likely was felt.

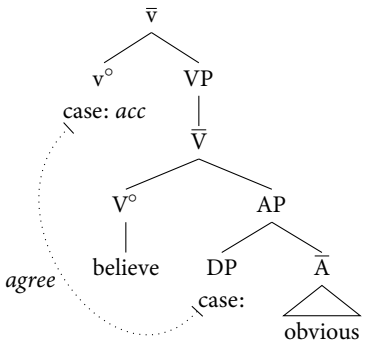
The verbs *hope*, *insist*, and *feel* do not c-select DP objects, as can be seen from the ungrammaticality of (194).

(194) a. * This outcome was hoped.
b. * This claim was insisted.
c. * This emotion was felt.

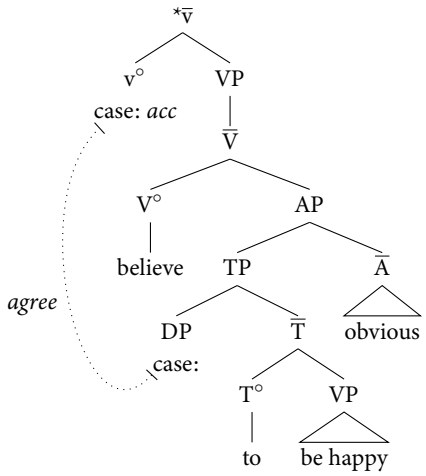
Imagine that c-selection is not for DPs, but is instead for phrases that have person and number features associated with them. Because all DPs are the only phrases that can, and must, have these features, this will come close to having the same empirical footprint as c-selecting for DP. Finally, imagine that person number features can be associated with CPs, but that CPs aren't required to have these features. Framing c-selection in terms of person/number features means that the verbs *hope*, *insiste* and *feel* don't allow their complements to be phrases with person/number features. The prevents them from combining with DPs (which must have person/number features) and with CPs unless those CPs don't have person/number features. But if a CP doesn't have person/number features, then it won't be able to stand in an AGREE relation with those unvalued features in T°. The only way to satisfy a • associated with these unvalued person/number features will be with an expletive. That is why (192) is grammatical, but (193) isn't.

Whether the AGREE relation between T° and an argument DP or CP involves •'d person/number features, or a •'d Case features, we'll have to abandon our present formulation of "klose," the locality condition on feature valuation when phrases are involved. We designed klose so that it would allow accusative Case assignment in (195), but not in (196).

(195)

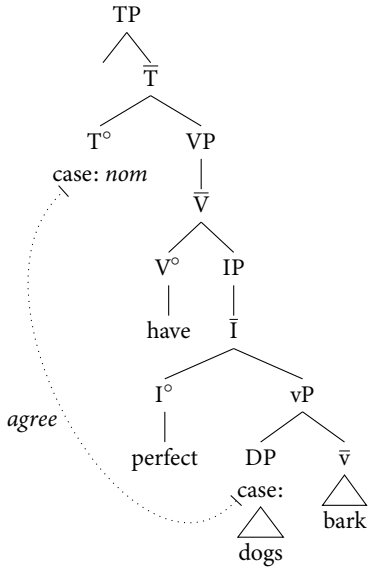


(196)



We now need it to allow (197) as well.

(197)



We cannot define klose in such a way that it legislates against too many phrases between the Case assigner and the DP that gets assigned Case, as we did originally.

One direction that some have taken is to make reference to the geometries of the phrase markers. In the good examples, both (197) and (195), the DP is embedded in phrases that are either complements or X°s. In the bad example, by contrast, the DP is embedded within a phrase that stands in Specifier position. A wide array of movement, and other, relations seem to be blocked when they reach into a phrase in a Specifier position. For instance, neither *wh* Movement nor Complex NP Shift are possible out of such phrases:

(198) a. * Who have you said that a book about – arrived.
b. * That she eats – is obvious lukewarm natto rice.
compare:
She eats fervently lukewarm nattoo rice.

It appears that there may be something quite general about phrases in Specifier positions that makes them incapable of engaging in syntactic relations outside those phrases. Ross (1967), who charted out many such cases, calls these regions "islands." One hypothesis about the ungrammaticality of (196), then, is that this has nothing to do with proprietary constraints on AGREE but is the result of a global constraint against establishing relations into Specifiers. Let us adopt that idea. This means that we have not found any locality condition on the assignment of features to phrases. We change AGREE to (199).

The particular island involved here goes by the names: "Subject Condition" and "Generalized Left Branch Condition." For two popular approaches to defining this island, see Huang (1982), Lasnik and Saito (1984) and Chomsky (1986a), for one, and Kayne (1983) and Pesetsky (1982) for the other.

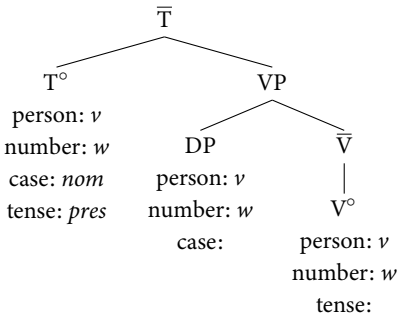
(199) AGREE (provisional)
Let X and Y be terms that dominate features, and let a feature in X have a value and those in Y be unvalued. A feature in Y can get the value of a matching feature in X just in case Y AGREEs with X. X and Y AGREE iff:
a. X c-commands Y, and
b. If X and Y are heads, there is no Z° such that X c-commands Z° and Z° c-commands Y.

Let's now turn to unpack agreement into the person and number features we've seen them consist in, and work out how AGREE can match the values of these features on T°, a verb and a DP.

4.7.3 Case and Agreement

Recognizing that the person and number features make up agreement, and that they are given the same values on both the finite verb and the nominative Case-marked DP, we want to distribute these features as indicated in (200).

(200)

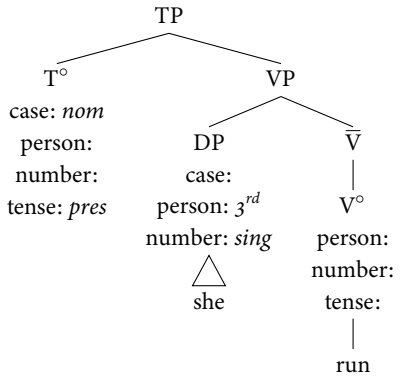


Let's consider now how AGREE would cause the values for the person feature (*v* in (200)) and the number feature (*w* in (200)) to be fixed.

It seems natural to think of these features as coming valued on the DP involved in the relation. These features have a role in the semantic interpretation of the DPs, though they don't obviously have any semantic role on T° or

the verb. For this reason, we should take the D-structure representation to be as indicated by (201), which maps onto the sentence *she runs*.

(201)



Under our present definition of AGREE, values are passed from one item, α, to another, β, only if α c-commands β. This means that the person and number values on DP cannot be assigned to T° in (201), though it can assign those features directly to the verb. This definition of AGREE, then, is not compatible with there being person and number features on T° if those features come valued on DPs. But if we do not put person and number features on T°, we will no longer be able to give these features the • attribute and employ the account for the subject inversion typology that we reviewed in the previous section.

In fact, there is another reason to think that T° has person and number features associated with it. There is a close connection between the features that figure in agreement—in English: person and number—and those that are responsible for Case. Features that can enter into agreement relations are called "ϕ features," and in addition to person and number, they also include gender and, perhaps, word class markers like those that play a role in Bantu languages and elsewhere. It is very common cross-linguistically that the DPs which bear nominative Case also agree in ϕ features with the verb. Moreover, the distribution of agreement and nominative Case within, and across, languages is very similar. Within English we see this in the fact that those DPs which bear (overt) nominative Case do so in the very clauses where they trigger (overt) agreement. Nominative Case marked DPs are exclusively found as "subjects" of finite clauses, and this is where agreement triggering DPs are found as well. This is extremely common: in those languages that have both nominative Case assignment and agreement, it is the nominative Case marked DP that enters into agreement. This correlation is also reflected in those languages that have only nominative Case assignment or agreement. Just as in English, the nominative Case, or agreement, targets certain DPs in finite clauses but not in other reduced clauses.

This relationship is found with respect to accusative Case as well. There is no agreement associated with accusative Case-marked DPs in English, but in many languages there is. This so-called "object agreement" has a distribution

that looks very much like that which accusative Case-marked DPs have in English: it depends on the choice of verb ("transitive" verbs trigger it and "intransitive" verbs don't) and it targets DPs that are roughly in the positions that accusative Case marked DPs are in English.

What these correlations suggest is that Case and agreement are always controlled by the same term, and that when both Case and agreement are involved, they are bundled together by that one term. I'll describe these correlations with (202).

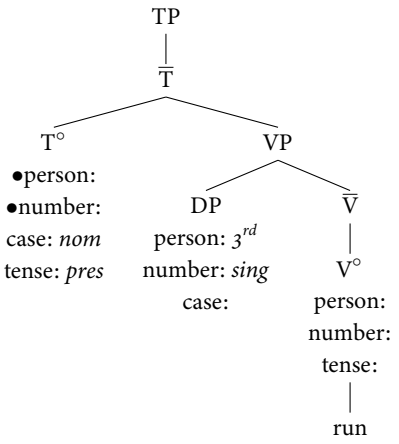
(202) If a language has an unvalued Case feature or unvalued ϕ features, then there will be some head in which they are all bundled.

This claims that if a language has both Case assignment and agreement, only one of the Case assigners will be forced to trigger agreement with the DP it assigns Case to. What we (probably) want is to force every Case assigner to control agreement with the DP it assigns Case to. We want to allow languages like English, in which one Case assigner (nominative) is tied to agreement but another (accusative) isn't, while still tying together the Case assignment and agreement phenomena.

ϕ features might well exist in languages where there is no agreement phenomenon, in which case they would only play a role in determining the meanings of the DPs they are in. In such languages, all the ϕ features would be valued. The first clause in (202) therefore restricts the constraint to those languages that have Case marking or agreement. It requires in such languages that there be a head which mediates the values of those features, thereby gluing them together. In English, this head is T°.

For these reasons, then, we should leave person and number features associated with T°. As noted above, this means that unless we decide against letting the valued versions of the person and number features come with the DP, we have a problem for our formulation of AGREE. We cannot restrict AGREE so that it only passes values down to c-commanded terms. In (201), repeated below, we need to let the values for person and number on the DP be passed up to T°.

(201)



To this end, AGREE becomes (203).

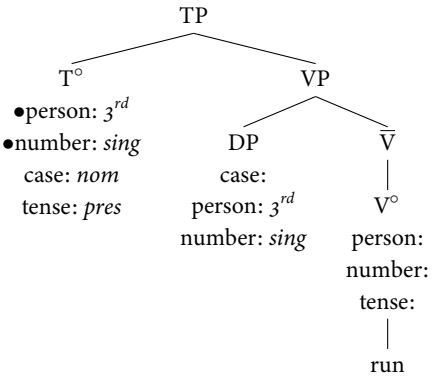
(203) AGREE (provisional)
Let X and Y be terms that dominate features, and let a features in X have a value and a feature in Y be unvalued. A feature in X or Y can get the value of the matching feature in the other just in case X and Y AGREE. X and Y AGREE iff:

a. one of X or Y c-commands the other, and

b. If Y is a head, there is no Z° such that one of X or Y c-commands Z° and Z° c-commands the other of X and Y.

This version of AGREE still requires that the two terms in an AGREE relation also be in a c-command relation. But it doesn't require that the values be passed from the c-commander to the c-commandee. This will have the desired effect, then, of letting the values for person and number features on a DP be assigned to a T° that this DP does not c-command. This will allow AGREE to convert (201) to (204).

(204)



The version of AGREE in (203) would also let the values that DP has for the person and number features be fixed on V° by putting these terms into an AGREE relation with DP. This isn't desired, however. If the DP could directly agree with verbs, and other terms, then our desire to tie agreement to Case would be lost. We would expect that the values for ϕ features on a verb could be valued by a DP that is not getting its Case from T°. What we see is something different, however. It is the very DP that bears nominative Case whose ϕ features arise on the verb. We must prevent the DP from entering into an AGREE relation with V°. Its values for person and number must come from T°. One way of ensuring this is to require that the c-commanding term in an AGREE relation be a head. This would prevent the DP in (203) from entering into an AGREE relation with the verb that follows. We therefore change AGREE once more.

(205) AGREE

If X c-commands Y, then the unvalued features in X or Y can get the value of the matching features in the other just in case X and Y AGREE. X and Y AGREE iff:

a. X is a head, and

b. If Y is a head, there is no Z° such that X c-commands Z° and Z° c-commands Y°.

4.7.4 A locality condition on Agreement

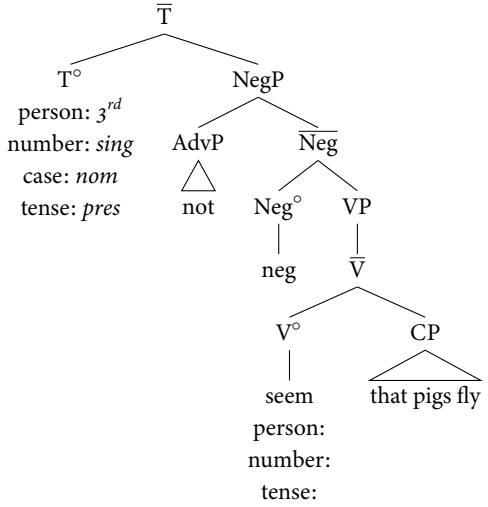
Now that the definition of AGREE has matured to the point where it applies correctly to the basic cases of agreement and Case assignment, we can investigate what locality condition, if any, there is on it. We started with the observation that Case assignment is constrained by a locality condition that we've now credited to a general ban against syntactic relations into phrases in Specifier positions.

For inflection feature assignment from T° to V°, we have relied on a locality condition that captures Lisa Travis's Head Movement Constraint. That condition prevents AGREE from holding from one head across another:

(206) If X° AGREEs with Y°, then there can be no Z° that c-commands Y° and is c-commanded by X°.

This is what is responsible for blocking AGREEment from T° across Neg° to a main verb, as in (207).

(207)



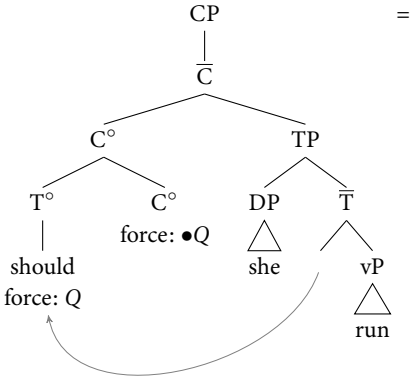
Before we'd developed the system of features and AGREE which now control head movement, we also saw that there was a locality condition that prevents deeply embedded I°s from moving to C°, as in (208).

(208) * Been Mary should have eating nattoo.

This is a scenario in which the C° heading this sentence has a silent question morpheme in it, and it is the need for that morpheme's stem that drives

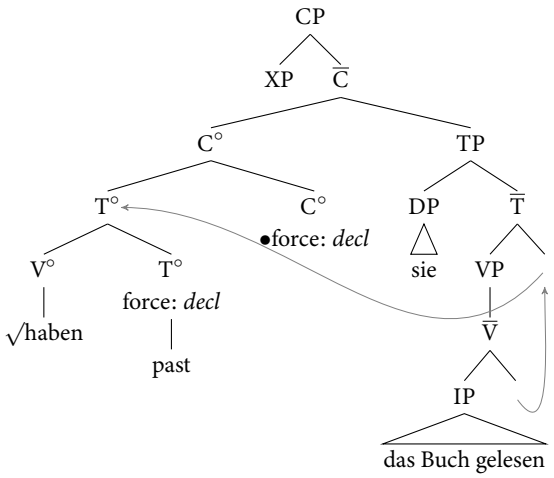
movement of I° to C°. Expressing this idea in terms of features, we would now say that this C° has a question feature in it that has the • attribute. In a grammatical instance of I° to C° movement, this will invoke derivations like that indicated in (209).

(209) = Should she run?



I've called the feature-type that Q is a value of, "force." Unvalued force features are foundz on inflectional heads, we might imagine, and therefore on the T°, *should*, in (209). We can translate our analysis of I° to C° movement in German, and the other languages displaying verb second word order, in a similar way. The CPs heading root declarative clauses in German can be headed by a C° that contains a declarative force feature with a • attribute. That will produce derivations like that indicated in (210).

(210)

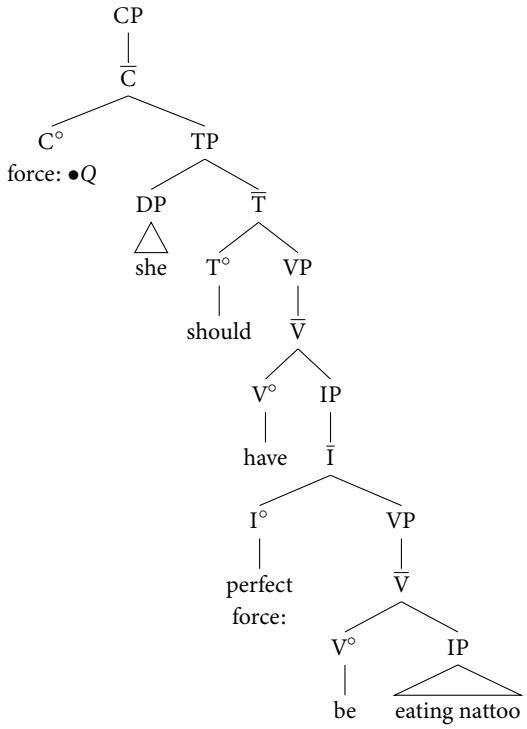


I have only given the relevant features on T° and V° here. I've suppressed tense and agreement features. XP represents some phrase which has been topicalized.

XP hat sie das Buch gelesen.
XP has she the book read
'She has read the book.'

In these context in which a feature in C° AGREEs with some inflectional head, we want to make sure that it correctly targets the highest of those inflectional heads. If it could target lower ones, then we should expect to find the ungrammatical (208). The D-structure for (208) would be (211).

(211)

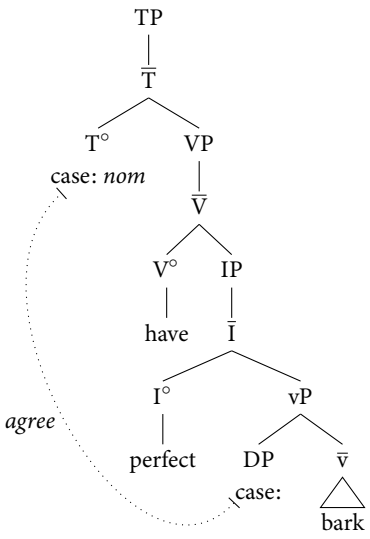


Note that there is no unvalued force feature in T°. We must allow the morphemes that are lexically inserted into T° to come without force features, for otherwise it wouldn't be possible to use, say, ***should*** in anything but a question.

If AGREE could allow the force feature in C° to value the force feature in I°, then the • attribute on Q will force *be* to move to *en*, and the resulting inflected verb to move to C°, creating the ungrammatical (208). This can be prevented if AGREE cannot hold from C° to I° in this scenario. The locality condition that we devise for AGREE should have these consequences.

In general terms, we want this locality condition to prevent some X° from AGREEing past another Y° in the way that the Head Movement Constraint (i.e. (206)) describes. But we don't want that locality condition to hold when X° is AGREEing with a phrase, as when ϕ or case features are involved. In those instances, AGREE can cross several heads, as we saw in (197), repeated here.

(197)



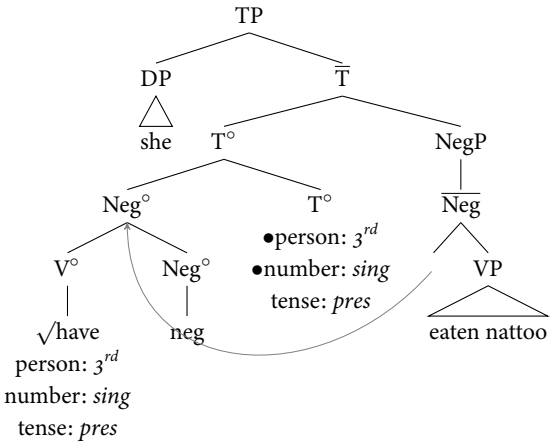
There is a popular way of framing the locality condition so that it has these consequences that builds upon work by Luigi Rizzi.[9] The central observation behind Rizzi's idea is that what distinguishes the cases where an intervening head matters from those where it doesn't is whether the intervening head could have satisfied the AGREEment relation or not. AGREEment from C° to I° in (211) is blocked because C°'s force feature could have agreed with a term in T°. That is why the grammatical sentence associated with this D-structure is (212).

[9] Rizzi, Luigi. 1990. ***Relativized minimality***. Cambridge, Massachusetts: MIT Press

(212) Should she have been eating nattoo?

And AGREEment between T° and the main verb across Neg° in (207) is blocked because T° can AGREE with Neg°, as indicated by the case of Head Movement involving negated auxiliary verbs:

(213)



By contrast, in (197), by contrast, it's the Case feature which is driving the AGREE relation, and Case features are not found on heads. They are found

only on argument DPs. The intervening heads do not matter in this instance to the AGREE relation, therefore, as it applies to Case. Rizzi's idea was to relativize the locality condition to the particular kind of AGREE relation involved. He called his condition "Relativized Minimality."

Rizzi framed his condition as a constraint on movement since the cases he examined were all ones in which movement was triggered by the AGREE relation. Translated into a condition on AGREE, it is often called "closest attract." I will frame the condition in the informal way in (214).

(214) Closest Attract
If AGREE holds between X and Y valuing f, then there can be no Z, such that AGREE for f could have held between Z and X and Y.

4.8 Summary

Here is a brief recap of the elements we have introduced to our grammar in this chapter.

(215) a. The EPP
Let α and β be features in an AGREE relation, and let A immediately dominate α and B immediately dominate β. If α or β have $\bullet$, then:
i. there must be a projection of A that dominates B, or
ii. there must be a projection of B that dominates A.

b. If a language has an unvalued Case feature or unvalued ϕ features, then there will be some head in which they are bundled.

c. AGREE
If X c-commands Y, then an unvalued feature in X or Y can get the value of the matching features in the other just in case X and Y AGREE. X and Y AGREE iff:
i. X is a head, and
ii. There is no Z that c-commands Y, and X could have AGREEd with that feature in Z.

d. Stray Feature Filter
The values of a feature must be morphologically expressed and every feature must get a value by S-structure.

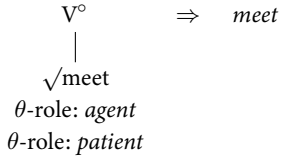
5
Complement Structure

In this chapter, we'll take a closer look at the principles that organize verbs and their complements. We'll see that there is evidence that complements are arranged in a more structured way than our present system claims. Let's start by reviewing how our present grammar works.

5.1 Accusatives Move

We have a semantics that derives the fact that verbs must combine with the arguments they select. If we restrict our attention to the special case where the arguments of verbs are DPs (and maybe PPs), this is done with the assistance of θ-roles. In chapter 4, we explored the idea that these θ-roles are always separate morphological parts that are associated with a verb. For instance, the verb *meet* has three morphological parts to it:

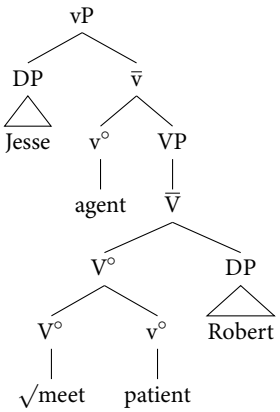
I have suppressed the c-selection information that comes with the meanings of these morphemes. Both Agent and Patient c-select DPs.

(1)


a. $[\![\sqrt{\text{meet}}]\!] = \lambda\epsilon\ \textsc{meet}(\epsilon)$
b. $[\![\text{agent}]\!] = \lambda P\ \lambda x\ \lambda\epsilon\ P(\epsilon) \wedge \textsc{agent}(x)(\epsilon)$
c. $[\![\text{patient}]\!] = \lambda P\ \lambda x\ \lambda\epsilon\ P(\epsilon) \wedge \textsc{patient}(x)(\epsilon)$

A D-structure that can house these morphemes and create a semantically well-formed phrase is in (2).

(2)



We derived the Theta Criterion from the requirement that vP must have a certain denotation—it must be a function whose domain is eventualities.

(3) Theta Criterion
   a. Every XP that must be interpreted as an argument must be "assigned" a θ-role.
   b. Every θ-role must be "assigned" to an argument.

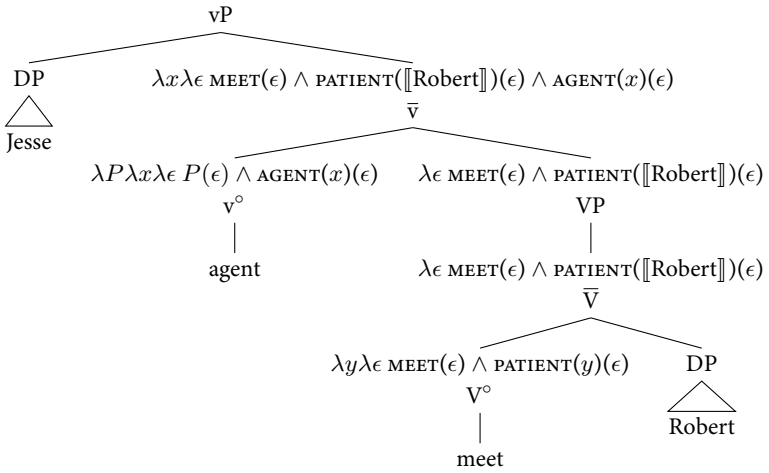
(4) A vP must have a denotation that makes it a function whose domain is eventualities and whose range is truth values.

If the rules of semantic composition are just those in (5), then the denotations we've assigned to the morphemes in (2) can be composed in a way that satisfies (4).

(5) a. Function Application
   If γ has daughters α and β, then $[\![\gamma]\!]$ is the result of applying the function $[\![\alpha]\!]$ to $[\![\beta]\!]$.
   b. Trivial
   If γ has only daughter α, then $[\![\gamma]\!] = [\![\alpha]\!]$

As shown in (6), for vP to have the denotation required of it by (4) it is necessary for the vP to contain the arguments that the two θ-roles require. If neither of those DP arguments were present, vP would be a function whose domain is things, not eventualities. Moreover, because the two DPs in (6) can only semantically combine if they are arguments of some function—this is what distinguishes them from DPs like *tomorrow*—they cannot be in a vP unless it contains θ-roles that apply to them. In this way, then, the Theta Criterion is derived.

(6) $\lambda\epsilon$ MEET(ϵ) $\wedge$ PATIENT([[Robert]])(ϵ) $\wedge$ AGENT([[Jesse]])(ϵ)



To ensure that when non-arguments are present inside VP, they appear higher than the arguments, we have the Projection Principle.

This is stated as a condition on Lexical Insertion in chapter 3. Recall that an adopted daughter of α is the daughter of α that α is not a projection of.

(7) Projection Principle
The c-selection requirements of the morphemes in X° must be satisfied by every projection of X°. A phrase, π, satisfies a c-selection requirement of X° if that requirement hasn't already been satisfied and the adopted daughter of π isn't different from what the c-selection requires.

This will correctly cause an argument to be closer to the verb than a non-argument linearly:

(8) a. She talked to me yesterday.
b. ?? She talked yesterday to me.

We also have an account of why when a verb has two internal arguments and they are PP and CP, there is a preference for the CP being last. CPs, we have discovered, move to the right edge of the clause.

(9) a. She explained to me that nattoo stinks.
b. ?? She explained that nattoo stinks to me.

What we lost in chapter 3, however, was an explanation for the fact that when the two internal arguments of a verb are DP and PP, it's the DP that comes first.

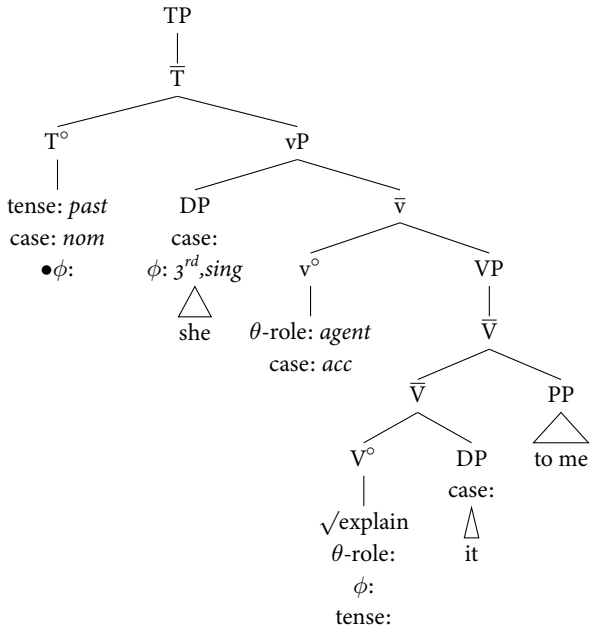
(10) a. She explained it to me.
b. * She explained to me it.

Stowell's proposal was that the property of DPs relevant for this fact was their susceptibility to the Case filter. Their positions are determined by where Case can be licensed. If we could make the position that accusative Case is licensed by the position adjacent to the verb, then we will be able to understand the

contrast in examples like (10). But we've now seen evidence that accusative Case is assigned by v°, not verbs, and this undoes Stowell's solution to the contrast in (10).

Or does it? If v° is the source of accusative Case, could we maintain Stowell's adjacency-based account of the placement of object DPs? Consider the D-structure our syntax assigns to (10).

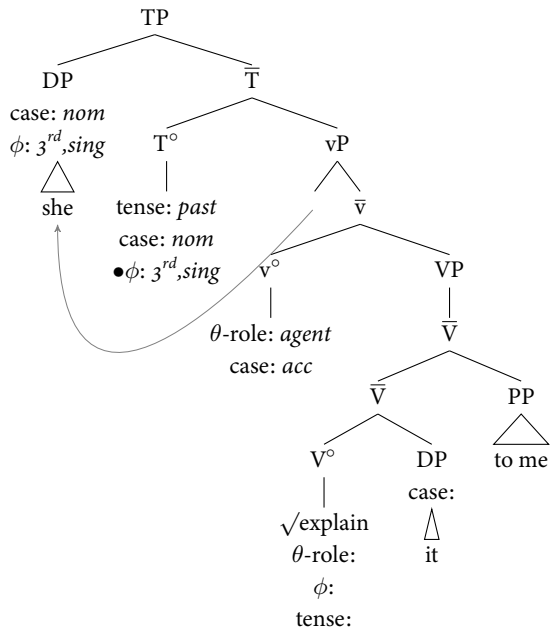
(11)



I adopt here the convention of bundling the person and number features together into one feature: ϕ.

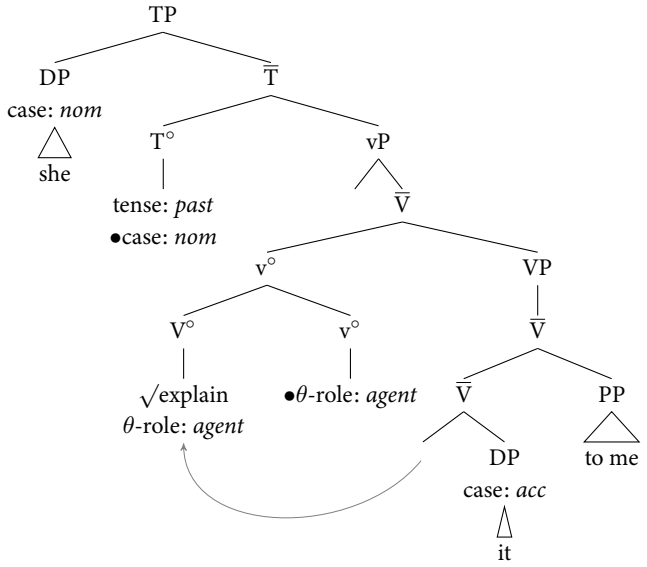
There are unvalued Case features on the two DPs, there are unvalued ϕ features on T° and V°, and there are unvalued θ-role and tense features on the verb. These can be valued by putting these features into the AGREE relation with the matching features in T° and v°. The • on ϕ will cause Argument Movement to bring the nominative Case marked DP into Specifier of TP. The derivation at this stage will look like (12).

(12)



Imagine that the θ-role feature in v° has a • as well. That is, imagine that movement of the verb to v° occurs. This will bring V° within range of T°, so that its ϕ and tense features can be valued. In the resulting structure, the accusative Case marked DP will be string adjacent to the accusative Case assigner: v°.

(13)

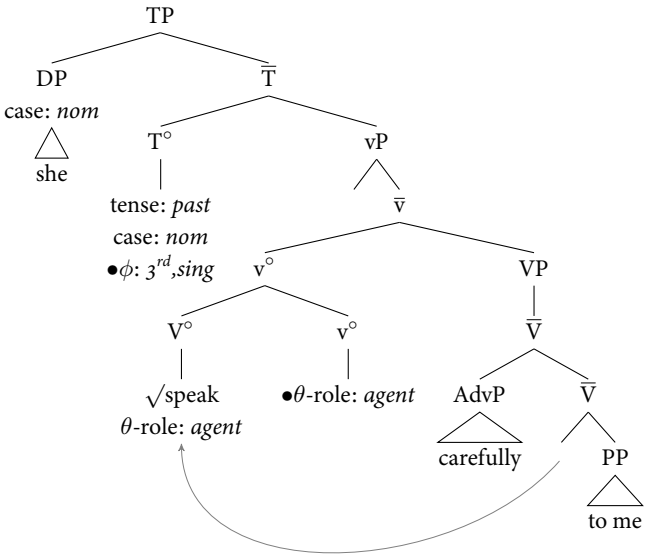


There are some reasons to think that the verb in English moves in the way indicated in (13). Recall that Pollock's argument for movement of verbs in French comes in part from the observation that the verb can be separated from its internal argument by an adverb. This is possible in English as well; it's only blocked for accusative Case-marked objects.

(14) a. She spoke carefully to me.
b. * She described carefully it.

One way we have seen that internal arguments can be separated from their verbs is by way of a rightward movement, Complex NP Shift. But it seems unlikely that this is the method by which the PP in (14a) manages to get separated from its verb, as *to me* is neither necessarily focused nor "heavy" in the way that we have seen necessary for Complex NP Shift to arise. If verbs move, we can give (14a) the syntax in (15).

(15)



A parallel derivation is prevented when the internal argument of the verb is an accusative Case marked DP, as in (14b), because the result would not cause that DP to be adjacent to the accusative Case assigner, v°.

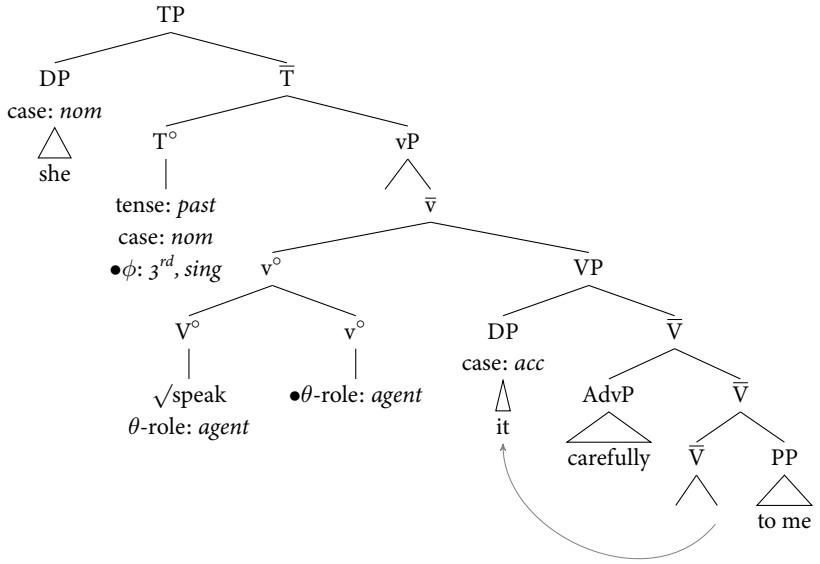
This account of (14) carries over to the ungrammaticality of (16a), since adjacency between the accusative Case marked DP and v° doesn't arise here either, but it doesn't explain the grammaticality of (16b).

(16) a. * She explained carefully it to me.
b. She explained it carefully to me.

I[1] argued that this word order involves Argument Movement of the accusative Case-marked DP into Specifier of VP, as in (17).

[1] Johnson, Kyle. 1991. Object positions. ***Natural Language and Linguistic Theory*** 9:577–636

(17)



My argument hinged on the behavior of a class of verbs which seem to have two parts. These are called "phrasal verbs" by Dwight Bolinger[2] and sometimes go by the name "particle verbs." Some of these verbs have a meaning that is not compositional; that is, their two parts do not have identifiable denotations whose meanings combine by a productive rule of semantic composition. Two examples are *look up* and *throw up*.

[2] Bolinger, Dwight. 1971. ***The phrasal verb in English***. Cambridge, Massachusetts: Harvard University Press

(18) She looked it up.
She threw it up.

The eventuality described by *look up* does not include the denotation associated with *up*, nor are the eventualities described by *throw up* a special kind of those eventualities described by *throw*. *look up* and *throw up* have the idiosyncratic meanings that individual words typically have. These expressions also undergo morphological processes which target words, rather than phrases. For instance, there is a morphological process that makes nouns from verbs by suffixing *ing*. The nouns in (19) are produced by this rule.

(19) a. the eating of cake
b. the drying of the towels
c. Shai's talking about Hebrew paradigms

This process is capable of making nouns out of particle verbs:

(20) a. the throwing up of cake
b. the looking up of telephone numbers
c. Shai's shouting out at Kiyomi

It's clear that to whatever extent these expressions are single verbs, they occupy different syntactic positions in (18). The cognate expressions in the verb

final Germanic languages are separated by Head Movement. The verb and particle are adjacent, and usually written as one word, when Head Movement has not relocated V° to second position. But when V° is moved into C° position in verb second contexts, the verb and its particle are separated. This is illustrated by (21).

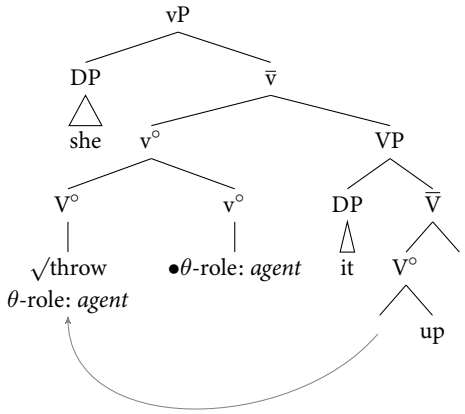
(21) a. Sie sagte dass Ich den Brief ausstellte.
She said that I the letter out.stood
'She said that I wrote the leter out.'

b. Ich stellte den Brief aus.
I stood the letter out
'I wrote the letter out.'

c. * Ich ausstellte den Brief.
I out.stood the letter
'I wrote the letter out.'

I suggested that Head Movement is also responsible for separating the verbal part from its particle in English as well. I suggested a D-structure representation that allowed Head Movement to target the verbal portion of the particle verb, as in (22).

(22)



An interesting feature of constructions with particle verbs is that though they c-select internal arguments of a variety of kinds, it is only DPs that can be positioned between the verbal part and its particle.

(23) a. * Claire shouted [$_{PP}$ to me] out.

b. * Aleksei pointed [$_{CP}$ that natto stinks] out.

c. * Nick turned [$_{AP}$ good] out.

compare:

d. Claire shouted out [$_{PP}$ to me].

e. Aleksei pointed out [$_{CP}$ that natto stinks].

f. Nick tuned out [$_{AP}$ good].

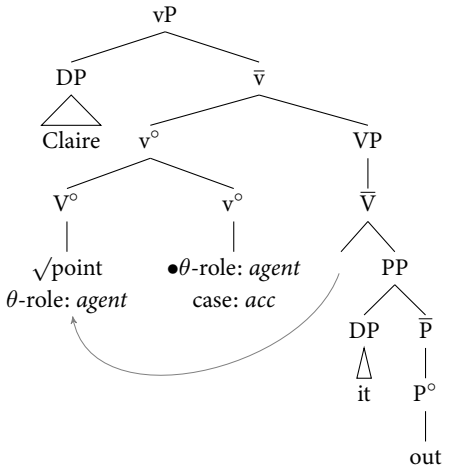
Richard Kayne[3] argued that this fact can be explained if the particle heads a small clause that the verbal part of the particle verb selects. Small clauses can only have DPs in their Specifier positions, as (24) indicates.

[3] Kayne, Richard S. 1984a. Principles of particle constructions. In ***Grammatical representation***, ed. Jacqueline Guéron, Hans G. Obenauer, and Jean-Yves Pollock, 101–140. Dordrecht, The Netherlands: Foris Publications

(24) a. She considers [[$_{\text{DP}}$ the woman] happy].
b. * She considers [[$_{\text{PP}}$ to me] obvious that natto stinks].
c. * She considers [[$_{\text{CP}}$ that natto stinks] obvious to me].

So whatever ensures this could be relied on to derive the same asymmetry with particle verbs. On his account, then, the representation for particle verbs would be (25).

(25)



Unlike (22), Kayne's representation in (25) correctly explains why the verb and particle cannot Head Move together. If that were possible, we would expect examples like *She pointed out it* to be grammatical. On the view in Johnson (1991), there would have to be another reason why the verb+particle unit cannot move by Head Movement.

On the view of particle verbs that involves movement of the accusative Case-marked DP into Specifier of VP, the fact that only DPs can come between the verbal part and its particle can be credited to Earliness, repeated here.

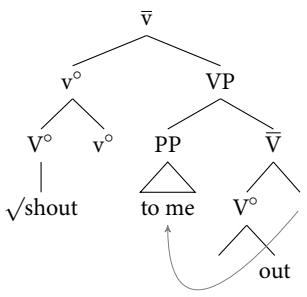
Recall that a sentence is the set of grammatical derivations which have the same D-structure.

(26) Earliness
Let the set of derivations, $\mathscr{D} = \{D, D_1, \ldots, D_n\}$, be a sentence. There can be no $D_i \in \mathscr{D}$ if there is a D_j in $\mathscr{D}$ such that $|D_i| > |D_j|$, where "$|D|$" is the number of elements in D.

If argument DPs are uniquely subject to the Case filter, and if we could manufacture a theory that let Specifier of VP be the only position that is assigned accusative Case, then movement of an argument DP into Specifier of VP will be required to meet the Case filter. Derivations that involve movement of an argument DP into Specifier of VP will therefore be permitted by Earliness—a derivation that didn't include that movement wouldn't be in the set of grammatical derivations that make a sentence – but movement of other phrases

would be blocked. Any derivation that includes the step in (27), for instance, will be longer than an equivalent, and equally grammatical derivation, that didn't include that step.

(27)

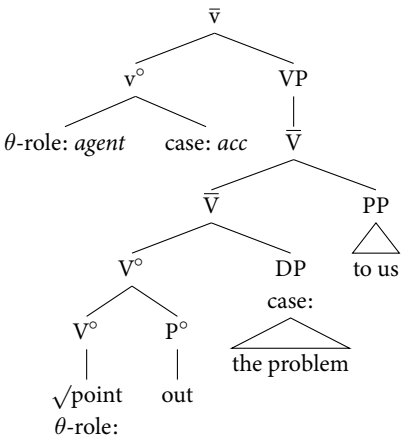


But if accusative Case is only assigned to Specifier of VP, then we don't have an understanding of how the word orders in (28) arise.

(28) a. She pointed out the mistake to us.
b. She threw up the burrito.

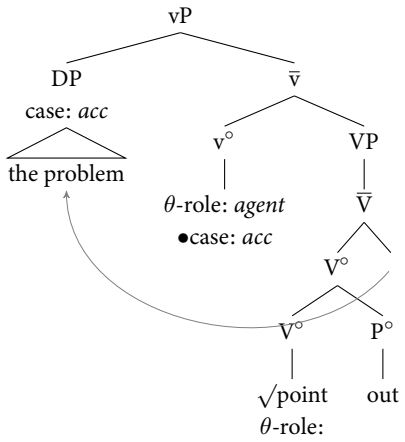
I'll sketch an account of these facts that closely follows Johnson (1991), but makes use of the framework we've developed in the last chapter for Case assignment. That system allows Case assignment—that is, AGREE—to hold between a head and a phrase that head c-commands across long distances. The locality condition we posited for AGREE merely prevented two terms from being in the AGREE relation if there was a third term between them that could be in that same AGREE relation. If v° is the source of accusative Case, then this theory should allow an accusative Case feature in v° to value an unvalued Case feature on an object DP in (29).

(29)



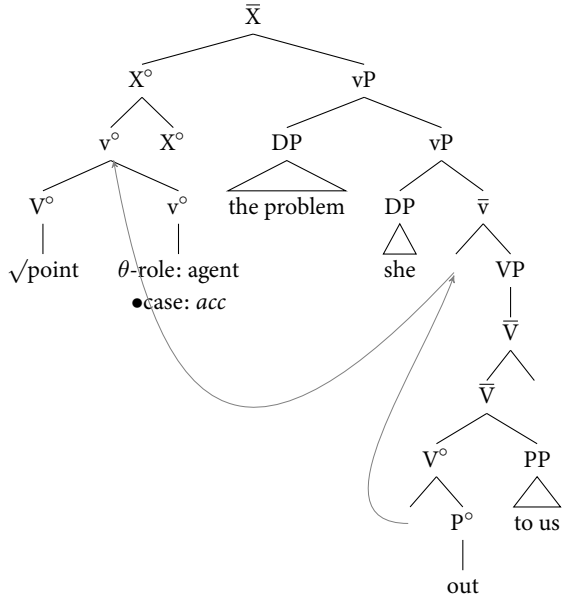
To force the accusative Case-marked DP to move, our present system equips one of the terms in the AGREE relation with a •. If we associated a • with the Case feature in v°, this would produce a parse like (30).

(30)



This would correctly put the object DP first among internal arguments, but it will require that the verb in English move even farther than v°, for otherwise accusative Case-marked objects would erroneously be forced to precede the verb. Let's call the mysterious position to which the verb would have to move on this account X°.

(31)



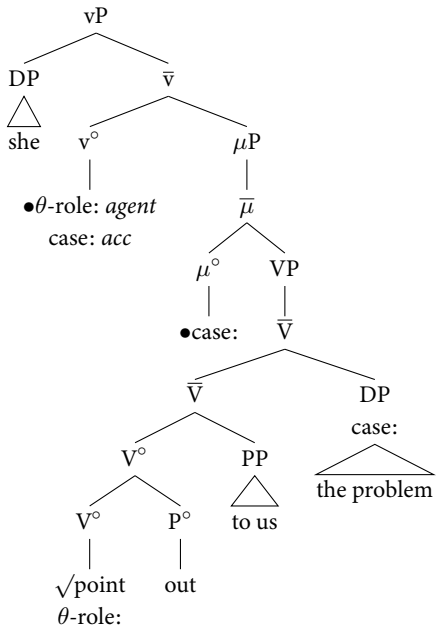
This is a popular approach, but I will present problems for its semantics later in this chapter.

A slight variant of this treatment adopts the view that the two pieces of the v° position that we have adopted to account for Burzio's Generalization can be syntactically separated in much the same way that French separates tense and agreement. We'll continue to adopt the solution to Burzio's Generalization that credits the external θ-role with being the type of word that comes with an accusative Case feature. But rather than let the D-structure assemble these two pieces into a single X° position, we will assume that the derivation

brings them together. At D-structure, let's put the accusative feature in a lower X° position; Johnson (1991) calls this position "μ."

I took the name from a paper that was circulating at the time by David Pesetsky. Pesetsky's paper had many of the same ideas as mine—we had independently arrived at very similar proposals. My paper got published first, robbing Professor Pesetsky of the credit he deserves. More common names for this head position found in the literature are Acc° for "accusative," and Agr_o°, for "object agreement."

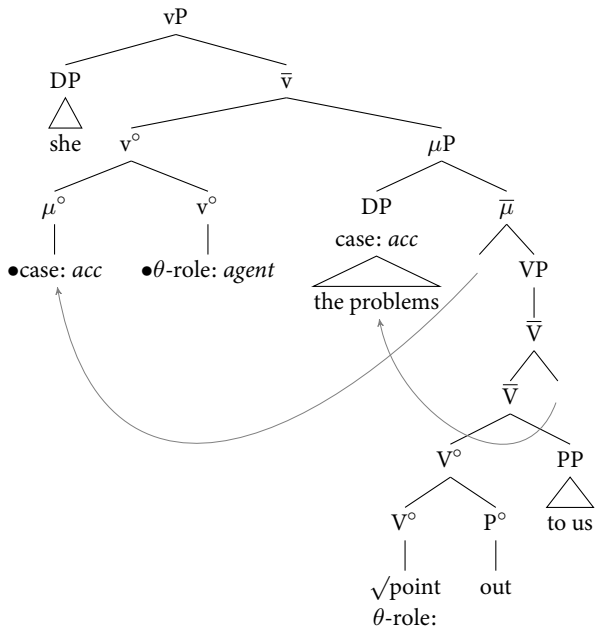
(32)



This view is sometimes called the "Split VP Hypothesis," and it is given an extensive defense in Koizumi (1995). From (32), repeated applications of AGREE can value the unvalued Case features on μ° and the DP, and because of the •s, this will trigger a movement of μ° and DP. The result is (33).

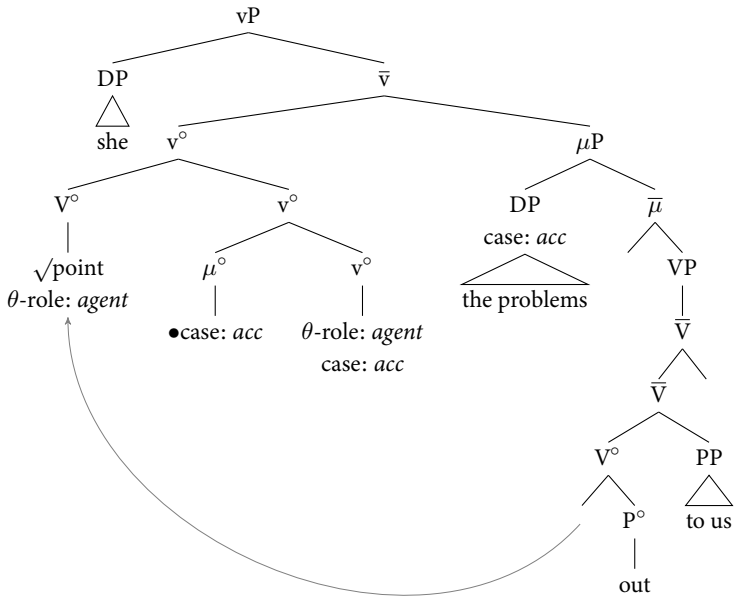
The proposal that accusative Case-marked DPs move in English interacts badly with the model of quantifier float that we reviewed in section 3.4. We would expect ungrammatical sentences like ***Meg pointed the problems out all to me*** to be well-formed. See Bobaljik (2003) for some discussion of this problem and ideas about its solution.

(33)



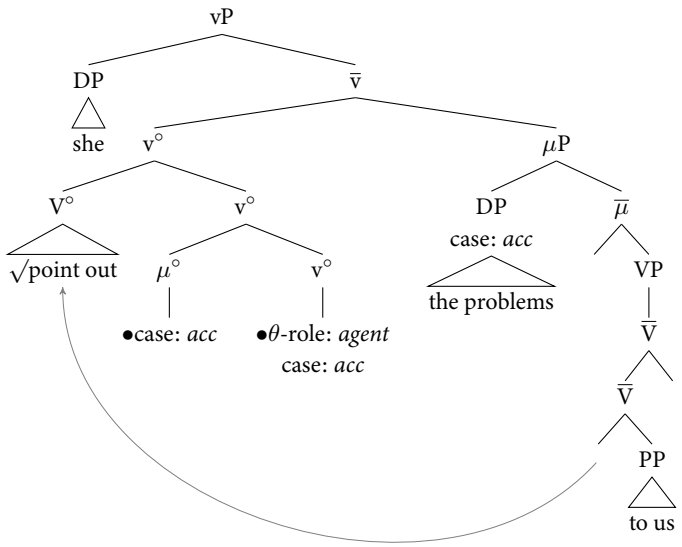
This representation allows AGREE to value the θ-role feature on the verb, and the • requires the movement shown in (34).

(34)



The • on the θ-role feature can also be satisfied with the movement in (35).

(35)



Note that because both of these instances of Head Movement satisfy the same requirement of the •, Earliness does not distinguish between them. Both outcomes contribute to the derivations that contain them an equal number of elements, and therefore both word orders are available.

The key to meeting the requirements of Earliness is the view that the verb and its particle make an X° that can move. In the cognate constructions in

the other Germanic languages, it is clear that Head movement of the verb and particle together is sometimes blocked. If the verb and particle were able to Head Move together to C°, for instance, sentences like (21c), where the verb+particle occupy second position, would be grammatical in German.

I suggested in Johnson (1991) that this might be connected to the fact that morphological processes generally resist taking the verb+particle as their host. For instance, the process that makes nouns from verbs by suffixing *ing* (illustrated by (36)) targets only the verbal part of the particle verb, as (37) indicates.

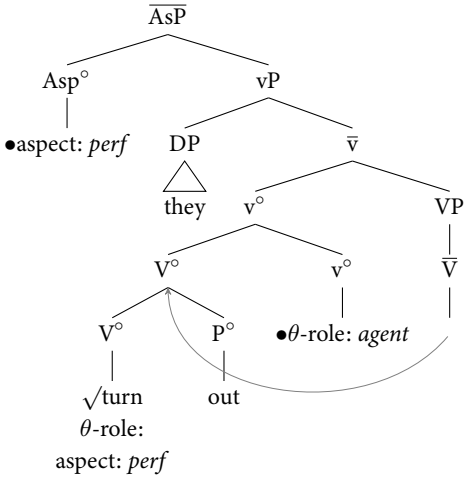
The possible exception to this trend is ***able***, where English speakers consistently judge ***pointoutable*** to be at least as grammatical as ***pointableout***.

(36) [$_{V°}$ read] (e.g., *read the book*) ⇒ [$_{N°}$ reading] (e.g. *the reading of the book*)

(37) [$_{V°}$ point out] ⇒ [$_{N°}$ point outing]
[$_{V°}$ point out] ⇒ [$_{N°}$ pointing out]

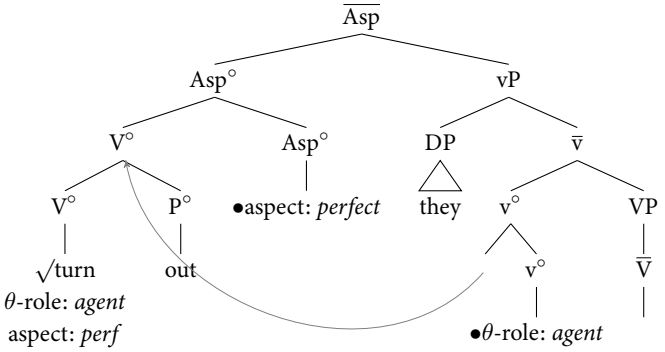
Imagine that this force influences how the morphemes in an X° that expresses verbal morphology can be arranged. For instance, consider how a particle verb that is embedded under the inflectional head that makes a perfect participle might deploy itself. There will be a parse in the derivation for the intransitive particle verb *turn out* that will look like (38).

(38)



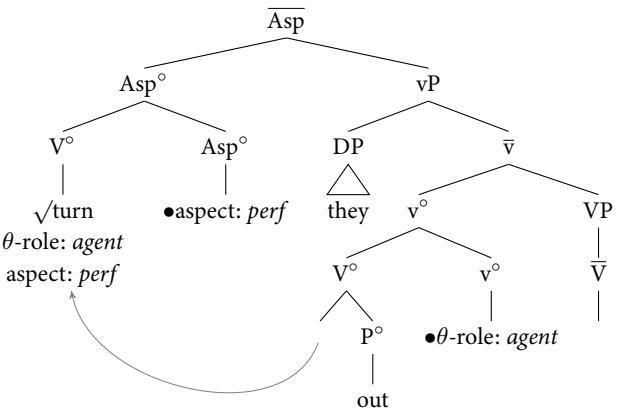
If Head Movement were to move the verb+particle pair to Asp° in (38), the result would put the aspect morphology outside the particle syntactically.

(39)



If the syntactic structure mirrors the morphological structure, this plausibly violates whatever is responsible for the contrast in (37). If, by contrast, Head Movement combines the verb and the aspectual morphology by moving just the verbal portion, the resulting syntax aligns with what (37) teaches is the right morphological structure.

(40)



If this representation is grammatical, there will have to be a way of matching a vocabulary item with the v° containing the particle and the agent θ-role. And, relatedly, the • will have to be satisfied by a representation in which the verbal part of the particle verb has been separated. Perhaps the agent θ-role feature should be associated with the V° that contains both the verbal part and its particle, rather than, as indicated in the representation here, with just the verbal part.

It is difficult to establish that this same kind of restriction on the morphology of particle verbs holds in German because the particle in German is a "prefix." But if there is a similar restriction, then this could be credited with ensuring that the verb and particle separate when the verb gains its finite morphology. This will account for why the verb+particle does not move as a unit to C° position in verb second contexts.

5.2 Split VP

In chapter 3 we saw reasons for thinking that the meaning one might naïvely have associated with one position—the position the verb occupies—is actually spread across two syntactic positions. In chapter 4 we developed a theory of how the syntax can assemble the morphological pieces of a word, and imported that theory to the task of assembling the two pieces of a verb's meaning. In this chapter, we've see that accusative Case assigning function of verbs is part of the morpheme that expresses the external θ-role, but it too is syntactically separated from that θ-role. Over the course of these chapters, what we've discovered is that a variety of attributes we associate with verbs—their external θ-role and their structural Case—are spread across different syntactic positions. What appeared to be a single projection of a verb—its VP—has turned into a variety of phrases, one embedded in the other.

There are other aspects of the meaning associated with a verb that have been argued to involve splitting the VP up further. We will look at a couple of these here, using the view of constituency that we get from the meaning of *again*.

5.2.1 *Hidden states and "again"*

An interesting feature of *again* is that its surface syntactic distribution is arguably determined entirely by its semantics. The position of many other modifiers appears to be at least partly determined by the morpho-syntactic category of the phrases they modify (adjectives modify projections of nouns and not projections of verbs, for instance) and partly by way of movement, allowing them to be spoken in positions distant from where they are semantically interpreted, as in: *Quickly, Trump should leave the news-cycle.*

To see this, consider the meanings of the sentences in (41), which are fashioned from a paper by James McCawley.[4]

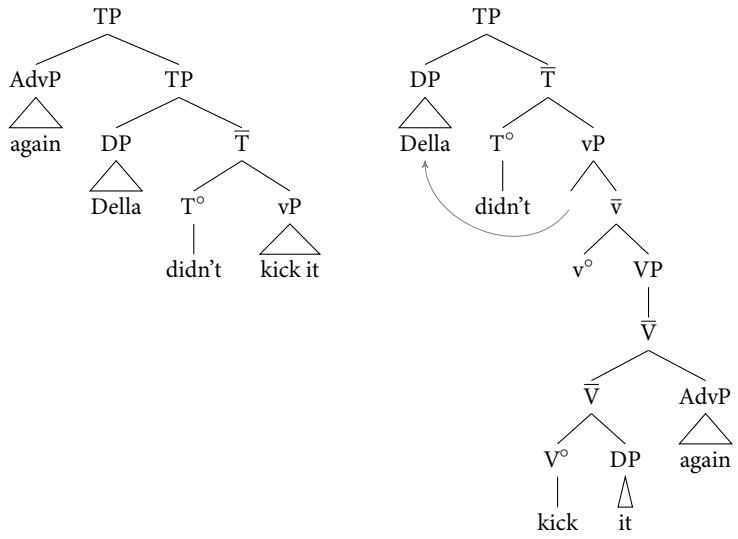
[4] McCawley, James D. 1973. Prelexical syntax. In ***Grammar and meaning: Papers on syntactic and semantic topics***, 343–356. Tokyo: Taishukan Publications

(41) a. Again Della didn't kick it.
≈ Della didn't kick it and she didn't kick it previously.

b. Della didn't kick it again.
≈ Della didn't kick it and it hadn't been kicked previously.

In (41a), *not* is part of what *again* semantically combines with, and in (41b), by contrast, *again* semantically combines with just *kick it*. If *again* cannot move, this ambiguity follows straightforwardly from the thesis of semantic compositionality, and the plausible parses for (41) in (42).

(42)



Note that the first of these parses requires that we allow D-structures to adjoin a phrase to a maximal projection. That is a form that, so far, we have only seen the output of Complex NP Shift to have.

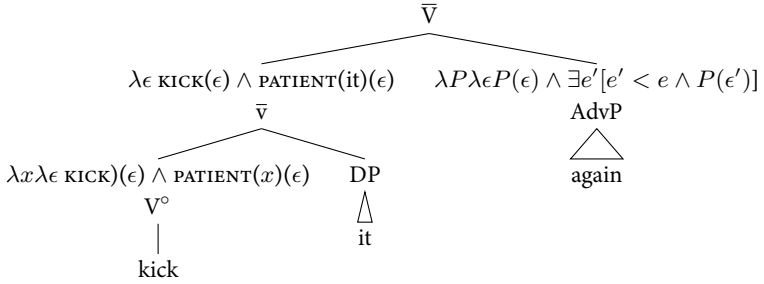
I'll provide a sketch of a semantics for *again* here that is based on proposals by Arnim von Stechow.[5] Roughly, the idea is that *again* combines with phrases whose denotations are descriptions of eventualities, and adds that those descriptions held of a previously occurring eventuality.

[5] von Stechow, Arnim. 1996. The different readings of ***wieder "again"***: A structural account. ***Journal of Semantics*** 13:87–138

(43) $[\![\text{again}]\!] = \lambda P \lambda \epsilon\, P(\epsilon) \wedge \exists \epsilon' \, [\epsilon' < \epsilon \wedge \; P(\epsilon')]$
"$<$" means "occurs previously to"

This will give to the $\overline{\text{V}}$s in the second parse of (42) the meanings indicated in (44).

(44) $\lambda\epsilon$ KICK(ϵ) $\wedge$ PATIENT(it)(ϵ) $\wedge \exists\epsilon'$ [$\epsilon' < \epsilon \wedge$ KICK(ϵ') $\wedge$ PATIENT(it)(ϵ')]



The denotation of the largest $\overline{\text{V}}$ makes it a description of eventualities that map to True when they are kickings whose patient is whatever *it* refers to, and there is another eventuality that held previously which also maps to True when it is a kicking whose patient is whatever *it* refers to. This matches the paraphrases given above, then.

Although this denotation captures the rough meaning of *again*, it misfires once we consider how it is put into the rest of (41b). Because *nt* corresponds to the logical operator $\neg$, the denotation we'll get from (44) for (41b)—ignoring the contribution made by tense—is roughly (45).

(45) $\neg\exists\epsilon$ AGENT(Della)(ϵ) $\wedge$ KICK(ϵ) $\wedge$ PATIENT(it)(ϵ) $\wedge$
$\exists\epsilon'$ [$\epsilon' < \epsilon \wedge$ KICK(ϵ') $\wedge$ PATIENT(it)(ϵ')]

A paraphrase of (45) is (46).

(46) There is no eventuality that has Della as its agent and is a kicking of it and has a prior eventuality that is a kicking of it.

This wrongly puts inside the scope of $\neg$ that there is a prior eventuality of kicking it. The meaning in (45) would describe a situation in which Della did kick it just so long as she didn't kick it previously.

What's needed is for the information that *again* supplies about a prior eventuality to not be part of the main assertion of the clause. Instead, this information must be "added" to the assertion of the rest of the sentence. This is the same kind of thing, perhaps, that we see in cases like (47).

(47) Della didn't kick Marcin, who teaches now in Canada.

The so-called appositive relative clause, *who teaches now in Canada*, isn't part of what *nt* negates. Like the information supplied by *again*, the information in the appositive of (47) adds information that is separate from the main assertion of the clause.

There are a variety of technical ways of expressing how the meaning of *again* is separated from the meaning of the rest of the sentence it is in. von Stechow (1996) suggests making the contribution *again* makes to the sentence a "presupposition." It won't matter for our purposes here how this is done. Let us simply assume that the information about the existence of a previous eventuality that *again* conveys is a separate message that accompanies the sentence that it is contained in.

An important difference between an appositive and ***again*** is that material in an appositive seems to be genuinely "outside" the sentence semantically. This can be seen by noticing that pronouns inside cannot be understood as variables bound to DPs outside the appositive. In ***Every woman met Marcin, who she'd heard about for years, she*** cannot be a variable bound to ***every woman***. That is possible for information that ***again*** conveys. In ***Every woman kicked her fridge again in front of me***, the previous kickings involve every woman's fridge. (See the discussion on the next page.)

(48) $[\![\text{again}]\!] = \lambda P \lambda \epsilon\ P(\epsilon) = T$.
Add to conversation: $\exists \epsilon'\ [\epsilon' < \epsilon \wedge\ P(\epsilon')]$
where $\lambda \epsilon P(\epsilon) = T$ is the denotation of the phrase that *again* combines with.

The denotation in (48) isn't quite sufficient to capture all of what ***again*** conveys. It isn't sufficient to simply require, as (48) does, that the previous eventuality merely exist. There are examples which indicate that ***again*** makes reference to a particular, salient, eventuality that has occurred previously. See Beck (2006), who cites Heim (1990). It also needs to be added that the previous eventuality and the one that is described by the constituent that ***again*** modifies, must lie between an eventuality in which the relevant property does not hold. In (44), for instance, the present kicking eventuality and the previous one that is being repeated must be separated by an eventuality which isn't a kicking. Without this additional element of meaning, a sentence like ***Paula is again crying*** would have a meaning that overlaps with that of ***Paula is still crying***. See, for instance, Bale (2007).

With this denotation, (45) becomes (49).

(49) a. $\neg\exists\epsilon$ AGENT(Della)(ϵ) $\wedge$ KICK(ϵ) $\wedge$ PATIENT(it)(ϵ)
$\approx$ there isn't an eventuality that Della is the agent of and is a kicking of it.

b. $\exists \epsilon'\ [\epsilon' < \epsilon \wedge$ KICK(ϵ') $\wedge$ PATIENT(it)(ϵ')]
$\approx$ there was a previous eventuality that was a kicking of it.

For this to provide the right meaning for the sentence in (41a), repeated here, we'll have to understand the TP that *again* combines with to be a description of eventualities.

(41a) Again Della didn't kick it.

We don't have enough of a semantics built up to include how tense makes its contribution, and so I won't attempt to work out how this can be achieved. What needs to emerge is for (41a) to get a meaning like that in (50).

(50) a. $\neg\exists\epsilon$ AGENT(Della)(ϵ) $\wedge$ KICK(ϵ) $\wedge$ PATIENT(it)(ϵ)
$\approx$ there isn't an eventuality that Della is the agent of and is a kicking of it.

b. $\neg\exists \epsilon'\ [\epsilon' < \epsilon \wedge$ KICK(ϵ') $\wedge$ PATIENT(it)(ϵ')]
$\approx$ there wasn't a previous eventuality that was a kicking of it by Della.

What's important here is that it's the phrase that *again* is the sister to that determines the information that *again* conveys. We can use this fact about *again* to see what the meaning of its sister is.

Let's consider one more contrast that supports this conclusion. Consider the examples in (51), based on ones in a paper by Alan Bale.[6]

Della kicked her broken refrigerator out of frustration while I was away. After I got home, …

(51) a. she kicked it again in front of me.

b. # she kicked it in front of me again.

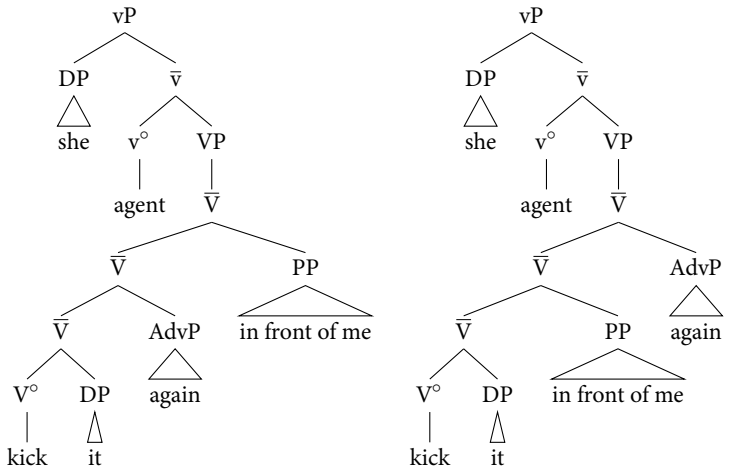
(Bale, 2007, inspired by (27): 462)

[6] Bale, Alan Clinton. 2007. Quantifiers and verb phrases: An exploration of propositional complexity. ***Natural Language and Linguistic Theory*** 25:447–483

The pronunciation of these sentences must not put pitch accent on ***again***. When ***again*** is focussed, which is what pitch accent signals in English, the meaning of ***again*** gets tangled up with the meaning of focus. See Beck (2006).

I've set up the context for these two sentences so that it's clear what the relevant preceding eventuality is. Potential parses for the VPs in the two sentences are given in (52).

(52)



Imagine that *in front of me* is a modifier that locates where the kicking eventuality takes place. That is:

(53) ⟦in the front door⟧ is a function that maps eventualities to True when those eventualities are in front of me.

The denotation of the $\overline{\text{V}}$ that *again* combines with in (51a) is (54a), and the denotation of the $\overline{\text{V}}$ that *again* combines with in (51b) is (54b).

(54) a. $\lambda\epsilon$ KICK(ϵ) $\wedge$ PATIENT(it)(ϵ)

b. $\lambda\epsilon$ KICK(ϵ) $\wedge$ PATIENT(it)(ϵ) $\wedge$ in_front_of_me(ϵ)

When *again* combines with these denotations, it introduces the meanings in (55).

(55) a. $\exists\, \epsilon'\ \epsilon' < \epsilon \wedge$ KICK(ϵ') $\wedge$ PATIENT(it)(ϵ')
$\approx$ there was a previous kicking of it

b. $\exists\, \epsilon'\ \epsilon' < \epsilon \wedge$ KICK(ϵ') $\wedge$ PATIENT(it)(ϵ') $\wedge$ in_front_of_me(ϵ')
$\approx$ there was a previous kicking of it in front of me.

In the contexts provided, only (55a) is true, and so (51b) doesn't fit the context (which is indicated by marking it with "#"). What we see here, then, is the same thing that was illustrated by (41). We can use *again* to see what the meaning of *again*'s sister is.

Consider now the meaning assigned to (56).

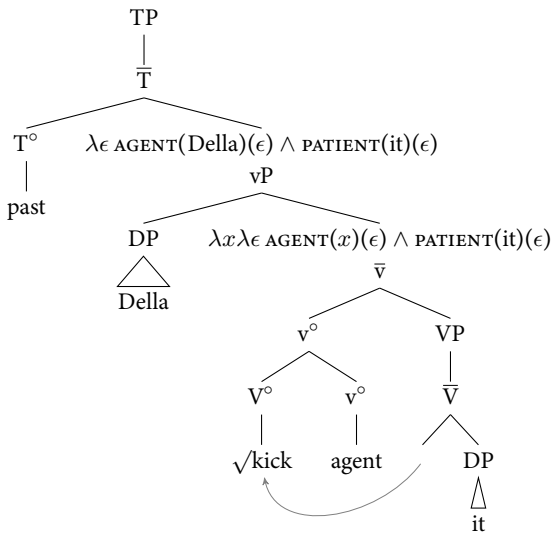
(56) Della again kicked it.

The information that *again* conveys in (56) is that Della kicked it previously. That is, (56) is synonymous with (57).

(57) Again Della kicked it.

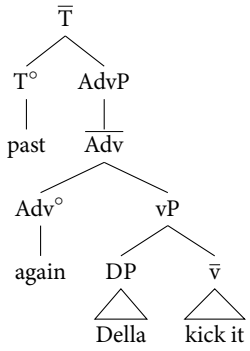
This is explained by the Derived Subjects Hypothesis, that is the idea that external arguments occupy a lower position at D-structure. If we set aside where *again* is, the D-structure representation for (56) is (58).

(58)



The constituent that *again* combines with must be a description of eventualities. That is, it must be a function from eventualities to True or False. The denotation of v̄ in (58) doesn't have this meaning, and we should expect that *again* cannot combine with it. The constituent that has the right meaning to be the sister to *again* is the vP. This means that there must be a position to the left of the vP that *again* resides in. Our present restriction on phrases provides only one way that can happen, and it's indicated in (59).

(59)



This gives to *again* the same position that we've given to Neg°. It separates the vP from T° by an embedding head, the one occupied by *again*. We might expect that this geometry would therefore have the same effects that the one in which Neg° occurs has. In chapter 4 we saw that when vP is embedded by an X° that separates it from T°, AGREE between T° and v° is blocked, and this results in ungrammaticality if there are features that need to get into an AGREEment relation:

(60) * Della not kicked it.

If this is correct, it's not clear why the similar representation in (59) shouldn't have the same consequence.

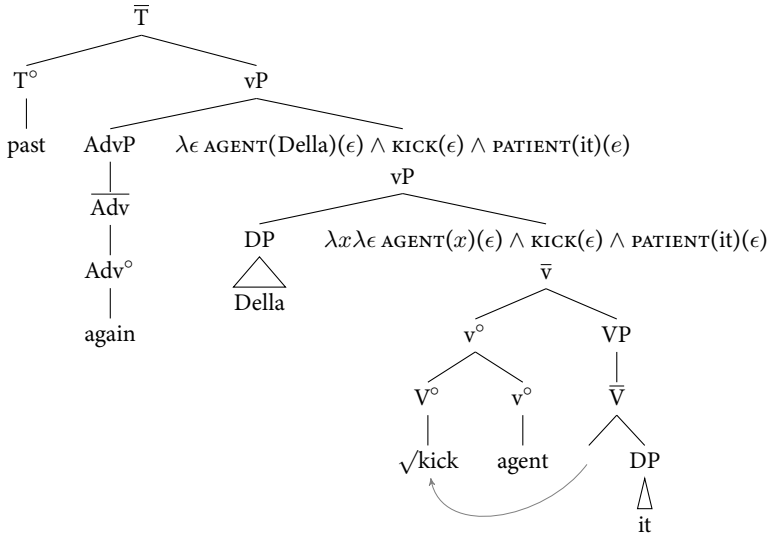
For this reason, let's allow phrases to have the shape in (61).

Recall that "{α,β}" means that α and β are sisters, with no linear order imposed on α and β.

(61) $\text{XP} \rightarrow \{\text{YP}, \text{XP}\}$

This isn't inconsistent with anything we've seen about English syntax so far, and it allows (56) to get the parse in (62).

(62)



This D-structure produces the correct interpretation. The information that *again* provides is (63).

(63) $\exists\, \epsilon'\ \epsilon' < \epsilon \wedge \text{AGENT}(\text{Della})(\epsilon') \wedge \text{KICK}(\epsilon') \wedge \text{PATIENT}(\text{it})(\epsilon')$

$\approx$ there was a previous kicking of it by Della

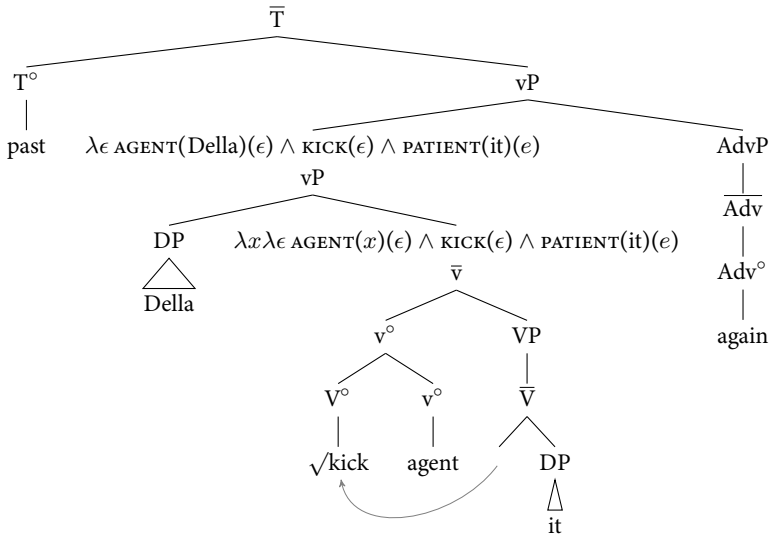
As expected on this account, (64) can get the same interpretation that (56) does.

> *Della, frustrated by her noisy refrigerator, kicked it. Her anger unquenched, …*

(64) Della kicked it again.

This interpretation emerges from the D-structure representation in (65).

(65)



Because *again* is combining with the same vP in (62) and (65), the contribution it makes to the meaning of the sentence, i.e., (63), is the same.

But (64) can have a meaning that (56) cannot have. To see this, consider the situation portrayed in (66).

> *Shai, frustrated with his noisy refrigerator, kicked it and left for the relative quiet of the neighborhood coffee shop. Awoken by the refrigerator's racket, …*

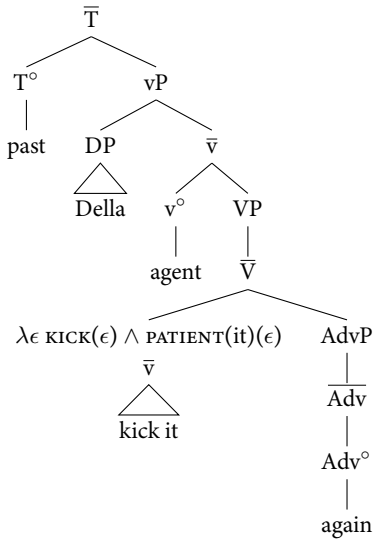
(66) a. # Della again kicked it.

b. Della kicked it again.

In this situation, the previous kicking was done by Shai, not Della. This is a context, then, which isn't consistent with the meaning in (63), and as expected the representation for (66a) given in (62) produces an anomalous result. That the word order in (66b) does not have a similar effect suggests that there is a representation for this sentence that is different from (65), which would produce the same anomalous meaning. Our parses for these sentences provide that representation; it is (67).

This observation originates with McCawley (1973). The conclusions from this difference that I describe here, come from Bale (2007), where it is also discovered that not all transitive verbs display this contrast. Bale uses this fact to argue that not all transitive verbs have a vP, and he reaches a similar conclusion about intransitive verbs. Indeed, his results are not entirely consistent with the portrayal of argument structure that has been developed in this book.

(67)



Our new syntax would also allow for ***again*** to combine with the VP headed by ***kick***, rather than its $\bar{V}$.

This representation causes the meaning that *again* contributes to the sentence to be (68), which is consistent with the context in (66).

(68) $\exists\epsilon'\ \epsilon' < \epsilon \wedge \textsc{kick}(\epsilon') \wedge \textsc{patient}(\text{it})(\epsilon')$

$\approx$ there was a previous kicking of it

This supports the hypothesis adopted in chapter 3 that the position the external θ-role occupies is different than the position the verb occupies. For these facts to emerge, it requires that there be a phrase that contains the verb that is distinct from the phrase that contains the external θ-role.

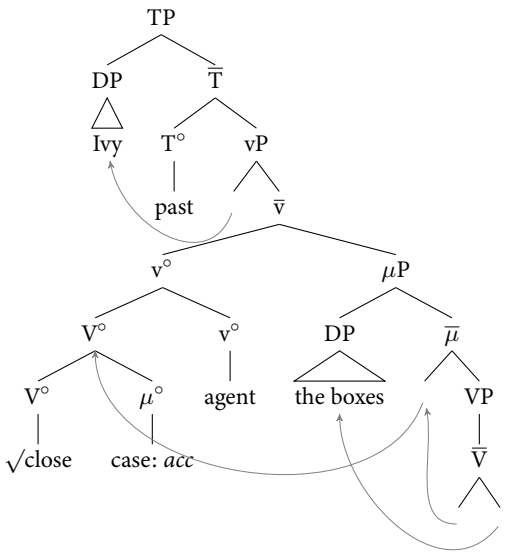
With this in mind, let's reconsider the alternation illustrated by (69).

(69) a. Ivy closed the boxes.

b. The boxes closed.

Our model of this is that the vocabulary item *close* spells out the root $\sqrt{\text{close}}$ with or without an accompanying agent θ-role. Because the agent θ-role is responsible for assigning accusative Case, its presence is also responsible for licensing the internal argument *the boxes* in a lower position. When agent isn't present, the internal argument gets nominative Case, and this leads to its movement into clause initial position. The derivations sketched in (70) illustrate.

(70)



As expected, the transitive version of *close* supports the ambiguity that arises when *again* is placed at the end of the VP. There is a meaning in which what *again* conveys is that there was a previous closing of the boxes by Ivy, as in (71), and there is a meaning which conveys merely that there was a previous closing of the boxes, as in (72).

> *Ivy is responsible for boxing up the diodes for shipment at the end of the line. She had just finished closing all the boxes, when she realized she'd forgotten to include the inventory sheets that are supposed to accompany each shipment. After opening them all up and including the sheets,…*

(71) Ivy again closed the boxes.
Ivy closed the boxes again.

> *Mako is responsible for boxing up the diodes for shipment at the end of the line. She had just finished closing all the boxes, when she realized she'd forgotten to include the inventory sheets that are supposed to accompany each shipment. She opened them all up and included the sheets, but had to leave before finishing the job. When the truck arrived to take them off,…*

(72) #Ivy again closed the boxes.
Ivy closed the boxes again.

These meanings arise because the D-structure for the transitive sentence has both a vP and a VP for *again* to combine with.

There is no vP in the intransitive use of *close*, however, and as expected therefore, *again* cannot produce a reading that would arise by combining with vP. Moreover, the meaning *again* conveys must include the object, as the contrast below indicates.

The company in an effort to increase efficiency had started using self-closing boxes. A timer was set at the beginning of each work day that would cause the boxes to automatically close and seal at the end of the day. Ivy, used to years of closing the boxes, frequently forgot about the new self-closing boxes, and would close them herself as they came into her station. Today was one of the days that Ivy mistakenly closed the boxes, but she caught her mistake before the inspector arrived and re-opened them all so that…

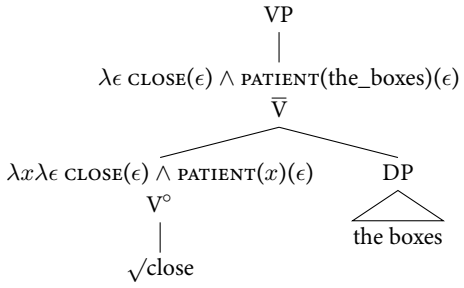
(73) the boxes closed again at the end of the day as planned.

The last of the self-closing boxes for the first run of the morning had just closed when the next group of boxes, on a delayed timer, arrived at Ivy's station. After being filled,…

(74) # the new boxes closed again.

A quick look at the D-structure for the intransitive case reveals why.

(75)



The only projection of V° that has the right meaning for *again* to combine with is V̄ and this constituent necessarily contains the object. To get the meaning necessary for the context in (74) would require *again* to combine with the verbal root $\sqrt{}$*close* directly, and this isn't semantically possible.

Interestingly, VPs headed by *close* support another meaning when *again* is present. To see this, consider the situation in (76).

The boxes arrived in the Receiving department as they usually do closed up ready for use. Robert opened them up in preparation for filling, and sent them to Ivy's station. After filling them,…

(76) Ivy closed the boxes again.

In this scenario, there is no previous event of closing the boxes. The boxes start this story in a closed state, and it is to this state that (76) says Ivy returned them. This meaning exists also for the intransitive use of *close*.

I was delighted when the tulips I'd planted the previous Fall finally bloomed. Their petals opened one warm afternoon to reveal a subtle multicolored pattern inside. But unaware of Nyctinasty, I was puzzled when…

(77) the tulips closed again that night.

In this story, the tulips have never closed previously. They start in a closed state, and it is to that state that (77) says they have returned. This meaning has a name. It is called the "restitutive" reading of *again*, and it is to be contrasted with the two meanings we have seen previously, which are known as "repetitive" readings.

Dowty (1979) contains perhaps the first extended discussion of this ambiguity. The account of the restitutive reading adopted here is the one provided in von Stechow (1995, 1996).

We can deduce from the presence of the restitutive reading that there is a constituent within the VPs of these sentences whose meaning describes the state of being closed. It is to that constituent that *again* must be combining to give rise to these meanings. That constituent has a meaning rather like that of the adjective *closed*, as we can see by considering (78).

(78) The tulips are again closed.

If we set aside the meaning of *closed* as a passive participle, and focus on the stative meaning it has when it is an adjective, then the message *again* conveys in (78) is the same that is conveyed in (77). Like (77), (78) says that the tulips are restored to a closed state. Note that the verb *closed* does not support that meaning when *again* precedes the verb, as in (79).

For the difference between passive participles and adjectives, see chapter 3.3.

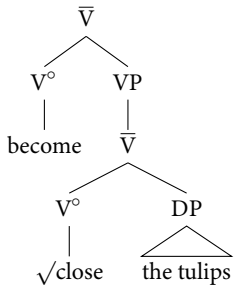
> *I was delighted when the tulips I'd planted the previous Fall finally bloomed. Their petals opened one warm afternoon to reveal a subtle multicolored pattern inside. But unaware of Nyctinasty, I was puzzled when…*

(79) # The tulips again closed that night.

The constituent that describes a state of being closed is not syntactically available to *again* in the position it occupies in (79).

All of this suggests that the VP associated with the verb *close* contains more syntactic structure than we have given to it. In addition to the optional presence of the external θ-role, there must be two phrases: one that denotes the state of being closed and another that describes the process of coming into that state. A parse that achieves that is (80).

(80)

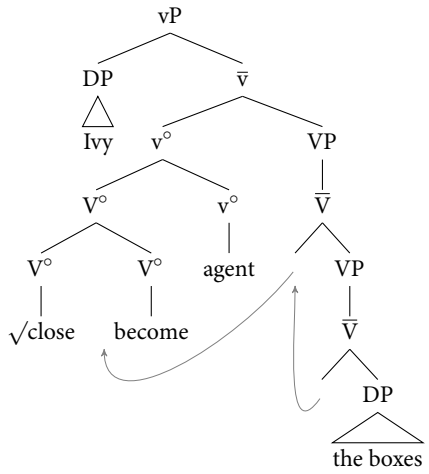


The intransitive *close*, on this view, is made of two syntactically distinct predicates. One, $\sqrt{close}$, describes states of being closed and takes, in this instance, *the tulips* as its argument. The other, here indicated by *become*, describes eventualities which lead to states. Roughly:

(81) $[\![\text{become}]\!] = \lambda P \lambda \epsilon\ \epsilon$ results in $\exists \epsilon'$ P(ϵ)=T

When the VP in (80) is embedded under the vP headed by the agent θ-role, we get the transitive version of *close*. Because the vocabulary item *close* can be matched to either the combination involving v° or not, we will want all of the morphemes that make up the meaning of *close* to be in an AGREE relation. If those morphemes come equipped with a •, it will require head movement to gather all of them into one X° position, as in (82) for instance.

(82)



The restitutive meaning for *again* arises when it is combined with a projection of $\sqrt{}$*close*.

Confirmation for this picture comes from considering how *again* interacts with modifiers that distinguish between the stative meaning of $\sqrt{}$*close* and the non-stative meaning of *become*. If we take the adjective *close* to represent the meaning of $\sqrt{}$*close*, then (83) shows us that *tightly* can modify it but *quickly* cannot.

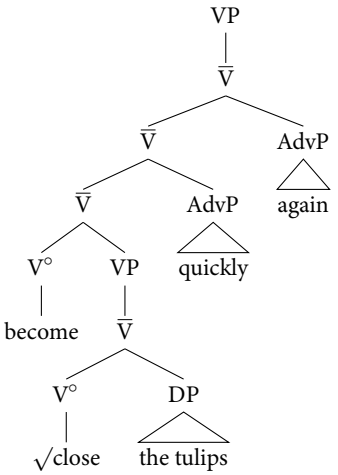
(83) a. The boxes remained closed tightly.
b. * The boxes remained closed quickly.

The model in (82) predicts correctly that (84) should not be able to support the restitutive reading because its syntax must be as indicated in (85).

> *I was delighted when the tulips I'd planted the previous Fall finally bloomed. Their petals opened one warm afternoon to reveal a subtle multicolored pattern inside. But unaware of Nyctinasty, I was puzzled when…*

(84) # The tulips closed quickly again that night.

(85)

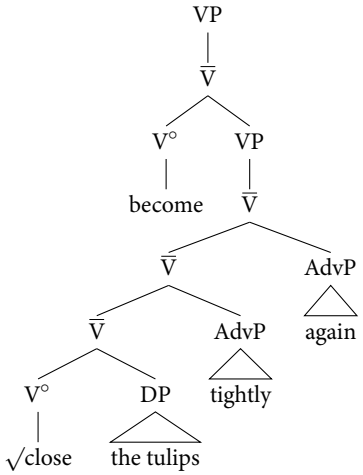


By contrast, (82) predicts that the restitutive reading should be found in (86) because the representation in (85) is available to it.

> *I was delighted when the tulips I'd planted the previous Fall finally bloomed. Their petals opened one warm afternoon to reveal a subtle multicolored pattern inside. But unaware of Nyctinasty, I was puzzled when…*

(86) The tulips closed tightly again that night.

(87)



The presence of the repetitive and restitutive reading with *again* is a phenomenon widely found in the world's languages, but there is considerable variation in how the readings correlate with the syntax. For instance, Minjeong Son[7] shows that *tasi*, the Korean equivalent of *again*, invokes the same restitutive/repetitive ambiguity that we have seen in English, as in (88).

[7] Son, Minjeong. 2006. Causation and syntactic decomposition of events. Doctoral Dissertation, University of Delaware

(88) Yenghi-ka koki-lul tasi el-le-ess-ta.
Yenghi-NOM meat-ACC again freeze-CAUS-PST-DEC
'Yenghi froze the meat again'
a. ≈ Yenghi froze the meat and it had been frozen previously, or
b. ≈ Yenghi froze the meat and someone had done that previously.
(Son, 2006, (49): 59))

Notice that in Korean the verbal morphology overtly signals that there is a stative part (*el*) and a "become" part (*le*, glossed as CAUS). A natural way of modeling the difference between English and Korean is letting the two V° that makeup the relevant vocabulary item map onto distinct exponents in Korean, which are brought together by Head Movement.

(89) Yenghi-ka [$_{VP}$ [$_{VP}$ koki-lul] [$_{V°}$ el le]]

Korean syntax allows object DPs to be moved to the left. This process allows Korean to have S-structures like that sketched in (90).

(90) Yenghi-ka [$_{XP}$ koki-lul [$_{VP}$ [$_{VP}$] [$_{V°}$ el le]]]

The ambiguity of (88), then, arises because (90) can be related to either of the D-structures in (91).

(91) a. Yenghi-ka [$_{VP}$ tasi [$_{VP}$ [$_{VP}$ koki-lul [$_{V°}$ el]] [$_{V°}$ le]]]
b. Yenghi-ka [$_{VP}$ [$_{VP}$ tasi [$_{VP}$ koki-lul [$_{V°}$ el]] [$_{V°}$ le]]]

In (91a), *tasi* combines with the VP headed by *le*, giving rise to a repetitive reading, and in (91b), *tasi* combines with the VP headed by *el*, producing the restitutive reading. While it is easier to see in the verbal morphology of Korean the two components of meaning that the repetitive and restitutive readings dependent on, it is harder to see from the word-order of Korean that the syntactic placement of *tasi* ("again") correlates with this ambiguity.A rapidly evolving project presently is understanding the cross-linguistic typology of adverbs like *again* and what they reveal about the hidden structure of VPs.

A challenge in the cross-linguistic study of ***again*** is distinguishing it from adverbs whose meaning is closer to ***back***. One can appreciate the closeness in the meanings of these two expressions by comparing ***Meg went back into the room*** and ***Meg went into the room again***. See Beck (2005) and Beck et al. (2008) for discussion.

5.2.2 Double Objects

Our present system derives the syntactic position of accusative Case marked arguments by putting them into an AGREE relation with μ° and forcing them to move to Specifier of μP. This predicts that there should never be two accusative Case-marked DPs dependent on the same μ°, and one expectation from that prediction is that no verb should have two accusative Case-marked objects. That expectation is frustrated by examples such as (92).

(92) a. Veena gave Rejean the answer.
b. Veena kicked Rejean the ball.
c. Veena sent Rejean a letter.
d. Veena baked Rejean some cookies.
e. Veena taught Rejean a song.
f. Veena promised Rejean the answer.
g. Veena denied Rejean his paycheck.
h. Veena envied Rejean his quick footwork.

These examples are sometimes described as involving the "double object" construction. The existence of the second DP is unexpected given our Case theory.

In the best known dialects of English, there is evidence that the Case that is borne by the second DP is not a structural Case. Recall that structural Case is what the Passive operation redirects. The contrast between (93) and (94) indicates that only the first DP bears structural Case.

See chapter 3.7 for a discussion of structural Case and the passive.

(93) a. Rejean was given the answer.
b. Rejean was kicked the ball.
c. Rejean was sent a letter.
d. Rejean was baked some cookies.
e. Rejean was taught a song.
f. Rejean was promised the answer.
g. Rejean was denied his paycheck.
h. Rejean was envied his quick footwork.

(94) a. * The answer was given Rejean.
b. * The ball was kicked Rejean.
c. * A letter was sent Rejean.
d. * Some cookies were baked Rejean.
e. * A song was taught Rejean.
f. * The answer was promised Rejean.
g. * His paycheck was denied Rejean.
h. * His quick footwork was envied Rejean.

Our syntax, then, should be able to recognize the different status of the two Cases involved.

The double object construction is common cross-linguistically and this asymmetry in the status of the two Cases is not always found. For instance in the Bantu language Kichaga either of the two arguments of a double object construction can switch to Nominative in the passive.

The "Nominative" Case relation is not signaled by morphology on the DP in Kichaga, but is instead signaled, as it is in Bantu generally, by "subject agreement" morphology. On the correspondence between subject agreement and nominative Case, see chapter 4.

(95) a. M̀-kà n-ā́-ī́-lyì-í-ò k-èlyâ.
1-wife foc-1S-pr-eat-ap-pas 7-food
'the wife is being affected by the food being eaten.'

b. K-èlyá k-ī́-lyì-í-ò m̀-kà.
7-food 7S-pr-eat-ap-pas 1-wife
'the bood is being eaten for/on the wife'

(Bresnan and Moshi, 1990, (5):p. 150)

The same is found in Norwegian.

These data originally from Tor Åfarli.

(96) a. Jon vart gitt ei klokke.
John was given a watch
'John was given a watch.'

b. Ei klokke vart gitt Jon.
a watch was given Jon.

(Woolford, 1993, (2):p. 682)

There are also dialects of English that permit the second object to convert to Nominative in the passive. In Manchester English, for instance, Haddican (2010) reports that when the object is a pronoun it may convert to Nominatives; examples like 97 are grammatical.

(97) a. It was sent him.

b. It was given her.

And Biggs (2016) reports that this is possible for definites in Liverpool English; examples like (98) are grammatical.

(98) a. The book was given the teacher.

b. The package was sent her nan's.

These double objects are known as "symmetric," and there are a variety of properties that correlate with the distinction between a symmetric and asymmetric double object construction. A complete account of the double-object construction would explain these correlations. This remains an area in which no consensus has emerged, and I will set this goal aside.

See Gary and Keenan (1977), Baker (1988), Bresnan and Moshi (1990), Woolford (1993), Anagnostopoulou (2003), Haddican and Holmberg (2019), and references cited therein, for some discussion.

Some of the verbs that allow for the double object construction permit another expression of their arguments, as (99) shows.

(99) a. Veena gave the answer to Rejean.

b. Veena kicked the ball to Rejean.

c. Veena sent the letter to Rejean.

d. Veena baked some cookies for Rejean.

e. Veena taught a song to Rejean.

f. * Veena promised the answer to/for Rejean.

g. * Veena denied his paycheck to/for Rejean.

h. * Veena envied his quick footwork to/for Rejean.

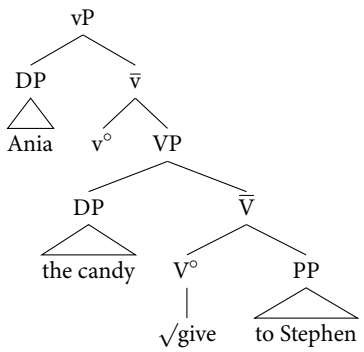
There is reason to believe that the relationship between these two ways of expressing the arguments of the relevant verbs is rule governed, that is, it's not merely that speakers of English have learned for each verb that they may express their internal arguments in these two frames. One reason for thinking this is that when new words that have the right properties are coined, speakers associate that word with both argument structure frames even if they've only been presented with one of them. For instance, Alec Marantz notes that the verb *shin*, can be introduced to a speaker with (100a), and the speaker will readily use (100b).

(100) a. Brian shinned the ball to Jason.
b. Jason shinned Aynat the ball.

There is experimental evidence that supports this conclusion as well. Gropen et al. (1989), which is a study of the acquisition of the double object construction by children, demonstrated for children that they knew that a double object frame was possible for a verb if they were taught something about that verb's argument structure. They presented children with scenarios where a character transports another character in a gondola car to some goal, with sentences like "The bear is pilking the pig to the giraffe." They were then presented with similar scenarios and asked to describe those scenarios. They found that children in the age range of 6 to 8 years old would respond with double object constructions (for example: *X is pilking Y Z*) 44% of the time. Knowing something about the meaning of a verb—perhaps its argument structure—confers knowledge about whether that verb can support the double object construction.

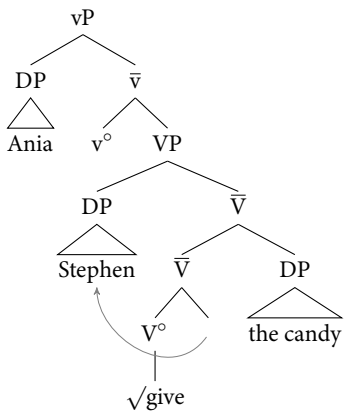
One way of capturing the relationship between the two argument structures is Transformational. We could imagine that there is a rule, or a series of rules, that convert a common D-structure representation into one or the other of these surface representations. This is the view taken by Larson (1988), who builds on Chomsky (1975). Larson proposes a rule that takes a D-structure like (101) to a surface structure like (102).

(101)



I am converting Larson's proposal into the framework developed here. Larson's work predated the adoption of vP, though one could argue that the vP hypothesis was nascent in his own syntax. He proposed that there are two VPs in the underlying structure of clauses, and that it is the verb heading the higher of these VPs that assigns structural (i.e., accusative) Case. This is exactly what the v° of vP is.

(102)



Note that he places the THEME argument in Specifier of VP at D-structure.

His proposed Transformation consists of two rules that work in concert. One deletes the preposition, here *to*, thereby depriving its argument of Case. The second moves the DP in Specifier of VP to a position that makes it sister to $\overline{V}$. The Case filter is satisfied by using Argument Movement to bring the DP that was dependent on the preposition for Case into Specifier of VP. There it can get Case from v°. The DP that is now sister to $\overline{V}$ gets Case from a special rule that interprets the $\overline{V}$ it is a sister to as a Case assigner.

As in the theory developed here, the D-structure representation on Larson's view is responsible for the semantic interpretation of the sentence. A way of understanding the alternation, then, is to posit that knowing that a verb takes the two arguments shown in (101) licenses this D-structure and the consequent transformations that give rise to the double object frame. When an English speaker learns that the meaning of *give* determines the D-structure in (101), the suite of transformations that the speaker has learned constitute part of the grammar of English are available to produce (102). Thus, learning the argument structure of *give* in English confers the knowledge that it will allow both frames, thus explaining facts like Marantz's.

For this to be workable, discrepancies in the meanings of some of the alternants will have to be accounted for. Consider, for instance, the pair in (103).

(103) a. Johannes baked cookies for Christine.
b. Johannes baked Christine cookies.

(103a) describes eventualities in which Johannes baked cookies on behalf of Christine, but (103b) does not. (103b) requires that Johannes baked cookies for a particular purpose: for Christine to have them. To capture this, Larson suggests that there is more than one *for*, and that they are distinguished by their meaning. The transformation that deletes *for*, thereby causing its argument to undergo A Movement, targets the particular *for* that denotes the "intends to have" type relation. For those verbs that do not have the frame for expressing their arguments that involve a preposition, Larson suggests that the process of preposition deletion is obligatory, perhaps because English doesn't have a vocabulary item that corresponds to that preposition. In

this way, then, *promise*, *deny*, and verbs like them obligatorily undergo the transformations that give rise to the double object frame. The transformations that are responsible for the alternation, then, will have to map one particular argument structure onto a D-structure that gives rise to either S-structure.

But the argument structure of a verb doesn't seem sufficient to predict whether a verb will be able to participate in the alternation. For instance, *donate* and *give* seem to involve the same θ-roles, and yet only *give* allows for the double object frame.

(104) a. Ania donated candy to Stephen.
b. * Ania donated Stephen candy.

If the productivity of the relationship is merely tied to argument structure, then the difference between *give* and *donate* would have to arise because they do not have the same θ-roles, or are in some other way different in their argument structure. But this seems difficult to maintain. Note that *X donates Y to Z* entails *X gives Y to Z* whereas *X lends Y to Z*, does not. And yet *lend* unlike *donate*, allows for the alternation. Pinker (1989) suggests that what distinguishes *donate* from *give* is that *donate* derives historically from Latin, while *give* has its roots in Germanic. One way of conceiving of this is that the process of learning involves the ability to subdivide verbs into arbitrary classes, and this process is sensitive enough to phonological patterns to be able to distinguish those verbs which inherit characteristics of Latin morpho-phonology from those which don't.One of the characteristics is length: English verbs of Latinate origin have, on average, more syllables than those of Germanic origin. What an English speaker might have acquired, then, is a subcategorization of verbs that includes criteria like their argument structure and phonological pattern, in addition to the transformational processes that deliver the two frames of the alternation from a common D-structure to one of those subcategorizations. These transformational processes would have to be keyed to the relevant subcategory of verb.

See Yang (2016) for a good discussion of this issue.

Or consider the problem posed by the contrast in (105), also discussed in Pinker (1989, pp. 110-1).

(105) a. Ania tossed Stephen a ball.
b. Ania threw Stephen a ball.
c. Ania flipped Stephen a ball.
d. Ania kicked Stephen a ball.
e. * Ania pulled Stephen a ball.
f. * Ania schlepped Stephen the balls.
g. * Ania lifted Stephen the balls.
h. * Ania hauled Stephen the balls.

Both *toss* and *pull* describe eventualities in which something is moved and something at whose location the moved term ends up. Both sentences in (106) entail that a ball moved and that its trajectory aims at Stephen.

(106) a. Ania tossed a ball to Stephen.
b. Ania pulled a ball to Stephen.

Our way of modeling this is to give *toss* and *pull* the same θ-roles. And yet this match in argument structure is not sufficient to allow *pull* to express its arguments in the double object frame. Both of these verbs are Germanic in origin, and they do not differ in the morpho-phonological ways that mark the distinction between Latinate and Germanic origin. Pinker suggests that what might distinguish these cases is in the structure of the eventuality that these verbs describe. He notes that *toss* describes a punctual eventuality: what makes an eventuality a tossing is the kind of action that puts an item into motion. By contrast, *pull* describes an eventuality that includes the trajectory of motion and cannot therefore be punctual. Pinker notes that many verbs which contrast along this dimension also differ in the way that (105) illustrates.To the extent that this observation holds, it suggests that there is more to the meaning of the verbs that participate in the alternation than just their argument structure. To account for facts like these, Larson's set of rules would have to be made sensitive to information that goes beyond the θ-roles associated with a verb.

See Krifka (1999) and Krifka (2003) for attempts to define this difference, and suggestions about how to use it to explain accessibility to the double object frame.

Richard Oehrle[8] argues against an account like Larson's that relies on putting the alternation in the set of rules that make derivations. Instead he suggests that the process belongs to rules that determine how the pieces that make up the denotation of a verb are mapped to vocabulary items. His evidence consists of observations about how the two frames differ semantically in ways that suggest that they do not have the same argument structure. To the extent that the argument structure of a verb is given by the shape of the VP hosting that verb, then these facts speak against giving both frames the same D-structure. He notes, for instance, that the θ-role assigned to the subject of *give* seems to differ depending on how its internal arguments are arranged. For instance, consider the pair of sentences in (107).

[8] Oehrle, Richard. 1976. The grammatical status of the English dative alternation. Doctoral Dissertation, Massachusetts Institute of Technology

(107) a. Nixon gave a book to Mailer.
b. Nixon gave Mailer a book.
(Oehrle, 1976, section 1.1, p. 19)

Oehrle notes that (107b) allows for an interpretation in which Nixon's behavior is responsible for Mailer writing a book. (107a), by contrast, cannot have that interpretation; instead, Nixon must be responsible for Mailer coming into possession of a book. This difference could be partly modeled by letting the external θ-roles be different. The *give* in (107a) might assign an AGENT θ-role, whereas the *give* in (107b) might assign a CAUSE θ-role. This shift in the external θ-role doesn't accompany the alternation with every verb. The external θ-role for *bake*, *send*, etc. remains AGENT in both frames.

For some verbs, there is a shift in the internal θ-roles assigned by the verb. Most famous, perhaps, are cases that involve certain kinds of motion verbs, including those in (108).

(108) a. Jon Ander sent the report to Aleksei.
b. Jon Ander kicked the report to Aleksei.
c. Jon Ander threw the report to Aleksei.
d. Jon Ander mailed the report to Aleksei.
e. Jon Ander tossed the report to Aleksei.

The *to*-phrase in these examples refers to a location, and so we can surmise that the θ-role assigned to this phrase calls for something with the denotation of a location. Let's call this θ-role LOCATION. We can see that the *to*-phrases refer to a location because these verbs allow this PP to be expressed with the locative pro-forms *there* and *where*.

(109) a. Jon Ander sent the report there.
b. Jon Ander kicked the report there.
c. Jon Ander threw the report there.
d. Jon Ander mailed the report there.
e. Jon Ander tossed the report there.

(110) a. Who sent the report where?
b. Who kicked the report where?
c. Who threw the report where?
d. Who mailed the report where?
e. Who tossed the report where?

In the double object frame, however, the argument that corresponds to the *to*-phrase is not a locative, as can be seen from the ungrammaticality of (111).

(111) a. * Jon Ander sent there the report.
b. * Who sent where the report?
c. * Jon Ander kicked there the report.
d. * Who kicked where the report?
e. * Jon Ander threw there the report.
f. * Who threw where the report?
g. * Jon Ander mailed there the report.
h. * Who mailed where the report?
i. * Jon Ander tossed there the report.
j. * Who tossed where the report?

The examples in (111) are grammatical if *there* or *where* is replaced by a pronoun that refers to an individual, like *him*. One way of describing the shift in meaning illustrated here is that the non-theme internal argument is a possessor, or perhaps goal. In saying *Jon Ander sent Aleksei the report*, one communicates that the sending eventuality was intended to result in Aleksei's having the report.

Indeed, as Georgia Green[9], and many following her including Oehrle, have noted, the double object constructions frequently lend themselves to this kind of paraphrase. This is, as noted above, a way of describing the difference between *bake X for Y* and *bake Y X* for example. In each of the cases in (113), a suitable paraphrase of the meanings is (112), where X is the subject, Y the second object, V the verb and Z the first object.

[9] Green, Georgia M. 1974. ***Semantics and syntactic regularity***. Bloomington, Indiana: Indiana University Press

The idea that the meaning of double object constructions involves "goals," in the way described by (112), comes from Koenig and Davis (2001). See Beavers (2011) for discussion.

(112) X V Y with the goal of Z having Y.

(113) a. Hotze sewed Jose a handkerchief.
b. Pius ironed Hotze a shirt.
c. Pat cooked Pius a biscuit.
d. Katy faxed Bernhard the money.
e. Bernhard painted Katy a picture.
f. Pius took Janina the gift.
g. Pat left Bernhard the report.
h. Jose brought Hotze a chalice.
i. Pius guaranteed Pat a position.
j. Jose granted Janina a pass.
k. Katy drew Jose a tesseract.
l. Hotze bought Jose a ring.
m. Janina advanced Jose a portion of his pay.
n. Jose promised Pat the answer.

The sense of *have* in these paraphrases is difficult to nail down, as it is sometimes not "possession" in the normal use of that verb. This is clear in examples like (114).

(114) a. Katy read Bernhard the report.
b. Janina sang Hotze a song.
c. Janina taught Jose a madrigal.
d. Pat assigned Bernhard the hardest problem.
e. Jose quoted Katy the price.
f. Janina showed Pius the answer.

What (114a) claims Bernhard possesses as a result of Katy's reading the report is not the report, but its contents. Similarly, what (114e) claims Katy has as a result of Jose's quote is not the price, but knowledge of the price. For discussion, see, in addition to Green (1974) and Oehrle (1976): Kayne (1984b), Harves and Kayne (2012), Manzini and Savoia (2011), Manzini and Franco (2016), Harley (2004), Harley and Kyoung Jung (2015), Beavers (2011), and especially Beavers et al. (2009). Because locations cannot "possess" things (**There has a ball*), the ungrammaticality of (111) will follow if the argument

structure of the double object construction requires the "have" predicate in (112) to relate the two objects.

Let's express Green's thesis, then, as the claim that all double object constructions have an argument structure that can be expressed with the formula (115).

(115) [Subject verb 2nd object] $\mathcal{R}$ [1st object have 2nd object]

For *buy*, *throw*, *mail*, etc., $\mathcal{R}$ is something like "with the goal of." The examples in (116) illustrate that $\mathcal{R}$ can also be something like "cause."

VPs that describe eventualities that have a goal are said to be "telic," and telic events can be identified by the ways in which they interact with certain modifiers. See Vendler (1967), and Dowty (1979) who is a good overall guide to the relevant phenomena. Jackendoff (1996) claims that double object constructions have the properties of telic events.

(116) a. Meredith lent Andrew the car.
b. Della fed Andrew dinner.
c. Andrew served Della lunch.
d. Meredith rented Della the car.
e. Andrew awarded Meredith the prize.
f. Della sold Meredith the car.
g. Andrew gave Della the picture.

What distinguishes these cases from (113) and (114) is that the possession relation between the two objects in (116) is entailed. This can be seen by comparing (117) with (118).

Which verbs support this entailment in the double object construction and which don't is not always clear. Green classed ***teach***, for instance, among those that do, but Oehrle argues otherwise.

(117) a. Pius ironed Hotze a shirt, but it got destroyed before Hotze got home.
b. Katy drew Jose a tesseract, but he never saw it.
c. Pat left Bernhard the report, but he never found it.
d. Jon Ander tossed Della the ball, but it got intercepted.

The "#" sign indicates infelicity, or semantic ill-formedness.

(118) a. # Meredith lent Andrew the car, but he never got it.
b. # Della fed Andrew dinner, but he didn't eat it.
c. # Della sold Meredith the car, but she didn't own it.
d. # Andrew gave Della the picture, but she never got it.

Hovav and Levin (2008) point out that this entailment also holds when these verbs are used in the other frame.

(119) a. # Meredith lent the car to Andrew, but he never got it.
b. # Della fed dinner to Andrew, but he didn't eat it.
c. # Della sold the car to Meredith, but she didn't own it.
d. # Andrew gave the picture to Della, but she never got it.

What the $\mathcal{R}$ is in the double object construction varies with the root, it appears.

Even (92g) might be able to be fit to this model if one of the ways of combining the two pieces of (112) can include negation.

(92g) Veena denied Rejean his paycheck.

An awkward, but not obviously incorrect, paraphrase of the meaning of (92g) is (120).

(120) Veena denied his paycheck and this caused not Rejean have his paycheck

Note that *deny* belongs to the class of verbs that support the entailment that we see in (118).

(121) # Veena denied Rejean his paycheck, but he got it anyway.

So we should build into its argument structure the *cause* component that is responsible for that entailment. And finally, we might capture the meaning of (92h) with an argument structure that corresponds to the paraphrase in (122).

(92h) Veena envied Rejean his quick footwork.

(122) [Rejean has his quick footwork] causes Veena envy

On this view, the possession of quick footwork by Rejean is the cause of Veena's state of envy. The causation relation reverses the two pieces of the meaning.

Green suggested that all versions of $\mathcal{R}$ included a *cause* component. She didn't countenance the goal oriented meaning that (112) uses. Capturing the differing entailment facts of (117) and (118) could be done by letting the $\mathcal{R}$ for the verbs that don't support the entailment have a meaning paraphrased by (123).

See Beck and Johnson (2004) for this view.

(123) $\mathcal{R}$ = cause becoming

On this view, the meaning of (124) matches the (terrible) paraphrase in (125).

(124) Ania threw Stephen the ball.

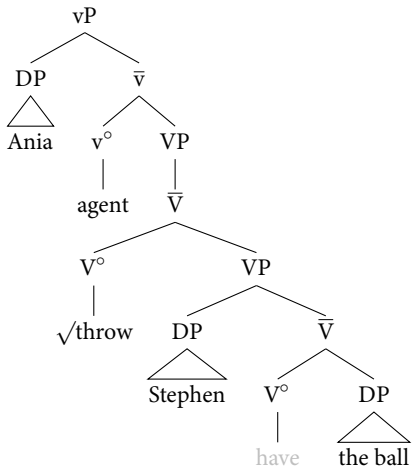
(125) Ania threw the ball which caused [Stephen to have the ball] becoming.

On this view, the eventuality of throwing the ball causes another eventuality which, if things progress normally, leads to Stephen having the ball. If $\mathcal{R}$ is restricted to flavors of *cause*, then this might reflect the fact that the double object construction is built upon an argument structure that necessarily includes a "cause X have Y" ingredient.

Richard Kayne[10] has argued that the meaning component associated with the *have* part of our paraphrases is found in the syntax of the double object construction. On his view the D-structure of a double object construction has the shape in (126).

[10] Kayne, Richard S. 1984b. Unambiguous paths. In ***Connectedness and binary branching***, 129–164. Dordrecht: Foris Publications

(126)



I am fitting Kayne's proposal into a syntax which gives the external θ-role the position v°.

Kayne's argument involves comparing the argument structure of verbs with the argument structure of nouns morphologically related to those verbs. I postpone looking at his argument until the next section.

If (126) is correct, there is a phrase within the double object construction that describes the possession relationship between the two objects. We should be able to detect the presence of that phrase using *again*. Sigrid Beck[11] has argued that this expectation is fulfilled. To see this, consider the contrast between (128).

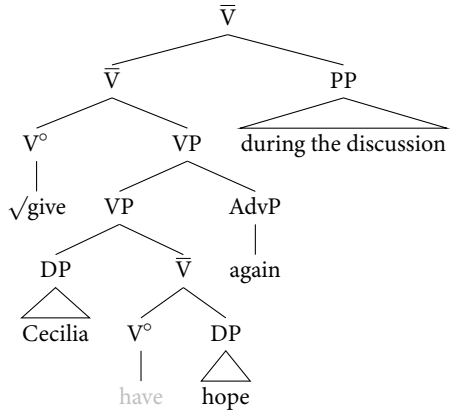
(127) *Cecilia lost her job and for the first time in her life, became despondent about her future. After he was introduced to Cecilia at the meeting, …*

(128) a. Marcin gave Cecilia hope again during the discussion.

b. # Marcin gave Cecilia hope during the discussion again.

The word order in (128a) permits the parse in (129), but the word order in (128b) forces a parse like that in (130).

(129)

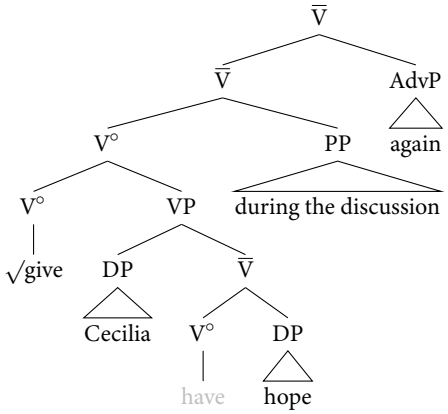


Beavers et al. (2009) credits McCawley (1974) with a similar proposal about making the ***have*** component part of the syntax of double object constructions.

Harley (1996) argues that only languages in which the possession relation is lexicalized into a verb like English's ***have*** are capable of hosting the double object frame. In her (small) sample of languages, there was a correlation between the languages that permit the double object construction and those that use a verb to express possession. (A common alternative way of expressing possession is with the copula (e.g., English ***be***) and a goal preposition or case.) She suggests this supports Kayne's proposal to the extent that the structure in (126) requires an X° that can express possession.

[11] Beck, Sigrid, and Kyle Johnson. 2004. Double objects again. ***Linguistic Inquiry*** 35:97–123

(130)



In the context provided, there is no prior eventuality of giving Ceclia hope, and that is the meaning that would be required of (130). Instead, the context supports the inference that Cecilia was in a prior state of hopefulness, and that is the meaning our semantics for *again* would deliver to (129). This is just what is expected if there is the *have*-VP within the double-object construction that Kayne proposes.

We can see the same confirmation by considering how certain duration modifiers work. For instance, the meaning of (131) has a meaning in which *for the week* specifies how long the possession state lasts.

See Beavers et al. (2009), and references cited therein, for this argument.

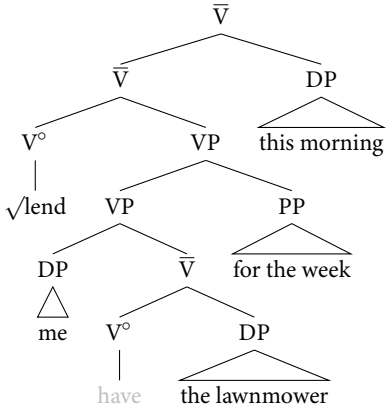
(131) She lent me the lawnmower for the week this morning.

Note that the two temporal modifiers do not lead to a conflict in (131), like they do in (132).

(132) # She hugged me for the week this morning.

In (132), both temporal expressions must describe the time at which the one eventuality of *hugging* transpires, leading to the conflict. But in (131), by contrast, there are two descriptions of eventualities that the temporal modifiers can combine with, as shown in (133).

(133)

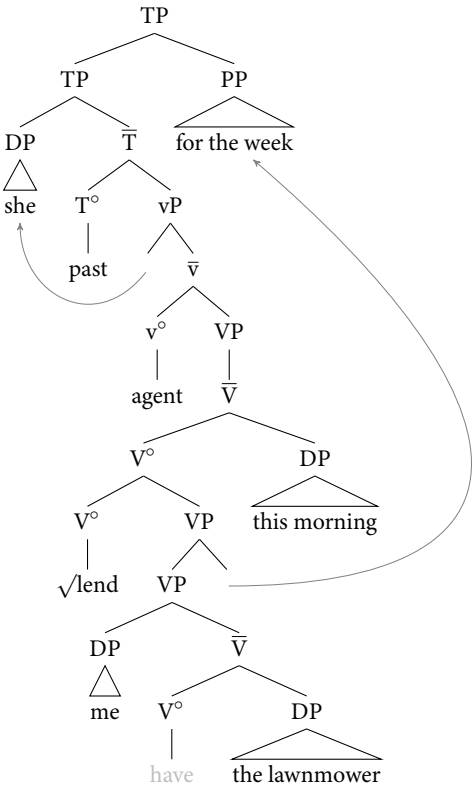


Unlike *again*, however, the word order doesn't correlate with the availability of this reading.

(134) She lent me the lawnmower this morning for the week.

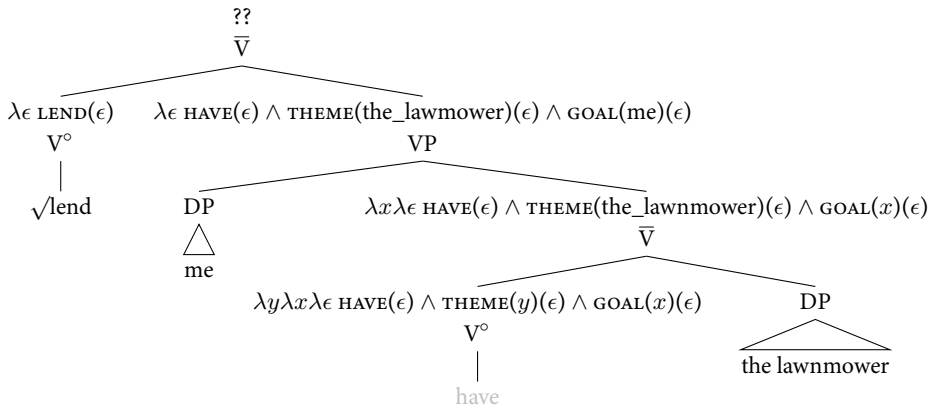
Perhaps this is because, unlike *again*, *for the week* can undergo Complex NP Shift.

(135)



Kayne's syntax does not combine with our semantics to deliver the right meaning. If the meaning of the root is a simple description of eventualities, then our present semantics doesn't include a way of putting the pieces together.

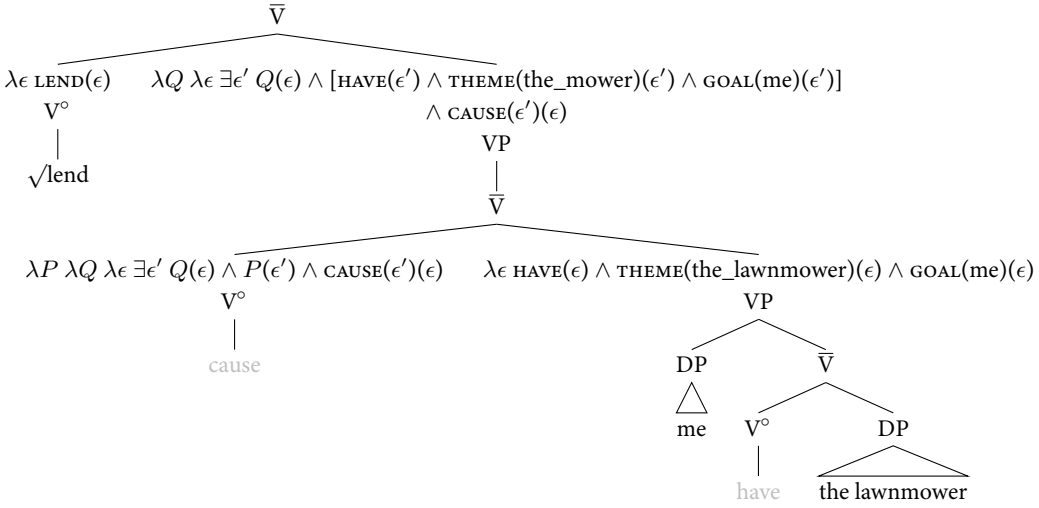
(136)



What is missing here, of course, is the $\mathcal{R}$ that connects the lending eventuality with the having eventuality. As we've seen, the content of $\mathcal{R}$ varies with the lexical item chosen. How these two pieces are semantically composed

should therefore be part of the information that comes with the lexical item. One way of doing that would be to build into the denotation of the root this meaning. See Beavers and Koontz-Garboden (2017, 2020) for that direction. Alternatively, there could be another syntactic piece to the VP which, when assembled with the others, is matched to the relevant lexical item. For *lend*, that could be achieved with the D-structure in (137).

(137) $\lambda\epsilon\exists\epsilon'$ LEND(ϵ) $\wedge$ [HAVE(ϵ') $\wedge$THEME(the_mower)(ϵ') $\wedge$ GOAS(me)(ϵ') $\wedge$ CAUSE(ϵ')(ϵ)]



Other verbs—*deny*, *toss*—could express the alternative $\mathcal{R}$s put into the V° position that *cause* occupies in (137).

What these implementations of Green's thesis miss is how to capture the fact that the second object of the double object construction is an object of the root as well. That seems semantically to be the case, as we can see from (138).

(138) a. # Laura melted Amalia a puddle.
b. Laura melted Amalia the ice.

The anomaly of (138a) is the same as the anomaly of *Laura melted a puddle*: puddles cannot be the patient of a melting eventuality.

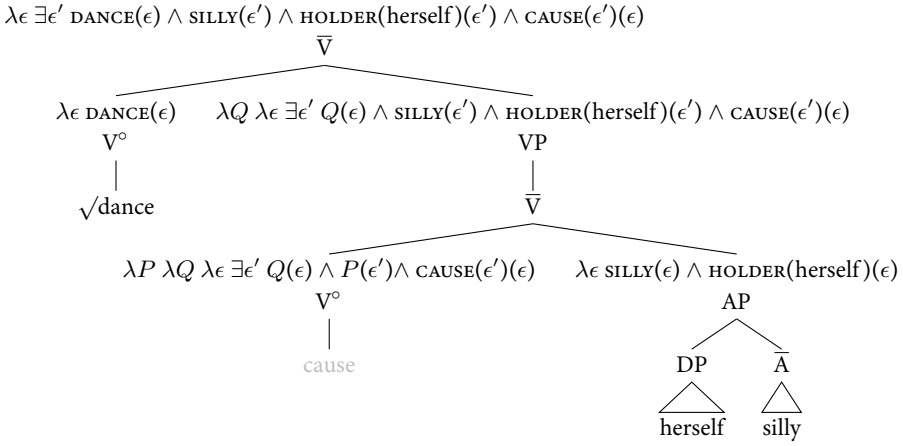
Note also that obligatorily transitive verbs can fit into the double object construction.

(139) a. Tohru sent (Suzanne) the letter yesterday.
* Tohru sent yesterday.
b. Tohru rolled (me) the marbles yesterday.
* Tohru rolled yesterday.
c. Tohru brought (me) the marbles yesterday.
d. * Tohru brought yesterday.
e. Tohru mixed (me) a drink yesterday.
f. * Tohru mixed yesterday.
g. Tohru bought (me) a new map yesterday.
h. * Tohru bought yesterday.

And no obligatorily intransitive ones can, even though intransitives can sometimes figure in resultative constructions which have many of the same features of the syntax and semantics of (137), as (140) illustrates.

See Hoekstra (1988) and Kratzer (2005) for a syntax and semantics of this kind for resultatives.

(140) Suzanne danced herself silly.



(137) would lead us to expect that something similar would be possible with the double object construction. But all obligatorily intransitive verbs are, like (141), ungrammatical in the double object construction.

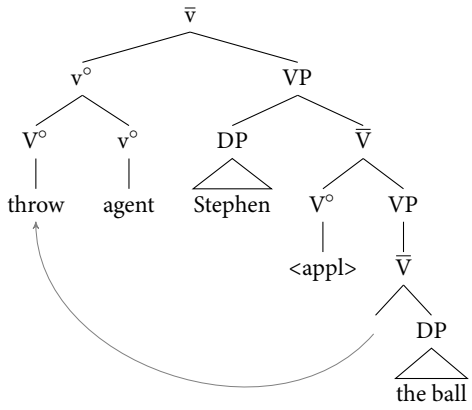
(141) * Suzanne danced herself two blisters.

This all makes sense only if the second object is a genuine object of the verbal root.

This has led Bruening (2010a,b), among others, to a syntax like (142).

Marantz (1993) might be the first to propose this structure. See also Pylkkänen (2008).

(142)

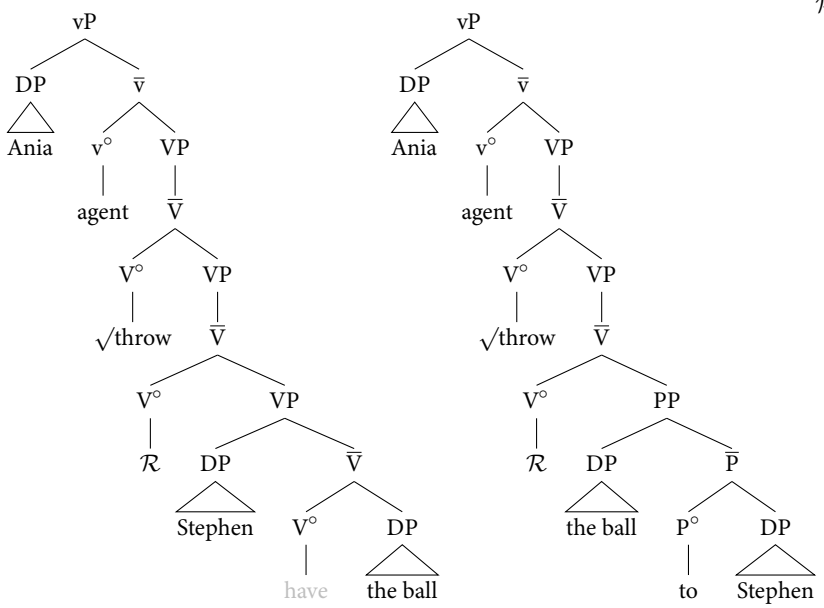


Here the <appl> head must somehow encode what I've expressed as HAVE. But this syntax does not allow for there to be a constituent that is made up of the two objects and HAVE (or its equivalent) that does not include the root. That is what we seem to require to get the facts about temporal modifiers and *again*. What seems to be required is for the second object to be both an argument of HAVE and, independently, an argument of the root. Our syntax does not presently provide a means for doing that.

If we set this problem aside for a moment, we can now ask how to capture the relationship between the two ways of expressing arguments of the verbs that participate in the alternation. If the argument structures of the two variants are not (always) identical, an approach that credits the regular relationship between them to rules that make a derivation seems wrong. As Oehrle concluded, what we look for is a process that relates different D-structures. We need a process that says that if a vocabulary item expresses such-and-such an argument structure, then it will express such-and-such other argument structure too. Harley (2002) suggests that the two D-structure representations we want to relate to a motion verb, like *throw*, are (143).

See Harley (2012) and Harley and Kyoung Jung (2015) for some modifications. I change the particulars of her proposal to give a place for $\mathcal{R}$, and fit our constraints on phrase markers.

(143)



Recall that the $\mathcal{R}$ which accompanies *throw* is the "cause becoming" or "with the goal of" sort; it does not entail that the state denoted by its complement comes to exist. And recall that the *to* which shows up with *throw* is a locative; its meaning can be paraphrased by "is at the location of." On Harley's view, then, both instantiations of *throw* involve a root that describes a throwing eventuality and a state that is the target result of that eventuality. In the case of the double object construction, that state is possession (i.e., *have*), and in the PP frame it is a thing residing in a location (i.e. *to*). That there is a phrase that denotes a locative state in the PP frame is confirmed by *again*.

> *The coach put the balls in the endzone before the game. When one of the balls rolled into the field,…*

(144) The waterboy threw [[it to the endzone] again].

In the context provided for (144), there is no previous throwing eventuality, and so *again* cannot be combining with the VP headed by *throw*. Instead, *again* attaches to the PP headed by *to*, conveying that the state of the ball being in the endzone is being restored.

If these were the correct D-structures for the two frames, then the relationship between them could be expressed with a rule like (145).

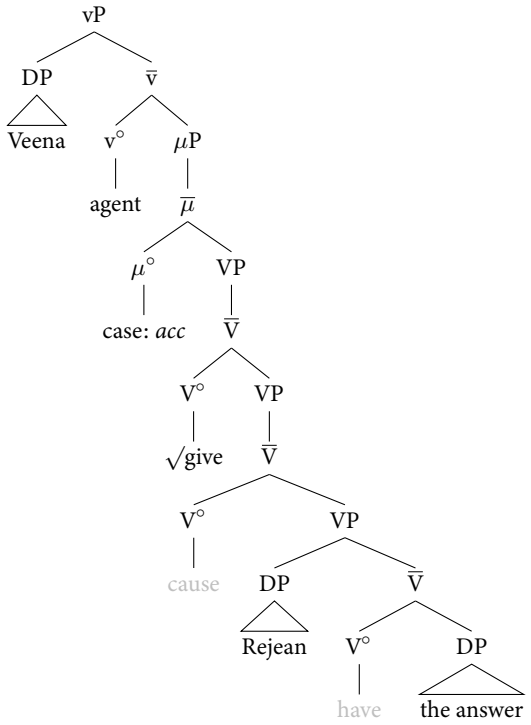
(145) If α is a vocabulary item that expones $\sqrt{\alpha}$+$\mathcal{R}$ + *have*,
then α also expones $\sqrt{\alpha}$+$\mathcal{R}$+*to*

What an English speaker has done is learn which verbs can be α.

The picture of the alternation between the PP and double object frames given here harbors several problems. In addition to the problem of understanding how the second object of the double object frame is an argument of the root,there is also the problem of understanding how rules like (145) can be made part of a natural class of processes, rather than simple *ad hoc* descriptions of what is needed. Nonetheless, if something like Kayne's proposal for the double object construction can be maintained, it does provide an account of how two structural cases arise in these constructions. Consider how our system of accusative Case assignment works in a Kaynian model of the double object construction.

Note that (143) also (wrongly) makes the first object of the PP frame not an argument of the root.

(146)



Suppose that *have* comes with an accusative Case feature. Perhaps this is telling us that the category of *have* is actually P°,a category that English otherwise imbues with an accusative Case feature. The second DP is in a position that allows its unvalued Case feature to be valued by the accusative Case feature on *have*. But it is too distant to have its Case feature valued by μ, since there is a another DP—the first object—between them. The first object's unvalued Case feature can get valued by the accusative feature in μ, however. Since it is the Case feature that is found in μ that is redirected in the Passive,

See Kayne (1993) for the idea that ***have*** is a preposition in the double object construction. Occasionally, the idea that what I've represented with "***have***" would be lexicalized as the preposition ***with*** is suggested. Expressions like ***the woman with a hat/all the answers*** involve a meaning for ***with*** that is close to what's needed.

we have an account for why in (some dialects of) English, the first DP, but not the second, converts to Nominative when the double object verb is passivized.

Kayne's thesis that the double object construction houses a small clause aids in capturing the facts about *again* and it sheds light on the availability of two accusative Cases in these constructions. His argument for the small clause, however, involves a deduction he makes from the fact that the double object construction does not show up in the NPs headed by nouns derived from verbs. Let's now examine this argument.

5.3 Argument Structure of Nominalizations

To appreciate Kayne's argument, we'll first have to learn a few things about the argument structure of nouns. We'll focus our discussion on nouns that are morphologically related to verbs. If we imagine that the nouns and verbs which are morphologically related share a root—that is are built from the same underlying morphological piece—then what we can learn by studying their different argument structures is something about how nouns and verbs differ. This is what is relevant to Kayne's argument.

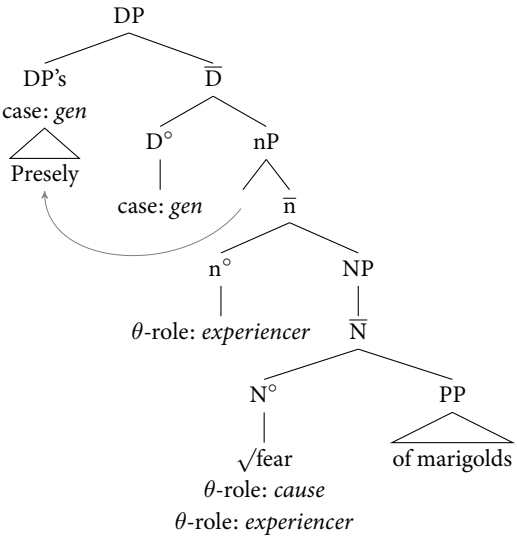
5.3.1 How the Theta Criterion applies to nominals

When nouns are related to verbs, the arguments they can combine with are often parallel to those found with the verbs. Rather than nominative Case, the external arguments of nouns appear in the genitive Case, and in the Specifier of DP position. Some examples are in (147).

(147) a. Presley's fear of marigolds
b. Meg's dislike of MaxElide
c. Noah's run to the store
d. Kathryn's tumble down the stairs
e. Michael's discussion of squirrels
f. Maribel's observation of the star
g. Aynat's advocacy of your proposal
h. Martin's toss of the baseball to you
i. Luis's exchange of coins with you
j. Jan's placement of the book on the table
k. Wendell's gift of the book to you
l. Cherlon's proposal that we should leave
m. Anna's suggestion that natto kills
n. Karen's attempt to fold the origami
o. Ilaria's promise to stay

The θ-roles and c-selection requirements of these nouns match those of their cognate verbs, with the exception that there are no DP internal arguments. This flows from the observation that accusative Case is not licensed in nominals. Instead, the argument that would appear as a structurally case-marked object of a verb appears with the preposition *of* in the noun version of that verb. If we import the model we've developed for verbs to nominals, we'll have a syntax like that in (148).

(148)



The external θ-role resides outside the projection of the noun, just as it does with verbs. I've called the position that the external θ-role resides in here n°, to designate that this is to be understood as part of a nominal. As with verbs, the external θ-role that resides outside of the projection of the word that expresses that θ-role is in an AGREE relation with that word. The external argument resides in the Specifier of nP at D-structure, and when that argument is a DP it comes with an unvalued Case feature. That feature is valued by the genitive Case that comes with the determiner heading the DP. This Case must have a • associated with it, as movement of the Case-marked DP into the Specifier of DP is obligatory.

Interestingly, the external θ-role in nominals is not able to assign accusative Case; if it were, of course, we'd expect to see accusative Case-marked internal arguments of nouns. We could assume that the structural Case which nominalized external θ-roles assign is the one expressed by *of*. But if that were correct, it would require that the *of* which shows up with nouns that do not support an external θ-role, as in (149), is different.

(149) a. the sister of Amy Rose
b. the piece of paper
c. the cup of Harissa

Indeed, even with the nominals in (147), the external θ-role bearer can be absent and the *of* remains.

(150) a. the fear of marigolds
b. a dislike of MaxElide
c. a run to the store
d. the tumble down the stairs
e. a discussion of squirrels
f. an observation of the star
g. the advocacy of your proposal
h. the toss of the baseball to you
i. the exchange of coins with you
j. the placement of the book on the table
k. a gift of the book to you
l. the proposal that we should leave
m. the suggestion that natto kills
n. an attempt to fold the origami
o. every promise to stay

On the other hand, *of* does not seem like a semantically contentful preposition. The DPs with *of* in (150) seem to combine with the nouns they are complements to in the same way those DPs would to the related verbs. It is the θ-roles that these nouns and verbs assign that is responsible for their semantic composition. The idea that *of* is a kind of Case morpheme, and moreover a semantically contentless one—a "structural" Case—seems appropriate. If all of this is correct, then the examples in (150) are situations where an internal argument is getting a structural Case from a term that has no external θ-role. These are violations of Burzio's Generalization. An explanation of Burzio's Generalization should include why it doesn't hold in (English) NPs. For concreteness, I will assume that the term that assigns the structural Case *of* is the noun itself.

(150) indicates that the arguments of nouns are not as obligatory as they are for verbs. This is true of internal arguments as well, as (151) shows.

(151)

a. the fear	i. the exchange
b. a dislike	j. the placement
c. a run	k. a gift of the book to you
d. the tumble	l. the proposal that we should leave
e. a discussion	m. the suggestion that natto kills
f. an observation	n. an attempt to fold the origani
g. the advocacy	o. every promise to stay

If the examples in (147) suggest that the argument structure of nouns is the same as the argument structure of verbs then there must be something that allows nouns to avoid the effects of Theta Criterion. That is, if the denotations of nouns like *exchange* is roughly the same as that of the verb *exchange*, then

we need an explanation for why the NP headed by *exchange* is grammatical without its internal argument, but a VP headed by *exchange* is not.

(152) a. Luis's exchange
b. * Luis exchanged.

And we also need an explanation for why the external θ-role for the verb is obligatory, but not for the noun.

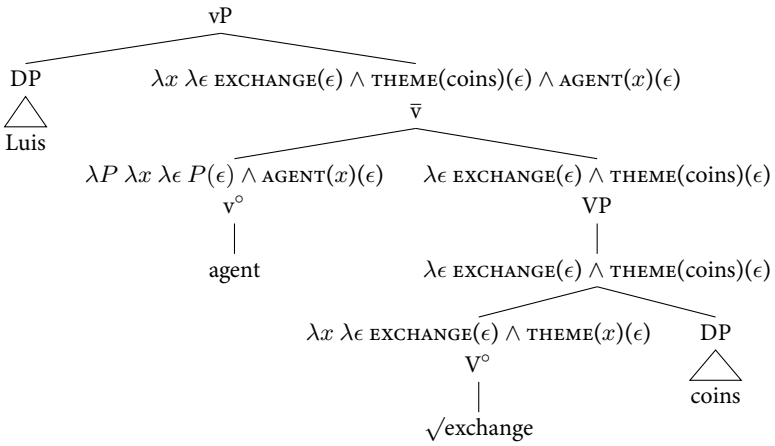
(153) a. the exchange (of coins)
b. * It (*expletive*) exchanged.

Our account for the Theta Criterion,relies on the stipulation that vP must be of a particular semantic type, and a combination of denotations assigned to the verb and its θ-roles along with a theory that limits the ways in which terms can be semantically combined. These pieces of the explanation are assembled in (154), and their application to a vP headed by *exchange* is shown in (155).

See the discussion in chapter 3.6.

(154) a. $[\![\text{vP}]\!]$ must be a function whose domain is eventualities and range truth values.
b. $[_{\text{V}^\circ}$ exchange$] \Rightarrow \textsc{agent}+\sqrt{\text{exchange}}$
c. $[\![\sqrt{\text{exchange}}]\!] = \lambda x\ \lambda\epsilon\ \textsc{exchange}(\epsilon) \wedge \textsc{theme}(x)(\epsilon)=\text{T}$
d. $[\![\textsc{agent}]\!] = \lambda P\ \lambda x\ \lambda\epsilon\ P(\epsilon) \wedge \textsc{agent}(x)(\epsilon)=\text{T}$

(155) $\lambda\epsilon\ \textsc{exchange}(\epsilon) \wedge \textsc{theme}(\text{coins})(\epsilon) \wedge \textsc{agent}(\text{Luis})(\epsilon)$



If the DPs *Luis* and *coins* had not been present, it would not have been possible ot use function application—our only semantic rule of composition—to produce from the denotations that *exchange* introduces a vP with the requisite meaning. Something in nominals must interrupt this effect.

Let's focus first on the absence of the external argument in nominals.We have a mechanism that is designed for removing the Theta Criterion's effects by existentially closing the function that otherwise requires an argument. This process is designed so that it only applies to internal arguments, capturing the fact that external arguments of verbs are required whenever the external θ-role associated with that verb is present. However, we also saw that this

My discussion here is led by Moulton (2014), with some tweaks to fit to our syntax. Moulton builds on Grimshaw (1990) and Borer (2005b). All of these approaches to the semantics of nominalizations have their roots in Higginbotham (1983).

process is capable of applying to an external θ-role's argument in the Passive. In the Passive, the external argument appears in a PP headed by *by*, or it is existentially closed:

(156) a. Coins were exchanged by Luis.
b. Coins were exchanged.

Perhaps the process of existential closure can apply to external θ-roles within nominals more liberally. One way of detecting the presence of an external θ-role in Passives is with adverbs that require them. For instance, the adverb *intentionally* can describe eventualities only if those eventualities have a suitable external θ-role.

See Jackendoff (1972).

(157) a. Luis intentionally exchanged the coins for buttons.
b. # Luis intentionally seemed happy.
c. # Luis intentionally existed.
d. # Coins intentionally equal buttons in value.
e. # The election intentionally affected us.
f. # We were intentionally affected by the election.

The grammaticality of (158) suggests that there is an external θ-role in it.

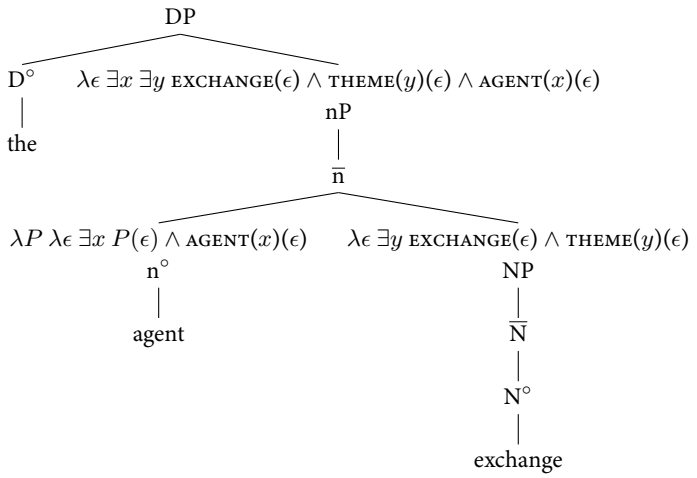
(158) Coins were intentionally exchanged for buttons.

The same reasoning suggests that an external θ-role remains in (159) as well.

(159) The intentional exchange of coins for buttons (was shocking).

A working hypothesis, then, is that nouns allow for a wider application of existential closure than verbs. This would allow for DPs like (160).

(160)



From this meaning for nP must be derived a denotation that allows the DP to refer to something of the appropriate sort. This is what is achieved by the meaning of the determiner *the*. The denotation of *the* is complex and must allow for various outcomes, it appears. We needn't engage the semantics of *the*

fully here. We can instead work with an informal sketch. Roughly, what the denotation of *the* does is use the meaning of its sister to identify an object to which the DP it heads refers. Because the meaning of its sister is a function whose range is True or False, we can use that function to identify the relevant objects by paying attention to those that the function maps to True. Our definition of *the* will be (161).

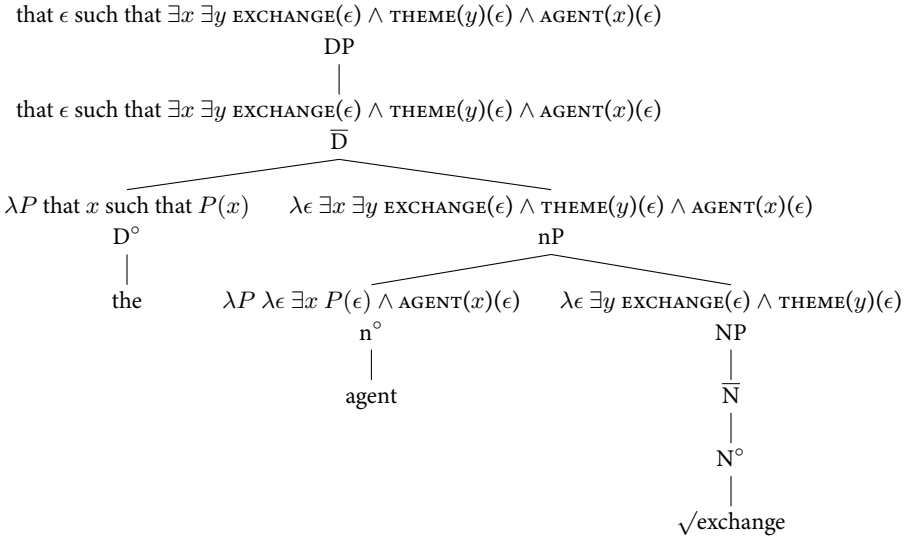
(161) $[\![\text{the}]\!] = \lambda P$ refer to ϵ if $P(\epsilon)$=T

There are other aspects to the meaning of *the* which are responsible for making sure that the object referred to is salient and "definite." These are the parts to the meaning of *the* that make it complex. I will try to evoke these parts by describing the denotation with English words that have a similar flavor. So, in place of (161), I'll use (162).

(162) $[\![\text{the}]\!] = \lambda P$ that ϵ such that $P(\epsilon)$=T

This gives (163) from (160).

(163)



These are sometimes called "event" nominalizations.

Notice that in this example, the DP refers to an eventuality. Eventualities have durations (sometimes), and it is for this reason that sentences like (164) are sensible.

(164) a. The exchange took over an hour.
b. The exchange began in the morning.
c. The exchange got interrupted.

Not all DPs whose nouns have verbal correlates refer to eventualities, however. For instance, *gift* refers to things, and not events, and for this reason sentences like (165) are odd.

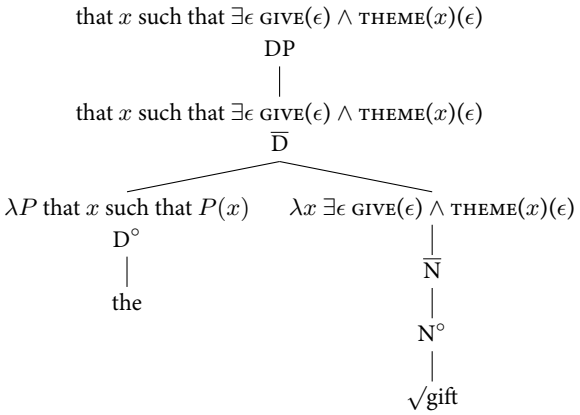
(165) a. The gift took an hour.
b. The gift began in the morning.
c. The gift got interrupted.

What *the gift* refers to is an object that was given, not the event of giving. Moreover, it may well be that there is no external θ-role present in NPs headed by *gift*. At least, I find awkward the use of modifiers that depend on the presence of an external θ-role.

(166) # The intentional gift was in the mailbox.

What this suggests is that *gift* occurs without the external θ-role that is required of the verb *give*, and that only the meaning associated with $\sqrt{give}$ attaches to the noun *gift*. To get the fact that *gift* refers to the thing given, we can let Existential Closure bind off the event variable and leave the Theme variable open. The result is portrayed in (167).

(167)



These are sometimes called "object" nominals, as the thing they refer to is the object of the noun.

This account predicts that these nouns cannot combine with an *of*-phrase that gets the internal θ-role.That seems to be a correct prediction, as can be seen from (168).

(168) the gift of this basket

To the extent that (168) has a grammatical interpretation, it refers to a gift that the basket, somehow, is responsible for. This is just what the analysis illustrated in(167) predicts.

If this is correct, it requires that *the* be flexible enough for it to combine with NPs or nPs. Further, the denotation of *the* must be flexible enough for it to combine with a phrase whose meaning is a function with a domain of things (as in (167)) as well as with a phrase whose meaning is a function with a domain of eventualities (as in (163)). In recognition of this, let's refine the definition of ⟦the⟧.

See Zucchi (1989) for this observation. There are apparent counter-examples, as for example: ***the gift of grace*** and ***the gift of apples***. These arise, so far as I can tell, when the object of the noun is indefinite or abstract. There are modes of semantic composition that are reserved for objects of this semantic type, and it seems to me likely that this is what's involved in these apparent counterexamples. This mode of semantic composition is consistent with the account sketched here. If this is how these cases arise, they are not counterexamples then. See Chung and Ladusaw (2004) and Dayal (2011) for some discussion.

(169) $[\![\text{the}]\!] = \lambda P$ that a such that $P(a)$=T, where
  a. a is a thing or an eventuality, and
  b. P is a function with domain a.

Recall that our convention is to use $\ldots, x, y, z$ to represent objects drawn from the domain of things, $e, e', e'', \ldots$ to represent objects drawn from the domain of eventualities, and P, Q, R to represent functions whose domains are either things or eventualities. We now add the convention that $a, b, c, \ldots$ will be used to represent objects from the domains of things or eventualities.

Nouns related to verbs can also refer to things that are reifications of the events they describe. For instance, the noun *report* can be used to make a DP that refers to an eventuality, as in (170).

(170) The report took an hour.

But it can also be used to build a DP that refers to a thing, and not an eventuality, as in (171).

(171) The report is several pages long.

In neither of these cases is *report* making a DP that refers to a thing that would be the object of the verb *report*, as can be seen from (172).

(172) I reported the violation to the authorities.
  a. The report took an hour.
  b. The report is several pages long.

(172a) does not claim that the violation took an hour, and (172b) does not report that the violation is several pages long. The thing that *the report* refers to in (171) corresponds in some fashion to an eventuality of reporting. The thing it refers to carries the same information that would be needed to individuate a reporting eventuality. We can imagine that nouns with this meaning are built from those that describe eventualities from a process that converts those descriptions of eventualities to descriptions of things with the same individuating content. However, unlike nouns that describe eventualities, the ones that describe things that correspond to eventualities do not seem to have an external θ-role associated with them. They do not support the use of modifiers that depend on the presence of an external θ-role:

(173) # The intentional report (of the infraction) is several pages long.

Here again, the external θ-role does not seem to accompany the root $\sqrt{report}$ in its nominal form, at least when it refers to things. The process that converts a description of eventualities to a description of things must correlate with this suppression of the external θ-role. Nouns that describe things that are related to eventualities are sometimes called "concrete" nouns. They are also sometimes called "result nominals."

We've seen now two ways in which nouns that are related to verbs can build DPs that refer to things, and one way in which nouns refer to eventualities. For object nominals like *gift*, the method involves manipulating the meaning of the noun so that it ends up describing things that are given. For event nominals like *report*, there is a process that takes the event-referring meaning of *report* and makes a noun that refers to eventualities of that kind.

And, finally, for concrete nominals like *report*, there is a process that takes the event nominal *report* and converts it to a thing-referring meaning. As these examples indicate, sometimes nouns are ambiguous, as for example *report*; they are able to undergo more than one of these three processes. But some nouns are not, as for example *gift*, which can only undergo one of these processes.

We'll look at one last way in which nouns with verbal meanings can be manipulated to refer to things. In this last example, illustrated by (174), the noun refers to a thing that would be the external θ-role bearer of the related verb.

(174) a. The baker arrived late.
b. The worriers fretted all night.
c. The rollers flattened the dough.

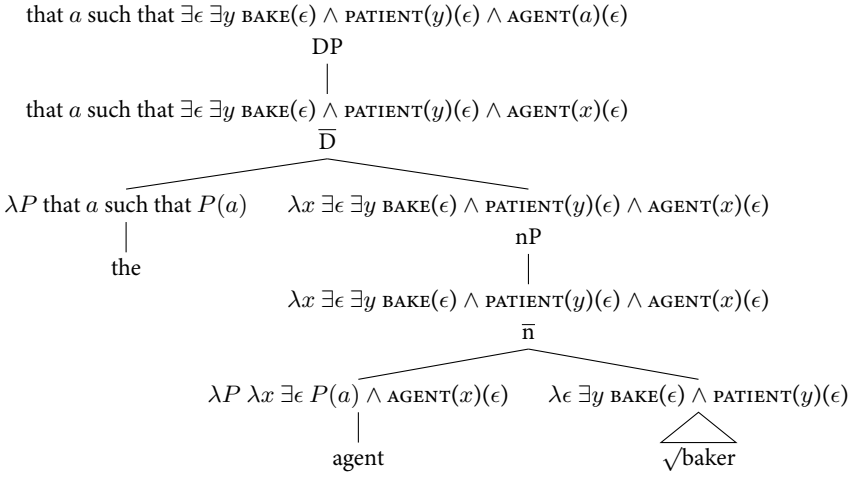
This meaning correlates with the presence of the nominalizer, *er*. Note that nouns built with *er* do not refer to the things that can more generally be understood to bear the relation that genitives have to the noun. For instance, whereas the genitive DP in (175a) refers to the owner of the painting, the referent of (175b) does not.

(175) a. The Vatican's painting of God touching Adam has thousands of visitors a day.
b. The painter of God touching Adam has thousands of visitors a day.

The referent of *painter of X* is to the person who painted X. In general, it is the θ-role of the root that determines what a noun with *er* refers to.

To capture this, we can let the presence of *er* require that the external θ-role function not be existentially closed, but allow for all other functions to be existentially closed. This produces representations like (176).

(176)



I'll call these "subject nominalizations."

See Rappaport Hovav and Levin (1992) for a discussion of these nominalizations. There are apparent counter-examples to the generalization that nominals built with the suffix ***er*** refer to the "subject" argument of the verb—"bestseller" is one. See Alexiadou and Schäfer (2010) for an analysis.

Subject nominalizations allow us to see that the meaning associated with the genitive Case-marked DP can have more than one source, a conclusion that was first, perhaps, defended by Mona Anderson.[12] In (177), the genitive Case-marked DPs cannot be understood as the external θ-role bearers of the nouns.

[12] Anderson, Mona. 1983-1984. Prenominal genitive NPs. ***Linguistic Review*** 3:1–24

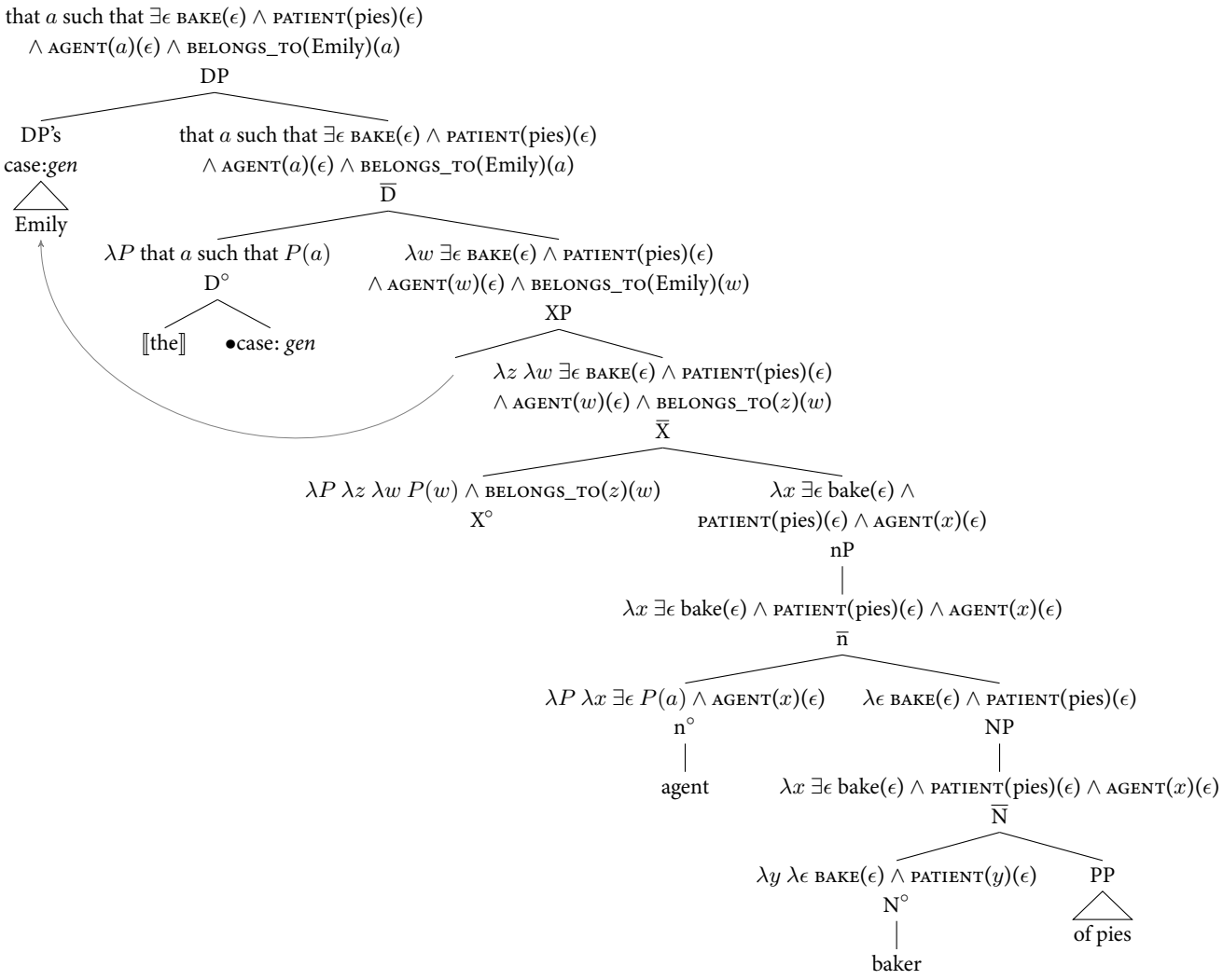
(177) a. Emily's baker (of pies) was late.
b. Jason's worriers (about the election) fretted all night.
c. The cylinder's rollers (of the dough) did their job.

This is what is expected if to get the meaning associated with the external θ-role of the verb related to the nouns in (177) requires that it be assigned to the DP in genitive Case. Because nouns built with the *er* morpheme require that θ-role to be bound by the determiner *the*, it cannot be assigned to a DP. Instead, the genitive DPs in these examples must be arguments of another predicate, one that leads us to understand that the baker in (177a), for instance, "belongs" to Emily. If we adopt the view that the determiner responsible for assigning genitive Case has the denotation of *the*,and we represent the term that gives the genitive DPs in (177) their meaning with $\mathcal{P}$, then (177a) could get the representation in (178) on the next page.

See chapter 2.

We've seen, then, a variety of ways in which the Theta Criterion is satisfied by projections of nouns. First, unlike verbs, it seems that the nominal versions of a verb's root can expone the root without its accompanying external θ-role. Second, it looks like Existential Closure can apply in a less constrained way to nouns and their θ-roles, including the external θ-role. And finally, it appears that the projection of a noun and its external θ-role need not be a function whose domain is eventualities. It is possible instead for the projection of a noun, or its external θ-role, to be a function whose domain is things. This last option allows for result nominals, and also for DPs that refer to a thing that is the bearer of a θ-role that comes associated with a noun: object and subject nominalizations. This model of how the nominal and verbal versions of a shared root can differ is summarized in (179).

(178)



(179) a. A root need not be accompanied by its External θ-role in nominal projections even when it must in verbal projections.

b. Existential Closure is more widely available in NP/nP than it is in VP/vP.

c. VP/vP must be a function with a domain of eventualities, but NP/nP may also be a function with a domain of things.

There are event nominals which behave more like verbal projections with regard to how the Theta Criterion applies to them. Jane Grimshaw[13] discovered one that involves putting into the NP a modifier that depends on the so-called aktionsart of the root. Aktionsart refers to ways of characterizing certain logical properties of the eventualities that verbs, and nouns, are descriptions of. When we first adopted a semantics of verbs built around eventualities, I noted that there are two "kinds" of eventualities: states and events. These are two Aktionsarten. Eventualities that are states have a kind of

[13] Grimshaw, Jane. 1990. ***Argument structure***. Cambridge, Massachusetts: MIT Press

homogeneity relative to time that other eventualities don't have. Like all other things that natural language describes, eventualities are situated in a temporal matrix. Let's call the duration of an eventuality, its "run time." The eventuality referred to in *Satoshi walked* occurs at a particular (previous) slice of the time line—that slice is its run time.If we examine the eventuality of Satoshi's walking moment by moment, we can observe that there will be slices of time within the run time that do not have the required attributes to be a walking eventuality. There will be moments, for instance, in which Satoshi is in a two-feet-on-the-ground stance that we might also find in portions of the run time of the eventuality described by *Satoshi stood* or *Satoshi fell*. There is a minimum portion of the run time of the eventuality described by *walk* below which we no longer have an eventuality that can be individuated as a walking. Eventualities that are states differ in this regard. The eventuality referred to in *Satoshi loved nattoo*, for example, has a run time all of whose moments are lovings of nattoo.Eventualities that are like those described by *love* are states. Other state descriptions are *know, fear, be, sad, blue*. Eventualities that are like those described by *walk* are events. Other verbs that describe events are *sneeze, arrive, bake*.

Sometimes the distinctions I'm describing here with the term "aktionsarten" are instead said to involve "aspect." In general terms, "aspect" refers to ways in which language relates an eventuality to a time. We can distinguish two ways in which languages do this. One is built into the denotation of the verbs, adjectives, and nouns that describe eventualities. The other is determined by morphological means that are somewhat independent of the lexical item that describes the eventuality. The difference between the perfect (e.g., ***Satoshi had walked***) and the progressive (***Satoshi is walking***) is an example of the latter type. I will reserve the term "aspect" for this type, and use aktionsart for the kind that is built into the denotation of a root. Sometimes aktionsart is called "lexical aspect" or "inner aspect."

This way of characterizing the difference between states and events comes from Dowty (1979). There are other properties that distinguish events and states. A better definition would explain why those properties correlate.

Events can be divided further into two types of aktionsarten. There are events that have an endpoint built into their meaning. For instance, the eventuality referred to in *Satoshi baked something* necessarily contains a moment at which something is baked. To see this, compare the eventuality referred to in *Satoshi baked something* to the eventuality referred to in *Satoshi heated something*. What distinguishes these eventualities is their results. The baking eventuality includes an eventuality that could be described by *heat*. A baking is a heating that culminates with something being baked. Eventualities that are individuated by virtue of their culminations—the properties of the final moment in their run time—are known as "telic."Other telic predicates are *die, make, build*, and *appear*. Events which aren't telic are called "atelic."

The term "telic" comes from Garey (1957), though the distinction is found in Aristotle's ***Metaphysics***.

It may seem counter-intuitive to think of telic events as requiring endpoints because it is possible to use telic predicates to describe eventualities which don't have endpoints. This is possible, for instance, in the so-called "progressive" construction in English. As (180) indicates, there need be no resultant baked thing when using the verb *bake*.

(180) Satoshi was baking a pie when the house caught fire and burnt to the ground.

What the progressive does in (180) is refer to a portion of the baking eventuality whose run time doesn't include the endpoint. What matters for the purpose of determining the aktionsart of an eventuality is how it is individuated. That is, what portions of its run time are necessary in distinguishing one kind of eventuality from another kind of eventuality. Telic eventualities require that their run time have a last moment by virtue of which they are

individuated, and atelic ones don't. Once individuated, language has ways of referring to portions of an eventuality. The progressive is one of those ways.

Note too that the syntax can affect the aktionsart of an event description. For instance, though the verb *walk* describes an atelic event, the VP *walk a kilometer* describes a telic event. That is, whether *walk a kilometer* is true of an event depends on properties of the last moment in its run time.

Many grammatical phenomena are sensitive to these distinctions in aktionsarten. One of interest to the task at hand is that certain temporal expressions which measure the duration of an event are compatible with atelic events but not telic ones. A classic example of this is *for X*, where "X" is an expression that refers to an interval of time. We can see this in the contrasts in (181).

Vendler (1957) divides telic events further into ones that are punctual (known as "achievements") and ones that are gradual (known as "accomplishments"). The event described by ***die*** is an achievement; the one described by ***bake*** is an accomplishment. Comrie (1976) similarly divides atelic events into ones that are punctual (known as "semelfactives") and ones that are aren't (known as "activities"). The event described by ***cough*** is semelfactive, and the event described by ***walk*** is an activity.

Note that the use of ***for***-phrases with telic predicates is "wrong" if they are meant to tell the duration of the event described by that predicate. They are grammatical under other interpretations. For instance (181g) can have the meaning that Jesse was dead for a full second; that is, it isn't ill-formed if Jesse isn't dead, but was previously.

(181) *atelic:*

a. Eva ran for an hour.
b. Jesse danced for just three minutes.
c. Eva sneezed for a full second.
d. Jesse observed the experiment for an hour.

telic:

e. # Jesse built it for an hour.
f. # Eva arrived for just three minutes.
g. # Jesse died for a full second.
h. # Eva destroyed the experiment for an hour.

There is something semantically anomalous about using these temporal modifiers with predicates that describe telic events. By contrast, the modifier *in X*, where "X" is an expression that refers to an interval of time, is compatible with predicates that describe telic events, but anomalous when used with predicates that describe atelic ones:

(182) *atelic:*

a. # Eva ran in an hour.
b. # Jesse danced in just three seconds.
c. # Eva sneezed in a full second.
d. # Jesse observed the experiment in an hour.

telic:

e. Jesse built it in an hour.
f. Eva arrived in just three seconds.
g. Jesse died in a full second.
h. Eva destroyed the experiment in an hour.

There is something about the meanings of these modifiers that interacts with the properties of eventualities to produce these results.

What's of interest to us is that nouns that share a root with a verb can also be divided with respect to this classification of eventuality types. We can see this from the contrasts in (183).

(183) *atelic:*

a. # Only observation of the patient in several weeks (can determine the most likely course of action).

b. Only observation of the patient for several weeks (can determine the most likely course of action).

telic:

c. The total destruction of the city in two days (appalled everyone).

d. # The total destruction of the city for two days (appalled everyone).

(Grimshaw, 1990, (28-9): 58)

Grimshaw discovered that nominals with modifiers that require a certain aktionsart must contain their object arguments. This is why there are the contrasts in (184).

(184) a. Only observation (can determine the most likely course of action).

b. * Only observation for several weeks (can determine the most likely course of action).

c. The total destruction (appalled everyone).

d. * The total destruction in two days (appalled everyone).

Note that the subject argument isn't required; it's hard to tell whether that is because the external θ-role need not accompany the noun or that is because existential closure is available for the external θ-role. What we learn from these examples is that the aktionsart of the event described by some term is tied to the presence of that term's object. As we've seen, it's possible for a noun to describe an eventuality without its object arguments being expressed overtly. But if those eventualities are to have the properties of aktionsarten that support modification, they must come with an overt expression of the object. This suggests that the denotation of the object contributes to determining the aktionsart of an event description.

See Borer (2005a) for an account that would have this consequence.

One scenario, then, in which nouns behave more like verbs with respect to the Theta Criterion is when they match the aktionsart of the verb they are related to. Another situation that has this effect on nouns arises, Grimshaw argues, when nouns bring with them their external θ-role. We've seen that the "subject" arguments of nouns can get a θ-role that is not the one characteristically associated with the root. In those scenarios, I suggested that there was hidden inside the nominal another term that assigned that θ-role (see (177)).

In order to see Grimshaw's effect, it is therefore necessary to create examples in which the subject argument is getting the θ-role associated with the root, and not this other one. If we pick roots that come with an Agent θ-role, we can do this by using the modifiers that are licensed by the presence of an Agent θ-role. Grimshaw uses the modifier *deliberately* to do this.

(185) a. The instructor's deliberate examination of the patient (took a long time).

b. * The instructor's deliberate examination (took a long time).

(Grimshaw, 1990, (11): 51)

The same fact emerges if we use *intentional*. Grimshaw didn't determine whether it was the presence of the external θ-role, or the presence of the external argument, that has this effect. If it is the presence of the external argument that forces the expression of the object argument, then the explanation for the effect must come from how the Theta Criterion is satisfied. In our system, we force the presence of arguments through our semantics by constraining Existential Closure. To force the presence of an object argument in these situations, then, will require that Existential Closure be denied of the internal θ-role when the external θ-role is present and Existential Closure does not apply to it. If, by contrast, the effect illustrated by (185) holds when the external θ-role is present, but not expressed with an argument, then Existential Closure of the internal θ-role is blocked when there is an external θ-role. Whether that external θ-role is existentially closed or not doesn't matter on this second scenario. We can tell which is the correct generalization by looking at examples where there is no external argument, but we can detect the presence of the external θ-role in another way. Our method for detecting the presence of an external θ-role is to use the modifiers that are sensitive to their presence. So, what's at stake is whether the contrast in (185) shows up without the external argument, but with the agent oriented modifier. Are the contrasts in (186) mirrored in (187)?

(186) a. her deliberate examination of the problem

b. * her deliberate examination

c. his deliberate construction of the building

d. * his deliberate construction

(Moulton, 2014, (8): 4)

e. your intentional observation of the conflagration

f. * your intentional observation

g. my intentional destruction of these theories

h. * my intentional destruction

(187) a. the deliberate examination of the problem
b. ? the deliberate examination
c. the deliberate construction of the building
d. ? the deliberate construction
e. the intentional observation of the conflagration
f. ? the intentional observation
g. the intentional destruction of these theories
h. ? the intentional destruction

I am frankly unable to tell.

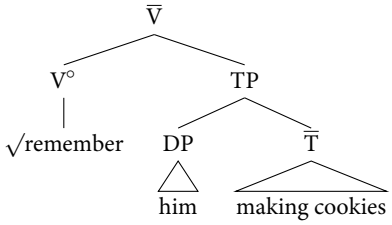
The last way to make nominals behave more like the verbs they are related to with respect to the Theta Criterion is to use the so-called "gerund." This term is used to refer to verbal forms that appear with the suffix *ing*, but are not progressive. We can distinguish three cases, illustrated by the bracketed portions of (188).

Abney (1987) has a good introduction to English gerunds, and see also Milsark (1988).

(188) a. Della remembered [him making cookies].
b. Della talked about [his making cookies].
c. Della talked about [his making of cookies].

The form in (188a) is a clause, whose accusative Case-marked subject comes from the μP embedding *remember*, as we can see from the effect that the Passive has on this kind of sentence: *He was remembered making cookies*. (189) sketches its syntax.

(189)



Chomsky (1970) suggested that gerunds are related to what we would now call vPs by a syntactic process that is part of the derivation. Gerunds, in other words, are built syntactically from projections of verbs, unlike other nouns which are constructed morphologically from a root that is shared by the nominal and verbal subclasses. Fraser (1970) has the first defense of this view of gerunds.

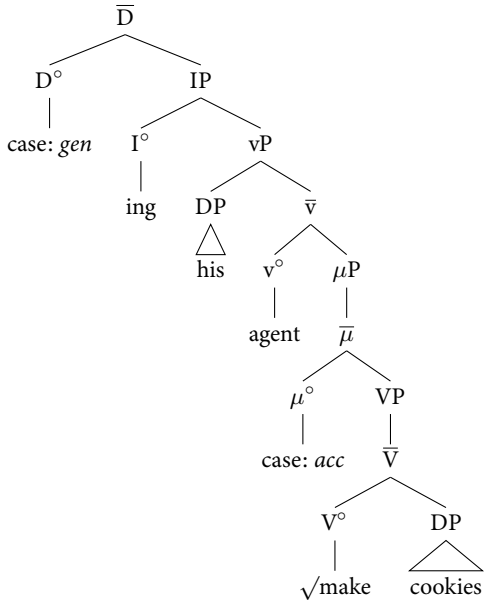
The examples in (188b) and (188c) are DPs, and of more interest to us. The nominal in (188c) is eventive, and has the properties we have already seen of eventive nouns.Existential Closure can apply to make either of its external or internal arguments non-obligatory.

(190) a. the making of cookies
b. ? the making
c. * his making

The only θ-role available to the genitive DP in (190c) is the external θ-role that comes with the root $\sqrt{}$***make***. For some reason, $\mathcal{P}$, the term that provides the general relation that genitives can have to the NP that follows them, is absent in these gerunds. For this reason, the internal argument is obligatory in this configuration.

In (188b), however, we have a DP with what appears to be a VP inside. The object of *make* here bears accusative Case—the structural Case associated with μ. A preliminary sketch of its syntax is (191).

(191)



As expected, because the accusative Case assigner comes into a D-structure with the external θ-role—Burzio's Generalization correlates them—the v° that carries the external θ-role is found in these nominals. Like external θ-roles outside nominals, Existential Closure cannot apply, and the external argument is obligatory, as (192) indicates.

(192) * Della talked about [the making cookies].

These kinds of gerunds might have no noun in them at all. As expected, the Theta Criterion applies to these cases in the same way that it applies to VPs.

5.3.2 A gap in clausal complementation

As we've seen, when nouns that are related to verbs take arguments, they match the θ-role and c-selection of the verbs. We've explained this by letting these nouns be built from the same root that the related verbs are built from, and therefore the θ-roles and c-selection requirements that travel with these roots are present in the nominals as well. But there are some systematic ways in which the argument structure of nominals is not like that of the verbs that they are related to. It is this fact that Kayne (1984b) uses to argue for the structure of double object constructions we were examining at the end of the previous section. We're now ready to examine this argument.

One of these differences concerns the kinds of clauses that nouns can take as objects. As expected, nouns that are related to verbs can select clausal complements if the verbs do. Thus, we have pairs like (193).

(193) a. i. Amalia's saying that nouns are interesting (puzzled us).
ii. Amalia said that nouns are interesting.
b. i. Amalia's declaration that nouns are interesting (puzzled us).
ii. Amalia declared that nouns are interesting.
c. i. Christine's attempt to understand nouns (amazed us).
ii. Christine attempted to understand nouns.
d. i. Christine's desire to understand nouns (puzzled us).
ii. Christine desired to understand nouns.
e. i. Susanne's trying to untangle nominalizations (took all day).
ii. Susanne tried to untangle nominalizations.
f. i. Amalia's question whether to let roots be shared (wasn't seriously entertained).
ii. Amalia questioned whether to let roots be shared.

But interestingly, verbs and adjectives that select raising or exceptional Case marking infinitives are not matched by nouns that can. This is what we see in the contrasts in (194).

(194) a. i. Wendell considers Tanja to be strong.
ii. * Wendell's consideration of Tanja to be strong
b. i. Wendell appears to be clever.
ii. * Wendell's appearance to be clever
c. i. This proposal is likely to be untenable.
ii. * This proposal's likelihood to be untenable

Not all speakers seem to find fully acceptable the gerunds that include VPs within them, instead of NPs, like that illustrated in (191). But for those that do, we have minimal pairs of the kind in (195).

(195) a. i. ? Wendell's considering Tanja to be strong (amuses him).
ii. * Wendell's considering of Tanja to be strong (amuses him).
b. i. ? Wendell's believing Tanja to be clever (amuses him).
ii. * Wendell's believing of Tanja to be clever (amuses him).

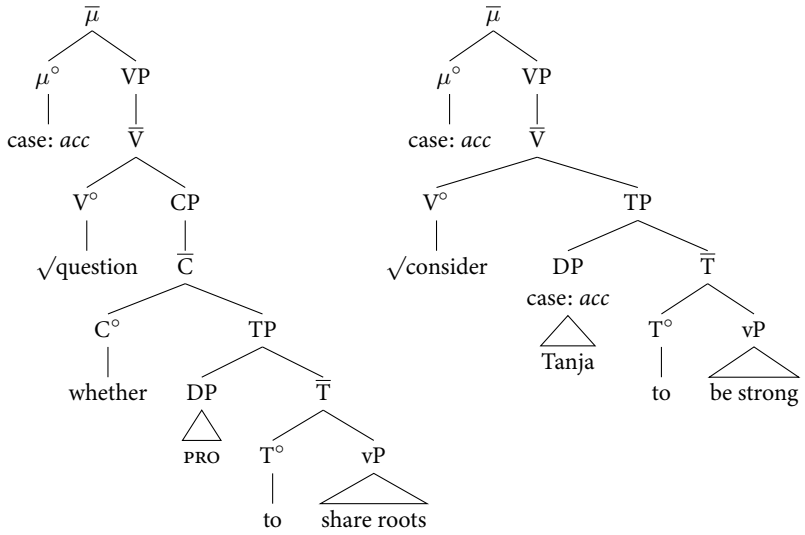
Infinitival clauses to nouns are possible if they are not of the raising or exceptional Case marking type. In particular, infinitival clauses that have PRO subjects are fine.

(196) a. Nick's eagerness to please
b. Helen's promise to be on time
c. Noah's attempt to leave
d. Suzi's refusal to give up
e. Claire's desire to stay
f. Aynat's offer to continue
g. Robert's demand to run

There is something about the properties of raising and exceptional Case marking infinitives that makes them incapable of being complements to nouns. Kayne discovered this pattern, though the cases that support the generalization can be found in Chomsky (1970).

Kayne's suggestion about what that might be involves the fact that in complement clauses of the sort that are blocked in nominals, there is a relationship between an argument in that complement clause and something outside of the complement clause. To see this, consider the difference between infinitival complements of verbs that house PRO, for instance, and infinitival complements that house an accusative Case-marked DP.

(196)

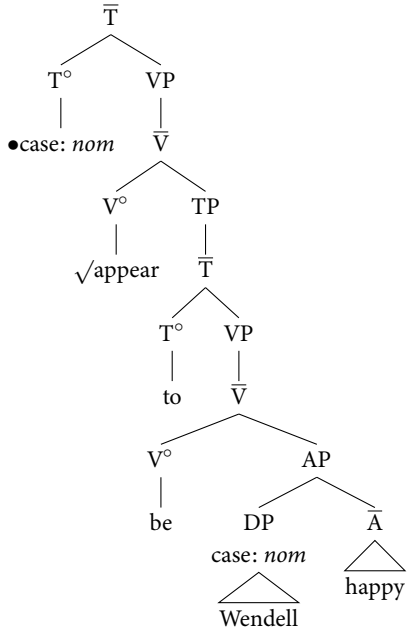


In the case of *consider*, the subject argument, *Tanja*, receives its Case from the μ that belongs with the VP headed by *consider*. Nothing like that is happening with the complement to *wonder*, however. In raising, there is a similar relationship between an argument in the complement clause and a Case marking term outside the complement, as can be seen in (197).

If the pro in the infinitival is "controlled" by an object, however, these infinitives seem to be blocked in nominals.

i. a. * Mary's compulsion of her husband to quit his job
b. * her encouragement of John to take up linguistics
c. * their obligation of John to join the army
d. * its impulsion of Mary to sue for divorce
e. * her permission of John to buy himself a new car
f. * their allowance of John to smoke
(Kayne, 1981, (84): 170)

(197)

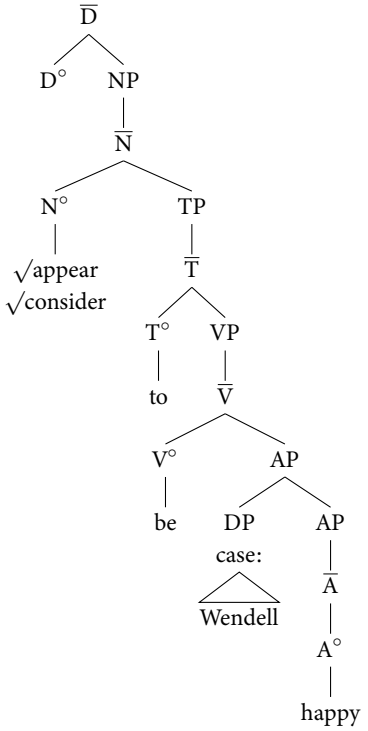


In (197), *Wendell* gets its Case feature valued by the nominative in T°.

Kayne suggests that the clausal complement to nouns is opaque for the relations that need to occur across infinitives of these kinds. Consider the form that nominal versions of these configurations will have.

Kayne's own proposal was expressed in terms quite different from those being used here. He suggested that government across the infinitive is blocked in nominals, where "government" is a relation akin, and ancestral, to our present conditions on Case assignment.

(198)



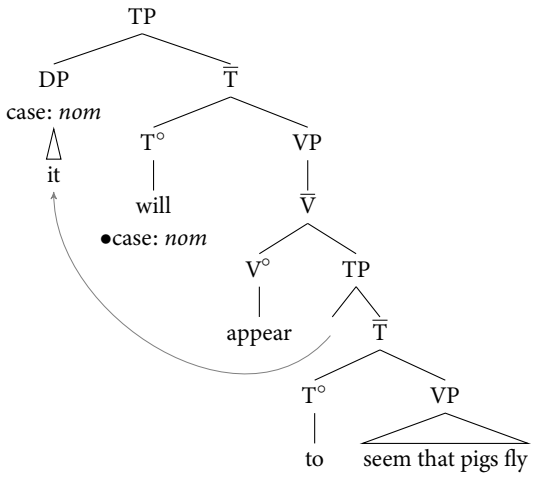
The DP, *Wendell*, must get its Case feature valued. If D° in (198) is equipped with genitive Case, then that is one source for its value. Alternatively, the

structural Case expressed with *of*, which I have assumed is assigned by a noun, could also be the source for its value. If TP, however, is opaque for AGREE, then neither of these will be able to value the Case feature on *Wendell*. Kayne's hypothesis, then, amounts to the claim that cases such as these are blocked by the Case Filter.

This account requires that clauses always have a DP within them that is subject to the Case filter. For only then can the ungrammaticality of this complementation structure be credited to a failure to satisfy the Case filter; there must always be a DP within these infinitives that requires Case form outside the infinitive if all instances of these infinitives are to be blocked. When the verb of the embedded clause c-selects one or more DP arguments, this will occur as long as one of those DPs relies on Case from a term outside the clause. Because these clauses are infinitives, the nominative Case that is typically capable of satisfying the Case needs of one DP will be absent. If the verb of the embedded clause c-selects a DP external argument, this DP will have to get its Case from a term outside the clause, and the failure of Case assignment that Kayne's proposal entails will lead to a violation of the Case filter. If the verb of the embedded clause doesn't c-select an external argument, but does c-select a DP internal argument, then that internal argument will not get structural Case from the verb. Burzio's generalization requires of a verb that comes equipped with structural Case that it also assign an external θ-role, so verbs that don't c-select an external argument won't deliver Case to their object DPs. In this case too, then, Kayne's proposal would prevent these clauses from being embedded within a TP.

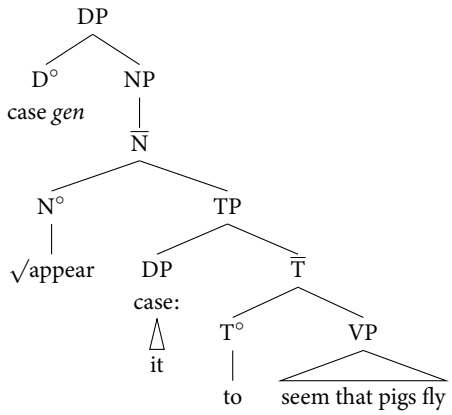
But what happens when the embedded clause contains a verb that does not c-select a DP? If the embedded TP is subject to the EPP, and the EPP is satisfied when a DP occupies the Specifier of the relevant TP, then there will have to be some DP that occupies its Specifier in one of the parses in its derivation. If this DP is subject to the Case filter, then it will be that DP which violates the Case filter. The syntax that this view requires is shown in (199).

(199)



The expletive *it* must occupy the Specifier of the embedded infinitival TP, and in that position AGREE with the nominative Case assigner, *will*, in the higher clause. Because of the •, this AGREE relation triggers the movement that is shown. If that syntax could be enforced, then Kayne's proposal will extend to explain why these infinitives with this argument structure are not found in nominals. Their syntax would require AGREE to hold across the TP as indicated in (200).

(200)



If the TP complements to nouns block AGREE, then this requirement will be frustrated in (200).

Finally, let's consider scenarios in which the infinitive contains no DP arguments that require Case, and the EPP is satisfied without the use of an expletive. The EPP can be satisfied by CPs, as (201) shows, and so we might expect on Kayne's proposal that infinitives that contain no DPs whatsoever can show up in nominals.

(201) That pigs fly seems to be unlikely.

These infinitives would contain CPs in their Specifiers. Unexpectedly, however, CPs cannot surface in the Specifier of infinitival TPs, as the ungrammaticality of (202) shows.

(202) a. * She believes [$_{TP}$ [$_{CP}$ that pigs fly] to be unlikely].
b. * She considers [$_{TP}$ [$_{CP}$ that pigs fly] to be unlikely].
c. * She finds [$_{TP}$ [$_{CP}$ that pigs fly] to be unlikely].

Instead, the word-order that is preferred employs an expletive:

(203) a. She believes it to be unlikely [$_{CP}$ that pigs fly].
b. She considers it to be unlikely [$_{CP}$ that pigs fly].
c. She finds it to be unlikely [$_{CP}$ that pigs fly].

In some of these cases, it is also possible to put the CP in the final position without an accompanying expletive.

(204) a. ? She believes to be unlikely [$_{CP}$ that pigs fly].
b. ?* She considers to be unlikely [$_{CP}$ that pigs fly].
c. * She finds to be unlikely [$_{CP}$ that pigs fly].

This word order might arise by extraposition of the CP. If so, we might expect these cases to arise in nominalizations as there is no obvious term within the infinitive that requires an AGREE relation to a term outside of the infinitive. This expectation is frustrated, however:

(205) * Her belief to be unlikely that pigs fly.

The particulars of Kayne's account did extend to account for the ungrammaticality of (205). This is because his syntax for movement required that they left silent pronouns, and these could be viewed as subject to his condition. We will encounter this account of movement in a later chapter.

If the reason infinitives of this kind are not possible complements to nouns is that the complements to nouns are opaque to AGREE, then the ungrammaticality of (205) is unexpected.

There are other systematic absences in the complement types to nouns. Another that Kayne proposes is that "small clauses" cannot be found as complements to nouns. There are no instances where a verb that selects a small clause complement can continue to do so in its noun version. The pairs in (206) illustrate this generalization.

(206) a. I believe this fact interesting.
b. * my belief of this fact interesting
c. I consider this sentence instructive.
d. * her consideration of this sentence instructive
e. Some find these facts remarkable.
f. * your finding of these facts remarkable

In each case, a verb that selects a small clause is matched with a cognate noun, and the subject argument of that small clause is assigned Case from a term outside the small clause: either accusative Case in the situations with verbs, or the Case expressed with *of* in the situations with nouns. The same kind of contrast is found when the Case that the subject of the small clause is assigned comes from a higher position, as (207) shows.

(207) a. This fact is believed interesting.
b. * this fact's belief interesting
c. This sentence was considered instructive.
d. * this sentence's consideration instructive
e. These facts were found remarkable.
f. * these facts' finding remarkable

When the small clause's argument is assigned nominative Case, and therefore moves, the result is grammatical. But when that argument is assigned genitive Case, again triggering movement, the result is ungrammatical. Like infinitival TPs, small clauses cannot be complements to a noun. If small clauses become

opaque to AGREE when they are complements to nouns, the ungrammaticality of these examples is understood.

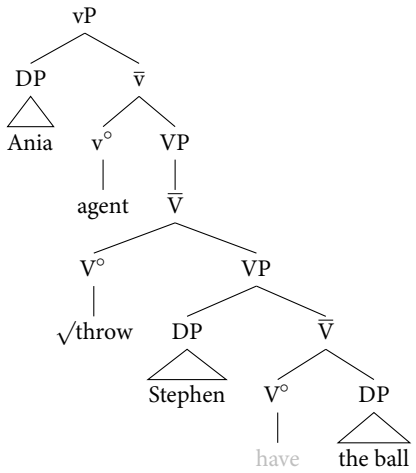
Kayne (1981)'s argument for the existence of a small clause in the double object comes from the observation that double objects are also not possible in nominalizations. There are contrasts parallel to those in (206) and (207) which involve double object constructions.

(208) a. Her teacher gave Mary the letter.
b. * Her teacher's gift of Mary (of) the letter
c. Her classmate offered her a crayon.
d. * Her classmate's offer of her (of) a crayon
e. This salesman sold us that defective car.
f. * This saleman's sale of us (of) that defective car
g. A friend of his rented John office space.
h. * A friend's rental of John (of) office space

(Kayne, 1981, based on (61)&(62): 161)

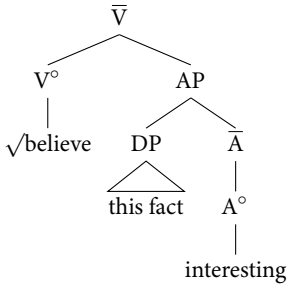
If what makes small clause complements incompatible with nouns is that they produce a barrier for AGREE, and the source of the ungrammaticality of the double object frame in nominals has the same source, then this is reason to give them parallel structures. This is the argument Kayne (1984b) provides for giving the double object frame a structure like that in (126), repeated below.

(126)



This will make it parallel to the structure of small clause complements—like that in (209).

(209)



In both structures, there is a clause-like phrase that separates the DP in the Specifier of that phrase and the term outside the phrase that is responsible for providing that DP with Case. This clause-like phrase is, by hypothesis, a barrier for Case assignment when it's the complement to a noun, and these structures are then all violations of the Case filter.

Interestingly the other complement structure that these verbs support is not similarly incompatible with nominals.

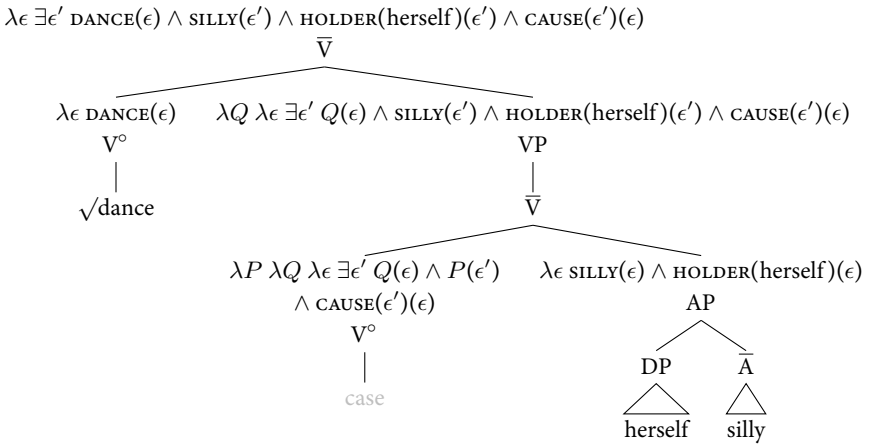
(210) a. John's presentation of a medal to Mary
b. his offer of a book to Mary
c. his recommendation of Mary to the Board
d. their surrender of the city to the enemy
e. her explanation of the problem to the students

(Kayne, 1981, (91): 171)

If Kayne's argument is correct, this suggests that the DP+PP frame doesn't make use of a small clause structure. This is inconsistent with Harley's proposal about how to relate the two frames, since her account requires that the feature which determines whether a verb participates in the alternation is its ability to c-selecte PP small clauses in both frames (see (143)). As far as I know, these conflicting pieces of evidence have not been reconciled.

There are other problems with Kayne's proposal. Borer (2013) discovered that there are certain kinds of small clauses that can be found in nominals, casting doubt on Kayne's method. There is evidence that so-called resultatives have a small clause structure, as we briefly touched upon when considering (140).

(140) Suzanne danced herself silly.



But Borer points out that these constructions can survive in nominalizations.

(211) a. Kim's hammering of the metal flat
b. the hammering of the metal flat (by Kim)
c. your burning of the kitchen walls black
d. the burning of the kitchen walls black (by you)

(Borer, 2013, (66): 108)

Borer's own account of the absence of the double object frame in nominals is that they do not have the resources for assigning two object Cases in English, and this is what distinguishes nominals from verbs. Pesetsky (1995) also has an account for the absence of the double object frame in nominals that divorces it from the fact that nominals don't permit many small clauses, raising and exceptional Case marking infinitives. His proposal targets the hidden morphemes that the verb expones in the double object frame. He suggests that these interact fatally with the morphology that creates nouns from verbs. If (140) is correct, then the grammaticality of (211) is also problematic for Pesetsky's proposal. I do not know of an account for the full gamut of effects described here about how the argument structure of nouns differs from the argument structure of the verbs they are derived from.

6
Theories of Movement

Our present grammar defines a sentence as a series of parses.

(1) $\mathcal{D}$ = (P_1, P_2...,P_n) is a derivation iff:
   a. P_1 is formed by the $\overline{X}$ Skeleton and Lexical Insertion, and it is semantically interpreted,
   b. Each $P_{i>1}$ is formed from P_{i-1} by a rule,
   c. P_n is subject to the linearization algorithm.

The cardinality of a derivation is the number of Ps in it.

(2) A sentence is the set of well-formed derivations which have the same P_1 (=D-structure). A grammatical sentence is a derivation whose cardinality isn't greater than the cardinality of any other derivation for that sentence.

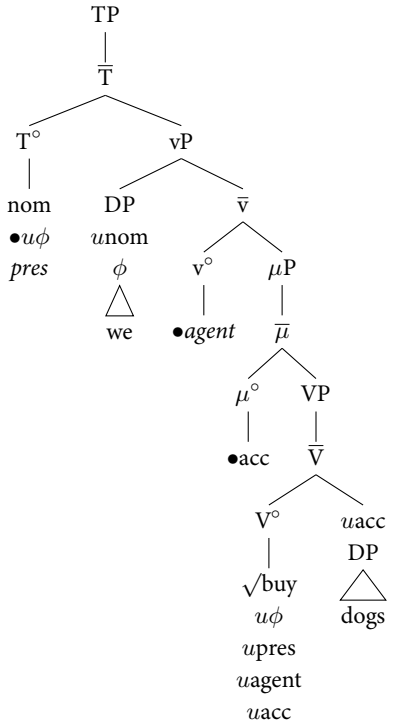
One of the rule types that creates parses is movement:

(3) α, a node in a phrase marker tree, moves to β, a position in a phrase marker tree, means:
   a. α is put in β, and
   b. α is removed from its original position.

We've developed a system for movement rules that characterizes the changes of position that move creates in solely morphological terms. Argument Movement arises, for instance, because the X° that assigns Case to a DP has a • associated with it. That is, the agreement relation that establishes Case assignment comes with the instruction that the Case marked DP should be pronounced in proximity to the Case assigner. And Head Movement arises under similar circumstances. When a tense morpheme, for instance, is in an agreement relation with a verb—causing that verb to expone the tense morpheme—and one or the other of the terms in that agreement relation has a • associated with it, Head Movement is required to bring the two heads together into one head.

Let's reconsider how this system works by way of (4).

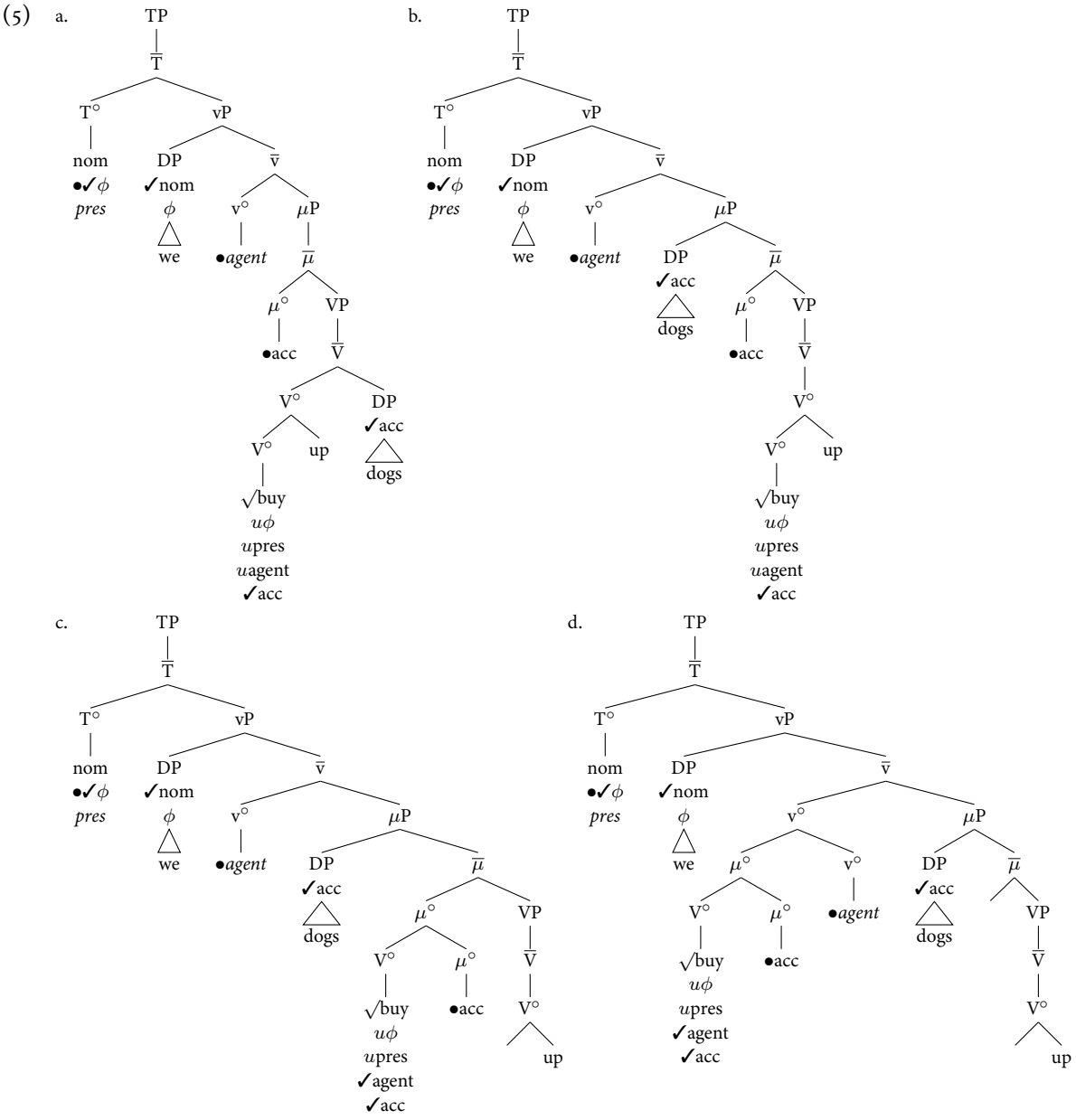
(4)



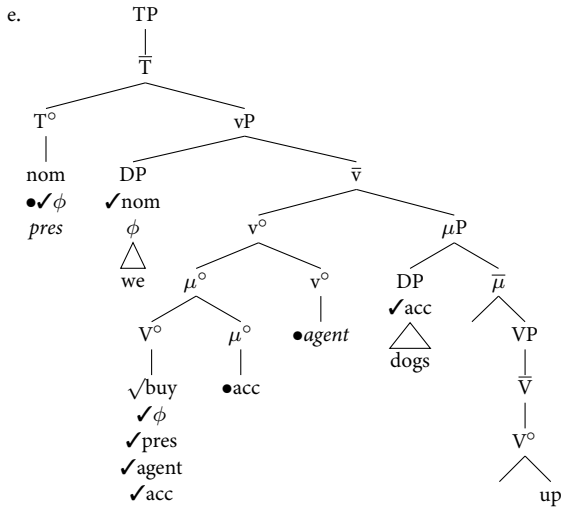
This structure can be created by the $\overline{X}$ Skeleton, and it can be semantically evaluated. It isn't a well-formed representation otherwise, however, as it does not meet the requirement that features AGREE, nor does it honor the requirements imposed by •.

To meet these requirements, a derivation must be created which includes parses that do meet these requirements. This produces the steps in (5). (I've omitted from this derivation some intermediary parses that don't create useful parses.)

(5)


a.
TP
T
T°
vP
nom
•✓φ
pres
DP
✓nom
φ
we
v̄
v°
•agent
μP
μ̄
μ°
•acc
VP
V̄
V°
V°
up
√buy
uφ
upres
uagent
✓acc
DP
✓acc
dogs
b.
TP
T
T°
vP
nom
•✓φ
pres
DP
✓nom
φ
we
v̄
v°
•agent
μP
DP
✓acc
dogs
μ̄
μ°
•acc
VP
V̄
V°
V°
up
√buy
uφ
upres
uagent
✓acc
c.
TP
T
T°
vP
nom
•✓φ
pres
DP
✓nom
φ
we
v̄
v°
•agent
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√buy
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upres
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we
v̄
v°
μ°
V°
√buy
uφ
upres
✓agent
✓acc
μ°
•acc
v°
•agent
μP
DP
✓acc
dogs
μ̄
VP
V̄
V°
up

e.



These four instances of movement and AGREE have satisfied the • attribute on *acc*, and have brought √buy into a position which allows it to value the person/number/tense features and produce a terminal that corresponds to *buy*. The only thing left to do to make this a well-formed s-structure is to satisfy the • attribute on the ϕ feature in T°. This is done by moving the subject so that it merges with a projection of T°.

I pointed out that (4) is a structure that our semantics can interpret, and which gives us the correct interpretation. But none of the representations produced from (4) can be semantically interpreted however. This is a general property of the derivations we've considered so far. A reasonable hypothesis is that this defines derivations. That is, that the first parse in a derivation is the one that is semantically interpreted, and no other parse is. The D-structure parse is the one that represents how the morphemes in a sentence are grouped semantically, and the S-structure parse is the one from which the morphological and prosodic grouping is determined. All the other parses in a derivation are just are model of the function that maps a D-structure onto an S-structure.

That derivations are defined in this way was a popular view in the 1960's. In 1964, Jerrold Katz and Paul Postal, formulated what's come to be known as the Katz-Postal Hypothesis. That thesis maintains that the denotations of the elements that make up a sentence are semantically composed in a way that the syntax of what we're calling D-structures corresponds to. The rest of the forms in a derivation, including the S-structure, do not reflect the way in which the denotations of the component parts are composed. Syntax furnishes, on the one hand, a way of structuring how the denotations are combined and, on the other hand, a way of structuring the morphology: bringing together the information that make up lexical items. A theory of derivations is a theory of the relation between these two representations. I don't know where the • attribute's requirements fit into this picture, but since their requirements are ones that hold of the morphological connection that

is established by AGREE, it doesn't seem too wild to hope that it too will come from some aspect of morphology. Syntax, then, could be seen as a way of mediating the independent needs that the morphology and the semantics place on the items that are being arranged into sentences. Derivations are models of the solution space to the open equations that the morphological and semantic requirements produce. It's this conception of syntax that we've been using up to now.

In the derivation we've just looked at, the movement operations are all ones that cannot feed the semantic interpretation. In this chapter we're going to confront evidence that this is incorrect. In particular, we'll see that movement can create structures that are semantically interpreted. As it stands, our definition of movement does nothing more than change the syntactic position a term has in order to make sentences comply with morphological requirements. But there are examples which seem to require that movement can itself create meanings. We'll need to develop two things together, then. We'll need a new definition of derivation, one that allows more than just the D-structure to be semantically interpreted. And we'll need a new definition of movement, one that weds a change in meaning to the change in position that it presently expresses.

6.1 Trace Theory

To see that parses other than D-structures are semantically interpreted, we're going to have to beef up our tools for seeing what the meanings of sentences are. There's a kind of anaphora that provides one such tool, and I'll use it in the following sections to give us a better picture of the meanings of sentences that involve movement. There are also argument DPs whose meanings are not as simple as those we've been considering up till now. These too provide us a better image of how the syntactic parts of a sentence relate to the meaning that sentence has. I'll show how the picture that results from looking at these cases can be modeled by a theory of movement that allows it to insert variables which are bound by the moved term. Those variables are called "traces," and the theory of movement that makes use of them is called "Trace theory." But first, we must discover what variables are, and how they are expressed in syntax. Variables are part of the system of anaphora that natural languages use. We'll begin, then, by introducing some preliminaries about anaphora.

6.1.1 Deep and Surface anaphors

In one kind of anaphora, the denotation some expression has is matched by the denotation of another linguistic expression. Hankamer and Sag (1976) calls this "surface anaphora," and it is to be distinguished from "deep anaphora." In deep anaphora, the term that the anaphor refers to is determined by virtue of how another linguistic expression refers. For example, the

pronoun *she* in (6) is most naturally understood to refer to the same individual that *Rodica* refers to.

(6) Rodica said that she would explain this.

Personal pronouns refer to individuals which the context has made salient. In sentences that have been isolated from a context of use—like (6)—this is typically done linguistically. In (6), the presence of the name *Rodica* has made salient in the relevant sense an individual with that name, and *she* is most naturally understood to refer to that individual. We can describe this situation as one in which *Rodica* and *she* corefer, and we'll call the term that establishes the requisite saliency in these cases the pronoun's antecedent. Coreference is an instance of deep anaphora. The anaphoric term, the pronoun, has its reference established with the aid of an antecedent. In (6), the anaphora arises indirectly; the DP *Rodica* makes an individual named Rodica salient, and this makes Rodica a possible referent for *she*.

Another instance of deep anaphora arises when pronouns are understood to be bound variables. In these kinds of examples, illustrated by (7), the anaphoric connection between a pronoun and another DP is more direct.

(7) Every woman remembers her first paper.

As in (6), the most natural interpretation of (7) gives the pronoun *her* an interpretation that depends on the meaning of *every woman*. But in this case, the pronoun doesn't corefer with *every woman*. In fact, the meaning of *every woman* is complex; it doesn't refer. Instead, it says something more general about the relation between women and rememberers. One can paraphrase the effect that *every woman* has on the meaning of (7), and include the way in which *her* is connected to that meaning with (8).

(8) For all the women, 1 …n, under consideration:
   a. woman #1 remembers woman #1's first paper, and
   b. woman #2 remembers woman #2's first paper, and
   c. woman #3 remembers woman #3's first paper, and
   ⋮
   d. woman #n remembers woman #n's first paper.

An interesting property of the meaning of sentences like (7) is that they can be matched to a family of sentence meanings, like that in (8). Each of the sentences in the family varies by virtue of a value given to certain terms within. One of these terms in (7) is *her*, and the other is the external argument, *every woman*. The pronoun's interpretation varies with the values given to *every woman*; it is a variable bound to *every woman*.

Let's unpack what's happening in (7). I'll begin by considering how to model the meaning (7) gets when the pronoun *her* is not interpreted as a variable bound by *every woman*. Let's imagine, for instance, that (7) is uttered as

The determiner ***every*** is called a "quantifier," and the DP it heads is called a "quantificational DP." Other quantificational determiners in English are ***some***, ***a***, ***no***, and ***each***. The semantics I will adopt for quantifiers comes from what is known as "generalized quantifier" theory. The central idea of this theory is that quantifiers denote relations between sets of things that meet certain restrictions. In (7), those sets of things are (a) women and (b) the rememberers of her first paper: ***every*** says that the first set is a subset of the second. This way of thinking about quantificational DPs originates with Frege (1879), and is formalized in Mostowski (1957) and Lindström (1966). Good introductions to the linguistics of generalized quantifiers are Barwise and Cooper (1981) and Keenan and Stavi (1986).

part of a conversation about Thui. In that context, (7) gets an interpretation that can be paraphrased by (9), where *her* refers to Thui.

(9) For all the women, 1 ...n, under consideration:
   a. woman #1 remembers woman her first paper, and
   b. woman #2 remembers woman her first paper, and
   c. woman #3 remembers woman her first paper, and
   ⋮
   d. woman #n remembers woman her first paper.

What's required is to find a way of giving a denotation to *every woman* that allows (7) to get a meaning like that (9) expresses.

One way of achieving this meaning is to let the determiner *every* have a denotation that causes it to relate the denotation of *woman* to the denotation of *remembers her first paper* in the way that (9) describes. Suppose that we take the denotation of *woman* to describe things which are women. We can do this by letting *woman* be a function that maps things to truth just in case they are women. Using the λ operator, we can model that with (10).

(10) $[\![\text{woman}]\!] =_{\text{def}} \lambda x\ \text{woman}(x)=\text{T}$

Remember that $[\![\alpha]\!]$ is the denotation of α.

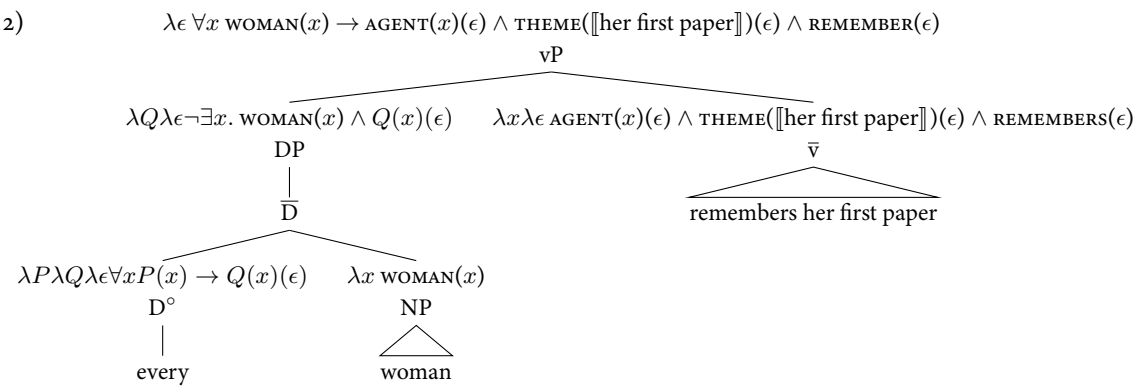
We can now capture the meaning of (7) with the denotation for *every* in (11).

(11) $[\![\text{every}]\!] =_{\text{def}} \lambda P\ \lambda Q\ \lambda\epsilon\ \forall x[P(x) \rightarrow Q(x)(\epsilon)]$.

The "$\rightarrow$" in (11) is a logical connective that comes close to the meaning of the word "implies." $\alpha \rightarrow \beta$ is True unless α is True and β is False.

This gives the vP of (8) the denotation in (12).

(12)



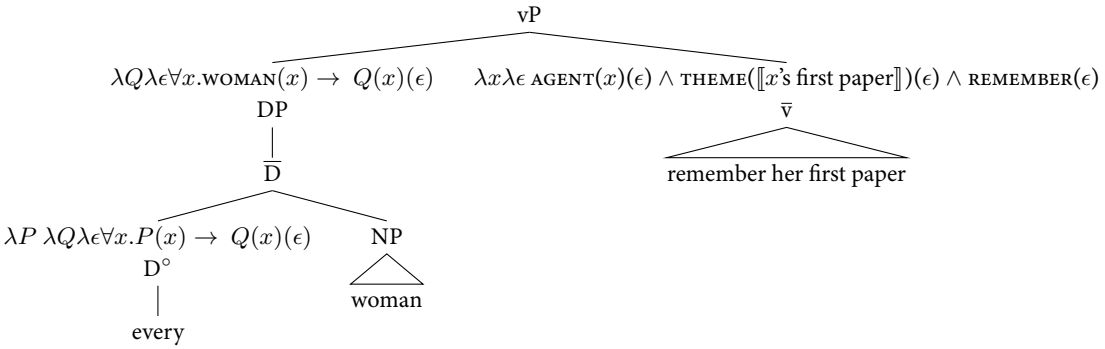
When *every* semantically combines with *woman*, a denotation is formed that is a function whose domain is relations between entities and events. Our semantics makes the denotation of v̄ a function of that sort, and so the function that the DP denotes can take the denotation of the v̄ as its argument. The result is [[vP]], which is a function from events to the statement that for every x if x is a woman then x is the agent of that event, where that event is a remembering whose theme is the thing that *her first paper* refers to. The vP describes, then, events which are rememberings of her (=Thui's) first paper by x if for every x, x is a woman. This is a very close approximation of the

meaning expressed by (9). An event of the kind that ⟦vP⟧ describes requires that the sequence of conjunctions in (8) hold of that event. That is, for an event to make the function that ⟦vP⟧ is map to True, it must be an event in which woman #1 remembers Thui's first paper and woman #2 remembers Thui's first paper, and woman #3 remembers Thui's first paper, etc., until every woman (in the context) is listed among the rememberers.

We'll see later that there is a subtle, but important, mismatch between what this semantics delivers and the meaning that (7) actually has. Our semantics will require some tweaking.

The bound variable interpretation of the pronoun *her* in (8) can be captured if it is possible to make the denotation of *her* one of the things that the "∀" binds in the denotation delivered to the vP. That is, we need to find a procedure that allows *her* to get interpreted as the variable x bound by ∀ in the way that is indicated in (13).

(13) $\lambda\epsilon\forall x\ \textsc{woman}(x) \rightarrow \textsc{agent}(\epsilon)(x) \wedge \textsc{theme}(\llbracket x\text{'s first paper}\rrbracket)(\epsilon) \wedge \textsc{remember}(\epsilon)$



When the domain of the function denoted by *woman*, i.e., λx $\textsc{woman}(x)$=T, is restricted to just those individuals about which the sentence is making a claim—that is, it includes only those women under consideration—(13) is a model of the meaning for (7) described in (8). One of our aims, then, will be to formulate a syntax that produces a way of interpreting a pronoun as a variable bound in the way that (13) indicates.

These are instances of deep anaphora. Surface anaphora arises when the anaphor's denotation is equated with the denotation of another term: its antecedent. An example of this is in (14).

(14) Rodica read a bad paper after she read a good one.
Rodica read a bad book after she read a good one.
Rodica read a bad poem after she read a good one.

The meaning of *one* in (14) matches the meaning of the underlined noun. In these cases, *one* is not referring to the same thing that the underlined noun is referring to. Instead, the semantic contribution *one* makes to the DP that it stands within is the same as the semantic contribution that the underlined noun makes to the DP it's within. We can capture this by ensuring that the denotation of *one* is identical to the denotation of the underlined noun. Whereas in Deep Anaphora, what's at stake is how the anaphor and antecedent refer to things, in Surface Anaphora, the relationship between antecedent and anaphor concerns the denotations of the two terms. Surface

anaphora is of interest to us because it provides a way of determining the denotation of expressions. We can use it to see if the syntax we have given to phrases correctly corresponds the way the denotations of terms inside that phrase semantically compose.

The surface anaphor we'll use is expressed prosodically. It arises when one phrase is offered as a partial recapitulation of another phrase. (15) illustrates.

(15) A: Thuy eats natto.
B: <Thuy eats> SPINACH (too).

In this short conversation, B can be understood as offering a response to A's sentence that either corrects it, or adds to it, depending on whether *too* is included. In those circumstances, B's utterance carries a particular prosody. The word *spinach* is accented, and the pitch range of *Thuy eats* is compressed. Pitch accent on *spinach* signals that the DP it makes is focused. Where relevant, I will mark portions of a sentence that have the compressed pitch range indicative of this prosody with angled brackets. These strings are said to be "de-accented." I will also signal the term that bears accent by putting it in all caps. If the focus marked term is different from the accented term, which happens when a multiword phrase is focused, I will mark that phrase with "$[\ ]_F$." Phrases that contain this pairing of de-accented string and focus-marked term are a kind of surface anaphor. Their denotation must match—in the same kind of way that a surface anaphor does—the denotation of a phrase uttered elsewhere.

This can be seen by considering the similar conversation in (16).

(16) A: When Rodica cooks, Thuy eats natto.
B: <She eats> SPINACH (too).
B′: Did you know that about her?

In B′'s reply, the referent of *her* can be either Rodica or Thuy, making this question awkwardly ambiguous. This establishes that A's remark makes both Rodica and Thuy salient enough for the pronoun to find them as a referent; either *Rodica* or *Thuy* can be the antecedent for *her*. The response in B, however, does not have this freedom of interpretation. Here *she* can only refer to Thuy. The prosody attendant with B's reply requires that its meaning correspond to the meaning of *Thuy eats natto*, and a consequence of this is that *she* and *Thuy* must corefer.

A similar thing is demonstrated by (17).

(17) A: Thuy will eat natto.
B: * <Thuy will> COOK <spinach> (too).
B′: <Thuy will> COOK <natto> (too).

The prosody in the replies by B and B′ in this conversation require that the meaning of the de-accented portion match the meaning in the corresponding

positions of A's remark. The ungrammaticality of B's response traces back to the fact that *spinach* doesn't have the same meaning that *natto* has; when that is corrected, the result is grammatical, as B′'s response demonstrates.

Let's try to characterize how the surface anaphora that involves this prosody works. Assume that both the de-accenting and the accent that marks focus expone the anaphor. Let's call a phrase that contains both of these prosodic cues a "focus anaphor." The phrase that a focus anaphor must match will be called a "focus antecedent." In (15)-(17), it is the entire TP that B utters that is a focus anaphor, and A's remark that is its focus antecedent. Other phrasal types can be focus anaphors and focus antecedents as well, as (18) indicates.

The account I am describing here is roughly that of Rooth (1992). Much of the foundation for this theory, including the relationship between focus and prosody, is in Jackendoff (1972). The idea that these prosodies expone anaphors can be found in Ladd (1980). Selkirk (1984) is a good introduction to the complex relationship between the prosody of focus marking and the item that is focused.

(18) [$_{DP}$ Thuy's dissertation] will be finished in July
and [$_{DP}$ RODICA'S <dissertation>] arrives soon thereafter.

Here, the DP *Rodica's dissertation* has the prosody of a focus anaphor, and it finds its focus antecedent in the DP that starts the sentence: *Thuy's dissertation*. I'll assume that any phrasal type can be a focus anaphor or a focus antecedent. A description of the relationship between anaphor and antecedent that the examples we've seen illustrate is (19).

(19) Let XP be a focus anaphor, and α be the focus-marked term within XP. A focus value of XP (=$[\![\text{XP}]\!]^F$) is created by semantically interpreting the XP that results from replacing α with a focus alternative of α (=FA(α)). β is a FA(α) if it is the same syntactic type as α, and contextually salient.

If XP is a focus anaphor, then there must be a YP and a $[\![\text{XP}]\!]^F$ such that $[\![\text{YP}]\!]=[\![\text{XP}]\!]^F$.

This gives an account of (15)-(17), as indicated below.

(20) Because *natto* is a FA(*spinach*),
$[\![\text{Thuy eats SPINACH}]\!]^F=[\![\text{Thuy eats natto}]\!]$, and therefore (15) works.

(21) Because *natto* is a FA(*spinach*),
$[\![\text{she eats SPINACH}]\!]^F=[\![\text{Thuy eats natto}]\!]$ when *she* and *Thuy* corefer, and therefore (16) works.

(22) Because *eat* is a FA(*cook*),
$[\![\text{Thuy will COOK natto}]\!]^F=[\![\text{Thuy will eat natto}]\!]$, and therefore (17) works.

In the cases we've reviewed, a focus anaphor is indicated by giving a phrase in a sentence a certain prosody, a prosody that combines pitch accent with de-accenting. There is another way that a focus anaphor can be indicated, that is slightly different. To see this, consider first the cases of focus anaphora in (23), which deploys the combination of de-accenting and pitch accent.

(23) a. Rodica eats natto when THUY <eats natto>.
b. Rodica eats natto when <she eats> RICE.

The TP in the *when* clause is a focus anaphor, and meets the conditions of (19), as (24) shows.

(24) a. Because Rodica is a FA(Thuy), $[\![\text{THUY eats natto}]\!]^F$ = [[Rodica eats natto]], and (23a) is grammatical.
b. Because natto is a FA(rice), $[\![\text{she eats RICE}]\!]^F$ = [[Thuy eats natto]], when *she* corefers with *Thuy*, and therefore (23b) is grammatical.

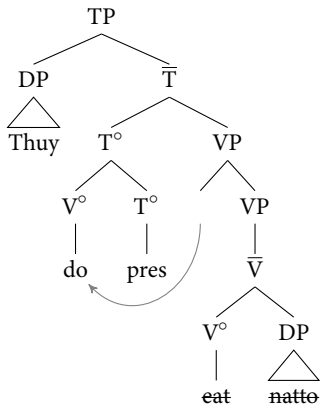
There is an alternative pronunciation of (23a)—it's in (25a)—but not of (23b).

(25) a. Rodica eats natto when THUY does.
b. * Rodica eats natto when RICE.

What's happened in (25a) is that the de-accented words have not been pronounced. That isn't possible for (23), however. A common conclusion about what makes these cases different is that the de-accented words form a phrase in (23a), but not in (23b). We should engineer the process that allows de-accented words to go unspoken so that it applies only to those words that form a constituent.

A popular way of characterizing this gives to (25a) nearly the same syntax as (23a), but allows for a way of preventing the words in the VP from showing up in the string that is associated with that syntax. The difference in the syntax between (23a) and (25a) is connected to the requirement that the tense feature in T° be exponed. If we understand the requirement that the AGREE relation which tense has with the morphology on a verb be satisfied only if the verb with that morphology is pronounced, then the failure to pronounce the verb in (25) will result in ungrammaticality. Instead, then, a sentence that has both tense and the process that unpronounces the verb must also provide a way for the tense feature to be exponed. This is achieved in (25a) by including the auxiliary verb *do* in the syntax. The syntactic representation for (25a) approximates (26).

(26)



I've used ~~strikeouts~~ to indicate the words that haven't been pronounced. These words are said to be "elided," and the process that achieves this in (26) is called "VP Ellipsis." This is an unfortunate name for the process, as it can elide the words in phrases that aren't VPs, as (27) demonstrates, and being a VP is not sufficient to allow the process, as (28) shows.

There are several ways of modeling ellipsis. We might imagine, for instance, that there is a designated anaphor—like a personal pronoun—for VPs, and that its phonology is silence. Hardt (1993, 1999) lays out an account of this type. But we shall see evidence later that supports the view that an ellipsis contains syntactic material that is unspoken.

(27) Thuy must be [$_{AP}$ happy] and RODICA must be [$_{AP}$ ~~happy~~] too.

(28) * She made Thuy [$_{VP}$ work all night] because he made RODICA [$_{VP}$ ~~work all night~~].

I'll persist in using this name, as it's what you're most likely to find in the literature, but you should remember that the expression "VP Ellipsis" doesn't have a compositional meaning.

If the presence of *does* adds no meaning to these sentences, then the conditions on focus anaphora are still met in (25). We can describe VP Ellipsis with (29).

(29) VP Ellipsis
If α is the complement to T° or an auxiliary verb, then a word in α can go unspoken iff:
a. every other word in α is unspoken,
b. the word is not part of something that is focus marked, and
c. α is dominated by a focus anaphor.

I've framed the conditions under which the words in a phrase can elide so that the cases in (27) and (28) are captured.

There is an interesting feature of the condition on focus anaphora in (19) that distinguishes instances where VP ellipsis has applied from those where it hasn't. To see this difference, consider first (30).

This contrast was first discussed in Lakoff?

(30) Thuy is an accomplished syntactician and RODICA <is smart> too.

The conditions in (19) require of the focus anaphor in the second conjunct that its meaning match that of the first conjunct.

(31) Because *Thuy* is an FA(*Rodica*), (19) requires that
$[\![\text{RODICA is smart}]\!]^F = [\![\text{Thuy is an accomplished syntactician}]\!]$

Because *Thuy is smart* (which is what *RODICA is smart* means when the FA(Rodica) is Thuy) does not have the same denotation that *Thuy is an accomplished syntactician*, it is not clear why the focus anaphor is allowed in (30). To account for this, Chris Tancredi suggested that we think of (19) as a condition which can be used by a speaker to invite inferences.[1] By making the second conjunct a focus anaphor whose antecedent is the first conjunct, the speaker leads the hearer to infer that being an accomplished syntactician is equivalent to being smart. The proposal, then, is that (19) is not a condition that distinguishes the grammatical from the ungrammatical sentences by

[1] Tancredi, Christopher. 1992. Deletion, deaccenting and presupposition. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge

virtue of their meaning. Instead, it is a condition that shapes the interpretation of the message being conveyed. We won't develop a framework in which conditions shape meanings, rather than factor sentences into the grammatical and ungrammatical bins. It will be sufficient for us to have an informal understanding of this.

See Fox (2000) for a sketch. A slightly different way of expressing the difference between ellipsis and de-accenting is offered in Merchant (2001).

Interestingly, this function of (19) collapses when the material in the focus anaphor relevant for drawing the inference is elided. An alternative pronunciation of (30) cannot involve VP ellipsis.

(32) * Thuy is an accomplished syntactician and RODICA is ~~smart~~ too.

compare:

Thuy is an accomplished syntactician and RODICA is ~~an accomplished syntactician~~ too

Using cases in which the focus anaphor employs ellipsis has a certain utility for us then. It allows us to see what the denotation of the antecedent of the focus anaphor is without the haze that arises when inferences of the sort demonstrated in (30) are tolerated.

Focus anaphora, especially in its ellipsis guise, gives us a tool to investigate the denotations of parts of sentences. In this chapter, we will use that tool to investigate the meanings of sentences that involve movement. We start by investigating a particular way in which focus anaphora interacts with the bound variable interpretation that we've seen pronouns can receive.

6.1.2 Sloppy Anaphora

Recall that one kind of deep anaphora arises when a personal pronoun gets an interpretation that depends on the value given to some other DP in the same sentence. This is to be contrasted with the interpretation a personal pronoun can get that depends on a context making some individual salient enough to be the referent of that pronoun. (33) illustrates.

(33) As they lunched in the Wellesley cafeteria, Carolyn told Thuy that no student understood her paper.

To appreciate the range of meanings that (33) gets, it is helpful to know a few things about Carolyn and Thuy. They are both women. They are both published linguists. Carolyn teaches at Wellesley, a college in Cambridge, Massachusetts, and Thuy teaches at Nguyen Tat Thanh University in Ho Chi Minh City. The pronoun *her* in (33) can be understood in one of three ways. It can refer to the same individual that *Carolyn* does: it can corefer with *Carolyn*. It can also refer to the same individual that *Thuy* does: it can corefer with *Thuy*. Finally, it can be understood as a variable bound by *no student*. On the first interpretation, (33) entails that Carolyn's report was that no student understood Carolyn's paper. And on the second, (33) entails that Carolyn's report was that no student understood Thuy's paper. On the final

interpretation, (33) says that Carolyn's report was that no student understood her own paper.

Consider now (34), which includes a focus anaphor. (I give the focus anaphor in both its ellipsis and non-ellipsis versions.)

(34) a. As they lunched at the Wellesley cafeteria, Carolyn told Thuy that no freshman understood her paper and that <no> SOPHOMORE <understood her paper> either.

b. As they lunched at the Wellesley cafeteria, Carolyn told Thuy that no freshman understood her paper and that <no> SOPHOMORE did either.

There are two instances of *her* in this sentence, but because the second is de-accented, it is part of a focus anaphor whose meaning is going to be determined by the clause that has the first instance of *her*. What we should expect to find, then, is that the interpretation of the two instances of *her* should match. This is almost true. It's completely true when the interpretations of the pronouns involve coreference. When the first *her* corefers with *Carolyn* then so must the second *her*; and when the first *her* corefers with *Thuy*, then so must the second *her*. (34) can have one of the two interpretations paraphrased by (35a,b), but neither of the interpretations paraphrased by (35c,d).

(35) As they lunched at the Wellesley cafeteria, Carolyn told Thuy that…

a. no freshman understood her (=Carolyn's) paper and no sophomore understood her (=Carolyn's) paper either.

b. no freshman understood her (=Thuy's) paper and no sophomore understood her (=Thuy's) paper either.

c. * no freshman understood her (=Carolyn's) paper and no sophomore understood her (=Thuy's) paper either.

d. * no freshman understood her (=Thuy's) paper and no sophomore understood her (=Carolyn's) paper either.

We can see why this is expected from the conditions on focus anaphora in (19) by considering how they apply to the last TP—the focus anaphor—in (34).

(36) Because freshman is a FA(sophomore),
$[\![$no SOPHOMORE understands her paper$]\!]^F$ =
$[\![$no freshman understands her paper$]\!]$ when *her* refers to the same individual in both.

What's unexpected in (34) is how the interpretation of the pronouns arises when they are understood as bound variables. We might expect that when the first *her* is interpreted as a variable bound by *no freshman*, so must the second *her*. In fact, this interpretation is not possible. It isn't possible to interpret (34) in the way that (37) describes.

(37) As they lunched in the Wellesley cafeteria, Carolyn told Thuy that…
no freshman, x, understood x's paper and
no sophomore understood x's paper either.

It's useful when judging the meanings of complicated sentences to frame those sentences with a context that biases towards the target meaning. A context that favors giving (34) the interpretation described by (37) is provided in (38).

In Carolyn's last class, her students were to report their interpretation of each other's final papers. She noticed an interesting pattern. If none of the sophomores understood a freshman's paper, then even the freshman author of that paper didn't either.

(38) # As they lunched that afternoon, Carolyn told Thuy that no freshman understood her paper and no SOPHOMORE did either.

The symbol "#" indicates that the sentence has an anomalous semantic interpretation.

If Carolyn has correctly reported the situation in her class to Thuy, then (38) is not the right way of describing that. Unexpectedly, (34) cannot have the meaning in (37).

This turns out to be due to a condition on the bound variable interpretation that limits the syntactic positions that the binder may have relative to the variable. Let's take a moment to learn about this condition.

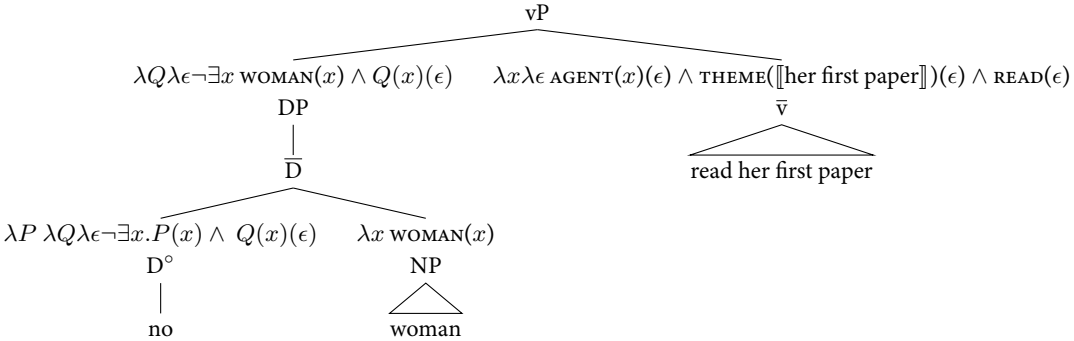
The method we adopted for providing the semantics of a bound variable interpretation crucially involved the meaning of the determiner responsible for introducing the "family of sentences" part of the meaning. We saw how that worked for pronouns that get bound by DPs headed by *every* earlier, but it's true in a parallel way for pronouns that get bound by DPs headed by other quantificational determiners as well. For instance, in (39), it is crediting *no* with a denotation that involves variable binding that is responsible for giving *her* its bound variable interpretation.

(39) No woman reads her first paper

The denotation of the quantificational determiner *no* (given in (40)) takes the NP it combines with, and the $\bar{\text{v}}$ that follows, and creates a meaning from the denotations of these two constituents, as shown in (41).

(40) $[\![\text{no}]\!] = \lambda P\, \lambda Q\, \lambda\epsilon\, \neg\exists x\, P(x) \wedge\ Q(x)(\epsilon)$

(41) $\lambda\epsilon\neg\exists x\ \textsc{woman}(x) \wedge \textsc{agent}(x)(\epsilon) \wedge \textsc{theme}([\![\text{her first paper}]\!])(\epsilon) \wedge \textsc{read}(\epsilon)$



Informally, *no* creates a meaning from these two parts which describes reading-eventualities in which the things that are the agents of that eventuality are not things which are a woman. The quantificational determiner *every* has a different meaning, as we've seen, but it binds a variable in a parallel way. The quantificational determiner *some* also has a meaning that can be expressed as a family of sentences, and it too can be given a denotation that achieves this meaning through variable binding. I've given examples of *every* and *some* below, so that all three cases—the situation with *no*, *every*, and *some*—can be compared.

(42) No woman reads her paper.

For all women under consideration, 1, …,n, it's the case that:

i. woman #1 doesn't read woman #1's paper, and

ii. woman #2 doesn't read woman #2's paper, and

iii. woman #3 doesn't read woman #3's paper, and

⋮

(43) Every woman reads her paper.

For all women under consideration, 1,…,n, it's the case that

i. woman #1 reads woman #1's paper, and

ii. woman #2 reads woman #2's paper, and

iii. woman #3 reads woman #3's paper, and

⋮

iv. woman #n reads woman #n's paper.

(44) Some woman reads her paper.

For some woman under consideration, 1,…,n, it's the case that

i. woman #1 reads woman #1's paper, or

ii. woman #2 reads woman #2's paper, or

iii. woman #3 reads woman #3's paper, or

⋮

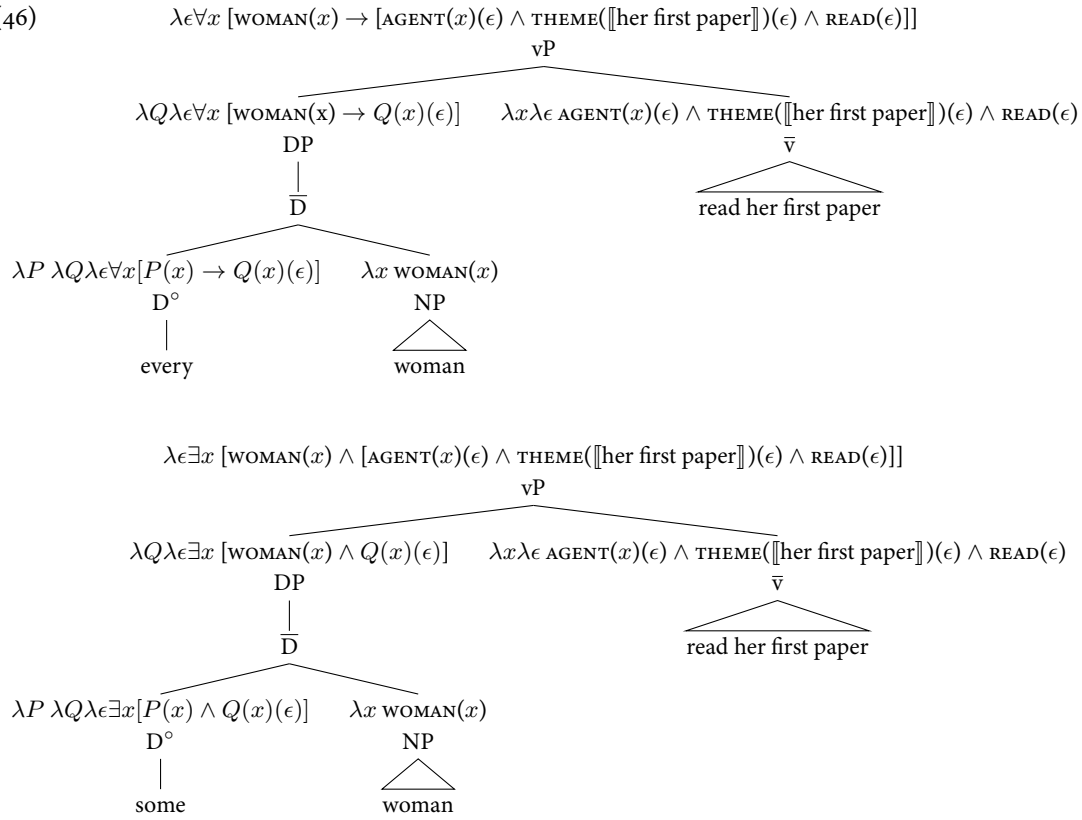
iv. woman #n reads woman #n's paper.

The meaning for *no* is captured by giving *no* the denotation in (40), as (41) shows. The meaning for *every* can be captured by giving *every* the denotation repeated in (45a). And giving *some* the denotation in (45b) provides a way of modeling its meaning too. The compositions in (46) show how.

(45) a. $[\![\text{every}]\!] = \lambda P \lambda Q \lambda \epsilon \forall x [P(x) \rightarrow Q(x)(\epsilon)]$

b. $[\![\text{some}]\!] = \lambda P \lambda Q \lambda \epsilon \exists x [P(x) \wedge Q(x)(\epsilon)]$

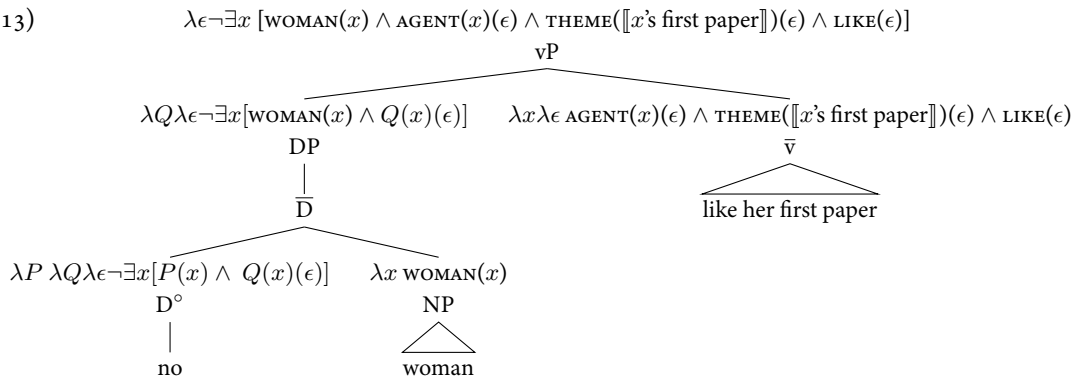
(46)



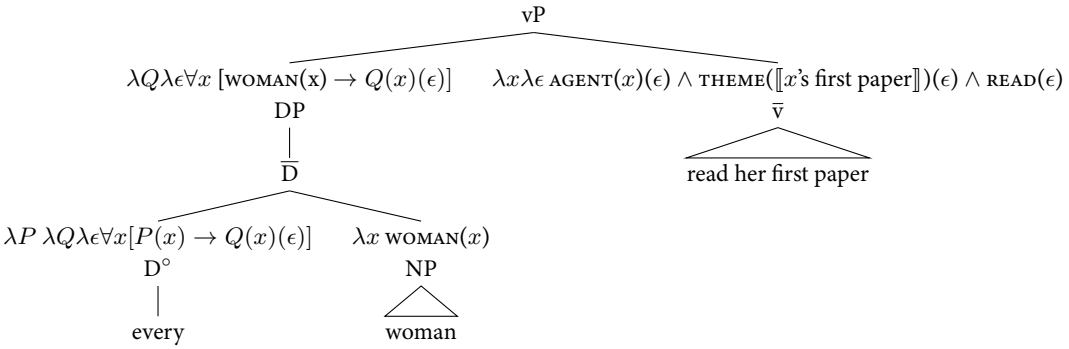
In each case, it is the fact that the second argument of the determiner's denotation is a $\bar{v}$ whose meaning binds a variable in the AGENT position that is responsible for causing the determiner to end up binding a variable there.

To get the bound variable interpretation of the pronoun *her* in these cases involves finding a way of letting those pronouns also be variables bound in the denotation of the $\bar{v}$. As we saw earlier, the bound variable reading for *her* can be achieved for (12) if we can produce the meaning for the $\bar{v}$ indicated in (13). And the same change to the denotations of the $\bar{v}$s in (46) will bring the bound variable interpretation to the pronoun, as (47), and (48) demonstrate.

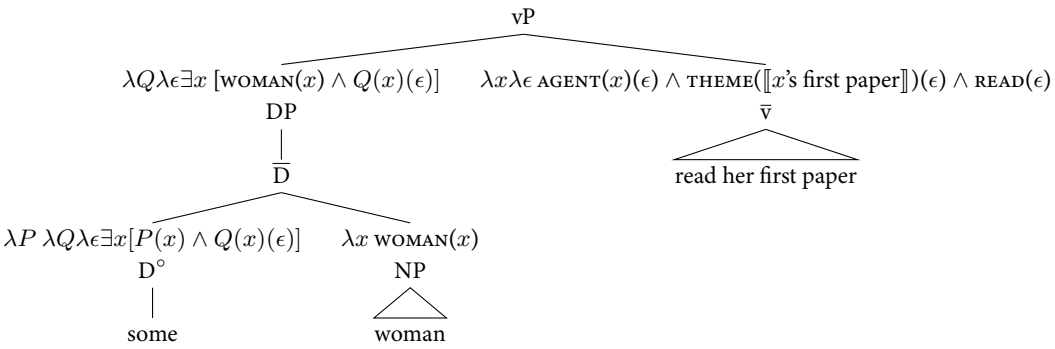
(13)



(47) $\lambda\epsilon\forall x\ [\textsc{woman}(x) \rightarrow [\textsc{agent}(x)(\epsilon) \wedge \textsc{theme}([\![x\text{'s first paper}]\!])(\epsilon) \wedge \textsc{read}(\epsilon)]]$



(48) $\lambda\epsilon\exists x\ [\textsc{woman}(x) \wedge [\textsc{agent}(x)(\epsilon) \wedge \textsc{theme}([\![x\text{'s first paper}]\!])(\epsilon) \wedge \textsc{read}(\epsilon)]]$



What these examples indicate is that for a personal pronoun to be interpreted as a variable bound by a quantificational DP, it must be part of the constituent that the quantifier takes as its second argument. Framing this in syntactic terms, this means that for a quantificational DP, α, to bind a variable, β, β must be part of the phrase that is the sister to α. This relation is usually expressed in terms of c-command, as in (49).

(49) For β to be a variable bound by α, α must c-command β.

(50) α c-commands β iff:

a. every phrase that (non-reflexively) dominates α, dominates β too, and

b. α does not (reflexively) dominate β.

The law in (49) is a prediction made by the semantics we have adopted for expressing the meanings of the quantifiers *no*, *some*, and *every*. To a certain extent, (49) is confirmed. The bound variable interpretation of *her* in (51) is, as (49) predicts, unavailable.

(51) a. That no woman [left] bothers her friend.

b. People that no woman [has talked to] like her friends.

c. Her friends [like] no woman.

The quantificational DP *no woman* doesn't c-command *her* in any of these cases. The sister to *no woman* is marked with square brackets, and *her* is not

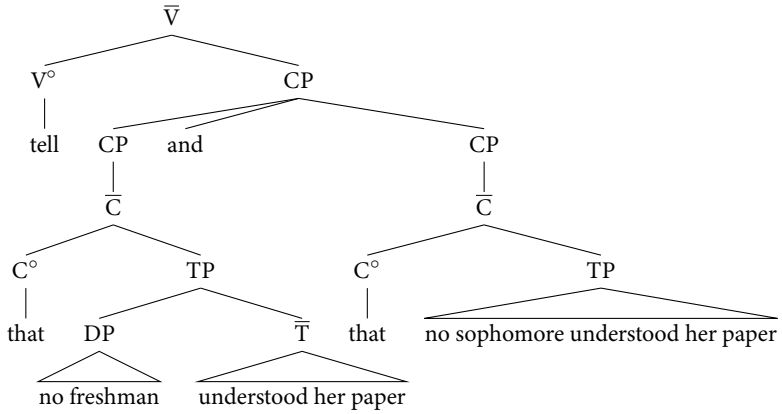
found within this constituent. There are apparent counterexamples to (49) too, however, and so there are reasons to remain skeptical that we have found the correct method of expressing the meaning of quantificational DPs. We will examine these apparent counterexamples in this chapter. One method of addressing them involves the new theory of movement that will be developed here.

But we can see that to the extent that (49) is true, it provides an account for the fact that an interpretation for both *her*s as a variable bound to *no freshman* in (34) isn't possible. The ellipsis version of (34) is repeated here.

(34) As they lunched at the Wellesley cafeteria, Carolyn told Thuy that no freshman understood her paper and that <no> SOPHOMORE did either.

The DP *no freshman* doesn't c-command the second *her*, as the rudimentary parse in (52) shows.

(52)



If we consider examples which minimally differ from (34), but put the second pronoun in a position that is c-commanded by *no freshman*, the bound variable interpretation emerges. Consider, for example, the interpretation of (53).

(53) Carolyn told Thuy that no freshman praised her paper if <no> SOPHOMORE did.

A possible meaning for (53) gives to both *her*s an interpretation in which they are bound by *no freshman*; that meaning can be paraphrased with (54).

(54) Carolyn told Thuy that there is no x, x a freshman, who praised x's paper if no sophomore praised x's paper.

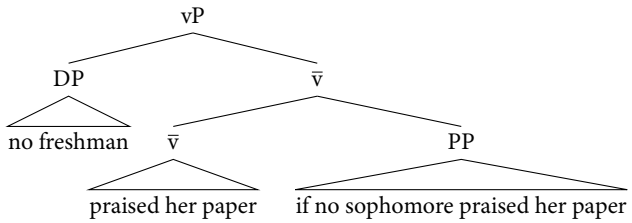
The context in (55) helps make that meaning salient.

In Carolyn's last class, the students gave short assessments of each others' final papers. Carolyn noticed a pattern. If no sophomore praised a freshman's paper, then that freshman didn't praise her own paper either. At lunch that afternoon,…

(55) Carolyn told Thuy that no freshman praised her paper if no SOPHOMORE did.

What makes (53) relevantly different from (34), is that there is a parse—it's sketched in (56)—that causes the second pronoun to be c-commanded by *no freshman*.

(56)



So far, then, the condition on focus anaphora correctly captures the meanings of the pronouns in (34), repeated here.

(34) a. As they lunched at the Wellesley cafeteria, Carolyn told Thuy that no freshman understood her paper and that <no> SOPHOMORE <understood her paper> either.

b. As they lunched at the Wellesley cafeteria, Carolyn told Thuy that no freshman understood her paper and that <no> SOPHOMORE did either.

It accounts for the fact that the second *her* is coreferent with whatever the first *her* is coreferent with. And it predicts, with the aid of the c-command condition on bound variables, that when the first her is a variable bound to *no freshman*, the second *her* cannot similarly be bound to *no freshman*. Interestingly, though, when the first *her* is bound by *no freshman*, there is a possible interpretation for (34). When the first pronoun *her her* to *no freshman*, the second *her* can be bound to *no sophomore*. (34) can have the interpretation expressed in (57).

(57) As they lunched at the Wellesley cafeteria, Carolyn told Thuy that no x, x a freshman, understood x's paper and that no y, y a sophomore, understood y's paper.

This meaning is brought out in the context provided below.

In Carolyn's last class, the students were responsible with reporting their interpretation of each of the final papers. Carolyn noticed a pattern. No one seemed to understand their own paper, though they understood everyone else's. Later that afternoon, …

(58) as they lunched at the Wellesley cafeteria, Carolyn told Thuy that no freshman understood her paper, and <no> SOPHOMORE did either.

John Ross[2] called this a "sloppy interpretation" of the pronoun. It too follows from the condition on focus anaphora in (19).

To see how, consider the denotations of the first and second conjuncts of (34).

(59) a. ⟦no freshman understood her paper⟧ =
$\lambda\epsilon\ \neg\exists y$ [FRESHMAN(y) ∧ AGENT(y)(ϵ) ∧ THEME(⟦y's paper⟧)(ϵ) ∧ UNDERSTAND(ϵ)]

b. ⟦no sophomore understood her paper⟧ =
$\lambda\epsilon\ \neg\exists y$ [SOPHOMORE(y) ∧ AGENT(y)(ϵ) ∧ THEME(⟦y's paper⟧)(ϵ) ∧ UNDERSTAND(ϵ)]

Because *freshman* is a focus alternative to *sophomore*, the focus value of (59b) can be:

(60) ⟦no SOPHOMORE understood her paper⟧^{F} =
$\lambda\epsilon\ \neg\exists y$ [FRESHMAN(y) ∧ AGENT(y)(ϵ) ∧ THEME(⟦y's paper⟧)(ϵ) ∧ UNDERSTAND(ϵ)]

Since (60) is the same as (59a), the conditions on focus anaphora are met. And since *her* in each conjunct is c-commanded by the DP that binds it, the c-command requirement on the bound variable interpretation is met as well.

Sloppy anaphora plays a role in the new theory of movement developed here.

6.1.3 Antecedent Contained Anaphora

We now have enough background to confront the problems that suggest our present theory of movement is wrong. The first that we will examine is in (61).

(61) a. Thuy will visit every student who [RODICA <will visit>].

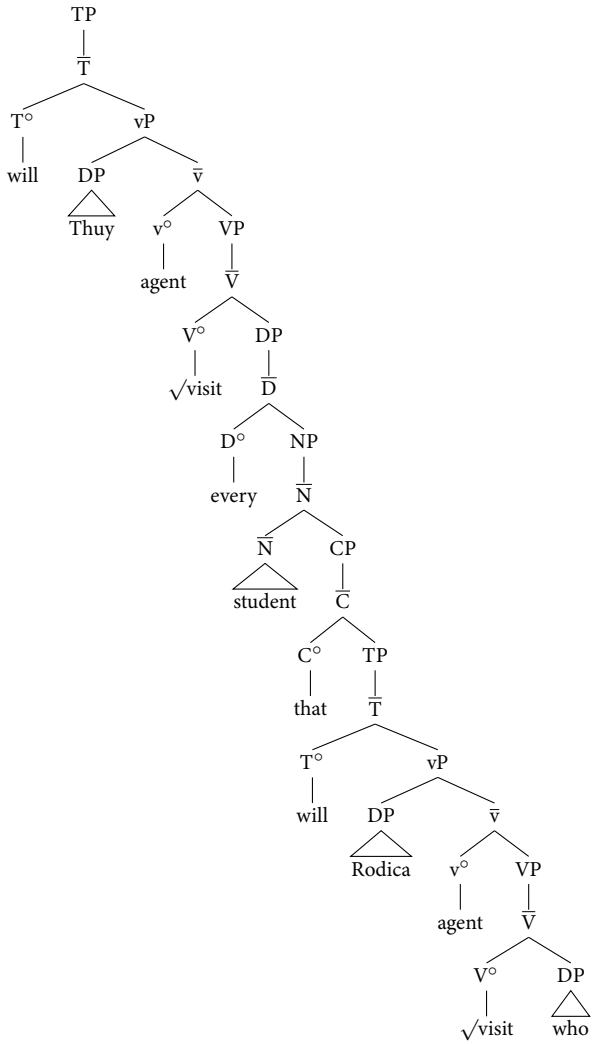
b. Thuy will visit every student who [RODICA <will>].

The material in brackets is a focus anaphor; (61b) uses ellipsis. A problematic property of these examples is that the focus anaphor must find an antecedent from the very VP that contains it. The focus anaphor must therefore have a meaning that matches the meaning of some other constituent in this sentence, where by "match," I mean the condition on focus anaphora that (19) expresses. Our present syntax makes D-structures the sole parse in a derivation from which meanings are derived. But if we consider The D-structure parse that our present syntax provides for (61), there is no constituent that has the meaning that the focus anaphor requires. These sentences include a construction-type that we have not yet examined, a kind of modifier of nouns that is called a "relative clause." The relative clauses in (61) are what contain the focus anaphor: they are the *who Rodica will visit* part. To see what the D-structures are for these sentences requires understanding how relative

[2] Ross, John Robert. 1967. Constraints on variables in syntax. Doctoral Dissertation, Massachusetts Institute of Technology

clauses work. We'll turn to that task immediately below, but in brief, they are CPs headed by the complementizer *that*, and their surface form involves movement of *who* plus deletion of the complementizer *that*. The D-structure representation of (61) is (62).

(62)

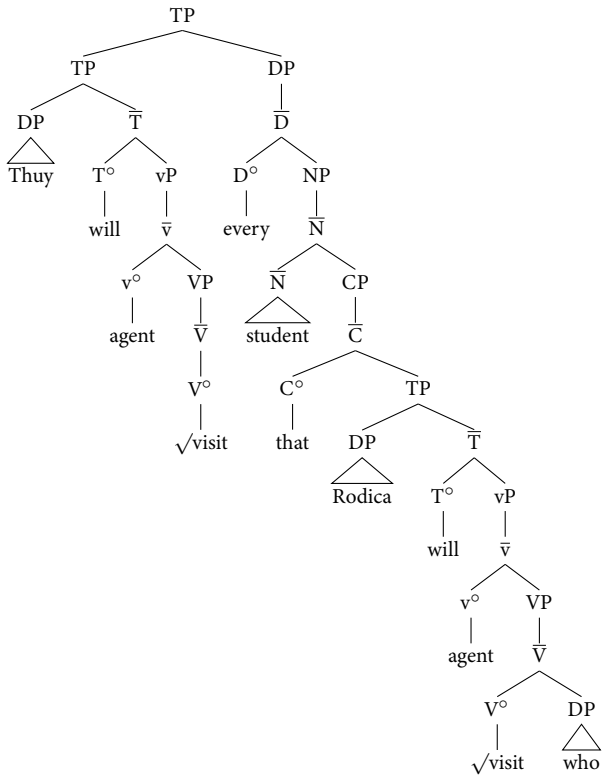


The problem is simply that any constituent that includes *Thuy will visit*, the focus antecedent, also includes the focus anaphor, *Rodica will visit*. Bouton (1970), who discovered the ellipsis version of these cases, called them instances of Antecedent Contained Deletion.

It is very difficult to see how the theory that explains what the antecedent for a focus anaphor is could be manipulated to allow the parse in (62) to get the interpretation it does. As long as the focus anaphor is part of the constituent whose meaning provides the antecedent meaning, there is the paradox of making the interpretation of the focus anaphor depend on its own meaning. The problem seems to be with our syntax. It forces (62) to be the parse from which the denotation is derived, and this leads to the paradox

of antecedent contained deletion. Interestingly, our syntax does allow this sentence to be associated with another parse, but it isn't a parse that the semantic interpretation is read off of. That parse employs Complex NP Shift, and is roughly (63).

(63)



This parse also doesn't provide the material necessary to satisfy the conditions on the focus anaphor. Indeed, this parse doesn't provide the right configuration for our semantics to give it an interpretation at all. In particular, the verb *visit* in the root clause cannot semantically compose with the object DP, and so this parse does not allow the object DP to get the θ-role we understand it to have. But at least this syntax does not put the focus anaphor inside the material that could furnish it with an antecedent. It overcomes the largest problem with deriving the requisite meaning from (62). The solution I will lay out here seeks a way of allowing (63) to get a semantic interpretation, and thereby furnish the antecedent for the focus anaphor.

This will require a change to our syntax, but also a change to how our syntax interacts with the semantics. It will no longer be the case that only the D-structure parse in a derivation gets a semantic interpretation. Instead, we will have to allow parses that are created from the D-structure by a transformational rule be able to give rise to a semantic interpretation. Let's change now our definition of derivation in anticipation of this new architecture.

(64) Lebeaux Model
$\mathcal{D}$ = (P_1, P_2...,P_n) is a derivation iff:

a. P_1 is formed by the $\overline{X}$ Skeleton and Lexical Insertion,

b. Each $P_{i>1}$ is formed from P_{i-1} by a rule,

c. Some P_i is semantically interpreted, and

d. P_n is subject to the linearization algorithm.

A sentence is the set of well-formed derivations which have the same P_1 (=D-structure). A grammatical sentence is a derivation whose cardinality isn't greater than the cardinality of any other derivation for that sentence.

The cardinality of a derivation is the number of Ps in it.

This definition of derivation is identical to the one we started this chapter with, except that the parse relevant for providing a semantic interpretation has not been restricted to D-structure. Instead, this definition allows any parse in a derivation to supply the semantic interpretation. We'll investigate other possible ways of giving up D-structures as the semantically interpretable parse; another popular option is to associate another specific element of the derivation—like the S-structure—as the semantically interpretable parse. The view expressed in (64) is often identified with David Lebeaux,[3] who argued for it in his dissertation. I'll call it the Lebeaux model.

[3] Lebeaux, David. 1988. Language acquisition and the form of the grammar. Doctoral Dissertation, University of Massachusetts, Amherst; and Lebeaux, David. 2009. ***Where does binding theory apply?***. Linguistic Inquiry Monographs. Cambridge, Massachusetts: MIT Press

The move to the Lebeaux model removes constraints on D-structures that our former model had. If only D-structures are semantically interpreted, then they will have to have a form that delivers a meaning. Therefore the requirements our semantics has will play a role in determining which D-structures are well-formed and which aren't. The Lebeaux model changes this. If we persist with the assumption that a grammatical sentence must provide a parse that can be semantically interpreted, the Lebeaux model lets a sentence satisfy this if any of the elements of its derivation is semantically interpretable. On the Lebeaux model, the semantics requires of D-structures, then, merely that they, or some parse derivable from them, be semantically interpretable. It's presently an open question whether this weakening of the conditions on D-structures brings unwelcome consequences.

To see the kind of issues involved, consider for instance how the Theta Criterion applies to derivations.

(65) Theta Criterion

a. Every phrase that is interpreted as an argument must be in a θ-marked position, and

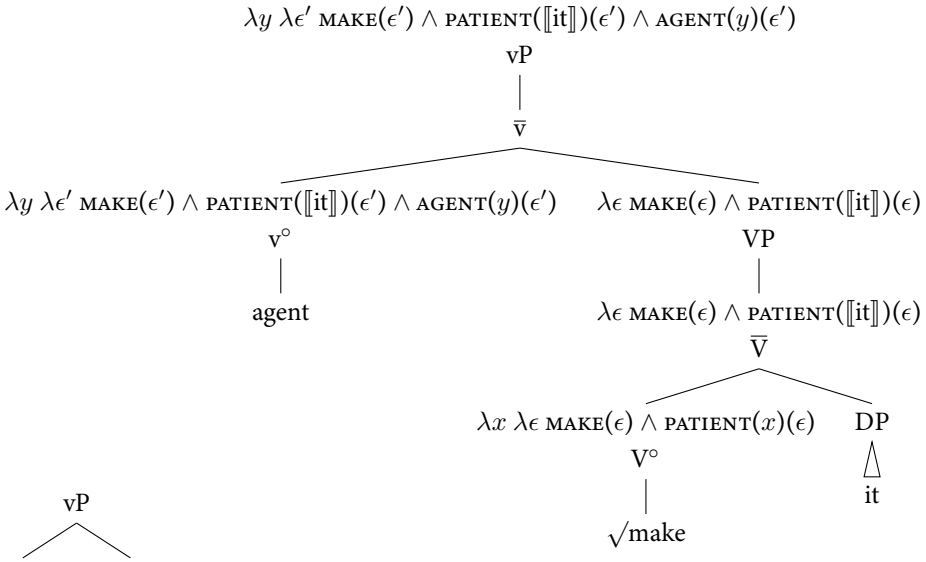
b. Every θ-marked position must contain a phrase that serves as an argument.

Our present model derives that the Theta Criterion is a generalization about D-structures from (66) and the system of semantic rules and denotations that were developed in chapter 3.6.

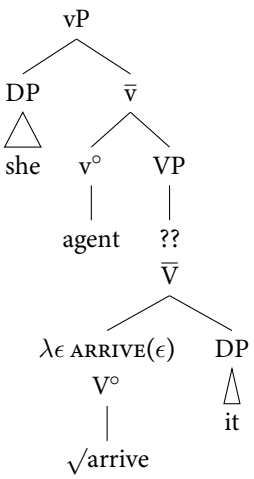
(66) A vP must have a denotation that makes it a function whose domain is eventualities.

Consider, for instance, what would happen if a verb were not supplied with enough arguments by the D-structure (as in (67)) or more arguments are put into a vP than there are θ-roles (as in (68)).

(67)



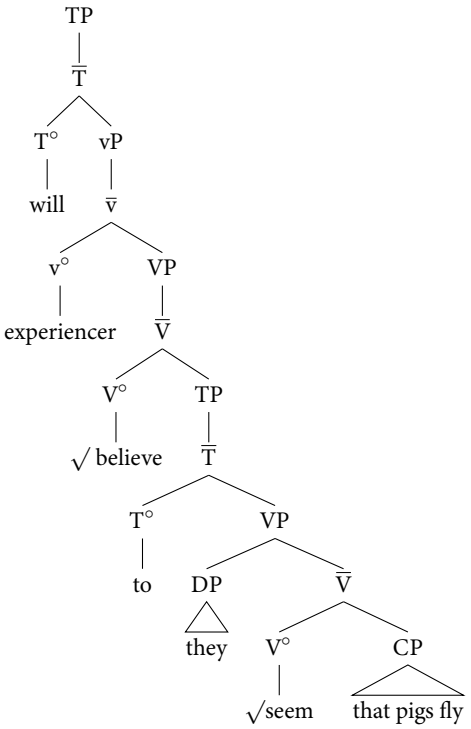
(68)



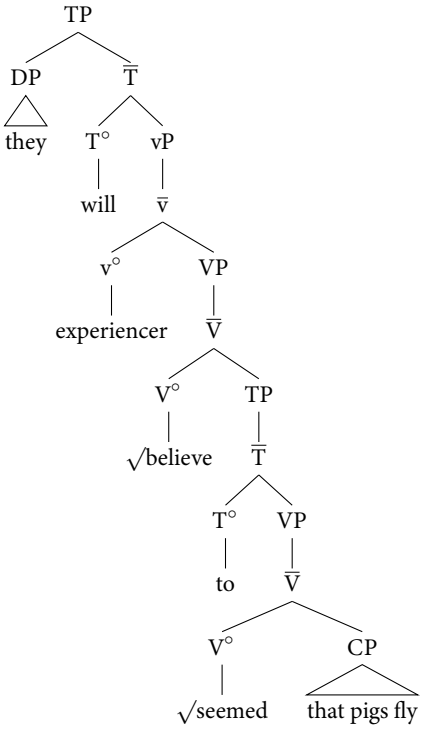
In (68) no denotation is delivered to the $\overline{\text{V}}$, and though our semantics delivers a denotation to vP in (67), it isn't one that satisfies (66).

If D-structures are no longer the parse in a derivation that must satisfy (66), there are new derivations to consider in which (66) is satisfied by another parse. For instance, we might consider a derivation whose D-structure is (69) and whose S-structure is (70).

(69)

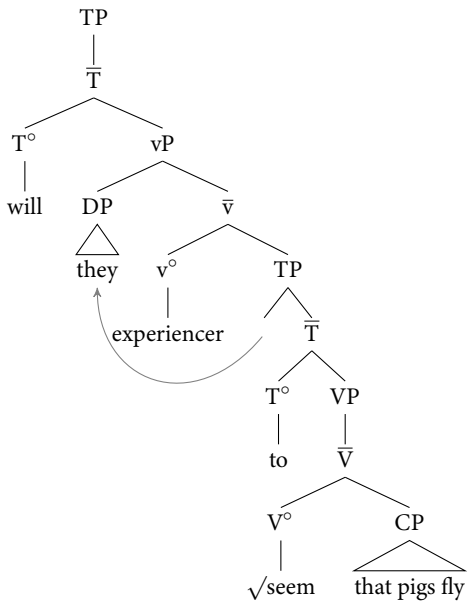


(70)



This derivation can have the parses indicated in (71).

(71)



This is a grammatical derivation, and the parse created by movement in (71) is semantically interpretable. But the sentence it creates—*They will believe to seem that pigs fly*—is ungrammatical. By adopting the Lebeaux model, we will have to ensure that the consequences of letting D-structures be freed from the requirement of semantically interpretability doesn't have untoward results, like admitting the derivation in (69)–(71)–(70).

To exploit the Lebeaux model, we'll change the definition of movement so that the structures created by movement operations have enough information in them to allow the semantics to assemble their meanings correctly. In particular, we'll need a way of allowing the verb *visit* and its object, *every student* …, to semantically combine in (63). We'll also need to have a syntax of relative clauses. And finally, we should make these changes in a way that helps close the problems that arise from letting D-structures be free of the conditions that result from forcing them to be semantically interpreted.

6.1.3.1 Traces, Referential Indices, and Quantifiers

Let's begin by giving to representations like (63) the ability to semantically combine the verb with the DP that has moved. The method for doing this that we will start with involves adding to the movement operation two additional processes that accompany the change of position function that we've been relying on up to now. The first of these processes inserts into the position that a moved term vacates a term that will be semantically interpreted as a variable. This term is called a "trace," and it is standardly represented by "*t*." The second process makes the term that has moved bind that variable. One method of expressing this is to let the movement operation change the meaning of the largest phrase that contains the trace but not the moved phrase, so that it becomes a function whose λ operator binds the trace.

I believe the trace-based system for movement was introduced in Chomsky (1977a), and developed in Chomsky (1977b). The method of expressing how traces are bound that I will sketch here is from Heim and Kratzer (1998).

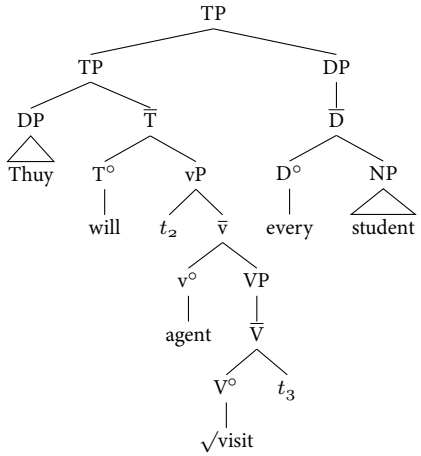
To do this will require a syntax that regulates the connection between variables and their binders. Up to now, the connection between a variable and its binder has been part of the denotation of the morphemes that populate syntactic representations. The fact that pronouns can also be interpreted as variables shows us that we will need a syntax that links expressions that are interpreted as variables to their binders as well. The idea behind Trace theory is that movement also invokes this syntax. We'll add to the morphemes that can be part of our syntactic representations, terms whose job it will be to express the variable-binder connection. These terms are known as "referential indices," and they will be represented as numerical subscripts. To begin with, we will let indices be parts of traces, and nothing else.

With this background, we can now redefine movement with (72).

(72) α moves to β =$_{\text{def.}}$ α takes position β and if α is an XP:
   a. α is replaced by t_n, where n is a unique referential index, and
   b. The denotation of the sister of β, γ, becomes: $\lambda n[\![\gamma]\!]$.

This will give to an S-structure that has been created with Complex NP Shift a representation like that illustrated by (73).

(73)

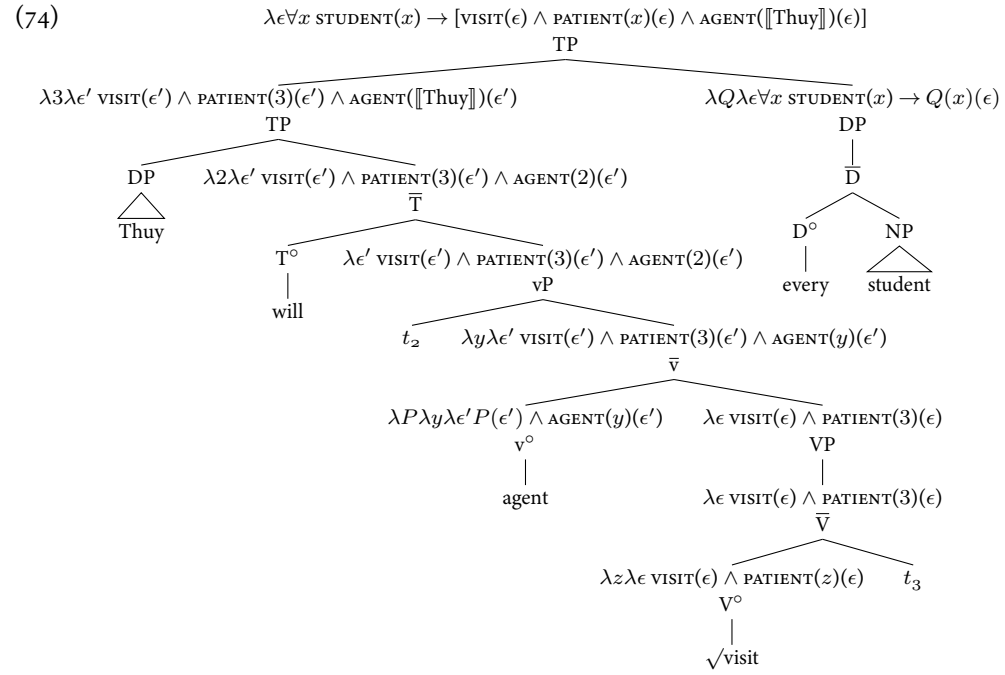


Note that both Argument Movement of *Thuy* and Complex NP Shift of *every student* have invoked traces, and that their referential indices are distinct. The requirement in (72a) that the referential index on a trace be unique is responsible for ensuring that each trace in a syntactic representation be associated with independent referential indices. To see the effect that (72b) has, we will look at how the semantics assigns denotations to the pieces of (72). We have not developed a semantics for *will*, however, and so we are not equipped with everything necessary to do that. In general terms, we have not developed a semantics for modals or tense—the two kinds of terms that occupy T° position in finite clauses—and this is why we've not endeavored to give meanings for anything larger than vP in our parses. The meanings of these expressions requires a dive into purely semantic issues that will not, in

my estimation, further our understanding of syntax. Let's therefore ignore entirely the semantic contribution that the contents of T° makes.

If we do this, the semantic composition of the pieces of (73) that we have previously determined is (74).

(74)



Note that the denotation of traces is just the variable that their referential index is. When a trace is the argument of a function it contributes its value to the argument position(s) in that function. Thus, when the denotation of $\sqrt{visit}$ combines with the trace in its complement position, the result puts the value of the variable 3 into the relation that the patient θ-role denotes.The same thing happens with the trace in the Specifier of vP. That trace is the argument of the function that $\bar{v}$ denotes, and when they compose, the value of that variable is inserted into the agent θ-role relation.

This means that the phrases which contain the trace, but not its binder, are technically without a denotation, since we have not given a denotation for a variable except when bound. A more common approach to the semantics of variables lets them have denotations even in contexts where they are not bound.

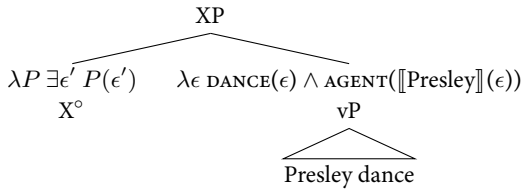
The action of (72c) can be seen in the denotations that $\overline{T}$ and the lower TP have. If we ignore the semantic contribution made by *will*, the denotation of $\overline{T}$ should be the same as the denotation of vP. They contain all the same meaningful items. But the denotation of $\overline{T}$ differs from vP by virtue of (72c). Because $\overline{T}$ is the largest phrase that dominates t_2 but does not dominate the phrase that t_2 replaces, it is the target for (72c). The rule in (72c) adds to the beginning of the denotation that $\overline{T}$ would otherwise have, a λ operator that binds the trace; in this case, that is "$\lambda 2$." A similar step is responsible for the "λ" that starts the denotation for lower the TP as well. That phrase is the largest phrase that contains the trace t_3, but not the phrase that the trace replaced, and so it is also the target for (72c).

In this way, giving movement the ability to change the meanings of the sentences allows the moved DPs in (74) to be interpreted in the positions they move to. But now that we are looking at the meanings of parts of sentences

that are larger than vP, a problem to our semantics is exposed. The denotation of vPs in our semantics is a function whose domain is eventualities: the denotation of vP is a description of eventualities. The denotations of clauses constitutes a claim about the existence of an eventuality of the sort that its vP describes. There must consequently be something in a clause that is responsible for changing the denotation of a vP into this claim. If we assume that this change is effected by the denotation of some term in a clause, then we can schematize what is needed with (75).

See Higginbotham (1985) for an early discussion of this issue.

(75) $\exists\epsilon'\ \textsc{dance}(\epsilon') \wedge \textsc{agent}([\![\text{Presley}]\!])(\epsilon')$



What the X° in (75) does is change the description of eventualities that vP denotes into a sentence that claims that there is such an eventuality. This is the type of thing that we want (non-negated) sentences in general to do. Our semantics so far has been focused on the issue of combining DP arguments and the verbs or adjectives that they are arguments of, and this has allowed us to restrict attention just to how vPs are semantically composed. Now that we must consider the semantics of larger parts of sentences, we will have to acknowledge the existence of something within them that converts the description of eventualities that vPs are into expressions that say that such eventualities exist. Somewhere within a sentence, there must be something that has the function that X° has in (75).

We can see that this term must be lower than negation in English from (76).

(76) Presley did not dance.

The denotation for this sentence should make it deny the existence of a dancing eventuality that has Presley as its agent. The existential quantification over events should be part of what *not* negates.

A popular speculation is that X° should be equated with a head that determines the "aspect" of a clause. Aspectual morphemes have meanings that relate the event described by a vP to tense. By way of illustration, consider the sentences in (77), all of which have present tense.

See Pancheva (2003) for an accessible introduction.

(77) a. She sings.
b. She is singing.
c. She has sung.

Semantically, these sentences differ in a way that can be described in terms of how the singing eventuality is related to the time at which these sentences are produced. Thus, for instance, (77b) conveys the information that the time at which the singing eventuality is happening includes the time at which

(77b) is produced. By contrast, (77c) claims that the time at which the singing eventuality occurs is prior to the time at which (77c) is produced. Finally, (77a) indicates that the singing eventuality is a kind of state that holds at the time when (77a) is produced. In each case, the tense provides a time to which the eventuality is related. This becomes clearer if (77) is compared to the meanings these sentences have when they contain past tense.

(78) a. She sang.
b. She was singing.
c. She had sung.

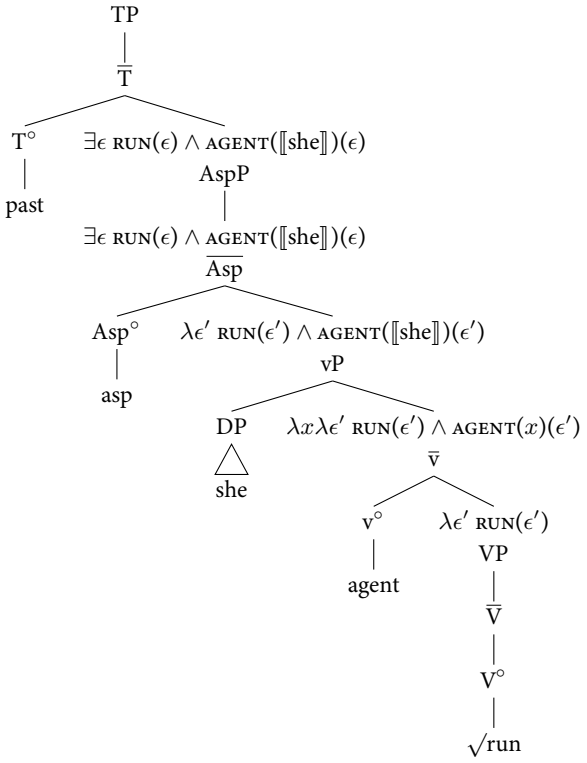
Just as in (77), these sentences differ in how the singing eventuality is situated temporally with respect to tense. But rather than that relation holding between the time of the eventuality and the time at which the sentences are produced, the relation holds between the time of the eventuality and a time prior to when the sentences are produced. Thus, for instance, (78b) says that the singing eventuality occurs at a time that includes some time prior to when (78b) is produced. And (78a) conveys that the singing eventuality should be regarded as a kind of state that occurred at a time prior to the time at which (78a) is produced.

We shall not investigate how this semantic information is conveyed by the material in a syntactic representation. As the examples above indicate, these distinctions in English involve the use of auxiliary verbs as well as morphology on the verbs those auxiliaries embed. Because neither of these ingredients are present in (77a)/(78a), it is not obvious that aspectual information is present in these examples. The evidence for an aspect morpheme in these examples comes largely from a semantic theory of aspect. It is also in the semantics of aspect that we find evidence for it being responsible for introducing the existential binder of an event variable. Because I will not develop this semantics, I will not be able to provide you with the evidence for equating X° with Aspect. Instead, I will rely solely on the evidence provided by (76) for the conclusion that X° is situated in a position syntactically lower than T°—a position I will call Asp°. For our purposes, we need not reveal the full semantics of the morpheme that occupies Asp°. Instead, we can let Asp° have just the semantics of X°. That is, we will focus on just the part of its denotation that existentially closes the event variable.

If we adopt the denotation for *asp*, the content of Asp°, indicated in (79), then *she ran* will have the denotation indicated in (80).

(79) $[\![asp]\!] = \lambda P\, \exists \epsilon\, P(\epsilon)$

(80) $\exists \epsilon$ RUN$(\epsilon) \wedge$ AGENT$([\![$she$]\!])(\epsilon)$



I continue to ignore the contribution made to the denotation of a sentence that comes from tense.

This gives to TP a denotation that claims that there exists an eventuality that has the properties that vP describes. In the case of (80), the TP's meaning claims that there is an eventuality of dancing whose agent is whatever *she* refers to.

Putting the term that existentially closes the eventuality in the position that (80) does correctly puts it inside the constituent that *not*, and the other exponents of *neg*, negates. Assume that *neg* is a function which takes the denotation of its complement and returns that denotation negated; this is what (81) does.

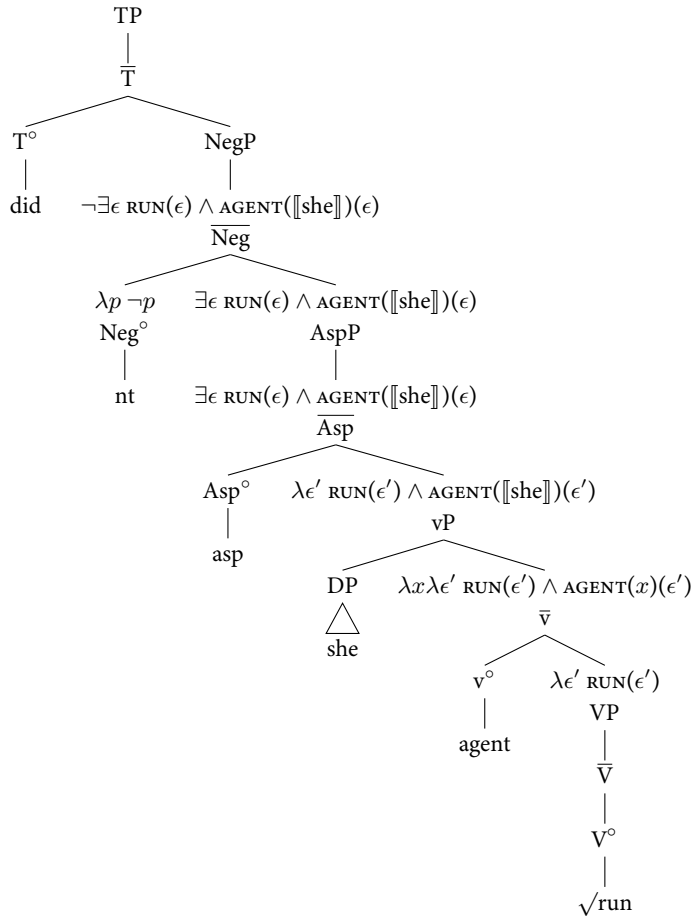
Understand the meaning of "$\neg p$" to have the paraphrase given by "not p." That is: if p is true, $\neg p$ is false.

(81) $[\![\text{neg}]\!] = \lambda p\, \neg p$

(82) illustrates how this will deliver the correct meaning to a sentence with *neg* in it.

she didn't run ≡

(82) $\neg\exists\epsilon\ \textsc{run}(\epsilon) \wedge \textsc{agent}([\![\text{she}]\!])(\epsilon)$

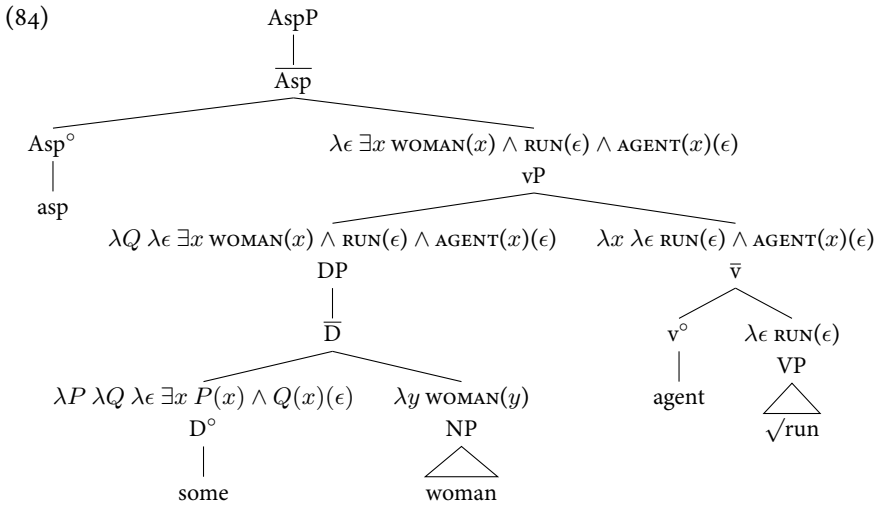


This change has consequences for the denotations we have adopted for the quantificational determiners *no*, *some*, and *every*. Those denotations, repeated in (83), are designed so that they can be interpreted in Specifier of vP and complement of V° positions.

(83) a. $[\![\text{no}]\!] = \lambda P\ \lambda Q\ \lambda\epsilon\ \neg\exists x[P(x) \wedge Q(x)(\epsilon)]$
b. $[\![\text{some}]\!] = \lambda P\ \lambda Q\ \lambda\epsilon\ \exists x[P(x) \wedge Q(x)(\epsilon)]$
c. $[\![\text{every}]\!] = \lambda P\ \lambda Q\ \lambda\epsilon\ \forall x[P(x) \rightarrow Q(x)(\epsilon)]$

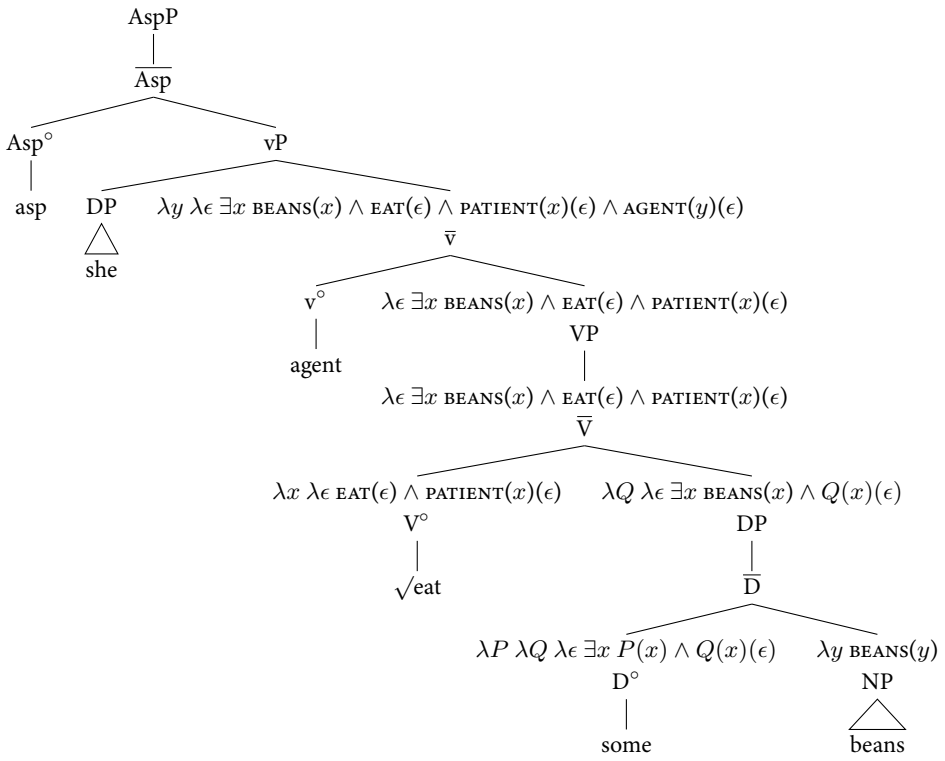
The second argument for these denotations calls for a term that relates things to eventualities. That is the right semantic type for $\overline{\text{v}}$s, and this is why these denotations allow a DP headed by one of these quantifiers to be semantically interpreted in Specifier of vP; (84) illustrates.

(84)



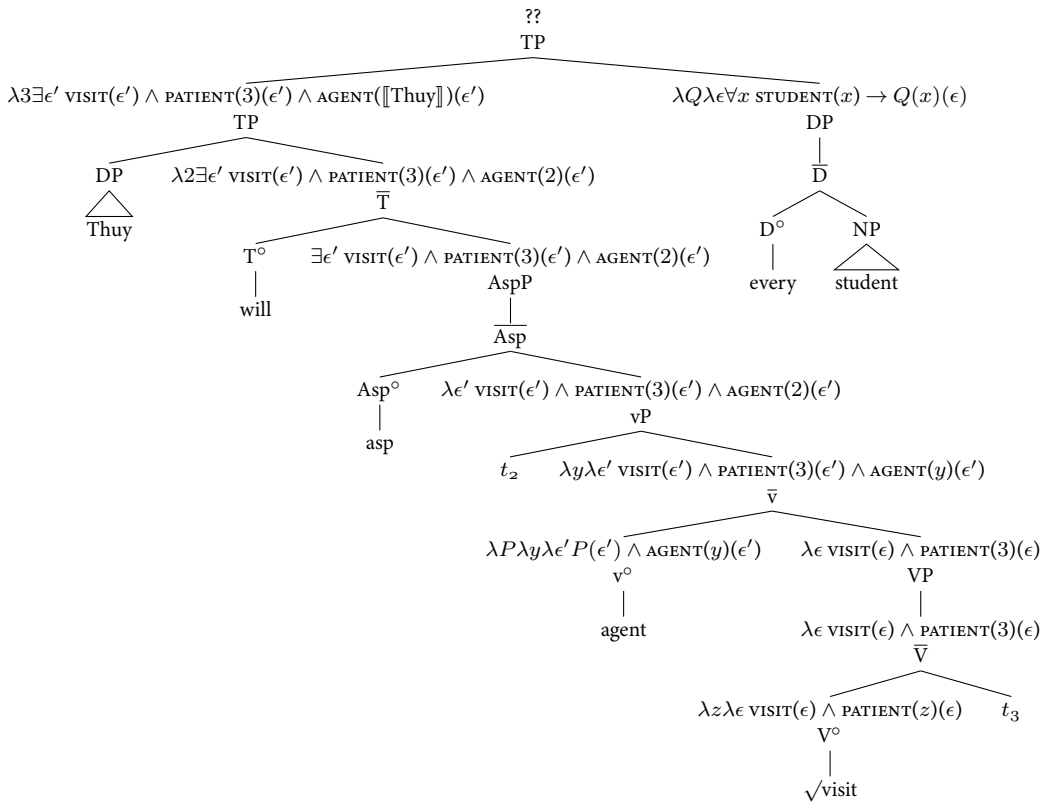
Because transitive verbs are also of this semantic type, DPs headed by one of these quantifiers can also combine with transitive verbs, as (85) illustrates.

(85)



But by introducing AspP, and thereby changing the denotations of $\overline{\text{T}}$ and TP, the position that DPs move to will no longer put them in a position that supplies an argument of the right semantic type for a quantifier. This can be seen by considering how (74) is changed by this semantics. The TP that the moved object DP is a sister to is no longer the right kind of semantic term, as (86) illustrates.

(86)



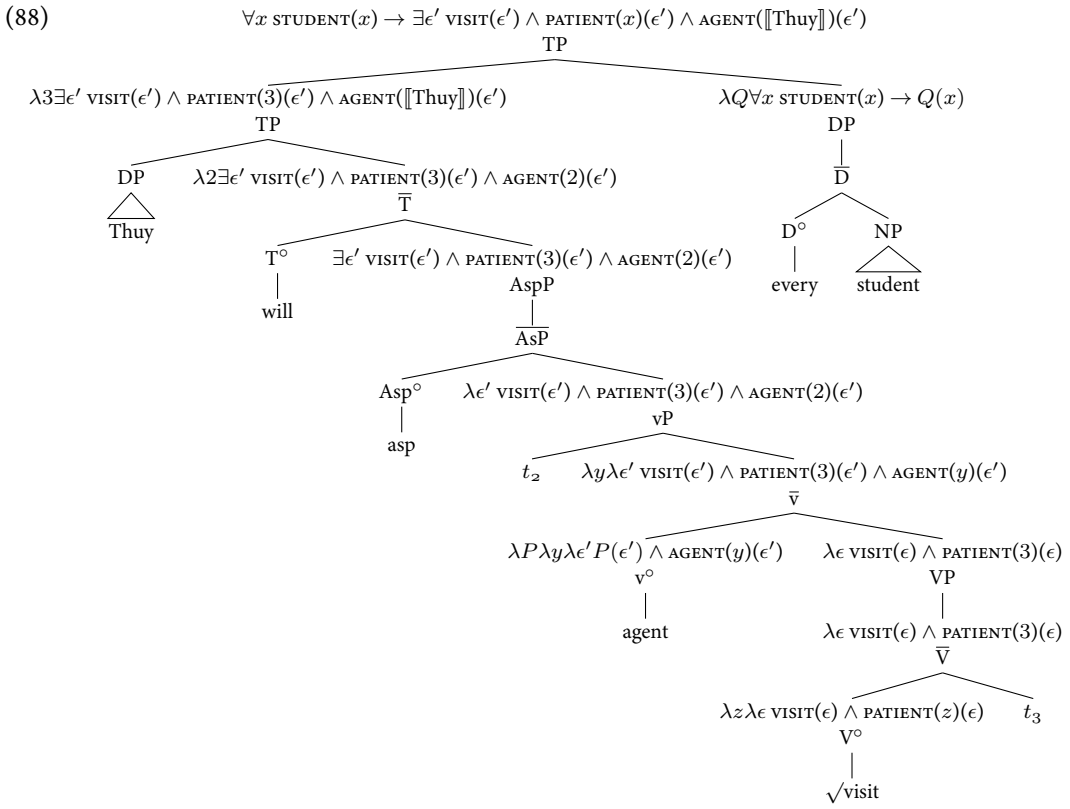
But this is just the syntactic representation that we need to find a denotation for, in order to solve the problem posed by antecedent contained anaphora.

Let's solve this problem by designing the denotations of *no*, *some*, and *every* so that they can be interpreted in their moved positions, but not in their underlying position. The new denotations are (87).

(87) a. $[\![\text{no}]\!] = \lambda P\ \lambda Q\ \neg\exists x[P(x) \wedge Q(x)]$
b. $[\![\text{some}]\!] = \lambda P\ \lambda Q\ \exists x[P(x) \wedge Q(x)]$
c. $[\![\text{every}]\!] = \lambda P\ \lambda Q\ \forall x[P(x) \rightarrow Q(x)]$

The second argument of these denotations now requires something that is a function whose domain is things and whose range is a truth value. That is the right fit for the position that the moved DPs occupy in (86), as can be seen from (88).

(88)



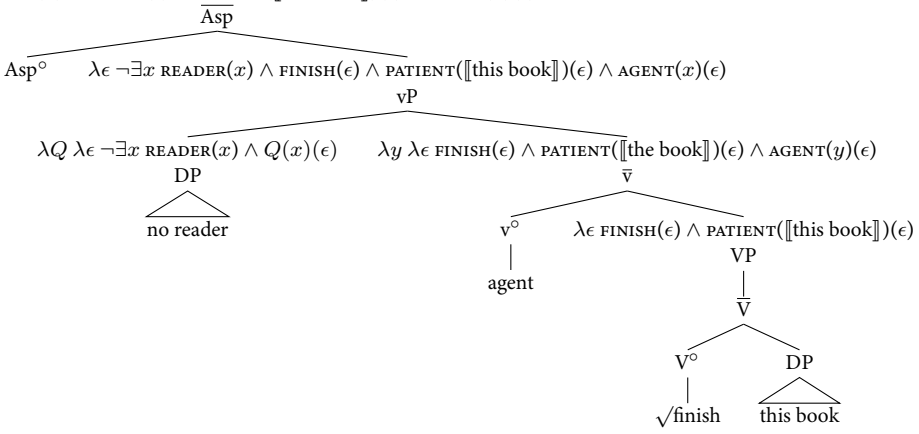
This change to the semantics of quantifiers is independently required. The semantics I introduced originally for quantifiers harbors a problem, which can be seen by considering (89).

(89) No reader finishes this book.

On the denotation for *no* that I gave originally (=(83a)), (89) should get the meaning indicated in (90).

Because we are ignoring the semantic contribution made by tense, the denotation of (89) is the same as the denotation of $\overline{\text{Asp}}$ in (90).

(90) $\exists\epsilon\ \neg\exists x\ \text{reader}(x) \wedge \text{finish}(\epsilon) \wedge \text{patient}(\llbracket\text{this book}\rrbracket)(\epsilon) \wedge \text{agent}(x)(\epsilon)$

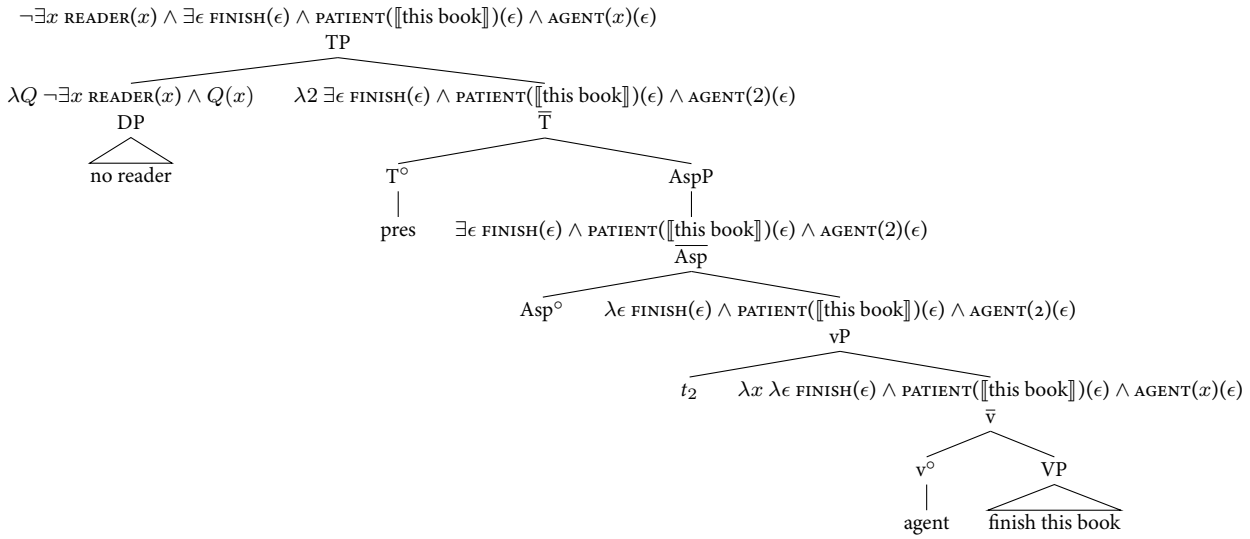


Informally, what (89) claims is that no one who is a reader is also someone who finishes this book. If we understand *reader* to be restricted to just those

individuals who are readers of this book, then (89) amounts to the claim that this book is never read to completion: there are no readers of this book who finish reading it. But that doesn't correspond to the denotation of $\overline{\text{Asp}}$ in (90). The denotation of $\overline{\text{Asp}}$ in (90) is that there is an event of finishing this book and that there are no readers who are the agent of that finishing event. It implies that there is a way of finishing this book without being a reader of it, and that, dear reader, is (regrettably) merely wishful thinking.

This problem is remedied by adopting the denotation for *no* in (87a). That denotation prevents *no reader* from being interpreted in its D-structure position, and instead requires that it be semantically interpreted in its S-structure position, as in (91).

(91)

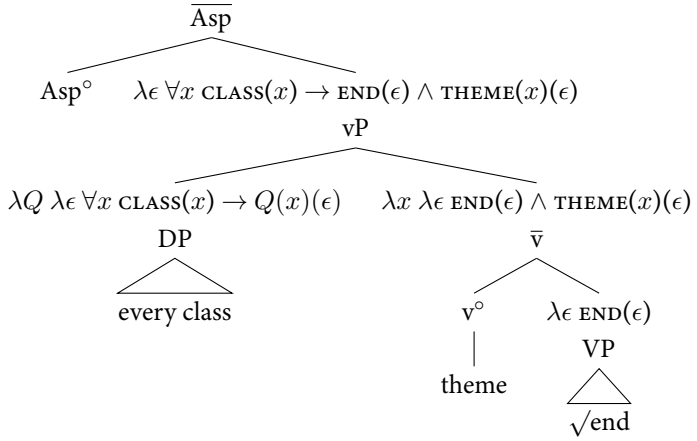


This delivers the right meaning. It produces a sentences that says there is no x which is a reader for which there is an event of finishing this book that has x as agent.

The same problem arises for our earlier denotation for *every* as well (i.e., (83c)). This can be appreciated by considering (92), whose meaning according to (83c) would be (93).

(92) Every class ended.

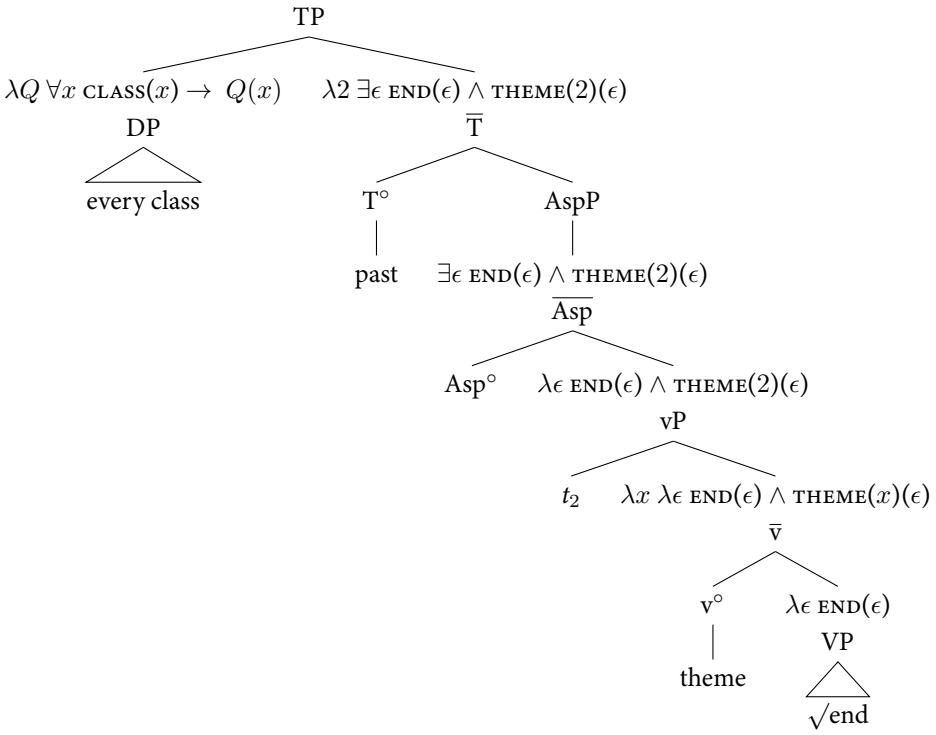
(93) $\exists\epsilon\ \forall x\ \textsc{class}(x) \rightarrow \textsc{end}(\epsilon) \wedge \textsc{theme}(x)(\epsilon)$



This D-structure assumes that ***every class*** is the external argument of ***end***. It could well be, however, that ***end*** is an unaccusative verb, and ***every class*** is its internal argument. This doesn't affect the point being made.

The denotation in (93) says that there is a single ending event that every class is the theme of. Instead, what we want is each class to be able to have its own ending. If we adopt the new denotation for *every* in (87), however, the meaning in (93) is blocked, and instead the denotation in (94) is derived.

(94) $\forall x\ \textsc{class}(x) \rightarrow \exists\epsilon\ \textsc{end}(\epsilon) \wedge \textsc{theme}(x)(\epsilon)$



As desired, (94) says that for all x if x is a class then there is an end-eventuality that it is the theme of.

These facts about the meaning of quantifiers are another indication that we don't want to force D-structures to be the only parse that is semantically

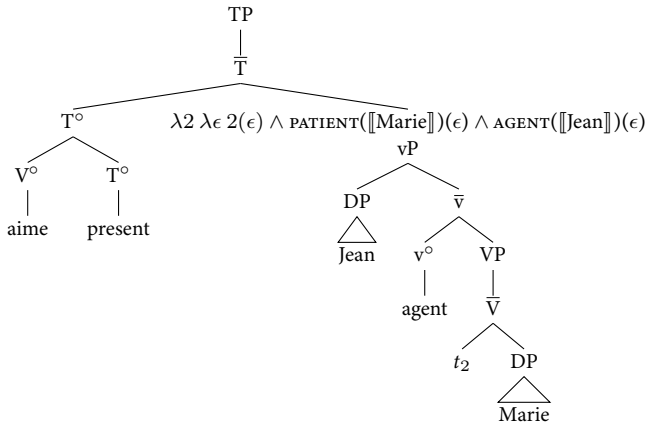
interpreted. DPs headed by quantifiers shouldn't be semantically interpreted in their D-structure positions, as the cases in (89) and (92) indicate.

Summarizing, then, we've done three things to allow movement to make new meanings. We've let movement create something—a trace—in the position that the term being moved resides in. That trace employs a new syntactic formative, an index. This has been wedded to a semantic rule that changes the meaning of the phrase that is the sister to the moved term's new position. That change introduces a λ-expression that binds the trace. And finally, we've adjusted the denotations of quantificational determiners so that they can be interpreted in the positions they move into. This adjustment now prevents the DPs that are headed by quantifiers to be interpreted in their D-structure position. Thus, this last amendment has made D-structures that have quantificational DPs in them incapable of delivering a meaning. Movement is forced to interpret these DPs.

This method of controlling where quantificational DPs are interpreted can be found in Chung and Ladusaw (2004).

Note that the revision to movement in (72) puts traces into the syntax only when phrases move. This is because the syntax we've adopted for Head Movement is incompatible with traces. Consider, for instance, the result of moving the verb *aime* to a tensed T° position in French. If this movement included traces, it would create a representation like that in (95).

(95)

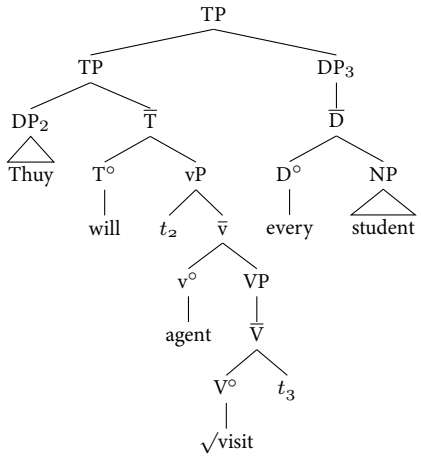


The (72c) part of the new definition of movement causes vP to have a denotation that forces it to take the *aime*+present=T° term as its argument. I have hidden from our view the meaning of tense, but there is compelling evidence that its denotation does not allow it to combine semantically with the denotation we have given to verbs. And even if there was a means of combining these two pieces, the resulting meaning should not be placed "into" the position marked by the variable "2." For these reasons, I have prevented the new rule of movement from allowing a head to be semantically interpreted in any except its D-structure position. We will have an opportunity to revisit this decision before this chapter is over.

It is useful to have a way of representing in our syntactic parses which phrases have undergone the change in meaning that (72c) describes. That change has the effect of making the phrase that has been replaced by the

trace, be the argument of the function that (72c) creates. Although technically not correct, this phrase is said to be the trace's binder. The standard way of representing which phrase has undergone the change that (72c) describes is to give the "binder" a subscript that matches the referential index on the trace. With this notation in place, (73) becomes (96).

(96)



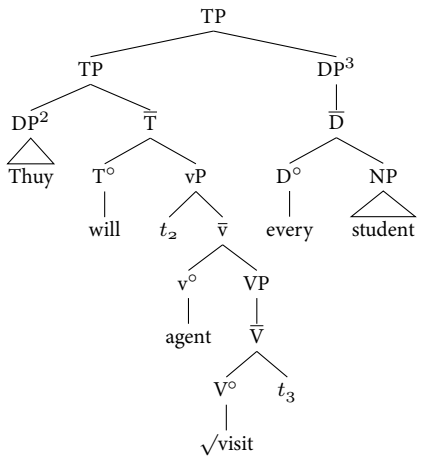
The subscript on *Thuy* indicates, then, that the movement rule has changed the denotation for $\overline{\text{T}}$ into a function that takes as its argument This notation gives us a compact way of indicating which traces are related to which moved phrases.

Although this notation is standard, I find it confusing and will modify it in these lectures. We will see that DPs other than traces can have referential indices as part of their meaning. When we encounter these cases, it will be convenient to be able to represent them with the same subscript notation that we are using now with traces. Some of these terms can also be binders; that is, we will want to be able to indicate that one term can have a referential index that is bound and simultaneously be a binder of another referential index. We will therefore want to have a different way of indicating what a term binds and what a term is bound by. The notation illustrated in (96) to illustrate that *Thuy* is a binder looks just like the way used to indicate that *Thuy* has a referential index and therefore is bound. We should keep these distinct. We need different ways of indicating where a referential index is and where a binder is. For this reason, I will adopt the convention of indicating where the binder of a trace is by appending to it a superscript that matches the referential index on the trace. Instead of (96), I will use (97).

Wherever convenient, I will suppress the presence of AspP in parses. I will reveal their presence only when I judge that doing otherwise would confuse.

This convention is introduced in Heim (1993). That terms can be both variables and binders of variables is first discussed in Higginbotham (1980).

(97)



Keep in mind that the superscripts on binders are not part of the syntax. They are merely a typographical diacritic that serves as a guide to the underlying semantics. In this respect they differ from referential indices, which are morphemes in a syntactic representation.

The machinery we've just reviewed for letting movement create semantically interpretable parses can now be exploited to give a treatment of pronouns when they get a bound variable interpretation. Let pronouns resemble traces in being capable of containing a referential index. We can think of pronouns as consisting of the features that express their person, number and, sometimes, gender information; when they are interpreted as bound variables, they will have a referential index as well. When pronouns are interpreted as referring, we can imagine that their person, number and gender features steer that reference. Informally, then, we can think of pronouns in the way that is portrayed by (98).

(98) ⟦DP⟧ = the x such that x is singular and female and not the speaker or hearer.

D̄
D°
sing
3rd
fem

When a pronoun has an index, we can let the value delivered to that index have the role that *the x* does in the formula in (98).

(99) ⟦DP⟧ = 3 and 3 is singular and female and not the speaker or hearer.

D̄
D°
sing
3rd
fem
3

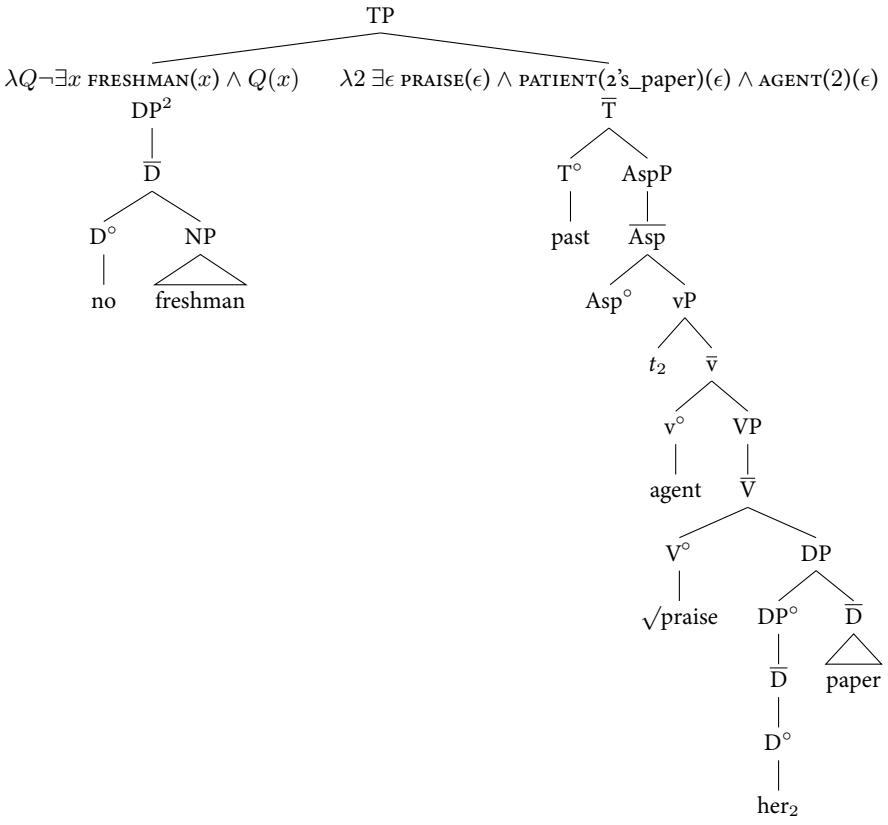
I will adopt the convention of indicating when a pronoun contains a referential index by appending that referential index as a subscript to the pronoun, parallel to the representation that expresses traces. Thus, (99) can be represented by:

(100) her_3

Let's take a look at how this will provide a way of deriving the bound variable interpretation of pronouns in (101). If we let *her* come with a referential index that matches the one created by Argument movement, the S-structure representation for (101) will be (102).

(101) No freshman praised her paper.

(102) $\neg\exists x\ \textsc{freshman}(x) \wedge \exists\epsilon\ \textsc{praise}(\epsilon) \wedge \textsc{patient}(x\text{'s_paper})(\epsilon) \wedge \textsc{agent}(x)(\epsilon)$



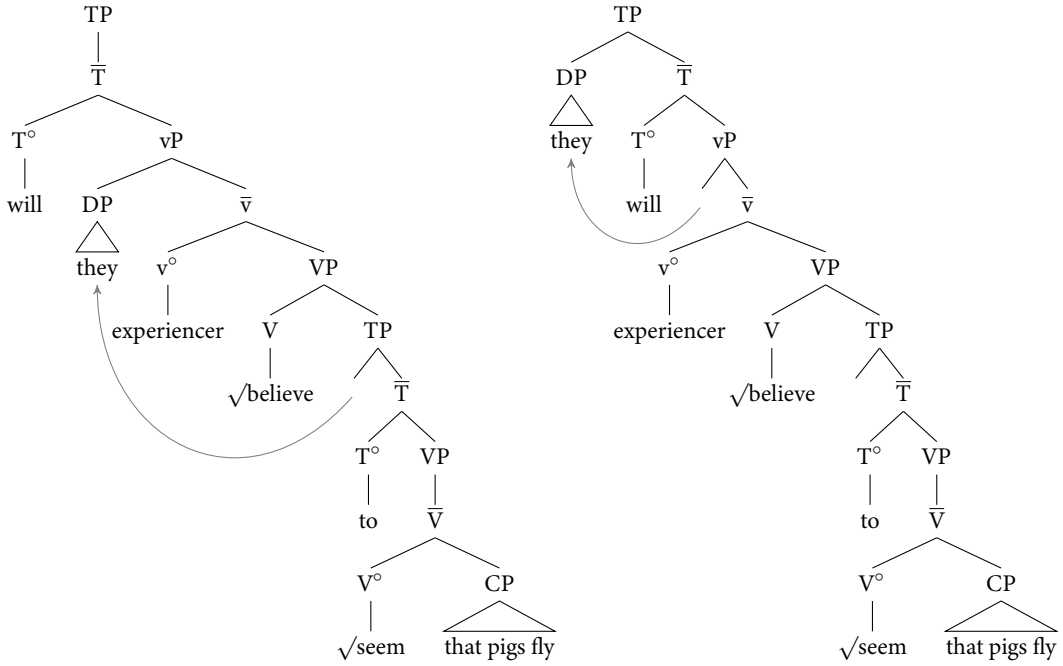
Because movement of *no freshman* has changed the meaning of $\overline{\text{T}}$ into a function that binds variables with the index "2," the pronoun *her* is construed as one of those variables when it includes that index. The fact that pronouns can be understood as variables corresponds to the proposal that pronouns can include indices as part of the ingredients that make them up. Those indices can be interpreted when movement has created a function that is positioned so that it binds that index.

Trace theory, then, has the useful consequence of providing a syntax for determining how, and where, pronouns can get their bound variable inter-

pretation. Because the rule of movement changes the meaning of the phrase that the moved term becomes a sister to, it creates a meaning that can bind pronouns in that phrase. The movement rule not only creates a meaning for this phrase that allows it to bind the trace, it also creates a meaning that allows it to bind any pronoun that shares the referential index of that trace as well. Independently of providing an ingredient in the still-to-be-given solution to the problem of antecedent contained anaphora, Trace theory gives a model of how pronouns can get their bound variable interpretation.

Trace theory also constrains derivations in a way that removes the problems that arise by adopting the Lebeaux model, and its abandonment of making D-structure the only semantically interpretable element of a derivation. Recall that by removing the requirement that D-structures be the only element of the derivation that is semantically interpreted, the Lebeaux model allows D-structures to violate the Theta Criterion. This wrongly allows derivations like that sketched in (103).

(103)

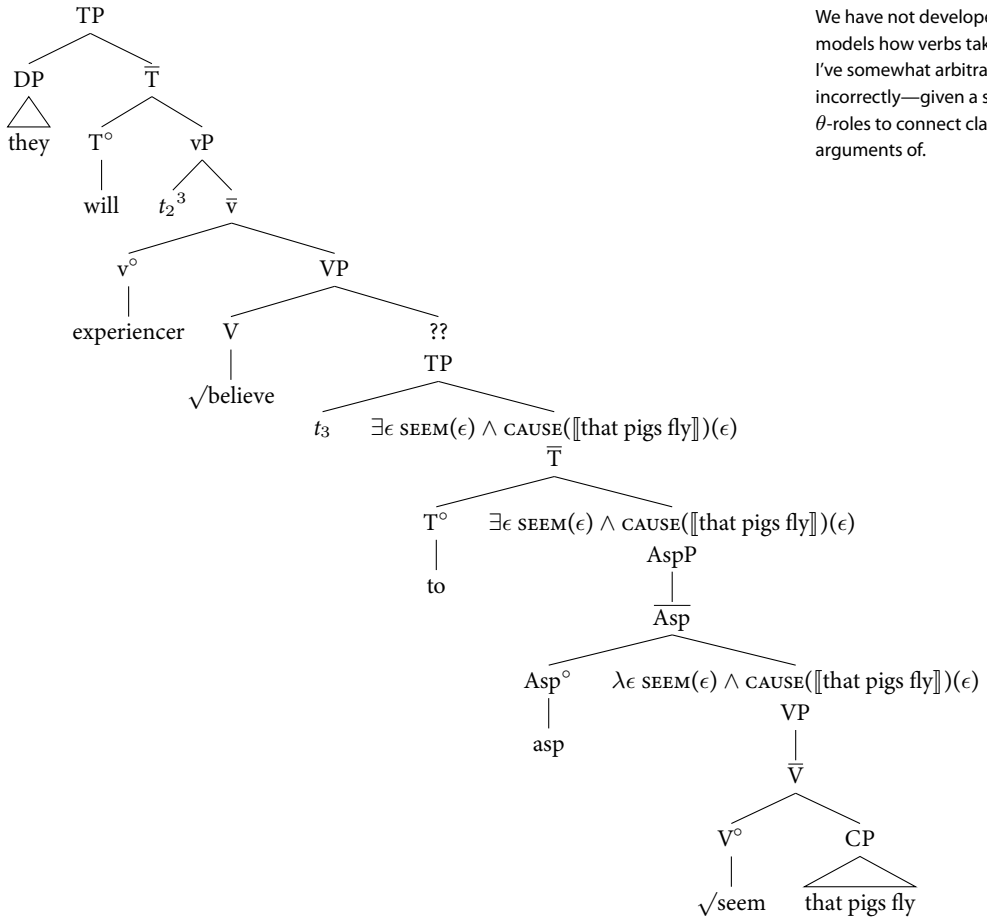


Without Trace theory, the derivation in (103) provides a parse—the one created by the first movement of *they*—that can be semantically interpreted. All the other conditions on derivations, e.g., Upwards, the requirements of bullets and AGREE, are satisfied as well. Unless this derivation is blocked, our system will characterize *They will believe to seem that pigs fly* as grammatical.

Trace theory closes this problem. Because it requires movement of phrases to leave a trace, and because traces can only be semantically interpreted as variables bound by the λ-operator that movement invokes, derivations like that in (103) will be blocked. The reason is that the trace left by the movement shown in the first step of (103) cannot get a semantic interpretation. Consider, for instance, what would happen if we let the S-structure, roughly

equivalent to the output of the second movement in (103), be subject to our semantic rules.

(104)



We have not developed a semantics that models how verbs take clausal complements. I've somewhat arbitrarily—and probably incorrectly—given a semantics here that uses θ-roles to connect clauses to the verbs they are arguments of.

The complement TP to √*believe* has no denotation, as there is no semantic rule that can compose the meaning of the trace in Specifier TP with the denotation of its sister: $\overline{\text{T}}$. This problem will arise in every parse of the derivation; therefore, no parse in this derivation will be semantically interpretable. The ungrammaticality of this derivation, then, derives from the fact that Trace theory blocks a semantic interpretation of the sentence the derivation models.

In general terms, Trace theory requires every position a moved XP occupies in a derivation to be a position in which it, or its trace, can be semantically interpreted. This requires that the XPs in a D-structure are either semantically interpretable in those positions, or could be replaced by traces that are semantically interpretable in those positions. That is sufficient to ensure that the Theta Criterion holds of D-structures. To my knowledge, it is sufficient to ensure that the constraints on D-structures that trace back to the semantics are preserved in the Lebeaux Model.

Trace theory has three useful ingredients, then. It allows more parses in a derivation to be relevant to the semantic interpretation of a sentence without losing the effects of restricting semantic interpretability to D-structures.

It provides a method for allowing pronouns to be understood as bound variables, by allowing movement to change the meaning of the phrase that contains a pronoun. And when this is wedded to the existence of traces, it provides a way of interpreting structures that are formed by movement.

This last effect brings us a step closer to understanding how to defease the problem posed by cases of antecedent contained anaphora. Our syntax now countenances a syntax for those examples in which the focus anaphor is not inside its focus antecedent, but is semantically interpretable. Those syntaxes arise by way of Complex NP Movement in the way that (88) illustrates. The second step in accounting for antecedent contained anaphora requires understanding the syntax of relative clauses. As it happens, relative clauses independently illustrate the need for a syntax that allows movement to create interpretable structures. We turn now to that task.

6.1.3.2 Relative Clauses and Wh Movement

Let's start with (105) as our model for building an analysis of relative clauses.

(105) No woman who you talked to will leave.

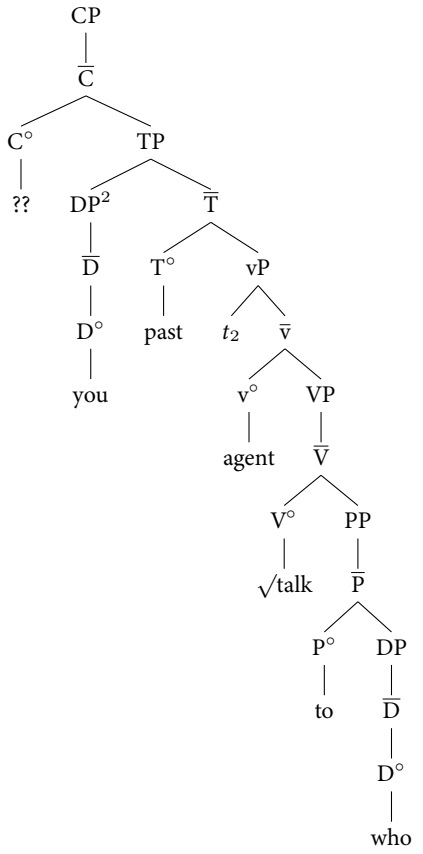
The relative clause is the portion of this sentence that corresponds to the string *who you talked to*. This expression has a form that is parallel to the "object" argument of the verb *know* in (106).

(106) She knows who you talked to.

At present, our syntax cannot form either of these types of clauses. Because their form is so close, it makes sense to develop a syntax that takes into consideration both relative clauses and complement clauses of the form in (106).

If we assume that the word *who* is a kind of pronoun, our phrase structure rules can create a clause like that in (107).

(107)



I will suppress the existence of AspP in parses where its existence doesn't matter, as in (107).

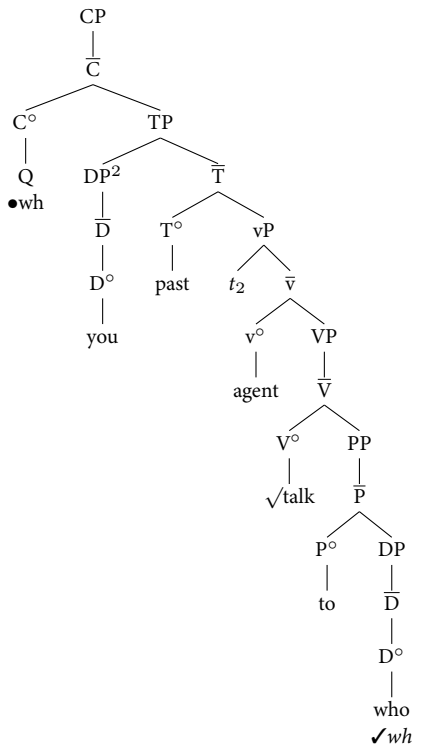
Let's assume that this is the D-structure representation for the expression *who you talked to* in both (105) and (106). To achieve the word order that we see in (105) and (106), we'll formulate a movement rule.

When this CP is the complement to verbs like *know*, it has a semantics that relates it to what a question is. The semantics for (106) should cause *who you talked to* to characterize a question that (106) says she knows the answer to. Similarly, in (108) the complement should characterize a question that she wonders about, asks, explains, etc.

(108) a. She wonders who you talked to.
b. She asked who you talked to.
c. She explained who you talked to.

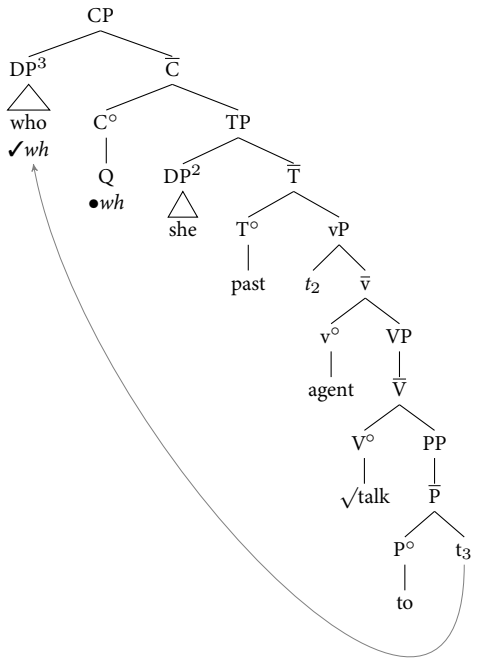
This isn't the meaning associated with relative clauses. It must be associated, then, with something in these clauses that is not entirely contained in the meaning of the words *who you talked to*. A standard hypothesis is that there is a morpheme in the C° position that is responsible for giving these CPs their question-like meaning. I'll represent this morpheme with "Q," and I'll assume that it comes with a feature that is an agreement relationship with the phrase that moves. If we call this feature "wh," then the non-relative clause form of (107) is (109).

(109)



If there is a • associated with the wh-feature, then our syntax requires a derivation that leads to the correct surface form. Agree will hold between C° and *who*, and the •'s requirements can be satisfied if this is paired with the movement shown in (110).

(110)



This is a way of using the methods developed here to deliver a syntax for these CPs.

The class of phrases that can host the wh-feature, and therefore satisfy the bulleted wh-feature in C° position, all contain a specialized lexical item. In addition to the pronoun *who*, there are other pro-forms, the determiner *which* and the degree word *how*. Some examples are in (111).

(111) She knows …

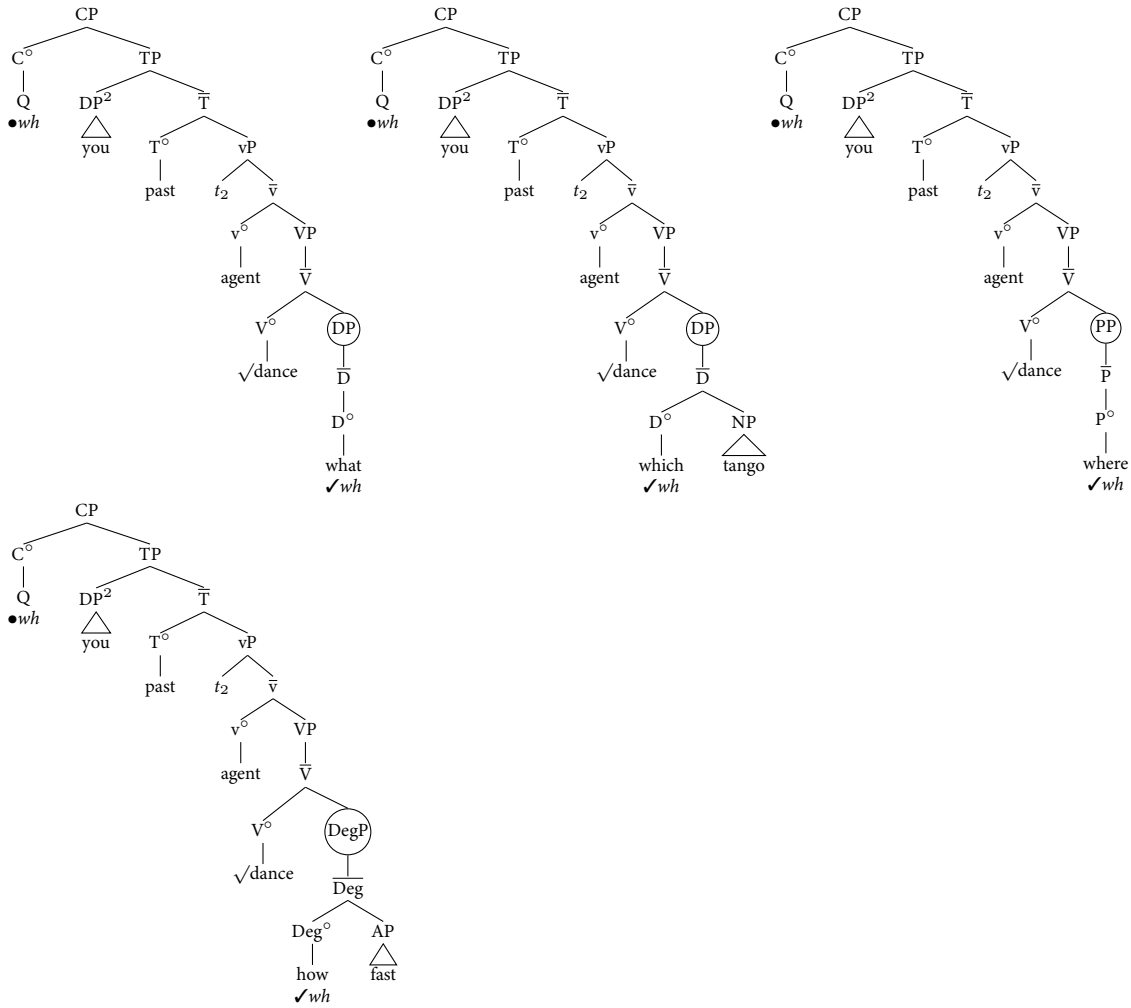
a. what/where/when/why you danced.

b. which tango you danced.

c. how fast you danced.

Because it is the existence of these words that correlates with the question-like and relative clause constructions, the wh-feature should reside in them. Thus, the D-structures for these constructions should have the form in (112).

(112)



The AGREE relation holds between C° and either D° or Deg°. The movement relation, however, never involves movement of the heads, but instead always affects a phrase, in (112) they are circled, that contains that head.

In the cases we've seen so far, the phrase that moves is always an $\overline{X}$ projection of the word that has the *wh*-feature. But this isn't always the case. There are examples—(113) illustrates some—in which the phrase that moves is not a projection of the word containing the *wh*-feature.

(113) She asked
- a. [$_{PP}$ to whom] you talked.
- b. [$_{DP}$ how long a race] you ran.
- c. [$_{DP}$ whose mother] you talked to.
- d. [$_{DP}$ whose mother's son] you talked to.

Ross (1967) dubbed this phenomenon "pied-piping." The phrases that pied-pipe are known as "*wh*-phrases" and the movement operation responsible for relocating *wh*-phrases is called *Wh* Movement.

Ross used the term "pied-piping" just for examples in which the phrase that is moved is not an $\overline{X}$-projection of the term that has the ***wh***-feature. I'll extend this term to all cases in which what moves is a phrase containing the term with the ***wh***-feature.

Relative clauses employ *Wh* Movement, just like the question-like CPs we've been examining, but there are some differences with respect to the shape of the *wh*-phrases involved. In general, with the exception of *what*, pro-forms can be found in relative clauses (see (114)).

(114)
- a. the guy who you talked to
- b. the place where you danced
- c. the time when you left
- d. the reason why you left
- e. * the thing what you bought

Pro-forms can pied-pipe in basically the same way in relative clauses as they do in question CPs.

(115)
- a. the guy to whom you talked
- b. the lady whose mother you visited
- c. the child whose mother's hat you stole

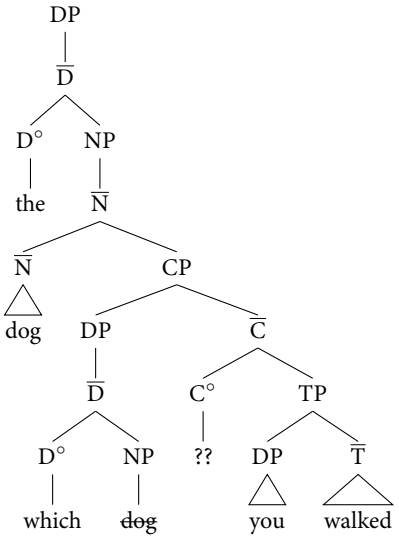
But degree words and the phrases they pied-pipe cannot be involved in relative clauses at all.

(116)
- a. * the distance how far you ran
- b. * the race how long a trail you ran

And when DPs headed by *which* are pied-piped, the NP portion of that DP must be silent and understood to be the same as the NP to which the relative clause is attached. So whereas (117) is ungrammatical, the form in (118) is its grammatical variant.

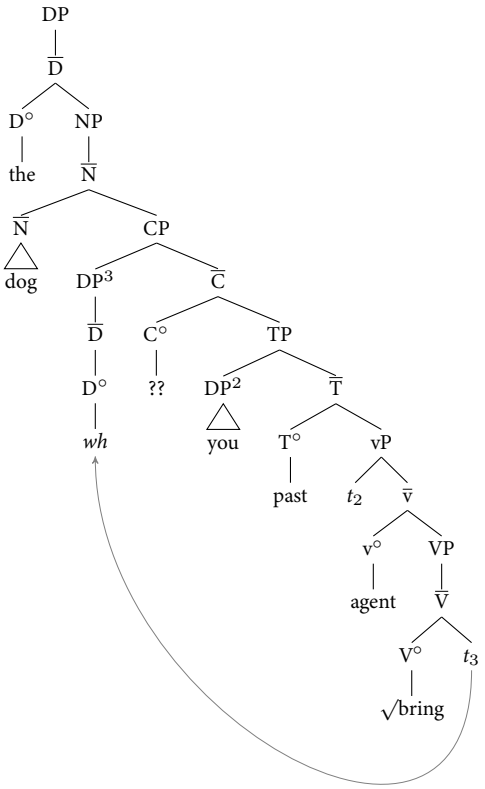
(117) * the dog which dog you walked

(118)



Interestingly, relative clauses have recourse to a wh-word that is not found in the question-like CPs that use *Wh* Movement. This wh-word has the behavior of a pro-form, but is phonologically silent. I'll represent it with "*wh*." An example is in (119).

(119) the dog you brought



This silent pro-form can be found in forming relative clauses that correspond to PPs as well.

(120) a. the time wh^3 you came t_3
b. the place wh^3 you left t_3
c. the reason wh^3 you left t_3

Strangely, in many Englishes it cannot be moved from the most local subject position, as in (121).

Sentences such as these are grammatical in Appalachian, African American, and Hiberno Englishes. See Sistrunk (2012) and Doherty (1993).

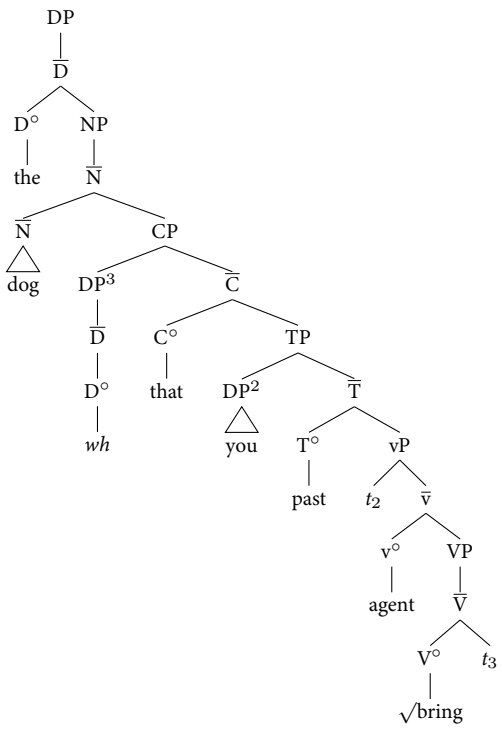
(121) * the dog is barking

compare:

(122) the dog which is barking

In contexts where the silent *wh* is involved in building a relative clause, the up-till-now mysterious identity of the clause's head is revealed. Alongside (119) is (123).

(123) the dog that you brought



The complementizer *that* has the option of being unpronounced, as can be seen from the alternation in (124).

(124) a. Katia knows that Petr loves his family.
b. Katia knows Petr loves his family.

And this option is enforced, it seems, when there is spoken material in the Specifier of the phrase that *that* projects:

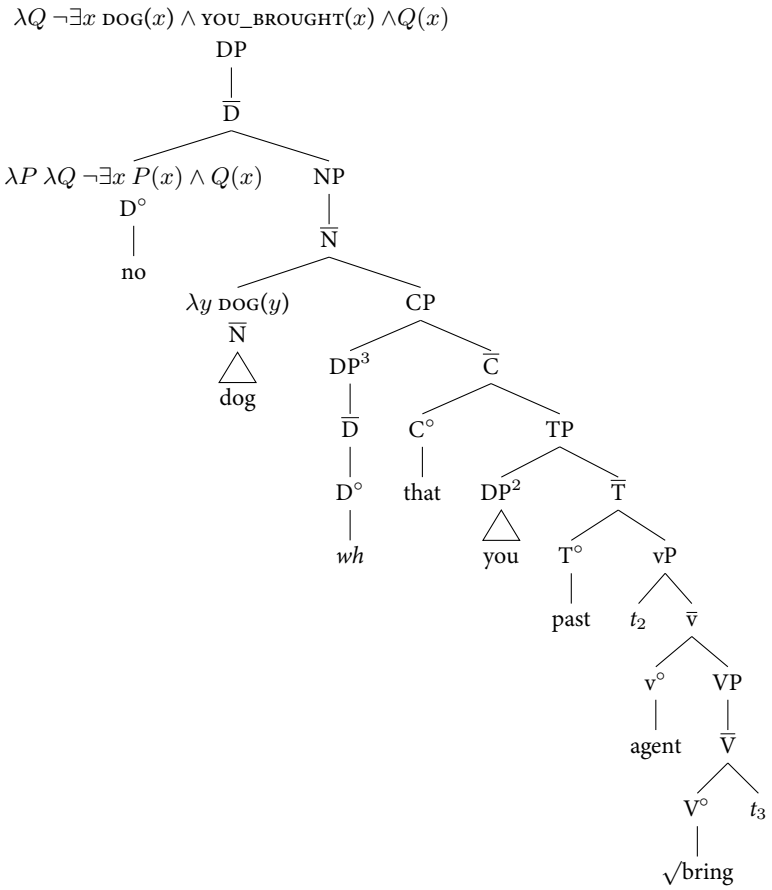
(125) a. * the dog which that you brought
b. * the woman who that left
c. * the time when that you left

Chomsky and Lasnik (1977) called this generalization the "Doubly Filled Comp filter."

(126) Doubly-filled Comp Filter
The Specifier and head of a CP cannot both contain phonological material.

Relative clauses can be seen as another case that illustrates the necessity of letting movement create meanings. To see this, consider the pieces of the DP in (127) whose denotations are identified.

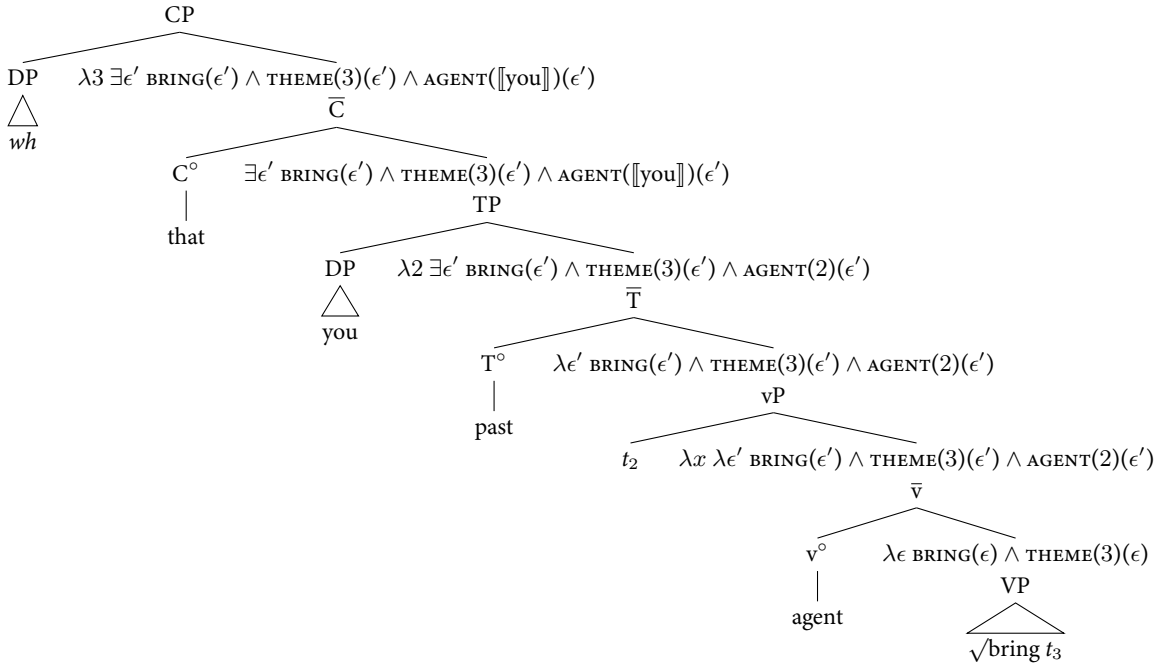
(127) no dog that you brought



We have taken the denotation of common nouns to be functions from things to truth values, and this informs the denotation I have assigned to the $\overline{\text{N}}$ that contains the common noun *dog*. The denotation for *no* is a relation between the functions that the NP it combines with is, and the denotation of the constituent that follows the DP *no* heads. As the aspirational denotation

assigned to the DP in (127) indicates, we want to be able to semantically combine the denotation of *dog* with the denotation of the relative clause so that it forms a meaning for the NP that they together comprise, so that the first argument for *no* is created. Let's take a closer look at how the relative clause's meaning is derived from the pieces that we presently have.

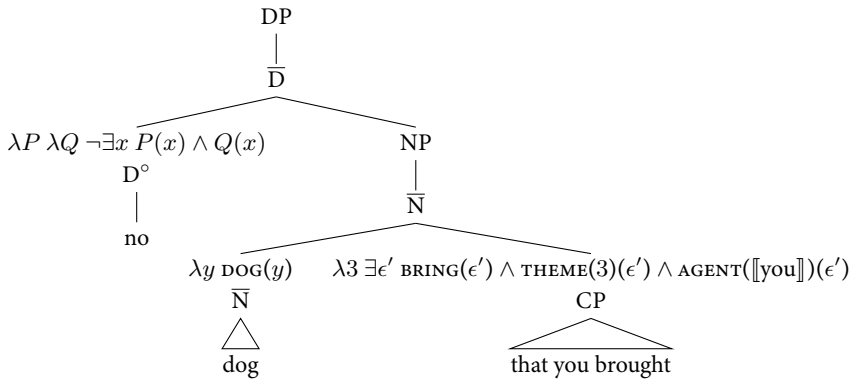
(128)



I've continued to ignore the contribution that tense makes, and I've made the assumption that the complementizer *that* has no denotation either.

Note that because of the meaning that movement creates, movement of the silent *wh* phrase causes $\overline{\text{C}}$ to be a function from things to a sentence meaning. It maps things to Truth just in case those things are brought by you. Suppose that this is also the denotation of the relative clause; that is, suppose that the silent *wh* phrase also has no denotation. This would give (127) the pieces in (129).

(129)

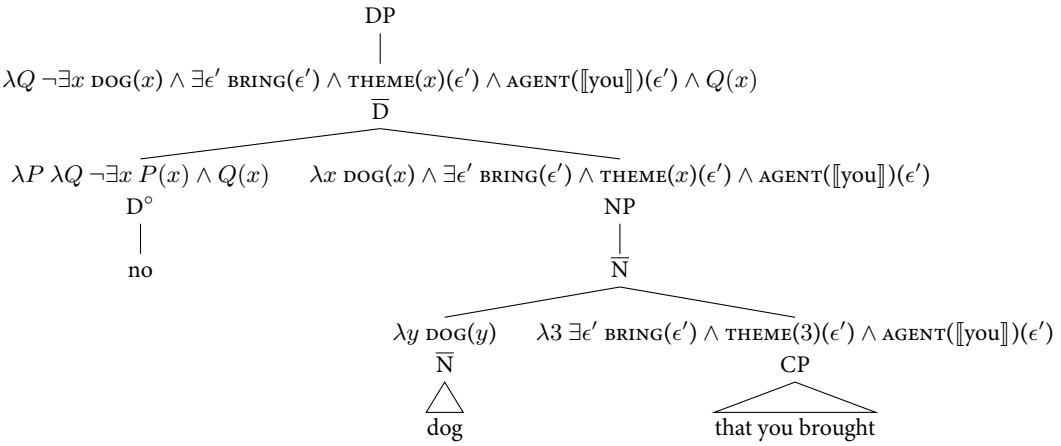


To bring the $\overline{\text{N}}$ and CP together in the necessary way, we use a rule of semantic composition rule called Predicate Modification.

(130) Predicate Modification
If $[\![\alpha]\!]$ is $\lambda z P(z) = T$ and $[\![\beta]\!]$ is $\lambda y Q(y) = T$, then the denotation of their immediate mother is $\lambda x P(x) \wedge Q(x) = T$

This produces (131).

(131) $\lambda Q\ \neg\exists x\ \textsc{dog}(x) \wedge \exists \epsilon'\ \textsc{bring}(\epsilon') \wedge \textsc{theme}(x)(\epsilon') \wedge \textsc{agent}([\![\text{you}]\!])(\epsilon') \wedge Q(x)$



This is the meaning we are after.

If these steps are correct, what we see is that movement has created a meaning that allows CPs to combine with $\overline{\text{N}}$s in the way required to derive the meaning of the constituent that they form. It's in this sense that relative clauses provide evidence that we want to let movement create new meanings. It is because movement is able to create a function from things to truth values that the CPs which are relative clauses have a denotation that allows them to combine with nominal projections. Trace theory makes movement creates responsible for converting CPs into functions of the right kind to compose by Predicate Modification. Trace theory, then, explains why movement is found in relative clauses.

At least, Trace Theory does this in simple relative clauses. It doesn't generalize to relative clauses in which a more complex *wh*-phrase has moved, however. In cases like (132), for instance, the moved *wh*-phrase makes a semantic contribution, , unlike what we see for the silent wh-phrase in (128).

(132) the dog whose owner^2 you met t_2

The meaning of the moved wh-phrase should make the relative clause describe properties that the dog has—namely, that it has an owner that you met—while also describing a property that the owner has—namely, being met. The Trace theory of Movement doesn't do that in (132). The conclusion that movement creates new meanings remains. The fact that *whose owner* has moved in (132) that is responsible for allowing the CP to combine semantically with *dog*. And it delivers the fact that the owner is understood to be met. But it doesn't capture how $[\![\text{dog}]\!]$ is modified by the relative clause. Our present syntax and semantics doesn't yet deliver the right outcome for (132).

A solution to the problem posed by (132) can be found in the Copy theory of movement, section 6.2.2.2.

Let's set aside these cases for the moment, and return to the job at hand: solving the problem of antecedent contained anaphora.

6.1.3.3 Antecedent Contained Anaphora Resolved

The puzzle we're in pursuit of solving is how the focus anaphor in (61), repeated here, can be resolved.

(61) a. Thuy will visit every student who [RODICA <will visit>].
b. Thuy will visit every student who [RODICA <will ~~visit~~>].

The bracketed portion of these sentences is a focus anaphor, whose antecedent is found in the sentence that seemingly contains the focus antecedent. Focus anaphors, recall, are subject to the requirement in (19).

(19) Let XP be a focus anaphor, and α be the focus-marked term within XP. A focus value of XP ($[\![XP]\!]^F$) is identical to $[\![XP]\!]$, except that α is replaced by a focus alternative of α (=FA(α)). β is a FA(α) if it is the same syntactic type as α.

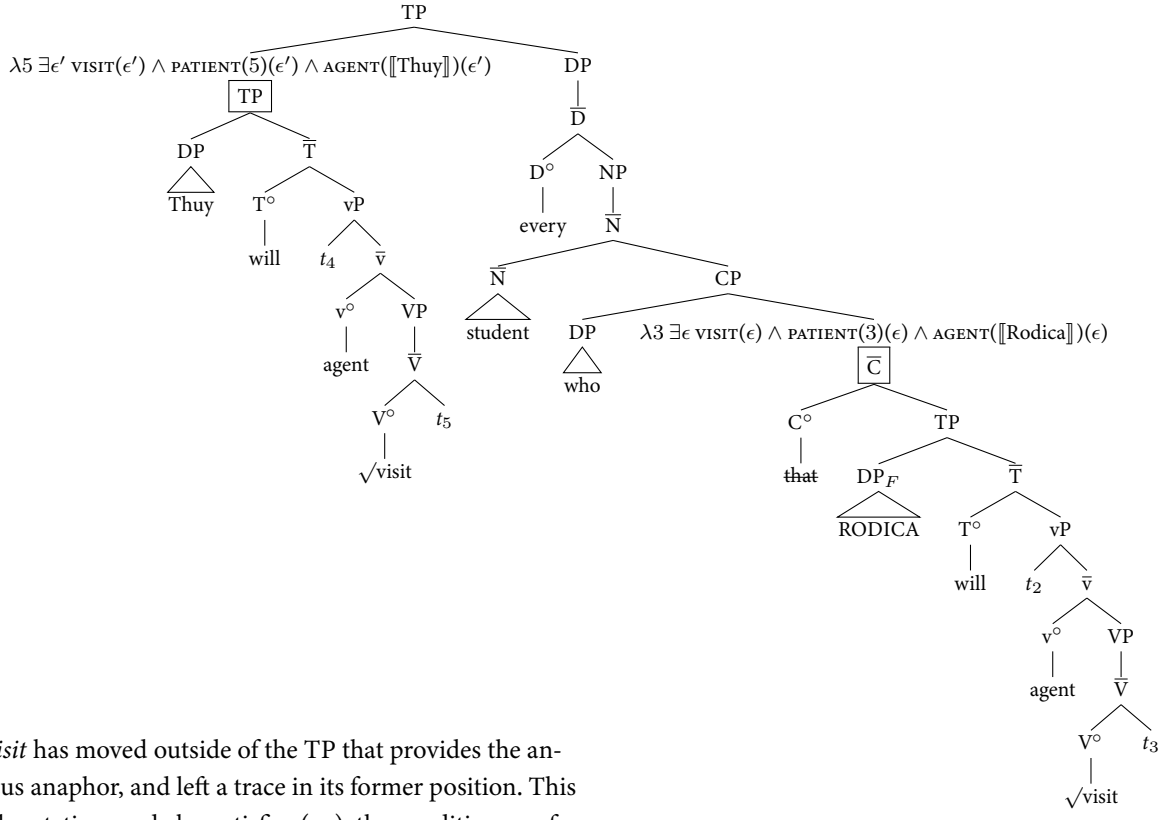
If XP is a focus anaphor, then there must be a YP and a $[\![XP]\!]^F$ such that $[\![YP]\!]=[\![XP]\!]^F$.

The puzzle, then, is giving (61) a syntax that allows (19) to be satisfied.

The Trace theory of movement does that. With Trace theory, the S-structure representation for (61) is (133).

(133)

$\forall x$ [student(x) $\wedge$ $\exists\epsilon$ [visit(ϵ) $\wedge$ patient(x)(ϵ) $\wedge$ agent([[Rodica]])(ϵ)]] $\rightarrow$ [$\exists\epsilon'$ visit(ϵ') $\wedge$ patient(x)(ϵ') $\wedge$ agent([[Thuy]])(ϵ')]



The object DP of *visit* has moved outside of the TP that provides the antecedent for the focus anaphor, and left a trace in its former position. This provides the right denotation, and also satisfies (19), the conditions on focus anaphors. The boxed $\overline{\text{C}}$ is the focus anaphor, and the boxed TP is its focus antecedent. There is a focus-alternative to the denotation of $\overline{\text{C}}$ that is identical to the denotation that TP has. In particular, replacing the focus-marked *Rodica* with *Thuy* produces something from the denotation of $\boxed{\overline{\text{C}}}$ that is equivalent to $\boxed{\text{TP}}$. Note that this is partly due to the fact that the traces t_5 and t_3 are sloppily bound. That is, they have the same semantics that we've seen sloppily-bound pronouns have in section 6.1.2.

This solution to the puzzle of antecedent contained focus anaphora predicts that it will arise only in cases where movement can create a trace bound by the term that contains the focus anaphor. This in turn entails that the focus anaphor in such cases must contain a variable in a position that corresponds to that trace, and that this variable and the trace meet the conditions of sloppy binding. I don't know that this prediction has been thoroughly examined, but it appears to be correct. There is a contrast, for example, between (61) and (134).

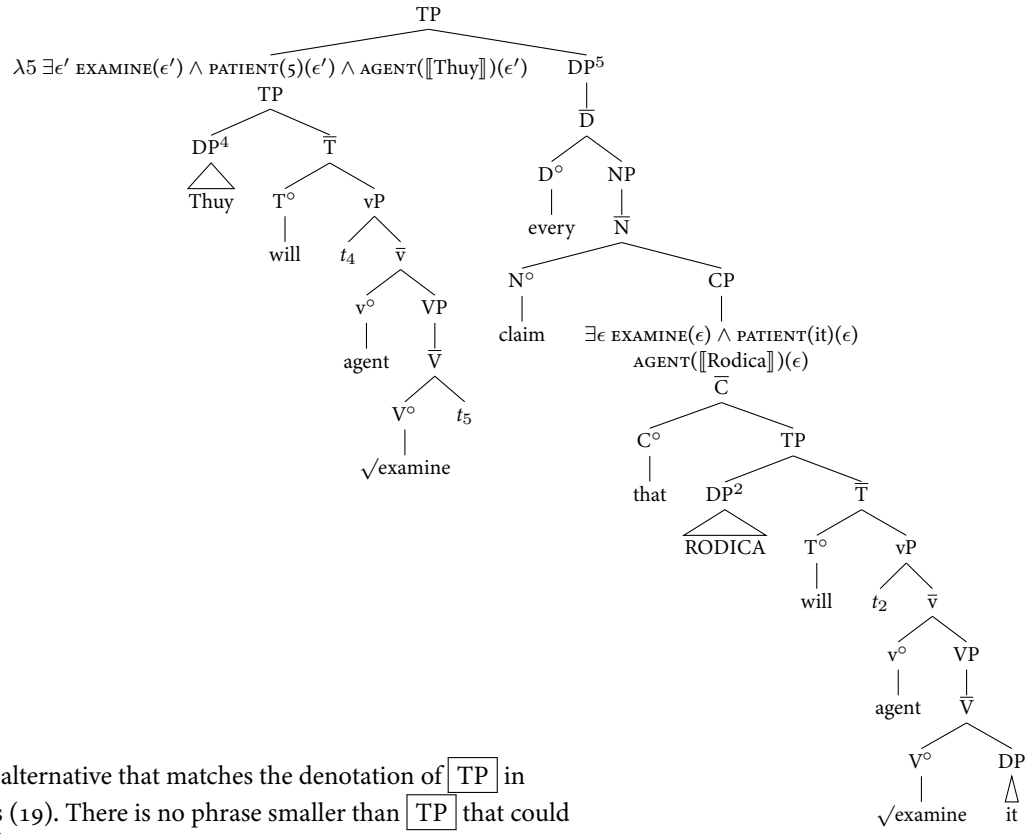
(134) * Thuy will examine every claim that [RODICA <will examine it>].

This is because in (134), the focus anaphor has a pronoun in the position that corresponds to where the trace of moving the *every claim*-DP resides in the antecedent. This pronoun and the trace do not obey the conditions on sloppy

binding, and as a consequence the condition in (19) is not satisfied. To see this, compare (135), the representation that corresponds to (134), with (133), the representation that corresponds to the successful instance of antecedent contained anaphora.

(135)

$\forall x$ [STUDENT(x) $\wedge$ $\exists\epsilon$ [EXAMINE(ϵ) $\wedge$ PATIENT(x)(ϵ) $\wedge$ AGENT([[Rodica]])(ϵ)]] $\rightarrow$ [$\exists\epsilon'$ EXAMINE(ϵ') $\wedge$ PATIENT(x)(ϵ') $\wedge$ AGENT([[Thuy]])(ϵ')]



$\boxed{\overline{C}}$ doesn't have a focus alternative that matches the denotation of $\boxed{TP}$ in (135), and therefore fails (19). There is no phrase smaller than $\boxed{TP}$ that could be an antecedent for $\boxed{\overline{C}}$, because $\boxed{TP}$ is the smallest phrase that contains a binder for the trace left by moving the object DP. Our semantics doesn't allow phrases with unbound traces in them to have a denotation. No phrase larger than $\boxed{TP}$ can be an antecedent because it will contain the focus anaphor. Consequently, the only focus anaphors that $\boxed{TP}$ can be the antecedent for are those that are also functions from things to truth values. In general, a focus anaphor whose D-structure position puts it within it antecedent will have a contain a trace that corresponds to the trace left when that focus anaphor moves out of its antecedent.

This solution also makes predictions about the position that the DP containing the focus anaphor will be semantically interpreted in. That DP will have to be interpreted in a position that puts it outside the antecedent for the focus anaphor. When the DP is headed by a quantifier, this can be tested by considering where that quantifier is interpreted. This prediction is examined in Williams (1977) and Sag (1976), and has been confirmed. Let's look at one of the cases that Williams discusses.

First, consider (136).

(136) [$_a$ Cherlon wants [$_b$ to beat every contestant that

a. i. MARIBEL <wants to beat>.

ii. MARIBEL <does ~~want to beat~~>.

b. i. MARIBEL <beat>.

ii. MARIBEL <did ~~beat~~>.

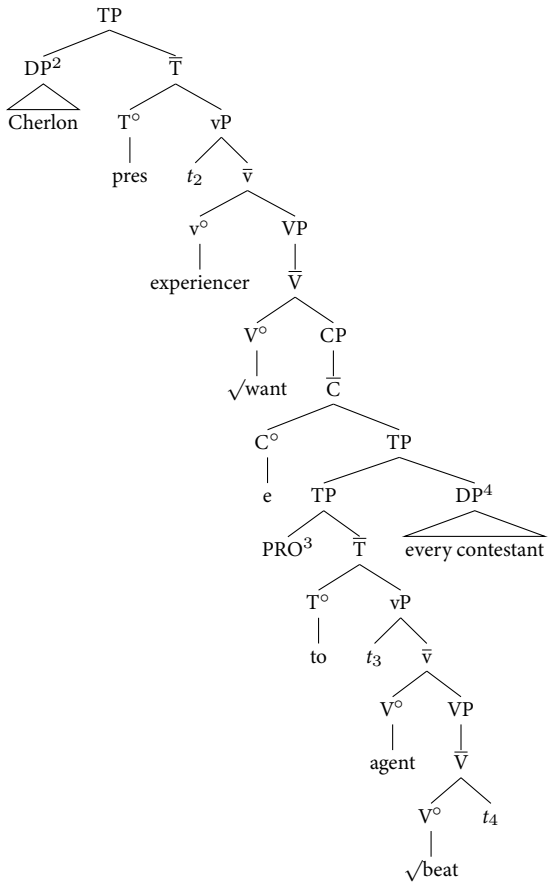
What (136) shows is that the sentence *Cherlon wants to beat every contestant* makes available two antecedents for a focus anaphor. One of those antecedents is marked with an opening bracket labeled a, and the other is marked with a bracket labeled b. The focus anaphors in (136a) make use of the a-antecedent, and the focus anaphors in (136b) make use of the b-antecedent.

Next, consider the simpler (137).

(137) Cherlon wants to beat every contestant.

This sentence has two interpretations that depend on where *every contestant* is found in the syntactic representation. If the syntax puts *every contestant* in the position indicated in (138), it will be interpreted within the infinitival complement clause.

(138)



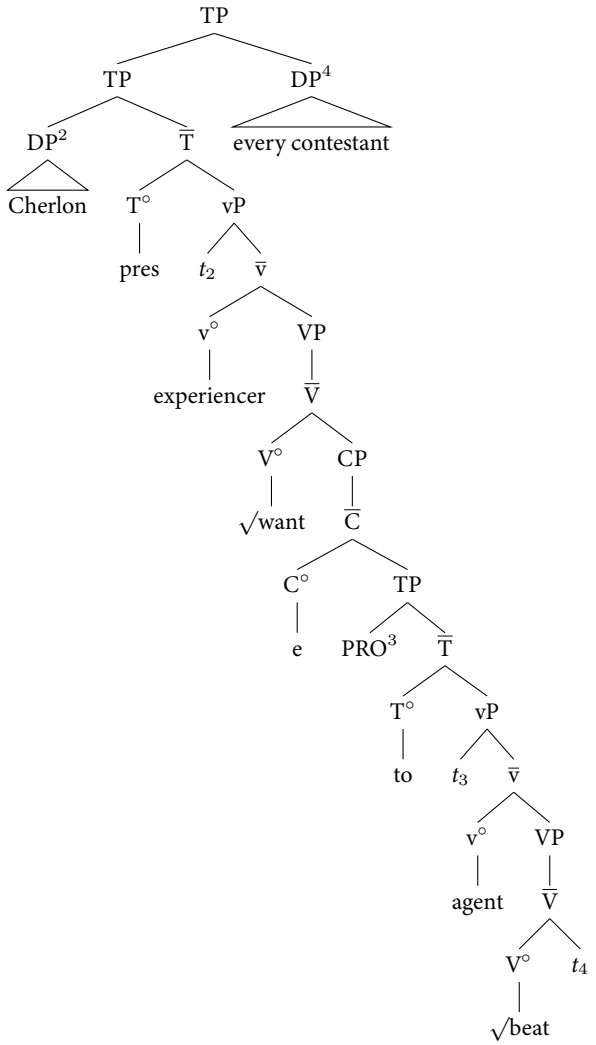
The meaning that (138) has corresponds to (139).

(139) It's to beat every contestant that Cherlon wants.

Note that the desire that (139) ascribes to Cherlon is consistent with her not wanting to beat each particular contestant. If her beloved sister is among the contestants, she may not want to beat that sister, for instance. (**??**) merely says that what Cherlon desires is to be the overall winner. It could describe a scenario in which Cherlon has conflicting desires. Let's call this the "narrow scope" interpretation of *every contestant*.

If, by contrast, the syntax puts *every contestant* in the position indicated in (140), then it will be interpreted outside the complement clause.

(140)



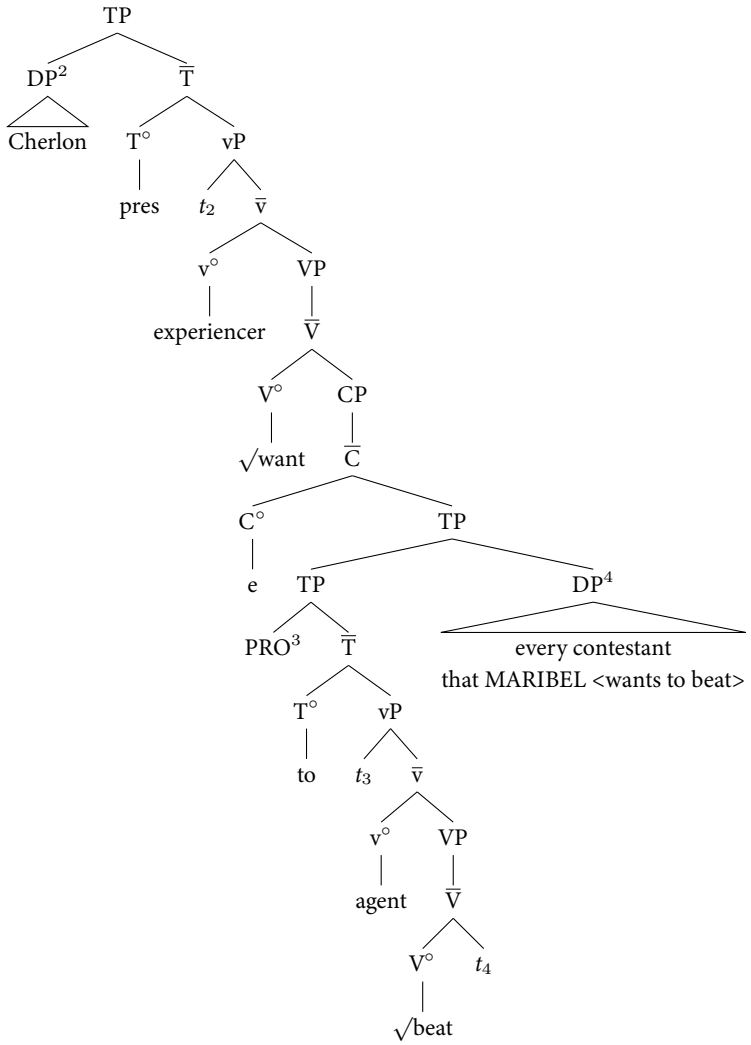
The meaning delivered by (140) can be paraphrased by (141).

(141) Every contestant is one that Cherlon wants to beat.

On this meaning, Cherlon has a desire regarding each contestant, and that desire is to beat them. If the contestants include Cherlon's sister, then the

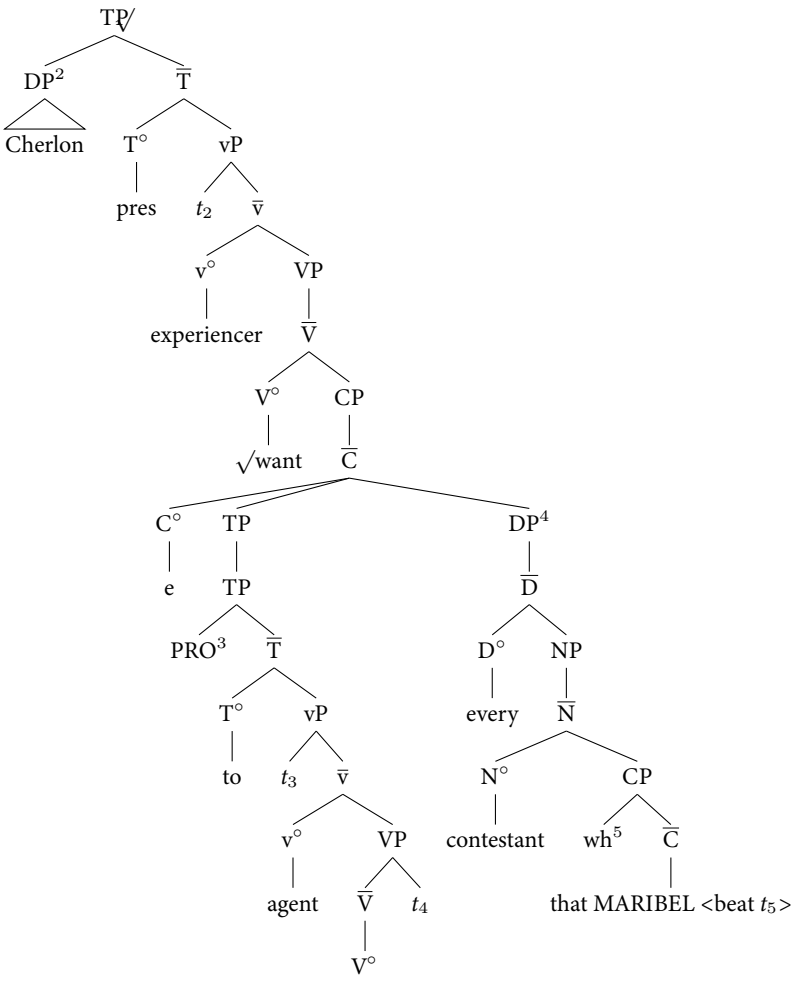
meaning derived from (140) entails that Cherlon has the desire to beat her as well. Let's call this the "wide scope" interpretation of *every contestant*. The account of antecedent contained anaphora provided by Trace theory predicts that (136) cannot simultaneously have the narrow scope interpretation of *every contestant* and resolve the focus anaphor so that it takes the larger a-antecedent. This interpretation would be mapped onto the syntax in (142).

(142)



The problem with (142) is that the focus anaphor lies within its antecedent. When *every contestant* takes narrow scope, the focus anaphor can only correspond to the lower clause. (143) shows how. The denotations aren't provided in (143), but the relationship between $\boxed{\overline{\text{C}}}$ and $\boxed{\text{TP}}$ is parallel to the $\boxed{\overline{\text{C}}}$ and $\boxed{\text{TP}}$ in (135). Because a focus alternative to *Maribel* is PRO, $[\![\text{TP}]\!]$ is the same as $[\![\overline{\text{C}}]\!]^F$. This makes $[\![\text{TP}]\!]$ in (143) an antecedent for $[\![\overline{\text{C}}]\!]$.

(143)



These predicted outcomes seem to be correct. In a context that limits the interpretation to a narrow scope one for *every contestant*, like that provided below, only the smaller b-antecedent is available for a focus anaphor.

> *Maribel and Cherlon are fierce rivals. Neither of them expects to win the tournament, but they each expect that they can outperform the other. So fierce is their rivalry, that Cherlon would even try to beat her sister, who is also a contestant in the tournament, if it would mean besting Maribel. This is surprising, because if Maribel weren't in the tournament, Cherlon would let her sister beat her. Cherlon hates beating her sister. I tell Cherlon's sister:*

(144) [$_a$ Cherlon wants [$_b$ to beat every contestant that

a. # MARIBEL <wants to beat>

   # MARIBEL <wants to>

b. MARIBEL <beats>

   MARIBEL does

"#" marks a sentence whose meaning is anomolous.

It's only in a context which allows the wide scope interpretation for *every contestant* that the a-antecedent for the focus anaphor becomes available. The context below illustrates.

> *Maribel and Cherlon are sisters. When they play a tournament together, they discuss who they hope to beat. It turns out that:*

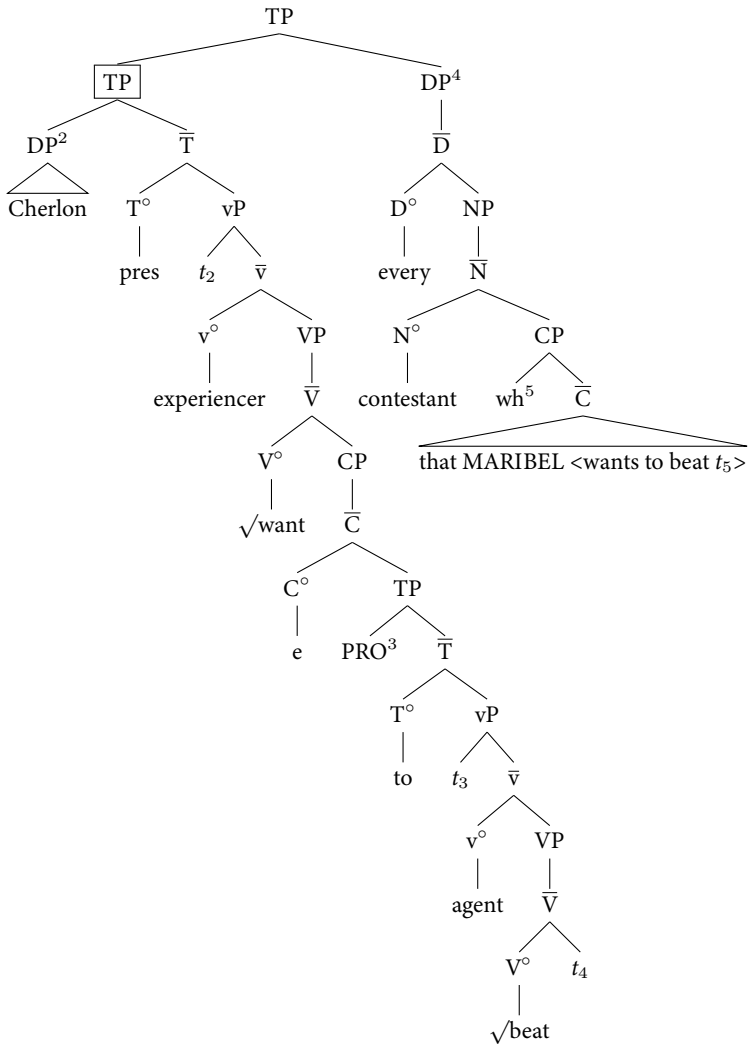
(145) [$_a$ Cherlon wants [$_b$ to beat every contestant that

a. MARIBEL <wants to beat>.

b. MARIBEL <wants to>.

The syntax for a wide-scope interpretation of *every contestant* is (146).

(146)



The a-antecedent is available here because $[\![\boxed{\text{TP}}]\!]$ is equivalent to a $[\![\overline{\text{C}}]\!]^F$, because *Cherlon* is a focus alternative to *Maribel*.

The solution to antecedent contained anaphora that Trace theory provides gives a good account of the conditions necessary for antecedent contained

anaphora to show up. Trace theory's requirements narrowly limit antecedent contained anaphora to environments where movement can put traces in positions where they qualify to be sloppily bound. Because Trace theory also gives a straightforward account of the semantics of relative clauses and the semantics of DPs with quantificational heads, it's widely regarded by syntacticians as a successful theory. It has known problems as well, as I'll show you in 6.2, but it has nonetheless become the standard model of movement.

Adopting Trace theory, and the connected idea that parses which movement creates can be ones that provide the denotation for a sentence, complicates the relation between form and meaning that derivations are intended to model. In the simpler models that we've used up till now, we can impose all the requirements that the semantics imposes on a single parse in the derivation: the D-structure. In the new model, a derivation created by movement can meet those requirements and therefore become the parse relevant for a semantic interpretation. This will have consequences for the ordering of the operations which give rise to those parses. Because some sentences will require a non-D-structure parse to provide the denotation, it will also require that derivations be steered so that those parses are created. This means that some derivations will require a specific order of operations to bring about the right parse. We look at this effect in the next section.

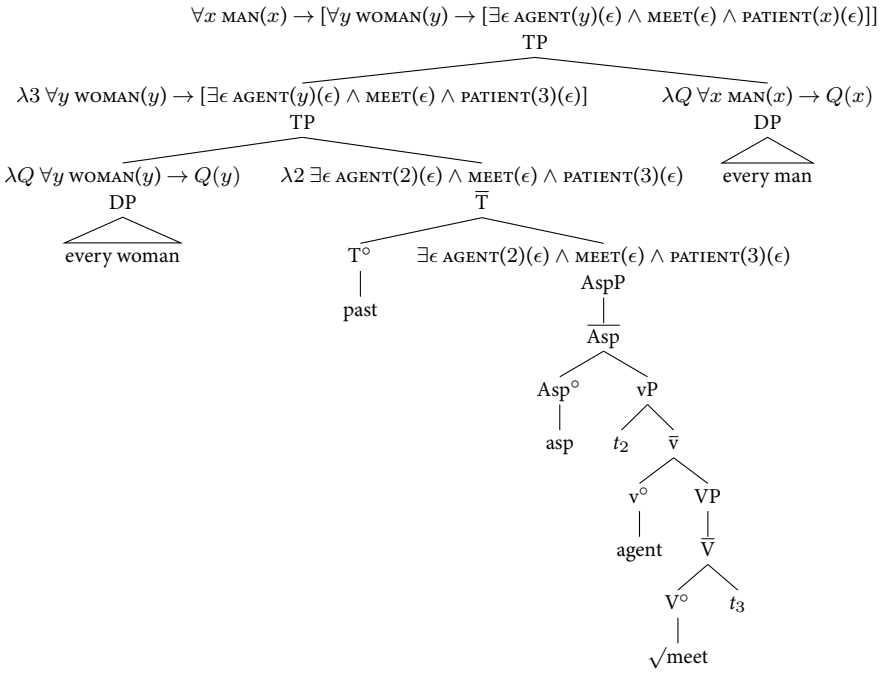
6.1.4 The Syntax of Quantificational DPs

One of the consequences of building a denotation for quantificational determiners that allows them to be interpreted in positions outside of vP, is that they cannot be interpreted in positions inside vP. The denotation we've given for vP is as functions whose domains are eventualities. There is some phrase containing that vP, however, where the denotations are no longer functions of this kind. Instead, they are closed formulas—no longer functions. Because quantifiers cannot be interpreted unless the DP they head is a sister to a closed formula, this means that all DPs headed by quantifiers must be semantically interpreted outside vP. This section investigates consequences of this decision.

To put the details back into view, we adopted the denotation for quantificational determiners like that in (147), which allows the quantificational DPs to be interpreted in their moved positions, as (148) illustrates.

(147) $[\![\text{every}]\!] = \lambda P\, \lambda Q\, \forall x\, P(x) \rightarrow Q(x)$

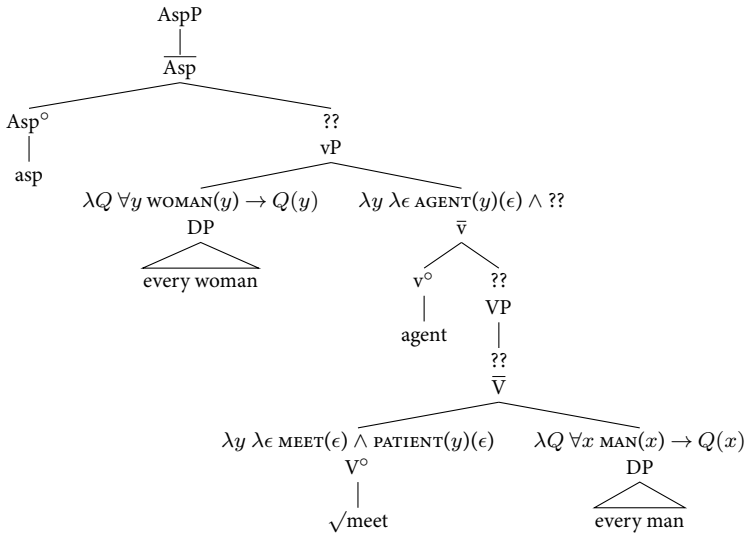
(148)



This correctly captures the semantic fact that the event variable is existentially bound within the scope of *no* and *every*. The denotation for (147), for instance, allows there to be a different meeting event for each of the men and women involved. It's also this syntax, and semantics, that provides the account of Antecedent Contained anaphora sketched above.

But this denotation for quantificational determiners does not allow the DPs they project to be semantically interpreted in their D-structure positions, as (149) indicates.

(149)

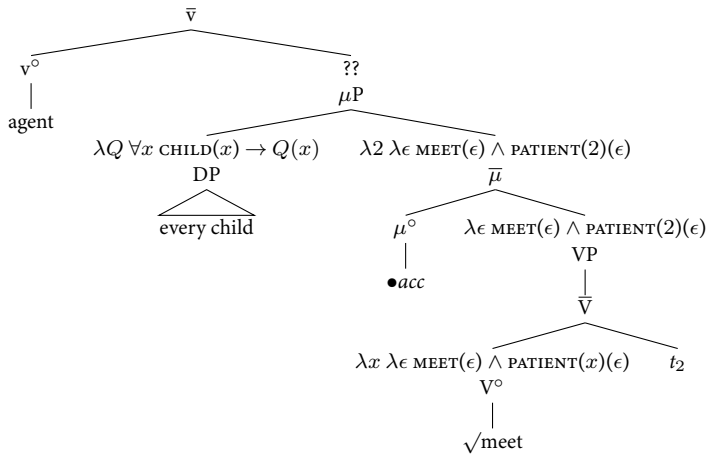


As noted, this means that quantificational DPs, unlike DPs that simply refer, must move out of vP to be semantically interpreted.

Our syntax presently independently requires that DPs which bear nominative Case move, and therefore our derivations already include a parse that will allow nominative Case-marked quantificational DPs to be semantically interpreted. The movement of *every woman* in the derivations sketched above exemplifies the general case.

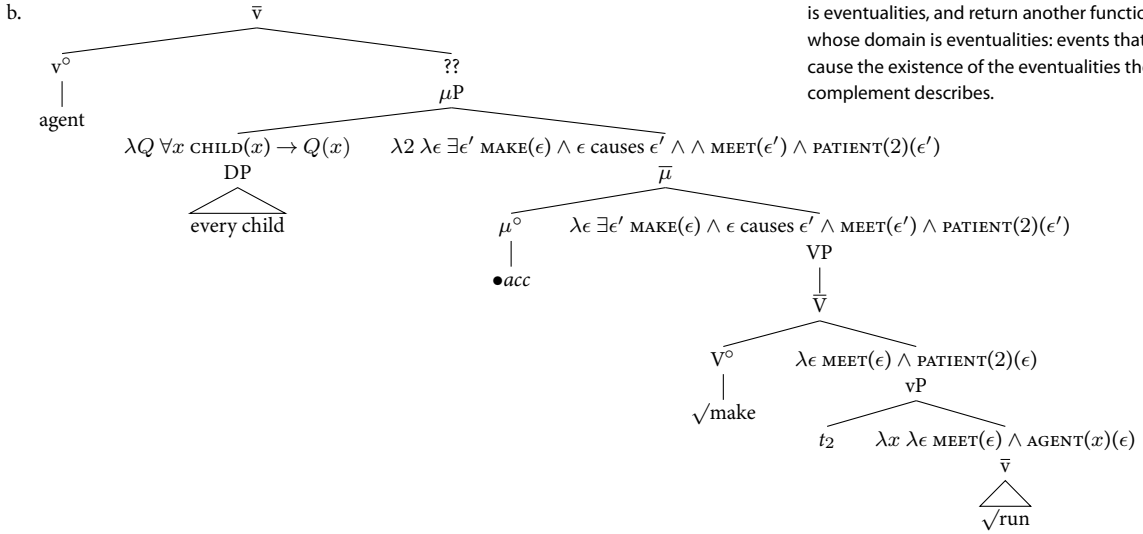
We also have a syntax that allows accusative Case-marked DPs to move, but this doesn't always provide for a derivation in which accusative Case-marked quantificational DPs will be able to be semantically interpreted. If we adopt the Split VP Hypothesis described in chapter 6.2.1.2, for instance, then Accusative Case-marked DPs move in the way indicated in (150). But this doesn't move these DPs far enough for them to be semantically interpreted if they are headed by quantificational determiners, as the denotations in (150) indicate.

(150) a.



Assume that the contents of μ° have no denotation, and therefore that $\overline{\mu}$ inherits its denotation unchanged from VP. The semantics of the new movement rule change this denotation in (150)

b.



I adopt here a denotation for ***make*** that allows it to combine with a function whose domain is eventualities, and return another function whose domain is eventualities: events that cause the existence of the eventualities their complement describes.

The only movement rule we have for accusative Case-marked DPs that will put them in a position where they can be semantically interpreted is Complex NP Shift.

Our semantics, then, predicts that DPs which are headed by quantifiers will necessarily undergo Complex NP Shift when they are accusative Case-marked. And yet, the positions that quantificational DPs can be pronounced in does not fit this prediction. Accusative Case-marked quantificational DPs need not be pronounced in the positions that Complex NP Shift would put them, as (151) shows.

(151) a. She will read every story to you.
b. She will make every child laugh.
c. She will look every number up.

Something in our system is wrong.

But what? There is evidence that the syntax which delivers the word order in (151) should allow the quantificational DPs to be interpreted outside of the vP. That evidence derives from the relative interpretations of the quantifiers in (152).

(152) a. A hired actor will read every story to you.
b. A funny actor will make every child laugh.
c. A different student will look every number up.

These sentences have an interpretation which allows for the actors to vary with the stories or children involved. These meanings are roughly those paraphrased in (153).

(153) a. Every story is such that a hired actor will read it to you.
b. Every child is such that a funny actor will make them laugh.
c. Every number is such that a different student will look it up.

To get that interpretation requires that the DPs headed by *every* in (152) occupy a syntactic position higher than the subject DPs. That is not where our present syntax puts these DPs. Facts like these, then, indicate that at least one flaw our system has is in how it makes the position that quantifiers are semantically interpreted in correspond to the position they are spoken in.

We need a semantics which allows a quantificational DP to be semantically interpreted so that it has access to information "higher" in the phrase marker relative to it. Our present semantics lets the quantifier only "see" the material that is a sister to the DP it heads. Our syntax for (152), for instance, puts *every* inside a DP that is in Specifier of μP. In this position, the phrase that *every* has as its second argument is the sister to this DP; that is: $\overline{\mu}$. What (152) shows is that *every* should instead have a constituent that includes the *actor* DP as its second argument. There are ways of doing this that de-couple the syntax and the semantics, allowing a quantificational DP to combine semantically with material that it doesn't syntactically combine with.To pursue this direction, we would need to revisit the relationship we have established between quantificational DPs and the variables they bind, among other things.

See Cooper (1983), for an example.

But there is some evidence that the solution to the problem posed by (153) should employ the very syntax/semantics we have developed. Perhaps the most direct evidence for preserving the syntax we now have is that antecedent contained anaphora can be resolved for DPs which are not in the positions that Complex NP Shift would give to them. (154) illustrates.

(154) a. She will read every story that I didn't to you.
b. She will make every child that I didn't laugh.

The grammaticality of these sentences rests on their prosody. They require a pronunciation that puts a rising tone on *didn't* followed by a pause and falling tone on the material that follows *didn't*. We'll return to that prosody, but setting it to the side for now, the grammaticality of (154) encourages the view that we need a syntax that places the accusative Case-marked DPs outside of the vP they seem to reside in. As we've seen, this is the only way to produce an antecedent for the surface anaphor in (154). I don't know how to avoid the conclusion that the sentences in (154) should have a syntax like that indicated in (155).

The non-elliptical versions of these cases are less pretty, but I think similarly grammatical.

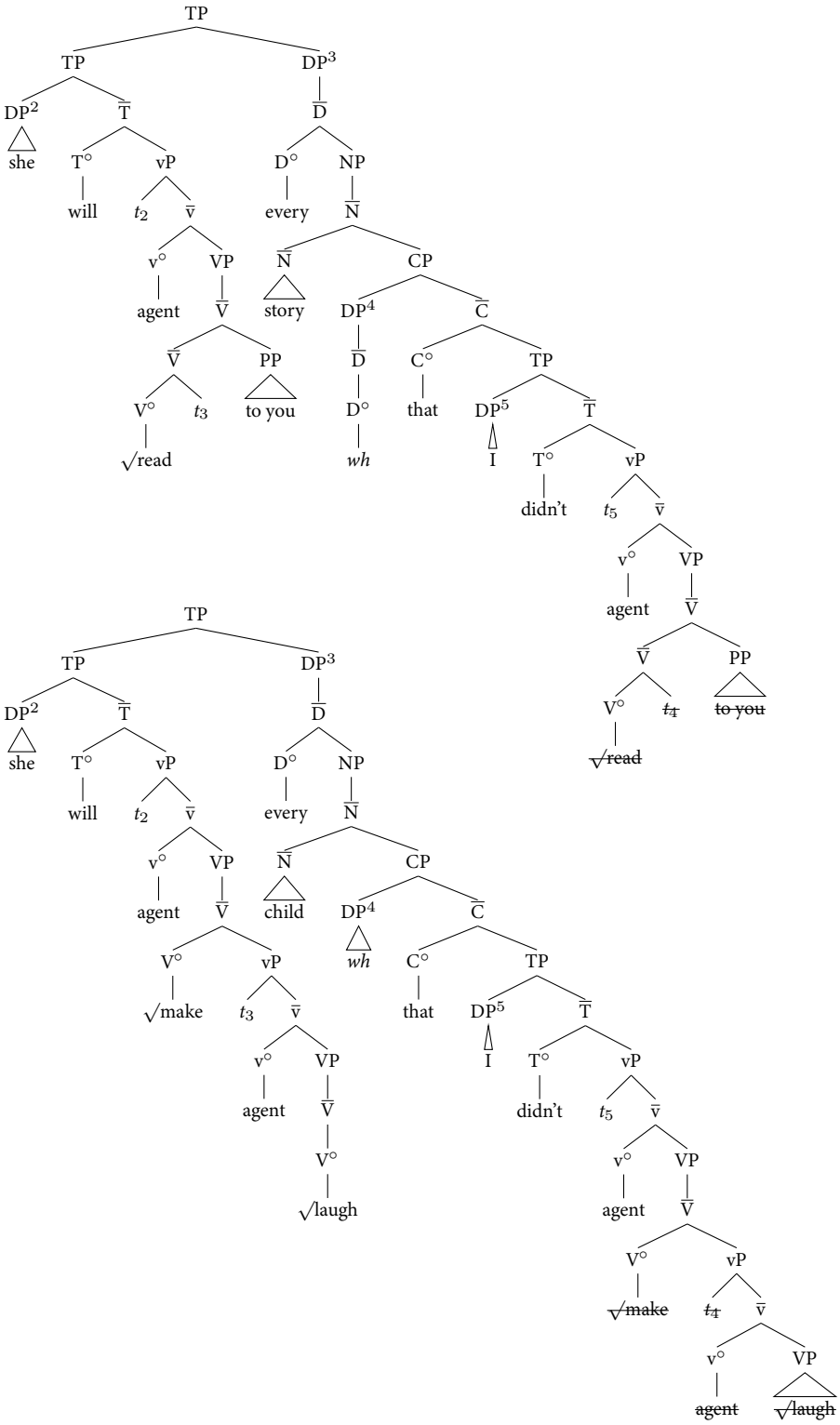
i. She will read every story that I didn't <read to you> to you.
ii. She will make every child that I didn't <make laugh> laugh.

My suspicion is that what ails these cases is the availability of the equivalent, but less repetitive, (154). I am encouraged to have this suspicion by the improved status that examples like (i)-(ii) have when there isn't a elliptical competitor. For example, the following cases sound better to me than (i)-(ii).

iii. She will read every story that I didn't <read to HIS child> to YOUR child.
iv. She will make every child that I <made CRY> laugh.

The focus anaphors in (iii) and (iv) contain focus-marked material which prevents them from having an elliptical version.

(155)


TP
TP
DP3
DP2
T̄
D̄
she
T°
vP
D°
NP
will
t2
v̄
every
N̄
v°
VP
N̄
CP
agent
V̄
story
DP4
C̄
V̄
PP
D̄
C°
TP
V°
t3
to you
D°
that
DP5
T̄
√read
wh
I
T°
vP
didn't
t5
v̄
v°
VP
agent
V̄
V̄
PP
V°
t4
to you
√read
TP
TP
DP3
DP2
T̄
D̄
she
T°
vP
D°
NP
will
t2
v̄
every
N̄
v°
VP
N̄
CP
agent
V̄
child
DP4
C̄
V°
vP
wh
C°
TP
√make
t3
v̄
that
DP5
T̄
v°
VP
I
T°
vP
agent
V̄
didn't
t5
v̄
V°
v°
VP
√laugh
agent
V̄
V°
vP
√make
t4
v̄
v°
VP
agent
√laugh

These representations correspond to the strings in (156).

(156) a. She will read to you every story that I didn't.
b. She will make laugh every child that I didn't.

These are grammatical; indeed, they don't require the very particular prosody that (154) does, and they have a more natural cast to them. The puzzle is to understand how these representations can be expressed with the strings in (154).

Robert May[4] developed a solution to these problems that rethinks the relationship between a derivation and the procedure that interprets a syntactic representation as a string of words. We've just developed a view of derivations which allows for representations created by movement to be semantically interpreted. So, our present syntax characterizes sentences with (64), repeated here.

[4] ; and May, Robert. 1985. ***Logical form: Its structure and derivation***. Cambridge, Massachusetts: MIT Press

(64) $\mathcal{D} = (P_1, P_2 \ldots, P_n)$ is a derivation iff:
a. P_1 is formed by the $\overline{X}$ Skeleton and Lexical Insertion,
b. Each $P_{i>1}$ is formed from P_{i-1} by a rule,
c. Some P_i is semantically interpreted, and
d. P_n is subject to the linearization algorithm.

A sentence is the set of well-formed derivations which have the same P_1 (=D-structure). A grammatical sentence is a derivation whose cardinality isn't greater than the cardinality of any other derivation for that sentence.

The cardinality of a derivation is the number of Ps in it.

May's proposal is to let the linearization algorithm apply to a non-final element of a derivation. A way of fitting May's proposal to the particulars of our syntax is with (157).

(157) $\mathcal{D} = (P_1, P_2, \ldots, P_n)$ is a derivation iff:
a. P_1 is formed by the $\overline{X}$ Skeleton and Lexical Insertion,
b. Each $P_{i>1}$ is formed from P_{i-1} by a rule,
c. P_j is subject to the linearization algorithm,
d. P_n is semantically interpreted.

A sentence is the set of well-formed derivations which have the same P_1 (=D-structure) and P_n (=Logical Form). A grammatical sentence is a derivation which has a $(P_1, \ldots, P_j)$ sub-sequence that is not longer than the $(P_1, \ldots, P_j)$ sub-sequence of any other derivation for that sentence.

May called the final parse in a derivation, "Logical Form" (LF), after the hypothesis that it is the representation that is semantically interpreted. I've not adopted that view in (157), as our syntax of Head Movement won't allow some LFs to be semantically interpreted. (157) defines S-structure (=P_j)

See section 6.2.1.2.

as the element of a derivation that meets the well-formedness conditions necessary to be subject to the linearization algorithm. Chief among those are the conditions that determine where Vocabulary Insertion puts the words that will be linearized. The requirements of AGREE and • are among those conditions, as they determine where two pieces of a word must be. (157) requires that an S-structure be as close to the D-structure as possible.

The role that semantic well-formedness plays in determining whether a structure can be linearized is no longer direct. An S-structure need not be the final parse in the series that contains something that is semantically interpretable. If it's not the final parse in that series, though, it must be possible to create an interpretable representation from an S-structure with the rules that define a derivation. Roughly speaking, what this model does is cause those rules which are unnecessary for forming a linearizable string, but necessary for forming a semantically interpretable string, to form the post-S-structure portion of the derivation.

By separating the parse in a derivation that determines the string from the parse that determines the meaning, May's proposal solves the problem posed by accusative Case-marked quantificational DPs. The parses that determine the semantic interpretation must be related to those that determine the string. But they need not be identical, and that, in May's system, is precisely what arises with accusative Case-marked quantificational DPs.

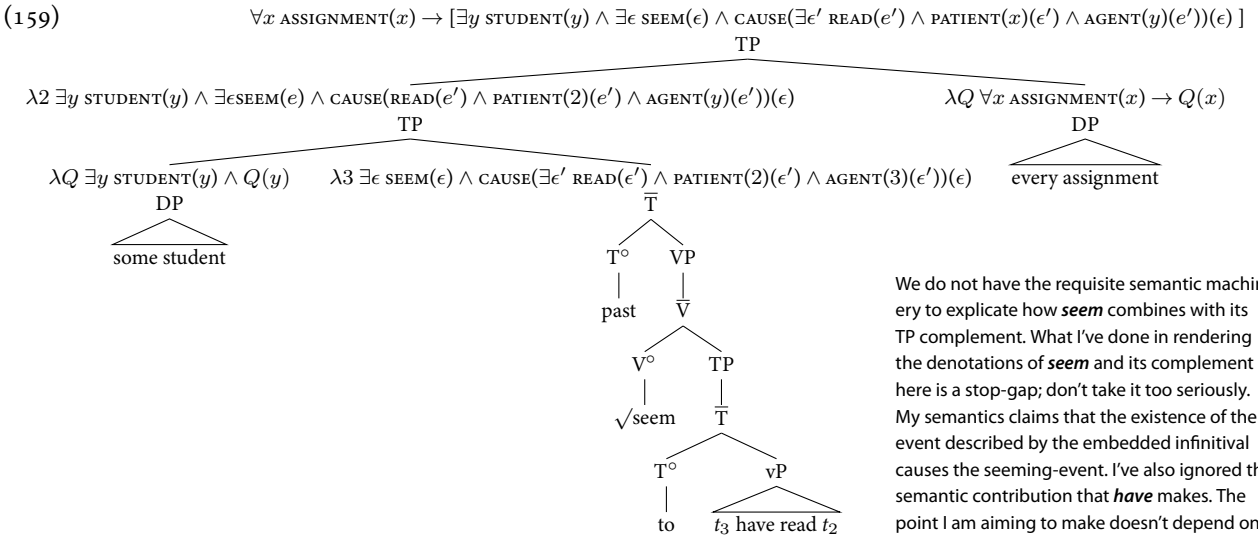
A feature of this model is that it allows one S-structure to correspond to more than one LF. That is, it allows the structure that determines the string to be related to more than one semantically interpreted structure. Thus, it predicts that in principle, a single S-structure should be able to be structurally ambiguous. There is evidence that suggests this is a correct prediction, though for a variety of semantic reasons the evidence is less direct than desirable.

To appreciate the difficulty in evaluating this prediction, consider what this syntax predicts for (158).

(158) Some student seemed to have read every assignment.

One LF that is expected for this sentence is (159).

(159)



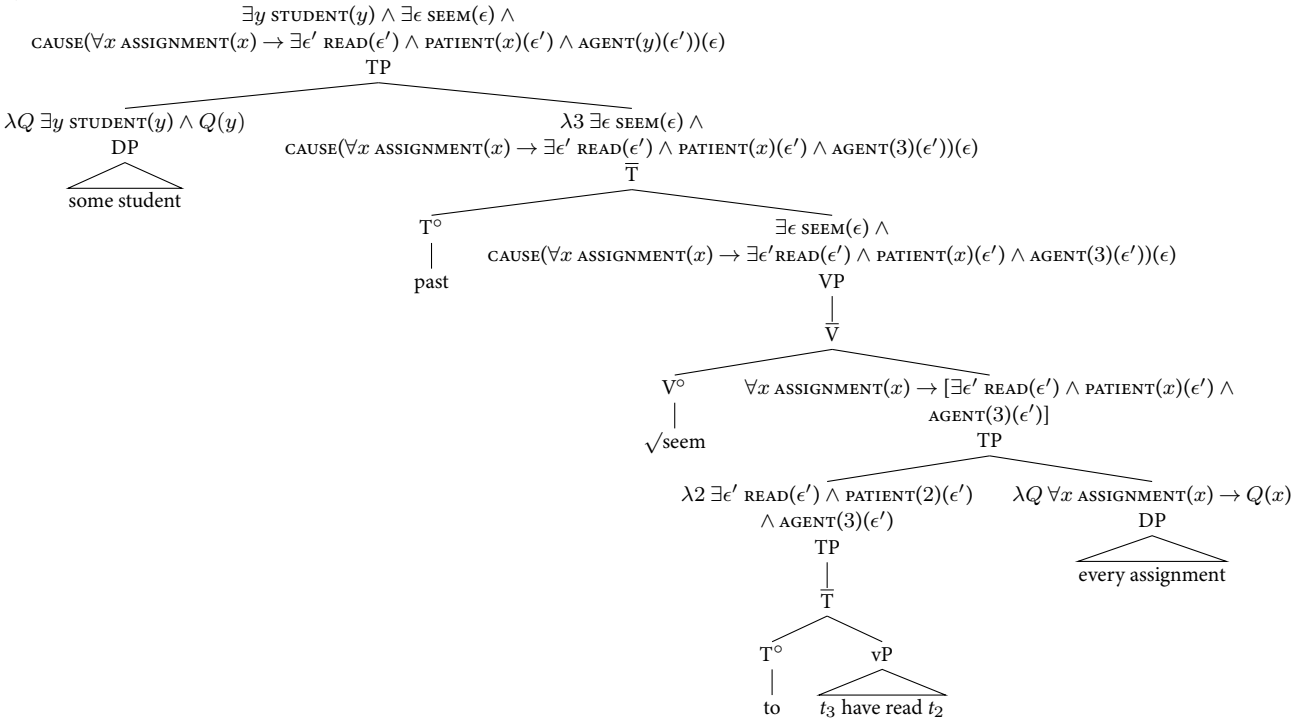
We do not have the requisite semantic machinery to explicate how ***seem*** combines with its TP complement. What I've done in rendering the denotations of ***seem*** and its complement here is a stop-gap; don't take it too seriously. My semantics claims that the existence of the event described by the embedded infinitival causes the seeming-event. I've also ignored the semantic contribution that ***have*** makes. The point I am aiming to make doesn't depend on these shortcuts.

This LF provides an interpretation that allows the students who seem to have read assignments to vary with those assignments. A rough paraphrase of the denotation derived for (159) is (160).

(160) For every assignment there is a (possibly different) student who seemed to have read it.

Our syntax also predicts that the LF in (161) should be related to (158).

(161)



This LF provides an interpretation that doesn't allow the students to vary with the assignments. A rough paraphrase of the denotation derived in (161) is (162).

(162) There is a student who seemed to have read every assignment.

If we could detect that (158) had these two denotations, then that would be evidence consistent with the predicted LFs.

Unfortunately, these denotations are not sufficiently distinct to know if they exist. In particular, the denotation delivered by (159) describes a state of affairs that includes as a special case those described by the denotation delivered by (161). This is because (163) holds.

(163) If it is true that there is a student who seemed to read every assignment, then it is true that for every assignment there is a student who seemed to read it.

The denotation delivered by (161) entails the denotation delivered by (159). Because we are not able to detect whether the denotation delivered by (161) exists, we cannot determine whether the LF that would deliver it does either. Insofar as the meaning of (158) is concerned, the LF in (159) is sufficient. The LF in (161) might exist, but we won't be able to detect it from the meaning of (158).

For reasons of this kind, testing the predictions of the LF model requires carefully curated examples. But it can be done, and to a large extent the predictions are confirmed. The string in (164) provides an illustration.

Katia said that …

(164) some professor seems to have read every assignment of hers to the class.

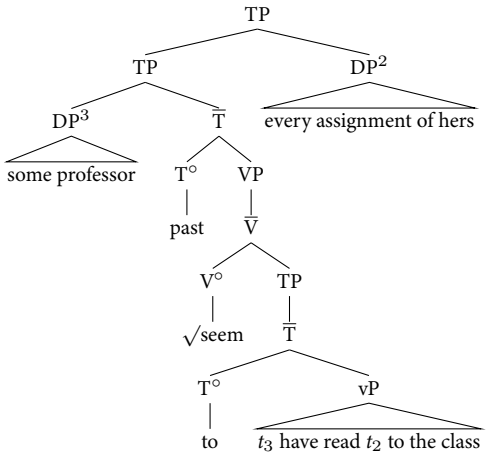
The pronoun in (164) can get one of two interpretations. It can be understood as coreferent with *Katia*, or it can be understood as a variable bound to *some professor*. When the pronoun is coreferent with *Katia*, an LF parallel to (159) (=(165)) is possible and the sentence has a denotation in which the professors can vary with the assignments.

It is also possible for ***her*** to refer to the entity made salient by ***some professor***. It turns out that an existential quantificational DP, like ***some professor***, cannot make an entity salient for the purposes of pronominal reference if it is in the scope of a universal quantifier. That can be discovered by considering the short conversation in (i).

i. A: Everyone read some paper.
   B: Yes, it was pretty good.

The pronoun ***it*** in B's response is felicitous, and capable of referring to the paper introduced by A's sentence, only if ***some paper*** is not understood to introduce different papers for every reader. For this reason, ***her*** can refer to the individual made salient by ***some professor*** in (164) only if there aren't different professors for each assignment. As a consequence, we can safely set aside this case. It is irrelevant to the question we are investigating.

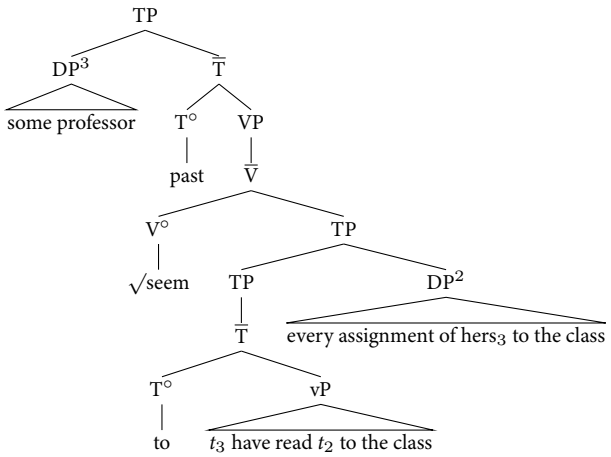
(165)



≈ for every assignment of hers (=Katia), there is some professor or other that seemed to have read it to the class.

But when the pronoun is understood as a variable bound to *some professor*, it will have to be c-commanded by *some professor*. This can be achieved if an LF parallel to (161) (=(166)) is also possible.

(166)



≈ There is some professor it seemed has read every assignment of hers (=that professor) to the class.

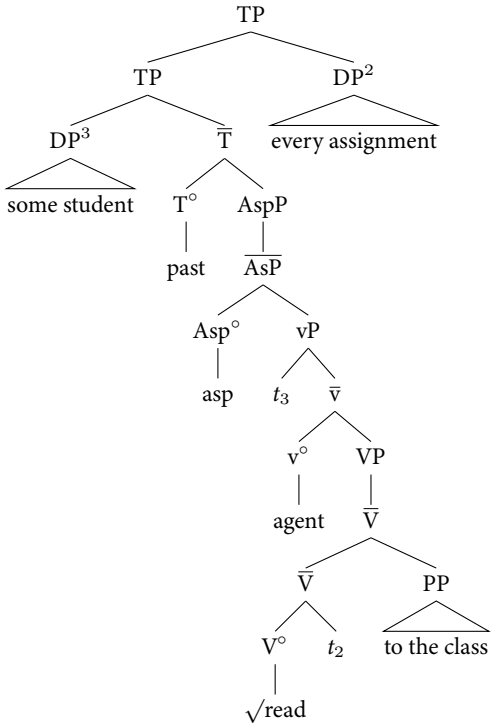
As expected, the LF in (166) does not allow a meaning in which the professors vary with the assignments. By adding the pronoun, then, we have separated the two meanings that are associated with how the quantificational DPs are interpreted, and revealed that they can be individuated in (164). This suggests, then, that there are two syntactic ways in which the quantificational DPs can be semantically composed, and a syntax which associates one surface parse with more than one LF does just that.

Another illustration of how to determine whether one S-structure is associated with more than one semantically interpreted syntactic representation is in comparing the sentences in (167).

(167) a. Some student read every assignment to the class.
b. Every student read some assignment to the class.

As we've seen, associating (167a) with the LF in (168) provides the right denotation, when it's coupled with the semantics of Trace Theory.

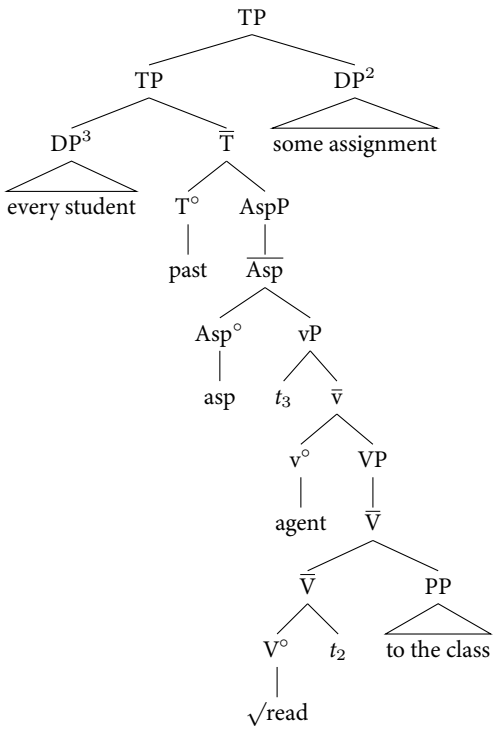
(168)



≈ For every assignment, there is some student or other that read it to the class.

But a parallel LF for (167b) does not provide the right denotation.

(169)

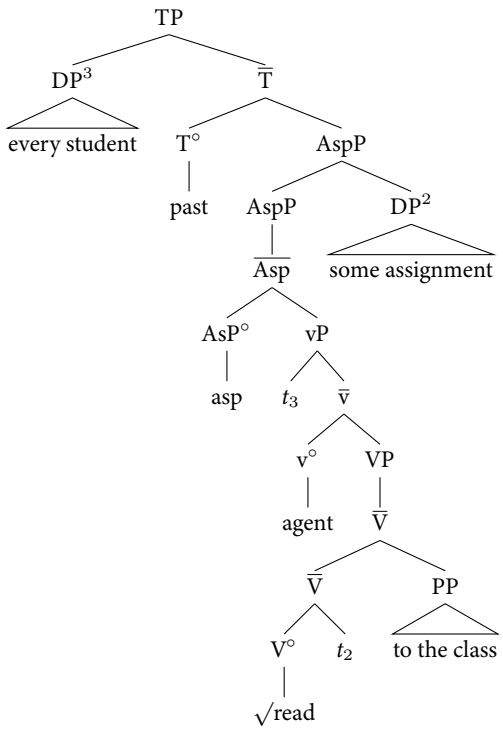


≈ It's the case that there is some assignment that every student read to the class.

This LF does correspond to an interpretation that (167b) has. That is, it is consistent with what (167b) means: it describes a scenario in which there is one assignment that was read by all of the students, and (167b) can be used to describe that scenario.

But (167b) is also a correct description of a situation in which every student read a different assignment to the class. While that meaning is not associated with the LF in (169), it can be achieved in our syntax by giving to (167b) the LF in (170).

(170)



≈ For every assignment, there is some student or other that read it to the class.

In a general way, then, the pair of sentences in (167) demonstrates that the S-structures they have must correspond to two different semantic interpretations. That can be achieved on May's proposal with our syntax and semantics.

6.2 Copy Theory

The Trace Theory of movement is perhaps the best known interpretation of movement, and it serves as the standard model. But it houses a variety of well-known shortcomings that we will examine in this section. At the time of writing, there isn't a successful solution to these shortcomings, but there is a rough set of ideas about how to solve them that has gained a certain following. In this section, I will present some of these shortcomings and sketch the ideas that are being explored to solve them.

6.2.1 Reconstruction

I'll start by considering a series of special cases that support the view that one S-structure can be related to more than one semantically interpreted syntactic structure. What makes these cases special is that a moved term is semantically interpreted, in part or in whole, in a position from which it moved. These cases are sometimes said to involve "reconstruction," as the moved term's meaning derives from "reconstructing" a parse in the derivation that feeds the S-structure. Instances of reconstruction pose problems for the Trace theory of

movement, and those problems have been an important source for building a better theory.

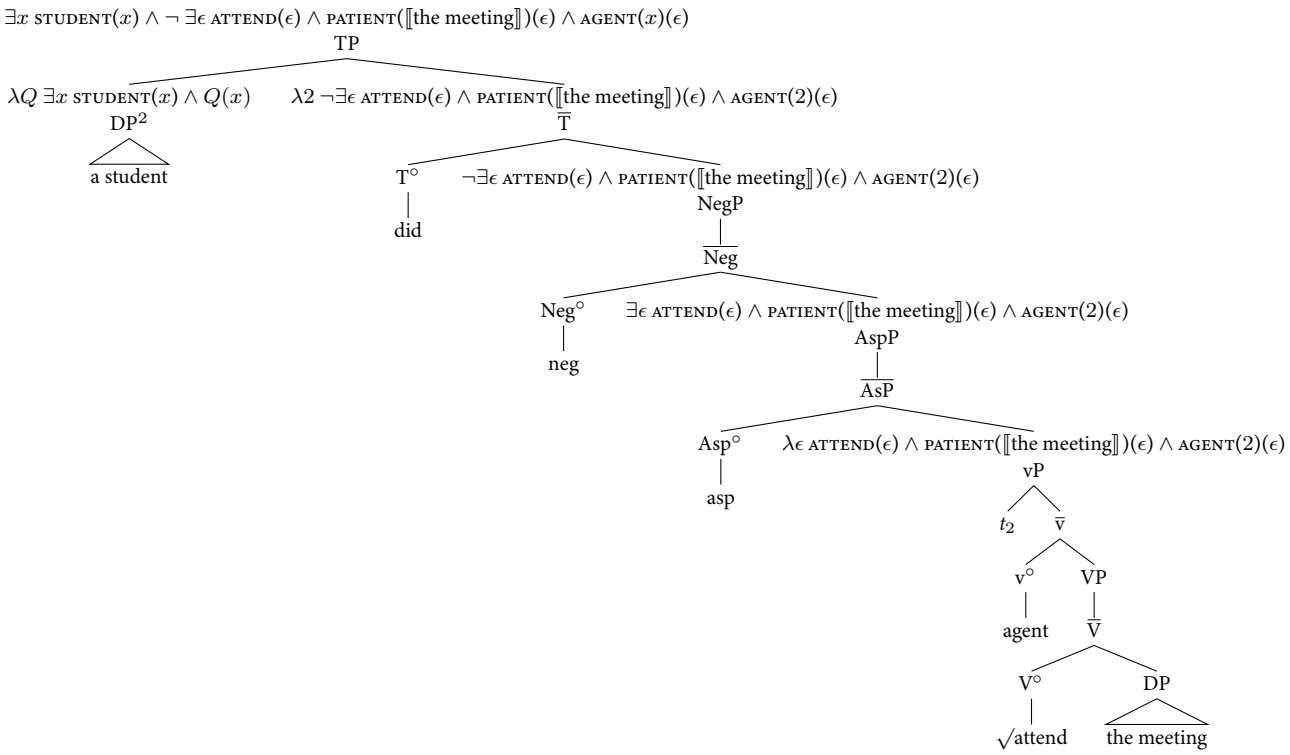
6.2.1.1 Quantifier Lowering

The first case of reconstruction that we'll consider is illustrated by the example in (171).

(171) A student didn't attend the meeting.

With the introduction of Aspect Phrase, our S-structure parse for (171) is (172).

(172)



This parse puts negation within the interpretation of the determiner *a*, and consequently the interpretation it delivers can be paraphrased by (173).

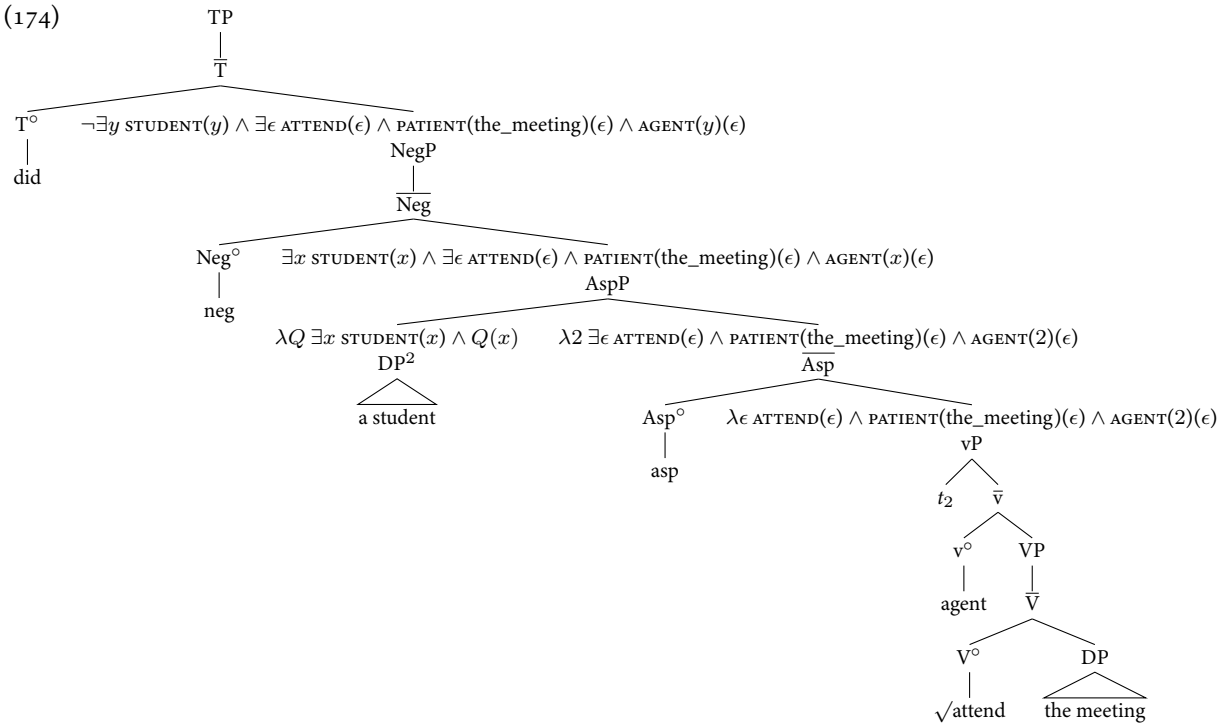
(173) There is a student who didn't attend the meeting.

But (171) has another interpretation in which *a student* is interpreted as part of what is negated.

Our present syntax provides this parse in the part of the derivation from which the S-structure is produced. A Movement can form (174) by moving *a student* into Specifier of AspP. From this parse, A movement can derive the S-structure position of *a student* by again moving it into Specifier of TP.

Alternatively, Complex NP Shift could create a parse parallel to (170), right-adjoining ***a student*** to AspP.

(174)



≈ It's not the case that there is a student that attended the meeting.

On the Lebeaux model, the presence of this parse in the denotation ensures that the meaning associated with it is available to the interpretation of (171). For this reason, (171) can also have the denotation indicated: what's negated includes *a student*.

Robert May suggested in May (1985) a different way of deriving this parse that is popular in the literature. His system abandons the Lebeaux model, as we've seen, and requires the final parse in a derivation—his "LF"—to be the structure that is semantically interpreted. Because May's system allows the S-structure to be a parse in the derivation from which an LF is derived, he is able to capture the fact that sometimes a DP is semantically interpreted in a position higher than where it is spoken. His proposals are not entirely successful, as we'll see in the next section, as they don't include derivations with Head Movement in them. But May does try to accommodate Reconstruction phenomena to his model. We'll look at his attempt to do so because it has features that might prove useful.

May's proposal has two components. One is to let movement only optionally leave traces when XPs are moved. The second is to abandon the Upwards constraint on derivations, and let the presence of traces be responsible for its effects. Putting these together with his goal of letting only LFs be semantically interpretable, we have what I'll call the May model.

By letting traces be optional, May's proposal also loses the use Trace theory has in ensuring that the Theta Criterion holds of D-structures. See the discussion on page 401.

(175) May Model

a. Only an LF is semantically interpreted, and

b. Leaving traces is not obligatory for moved XPs, and

c. Derivations aren't constrained by Upwards

The effects of Upwards, which is repeated here, can be derived from the syntax and semantics of traces when they are present.

See 207 for discussion.

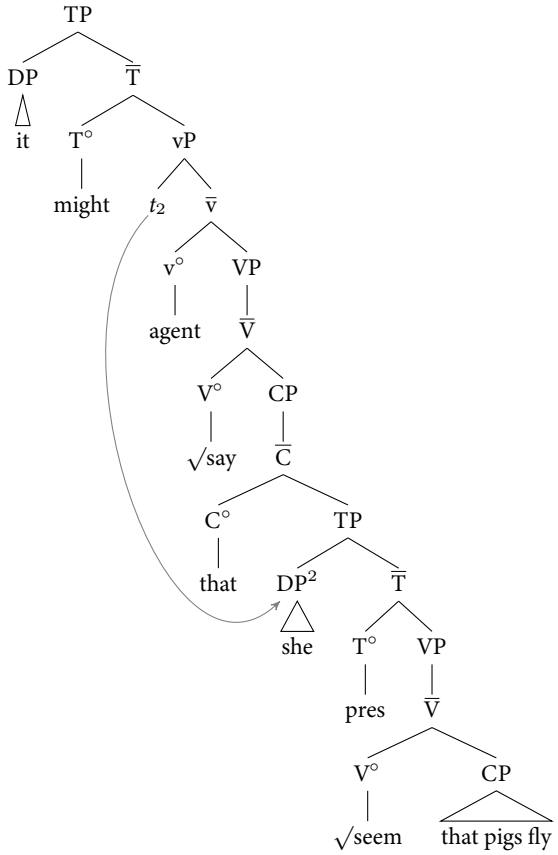
(176) Upwards
If α moves from position T to position P, then the sister of P must have a projection that dominates T.

The syntax and semantics of traces derives the generalization in (177) for movement that involves them.

(177) Let α be a term that is moved. α must c-command any trace that arises from its movement and is otherwise required.

It will therefore prevent derivations that would also be blocked by Upwards, like that shown in (178).

(178)

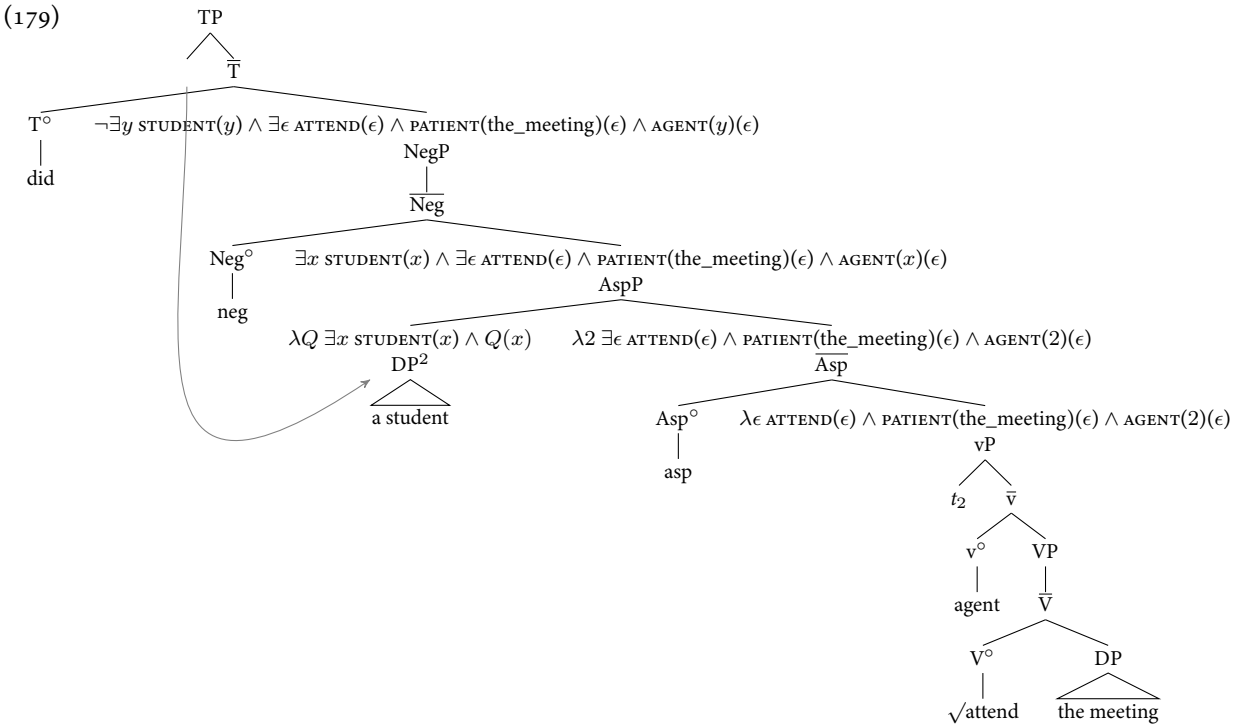


The trace cannot be interpreted as a variable in this representation, because it is not part of a phrase that combines with the DP that should bind it. It doesn't conform to the generalization in (177).

In (178), the option of not leaving a trace when *she* moves is not available, since the agent θ-role would not be assigned. Because the May Model doesn't allow D-structures, or any other non-LF parse, to be semantically interpreted, it requires that a parse created by moving *she* must be interpretable. These two ingredients, then, combine to require moved argument DPs to move to a position where they c-command the position they have at D-structure. For movement of argument DPs, then, the results of Upwards can be derived from the syntax and semantics invoked by Trace theory.

But because movement of DPs from non-argument positions doesn't require a trace to be left in that position, the effects of Upwards will not be present in those cases. This will give (171) a second LF, and this, May suggests, is why it has a second interpretation. Imagine that there is a rule that can produce (179) from (172), the S-structure of (171).

(179)



This could, for instance, be what Argument Movement looks like if Upwards were not present. (179) is the same as (174), and so provides the same semantic interpretation. On May's system, that parse can be achieved at LF, however, and this would be consistent with the May model's hypothesis that only LFs provide the parse that is semantically interpreted. May calls the operation illustrated by (179) "quantifier lowering." This is how these kinds of examples have come to be known.

May's analysis of quantifier lowering depends on the success of his method of deriving the effects earlier ascribed to Upwards. May's method derives a condition on derivations from the syntax and semantics of interpreting variables. It replaces Upwards with (177). (177) is narrower than Upwards,

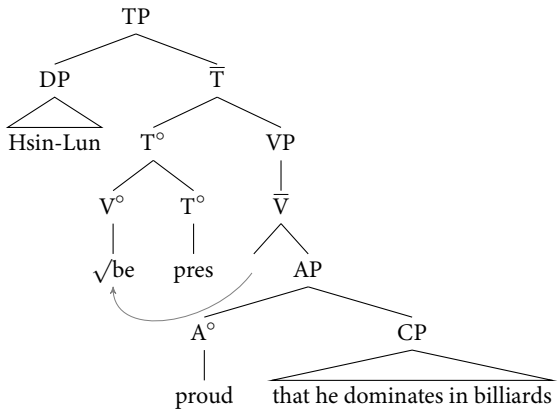
however. Upwards is tailored to apply to the syntax we've adopted for Head Movement which, as the contrast in (180) indicates, should fall under its purview.

(180) a. Hsin-Lun is proud that he dominates in billiards.

b. * Hsin-Lun must proud that he is dominant in billiards.

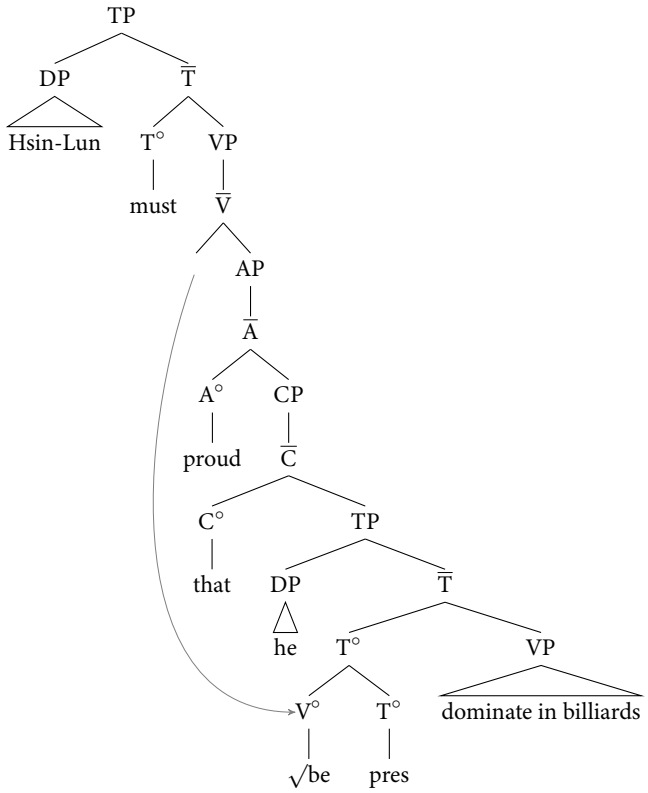
Upwards is obeyed in the syntax our system gives to (180a), sketched in (181).

(181)



The moved term is the verb *be*, and its sister—T°—projects T̄ and TP both of which dominate the position from which *be* moves. This isn't true of the syntax our system gives to (180b), however, as (182) shows.

(182)



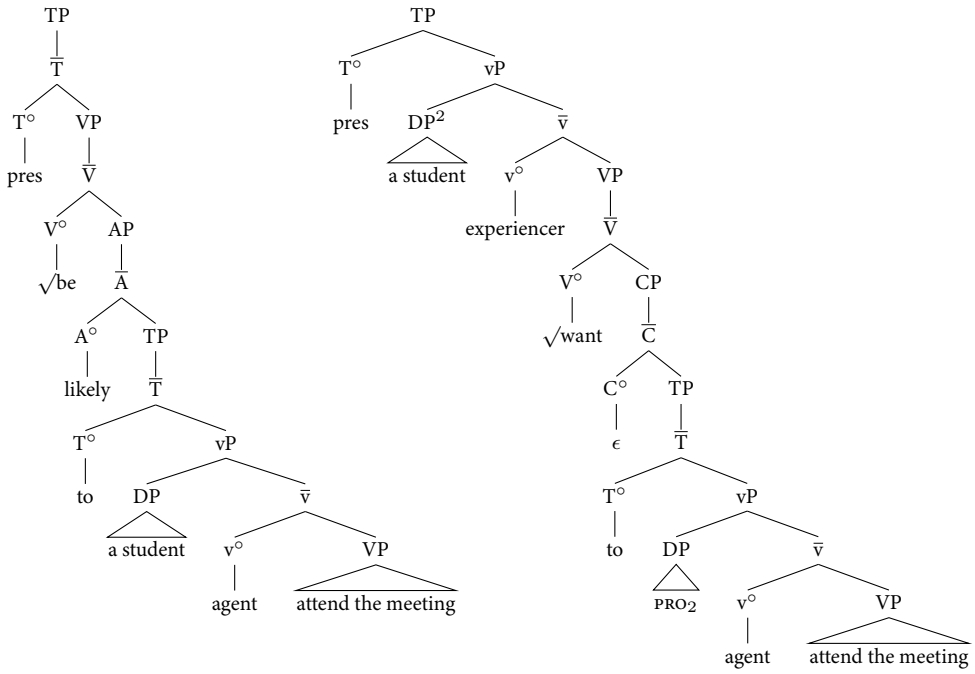
For these cases to be fit into (177), the syntax and semantics we've given for binding and traces requires revision. Because *be* doesn't c-command the position it moves from in either of the cases in (180), a trace in that position will not be bound by it. Either the syntax of Head Movement would require revision, or the syntax we've developed for movement would. We'll examine this issue in 6.2.1.4.

Quantifier lowering provides interesting confirmation of the analysis of infinitives described in chapter 3.5. On that analysis there are two related differences between the sentences in (183).

(183) a. A student is likely to attend the meeting.
b. A student wants to attend the meeting.

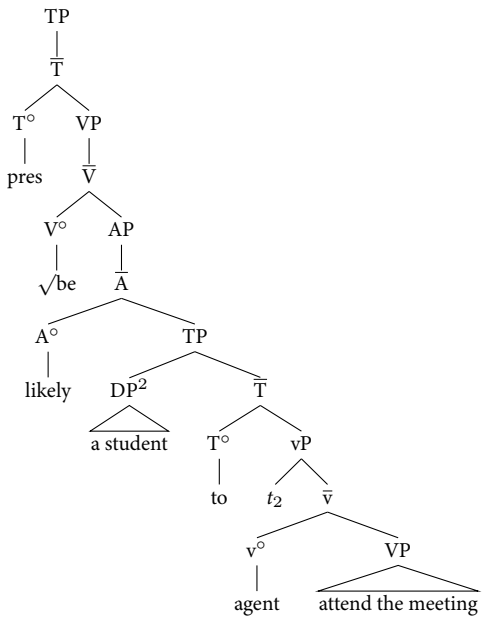
One difference is that *a student* is an argument of *want*, but not of *likely*. Connected to this is the status of the term that serves as an argument of *attend* in these examples. In (183a), the argument of *attend* is *a student*. But in (183b), the argument of *attend* is a silent pronominal that is bound by *a student*. If we use PRO to represent the silent pronoun, then the D-structure parses for (183) are shown in (184).

(184)



If we adopt the analysis of quantifier lowering that preserves Upwards and adopts the Lebeaux model, then from the D-structure for (183) can be derived (185).

(185)



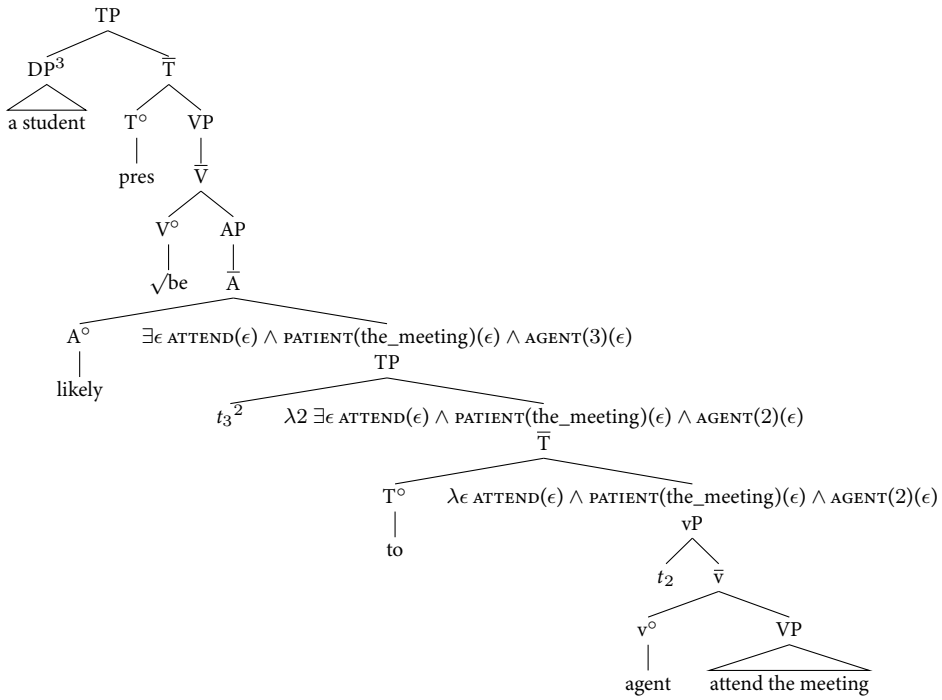
If this parse is semantically interpreted, it will deliver an interpretation in which *a student* is part of what the sentence claims is likely. That interpretation can be paraphrased with (186).

(186) It's likely that a student will attend a meeting.

This, indeed, is an interpretation that (183a) has.

From (185), the S-structure representation in (187) can be derived.

(187)



This parse can also be semantically interpreted, but it puts the denotation of *a student* outside the embedded clause and therefore outside what this sentence claims is likely. A paraphrase of this meaning is (188).

(188) There is a student who is likely to attend the meeting.

This is also an interpretation that (183a) has.

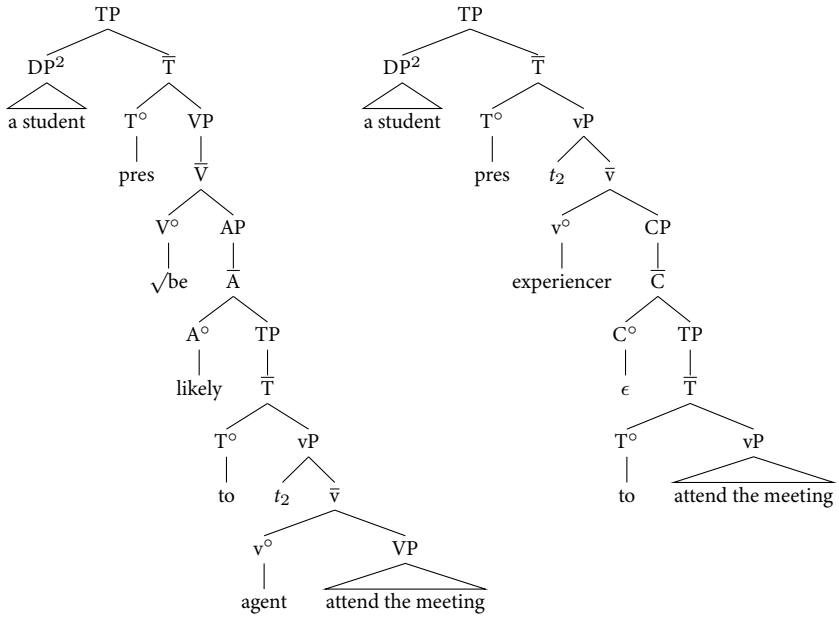
Note that the trace left by moving *a student* to the highest Specifier of TP is both a binder and a variable. It is a variable bound by *a student*, and it binds the trace in Specifier of vP. I've given a partial explication of the semantics for (187) to indicate how this works. (A full explication of the semantics of these sentences requires understanding how predicates combine with embedded clauses, and this is not in our toolkit.)

This situation for (183a) is not reproduced in (183b), because the D-structure for (183b) puts *a student* outside the embedded infinitive. On a theory that accepts Upwards, this prevents Argument Movement from building a parse in which *a student* is part of what (183b) claims is likely. This predicts, correctly, that (183a) is ambiguous in a way that (183b) is not.

(183) a. A student is likely to attend the meeting.
   i. $\approx$ It is likely that a student will attend the meeting.
   ii. $\approx$ There is a student that is likely to attend the meeting.
  b. A student wants to attend the meeting.
   i. $\neq$ It is wanted that a student will attend the meeting.
   ii. $\approx$ There is a student that wants to attend the meeting.

This same contrast emerges from a theory that abandons Upwards, and derives its effects from the semantics of Trace theory, as May suggests. On that model, the semantically interpreted parse is derived from S-structure; the S-structures of (183) are shown in (189).

(189)



Because the trace t_2 is required in the semantically interpreted parse, movement cannot bring *a student* to a position that doesn't c-command that trace. This will prevent movement from moving *a student* into the embedded infinitive from the S-structure for (183b) (the second graph in (189)). But because the trace is inside the embedded infinitive in (183a), this is a structure that can be derived from the S-structure for (183a) (the first graph in (189)).

What we learn from quantifier lowering, then, is that the parse that is semantically interpreted can differ from the parse that is pronounced in having a DP in a lower position. We've entertained two ways of modeling this. One allows movement to put the moved term in a position that doesn't c-command the position it moves from. It relies on the semantics of traces to limit where this can happen. This approach, the May model, raises the possibility that only LFs provide the semantically interpretable parse. The other model, the Lebeaux model, preserves Upwards, and doesn't specify which parse in a derivation provides the semantically interpretable parse.

We'll now look at other evidence that bears on choosing between these two views.

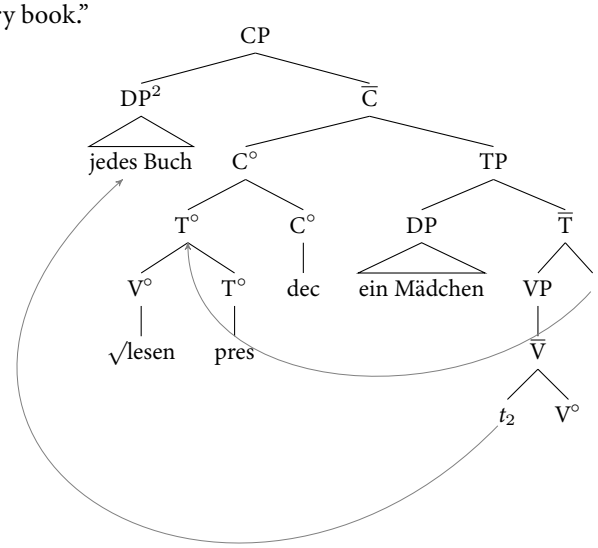
6.2.1.2 Head Movement and Derivations: a problem

Our present definition of derivation orders the parses that make up our representation of a sentence. If a parse contains a trace, for instance, the definition of a derivation requires that this parse not be a D-structure, but must be related to a D-structure by a series of parses that includes one in which the trace is created by movement. If we assume that the S-structure—which is visible—and the D-structure—which can be determined given the S-structure—are known, then the space of possible parses derivable from movement that fit those two parameters can be calculated. Any of those parses from which

the semantic rules can produce a denotation corresponds to a meaning the sentence should be able to have. A component in the account we have for the meanings that a sentence can have, then, is in what our syntax allows a derivation to be. The number of parses in a sentence's derivation that have a semantic interpretation should correspond to the number of denotations that sentence can have. If that number is zero, then we should expect the corresponding sentence to have no denotation. We could interpret this failure of the syntax to provide a denotation as one way a sentence could be judged ungrammatical. In this section, we're going to see that our syntax and semantics don't successfully furnish derivations for certain kinds of grammatical sentences that deliver a denotation. Our model fails to capture the grammaticality of a certain type of sentence. What causes the problem are derivations in which both a head and a phrase move.

Our present semantics for movement doesn't allow a head, a verb say, that has been moved to be semantically interpreted in the position it moves to. Thus, for instance, the S-structure parse for a German main clause can put the main verb into C° position in the way that (190) illustrates.

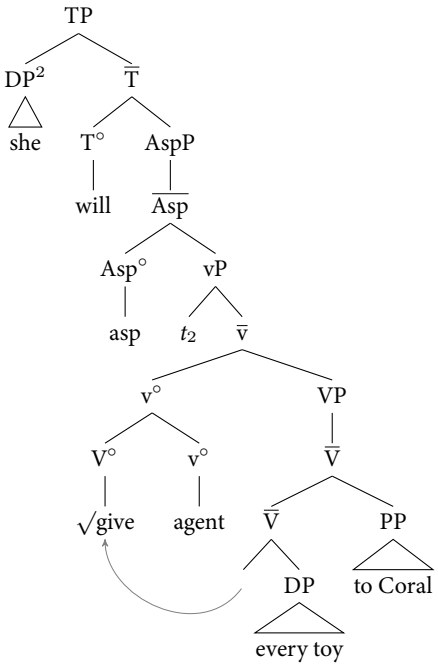
(190) Jedes Buch lesen ein Mädchen.
every book read.pres a girl
"A girl reads every book."



But the syntax of Head movement doesn't permit a verb to bind a trace in the position from which it's moved, and so our definition of movement doesn't allow for traces of verbs. As a consequence, the verb in (190) must be semantically interpreted in the position it resides in. But this isn't possible; the sister of $\sqrt{}$*lesen* is not the object its θ-role needs, but instead present tense. (190) is semantically anomalous.

A parallel situation arises in English, if verbs move to v°. The S-structure in (191), for instance, cannot be semantically interpreted given our present syntax/semantics, because the denotation of $\sqrt{}$*give* cannot compose with the denotation of *agent*.

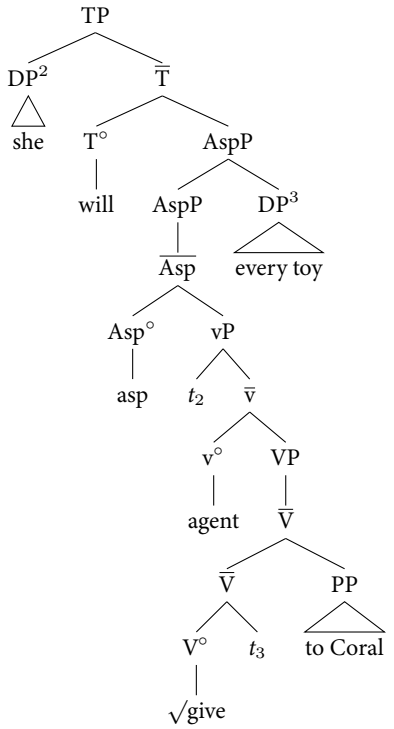
(191) She will give every toy to Coral.



In general terms, sentences that involve Head Movement must have derivations in which a non-S-structure parse provides the denotation. If Head Movement is governed by Upwards—as the evidence we have suggests—then this means that the parse which provides the denotation will have to be D-structure, or some parse between D-structure and S-structure that preserves the D-structure position of the head.

But consider now what's required to give a parse for (191) that allows *every toy* to be semantically interpreted. If our semantics for *every* is correct, then *every toy* cannot be interpreted in the position it occupies in (191). Instead, it will have to move to a position outside the term that binds the event variable; that term is *asp* in our syntax. A parse that meets these twin goals—putting $\sqrt{}$*give* and *every toy* in their semantically interpretable positions—is (192).

(192)



The derivation for this sentence must be able to contain this parse if it is to deliver the right denotation.

We get independent evidence for a parse of this kind being part of the derivation from the methods we've developed elsewhere to reveal the denotations of sub-constituents of a sentence. For instance, we can learn what the meaning of the VP is from *again*, which, in (193), can invoke a reading in which its presupposition doesn't include the subject.

Bale (2007) is a good introduction to the interaction of ***again*** and quantifiers.

(193) At the beginning of the day, the teacher gave every toy to Coral. But now the other kids have taken them from Coral, and she's unhappy, so someone should give every toy to Coral again.

The meaning of *again* in this sentence requires that it combine with a constituent that contains the meaning of $\sqrt{give}$ and its arguments, but not the meaning of *agent* and its arguments. (See chapter 5.2.1 for the details.) That constituent is provided by (192)—it's the VP—but there is no constituent of that type in (191).

Similarly, we learn from (194), that the *to*-phrase can fall within the scope of *every toy*.

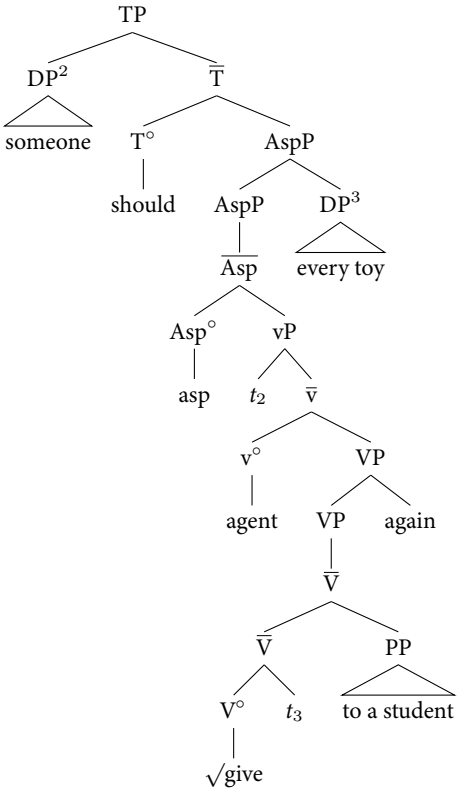
(194) She gave every toy to a student.

This requires that *every toy* be the sister to a constituent that contains *a student*. A parse like that in (192) would achieve that goal, but a parse like (191) wouldn't. A sentence that combines these two assays, like (195), shows that (196) must be available.

At the beginning of the day, the teacher gave every toy to a student. But now Coral has taken them all for herself, so

(195) Someone should give every toy to a student again.

(196)

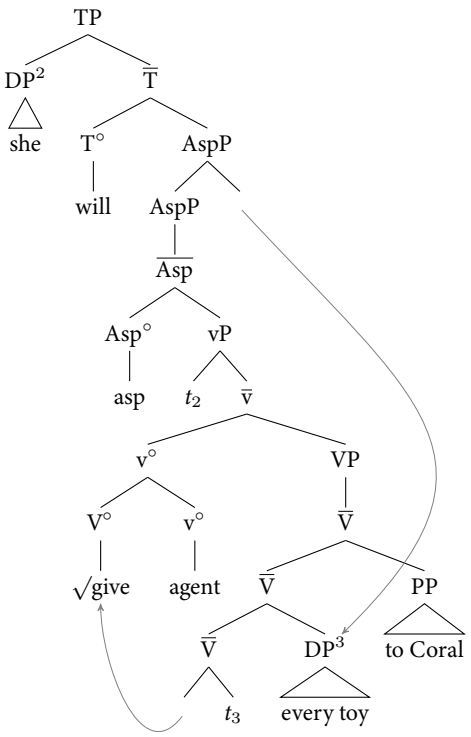


This parse leaves ***a student*** inside VP, but our semantics for ***a*** would require it to also be interpreted above ***asp***. A parse that meets that requirement would put ***a student*** in a position adjoined to AspP that is below ***every toy***. I simplify the picture here. The semantics given for ***again*** in chapter ?? does not include how it interacts with quantifiers and traces, so for a complete picture to emerge here would require examining how these interact.

The same problem arises for the German sentence in (190). Just as in the English (191), this sentence has argument DPs that are headed by quantificational determiners. If the semantics of these expressions is the same as in English—and they do seem to have the same denotations—then the derivation for (190) must provide a parse in which both subject and object DP have moved out of the vP. That parse must also, however, put the verb root $\sqrt{lesen}$ in its D-structure position. To give the derivation that parse, and also achieve the S-structure in (190) in a way that preserves Upwards isn't possible.

The challenge, then, is to see how to design a derivation for (191) that includes the parse in (192) but provides the S-structure in (191), and then generalize that solution to all the cases, like (190), that share the problem. If derivations obey Upwards, our present system doesn't allow a derivation of the requisite kind. But if the effects of Upwards are derived entirely from the semantics of traces in the way that May suggests, then there is a derivation that does what's required. In that derivation, QR or Complex NP Shift applies before Movement relocates the verb forming the parse in (196). The S-structure is derived from this by using Movement to move the verb into v°, and moving *every toy* to the lower position indicated in (197).

(197)



Abandoning Upwards, however, brings its own unwanted consequences. We lose the consequences of Upwards for any movement that doesn't have to leave a trace. Because only DPs can leave traces in our present syntax, this means we will have to find another way of ensuring Upward's consequences for moved heads, as well as moved VPs, PP, and APs.

This section, like the previous, then, shows that we need to allow for derivations in which a moved term is semantically interpreted in a position lower than where it is pronounced. The previous section discussed cases that involve subjects, and for these we have two options. One is to let a parse in the derivation that feeds the S-structure provide the semantic interpretation. The other is to allow for derivations that move terms downwards. The second option maintains the possibility that the parse which the semantic rules interpret is limited to either the S-structure, or a parse derived from an S-structure.

The cases examined in this section, however, involve interpreting a head in a lower position than it occupies at S-structure, and this complicates matters. If we let a parse that feeds the S-structure be semantically interpreted, we can avoid reworking the semantics of traces so that they fit the syntax of Head Movement. Heads can be semantically interpreted in a parse in which they have not moved. But the situation involving quantificational objects in English requires that movement downwards be allowed for moved phrases in that scenario. Alternatively, we can maintain the view that the semantics is only read off of parses derivable from an S-structure, but equip our syntax and semantics with a way of letting moved heads leave a trace. In this sce-

nario, not just phrases, but heads too, will have to be equipped with the ability to move downwards.

Together, then, these two sections suggest that we should find a way of giving a derivation the ability to create a semantically interpreted parse that derives from an S-structure but allows a term in that S-structure to be interpreted lower than where it sits at S-structure. I will use the term "reconstruction" for any scenario of this kind.

(198) Let an Upwards-obeying position for x (=UP(x)) be a position to which Upwards would allow a term in position x to move to, and let an S-structure position (="s-position") be the position that a term occupies in the S-structure parse of a derivation. α reconstructs to position β if α is semantically interpreted in β and:

a. β isn't the s-position of α and

b. β isn't UP(s-position of α)

Reconstruction describes cases in which a term is semantically interpreted in neither its S-structure position nor in a position that could be derived by an Upwards-obeying instance of movement from its S-structure position. Cases involving Quantifier Lowering can be alternatively described as situations in which the quantificational DP reconstructs. Similarly, the cases described in this section, where a verb is semantically interpreted in a position lower than its surface position, can be described as situations in which the verb reconstructs.

A law of Reconstruction seems to be (199).

(199) Law of Reconstruction
Let α occupy position β at D-structure. If α reconstructs to β', then either:

a. $\beta' = \beta$, or

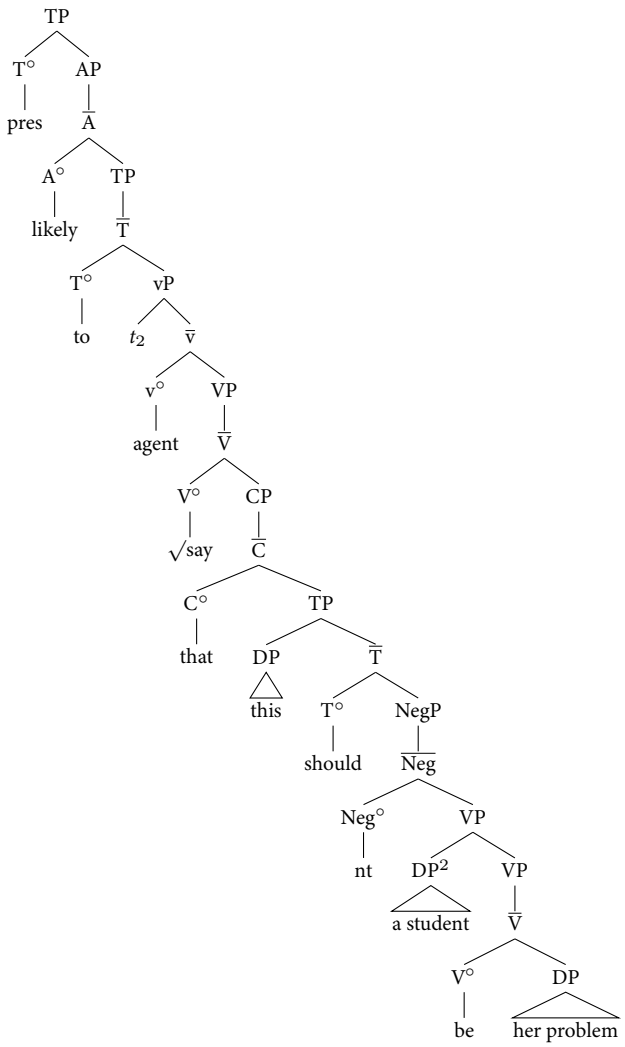
b. β' is UP(β).

This law says that Reconstruction is only possible to the D-structure position of the term reconstructing, or to a position that an Upwards-obeying instance of movement could take that term to from its D-structure position. To date, at any rate, I do not know cases of Reconstruction which violate (199). For instance, it is not possible to interpret (200) in such a way that puts *a student* within the scope of *shouldn't*.

(200) A student is likely to say that this shouldn't be her problem.

That would be a possibility if the S-structure for (200) could be related to the parse in (201).

(201)



But this violates the Law of Reconstruction, as the position that *a student* occupies in (201) is not UP(t_2): t_2 being the position *a student* occupies at D-structure.

The Law of Reconstruction is straightforwardly derived on the Lebeaux model of the syntax/semantics interface, if it is wedded to Upwards. By contrast, the May Model does not derive the Law of Reconstruction. The May Model also fails, as we've seen, to derive Upwards for terms that don't leave traces—like Verbs—and it wrongly allows D-structures to put arguments in non-θ-marked positions. The May Model is presently a live contender in the literature at large. But let us abandon it in these lectures.

Instead, let's provisionally adopt Upwards and the Lebeaux Model.

(202) Lebeaux+Upwards Model

a. Derivations are constrained by Upwards,

b. If P is a semantically interpretable parse in a derivation for S, then ⟦P⟧ is a possible denotation for S.

The Lebeaux+Upwards model also suffers from the ordering problems this section describes. It requires that a derivation include a parse in which quantificational DPs be interpreted in their moved position, but verbs (and other heads) be interpreted in their un-moved positions. If derivations are governed by Upwards, our present syntax does not let derivations have those parses. Unlike the May Model, however, there is an idea about how to fix these problems for the Lebeaux+Upwards model. The solution we'll investigate below involves changing the syntax and semantics of movement. Before looking at the solution, let's first look at another kind of environment where reconstruction arises.

6.2.1.3 Reconstructed Variables

In the previous two sections we have seen cases in which a term "reconstructs." We've seen that the denotation of the determiner *a* can be interpreted in a position lower than it's pronounced, and we've interpreted that as a situation in which the DP it heads reconstructs. We've also seen that the denotation of a verb can be interpreted in a position lower than where it's pronounced: verbs too can reconstruct. In this section we'll examine another case of reconstruction. Expressions that are interpreted as variables can reconstruct.

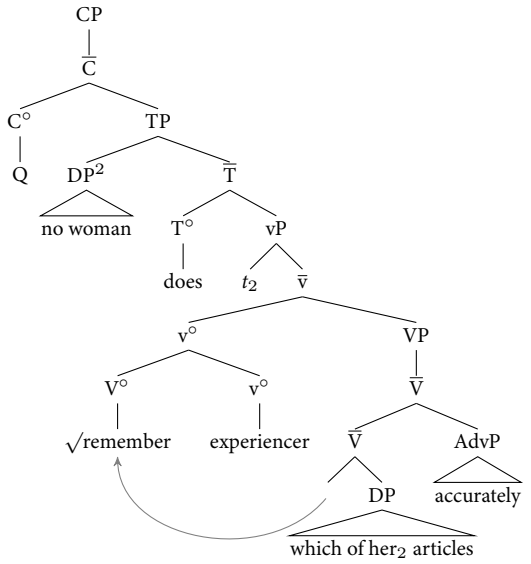
One case of this is illustrated by (203).

The first to examine these cases is, perhaps, Engdahl (1982). A pretty thorough examination of cases like this is in Barss (1986).

(203) Which of her articles does no woman remember accurately?

The derivation for (203) includes a parse in which *which of her articles* resides in the complement position of *remember*. Suppose that parse has been created by movement of the subject argument; an abbreviated picture of that parse could be (204).

(204)



If the pronoun is equipped with the same referential index that the trace of moving the subject has, then this parse allows the pronoun an interpretation in which it is a variable bound by *no woman*. This is indeed an interpretation (203) can get. That interpretation can be paraphrased with (205).

(205) What is the article such that for no x, x is a woman and x remembers accurately that article of x's.

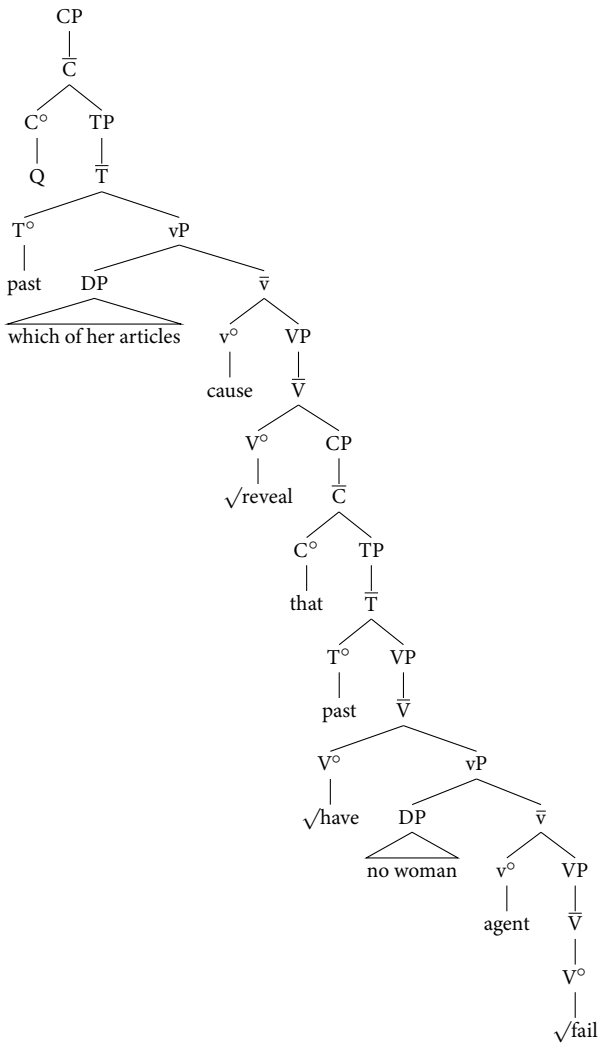
This question seeks the identity of articles that no woman has written and remembers accurately. The identity of the articles, then, varies with the value given to the possessive pronoun. That can be achieved by making the pronoun a variable bound to *no woman*.

Engdahl (1982) shows that the ability of the pronoun in sentences like these to get the bound variable interpretation depends on there being a parse in the derivation in which that pronoun is c-commanded by the binder. That a pronoun can only be a variable bound by a c-commanding DP derives, recall, from the semantics we have given to movement. When a phrase moves, like *no woman* in (204), then the phrase it becomes a sister to turns into a λ-expression. That λ-expression will bind every referential index within that the λ-operator is related to. For a pronoun to be a variable, it must bear one of those referential indices. It must, in other words, be within the phrase that is the sister to its binder. This condition holds in (204): *her* is within the phrase—$\overline{\text{T}}$—that is the sister to its binder. By contrast, there is no parse in the derivation of (206) in which that relation holds between *no woman* and *her*.

(206) Which of her articles revealed that no woman had failed.

The D-structure for (206) is sketched in (207).

(207)



The phrase *which of her articles* is not c-commanded by *no woman* in this parse, and because movement will only move *which of her articles* into higher positions, there is no parse derivable from this one in which *no woman* c-commands *which of her pictures*. The prediction then is that *her* cannot be interpreted as a variable bound by *no woman* in (206). That prediction is confirmed. If *her* could be a variable bound to *no woman* in (206), then it should have an interpretation like that paraphrased in (208).

(208) What is the article such that for no x, x is a woman and that article of x's reveals that x had failed..

That is not what (206) means. Instead, it gets an interpretation in which *her* refers to some female individual whose identity the context must supply. The resulting meaning is roughly:

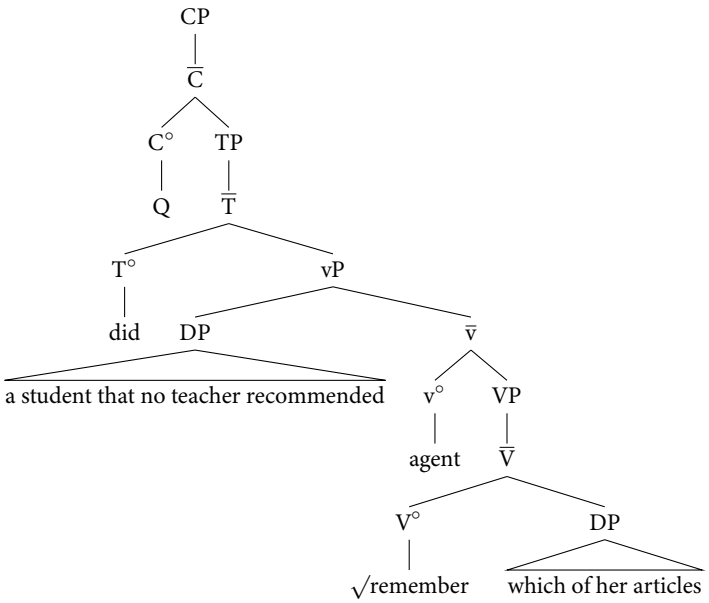
(209) What is the article such that that article of hers reveals that no woman had failed.

Another example that demonstrates the same thing is (210).

(210) Which of her articles did a student that no teacher recommended remember?

The D-structure for (210) is sketched in (211).

(211)



The pronoun *her* is not c-commanded by *no teacher* in (211), and movement is incapable of creating from (211) a parse in which that happens. As expected, then, (210) cannot have an interpretation like that described by (212).

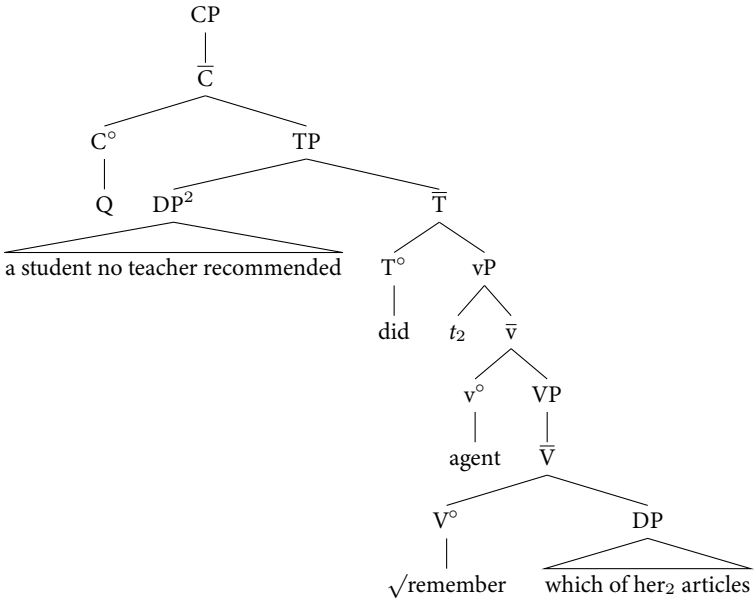
(212) What is the article such that for no x, x is a teacher there is a student that x recommended who remembers that article of x's.

This is the interpretation that would arise if *her* could be a variable bound to *no teacher*.

Instead, *her* can be interpreted as a variable for the expression *a student no teacher recommended*. That's because there is a derivation for (210) that includes the parse in (213).

If (210) is uttered in a context which made salient a female individual, then (210) could also get an interpretation in which ***her*** refers to that individual.

(213)



This parse puts *her* in a position that allows it to be bound by the moved phrase *a student no teacher recommended*. Consequently, it gives (210) the interpretation described by (214).

Note that ***no teacher*** still doesn't c-command ***her*** in this parse.

(214) What is the article such that a student no teacher recommended, y, remembers that article of y's.

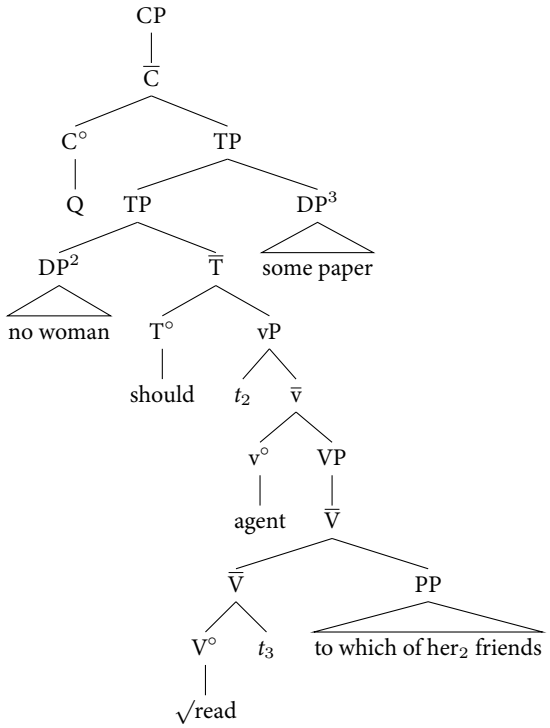
The pronoun in (210) can be a variable bound to *a student no teacher recommended*, but not to *no woman*. This is just what is expected if the pronoun's interpretation can be computed from a position it occupies in a non-s-structure parse in the derivation.

These cases of reconstruction can be captured by the Lebeaux+Upwards model, so they do not pose a problem for our present theory of movement. But they can be used to create the same kind of ordering problem that is described in the previous section. Consider, for instance, (215).

(215) Which of her friends should no woman read some paper to?

This can get an interpretation in which *her* is a variable bound by *no woman* and *some paper* is interpreted outside the scope of *no woman*. The parse that will deliver that interpretation puts *which of her friends* in its unmoved position, and *some paper* is a position higher than *no woman*. For instance, (216) has the right configuration.

(216)



There is no derivation that obeys Upwards which has this parse as well as an S-structure that corresponds to (215).

6.2.1.4 Achieving reconstruction with traces

These cases of reconstruction show that we need a syntax which allows a moved term to be semantically interpreted in a position it occupies before moving. In the previous sections, I've sketched how this is accounted for on the Lebeaux+Upwards Model, which allows the semantic interpretation of a sentence to be derived from any element in its derivation. Because the Lebeaux+Upwards Model allows one to determine the elements of a derivation from the meanings a sentence has, it provides a method for determining what a derivation consists of. If our model gives a sentence a derivation that consists of elements which are semantically interpretable, but those interpretations are not available to the sentence, then we can read this as evidence that we have the wrong model of a derivation. We should seek a model of derivations that prevents parses whose meanings are not associated with the sentence in question. Conversely, if a sentence has an interpretation that doesn't correspond to a parse in its derivation, then we should modify our model of derivations so that it includes that parse. On the Lebeaux+Upwards model, then, we can use the meanings a sentence has as a guide to the parses in its derivation.

A derivation is a syntactic object. It is a set of parses that represents a sentence. Reconstruction provides us a view of that set of syntactic parses through the meanings associated with those parses. The view reconstruction

gives us of a derivation therefore rests on how parses are semantically interpreted. As our model of the semantic interpretation of parses matures, so too will our view of the parses that make up a derivation. If, for instance, our model of semantic interpretation associates a denotation to the parse P but not the parse Q, then we will only be able to determine whether the derivations we design are correct with respect to P. Our model of derivations will predict that sentences that have P in their derivation will also have the meaning associated with P, and that sentences which don't contain P in their derivation won't have the meaning associated with P (unless that meaning can be derived from some other parse in the derivation). A model of semantic interpretation of this kind will not allow us to determine anything about the distribution of Q in the parses of a sentence, however. As a consequence, we will not be able to determine whether one theory of derivations is better than another by virtue of what those theories predict about Q. This is one place, then, where the model we have of semantic interpretation and the model we have of syntactic representation are intertwined. They cannot be developed independently.

This section illustrates this relationship between the semantic model used and the kinds of derivations it fits. I will do this by considering a popular extension of Trace Theory that allows traces to be variables of different semantic types. A consequence of this alternative semantics for traces changes the way in which reconstruction arises, and therefore gives a different image of how derivations fit to semantic interpretations. As we'll see, this extension to Trace Theory also affects how we can model the syntax/semantics interface. It will provide a way of modeling ambiguity that allows one parse in a derivation to provide two denotations that in our present theory can only be delivered by considering two parses in a derivation. The extension, then, not only gives us a different view of the parses that make up derivations, but also a different view of how those parses need to be sampled in order to get the correct meanings for a sentence.

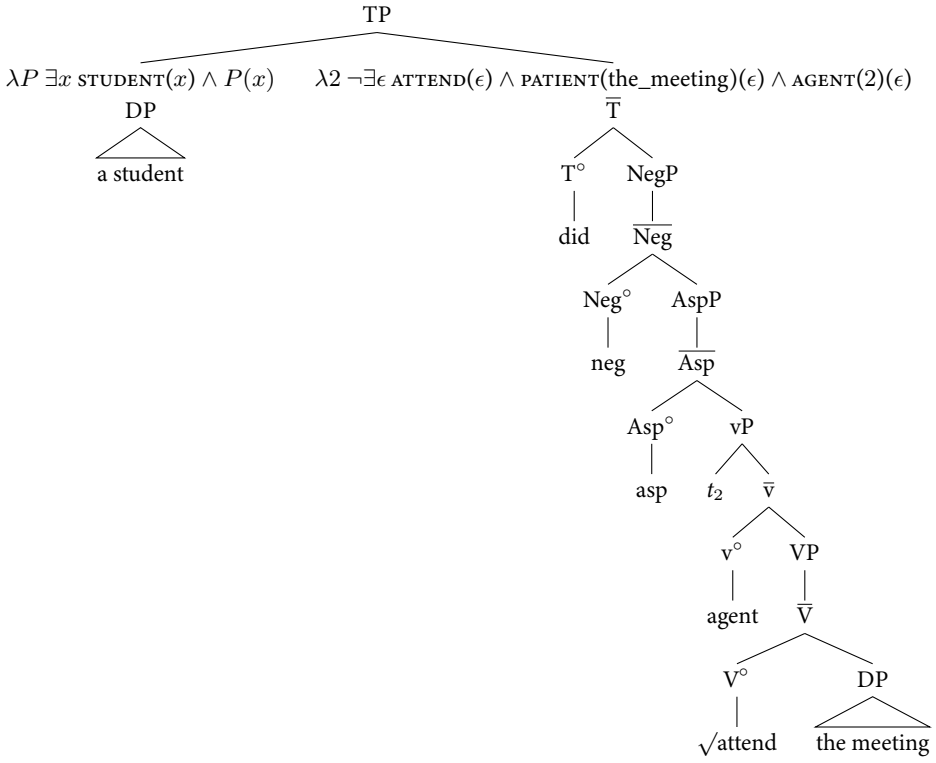
Perhaps the most extensive development of the technique that I'll describe here can be found in Cresti (1995).

In our present theory, traces are interpreted as variables over expressions that are entities. That is, the functions that movement creates has entities as its domain. Let's reconsider how this works in (171).

(171) A student didn't attend the meeting.

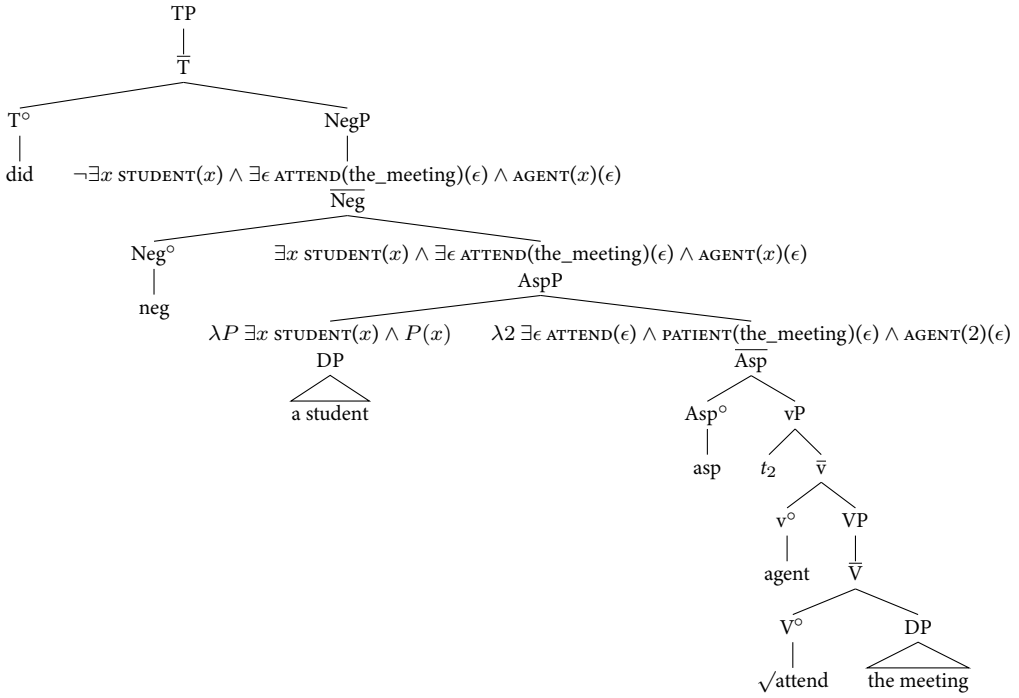
Argument movement of *a student* allows for a trace to be left in the Specifier of vP, from which it moves, and causes the $\overline{\text{T}}$ that *a student* ends being a sister to to be interpreted as a function with that trace as its variable. This is illustrated in (217), which we can think of as approximating the S-structure representation of (171).

(217) $\exists x$ student$(x) \wedge \neg\, \exists \epsilon$ attend$(\epsilon) \wedge$ patient(the_meeting)$(\epsilon) \wedge$ agent$(x)(\epsilon)$



This parse provides a denotation that puts *a student* outside the scope of negation, and its existence in the derivation for (171) is responsible for allowing (171) to have this meaning. Our syntax also allows the derivation for (171) to include (218). This parse, which corresponds to neither the S-structure nor the D-structure for (171), has a denotation that puts *a student* within the scope of negation. Its existence, then, is responsible for (171) having this denotation as well. The S-structure representation for (171) can be derived from (218) by using movement to bring *a student* into the Specifier of TP. And it is, in turn, derived by moving *a student* into Specifier of AspP, from its D-structure position in Specifier of vP. As in (217), the interpretation of (218) uses a semantics of movement that leaves a trace that is interpreted as a variable over entities. And for that reason, the function that $\overline{\text{Asp}}$ is converted into by movement is one whose domain is entities.

(218) $\neg\exists x$ STUDENT$(x) \wedge \exists\epsilon$ ATTEND(the_meeting)$(\epsilon) \wedge$ AGENT$(x)(\epsilon)$



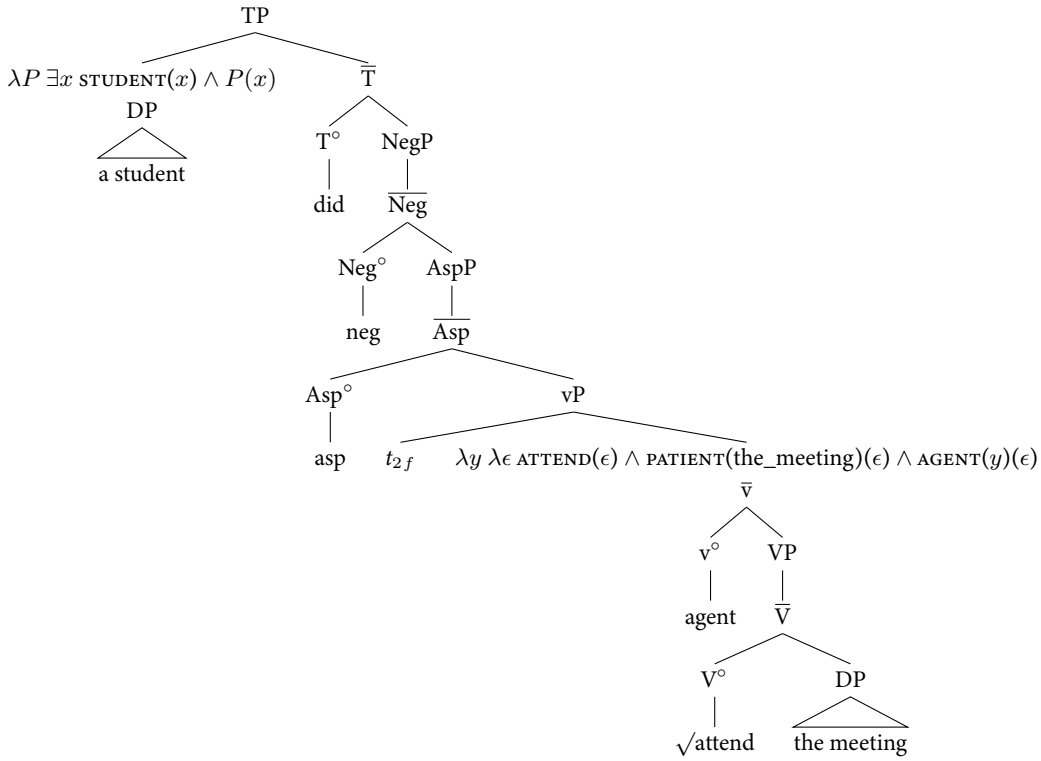
Consider now an alternative semantics that allows a trace to be a variable over things of various semantic types. In addition to being a variable over things, or "entities," for instance, imagine that a trace could be interpreted as a variable over objects of the same semantic kind of thing that the moved phrase is. For instance, if the moved term has a denotation which makes it a function, let the trace left by that term be a variable over the same functions. I'll indicate the semantic type of a trace with a subscript on the index it bears. The symbol in (219a), for instance, will indicate a trace which will be interpreted as a variable over things, or "entities." The symbol in (219b), by contrast, will indicate a trace whose denotation makes it a variable over functions of the same kind as the term whose movement creates the trace.

(219) a. t_{2e}

b. t_{2f}

Allowing this flexibility to the semantics of traces doesn't change how things work in (217) and (218), because in these parses the trace cannot compose semantically with its sister unless it is of entity type. To see this, consider (217), but with the type of the trace adjusted so that it matches that of the moved DP, *a student*.

(220)



Consider how the trace in Specifier of vP combines semantically with its $\bar{v}$ sister in (220). Because the domain of the function AGENT is entities, the function that $\bar{v}$ denotes also has that domain. If the semantic composition were to treat the trace as the argument of the function denoted by $\bar{v}$ the result would be (221), which gives *agent* an argument of the wrong semantic type.

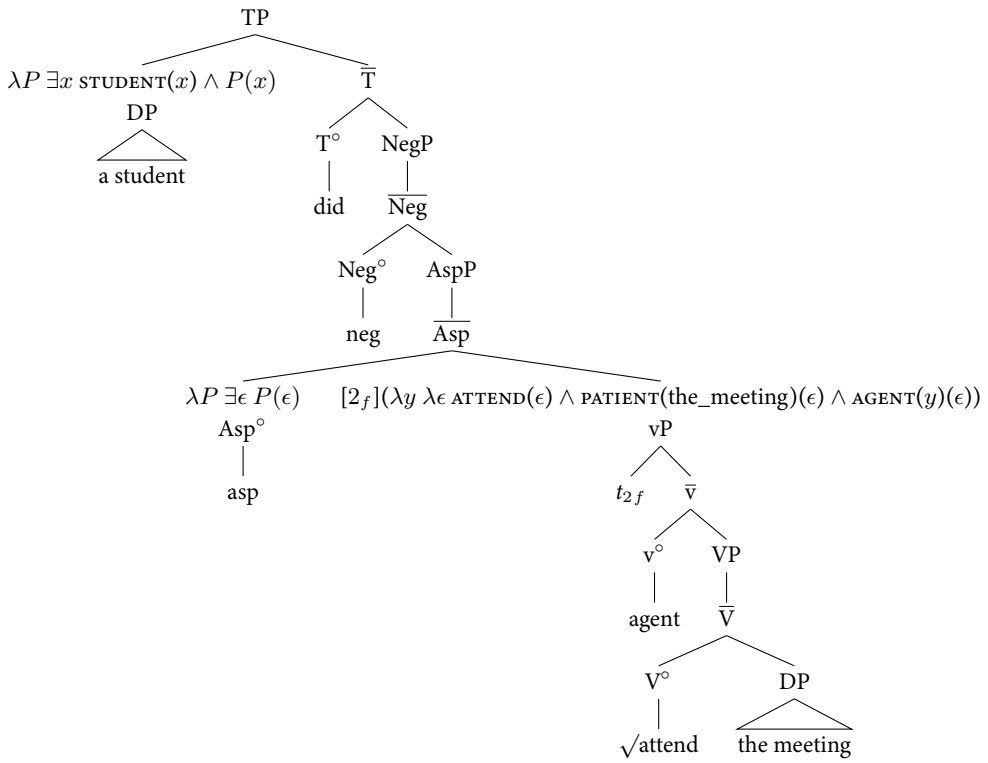
(221) $\neg\exists\epsilon$ ATTEND(ϵ) $\wedge$ PATIENT(the_meeting)(ϵ) $\wedge$ AGENT(t_{2f})(ϵ)

However, the trace is a variable over functions, and this is therefore also the semantic type of the domain of the function that the denotation of $\bar{v}$ is. Thus, the trace can take $\bar{v}$ as its argument. We can represent this with (222).

(222) $[\![\text{vP}]\!] = [2_f](\lambda y\ \lambda\epsilon$ ATTEND(ϵ) $\wedge$ PATIENT(the_meeting)(ϵ) $\wedge$ AGENT(y)(ϵ))

A snapshot of the parse in (220) with the relevant denotations revealed is (223).

(223)

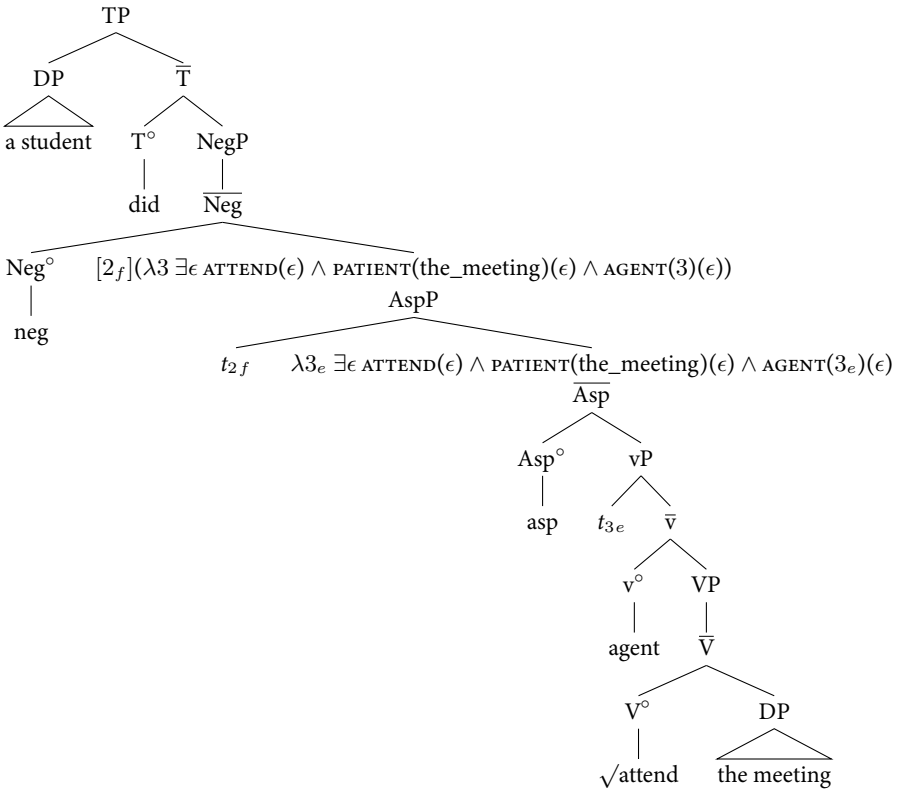


The domain of *asp* is functions from eventualities to truth values. It takes a description of eventualities and creates a sentence that says there is such an eventuality. The semantic type of vP in (223) isn't the right semantic type for *asp*. It isn't a description of eventualities, but is instead a complicated sentence: whatever value the index 2 gets applied to the argument enclosed in parentheses. For this reason, taking the $\bar{v}$ in (220) to provide an argument to the trace in Specifier of TP, also fails to provide a denotation of this sentence.

Our semantics, then, blocks an interpretation of (217), where the trace created by moving *a student* is interpreted as a variable over the same functions that *a student* is. For entirely parallel reasons, letting the trace in (218) have that semantic type will fail to yield a denotation. For these two cases, then, allowing traces to be anything but variables over entities has no consequences. Our semantics doesn't allow traces to exploit that flexibility in these parses. The denotations of the other material in the parse fits together only if the traces in (217) and (218) are interpreted as variables over entities.

Where allowing traces to be variables over functions has consequences is when traces aren't arguments of θ-roles. To see this, consider the parse that arises from movement applying to *a student* in (218). This movement operation can leave a trace in Specifier of AspP, and this trace can successfully be a variable over things of the same semantic type that *a student* is. (224) is a sketch of that parse.

(224)

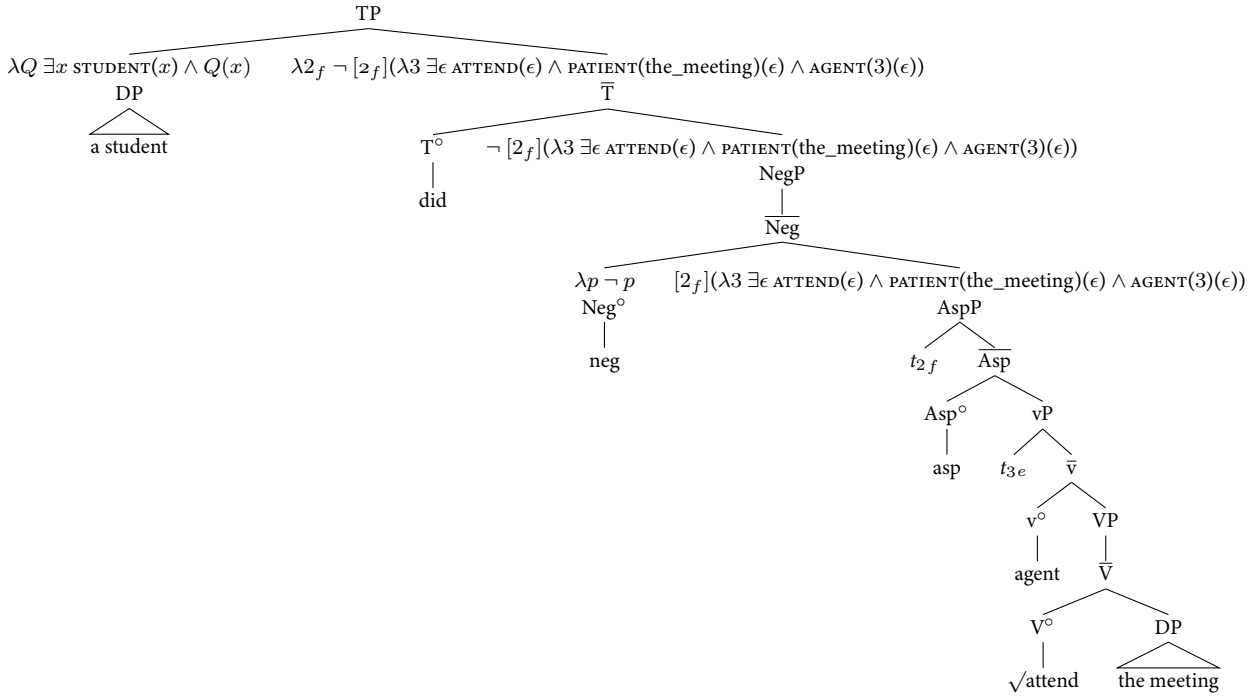


Note that movement of *a student* into Specifier of AspP from Specifier of vP has left a trace of entity type. This is going to give the function AGENT something of the type that is in its domain. When subsequent movement of *a student* into Specifier of TP occurs, it's this movement that leaves a trace of function type. If we assume that the denotation of *neg* is (225), then it's possible for the denotation of AspP and Neg° to semantically combine.

(225) $[\![\text{neg}]\!] = \lambda p \neg p$

This delivers the denotations indicated in (226).

(226)



The denotation of $\overline{\text{T}}$ in (226) is a function that can take the denotation of *a student* as its argument. Applying $[\![\overline{\text{T}}]\!]$ to the denotation of *a student* gives to TP the denotation in (227).

(227) $[\![\text{TP}]\!] = \neg\,[\lambda Q\ \exists x\ \textsc{student}(x) \wedge Q(x)](\lambda 3\ \exists \epsilon\ \textsc{attend}(\epsilon) \wedge \textsc{patient}(\text{the_meeting})(\epsilon) \wedge \textsc{agent}(3)(\epsilon))$

This formula can be simplified by applying the function in "[]" to the argument in "()." The result is (228).

(228) $[\![\text{TP}]\!] = \neg\ \exists x\ \textsc{student}(x) \wedge \exists \epsilon\ \textsc{attend}(\epsilon) \wedge \textsc{patient}(\text{the_meeting})(\epsilon) \wedge \textsc{agent}(x)(\epsilon)$

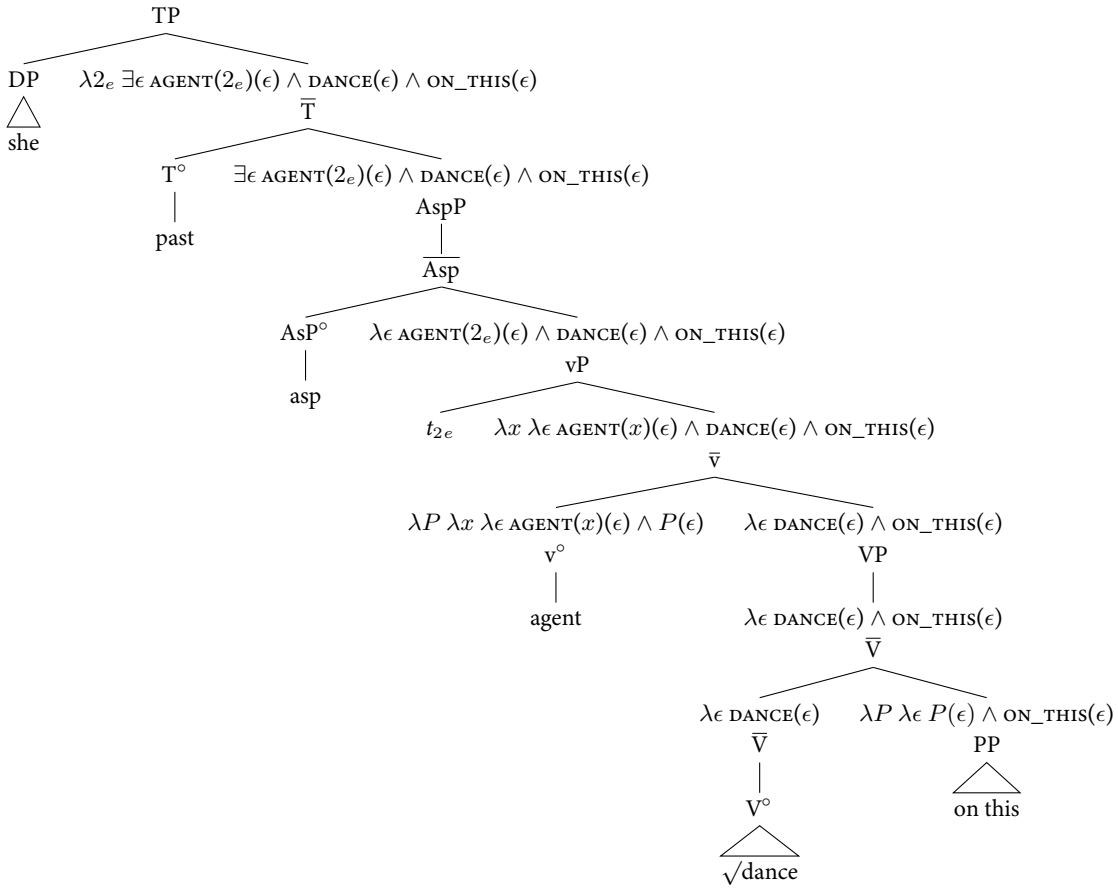
This is a denotation in which *a student* is part of what is negated; it is one of the meanings that we've seen this sentence can get.

Allowing traces to be interpreted as variables over functions of the same semantic type as the moved phrase, then, provides a way of semantically interpreting (226), which corresponds (roughly) to the S-structure of (171), and deliver the negation wide scope interpretation. This alternative semantics, then, changes which parses in the derivation we need to say are semantically interpreted. To capture the ambiguity of (171), we needn't require that more than the S-structure parse be semantically interpreted. The ambiguity arises as a function of the flexibility of the semantic type that traces may have. When the derivation puts a trace in Specifier of AspP, and that trace is interpreted as a variable over functions of the kind that *a student* is, then the denotation of *a student* falls within the scope of negation. When, by contrast, the trace in Specifier of AspP is interpreted as a variable over entities, or alternatively the derivation doesn't put a trace in Specifier of AspP, the result will put the denotation of *a student* outside the scope of negation.

The stepwise explication of how the semantics works if the trace in Specifier of AspP is interpreted as a varaible over entitites isn't provided here. I recommend it as an exercise for the student.

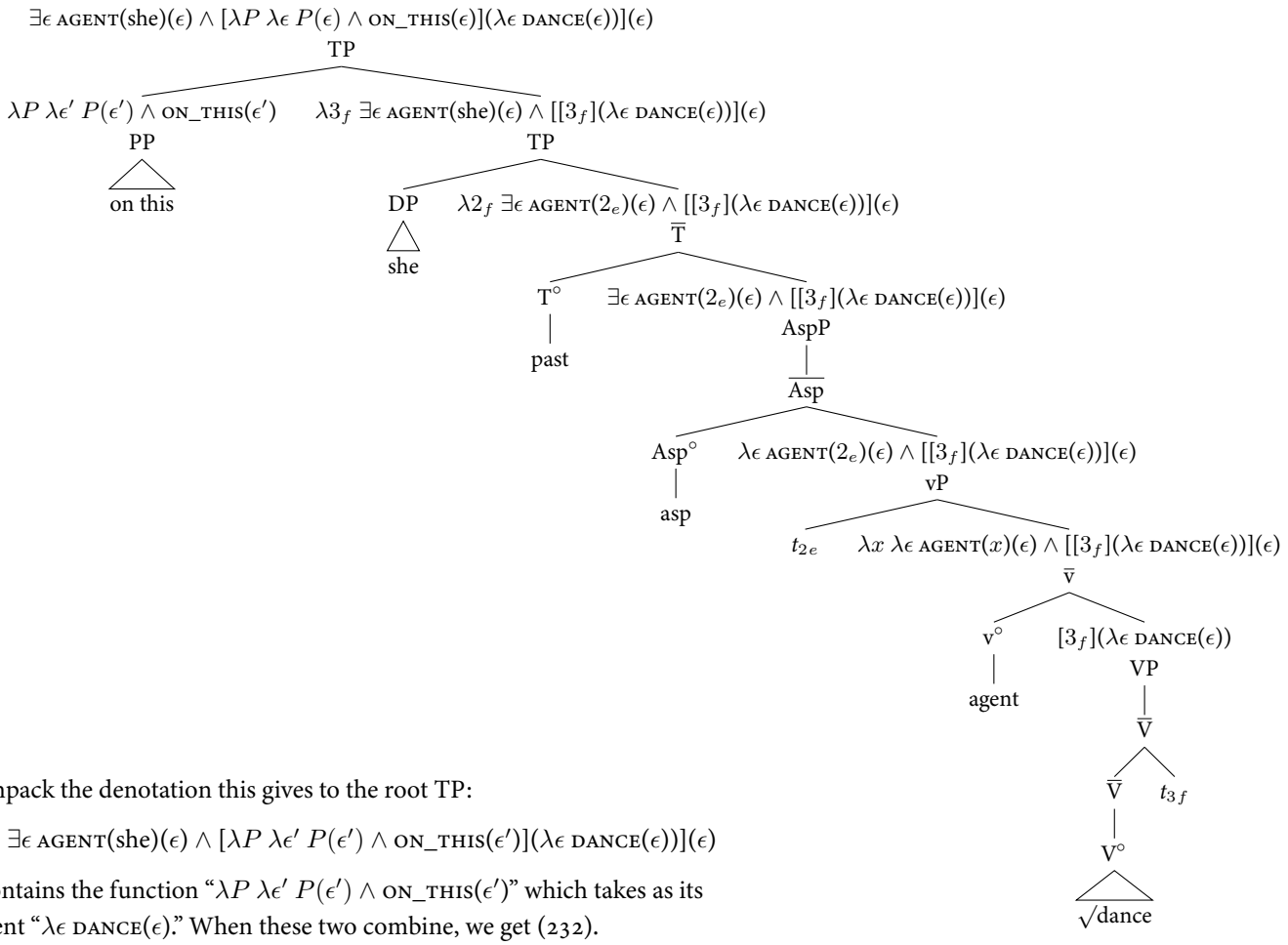
In our present syntax, only movement of DPs is capable of leaving a trace, and this corresponds to the decision to let traces be interpreted as variables over things. If the semantics of traces is broadened to allow them to be variables over other kinds of things, then the syntax of movement can be broadened to allow for traces in other contexts. Consider, for instance, the case of moving PPs that modify VPs. For concreteness, suppose that some PPs have denotations that allow them to combine with VPs in the way shown in (229).

(229) $\exists\epsilon$ AGENT(she)(ϵ) $\wedge$ DANCE(ϵ) $\wedge$ ON_THIS(ϵ)



Consider now the S-structure that could be created from movement of these PPs in a syntax that allows traces to be created and interpreted more broadly.

(230) On this, she danced.



Let's unpack the denotation this gives to the root TP:

(231) $\exists\epsilon$ AGENT(she)(ϵ) $\wedge$ [$\lambda P\ \lambda\epsilon'\ P(\epsilon') \wedge$ ON_THIS(ϵ')]($\lambda\epsilon$ DANCE(ϵ))](ϵ)

This contains the function "$\lambda P\ \lambda\epsilon'\ P(\epsilon') \wedge$ ON_THIS(ϵ')" which takes as its argument "$\lambda\epsilon$ DANCE(ϵ)." When these two combine, we get (232).

(232) $\exists\epsilon$ AGENT(she)(ϵ) $\wedge$ [$\lambda\epsilon'$ [$\lambda\epsilon$ DANCE(ϵ)](ϵ') $\wedge$ ON_THIS(ϵ')](ϵ)

Applying "$\lambda\epsilon$ DANCE(ϵ)" to "ϵ'" produces:

(233) $\exists\epsilon$ AGENT(she)(ϵ) $\wedge$ [$\lambda\epsilon'$ DANCE(ϵ') $\wedge$ ON_THIS(ϵ')](ϵ)

This expression contains the function "$\lambda\epsilon'$ DANCE(ϵ') $\wedge$ ON_THIS(ϵ')" which is given the argument "ϵ." These combine to form:

(234) $\exists\epsilon$ AGENT(she)(ϵ) $\wedge$ DANCE(ϵ) $\wedge$ ON_THIS(ϵ)

This is, arguably, the correct denotation. At least it puts the pieces together in such a way that (230) matches the meaning of its counterpart in (229).

Note that this alternative semantics doesn't change the model we need for derivations; at least not for the cases illustrated by (171). Both semantics require derivations that include a parse that has *a student* occupy Specifier of AspP. The old semantics requires a model of the syntax/semantics interface that lets that parse provide a denotation for the sentence. The new semantics allows for a model of the syntax/semantics interface that doesn't require

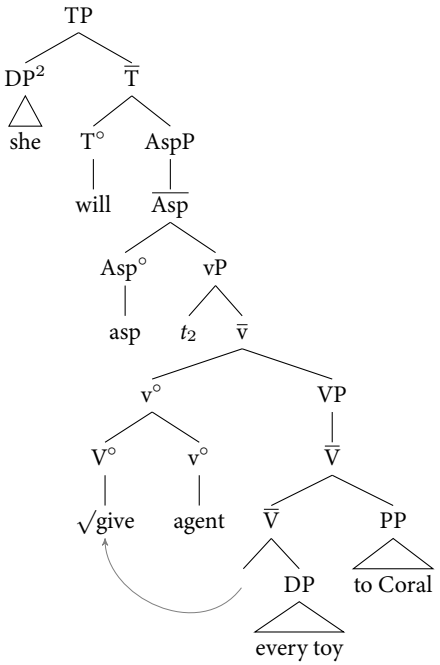
that parse to be semantically interpreted. Instead, the S-structure parse of the derivation has sufficient information in it to provide both interpretations. Letting the trace in Specifier of AspP be interpreted as a variable over functions of the right kind, when combined with the definition of λ-expressions, provides a semantic computation that evaluates the denotation of the phrase in Specifier of TP in Specifier of AspP. It provides for a method of expressing reconstruction that relies on this feature of the rules of semantic composition. This method is sometimes called "semantic reconstruction," and it is contrasted with "syntactic reconstruction," which hypothesizes instead that the ambiguities involved are provided by different syntactic parses. Note that because the syntax and semantics employed by these two treatments are very similar, it is difficult to find evidence that distinguishes them. They are not so much alternatives as different executions of the same project.

In section 6.2.1.2, we saw that our present syntax struggles to deliver a derivation for certain sentences that have both movement of a verb and a quantificational DP in them. Embracing this alternative semantics can be combined with a slightly different syntax of head movement to allow for derivations that make these cases work smoothly.

The problem is illustrated by (235), whose S-structure representation puts the direct object and verb in the positions indicated in (236).

(235) She will give every toy to Coral.

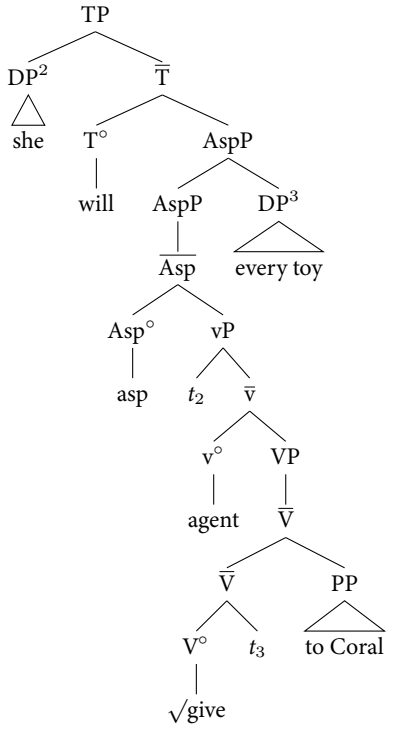
(236)



This S-structure isn't semantically interpretable for two reasons: the quantificational DP must be moved outside AspP for it to be interpreted and neither $\sqrt{give}$ nor [$_{v^\circ}$ agent] are in positions where they can be semantically combined with its surrounding material. Moreover, if derivations obey Upwards,

then there is no derivation that contains (236) which is semantically interpretable. Derivations that obey Upwards require the parse in (237) to be part of the derivation, for that is the one our semantics provides the right denotation from.

(237)



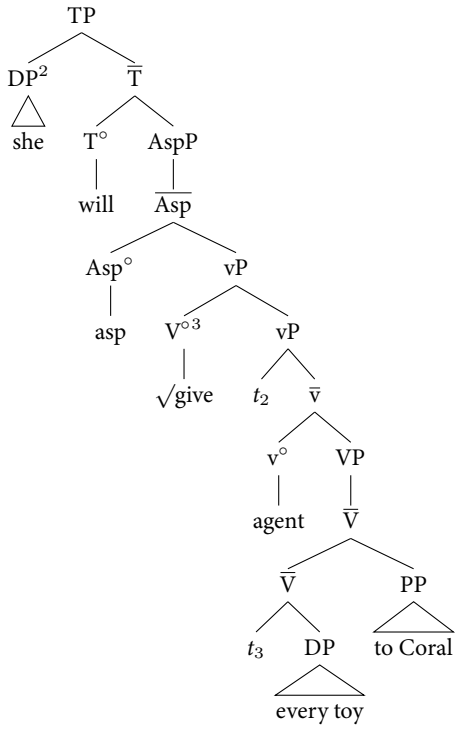
Our present rules do not make this a parse that can be achieved from the S-structure in (236), nor can it be part of the parses that lead to (236).

Consider now the parses that make up a derivation for this sentence in a theory that lets traces be able to be interpreted as variables of the same semantic type as the moved term that they are related to, and a syntax that includes the following changes.

(238) a. Moved heads can leave a trace.
b. Moved heads can adjoin to a phrase.

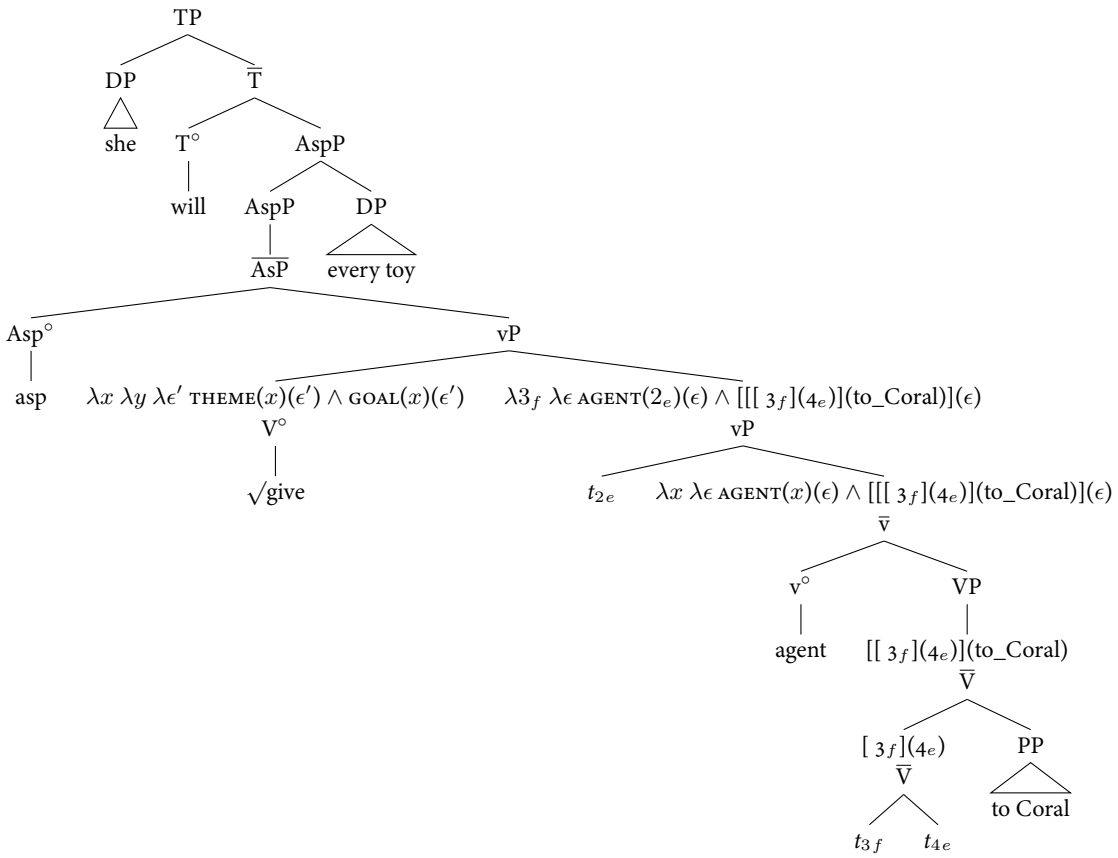
This syntax allows the S-structure of (235) to have the representation in (239).

(239)



The derivation that leads to this parse exploits the option of moving a head, here *√give*, to a phrase and leaving a trace. In this case, *√give* has moved to a position left-adjoined to vP, and left a trace in its D-structure position. Like (236), (239) isn't semantically interpretable because *every toy* isn't in a position where the rules of semantics allow it to be combined with its surrounding material. But unlike (236), it's possible to create a parse from (239) that preserves Upwards and puts *every toy* in a position that it can be semantically interpreted. That parse is (240).

(240)



Note that the denotation shown for the lowest vP in (240) includes the change that Movement has wrought; Movement causes the sister to a moved term to turn into a λ-expression that binds the variable created by the trace. In (240), that trace is left by movement of $\sqrt{}$*give*. The function that is created by this is one whose domain includes functions of the kind that $[\![\sqrt{}give]\!]$ is. Therefore, the way that this vP and $\sqrt{}$*give* combine semantically is by taking the denotation of $\sqrt{}$*give* to be the argument of the function that vP is. The result is (241).

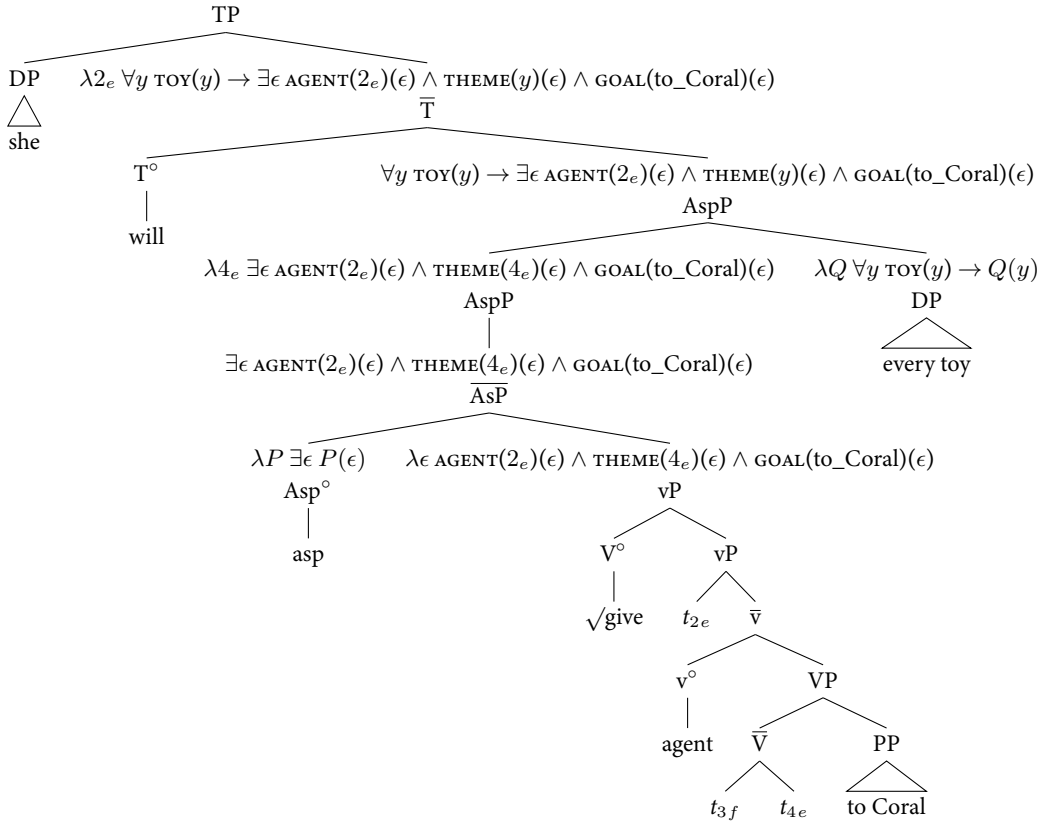
(241) $[\![\text{vP}]\!] = \lambda\epsilon\ \textsc{agent}(2_e)(\epsilon) \wedge [[[\lambda x\ \lambda y\ \lambda\epsilon'\ \textsc{theme}(x)(\epsilon') \wedge \textsc{goal}(y)(\epsilon')](4_e)](\text{to_Coral})](\epsilon)$

By applying the function "$\lambda x\ \lambda y\ \lambda\epsilon\ \textsc{theme}(x)(\epsilon) \wedge \textsc{goal}(y)(\epsilon)$" recursively to its arguments, we can reduce this formula to:

(242) $[\![\text{vP}]\!] = \lambda\epsilon\ \textsc{agent}(2_e)(\epsilon) \wedge \textsc{theme}(4_e)(\epsilon) \wedge \textsc{goal}(\text{to_Coral})(\epsilon)$

Plugging this into the syntactic representation gives (243).

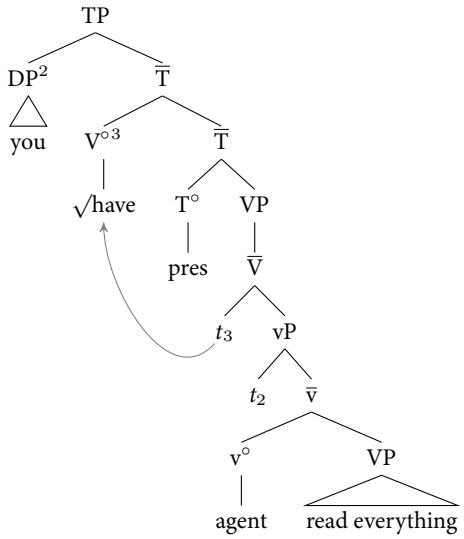
(243) $\forall y$ TOY(y) $\rightarrow \exists \epsilon$ AGENT(she) $\wedge$ THEME(y)(ϵ) $\wedge$ GOAL(to_Coral)(ϵ)



In calculating the denotation of $\overline{\text{T}}$, I've ignored the contribution that *will* makes, and this absence is therefore present in the denotation given for the resulting TP. But with that caveat, the syntax and semantics of this parse delivers the right result.

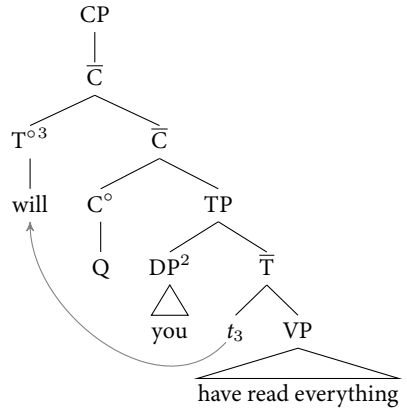
Adopting this alternative semantics, then, can be wedded to a different syntax of Head Movement that allows for derivations that obey Upwards and overcomes the problem posed by examples like (235). But the syntax it requires of Head Movement is problematic. For a moved verb to bind its trace, it must move to a position that c-commands that trace. In the cases we've just worked through, this involves moving V° so that it left-adjoins to a projection of v°. But the same kind of syntax would be required of moved verbs whenever there is a quantificational DP in the sentence. So for instance, it would require that an auxiliary verb in English move to some projection of T° in a finite clause, perhaps in the way sketched in (244).

(244)



We've also learned that in question contexts, the contents of T° move to C°. In a syntax that allows Heads to move to c-commanding positions, this might produce representations like (245).

(245)



We've seen that when these two operations are both present, the result should put the auxiliary verb in the position to which T° moves.

See chapter 4 for details.

(246) Have you read everything?

The requires a syntax that causes the moved auxiliary verb to combine with the tense features in T° so that they are one X°. That is, they must have the same status that the T° *will* has in (245). That is not what the syntax in (244) delivers. A syntax that requires a moved verb to c-command the position it moves from is in conflict with the need to form a new head from the moved verb and the X° whose needs are being satisfied by the movement. If the syntax/semantics sketched in this section is to solve the problem posed by head movement that is discussed in section 6.2.1.2, then we must find a way of resolving this incompatibility.

See Matushansky (2002) for a suggestion.

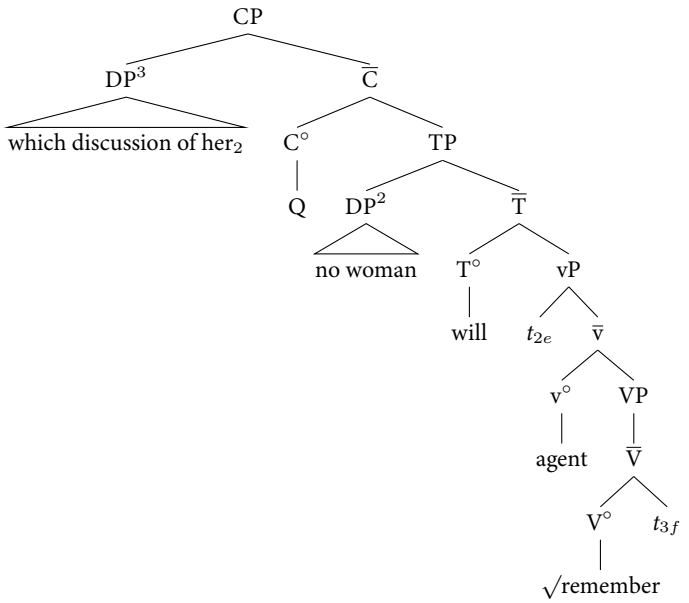
As it stands, the Trace theory sketched in this section also doesn't provide a straightforward solution to the cases of reconstructed variables, discussed in

section 6.2.1.3. These cases arise when a moved phrase contains a term that is interpreted as a variable bound by a lower phrase. (247) illustrates.

(247) Which discussion about her$_2$ will no woman^2 remember?

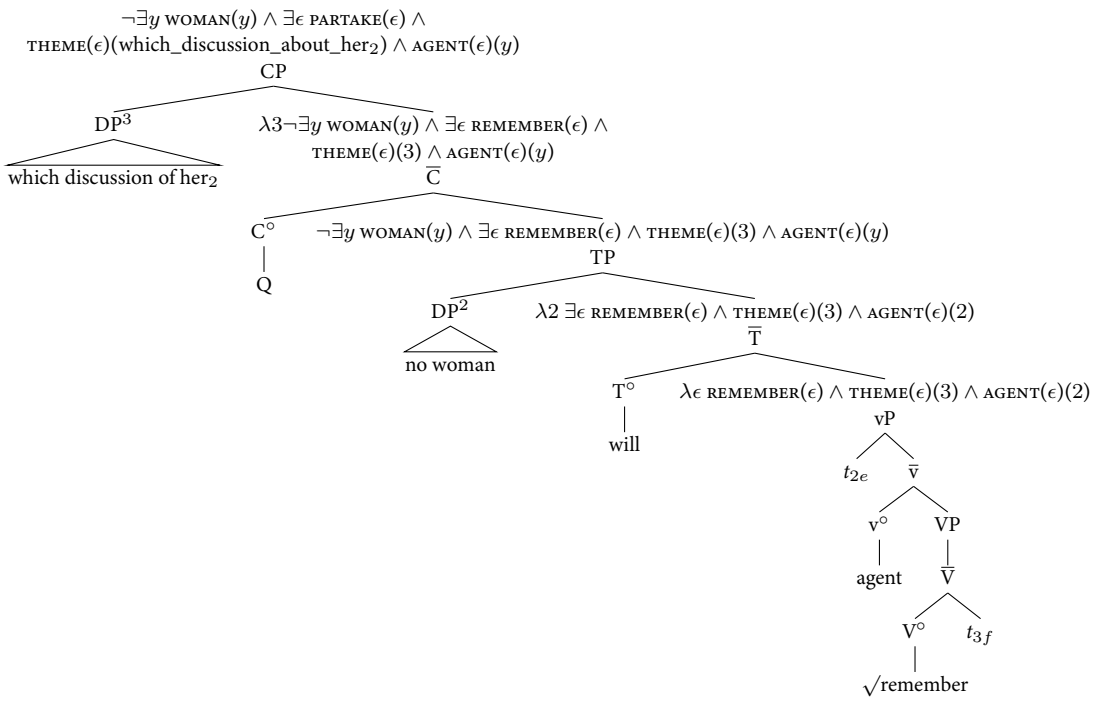
A sketch of the relevant parse for (247) is in (248).

(248)



We have not determined a semantics for wh-phrases like *which discussion of her*. Let us suppose, for the purposes of illustration, that it refers to an entity—it is of semantic type e. If so, even if its trace matches it in semantic type (as indicated in (248)), it will be of type e, and this gives us the semantics in (249).

(249)



Note that the index on *her* is not part of the constituent that is a sister of "$\lambda 2$." For this reason, it is not bound by $\lambda 2$, and so we do not get a well-formed result.

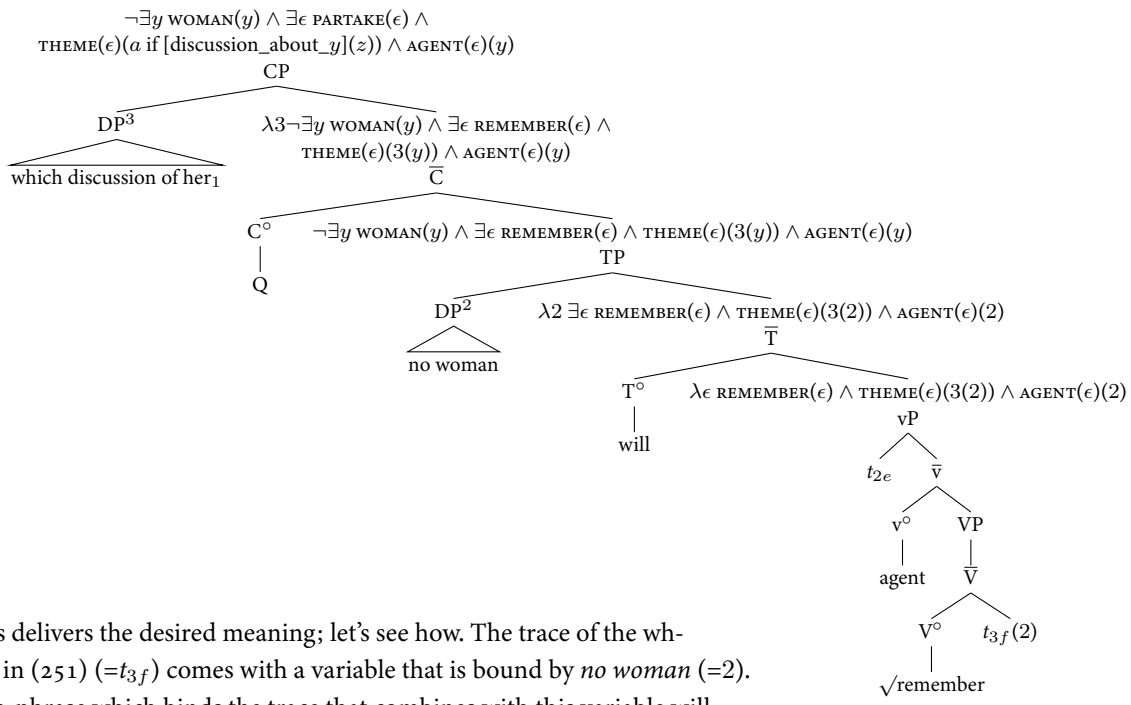
Moreover, the index on ***her*** is prevented from getting bound by "$\lambda 2$" because of the law that prevents variables inside the domain of a λ-operator from being bound by that λ-operator. See page 154 for discussion.

The way of bringing these cases in line with the Trace theory of this section is to give to wh-phrases a more complex semantics. Suppose, for instance, that wh-phrases could be functions from entities to entities. The entities of their domain are the values that a variable they contain has, and the entities of their range are the entities that they refer to. For instance, imagine that *which discussion of her*$_2$ could have the denotation in (250).

(250) $[\![\text{which discussion of her}_1]\!] = \lambda 1\ a$ if DISCUSSION_OF_1(a)

The function defined in (250) applies to entities, x, and returns an entity, a, just in case it is a discussion of x. Further, suppose that traces can be more than simple variables of various semantic types. Imagine that they can themselves take arguments when they are the type of a function. Suppose, that is, that $t_f(n)$, where n is a referential index, can be put into the position from which a phrase moves. This allows (247) to have the representation in (251).

(251)



This delivers the desired meaning; let's see how. The trace of the wh-phrase in (251) (=t_{3f}) comes with a variable that is bound by *no woman* (=2). The wh-phrase which binds the trace that combines with this variable will semantically combine with the variable, if the trace has the same semantic type as the wh-phrase. Schematically, then, (252a) yields (252b).

(252) a. which discussion of her$_2$ λ3...no woman λ2...remember 3(2)

b. ...no woman λ2...remember [λ1 z if DISCUSSION_OF_1(z)](2)

If we reduce the the "[λ1 z if DISCUSSION_OF_1](2)" expression by applying the function to the argument, we get from (252b) what's schematized in (253).

(253) ...no woman λ2...remember z, if DISCUSSION $_o$F_2

In this way, then, we can model cases of reconstructed variables in Trace theory by letting traces contain variables that can be bound by other terms in the sentence.

While this method technically brings reconstructed variables within range of the Trace theory of movement, it doesn't do so in a way that can capture an important fact about these cases. That fact is that the morphological form of the reconstructed variable determines the conditions under which it can be bound under reconstruction. First, let me illustrate that the morphological form of a variable can determine the conditions under which it is bound. This is illustrated by the contrast of interpretations that are available to the sentences in (254).

(254) a. No woman will remember someone's discussion of her.

b. No woman will remember someone's discussion of herself.

Both *her* and the reflexive pronoun *herself* can be interpreted as bound variables. In (254), for instance, *herself* can be interpreted as a variable bound by

someone. But only in (254a) can *no woman* bind the pronoun; that is, *her* but not *herself* can be interpreted as a variable bound by *no woman*. This is because reflexive pronouns, unlike non-reflexive pronouns, can only be bound by DPs that are sufficiently close. One way of describing that condition is with the requirement that a reflexive pronoun be bound by the closest DP that can bind it. Because *someone* can bind *herself* in (254b), this requirement would force *herself* to be understood as a variable bound by *someone* rather than *no woman*. It would also require *herself* to be interpreted as a variable bound by *someone* rather than finding a contextually salient person to refer to, as *her* can. That is also a difference between (254a) and (254b). And, finally, it would explain why both *her* and *herself* can be bound by *no woman* in (255).

(255) a. No woman will remember this discussion of her.
b. No woman will remember this discussion of herself.

Because there is no genitive DP *someone's* in (255b), the closest possible binder for *herself* is *no woman* and this is what *herself* can, and must, be bound by.

Consider now the contrast in (256).

(256) a. Whose discussion of her$_2$ will no woman^2 remember?
b. * Whose discussion of herself$_2$ will no woman^2 remember?

Parallel to (254), only (256a) allows the pronoun to be bound by *no woman*. In (256b), *herself* must be bound to *whose*, as this is the closest possible binder. Just as in (255), when *whose* is absent, *herself* can be bound by *no woman*:

(257) Which discussion of herself$_2$ will no woman^2 remember?

Paradigms like these are examined, perhaps for the first time, by Barss (1986).

These contrast aren't explicable under Trace theory, even when embellished in the way just described. What's needed is a theory that allows binding of reconstructed variables by way of a syntactic representation that puts the phrase containing the variable in the position it needs to be bound. In a Trace theory that allows a trace to contain a variable which, indirectly, does the job of binding the pronoun in the moved expression—as, for example, sketched above—there is no way of capturing the fact that the constraints on binding that come with the pronoun matter. We need a way of putting the pronoun itself into the relevant position for binding in order to capture this fact.

What we've seen in these sections, then, is that Reconstruction phenomena pose a kind of ordering problem for Trace theory. There is no way of capturing these effects with Trace theory that is consistent with Upwards. If derivations obey Upwards, they cannot provide the parses needed to capture certain cases of Reconstruction and still deliver the S-structures that represent those derivations. As we've seen in this section, it is possible to preserve Upwards using a version of Trace theory that allows the trace to have the same denotation as the binder. This allows more of the parses in a derivation to be

semantically interpretable and one of the consequences of this is a treatment of reconstruction that fits Upwards constrained derivations. But broadening the interpretation of traces in this way has its own problems. It doesn't fit well the syntax we've given Head Movement, which requires a head to move to a position from which it cannot bind a trace. And it doesn't capture the cases of Reconstruction of pronominal variables in a way that enforces the constraints imposed by the pronouns that are reconstructed.

What's needed to shore up these problems is a way of thinking of traces as having the same meaning as the term that moved, whether or not they are variables bound by that moved term. That will allow for Upwards obeying derivations in the way that the Trace theory of this section does, but fit the syntax of Head Movement better. And we need to understand trace to be able to have the same syntactic content as the moved phrase. That will capture the cases of Reconstructed pronominal variables in a way that preserves the conditions on binding that come with the choice of pronoun. It's providing these two attributes that the Copy Theory of movement is designed for.

6.2.2 Copies

Instead of "traces," this theory of movement creates "copies," and it is named after this feature of the theory.

That movement involves putting copies of the moved term in different positions is adopted in Ross (1967), and given perhaps its first defense in Ross (1969). But it is revived to handle these semantic problems in Chomsky (1993), and developed in Chomsky (1995). It's Chomsky's version of this theory that will be discussed here.

There are various ways of formulating the Copy theory of movement, but I'll use what strikes me as the most common of those ways. In this formulation, movement is factored into three parts. The first is an operation that creates the "copy."

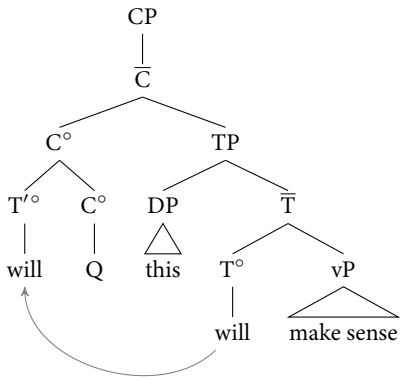
(258) Copy
For α a constituent in a phrase marker, copy(α)=α', where α and α' are syntactically, semantically, and morphologically exactly the same.

The copy operation is used by the move operation.

(259) Move
α moves to β means that copy(α) is put into the position β.

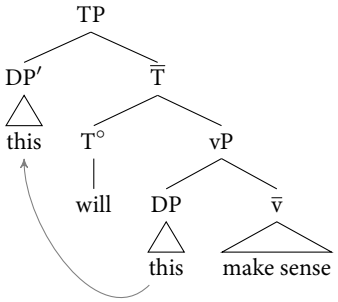
On this definition, Head Movement creates representations like (260).

(260)



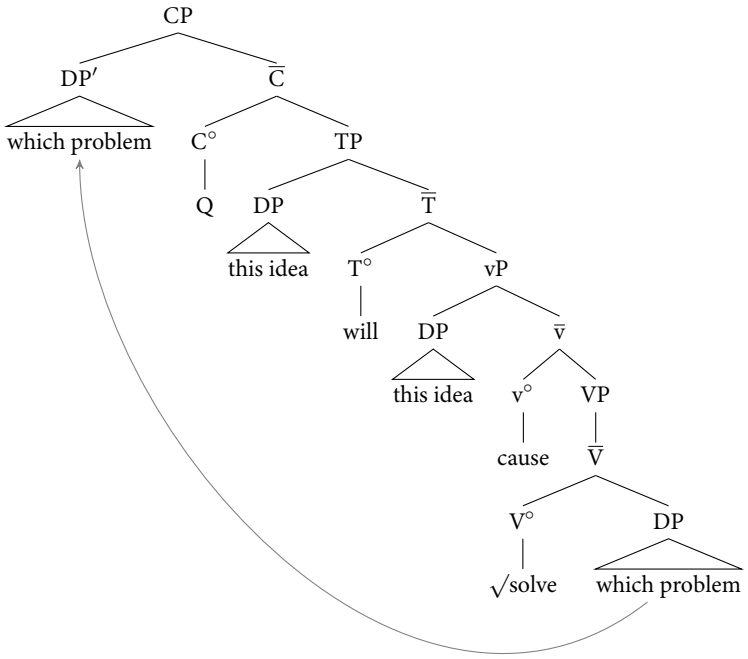
Argument Movement creates representations like (261).

(261)



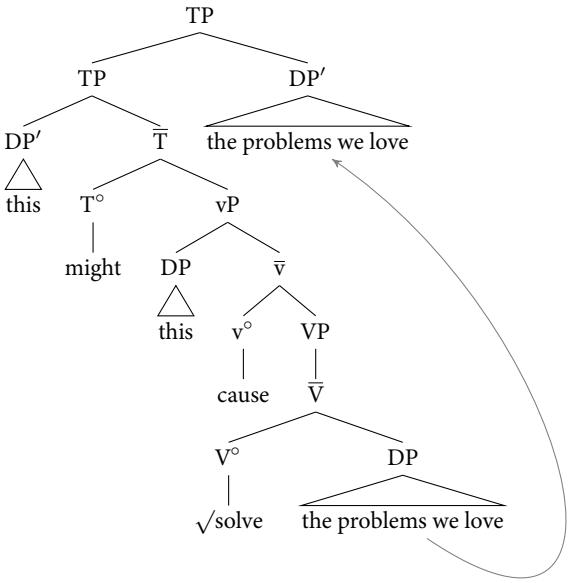
Wh Movement creates representations like (262).

(262)



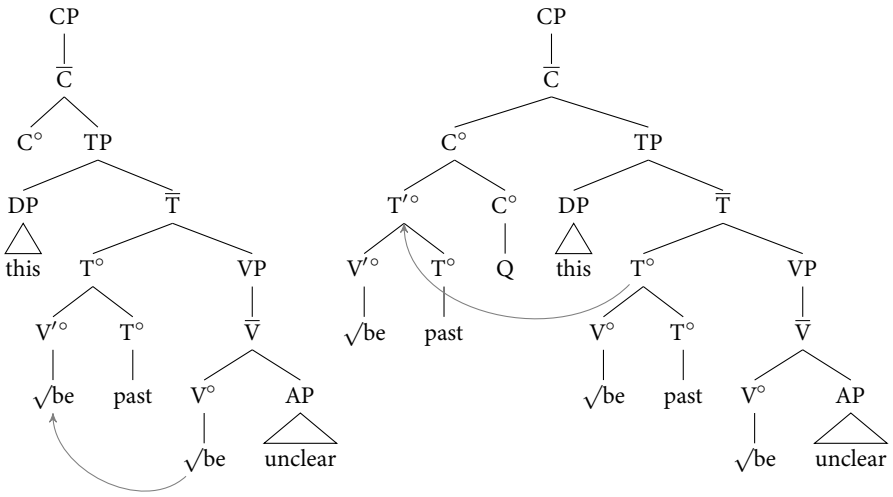
And, finally, Complex NP Shift creates representations like (263).

(263)



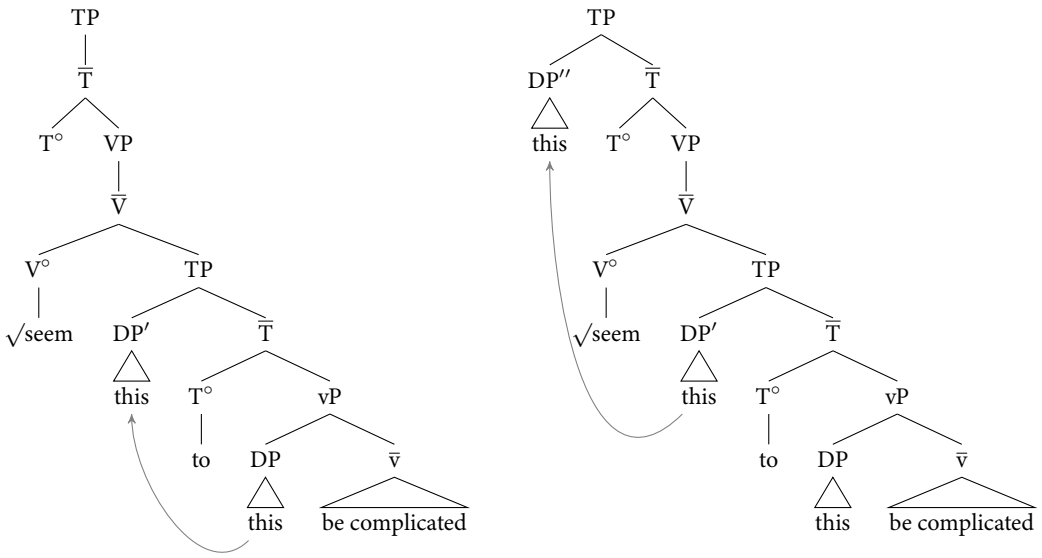
When Movement happens to a term more than once, it will invoke additional instances of COPY. Thus, for instance, Head Movement can produce derivations like (264).

(264)



And Argument Movement can produce derivations like (265).

(265)



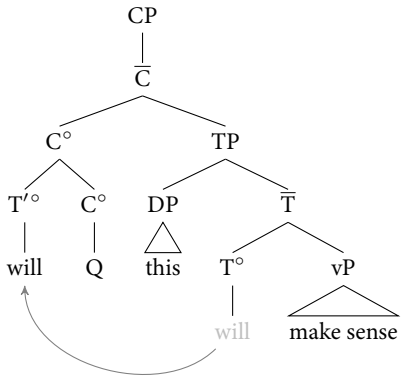
In what follows, I will not distinguish the copy from the thing it is copied from in parses of movement. (That is, I will represent "XP′" with "XP".) And I will adopt a correspondingly sloppy habit of speech: I will call all the phrases that are related by COPY "copies," even though technically one is a copy of the other.

Obviously, this rule incorrectly results in two pronunciations of the moved term. So, it must be wedded to a rule that prevents pronunciation of more than one copy. Let's know this rule as Suppress Copy.

(266) Suppress Copy
Exactly one copy can be in the linearization.

In all of the examples above, Suppress Copy is satisfied by suppressing all but the highest copy. It would be reasonable to strengthen Suppress Copy so that it captured this generalization; it could be expressed as allowing only the highest copy to be linearized. When we discuss QR, we will see reasons for adopting this weaker version. But in the end, a means for determining which copy is linearized will be necessary, as in the normal case, it doesn't appear to be free. In none of the examples above, for instance, can anything other than the highest copy be linearized. I will adopt the convention of putting the copies that are not pronounced into a shaded font. (260), for instance, shows the pronunciation in (267).

(267)



6.2.2.1 Copies and Reconstruction

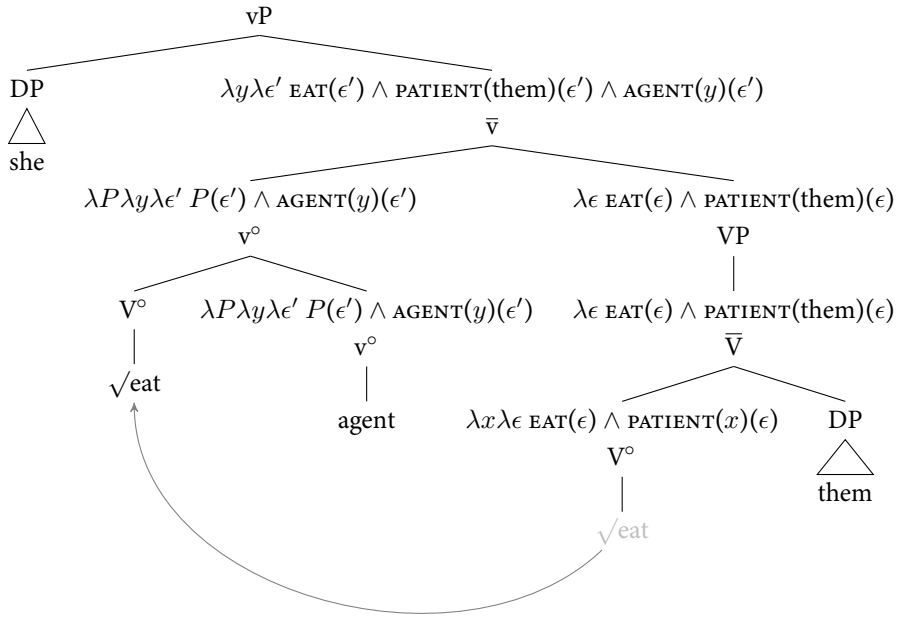
The Copy Theory of movement has features like that of Trace theory when it is extended in the way described in the previous section. That version of Trace theory allows traces to be variables over the kind of things that the moved term is. If a verb moves, for instance, the trace left by that verb can be a variable over verb meanings. And if a quantificational DP moves, the trace it leaves can be a variable over quantificational DP meanings. The Copy Theory of movement creates representations which have similar information in them. If a verb moves, for instance, the lower position of the moved verb remains occupied by that verb. And if a DP moves, again the lower position remains occupied by that DP. It's this feature of the Copy Theory that allows it to model reconstruction.

We'll need a new semantic interface procedure for the Copy Theory of movement to do so, however. To begin with, let us adopt the view that no more than one copy is required to be semantically interpreted. Like Suppress Copy, this lets copies be in principle invisible.

(268) No more than one copy needs to be semantically interpreted.

In a situation where verbs move—as in French finite clauses or English vPs—Head Movement creates representations that can be semantically interpreted. In those representations, the lower copy of the verb will be semantically interpreted, but the higher copy won't be. This is illustrated by the English case of verb movement in (269).

(269) $\lambda\epsilon'$ EAT(ϵ') $\wedge$ PATIENT(them)(ϵ') $\wedge$ AGENT(she)(ϵ')



The denotation of the v° that contains *eat* is the same as the denotation of the v° that doesn't. The meaning of *eat* has been ignored in computing the denotation of the larger v°.

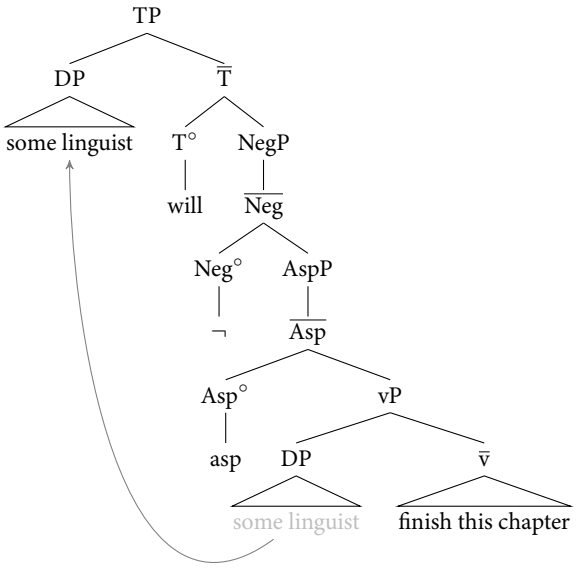
Note that (269) provides a way of capturing reconstruction that, unlike the use of traces in the previous section, does not require that the moved term c-command the position in moves from. In general terms, what the copy theory of movement does is allow a term to be put into a variety of positions in a phrase marker. That term can be pronounced in one of those positions and semantically interpreted in another. The positions that a term can be distributed in is given entirely by movement; they are the positions that Upwards and the other conditions on derivations permit. The parses in a derivation that are semantically interpretable grow as a consequence. For a parse derived by movement to be semantically interpretable does not require, as it does on Trace theory, that the moved term be able to bind the variable it leaves in the lower position. This allows the Copy theory to provide a less problematic model of reconstruction than does Trace theory. In particular, the problems associated with Head Movement that are chronicled in the previous sections are unproblematic on the Copy theory of movement.

To see this we'll need to add the semantic component that allows the Copy theory of movement to give the variable binding interpretation that the Trace theory of movement is designed to achieve directly. One place where that interpretation is needed is in (270), where Argument movement brings a quantificational DP into a position higher than negation.

(270) Some linguist won't finish this chapter.

The Copy theory of movement provides this sentence with the representation in (271).

(271)



The Trace theory of movement coupled movement with a set of interpretive rules that, in addition to inserting a variable in the lower position, changed the denotation of the phrase that the moved DP becomes a sister to. For that effect to be mimicked in the Copy theory of movement, we'll need a similar set of rules. I'll present here an approach to these rules that has been made popular by Danny Fox.[5]

To understand Fox's proposal, it will first be useful to know a few things about DPs that are headed by the determiner *the*—so-called, "definite descriptions." One popular conception of the meaning of *the* is that it creates a DP which, like a pronoun, refers to something salient in the context. For instance, *the man* and *he* are roughly interchangeable in the scenario described below.

(272) *My sister and I are watching people exit an event at school. After most people have left the building, a lone couple walks out: a man in a suit and a woman in a simple dress. My sister says:*

a. He used to be my neighbor.

b. The man used to be my neighbor.

Neither of these expressions would be appropriate in the scenario described below.

[5] Fox, Danny. 1999. Reconstruction, binding theory, and the interpretation of chains. ***Linguistic Inquiry*** 30:157–196

(273) *My sister and I are watching people exit an event at school. After most people have left the building, a lone couple walks out: a man in a suit and tie and a man in jeans and a t-shirt. My sister says:*

a. He used to be my neighbor.

b. The man used to be my neighbor.

The first context makes salient a single male, but the second doesn't. If a requirement of both *the man* and *he* is that their referent be contextually salient, then we have an explanation for the contrast in (272) and (273).

The conditions under which a definite description can be used are not identical to those that govern where pronouns can be used, but there is a kind of family resemblance to their distribution. Another way in which they are similar is that definite descriptions can be interpreted as variables, like pronouns can. For instance, (274a) can have an interpretation that is roughly like that of (274b).

There is a clear preference for (274b) over (274a). A complete account would include an explanation for this contrast; we'll focus merely on how (274a) can get the relevant interpretation.

(274) a. No man's partner should belittle the man.

b. No man's partner should belittle him.

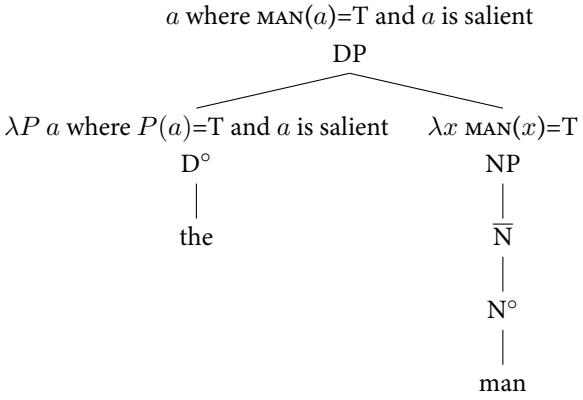
In our syntax, bound variable interpretations come by way of the presence of a referential index. Because pronouns can contain a referential index, for instance, pronouns are capable of being interpreted as bound variables. What (274a) suggests, then, is that definite descriptions can also be equipped with referential indices.

Let's put these observations together. The denotation of *the* must allow it to combine with an NP and produce something that refers to a salient individual. Let's take the denotations of NPs, like nouns, to be functions whose domains are things and whose ranges are the values True or False. They are functions, then, that characterize the sets of things that yield truth of the property named by the NP. For instance, the noun, and NP, *man* can be given the denotation in (275), which is a function that yields truth for any thing that it applies to which makes MAN true.

(275) $[\![\text{man}]\!] =_{\text{def}} \lambda x\ \textsc{man}(x)=\text{T}$

We want *the* to combine with this function and refer to a salient individual that, when fed to this function, yields True. A preliminary way of describing this is illustrated in (276).

(276) $[\![\text{the}]\!] =_{\text{def.}} \lambda P\, a$, where $P(a) = T$ and a is a salient individual.



I'm going to commandeer a symbol from the semantics literature to aid in abbreviating the denotation of *the*. That symbol, $\imath$, I will define informally with (277).

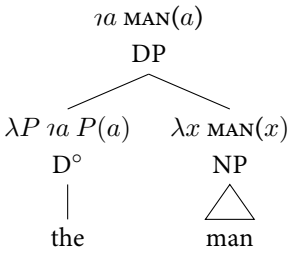
(277) $\imath a\ P(a) = a$, if $P(a)$ and a is made salient by the context of use.

Using this abbreviation, the denotation of *the* can be expressed by (278).

(278) $[\![\sqrt{\text{the}}]\!] =_{\text{def.}} \lambda P\ \imath a\ P(a)$

With this definition, (276) becomes (279).

(279)



Now we need to see how to build a definite description so that it includes a referential index. Let's adopt view suggested by Florian Schwarz[6]: a referential index can be part of the meaning of *the*. The ambiguity of definite descriptions, then, comes from the ambiguity of *the*. In addition to (278), we'll give $\sqrt{}$*the* the denotation in(280). Assume that any D° that contains $\sqrt{}$*the* is exponed by the word *the*.

(280) $[\![\sqrt{\text{the}}]\!] =_{\text{def.}} \lambda n\ \lambda P\ \imath a\ a = n \wedge P(a)$

Take "n" to be a variable that ranges over referential indices. The denotation in (280) allows *the* to combine first with a referential index and then an NP to form a definite description. How this works is illustrated in (281).

Understand a to be of semantic type entity: it is an arbitrarily chosen individual that satisfies P.

The symbol "$\imath$" was made popular by Whitehead and Russell (1910), where it is used to model the meaning of ***the***. The semantics in Whitehead and Russell (1910) is different than (277) however. They treated "$\imath$" as a quantifier, which, like other quantifiers, relates the meaning of the NP it combines with to the meaning of the function that the DP headed by that quantifier is a sister to. On their treatment, $\imath$ is an existential quantifier that imposes a requirement that there be only one thing of the kind that is quantified. This requirement is responsible for the contrast between (272) and (273). (i) exemplifies. (Note that Whitehead and Russell do not use events or θ-roles to describe the denotations of sentences.)

i. $[\![\text{The man sat}]\!] = \exists x\ \text{man}(x) \wedge \text{sit}(x) \wedge \forall y\ (\text{man}(y) \rightarrow y = x)$

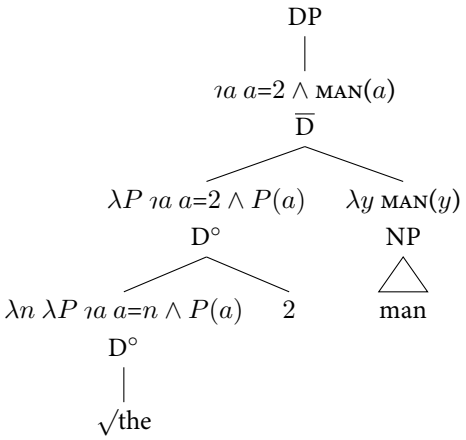
This approach to the meaning of ***the*** is broached in Russell (1905) and improved in Neale (1990). In (277), the uniqueness requirement is captured by the more general (and vague) requirement that x be salient.

Note that (278) gives the denotation for the root of the word that will be exponed by the vocabulary item ***the***. This is explained immediately below.

[6] Schwarz, Florian. 2009. Two types of definites in natural languages. Doctoral Dissertation, University of Massachusetts at Amherst

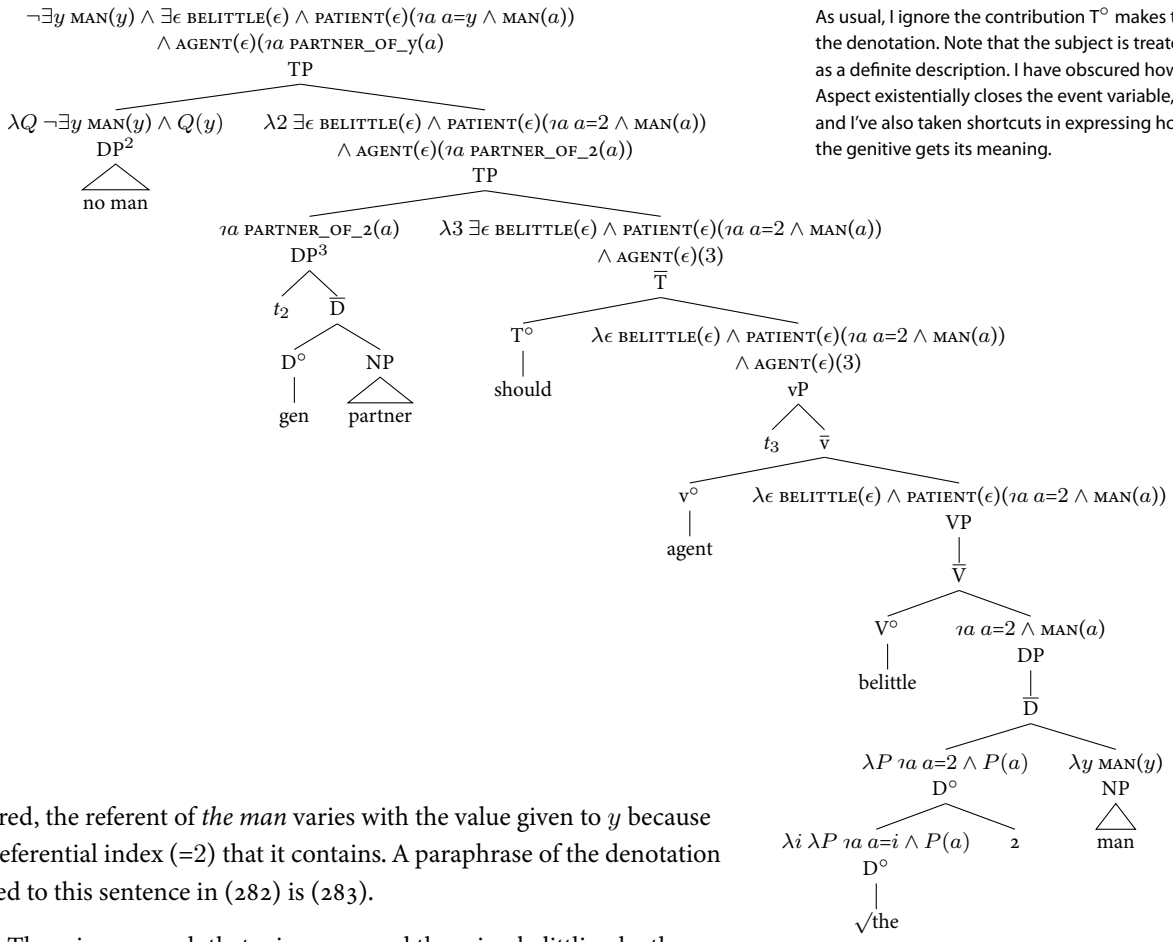
There are other ways of expressing how a referential index allows a definite description to get the interpretation of a variable. This is essentially the method advocated by Schwarz (2009), who uses it to give an account of the cross-linguistic variation that definite determiners can have. Elbourne (2005) also has a good discussion of the issues involved in how the referential index is put into definite descriptions. Hanink (2018) provides evidence from Washo that helps situate indices in definite descriptions.

(281)



The expression in (281) refers to the salient a which is both a man and has the value that 2 gets. I'll illustrate how this achieves the bound variable reading of *the man* in (274a), falling back on the trace theory of movement. To produce a syntax that gives the right semantic interpretation to (274a) requires a movement rule that we haven't previously encountered. That rule will move *no man* to the position indicated in (282). If this rule exists in English, it will have to be a rule that is trapped in that portion of the derivation that does not feed the S-structure. That such a rule exists, and why it is constrained to never produce S-structures, is discussed in Kennedy and Merchant (2000). This rule won't be examined in detail in these lectures, so I will do nothing beyond using it here in an *ad hoc* way.

(282)



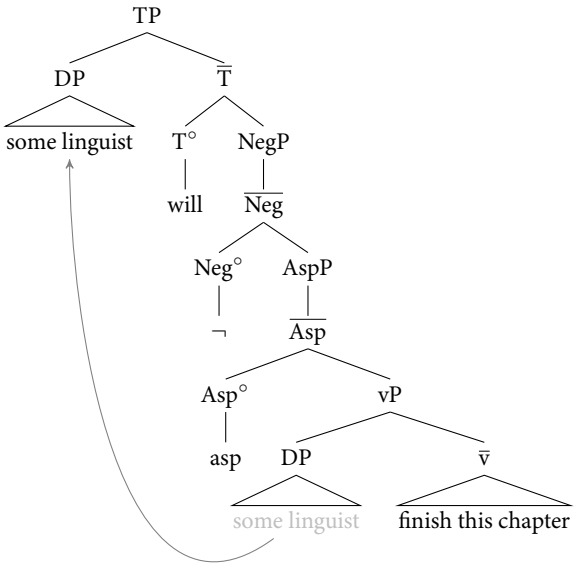
As usual, I ignore the contribution T° makes to the denotation. Note that the subject is treated as a definite description. I have obscured how Aspect existentially closes the event variable, and I've also taken shortcuts in expressing how the genitive gets its meaning.

As desired, the referent of *the man* varies with the value given to y because of the referential index (=2) that it contains. A paraphrase of the denotation delivered to this sentence in (282) is (283).

(283) There is no y such that y is a man and there is a belittling by the partner of y whose patient is the man which is y.

Now that we've taken this detour into how definite descriptions work, let's return to the central task: giving the Copy theory of movement a way of expressing the bound variable meanings that movement sometimes creates. If movement creates representations like (271), repeated below, then how do we extract from this representation the meaning in which *some linguist* binds a variable in the Specifier of vP?

(271)



Danny Fox's proposal is to let the phrase in the lower position be interpreted in the same way that definite descriptions with a referential index are. Movement will be defined with (284).

This proposal is in its essentials like a proposal in Engdahl (1980), whose theory of movement is an early variant of Copy theory. (See also Engdahl (1986), for a different implementation of the idea.)

(284) A term α moves to position β $=_{\text{def.}}$

a. β becomes occupied by COPY(α),

b. If α contains a DP, then:

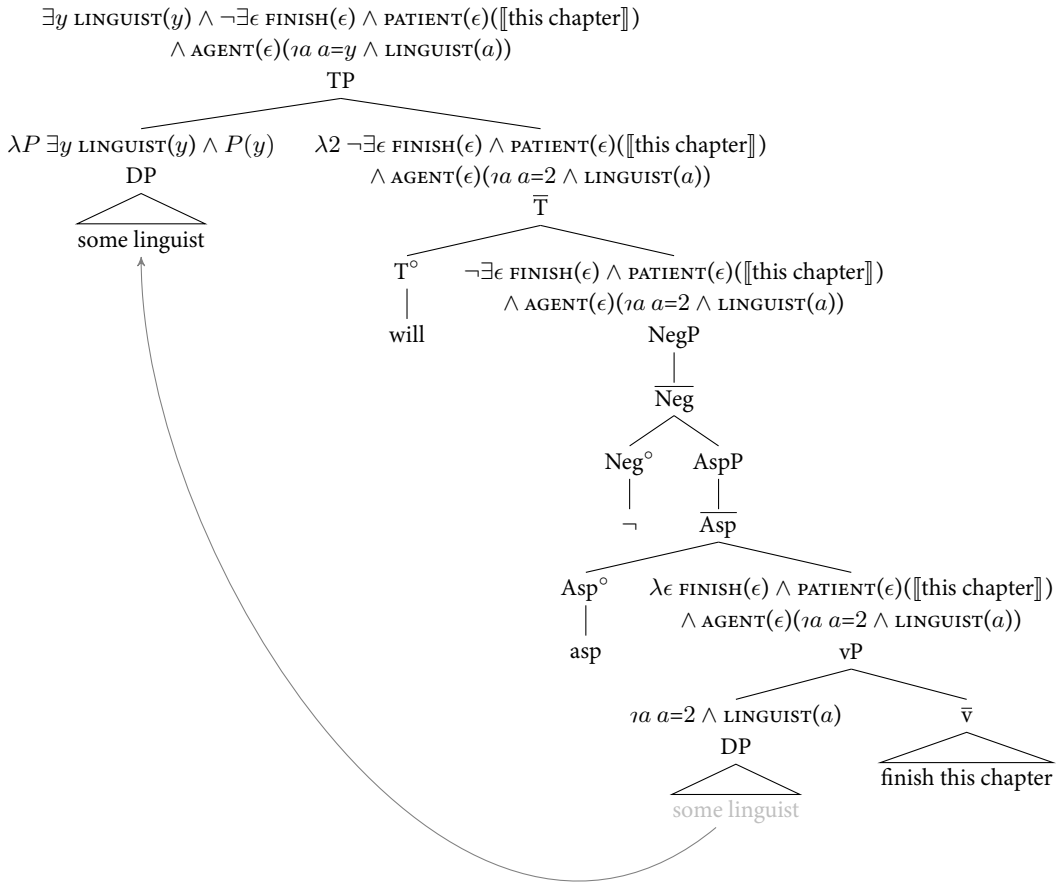
i. that DP's head can be interpreted as the *the* which contains referential index n, and

ii. The sister of β goes from $[\![\rho]\!]$ to $\lambda n[\![\rho]\!]$.

See Fox (1999, 2002, 2003). (284) deviates from Fox in requiring α to contain a DP for it to undergo Trace Conversion. Fox required α to be the DP that Trace Converts. My version extends to cases of Pied-Piping; see 6.2.2.2. Sauerland (1998, 2004) also provides a method for interpreting copies as variables that shares features with Fox.

Fox calls (296b) "Trace Conversion." It mimics closely the definition of traces in the Trace theory of movement, by allowing DPs involved in a movement relation to be interpreted as bound variables. It delivers to (270) the denotation shown in (285).

(285)



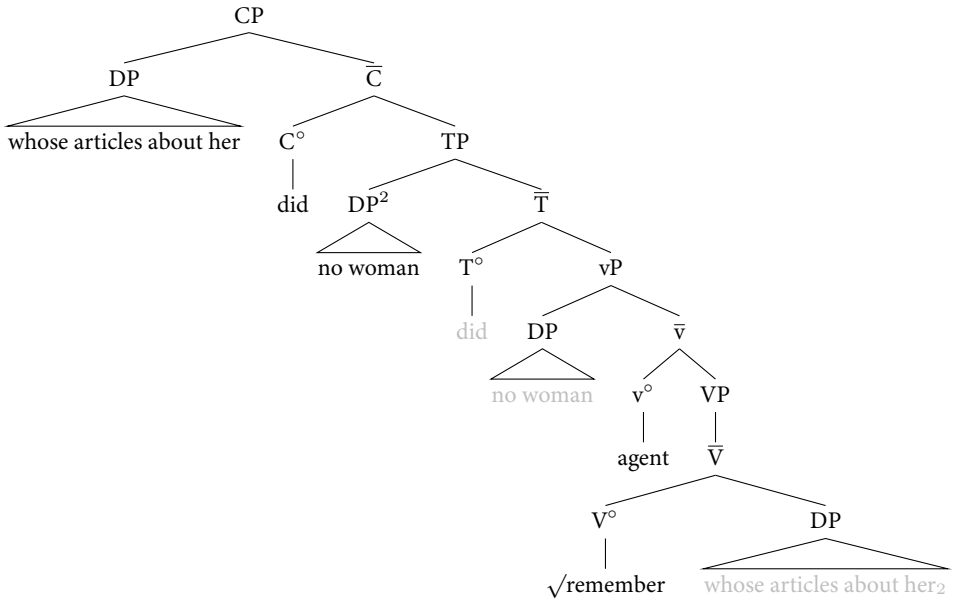
This is the desired meaning: it interprets *some linguist* outside negation, and causes it to bind a variable inside the scope of negation. That variable is now a definite description—something semantically equivalent to *the linguist*—rather than a trace.

By letting DPs that are in a movement relation become definite descriptions which get interpreted as variables, the Copy theory of movement provides an account for Reconstruction of pronominal variables that explains why the constraints imposed by the pronoun are preserved. To see how, consider (286).

(286) Whose articles about her did no woman remember?

The pronoun *her* in this example can be interpreted as a variable bound by *no woman*. As we saw in section 6.2.1.3, this is possible because *whose articles about her* has moved from a position that is c-commanded by *no woman*, and when the pronoun is in that position it is capable of being bound by *no woman*. The Copy theory of movement produces the representation in (287).

(287)



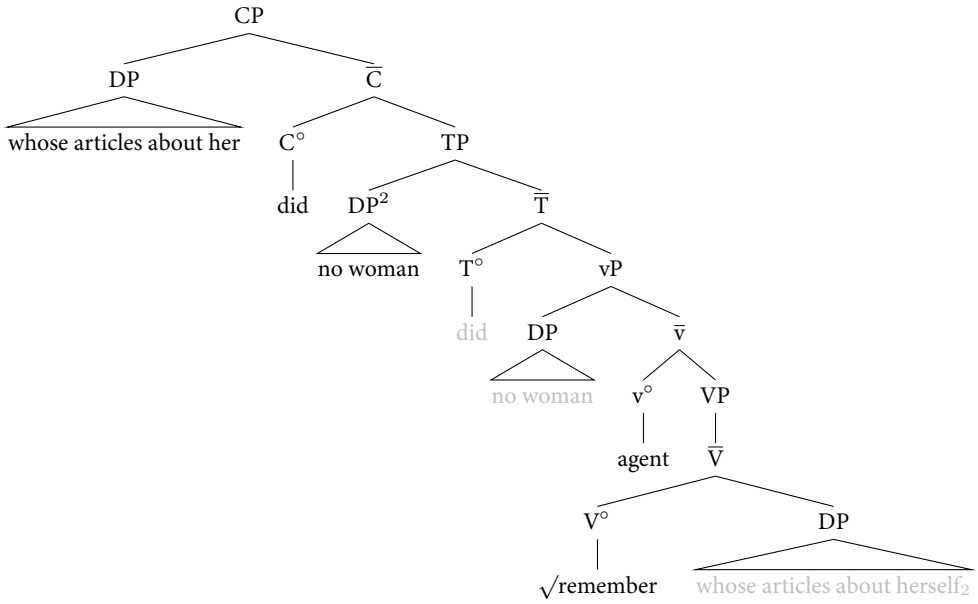
Note in particular that *her* is capable of being bound in the parallel (288).

(288) No woman^2 remembered my articles about her$_2$.

By contrast, the reflexive pronoun in (289) cannot be bound by *no woman*, because the representation the Copy theory of movement gives to it (in (290)) does not put the reflexive pronoun in a position that allows it to be bound by *no woman*, as the comparison with (291) indicates.

(289) * Whose article about herself$_2$ did no woman^2 remember?

(290)



(291) * No woman^2 remembered someone's article about herself$_2$.

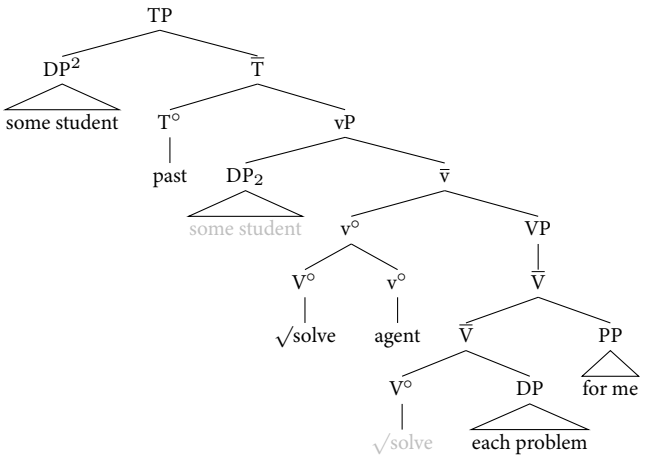
Note that as these representations make clear, we should adopt an account of the semantics of wh-questions that allows the wh-phrase to be semantically interpreted in just its lower position. We do not want the higher copy to be semantically interpreted, for that would wrongly put *her*, or *herself*, in a position in which it cannot be bound by *no woman*.

The Copy theory of movement is successful at addressing the problems chronicled in this section for the Trace theory of movement. It does so by allowing one parse in a derivation to have a lot of the information that previous parses in that derivation contain. This makes it possible to let one parse give the semantic interpretation mechanisms what is needed to get interpretations that were available only in different parses in the Trace theory of movement. In this way, the Copy theory of movement allows derivations to obey Upwards, but still deliver parses whose semantic interpretations were inconsistent with an S-structure on Trace theory. Let me illustrate that by considering how Trace theory and Copy theory apply to (292), which is akin to a case discussed in section 6.2.1.2.

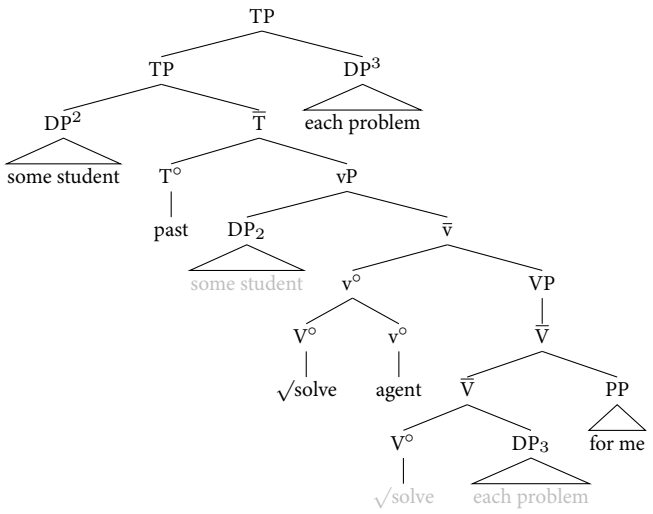
(292) Some student solved each problem for me.

The S-structure/interpreted parse pair that the Copy theory allows for (292) is (293).

(293) a.



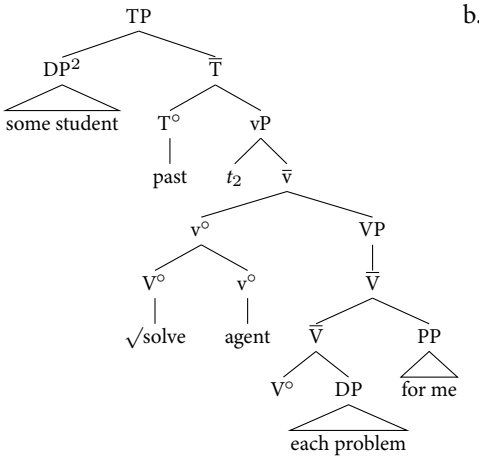
b.



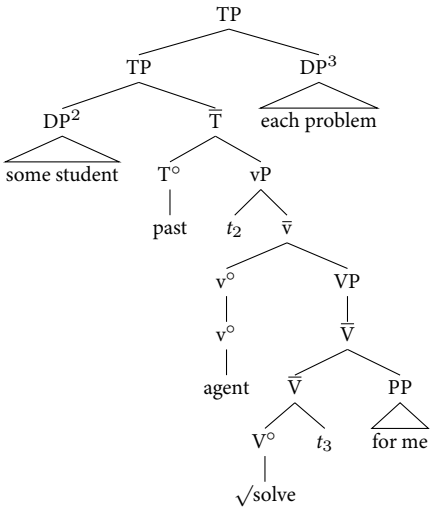
The higher copy of *solve* can go uninterpreted, as it must. And the lower copies of both DPs can be interpreted as variables, giving *each problem* higher scope than *some student*.

Trace theory provides the S-structure/interpreted parse in (294).

(294) a.



b.



The interpreted parse can be derived from an Upwards obeying movement of the object *each problem* from the S-structure in (293). But there is no way of putting both parses in (294) into an Upwards obeying derivation. The Copy theory of movement provides a solution for all of the problems this section chronicles for Trace theory. It is compatible with the Lebeaux+Upwards model, and inherits from that model its ability to derive the Law of Reconstruction.

6.2.2.2 Pied-Piping

The Copy theory of movement also has a means for addressing a problem we encountered in the discussion of relative clauses. That problem is illustrated

See section 6.1.3.2.

by the relative clause in (132), repeated here.

(132) the dog whose owner[2] you met t_2

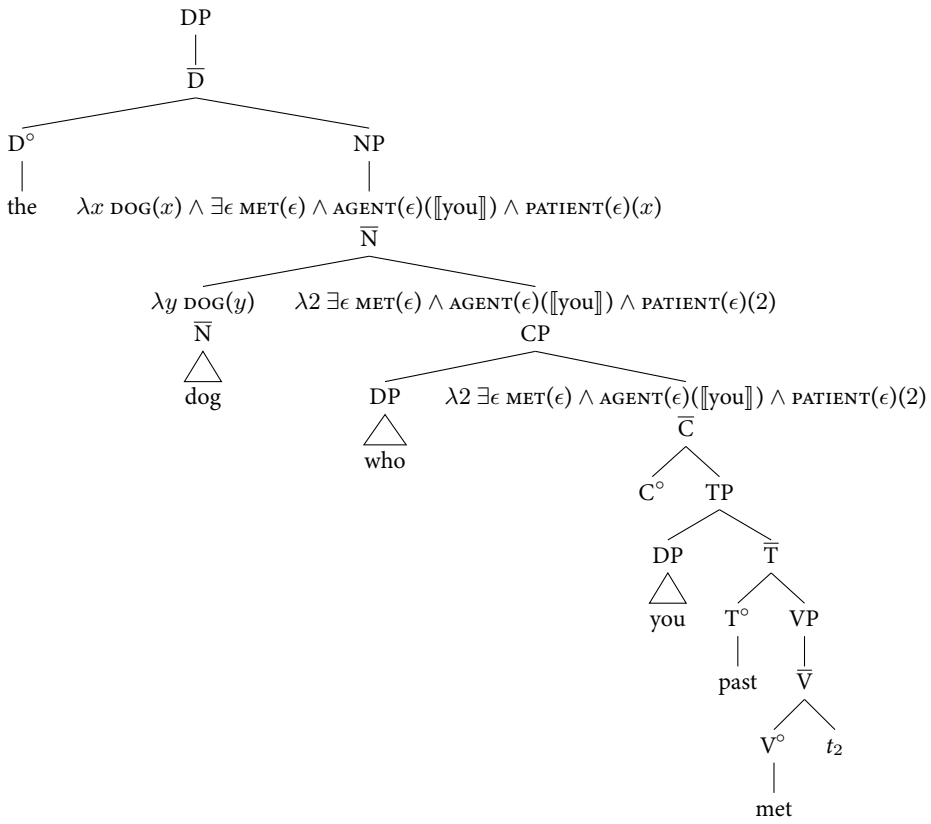
In (132), Wh Movement has moved a constituent larger than the wh-phrase. These are cases of what is known as "pied-piping": Wh Movement moves more than just the X° that contains the wh-feature. It instead moves some phrase dominating that X°. In (132), it has moved the DP *whose owner*, which contains the wh-feature in *who*.

Cases of pied-piping like this present a problem for our analysis of relative clauses because it doesn't deliver the right meaning from Trace theory. Relative clauses have a meaning that restricts the range of things that the $\overline{\text{N}}$ they combine with can refer to. In (132), we want the relative clause to restrict the range of things that *dog* can refer to. But Trace theory causes relative clauses to describe the things that could fit semantically into the trace position. In (132), this means that the relative clause describes things that you met. Trace theory therefore causes (132) to have the same meaning that *the dog who you met* has. Because the Copy theory of movement doesn't require the position from which a phrase moves to be a variable, it doesn't have the same consequences in this context. Let's see how.

Recall that relative clauses are formed with Wh Movement, and that the semantics of movement causes them to become functions from individuals to truth values. In the simpler relative clause of (295), for instance, a semantics like that in (296) is created on the Trace theory of movement. (I take a variety of simplifying shortcuts in (296).)

(295) the dog who you met.

(296)



Movement of *who* causes its sister $\overline{C}$ to become a λ expression in which the trace left by movement is interpreted as a variable. If relative pronouns like *who* are semantically vacuous, which is the analysis we adopted in section 6.1.3.2, then the denotation that $\overline{C}$ has will be the same as the denotation that CP has. This allows the CP and the $\overline{N}$ that is its sister to combine semantically in the way indicated in (296), forming the desired meaning. The relative clause in (132) does not get the right meaning on this theory because the variable in (132) is bound by *whose owner*, but what we want is, parallel to (295), a relative clause whose variable corresponds to dogs.

This can be remedied by the Copy Theory of movement, which is repeated below.

(284) A term α moves to position β $=_{\text{def.}}$

a. β becomes occupied by COPY(α),

b. If α contains a DP, then:

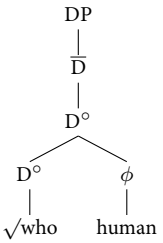
i. that DP's head can be interpreted as the *the* which contains referential index n, and

ii. The sister of β goes from $[\![\rho]\!]$ to $\lambda n[\![\rho]\!]$.

Let's first consider how the Copy theory applies to (295), which involves the movement of a wh-pronoun.

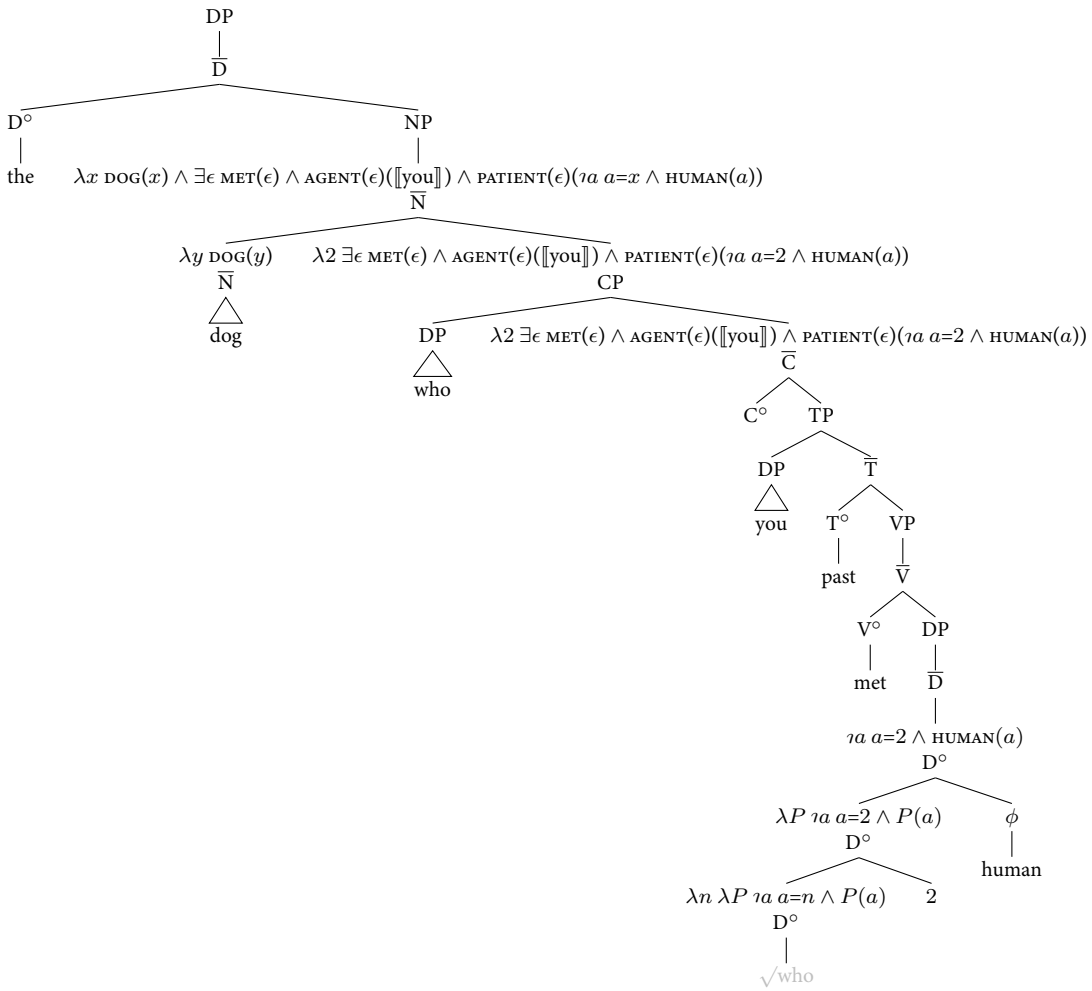
Recall that pronouns are determiners that include information about the thing that their DP refers to. The pronoun *she*, for instance, indicates that the DP it heads will refer to a singular female. The pronoun *who* has information that indicates that its referent will be human. Let's give this a syntactic expression by letting the DPs that *who* heads have the form in (297).

(297)



With this analysis of *who*, the Copy theory can assign (295) the representation in (298).

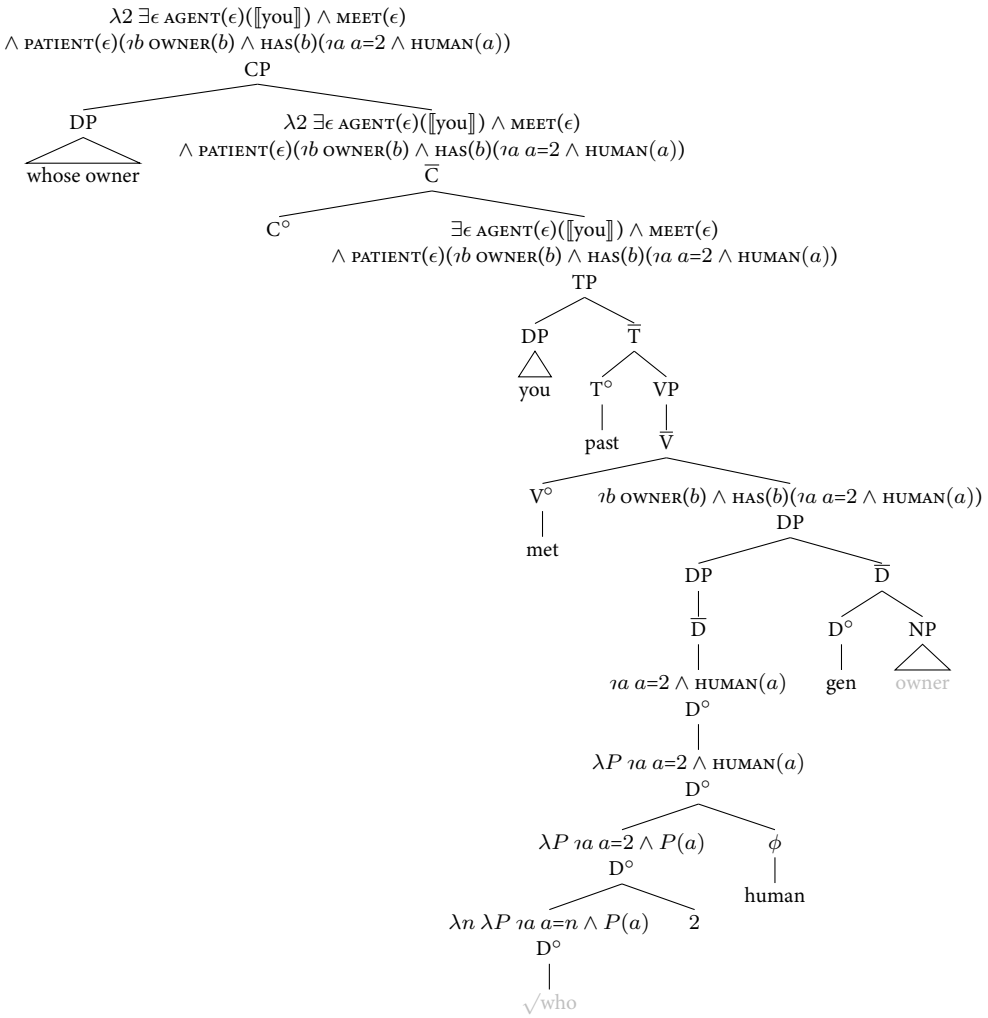
(298)



Note that the lower copy has undergone the "trace conversion" described by (296b). Trace conversion gives the D° occupied by $\sqrt{}$*who* the meaning that a definite determiner with an index has. This turns the lower copy of $\sqrt{}$*who* into a variable that is restricted in the same way that *who* is—that is, it ranges over humans. The result is equivalent, meaning-wise, to (296).

Consider now the representation that the Copy theory will give to the relative clause in (132).

(299)



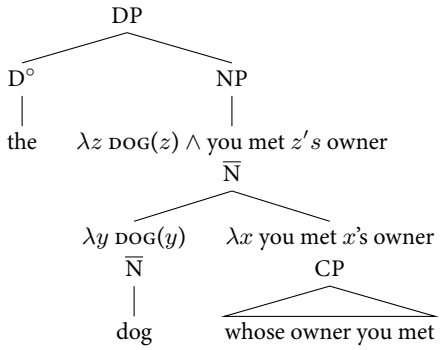
The object of *meet* in (299) refers to owners, as desired. But the variable bound by the λ expression that is the denotation of CP is buried within the object of *meet*: it corresponds to *whose*, which has been converted into a variable by the Trace Conversion part of the Copy Theory of movement. Simplifying out the irrelevant details of (299), the meaning delivered to the relative clause is (300).

(300) λx you met x's owner

And that is just the meaning needed for (132):

Pied-piping, on this account, requires a theory which determines which phrases dominating a feature can, or must, move. There is a large literature attempting to provide that theory, but at the time of writing, there is no consensus view that has emerged. A good introduction to the phenomena, and a development of one of the popular theories, is Heck (2008, 2009).

(301)



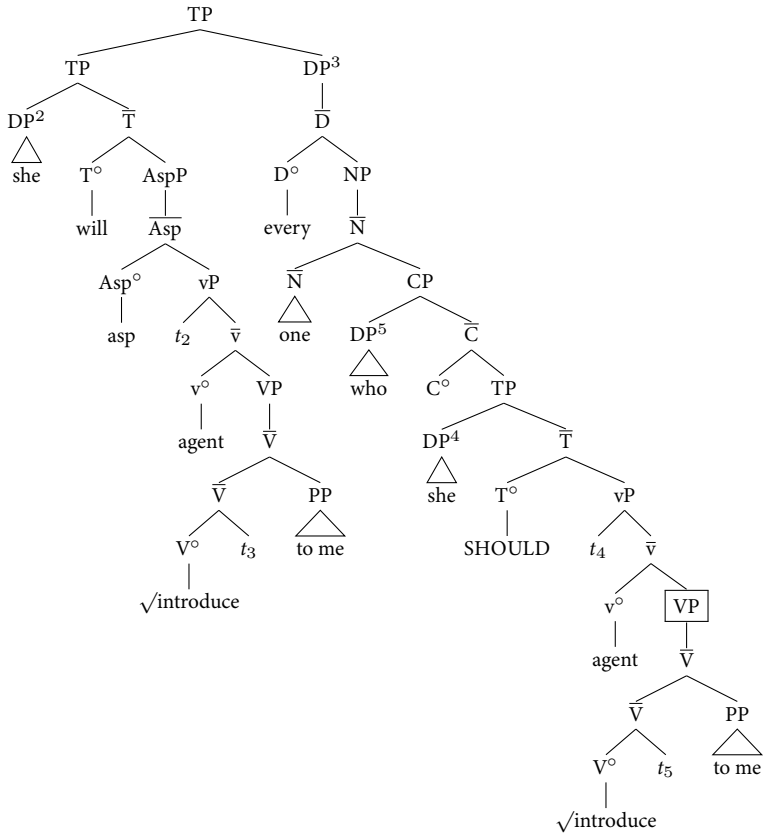
The Copy theory of movement can deliver the right semantics for relative clauses when they involve Pied-piping.

6.2.2.3 Problem: Antecedent Contained Anaphora

As successful as the Copy theory of movement is in solving the problems that reconstruction poses for Trace theory, it harbors a serious problem. It isn't well-equipped to handle the central problem that Trace theory solves: antecedent contained anaphora. An illustration of that problem, and its Trace theoretic solution, is in (303).

(302) She will introduce everyone who she should to me.

(303)

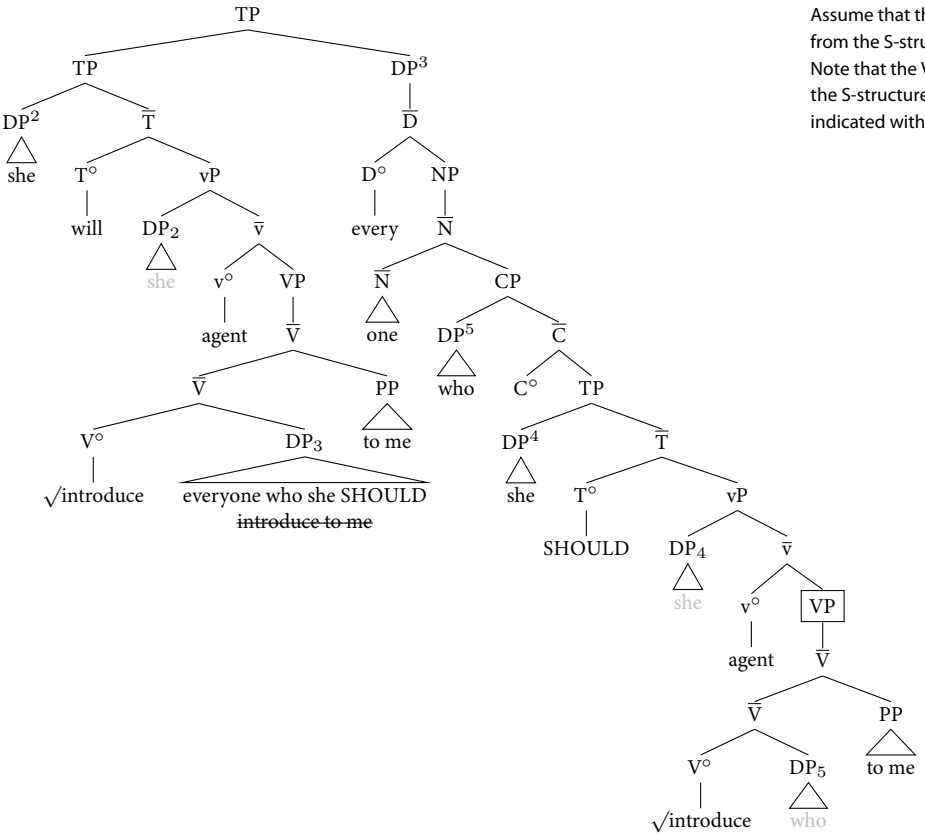


Movement of the verbal roots into v°, as well as other details, are ignored in this graph.

The S-structure that corresponds to (302) can be in a derivation that includes the parse shown in (303)—QR can form this parse from the S-structure—and this allows the boxed VP to elide. This is possible in part because the boxed VP matches the VP from which the object has QR'd. On the Trace theory of movement, movement is capable of creating an antecedent for the elided VP in (302). By contrast, the Copy theory of movement associates with (302) the representation in (304), and the VPs in this representation do not match in the necessary way.

The condition on ellipsis requires that the elided phrase be part of a constituent that matches the meaning of an antecedent phrase, except for those places where there is contrastive focus. See section 6.1.2 for details.

(304)



Assume that this representation is derived from the S-structure by Moving the object DP. Note that the VP ***introduce to me*** is silent in the S-structure because of VP ellipsis—this is indicated with ~~strikeouts~~.

The Copy theory of movement doesn't do the job of creating the antecedent for VP ellipsis.

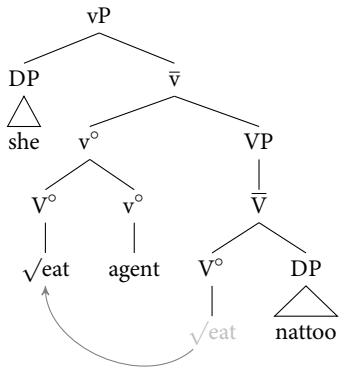
There is a conflict in what reconstruction phenomena require, and what antecedent contained anaphora require. Reconstruction phenomena seem to need the moved phrase to be in a position different from where it is pronounced. If movement is the process that relates a term to its two (or more) syntactic positions, then reconstruction phenomena recommend a theory of movement that keeps the moved term in its various positions syntactically. The Copy theory is designed for this effect. By contrast, antecedent contained anaphora seems to require that movement remove a term from one of its syntactic positions, and occupy that position with a variable instead. This is what Trace theory is designed for. At the time of writing, the resolution of these conflicting needs has not been fully achieved.

One direction presently under investigation extends that part of the Copy theory of movement that allows one of the copies to be ignored by the semantic component. In cases where a verb moves, for instance, it's necessary that only the lower copy of that verb be semantically interpreted: the higher copy of the verb's semantic contribution can be ignored. This is one of results of the instructions that determine how the syntactic representations created by movement are mapped onto a semantics and a string. Those instructions are repeated in (305).

(305) a. Only one copy may be linearized.
b. Only one copy must be semantically interpreted.

(305a) prevents more than one of the copies of some term from being part of the string that corresponds to the syntactic representation of movement. And (305b) allows copies created by movement to be semantically ignored, as long as just one of them is interpreted. Together, (305a) and (305b) mimic some of the information that derivations have. In conjunction with the notion of copies, they permit a term to be pronounced in a position different than where that term is semantically interpreted. In (306), for instance, the verbal root $\sqrt{}$*eat* is pronounced in the v° position, but semantically interpreted in its lower position.

(306)



Determining which copy is pronounced and which is semantically interpreted is equivalent to determining which parse in a derivation is pronounced and which is semantically interpreted. The solution to antecedent contained anaphora I will describe involves loosening (305) so that parts of copies are allowed to be semantically and phonologically interpreted.

In place of (305), then, consider (307).

This condition is modeled on a proposal in Sportiche (2019).

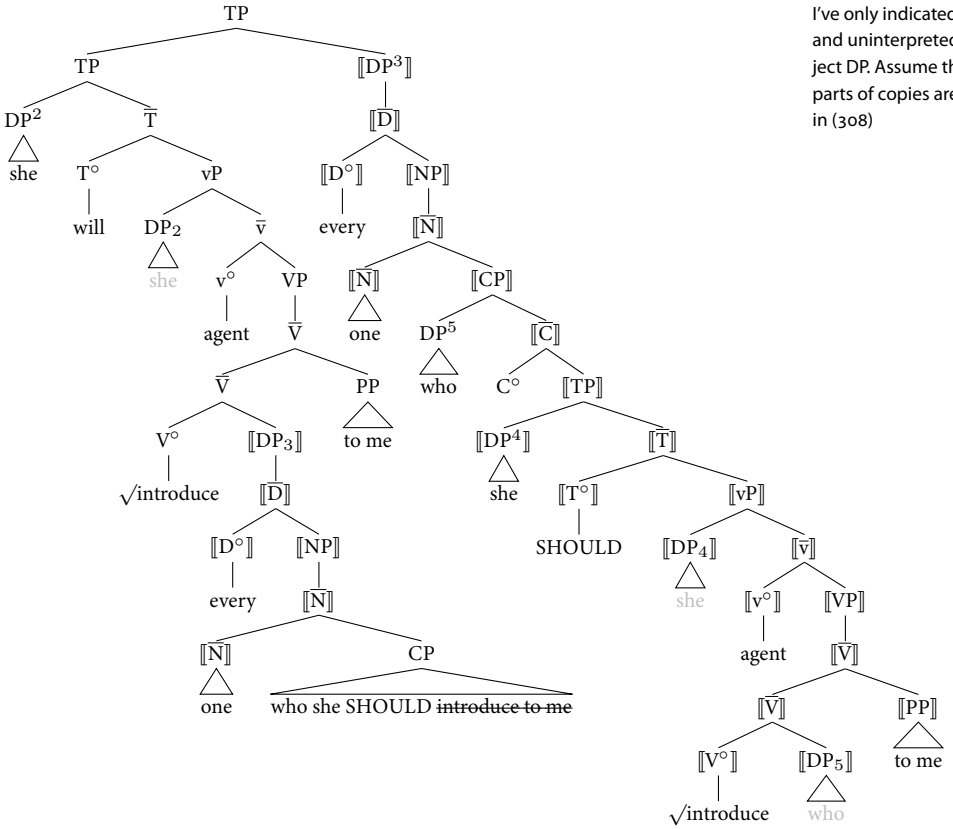
(307) Let $\{\gamma, \gamma', \gamma'', \ldots\}$ be matching phrases or heads dominated by copies $\{\alpha, \alpha', \alpha'', \ldots\}$.
a. Only one γ^n may be linearized.
b. Only one γ^n must be semantically interpreted.

This allows parts of a moved phrase to be semantically, or phonologically, interpreted. It takes away from movement the job of choosing which things

will be pronounced and which semantically interpreted. Instead movement only determines which phrases are eligible to be semantically/phonologically interpreted in different places, and where those places will be.

I'll adopt the convention of placing within denotation brackets (i.e., ⟦ ⟧) the terms that will be semantically interpreted, and I'll continue the practice of indicating copied material that isn't pronounced by putting it in shaded font. Reconsider now the representation that movement allows for the instance of antecedent contained anaphora in (302). That representation (in (304)) can now be interpreted along the lines allowed by (307) in the way indicated in (308).

(308)



I've only indicated the semantically interpreted and uninterpreted parts of the moved object DP. Assume that all the phrases that aren't parts of copies are semantically interpreted in (308)

Note that the relative clause is semantically interpreted in the higher copy but not the lower copy. This will give rise to the desired outcome; let's see how.

In the complement position of the VP headed by *√introduce* is spoken the DP *everyone who she should*. (The VP *introduce to me* is silent because of ellipsis.) But only the determiner *every* and the N̄ *one* are semantically interpreted in this position. Thus, the phrases that dominate these terms have meanings that are determined only by *every* and *one*: the DP in object position has the meaning that is assigned to *everyone*. It doesn't include the semantic contribution of the relative clause. Because *everyone* in this position is the lower of two copies, it can get the semantic interpretation that involves

Trace Conversion; it can be interpreted as equivalent to *the*$_3$ *one*. Thus, the smaller TP of (308) will have an interpretation like that sketched in (309).

(309) λx she will introduce the$_x$ one to me

The higher copy of the object DP has a meaning that derives from interpreting everything inside it, including the relative clause. The VP of that relative clause contains the lower copy of the moved relative pronoun *who*. Because it is the lower copy of a movement relation, it too can be given an interpretation by Trace conversion. Its Trace converted interpretation is *the*$_5$ *one*. This makes the TP of the relative clause have an interpretation like that sketched in (310).

(310) λy she should$_F$ introduce the$_y$ one to me

Because (309) matches (310), except where there is a focus-marked term, the conditions on focus anaphora and ellipsis are met. In this way, by ignoring the relative clause's semantic contribution in the lower copy, Copy theory is capable of essentially mimicking Trace theory.

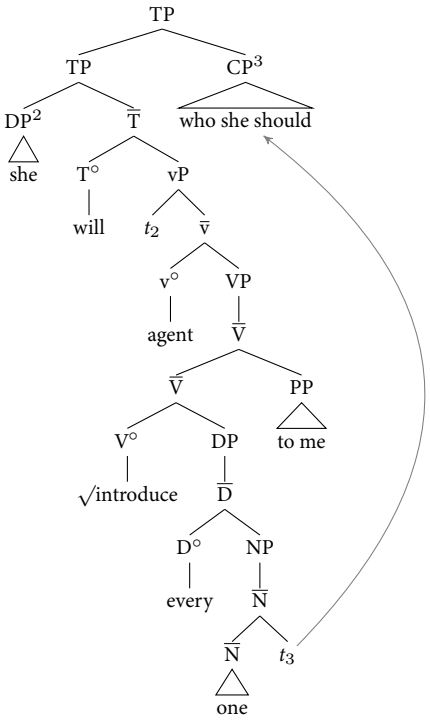
This method of accounting for antecedent contained anaphora showcases that part of (307) which allows just a part of a copy to be semantically interpreted. But (307) also allows just a portion of a copy to be phonologically interpreted as well. Jon Nissenbaum and Danny Fox[7] have argued that examples like (311) arise in this way.

(311) She will introduce everyone to me who she should.

The relative clause in (311) is not pronounced together with the DP it is semantically part of. One way of accounting for this case is to posit a Transformational rule that moves relative clauses to the right. Using the Trace theory, this rule—sometimes called "extraposition from NP"—would permit derivations to contain parses like those in (312).

[7] Fox, Danny, and Jon Nissenbaum. 1999. Extraposition and scope: a case for overt QR. In ***Proceedings of the West Coast Conference on Formal Linguistics 18***, ed. Sonya Bird, Andrew Carnie, Jason D. Haugen, and Peter Norquest, 132–144. Somerville, Massachusetts: Cascadilla Press

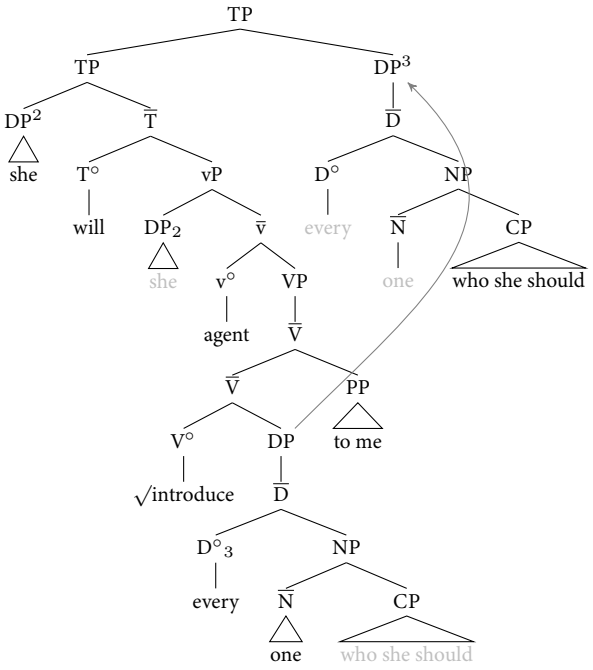
(312)



This is the analysis offered in John Ross's dissertation.[8] But there is compelling evidence that a better analysis is one that uses Copy theory, and makes use of that part of (307) that allows only portions of a copy to be pronounced. That view would give to (311) a representation like (313).

[8] Ross, John Robert. 1967. Constraints on variables in syntax. Doctoral Dissertation, Massachusetts Institute of Technology

(313)



A salient difference between the two accounts concerns what they say about the interpretation of the quantificational determiner. In the account

that (312) exemplifies, the position that the quantificational determiner is interpreted in is independent of the position of the relative clause. But in the account that (313) sketches, the relative clause is in the position that the DP headed by the quantificational determiner occupies. As a consequence, the quantifier must be interpreted in a position that matches where the relative clause is spoken in the second account, but not the first. It's this difference that Fox and Nissenbaum (1999) base their account on.They discover examples which suggest that the position a quantifier is interpreted in is dependent on the position that the relative clause is pronounced in the way that (313) predicts.

See Overfelt (2015) for another kind of argument for (313). Baltin (1987) is where the idea originates for taking the syntax of extraposed relative clauses to also be the syntax that resolves antecedent contained anaphora.

Let's look at one of their cases. Consider first (314).

(314) I must read every paper that I assigned before you do.

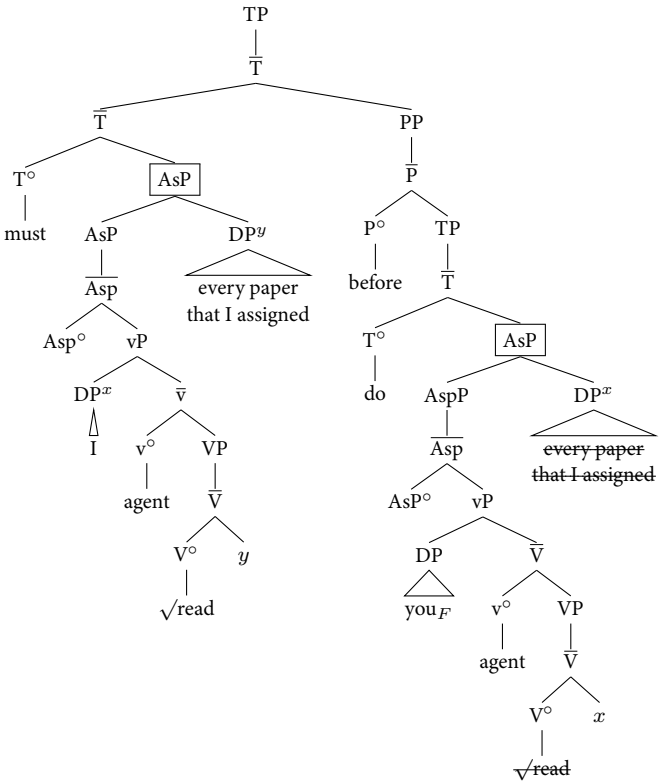
This example has an adjunct clause containing an elided VP whose meaning is determined by the meaning of the other VP in the sentence. We can use this fact to determine what the syntactic form of the other VP is. What we discover is that the VP which forms the antecedent to the ellipsis has two meanings, and therefore two syntactic forms. One of these meanings is, simply, the one associated with *read every paper that I assigned*. On that interpretation, (314) is synonymous with (315).

(315) I must read every paper that I assigned before you read every paper that I assigned.

What (315) expresses is the requirement that I finish reading the assigned papers before you finish reading the assigned papers. This requirement can be satisfied even if you read some of the papers before I do. As long as I have read every assigned paper before you have finished reading those papers, the requirement that (315) describes will be satisfied.

A parse that corresponds to this interpretation is (316).

(316)



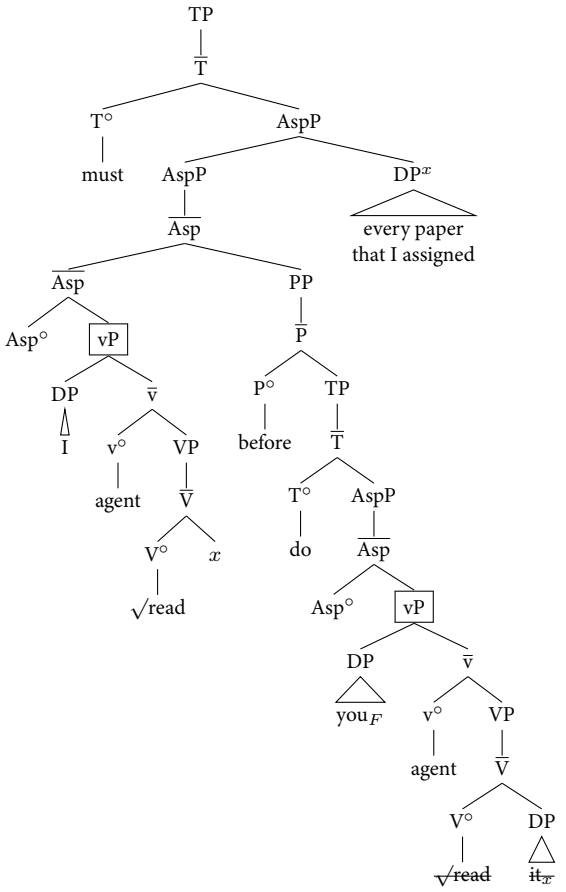
In (316), I've dodged the issue of what is in the position from which the DPs have moved by putting in those positions the variables x and y. What's key is that the object DP *every paper that I assigned* is inside a constituent—here AspP—that serves as the antecedent for the focus anaphor inside the PP headed by *before*. The boxed AspPs are identical, except for constituents that are focus-marked, and this is what is required for the VP inside AspP to elide. The parse in (316) can be derived from the S-structure parse for (314) by moving the two instances of *every paper that I assigned* outside the VP they are part of. Because they are part of those VPs in the S-structure, however, VP ellipsis will cause the second *every paper that I assigned* to be elided. The elided material is shown in strikeouts. The S-structure will have the subject DPs (i.e., *I* and *you*) in Specifier of TP, but they will be semantically interpreted as if they haven't moved from the positions they occupy in (316). I've taken the shortcut here of showing these DPs in the positions that they are interpreted. A more accurate parse would put them in their S-structure position, and use the semantics of movement to give the vPs they've moved from the meanings indicated in (316).

In addition to the meaning expressed by (315), (314) can also have the different meaning paraphrased by (317).

(317) Every paper that I assigned is such that I must read it before you read it.

On this interpretation, (314) expresses the requirement that every assigned paper must be read by me before it is read by you. Unlike the meaning paraphrased by (315), the requirement expressed by (317) is not satisfied if there is some assigned paper you read before I read it. (317) requires not just that I finish reading all of the assigned papers before you finish reading them all, but that I finish reading each of the assigned papers before you've read that paper. To achieve this meaning with our semantics, we need a syntax that puts the *before* PP inside a constituent that *every paper* quantifies over. A consequence of that syntax is that *every paper* won't be within VP that is elided; instead only a variable bound by *every paper* will be. A parse that represents this interpretation is (318).

(318)



As in (316), I've dodged specifying exactly what's in the position left by movement of the *every paper* DP by putting an x in that position, and I've put the subject DPs in their semantically interpreted positions, rather than their S-structure positions. Unlike (316), there is only one instance of *every paper that I assigned* in (318). In the object position of the elided VP is a pronoun that is bound by the the *every paper* DP, rather than another *every paper* DP. The boxed vPs have the same meaning, except for what is focused-marked, and should therefore meet the conditions on ellipsis.

In fact, the antecedence conditions on focus anaphors, and therefore ellipsis, are not met in this case. That is partly because the semantics I have given of bound variables does not give to them a denotation until the λ-operator that binds them is included. As a consequence, there isn't a denotation delivered to any phrase that contains a bound variable, but not the λ-operator that binds it. What is needed is a semantics that allows variables to have a denotation in phrases in which they are not bound, and then a modification of the antecedence conditions on focus anaphors that lets two variables bound by the same λ-operator have the same denotation. It would take us too far into the technicalia of variable binding to make these corrections, so I won't. See Heim (1993) for a direction.

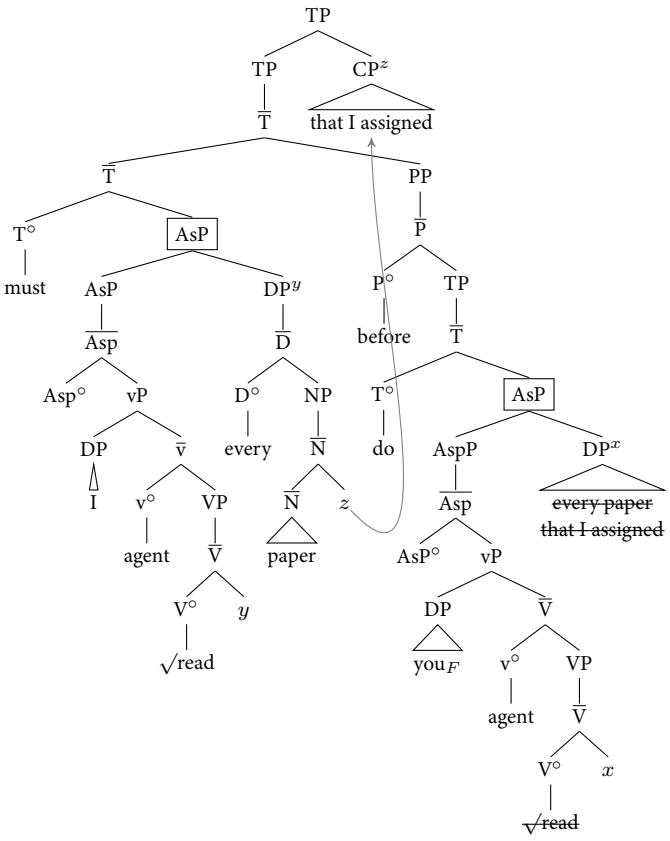
What Fox and Nissenbaum (1999) observe is that (316) can't be the source for an example that is identical to (314), but with the relative clause in extraposed position. The example in (319) has only the interpretation that goes with (318), not (316).

(319) I must read every paper before you do that I assigned.

This is what is expected if extraposition of a relative clause comes by way of selectively pronouncing parts of copies, but not if it comes by way of moving the relative clause.

To see this, consider what an account which allows relative clauses to move would say about (319)'s derivation. That derivation would include the parse in (320).

(320)



Note that (320) presents a problem of reconstruction: the moved relative clause must be semantically interpreted in the position it occupies before movement. However this problem is solved, the fact that the relative clause is interpreted as if it hadn't moved is why the antecedent in (320) behaves as if it contains the relative clause for the purposes of satisfying the conditions on the focus anaphor. That is, the antecedent AspP is equivalent to *I read every paper that I assigned*, and therefore eligible to be the antecedent to the anaphoric AspP. In other words, movement of the relative clause does not affect the ability of VP Ellipsis in (320). This can be confirmed by observing

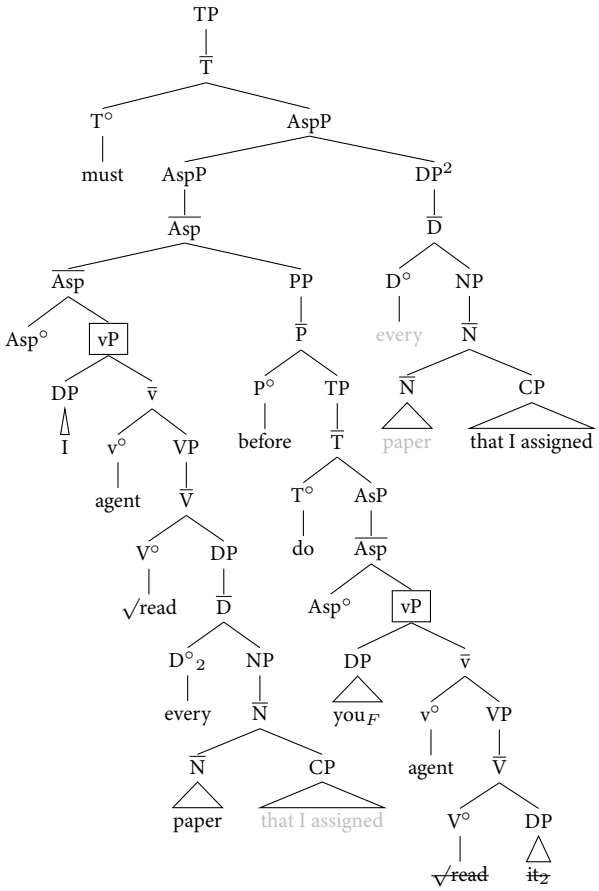
that the elided VP in (321) has the interpretation that would be given by (320) to (319).

(321) I must read every paper before this afternoon that was assigned in class and you must △ before tomorrow.
△ = read every paper that was assigned in class

The antecedent in (321) is *read every paper that was assigned in class*, even though the relative clause is pronounced outside that constituent.

An account of extraposed relative clauses that claims they move, therefore, wrongly predicts that (319), and its accompanying interpretation, should be available to (319). But an account like that of Fox and Nissenbaum's doesn't. That's because on that account, the relative clause is spoken inside the higher copy of the DP it modifies. When the relative clause is in extraposed position, it is in a higher position than the material that precedes it. In (319), that means that the *every paper* DP is outside the constituent that serves as antecedent to the focus anaphor; (322) illustrates.

(322)



The higher copy of the *every paper* DP is the one in which *every* is interpreted as the universal quantifier. This higher copy cannot be interpreted inside a constituent to the left of the PP, and cannot therefore be part of the antecedent for the ellipsis in that PP. This is how the Copy theory of movement prevents

(319) from having an interpretation in which the ellipsis contains an *every paper* DP, i.e., (316).

Instead, (322) produces the meaning that is associated with (318). Let's see how. The antecedent (boxed) vP in (322) is semantically parallel to (323a), and the elided (also boxed) vP is parallel semantically to (323b).

(323) a. I read that paper that I assigned

b. You$_F$ read it

The lower copy of the movement relation in (322) gets an interpretation as a definite description with a bound variable meaning, and I've tried to evoke that interpretation in (323a). Because the lower copy and the *it* in the elided vP are both bound by the higher copy of *every paper that I assigned*, they have the same meaning. As a consequence, the boxed vPs meet the condition on focus anaphora—their meanings are identical except for what is focus-marked—and the ellipsis indicated with strike-outs is permitted.

Again: this isn't technically true in the semantics provided for bound variables in this book. As noted in the previous sidebar, our semantics needs to be modified so that bound variables have an interpretation in phrases that don't include their λ-operator.

There is evidence, then, that extraposition of a relative clause involves the selective pronunciation of parts of two copies. And Antecedent Contained anaphora can be captured by the Copy theory of movement if it is possible to semantically interpret parts of two copies. Together, then, these cases support (307), repeated here.

(307) Let $\{\gamma, \gamma', \gamma'', ...\}$ be matching phrases or heads dominated by copies $\{\alpha, \alpha', \alpha'', ...\}$.

a. Only one γ^n may be linearized.

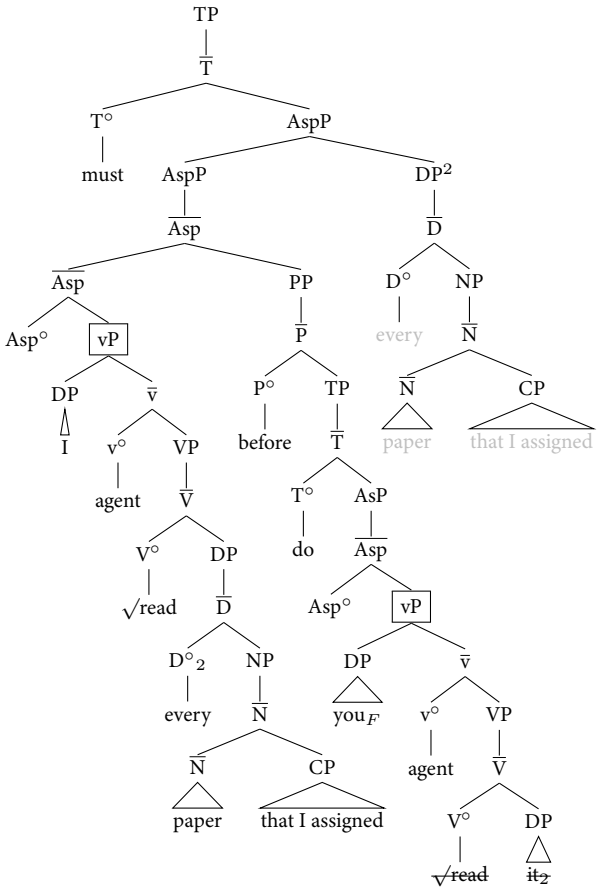
b. Only one γ^n must be semantically interpreted.

If Movement works in tandem with (307), then the derivations it defines have a different relationship to the sound-meaning mapping. Without (307), movement provides the range of parses from which a spoken form and a semantically interpreted form can be derived. With (307), each parse in a derivation has a variety of ways of furnishing a spoken and semantically interpreted form. Each parse in a derivation corresponds to a range of different sound-meaning relationships.

This makes it possible to consider rethinking the relationship between a derivation and the parse that represents its spoken string. It may be possible, for instance, to let S-structures always correspond to the final parse in a derivation. The present definition of derivations allows the S-structure—the parse that corresponds to the spoken string—to feed parses that are semantically interpreted. In many situations involving quantificational DPs, for instance, we have let a semantically interpreted parse be derived from the S-structure by Quantifier Raising. But if the approach to extraposed relative clauses described above is correct, this isn't necessary. The representation that gives rise to extraposed relatives could be interpreted so that all of the lower copy is pronounced and none of the higher copy is. For instance, an

alternative to (322) might be (324), where all of the lower copy is pronounced and none of the higher one is.

(324)



The S-structure in (324) has QR built into it.

The Copy theory of movement allows S-structures to have all of the information that the parses in the derivation leading up to that S-structure have. And amending the Copy theory of movement with (307) allows an S-structure to also contain the information that the parses which could be derived from that S-structure would have. S-structures, then, can have all the information needed for both the semantic interpretation and the pronunciation of a sentence. Jonathan Bobaljik[9] argues for this theory of derivations, and calls it a "single output" model. In a single output model there is one parse that contains all the information necessary to get the correct semantics and phonology. On these models the place for derivations is unclear. Derivations were originally conceived as a way of expressing how the semantic phrases in a sentence don't always correspond to a contiguous string—a phonological phrase. Derivations do this by letting a sentence be mapped onto a series of parses, one of which represents the string and another that represents the semantic composition. On a single output model, there is one parse that has both the words in their spoken positions and the words in their semantically

[9] Bobaljik, Jonathan. 1995. Morphosyntax: The syntax of verbal inflection. Doctoral Dissertation, Massachusetts Institute of Technology, Cambridge

interpreted positions. Derivations are no longer what represents the differing spoken and interpreted positions of phrases and their words. Instead, derivations are just a way of expressing the function that takes a set of morphemes into one syntactic representation. If a single output model should prove to be correct, we should reassess what is required of this function. Thinking of it as a series of operations—Transformations—may not be needed. We may find calling the model of syntax these lectures are charged with explicating "Transformational Grammar," to be misleading.

6.3 Summary

This chapter has sketched two theories of movement. Both let movement change the meaning of the parse they apply to. In both, the moved term can become a binder of a variable left in the position the term moved from. In Trace theory, this is achieved by letting the position a term moves from be occupied by a silent variable bound by the term from its new position. In the Copy theory of movement, this is achieved by letting a term be turned into a definite description with the meaning of a variable when a copy of it is put into a position from which it binds that definite description. The Copy theory of movement needs to be wedded to a theory that determines which parts of two copies can, and can't, be pronounced and semantically interpreted. The Trace theory of movement needs to be wedded to a theory that explains why a trace is silent and why it arises only in movement contexts.

The Trace theory of movement is demonstrably inadequate. It has not been possible to define derivations correctly under Trace theory. In particular, it's not possible to generate parses in a derivation that correspond to the meanings needed. At least this is not possible unless derivations are allowed to violate Upwards, and without Upwards it's not presently known how to prevent derivations from allowing too many possibilities.

The Copy theory of movement, on the other hand, does not straightforwardly account for antecedent contained anaphora. At the time of writing, the remedy for this has been to interpret movement as an operation that gives a term two positions from which parts of that term can be semantically and phonologically interpreted. On this view, the parses in a derivation created by movement don't uniquely determine one denotation and one string each, but instead proffer a range of possible denotations and strings. While this successfully captures the cases that are problematic for Trace theory, it allows too many possibilities. Without a model of which parts of the two copies can be semantically and phonologically interpreted, there are too many ways for even one instance of movement to provide a string-meaning correspondence. The Copy theory of movement relies on the success of a program that has yet to provide such a model.

One piece of striking evidence that the Copy theory of movement is on the right track comes from examples in which a moved term can be pronounced

in both of its positions. This arises in many languages in cases where a verb, or a verb phrase, are topicalized or clefted. Many languages have a rule that moves a constituent into a clause initial position when that constituent is intended to be contrasted, or otherwise highlighted. In some languages it appears that a verb can undergo this process, leaving its projection behind. In many languages that have this process the moved verb must be pronounced in both of its positions. Some examples are in (325).

Nunes (1995, 2004) pioneers an approach to this phenomenon which involves combining the Copy Theory of Movement with a theory that expresses how syntactic representations are mapped onto strings. See Landau (2006), Trinh (2011), and Hein (2018) for more.

(325) a. ká dɛ̄-kā àlī Àtìm dɛ̀ mángò-ku dɪ̄ɛ̄m
FOC eat-nmlz C Atim ate mango-def yesterday
"It's eating that Atim ate the mango yesterday."
(*Buli,* Hiraiwa, 2005, p. 262)

b. masi-ki-nun Chelsu-ka mayckwu-lul masi-ess-ta
drink-nmlz-top Chelsu-nom beer-acc drink-pst-dec
"As for drinking, Chelsu drank beer."
(*Korean,* Jo, 2000, p. 90)

c. doc thi no nen doc sach.
read top he should read book
"As for reading this book, I read."
(*Vietnamese,* Trinh, 2011, p. 60)

The verbs in these constructions can have different inflections, but they cannot otherwise be different. They look, in other words, like they are copies of the same verb, but with the morphology that is appropriate for the position they are spoken in. The conditions we normally see of movement—like Upwards—also govern the relationship between the higher and lower verb. This is precisely what would be expected on the Copy theory of movement, if the prohibition against speaking both copies was lifted. By contrast, it's not at all what would be expected on Trace theory.

7
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